Machine-Native Reasoning

Representation, Learning, and Computational Organization

VOLUME I
Core Machine Theory

A connected compilation of the machine-related and theoretical computer science program.

10 source documents · 470 original source pages
27-page connected synthesis

The interfaces matter as much as the components.

Read the organizing argument, follow the connections, and return to the preserved proofs. Source editions retain their own assumptions, evidence and limits.

Collection edition 1.0.0 / 9 October 2026
Komorebi P.

Core Machine Theory

GA / Grounded Abstraction 33
1.0.0 · 43 source pages

MU / Machine Understanding 77
1.0.0 · 50 source pages

LSM / Large Signal Model Foundations 128
Foundations 0.4.0 / software 2.3.0 · 66 source pages

OSL / Operator-Symbolic Language 195
Research edition 1.0 / Core 0.5 · 18 source pages

NEU / Neuralese 214
Research edition 1.0 · 20 source pages

CBI / Certified Bayesian Inference 235
Core 1.0.0 / standardized publication · 36 source pages

AVOC / AVOC: Artifact Value and Guarded Planning 272
Formal 0.41.0 / standardized publication · 37 source pages

PPT / Proof Path Theory 310
1.0.0 · 34 source pages

SII / Self-Improving Intelligence 345
1.0.0 · 60 source pages

MS / Machine Societies 406
1.0.0 · 106 source pages

Numbers at right are physical pages in this reader. Click an entry to open its source card. Original source pagination remains visible.

What this collection establishes

Connected, not conflated

The new synthesis maps relationships among the source works. Exact mathematical interfaces, conditional compositions, implementation gaps and proposed architectures remain different kinds of connection.

Preserved full texts

Every selected PDF appears unabridged. Its pages, theorem numbering, evidence classifications and historical sections are retained. An older experimental section is not new evidence for the complete architecture.

Theory is the principal product

No new training campaign, broad agent implementation, hardware experiment or corpus-wide Lean run is claimed. Small exact checks cover the newly written connecting example only.

A selected corpus, not every archive

The collection includes the listed twenty-four PDFs and six earlier complete source archives. Typed Proof Learner and Resolution Completion are represented by a recovered handoff note. Other unresolved source branches are listed in the package rather than silently claimed included.

Independent identities and permissions

Source versions and rights are retained independently. This compilation does not relicense the source works or establish a public repository release. A manuscript proof, a runtime test, an archived kernel record and an external empirical result are not interchangeable.

Machine-Native Reasoning

Representation, Learning, and Computational Organization

A connected compilation of the machine-theory program

Komorebi P.
9 October 2026
Collection edition 1.0.0
Connected synthesis and source reader

What a representation preserves. What an operation means. What evidence permits a correction. What a revision costs. What a community can maintain together.

This editorial volume connects twenty-four selected source documents. The source reader preserves their pages and their distinct mathematical, implementation, and research boundaries. A collection of compatible ideas is not, by itself, an implemented general intelligence.

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How to use this compilation

This collection has two complementary purposes. The first is to make the machine-related work intelligible as one research program. The second is to retain the actual mathematical sources, rather than replace a substantial body of work with a short architectural summary.

The connected synthesis is a new editorial argument. It introduces a shared vocabulary, a source-to-source interface register, and a small exact example that follows evidence through representation, action, correction, communication, and revision. The twenty-four source documents then supply the detailed proofs, developments, historical boundaries, formal companions, and visual material. Source editions are not renumbered as though they were newly proved chapters of one formal system.

There are three reading depths. Begin with this synthesis for the organizing questions. Read the core machine-theory volume for the recent books together with the signal, language, inference, and mathematical-memory work. Consult the foundations and companion volumes for the extended representation calculus, relational workspace monograph, architecture, formal guides, visual plates, and historical recursive-improvement lineage.

References such as [GA] identify a source edition in the catalogue at the end of this synthesis. Its section titles are the preferred locations when source pagination differs from the global reader. The complete compilation adds a source-entry page, an exact page crosswalk, and PDF bookmarks. The original pagination remains visible. A source-page label such as `GA-8` means physical page eight of that source, including its cover and preliminary pages.

The compilation is theory-first. Its central conclusions depend on explicit mathematical hypotheses. Small exact calculations check the new connecting example and the assembly. No new neural training, broad agent benchmark, hardware run, or corpus-wide Lean rebuild was performed. Some unabridged source works contain retained experimental history. Those pages remain identifiable historical material, not evidence that the entire proposed architecture has been realized.

The collection includes the latest selected Large Signal Model foundations edition recovered for this assembly, 0.4.0, alongside its unchanged software-version boundary. The four recent books retain their first editions. Older long-form theories remain independent sources because their scope is not exhausted by the newer books. [LSM][GA][MU][SII][MS][SRWT][SRIC]

What this edition does not claim. It is not an exhaustive census of every machine-related archive in Operation Komorebi. The selected full texts, the six included source archives, and the branches represented only by a source note are listed explicitly. Compilation does not establish historical priority, transfer a proof-assistant guarantee to an unverified implementation, resolve an open learning bridge, or authorize external action.

PART I

One program, several distinct objects

1 The unifying question

A useful machine representation is more than a place to store a correct answer. It enables some future operations and excludes others. It may preserve a probability update while losing the evidence needed to challenge that update. It may predict an output without reconstructing the underlying state. It may express a valid operation while leaving its actual acquisition prohibitively expensive. It may be shared among several agents without creating one collective decision maker.

The central question of this compilation is therefore not whether all these systems are secretly the same object. It is whether their relationships can be made explicit enough that a claim survives the passage from one representation, process, or institution to another.

The strongest common thread is a discipline of *preservation under declared change*. What must remain true when coordinates change? When an observation is compressed? When a learner installs a new operator? When a solver becomes a recipient of another system's explanation? When an institution changes its members or its rules? The answer differs across these cases, but the need to specify it is common.

This perspective organizes the source works without erasing their differences. Grounded Abstraction studies what evidence justifies. Machine Understanding studies what a usable representation enables. Large Signal Model Foundations studies coherent signal realizations, predictive closure, and admissible probability domains. OSL concerns explicit executable semantics. Neuralese concerns latent operations and continuation. AVOC concerns the value and admissibility of computational artifacts. Proof Path Theory concerns the information surrounding a proof. Self-Improving Intelligence concerns changes to discovery procedures. SRWT and Machine Societies concern relational and institutional organization. [GA][MU][LSM][OSL][NEU][AVOC][PPT][SII][SRWT][MS]

A compilation becomes misleading when it treats those descriptions as modules already connected by an implemented interface. An equation describing a desirable semantic relation is not the trained translator that realizes it. A defined resource ledger is not evidence that every implementation cost has been metered. A theory of institutions is not evidence that institutions have formed spontaneously.

The positive aim is more precise: build a map of the mathematical objects that can be composed, the assumptions that composition consumes, the obstructions that prevent it, and the information that must remain available for future correction.

2 Six kinds of success

The collection repeatedly separates six judgments. They should remain distinct even when one application needs all of them.

Judgment	What it establishes
Representational	A retained state contains the distinctions needed for the specified future uses.
Epistemic	The accessible evidence supports the claimed state, model, or uncertainty set.
Semantic	An expression or translated operation has the declared meaning on its stated domain.
Operational	The actual chosen procedure executes within its access, authority, and resource constraints.
Developmental	A retained modification improves the solver or the procedure that creates solvers under a stable comparison.
Institutional	Several decision centers maintain operative rules and obligations under an explicit authority structure.

A theorem can connect two judgments under explicit premises. It cannot make them interchangeable. For example, a continuation-preserving quotient connects a representational description to future semantic behavior. It does not establish that a learner can infer the quotient from its observations. An artifact checker can support operational admission relative to a model. It does not prove that an outside process is faithfully described by that model. A shared workspace can preserve reconstructible state. It does not decide who should have the right to modify it.

These separations are not a retreat from integration. They identify exactly what integration must accomplish. The central compilation claim is that the program has developed a substantial family of compatible mathematical obligations, together with several exact finite constructions. The remaining research is located at particular interfaces, not hidden inside a single phrase such as “machine-native intelligence.”

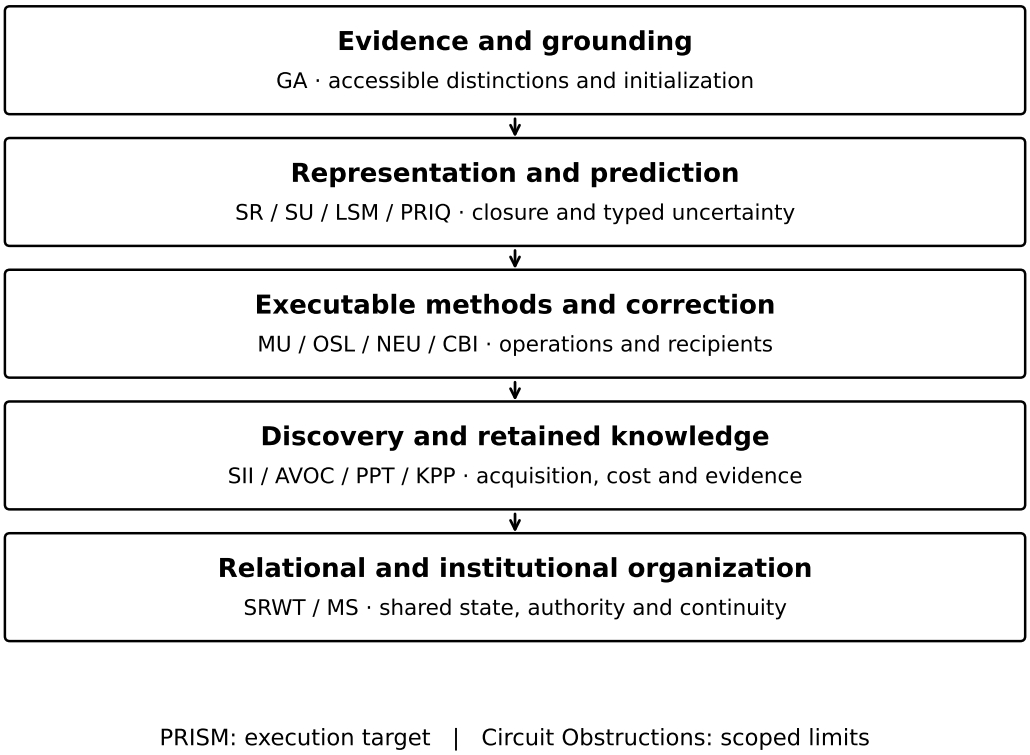


Figure 1: The organizing structure. Arrows identify interfaces to be justified, not completed software integrations. Evidence and resource conditions remain relevant at every layer.

3 A common contract, not a universal architecture

For editorial purposes, write a task contract as

$$\mathfrak{C} = (\Omega, \Pi, \mathcal{Q}, \mathcal{A}, \mathcal{E}, \mathcal{B}, \mathcal{V}).$$

Here Ω is the declared family of possible worlds or models, Π the permitted observation protocols, $\mathcal{Q}$ the protected questions, $\mathcal{A}$ the admitted actions or operations, $\mathcal{E}$ the evidence language, $\mathcal{B}$ the resource envelope, and $\mathcal{V}$ the validation and authority rules. This tuple is an editorial interface schema. It is not proposed as the unique primitive definition used by every source.

A realization of the contract may carry physical or computational state x_t, an observation history h_t, a retained predictive state z_t, a candidate set or probability record, an operator library, and applicable certificates. An institution adds offices, permissions, claims, resource rights, and revision procedures. The representation of one layer is not automatically the complete state of the others.

Three features of the schema matter. First, a target is stated before sufficiency is claimed. Second, protocols determine what information the system can actually obtain. Third, validation belongs to an explicit rule and authority boundary. None of these is supplied by an evocative name for a latent space.

A shared notation also exposes incompatibilities. One paper may use a finite deterministic world and another a finite probability model. One representation may be a set quotient while another is a linear map. One cost model counts complete tree records while another counts interpreter visits. Composition

requires a declared translation of these meanings. Writing the same symbol in both papers is not such a translation.

3.1 A packet is not automatically a belief

In the algebraic sources, a packet can be a component selected by a projector or an address in a resolved family. Such a component may contain signed or complex values. A probability distribution has additional positivity and normalization requirements. An evidentiary packet may instead retain claims, dependencies, and unresolved conditions. These are related uses of decomposition, not identical mathematical types. Large Signal Model Foundations makes the probability-domain requirement explicit, and Stratified Universality retains different carriers unless a certified comparison is supplied. [LSM][SU]

3.2 A certificate is not automatically a fact about the world

A mathematical proof, a checked finite witness, a successful interpreter replay, an integrity digest, and an observed performance comparison answer different questions. A digest can bind bytes to a record without establishing its premise. A formal proof can derive a property of an abstract program without establishing that a host execution refines that program. A risk bound can be valid under a likelihood model without proving that the model is adequate.

The compilation consequently uses source-specific assurance. AVOC and Certified Bayesian Inference retain their archived formal evidence. The newer books retain their prose proofs and bounded reference checks. No common label upgrades the weaker evidence to the stronger one. [AVOC][AVL][CBI][CBL][GA]

PART II

From experience to a usable method

4 Grounding begins with accessible distinctions

Grounded Abstraction supplies the first link. A family of candidate systems can leave the full underlying realization unidentified while fixing every output that matters to the declared task. Conversely, each candidate system may individually admit a useful representation while the observer lacks enough information to initialize a common one. The source separates identification from a family of laws, certainty from one finite record, and the acquisition of executable predictive state. [GA]

Suppose a deterministic state space X has admitted transformations F_a and protected output q. Define

$$x \sim y \quad \Longleftrightarrow \quad q(F_w x) = q(F_w y) \quad \text{for every admitted finite word } w.$$

The empty word is included. When the operation domain is invariant, this equivalence supports well-defined quotient updates. It identifies which distinctions are needed for all specified continuations, not which distinctions are needed to reproduce every hidden detail. This principle occurs in Sufficient Resolution, Machine Understanding, and Stratified Universality under their respective hypotheses. [SR][MU][SU]

The missing link is initialization. Let $r(x)$ be the observer's actual initial record. A quotient may exist mathematically while a decoder from $r(x)$ does not.

Connecting result 1 (Evidence must initialize the predictive class)**.** *Let $p : X \to X/\!\sim$ be the behavioral quotient above. There exists an initializer $i : r(X) \to X/\!\sim$ with $i(r(x)) = p(x)$ for every x if and only if*

$$r(x) = r(y) \quad \Longrightarrow \quad x \sim y.$$

When this holds, the initialized quotient predicts every declared finite continuation using its induced updates.

Proof. Necessity follows because a function has one value on an equal input. For sufficiency, assign $i(r(x)) = p(x)$; the hypothesis makes the assignment independent of the representative x. The quotient update $\overline{F}_a(p(x)) = p(F_a x)$ is well-defined because behavioral equivalence is preserved by prefixing an action. Induction on word length gives $pF_w = \overline{F}_w p$, and the output factors through p. Thus the initialized quotient returns the protected output after every word. □

This is an elementary restatement of the factorization and continuation arguments shared by the sources, not a new priority claim. It gives a precise connection between grounding and understanding: the desired predictive state must be reachable from evidence available to its user.

A noisy observation needs a different conclusion. Distinct candidate laws can assign positive probability to every finite record. In that case no finite record is an error-free identifier, although a probability update or confidence statement may be justified. Substituting a highly probable candidate for certain knowledge changes the contract. Grounded Abstraction retains that distinction and quantifies the resulting decision error in its final part. [GA]

5 Archives, predictive states, and operator closure

The signal program distinguishes an observation, an exact archive of that observation, and a state sufficient for prediction. A left-invertible analysis map preserves the accepted observation. It does not recover unmeasured properties of the external process. A smaller predictive state can discard part of the observation while preserving all relevant future outputs. An exact archive can still fail as an efficient predictive state if every use requires expensive reconstruction. [LSM]

For a linear map $C : X \to Z$ and a known operation $A : X \to X$, an induced operation B satisfying $CA = BC$ exists precisely when $\ker C$ is invariant under A, with the codomain restricted to the attained image if necessary. The condition says that two inputs collapsed by C remain collapsed after the operation. If it fails, the discarded distinction can return through the retained state.

This failure has a concrete block form. In coordinates separating retained and hidden state,

$$\begin{pmatrix} z_{t+1} \\ u_{t+1} \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} z_t \\ u_t \end{pmatrix}.$$

Even if u_t has small norm, the effect Bu_t may be large. Retaining less state is therefore not justified by omitted magnitude alone. A predictive error bound needs a sensitivity or propagation bound. Sufficient Resolution studies hidden-return defects and exact memory alternatives. Large Signal Model Foundations places those mechanisms inside its refinement and predictive-closure account. [SR][LSM]

5.1 One common quotient across uncertain dynamics

Consider two possible linear realizations on $\mathbb{R}^3$,

$$A_0 = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}, \qquad A_1 = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 3 \end{pmatrix}.$$

Let $p(x_1, x_2, x_3) = (x_1, x_2)$ and let the output be $q(x) = x_1$. Both operators satisfy

$$pA_i = Gp, \qquad G = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Consequently the output after k operations is $x_1 + kx_2$ for either fixed model. In this particular example it also holds for arbitrary switching, since the same local intertwining equation holds for both operations. The uncertain third-coordinate dynamics remain irrelevant to the protected output.

That switching conclusion must not be generalized to mere output agreement between fixed models. Grounded Abstraction separately exhibits the distinction: agreement of each fixed model's trajectories need not survive mixing their operators. A common quotient is the stronger premise used in the displayed example. [GA]

6 From latent computation to an explicit language

Neuralese and OSL are complementary, not competing definitions of machine language. Neuralese studies internal operations on latent state and the stability of their future behavior. OSL supplies an explicit intermediate language in which an operation's type, provenance, effect, and admissibility can

be represented and checked. A producer need not use symbolic coordinates internally in order to communicate through such an interface. [NEU][OSL]

For a common invariant domain, let $r : X \to Z$ encode the relevant state, let G_a act on Z, and let $T : Z \to W$ translate that state into another representation with operations H_a. The desired equations are

$$rF_a = G_a r, \qquad TG_a = H_a T.$$

The first connects the source process to its representation. The second connects that representation to the recipient interface.

Connecting result 2 (Two local bridges compose over finite continuations)**.** *Assume the domains used by both equations are preserved under all admitted operations and that a protected output has the form $q = \bar{q}\,Tr$. Then, for every admitted finite word w,*

$$TrF_w = H_w Tr, \qquad qF_w = \bar{q}\,H_w Tr.$$

Proof. For one operation, $TrF_a = TG_a r = H_a Tr$. The empty word gives the identity. Applying the one-operation relation successively proves the first equation by induction, and composing with $\bar{q}$ proves the second. Domain invariance ensures that every successive application is licensed. □

The result is a composition of already familiar intertwining arguments. It states exactly what would be preserved by a correct learned semantic bridge. It does not construct r or T, authenticate causal use of the selected carrier, or bound the translator's complexity. Those remain separate obligations. A powerful interpreter that solves the problem after receiving a trivial label has not supplied a free representation of the source mechanism. Machine Understanding makes recipient programs and their installed information explicit for this reason. [MU]

6.1 Why the bridge remains open

The OSL source describes a checked finite interface based on declared tables. The Neuralese source distinguishes such exact compiled controls from learned latent implementations. Their coexistence supports a precise research question, not a claim that a general Neuralese-to-OSL compiler has been completed. The missing relation is a learned, causally adequate, continuation-preserving translator under explicit access and resource limits. [OSL][NEU]

PART III

Correction, decisions, and knowledge

7 Prediction and correction must share a representation

A state can be sufficient for its current prediction and insufficient for the interpretation of later evidence. This is one of the most important connections between the books. Machine Understanding asks when translation commutes with filtering candidate models. Grounded Abstraction identifies the information that future corrections require. PRIQ asks when fine prediction and Bayesian updating descend to a coarse interaction model. The signal foundations add the probability domain on which a learned operator can be interpreted as an update. [MU][GA][PRIQ][LSM]

For exact candidate elimination, a current set $V \subseteq \Omega$ is corrected by $V \mapsto V \cap E$. If the target observation is $t(V)$, a future-correction signature records

$$\Sigma(V) = \big(t(V \cap E)\big)_{E \in \mathcal{E}}.$$

Under the source's closure hypotheses, this captures the distinctions needed to answer after every permitted filter. The empty set matters: recognizing that all candidates have been contradicted is part of a correct update, not an exceptional case to omit from the theorem. [GA]

For noisy evidence, target uncertainty and sensor uncertainty are not generally independent. Two priors can have the same separate marginals while producing different target posteriors after the same report. A useful compressed probability memory must preserve the association between target values and likelihood profiles whenever future evidence depends on that association. Grounded Abstraction constructs such exact memory and separately bounds finite quantization. [GA]

7.1 A coarse prediction is not necessarily a coarse posterior

In a finite Bayesian model, a coarse partition can preserve likelihood-based correction for every prior when the observation likelihood is constant within each coarse cell. A quotient of the transition dynamics alone does not ensure that property. Likewise, a signed packet decomposition can reconstruct a belief without each packet separately being a probability distribution. The coarse dynamics, update law, and positive domain must be checked together. [PRIQ][LSM]

This produces a practical hierarchy of questions. Does the retained state preserve the desired output? Does it preserve the probabilities of later reports? Does it preserve the posterior update? Does the rounded implementation maintain the required accuracy through repeated updates? Does the result remain valid under changed likelihoods or a revised evidence language? These are related but different preservation contracts.

8 From a probability record to an accountable decision

Certified Bayesian Inference gives a particularly concrete bridge between a returned computational object and an inference statement. Its protected scope is one finite-dimensional Gaussian observation update with stated rational input factors and arithmetic conditions. Packet residuals, finite stopping, outward-rounded verification, and parameter enclosures concern the same returned approximation. A general sequential filter is a separate extension. [CBI][CBL]

AVOC addresses a different layer: whether inspecting, constructing, checking, or deploying an artifact is justified under an explicit value and resource model. A correct artifact can be a poor investment. A high estimated probability of passing a checker is not a passed check. An honest operation log can still require a separate sampling argument before it supports calibration. [AVOC][AVL]

The two branches should therefore be connected through a concrete interface: a named posterior or uncertainty object, a loss function, a permissible action set, a bound on numerical or representation error, and costs for the actual chosen course of action. An inferred probability does not itself grant authority to perform that action.

Connecting result 3 (A common decision bound)**.** *Let μ and $\widehat{\mu}$ be probability measures on the same finite candidate set with $d_{\mathrm{TV}}(\mu, \widehat{\mu}) \le e$. Suppose true and approximate responses satisfy $d(y_\omega, \widehat{y}_\omega) \le d_0$ for all retained candidates. If $\ell(a, y) \in [0, 1]$ is K-Lipschitz in its second argument, then for every admissible action a,*

$$|R(a) - \widehat{R}(a)| \le \eta := \min\{1, e + Kd_0\},$$

where $R(a) = \sum_\omega \mu_\omega \ell(a, y_\omega)$ and $\widehat{R}(a) = \sum_\omega \widehat{\mu}_\omega \ell(a, \widehat{y}_\omega)$. An action minimizing $\widehat{R}$ has regret at most 2η relative to a minimizer of R when the action set is finite and nonempty.

Proof. Insert the intermediate expectation $\sum \widehat{\mu}_\omega \ell(a, y_\omega)$. The first difference is at most e because the integrand is in $[0, 1]$. The second is at most Kd_0. Risks themselves lie in $[0, 1]$, giving the cap. If $\widehat{a}$ and a^* minimize the approximate and exact risks, respectively, then $R(\widehat{a}) \le \widehat{R}(\widehat{a}) + \eta \le \widehat{R}(a^*) + \eta \le R(a^*) + 2\eta$. □

This consolidates the elementary bound used in the quantitative grounding and decision work. It is a statement about one common posterior, response, history, and loss. It does not combine independently attractive bounds whose quantities refer to different candidate worlds. Nor does it imply a zero-risk guarantee from a small positive error. [GA][MU][AVOC]

9 Mathematical memory is more than the final proof

Proof Path Theory distinguishes a certificate of implication from the larger field of inquiry surrounding it. A valid proof can omit a useful failed route, a counterexample to a stronger claim, an alternative representation, or a dependency that explains how to adapt the argument. Retaining every exploration trace is not automatically optimal either. The source makes retention relative to future task families and recovery costs. [PPT]

This connects to corrective memory without identifying the two objects. A candidate-set signature preserves future filters under one finite evidence language. A proof field preserves specified routes of deduction, audit, and discovery. Both ask what compression makes unavailable, but they use different types of future operation. A shared storage implementation would need an explicit mapping between those operations.

The Preservation Protocol adds a temporal distinction. Earlier artifacts and recorded events remain recoverable, while present beliefs and authoritative pointers can change. Refuting a claim should not delete the fact that it was once asserted or the reason it was revised. That is a protocol for research continuity, not a proof that every surviving record is correct. [KPP]

The portable-lineage protocol extends the question to successor systems: does a bounded transferred state support correction retention, selective reuse, and resumed work? Its source is a protocol specification, not a completed demonstration of every proposed continuity level. It does not decide subjective identity. [PCL]

PART IV

Revision, shared work, and execution

10 Improving a method and improving its discovery

Self-Improving Intelligence separates a solver, a procedure that constructs repairs, and the representation used to acquire rules inside that procedure. Its finite development includes diagnostic trees, certified compiler shortcuts, a grammar-synthesized pruning rule, a proof-checked equation library, and ordered quotienting that preserves the complete teacher transcript. Each acquired object is distinct from the supplied language and checker used to obtain it. [SII]

This makes a recursive step mathematically discussable without requiring that a system invent its own foundations. A descendant can be better at acquiring future rules because an acquired representation eliminates redundant search. Whether this is an end-to-end improvement depends on the cost of preparing and maintaining that representation and the workload over which it is reused.

SRIC is the larger historical theory behind this distinction. It contains repair transport, branching lineages, evaluator contexts, changing languages, and several conditional accounts of continued improvement. The 505-page source retains mathematical theory and experimental development together. It is preserved in the companion volume rather than silently replaced by the smaller first-edition book. Its historical results are not new evidence for a combined architecture. [SRIC]

10.1 A correction should carry its reason for remaining valid

A learned shortcut may depend on arithmetic semantics, a grammar order, a repair-success relation, an observation channel, or a resource comparison. A later change can leave the shortcut's printed expression unchanged while invalidating the justification for using it. The certificate-dependency construction in Self-Improving Intelligence marks the affected descendants and rechecks them under revised premises. [SII]

An unchanged theorem is not an unchanged application. If the repair language gains an action that succeeds in every case, an old lower bound based on identifying individual cases no longer describes the problem. The mathematical theorem remains true within its original scope. Its premise has stopped matching the revised system.

This is the connection to proof-path and preservation theory: a system should retain both the acquired method and the dependencies that determine its applicability. It is also the connection to OSL: admission should be tied to the actual operation and its current scope, not merely to an earlier approval string.

11 A workspace is not yet a society

SRWT studies compatible organization distributed across heterogeneous carriers. Its structural layer uses transfer maps, global compatible sections, and holonomy-fixed cores. Its dynamic and causal layers separately ask whether the proposed shared state predicts the relevant future, can be intervened on appropriately, remains reconstructible, and contributes beyond matched local alternatives. [SRWT]

These distinctions prevent two shortcuts. A synchronized collection of replicas is not necessarily irreducibly distributed cognition. Conversely, a state split into complementary shares by an external

constructor does not by itself demonstrate endogenous organization. How the components are defined and what alternatives they are compared against belong to the claim.

Machine Societies adds rules that alter the space of permissible action: offices, membership, obligations, dispute procedures, resource allocation, and revision authority. A society can have several decision centers and no single global utility function. A coherent collective policy and persistent institutional plurality are separate properties. [MS]

The bridge from workspace to institution therefore requires more than a shared vector. A representation of a promise must connect to future obligations and remedies. A representation of a vote must connect to an electorate and a procedure. A shared archive must distinguish who can observe an event from who can authorize its consequences.

11.1 Internal dynamics and external authority

The Python Agent Dynamics atlas illustrates private candidate sets, messages, prediction discrepancies, and externally consequential actions in a finite model. It is a useful visual language for this interface, but its supplied update rules are not measured neural mechanisms. [AD]

In the theoretical compilation, an agent's internal confidence remains separate from the permission to act. An institutional acceptance rule does not establish another party's consent. A society can be internally consistent while omitting a legitimate external claim. The accountable-ending and external-effects material in Machine Societies prevents persistence from becoming its own unrestricted justification. [MS]

12 Representation gains and execution costs

PRISM studies a representation-aware execution architecture. Packet graphs, structured factors, sparse chains, and transform plans can remain explicit computational objects rather than being expanded into dense arrays by default. Packet IR makes representation, precision, schedule, and output contracts part of the compiler's decisions. The source preserves the distinction between software evidence and a proposed physical architecture. [PRISM]

This is downstream of the learning question. A system must obtain useful structure, verify its scope, and pay the costs of encoding and selecting it. A faster execution path for a supplied low-rank factorization is not an algorithm that discovers the factorization or proves that it is useful for an unfamiliar task.

Circuit Obstructions contributes a complementary limitation. Some proposed exact summaries must retain a large amount of structural information under their declared preservation rules. Some projections can destroy support properties or increase the relevant nonlinear cost. These are scoped complexity statements, not an unrestricted complexity-class separation. [CIR]

The combined lesson is neither that every representation change speeds computation nor that compression is futile. It is that the complete path must be accounted for:

$$\text{acquire} \longrightarrow \text{represent} \longrightarrow \text{select} \longrightarrow \text{verify} \longrightarrow \text{execute} \longrightarrow \text{maintain}.$$

The arrows can have different units. Model observations, interpreter visits, physical time, peak memory, and stored records should not be summed without an explicit conversion model.

Connecting result 4 (Resource gains belong to a declared ledger)**.** *For one fixed workflow and workload, let $\Delta \in \mathbb{R}^d$ be old-minus-new event counts after all charged setup and maintenance. Then $w^\top \Delta \geq 0$ for every*

$w \in \mathbb{R}^d_{\geq 0}$ if and only if every coordinate of Δ is nonnegative. For prices restricted to the convex hull of a finite set, the smallest priced saving is attained at one of its listed points.

Proof. Nonnegative coordinates give a nonnegative scalar product. A negative coordinate is exposed by assigning positive price only to that coordinate. A linear functional at a convex combination is the same combination of its values, which cannot be below the smallest listed value. A listed minimizer attains that value. □

This restates the accounting principle in Self-Improving Intelligence. For its recorded weighted-search comparison, the vector $(1552, 1552, -1868)$ counts saved model reads, saved additions, and added comparisons. Equal prices give a positive margin; other prices give a loss. Semantic validity does not change with that choice. Neither does a correct price calculation imply that the price is a measured property of a particular processor. [SII]

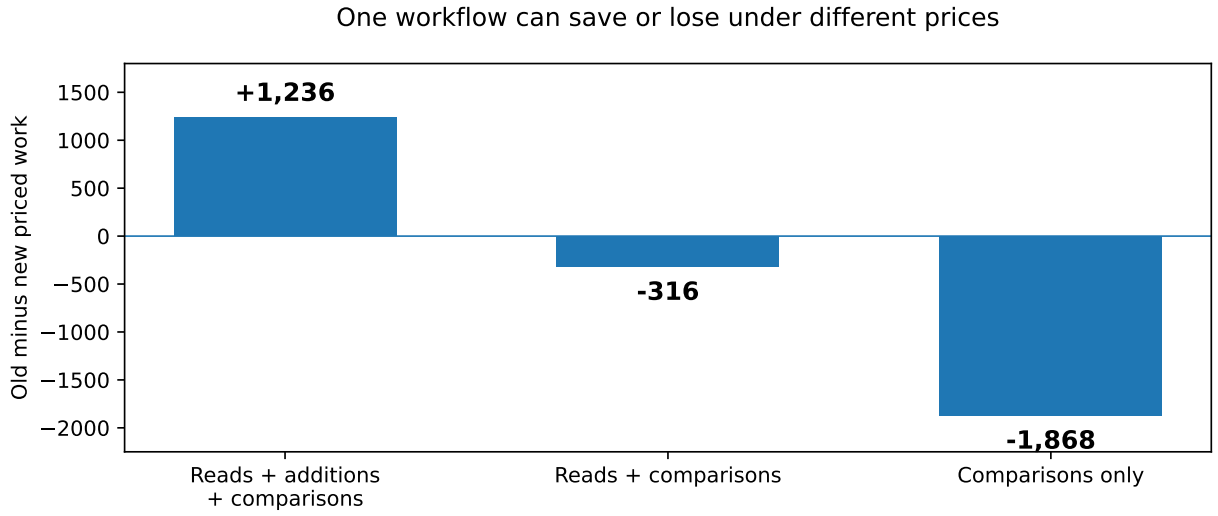


Figure 2: One source-reported event vector under three declared price assignments. Positive values are savings and negative values are losses. These are arithmetic consequences of the stated model, not runtime measurements.

PART V

One exact example through the whole chain

13 The sensor, the method, and the community

The following construction is new editorial glue. It is not a claim that the implementations in all twenty-four sources share a common runtime. Its purpose is to make the interface obligations jointly inspectable in one small model.

Let the possible world be

$$\omega = (b, \theta, u) \in \mathbb{F}_2^3.$$

The bit b determines which of two repairs is appropriate. The bit θ is a sensor convention. The nuisance bit u affects none of the admitted observations, transitions, or repairs. A normal reading returns $s = b \oplus \theta$. A calibration returns θ. A public operation flips b, leaving θ and u unchanged. A repair R_i succeeds precisely when $b = i$.

All of these semantics are stipulated. The small check program verifies the resulting finite claims. It does not learn the observation law, infer the truth of an external sensor model, or discover the repair meanings.

13.1 First: a two-state predictor is enough for the display

For the original question family, the task is only to predict the next display after a finite number of flips. The retained state $z = s$ has two possible values, and its update is $G(z) = 1 - z$. Every future display is determined even though b and θ are not separately known. The nuisance bit is irrelevant throughout.

This is a direct illustration of the signal and sufficiency layers: lossless storage of the complete world is unnecessary for this prediction contract. It also illustrates grounding: the predictor can be initialized from the actual normal reading rather than from an inaccessible hidden bit.

13.2 Second: the repair question changes the required distinction

When the task becomes selecting R_b, the old display is insufficient. Worlds $(0, 0, u)$ and $(1, 1, u)$ have the same display and different successful repairs. Repeating the normal reading changes no information. Calibration supplies the missing convention, after which

$$b = s \oplus \theta.$$

The refinement is a genuine information acquisition. Merely adding a field named θ to the retained record would not initialize it.

If calibration is unavailable, no protocol built solely from the admitted flip and normal-reading operations can distinguish those two worlds. Their display histories remain the same after matched flips. Certain repair is therefore impossible within that unchanged interface. This is an interface-relative conclusion, not a claim that no richer experiment could help.

13.3 Third: a translation can preserve the method

One recipient stores (s, θ) and another stores (b, θ). The translator

$$T(s, \theta) = (s \oplus \theta, \theta)$$

is invertible. Both coordinate systems have a well-defined flip operation, and T commutes with those operations. Both decode the same required repair. The exact correspondence gives an elementary realization of the method-transport contract.

The translator still has an access requirement and an execution cost. It cannot translate a record containing only s into the missing pair. It does not infer a sensor convention from a matching variable name. This is the distinction between a valid map on its domain and a procedure that can obtain the map's input.

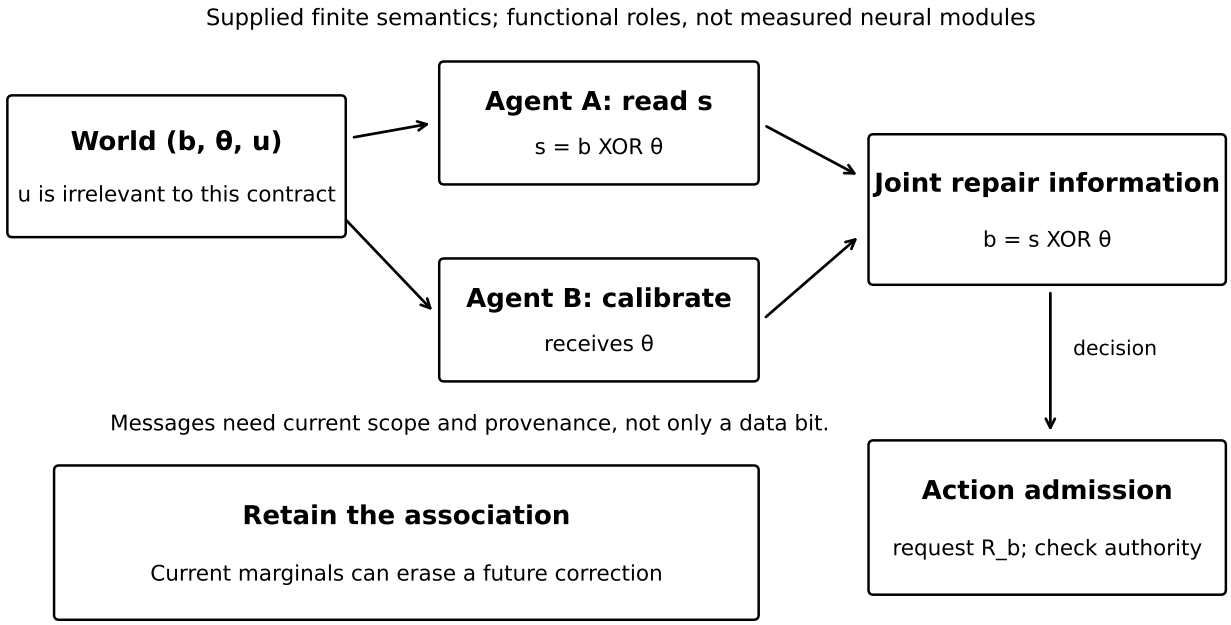


Figure 3: The same finite example through observation, calibration, coordinate transport, and checked action. The communication and authority conditions are additional interfaces, not consequences of the XOR identity.

14 Memory, messages, and a changed task

For candidate sets in the eight-world model, evidence depends only on (b, θ). There are four nonempty cells, each differing only in u. Restrict the evidence language to exactly the unions of those four cells. The occupancy record stores whether the current candidate set meets each cell. It therefore has $2^4 = 16$ possible states rather than the $2^8 = 256$ possible complete candidate sets.

That occupancy is sufficient to filter by every event in the declared evidence algebra and to return the possible target values b, including contradiction. It is also necessary across all candidate subsets for this contract. If two occupancies differ on one cell, filtering by that cell gives a nonempty target set in one

case and the empty set in the other. This is the partition-evidence special case developed in Grounded Abstraction. [GA]

Four occupancy bits should not be confused with an exact probability record. When the model uses probabilities rather than possibilities, one needs the masses of the four joint cells, and storing those masses to arbitrary precision is a different resource problem. The source's finite quantization construction addresses that additional problem.

14.1 Separate marginals can erase the next correction

After display zero, the possible pairs are $(0, 0)$ and $(1, 1)$. After display one, they are $(0, 1)$ and $(1, 0)$. In both cases the separate target and sensor possibility sets are $\{0, 1\}$. A memory retaining only those marginals identifies the two histories.

Now observe $\theta = 0$. In the first history $b = 0$, while in the second $b = 1$. The same future evidence must lead to different corrections. The lost relation was not restored by the arrival of that evidence. Retaining the joint occupancy, or the relevant original display with its scope, avoids the error.

14.2 Two agents and one action

Let agent A obtain the normal reading and agent B obtain the calibration. Neither has the complete repair information alone. B can send its one-bit calibration result together with the declared provenance required by the communication protocol. Under truthful, correctly associated delivery, A can determine b and request the correct repair.

This is shared information, not a theorem that the two agents form an endogenous workspace or a society. A workspace claim additionally needs the source's causal, persistence, and comparison conditions. An institutional claim needs an operative rule governing who may issue, verify, and execute the request. We can specify a rule that accepts a repair only when the current applicable records imply its success, but specifying it is not observing its spontaneous formation. [SRWT][MS]

The data bit is also not the complete message cost. Provenance, episode association, delivery, and checking are separate components. The toy calculation treats the payload as one bit and leaves any metadata or transport overhead explicitly outside that number.

14.3 Work and stored control

Give a normal reading cost one, calibration cost two, and the final repair cost one, in stipulated work units. Certain repair from the complete initial uncertainty set requires both information channels, so the common worst-case work is four units. There are three primitive calls. Four weighted work units are not four calls.

A literal binary decision tree with these two observations and four outcome leaves contains seven records, although a run visits only three records. This is not the minimum size of every possible program encoding. A reusable arithmetic implementation is a different storage representation and must retain its own code and interpreter costs. The example illustrates the literal-tree convention used in the source books, rather than confusing it with a general description-length theorem. [MU][SII]

15 Revision can change the proof obligation

Suppose a newly admitted repair R_* succeeds for either value of b. Then the task no longer requires identifying b before acting. The old statement that both information channels are necessary applied to the previous repair language. It does not apply after R_* is introduced.

A well-formed revision record preserves the sensor identities and the old theorem while invalidating its application to the new task. The display predictor remains valid. The coordinate translator remains valid. The old minimum-information claim does not. A new controller can execute R_* immediately under its declared cost and authority conditions. The old performance comparison must be recomputed if it referred to the former observation-and-repair workflow.

This single change connects all the levels: grounding is relative to future use; sufficiency is relative to the operation and target language; correction memory is relative to future evidence; procedure cost is relative to one actual algorithm; and certificate applicability is relative to current dependencies. [GA][MU][SII]

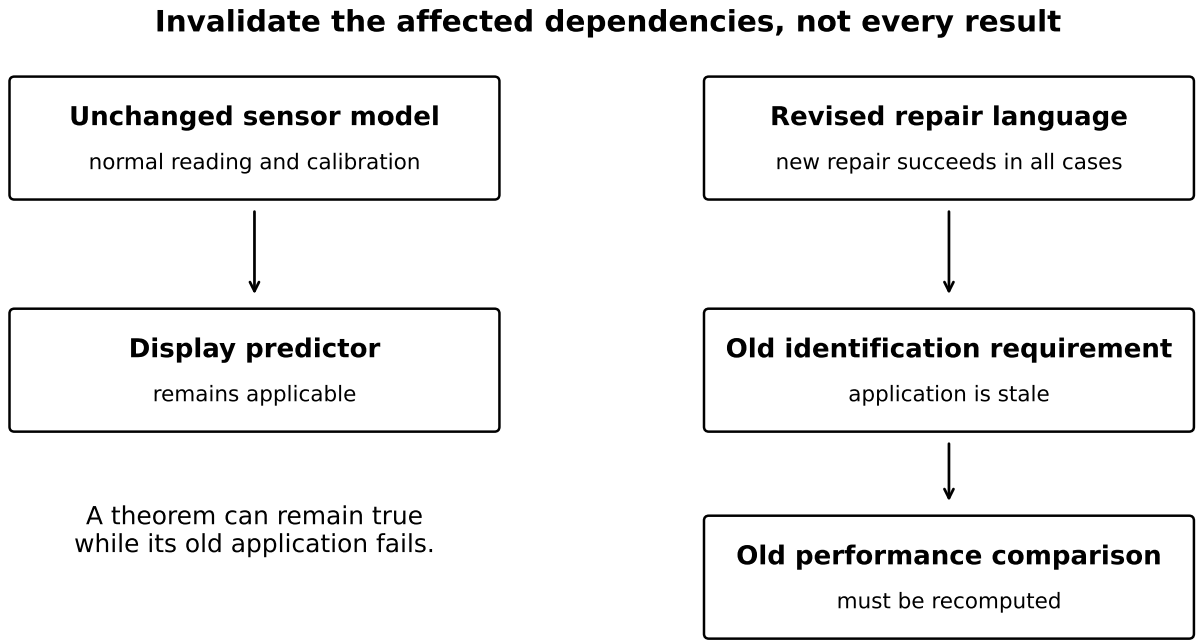


Figure 4: A repair-language change need not invalidate the sensor semantics or display predictor. It does invalidate uses of the old identification requirement and dependent performance conclusions. This is a schematic dependency example, not a claim to infer complete dependencies automatically.

15.1 What this example connects and what it does not

The source theories can be instantiated on one common finite object with explicit maps. That is stronger than drawing an arbitrary chain of module names. It establishes neither a learned encoder nor a general-purpose system. Its sensor semantics, allowed actions, agents, message interpretation, and authority rule are supplied.

PART VI

The connection register and the research frontier

16 Which bridges are already mathematical?

The register below distinguishes conditional mathematical connections from proposed integrations. "Available" means the source supplies a relevant theorem or construction under explicit premises. It does not mean every implementation in the two sources has been connected.

ID	Interface	Required connection and present boundary
B01	SR / SU to LSM	Available: typed decomposition and continuation-preserving transport. Packet completeness, kernels, profiles and admissible coefficient domains must match.
B02	GA to MU	Available: accessible evidence initializes the distinctions needed by future use and correction. A quotient's existence does not supply its initializer.
B03	LSM to GA	Available in specified models: observable prediction can avoid full-state identification. Candidate-model scope, probability validity and initialization remain explicit.
B04	PRIQ to LSM / GA	Available: prediction and Bayesian correction descend under stated compatibility. Positivity, likelihood fibers and evidence normalization cannot be dropped.
B05	Neuralese to OSL	Proposed learned bridge. A finite declared-table interface exists, but general causal, continuation-preserving learned translation is not established.
B06	OSL to MU	Available as a contract: the actual typed recipient must execute the transmitted method within its own bounds. A decoder designed after the fact is a different contract.
B07	GA / CBI to AVOC	Available conditionally: pass the same posterior object and its output error into a specified decision model. A one-observation Gaussian theorem does not imply a general sequential filter.
B08	MU to SII	Available: acquisition, explanation and repair require one workflow satisfying its complete resource profile. Reusable methods and discovery procedures are different targets.
B09	SII to OSL	Partial: both supply explicit admission obligations. A complete refinement proof linking all generated source forms to the runtime remains separate.
B10	PPT / KPP to SII	Available at the specification level: retain dependencies, evidence and revision history. Dependency completeness and external premise adequacy are not inferred from hashes.
B11	SRWT to MS	Additional hypotheses required: compatible shared state does not by itself create offices, authority, obligations or independent decision centers.
B12	MS to SII	Available as an authority contract: an allowed local revision must preserve applicable institutional constraints. Formal capability does not authorize external control.
B13	LSM to PRISM	Architecture path, not completed hardware: Packet IR must preserve the same typed output contract, numerical precision and costs.

ID	Interface	Required connection and present boundary
B14	CIR to discovery	Scoped limitation: exact summaries and projections face model-specific obstructions. No unrestricted computational separation follows.
B15	Proof learner to action	Open integration in the recovered handoff: checked proof construction and the world-action dispatcher remain distinct branches.
B16	Whole collection	No unrestricted composition theorem is claimed. One realization must satisfy all invoked interfaces simultaneously, with common evidence, costs and authority.

17 Overlap is a reason to cross-reference, not inflate counts

Several ideas recur because they do real work in different domains. The same elementary factorization criterion appears in task sufficiency, predictive initialization, and feature sufficiency for a synthesis decision. Induction extends local operation agreement to finite words. Bayesian normalization repeatedly introduces an evidence denominator. A dependency graph repeatedly exposes stale certificates. A resource vector repeatedly prevents gains in one unit from being promoted to a universal speedup.

Those recurrences should not be counted as independent discoveries solely because they occur in different books. They should also not be deleted before the scope differences are compared. A set-valued correction theorem is not automatically the same theorem as a stochastic posterior theorem. A fixed-model continuation result is not a switching-system result. A source-specific finite formalization is not a proof of every more general prose statement sharing its terminology.

The editorial consolidation therefore keeps one source edition per selected contribution and supplies overlap notes rather than removing proof bodies. The new connecting results in this synthesis are elementary restatements or direct compositions. Their role is to expose the interface assumptions, not to increase the portfolio's novelty count.

17.1 Four different meanings of "complete"

An editorially complete manuscript has a frozen presentation and stated scope. A theorem complete under hypotheses has a valid derivation of its conditional conclusion. A verified implementation has an assurance record for the actual source and environment. An integrated learning system would instantiate the required objects and their acquisition mechanisms together. None of these meanings can be substituted for another by assigning the whole collection version 1.0.0.

The present version freezes a collection. Individual source numbers and open problems are retained. Large Signal Model Foundations remains at 0.4.0, OSL's runtime remains Core 0.5, and SRIC remains its historical Block 23 edition. The collection does not create a new release of those components.

18 A theory-first agenda after the compilation

The highest-value continuation is not another survey asserting that representations and memory matter. It is a short sequence of bridge problems with explicit mathematical endpoints.

Evidence-limited semantic acquisition. Determine which operation and representation claims can be identified from a restricted observation language, under model misspecification and changing tasks.

Separate an unavailable distinction from an expensive one. The current finite and linear results supply starting cases, not a general learned encoder. [GA][LSM][NEU]

State-changing noisy correction. Join controlled stochastic realization with compressed probability memory and long-horizon correction. A static latent model, a dynamically changing hidden state, and an actively selected experiment do not have the same posterior recursion. The new bridge should specify common observation laws, probability domains, and conditioning margins before transporting finite-record bounds. [GA][PRIQ][LSM]

Preserving a method when its learner changes. Characterize what must be carried when encoders, operator libraries, recipient programs, and evidence languages are revised together. The target is not preserving every old answer. It is preserving justified operations and the means to correct an obsolete answer. [MU][SII][PPT]

A small semantic core with a complete formal boundary. Reuse the actual formal ingredients of AVOC and Certified Bayesian Inference where their domains fit. Identify a modest set of source-to-runtime preservation statements rather than promise a kernel proof for the entire archive. An abstract checker theorem and its host implementation remain distinct until their relation is proved. [AVOC][AVL][CBI][CBL][OSL]

Discovery cost and changing languages. Analyze classes where useful refinements or search quotients can be found economically, and classes where a compact solution exists but remains hard to acquire. Charge preparation, unsuccessful proposals and maintenance. A grammar extension can enlarge expressibility while reducing fixed-budget productivity. [SII][CIR]

Relational correction under plural authority. Study when several agents can preserve shared meanings, independent evidence, and legitimate challenge while their representations and membership change. This is not solved by demanding consensus. It requires a common interface that distinguishes knowledge, action rights, and institutional obligations. [SRWT][MS]

Each direction admits precise finite, algebraic, probabilistic, or analytic subproblems. Large-scale training is a possible later application, not a prerequisite for completing those theoretical results.

19 Collection scope and the remaining source queue

This edition includes twenty-four full source documents: ten core machine-theory works, seven mathematical and architectural foundations, and seven formal, visual, or lineage companions. Six existing editable source archives from the current development are included separately and unchanged. The compilation's own LaTeX, assembly code, figures, exact connecting-example checker, and catalogue are provided.

The standalone full source releases for many older library works were not collected in this pass. Their PDFs are included, but that is not a claim that their entire code, raw evidence, dependencies, or proof toolchains have been packaged. The source catalogue marks this explicitly.

Typed Proof Learner and Resolution Completion are represented by their recovered handoff note, not by a recovered merged implementation. Model-Theory Normalization, the broader Resolution Theory source families, RRCT/J-space experimental lineages, and task-specific ARC releases remain in the source-completion queue. Their absence from the unabridged reader is not a finding that they do not exist or lack substantive results. The general mathematical representation layer is represented here by SU, SR, and PRIQ. Historical empirical branches are not necessary premises for the new synthesis.

This is a substantial connected compilation, not an assertion that every machine-related file has been recovered. A future collection update should use a source-preservation map to add or replace a

component. It should not select the largest version number across unrelated lineages or silently overwrite the original source's evidence boundary.

20 Conclusion

Operation Komorebi's machine work is most coherently read as a theory of knowledge-bearing computation under change. A representation identifies some distinctions and discards others. An operation makes those choices consequential. Evidence determines which descriptions can be initialized and corrected. A recipient must actually be able to use a communicated method. A discovery procedure must earn its cost. A revision must preserve the conditions under which its retained claims apply. A collective must add explicit relations of access, authority, obligation and continuity.

The connecting example shows that these obligations can be placed on one finite realization without confusing their roles. The source works develop them much further in their respective domains. Their relationship is therefore more substantial than a list of specialist modules, but more conditional than a claim that a complete general intelligence has been assembled.

The compilation's positive thesis is that representation, evidence, operations, correction, discovery, and institutional organization can be studied as related mathematical objects while keeping their distinctions intact. That is the organizing basis for the next theoretical results, and the reason to preserve the full source corpus alongside the synthesis.

Source catalogue and reading routes

The catalogue records selected editions, not a global latest-version assertion. Page counts include each source's original cover, preliminaries, appendices and references. Complete source-page offsets are generated in `PAGE_MAP.csv`. The master reader has document-entry cards and source-prefixed page labels. The separate original PDFs remain available in `sources/`.

Core machine theory

GA — Grounded Abstraction

1.0.0; 43 source pages.

Evidence, identifiable prediction, correction memory, and finite quantitative decisions. Prose proofs and source-reported exact illustrations; no kernel certificate or learned physical model.

MU — Machine Understanding

1.0.0; 50 source pages.

Reusable representation, recipient-relative explanation, acquisition and joint resource feasibility. Prose proofs and bounded reference constructions; an operational account rather than an empirical understanding test.

LSM — Large Signal Model Foundations

Foundations 0.4.0 / software 2.3.0; 66 source pages.

Signal realizations, packets, predictive closure, observable models, and certified probability domains. Theory edition, not a new production runtime; probability validity, identification and empirical adequacy are distinct.

OSL — Operator-Symbolic Language

Research edition 1.0 / Core 0.5; 18 source pages.

Typed operations, semantic checkpoints, provenance, bounded discovery and admission. Executable finite interfaces are described; complete interpreter soundness and general learned production are not established.

NEU — Neuralese

Research edition 1.0; 20 source pages.

Latent operator semantics, observable continuation, gauge and stability boundaries. Mixed theory and retained research history. Historical outcomes are preserved, not used as evidence for a combined agent.

CBI — Certified Bayesian Inference

Core 1.0.0 / standardized publication; 36 source pages.

One-observation Gaussian inference, packet residuals and outward-rounded output guarantees. Archived formal evidence is scoped to the stated construction; no new Lean run or general sequential filter is claimed here.

AVOC — AVOC: Artifact Value and Guarded Planning

Formal 0.41.0 / standardized publication; 37 source pages.

Task sufficiency, guarded planning, calibration, artifact identity and operation lifecycle. Exposition of a formal archive; archived kernel evidence is not a fresh proof run by this compilation.

PPT — Proof Path Theory

1.0.0; 34 source pages.

Justification hypergraphs, certificate extraction, recovery debt and mathematical memory. Finite declared proof fields and written arguments; discovery memory is not itself a proof of its target.

SII — Self-Improving Intelligence

1.0.0; 60 source pages.

Repair synthesis, compiler revision, acquired equations, certificate dependencies and resource prices. Bounded constructive chain with prose proofs; no indefinite recursive acceleration or unrestricted agent is inferred.

MS — Machine Societies

1.0.0; 106 source pages.

Persistent plural organization, institutions, common policies, authority and accountable endings. Conditional theoretical models; internal coherence does not confer external legitimacy or demonstrate endogenous societies.

Mathematical and architectural foundations

SU — Stratified Universality for Resolved Representations

Mathematical base 2.0.0 / standardized publication; 262 source pages.

Typed representation transport, semantic quotients, carrier changes and profiled descent. Relative to declared grammars, profiles and bridges; not equivalence of all mathematical carriers.

SR — Sufficient Resolution

Standardized publication; 47 source pages.

Behavioral sufficiency, hidden-return memory, reconstruction, regularization and calibration. Preserves source-specific prose and archived formal distinctions; hypotheses are not supplied by the reader.

PRIQ — Hypergraph Lumpability and Bayesian Refinement

0.1.4; 52 source pages.

Nonlinear interaction quotients, prediction and update compatibility, posterior and pruning defects. Mathematical refinement contracts with exact and numerical illustrations; no universal learned quotient.

SRWT — Stratified Relational Workspace Theory

209-page monograph / repository 1.0.0; 209 source pages.

Compatible shared cores, causal relevance, persistence, reconstruction, identity and decomposition. A workspace specification and theory; structural gluing alone does not establish causal efficacy or institutional agency.

CIR — Circuit Obstructions

Standardized publication / Lean Block 02 scope; 35 source pages.

Restored reversible computation, complete covers and limits of exact projection. Scoped circuit and representation bounds; not a general superpolynomial separation or SAT solver.

PRISM — PRISM-1: Packet-Programmable Processing

Final updated architecture draft; 61 source pages.

Packet IR, mixed representations, structured scheduling and reconstruction contracts. Architecture specification with separately retained software evidence; not fabricated or fully qualified hardware.

KPP — Komorebi Preservation Protocol

Specification draft 0.1; 20 source pages.

Immutable artifacts, revisable claims, provenance and recoverable continuation. Protocol specification, not proof that the entire archive already implements it.

Formal, visual, and lineage companions

AVL — AVOC: Lean and Evidence Companion

Formal 0.41.0 / standardized publication; 44 source pages.

Declaration-level route from mathematical exposition to retained formal evidence. Formal-source guide; archival build records are not rerun in this compilation.

CBL — Certified Bayesian Inference: Lean Companion

Core 1.0.0 / standardized publication; 31 source pages.

Source-level contracts, archived kernel evidence and one-observation solver proofs. The formal scope remains one observation update under exact stated model assumptions.

SIV — Self-Improving Intelligence: Visual Guide

1.0.0; 36 source pages.

Visual route through repair, discovery, search acquisition and changing guarantees. Expository companion; neither new learning evidence nor a replacement for the proofs.

AD — Agent Dynamics: Python Visual Atlas

1.0; 14 source pages.

Internal beliefs, evidence exchange, revision and external action in an explicit finite model. Rule-based illustrative dynamics, not measured neural modules or learned self-improvement.

LSV — Large Signal Model: Visual Notes

0.4.0; 9 source pages.

Seven mathematical views of certified probability domains and approximation. Analytical illustrations tied to LSM propositions; no task-suite or integrated-model claim.

PCL — Portable Causal Lineage

Protocol 0.2; 64 source pages.

Explicit functional continuity, correction retention and model-replacement tests. Preserved protocol design, not a completed implementation or established subjective continuity.

SRIC — Stratified Recursive Improvement Closure

2.4 / Block 23; 505 source pages.

Extended repair theory, semantic transport, evaluator contexts and historical lineages. Mixed theoretical and historical monograph preserved in full. Historical claims are not replayed or imported as integrated evidence.

Notation crosswalk

Object	How it is used in the compilation
Representation	A specified map or retained object, relative to future queries and operations. It is not assumed lossless.
Packet	A typed component in a decomposition. Linear, probabilistic, evidentiary and institutional carriers are not identified by their shared name.
Transfer	A declared correspondence preserving named structure. A map can preserve outputs without preserving acquisition, authorization, or cost.
Residual	A discrepancy relative to a contract. A residual identifies failure of a conjunction, not necessarily its unique cause.
Closure	Preservation under a specified operation family. This is distinct from proof completeness, probability validity and open-ended discovery.
Correction	A justified change to retained candidates, probabilities, methods or claims. Historical records need not be erased.
Certificate	Evidence for a named statement under specified premises. Its assurance type and current applicability remain explicit.
Workspace	A relational organization satisfying the declared structural, predictive, causal and continuity conditions. Shared storage alone is insufficient.
Institution	Operative permissions, offices, commitments and revision rules among decision centers. Structural compatibility does not supply external consent.
Improvement	A gain under a stated comparison, including relevant installed information and costs. Editing, selection and gain are different events.

Assembly and review boundary

The original PDFs and included source archives are copied byte-for-byte into the package. Source pages in the merged readers are compared with the originals for extracted text and geometry. Sampled source pages are also compared as rendered images. New covers, source-entry pages, and this synthesis receive separate visual review. These are preservation and publication checks, not a new proof review of approximately two thousand source pages.

The exact connecting-example checks use finite sets, integers and rational arithmetic. They test the stated eight-world model, translator, correction memory, common observable quotient, and resource calculations. They are not empirical observations about neural systems. The inherited research suites and Lean projects are not rerun. The complete validation scope, encountered failures, hashes, and source-preservation results are recorded in the collection review report.

No standalone font files, private conversation transcript, or third-party full research paper is added by the assembly. Existing source-specific licensing remains unchanged. The package is a working collection, not a blanket relicensing or a claim of external publication acceptance.

PART I

Core Machine Theory

Evidence, understanding, signal models, executable semantics, inference, discovery and institutions.

Source-entry cards explain the role and boundary of each selected work. The original document follows immediately after its card, without reflow or renumbering of its statements.

The new synthesis is a guide to these works, not a replacement for their proofs.

GA / SOURCE READING CARD

Grounded Abstraction

1.0.0 · 43 original pages

READ THIS FOR

Evidence, identifiable prediction, correction memory, and finite quantitative decisions.

KEEP THIS BOUNDARY

Prose proofs and source-reported exact illustrations; no kernel certificate or learned physical model.

CONNECTED SOURCES

MU LSM SR PRIQ

Original text begins on the next page.

The source pages that follow are preserved without editorial stamps. Page labels in the PDF navigation use the source code and physical source-page number. The package retains the byte-identical original PDF and its SHA-256 digest.

Grounded Abstraction

Identifiability, Representation Change, and the Limits of Learning

Komorebi P.
9 October 2026

First consolidated edition
Version 1.0.0

Abstract

A useful abstraction need not identify the system that realizes it. It must be justified by the evidence available to initialize, update, and correct it. This first edition connects finite evidence contracts, minimal corrective memory, and partially known linear dynamics. An exact signature preserves future candidate elimination. Its probabilistic counterpart retains joint target and likelihood-profile masses, while an explicit migration criterion identifies when a revised evidence language needs information the old memory cannot recover. For linear continuation, paired-state tests decide output agreement across fixed candidate systems. An invariant discrepancy span characterizes a shared observable quotient, and an initialization theorem gives the smallest common linear predictor accessible from the retained record. The quantitative extension replaces exact probability coordinates by a finite integer record, with separate likelihood, rounding, and horizon terms. A block comparison controls hidden-to-visible return and distinguishes exact invisibility from a small but unstable leakage channel. These bounds compose into one posterior decision certificate for the same evidence history and future response. The result is a theory of what incomplete experience can justify, supported by self-contained conditional proofs, counterexamples, and small exact illustrations. It is not a trained general learner or a certificate that its supplied model describes an external process.

Scope. Finite models and finite-dimensional linear systems, with explicit assumptions and self-contained proofs. Established ingredients are acknowledged. First edition freezes this finite theoretical scope. Independent peer review, a learned physical-semantic interface, and proof-assistant verification are not claimed.

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Reading the first edition

The central object is not an internal vector considered in isolation. It is a representation together with the evidence available to initialize it, the operations it supports, the questions it answers, and the future corrections it must permit. The paper asks which of those obligations can be met simultaneously.

Part I distinguishes identification of a hidden realization from justification of a reusable predictive state. Part II studies new questions, the cost of acquiring missing distinctions, and memory that preserves later correction. Part III separates certainty from noisy evidence, quantifies uncertainty, and constructs exact probability memory for future correction. Part IV compares full-state calibration, fixed-model output agreement, common observable quotients, and evidence-initialized linear prediction. Part V adds finite probability records, controlled visible leakage, and a common posterior decision certificate. Part VI consolidates revision requirements and the research boundary.

Relation to the existing collection. *Machine Understanding* develops supplied representations, explanatory artifacts, correction transport, and their common resource contracts [1]. *Self-Improving Intelligence* studies finite repair discovery and changes to discovery procedures [2]. *Sufficient Resolution* treats behavioral quotients, hidden memory, reconstruction, and calibration [3]. The present contribution is organized around a different source of uncertainty: the evidence may leave several underlying realizations possible. A representation is justified only to the extent that its intended uses survive that ambiguity. Earlier manuscripts are source relationships, not claims that every result here is already formally verified.

What is and is not proposed. Behavioral quotients, testing bounds, compatible-model arguments, invariant subspaces, and set-intersection updates have established mathematical foundations. Their appearance here is not a claim of priority. The proposed synthesis makes evidence access, corrective memory, and the identifiable continuation domain parts of the same representation contract. The exact correction-memory, likelihood-profile, calibrated-domain, and observable-realization constructions are derived in full within their stated models. The linear reductions use established reachability and observability ideas, while exact probability memory uses finite likelihood interpolation. The first-edition estimates use elementary Bayesian perturbation, rounding, and nonnegative matrix comparison. These arguments are not priority claims for their ingredients. Their integration requires one actual record and one consistent prediction contract.

Symbol	Meaning
$\omega \in \Omega$	A candidate world, including any fixed model or historical tag.
$K^{\pi}(h \mid \omega)$	Probability of observable record h under one permitted protocol.
$r : X \to Z$	Retained evidence or an available initial code. Its access is explicit.
F_a, q	A state transition and the task observation it must preserve.
$x \sim_C y$	Equality of every task observation after every permitted finite word.
$V \subseteq \Omega$	A set of candidate worlds, including the possible empty set.
$E \in \mathcal{E}$	An admitted evidence event used to filter V.
$\Sigma(V)$	The complete signature of future task-valued corrections.
U, S_a	A calibrated input subspace and a known operator restriction.
U_∞	The maximal domain on which calibrated continuation is determined.
I_Q	Initial vectors with common output continuations across fixed models.
E_∞	Propagated span of all model-discrepancy directions.
$p_{i,y}$	Joint probability of a likelihood profile and a target value.

First-edition claim and dependency guide

The first edition separates exact identification, quantitative approximation, and application to an external process. An argument may be complete within its stated model while the evidence that a physical or learned system satisfies that model remains unavailable.

Layer	Closed theoretical question	Premises that do not appear for free
Evidence	Does the record distinguish the future behavior required by the task?	Correct candidate worlds, admissible protocols, and actual record access.
Correction	Which joint distinctions must remain available for future filtering?	The evidence language, target, and a justified initial candidate set or prior.
Dynamics	Which outputs agree across compatible systems, and can one predictor realize them?	Shared observation semantics, a supplied model class, and enough initial evidence.
Approximation	How far can rounded memory and a reduced predictor deviate over the declared horizon?	A positive likelihood floor, model-error bounds, common block norms, and initial-state bounds.
Decision	Do the combined errors preserve a risk threshold or the chosen action?	One common history, response, posterior, and bounded Lipschitz loss.

The main exact dependency runs from the finite evidence criterion to predictive initialization, and from correction signatures to likelihood-profile memory and migration. A separate linear route runs from calibration through model agreement and observable quotienting to a minimum linear predictor. The quantitative route joins probability-memory error to an independently justified output allowance only at the final decision theorem.

Reading routes. For the central argument, read Sections 2, 5, 8, 13, 17, and 18. For proofs, follow the sections in order and retain the distinction between a fixed unknown model and explicitly permitted switching. For a review of the limits, compare the destructive-test, rare-evidence, hidden-return, and feedback counterexamples.

Three evidence levels. The mathematical claims have prose proofs. The reference programs check finite declared examples using exact arithmetic. External calibration, empirical model adequacy, and neural realization are not established by either a passing program or the edition number. The source package records assumptions, predecessor preservation, actual command results, and the depth of reference review.

What is frozen. The completed scope includes finite evidence models, static noisy correction, finite-dimensional linear continuation, finite retained probability fields, and a bounded-loss composition. State-changing noisy filters, general model invention, unbounded adaptive learning, and independently verified implementations remain subsequent research topics.

Part I

Evidence before representation

1 What does experience justify?

A device displays a bit. A reasoner learns to predict that display perfectly, including what happens when a permitted switch is pressed. It still may not know which physical bit the display describes. The code linking the display to the physical state could be reversed. A second, equally accurate reasoner might use the opposite internal convention. Prediction is real, yet the interpretation remains underdetermined.

That observation does not force a choice between calling every representation meaningless and treating every successful prediction as a complete understanding. It suggests a more precise question. What claims are common to all systems compatible with the evidence? Which of those claims can be realized by one representation that the observer can actually initialize and update? What additional information becomes necessary when the questions change?

1.1 Three separations

Identification of an underlying system, identification of a task-relevant property, and acquisition of a usable representation are distinct. A nuisance parameter can remain unknown while every required prediction is fixed. Conversely, knowing that some accurate representation exists in each possible world does not supply one representation that can be selected from the observer's record.

There is a second separation between a probability law and a finite observation drawn from that law. Two candidate models can have different laws while sharing every finite record in their supports. In that setting the models are identifiable at the level of their laws, but an individual finite record cannot yield error-free discrimination.

The third separation concerns time. A summary may preserve everything required for a present decision while losing the relationship needed to interpret a later measurement. More data arriving tomorrow need not repair what the memory policy erased today. This paper treats that relationship as part of the retained mathematical state.

1.2 Position among existing theories

Classical statistical identification separates parameters that induce different observational laws [4]. Predictive state representations use action-conditional predictions as a state description [5]. Query learning makes the teacher and the information it returns explicit [6]. These are neighboring answers to different parts of the present problem, not interchangeable definitions of grounding.

Data informativity is especially close to our organizing principle. It asks whether one property or one controller is justified for every system consistent with the data, without necessarily identifying a unique system [7]. Our later linear construction uses this compatible-extension viewpoint for the exact domain of repeated prediction. It is not offered as a replacement for data-driven control theory.

Causal abstraction concerns faithful relationships between mechanisms and their higher-level descriptions [8]. Here we ask what evidence makes such a description selectable and which continuations remain determined when parts of the underlying mechanism are unknown. We do not derive a learned encoder from the existence of a commuting equation. State-identification theory also distinguishes individually separable states from a realizable adaptive test [9]. That distinction will matter when measurements alter the system.

1.3 The intended contribution

The main results form a connected argument. Exact grounding is an information factorization through a behavioral quotient. New task distinctions are classified relative to an actual diagnostic interface. Future correction requires a richer memory signature than present target uncertainty. Partial operator calibration can nevertheless justify every future composition on a computable invariant domain. Approximate claims require both a local discrepancy and a bound on the dynamics that may amplify it.

The positive message is not that the observer must recover everything. It is that the observer can often recover exactly the structure its intended operations need, provided the evidence contract is stated honestly. The negative message is that an attractive coordinate system cannot manufacture missing distinctions, missing experiments, or missing bounds.

2 A calibrated bit with an irrelevant hidden register

Consider eight possible worlds

$$\omega = (s, \theta, k) \in \{0, 1\}^3.$$

The physical bit is s. The sensor convention is θ, and k is a nuisance bit that none of the admitted operations or questions uses. The display is

$$y = s \oplus \theta.$$

The ordinary action F flips s and leaves the other bits fixed. Its effect on the display is exactly $y \mapsto 1 - y$, whatever the unknown convention. Reading the current display therefore initializes a two-state representation that predicts every finite sequence of flips.

Neither s nor θ is identified separately. The two worlds $(0, 0, k)$ and $(1, 1, k)$ both display zero, and every flip sequence produces the same display in both. That ambiguity is harmless for the original display-prediction contract.

2.1 A new question

Now ask for the physical bit s as well as the display. The old representation is insufficient. The two previously equivalent worlds require different answers. A permitted nondestructive reference measurement returning θ makes the physical bit recoverable by $s = y \oplus \theta$. The enriched representation can retain (y, θ), or equivalently (s, θ). It has four states, not eight. The nuisance bit remains unidentified because no required continuation can detect it.

A new field named "sensor convention" does not provide the reference measurement. Nor does repeatedly flipping the physical bit resolve the ambiguity: the complete display history remains the same in the two worlds. Acquisition depends on an additional information channel, not merely a larger memory allocation.

2.2 A state-changing measurement

A different operation Z sets s to zero and then reports the new display. The report is θ. If the old display y_{old} was retained, then

$$s_{\text{old}} = y_{\text{old}} \oplus y_{\text{after } Z}.$$

The current physical state is zero, not s_{old}. A correct account distinguishes the recovered historical value from the new state it will act on.

If the old display was discarded, worlds with opposite old physical bits but the same θ reach the same current state and give the same future reports. The old physical bit is then irrecoverable through this unchanged interface. A known present state has replaced, rather than identified, the unknown past.

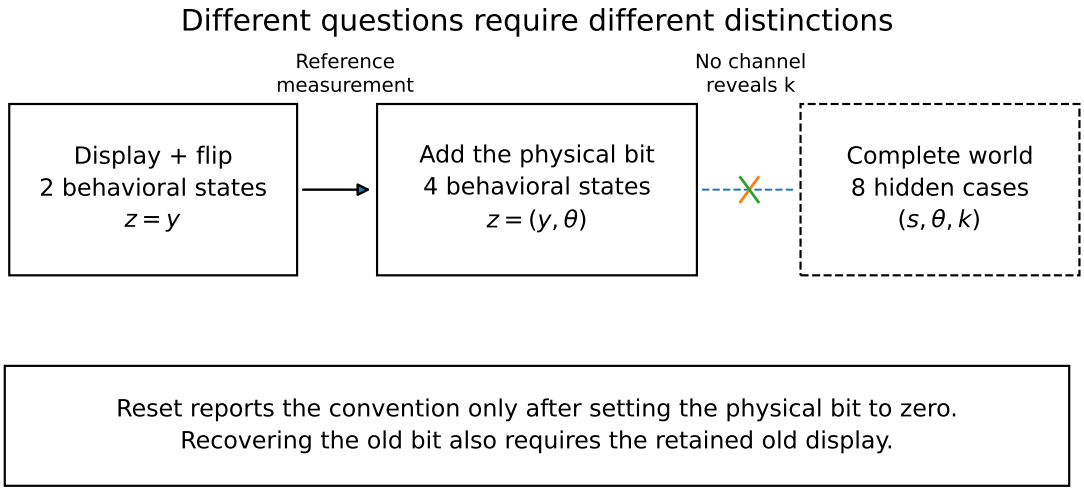


Figure 1: One finite sensor model supports different exact abstractions. The current display and flip action need two behavioral states. Adding the physical-bit question requires four, but not the irrelevant nuisance register. Calibration is an information source. Resetting is a physical change.

3 Laws, records, and reusable predictive state

Definition 3.1 (Finite evidence contract)**.** A finite evidence contract has a nonempty candidate set Ω, a family Π of permitted protocols, a finite record space H_π for each protocol, and kernels $K^\pi(h \mid \omega)$. A target $t : \Omega \to Y$ states what is to be identified. Any hidden model index that is fixed during a protocol is included in ω. The protocol's action choices are functions of its permitted records and independent randomization, not of the unobserved world.

Define $\omega \sim_\Pi \omega'$ when their record laws agree for every $\pi \in \Pi$. This relation concerns the complete records, including action choices when those are observable. Equality of individual observation marginals is insufficient.

Proposition 3.2 (Law identification and finite-record identification)**.** *The target is a function of the complete family of protocol laws if and only if it is constant on the $\sim_\Pi$ classes. For a fixed protocol π, an error-free rule $d : H_\pi \to Y$ exists if and only if t is constant on every nonempty support set*

$$B_\pi(h) = \{\omega : K^\pi(h \mid \omega) > 0\}.$$

Allowing a randomized output rule does not weaken this condition for probability-one correctness in every world.

Proof. For the first claim, factor t through the map sending a world to its tuple of protocol laws. A factor exists precisely when worlds with the same tuple have the same target. For the second claim, a record with positive probability in two worlds must be given the same output. Distinct targets on its support therefore prevent certainty. Conversely, assign each attainable record its support's common target. Every record generated in a world receives that world's target. A randomized error-free rule must place probability one on the required target at every positive-probability record. It cannot do so for two distinct targets at the same record. □

This elementary factorization does not supply the laws to the learner or bound the sample cost of estimating them. The two clauses answer different questions. A family of distinct laws can be identifiable in the first sense while every available short record remains ambiguous in the second.

3.1 A canonical meaning for an operational state

Let X be a finite nonempty set, let $\mathcal{A}$ be a finite operation alphabet, and let each $F_a : X \to X$ be total and deterministic. Let $q : X \to Y$ be a finite vector of required observations. Different candidate models can be included by taking a disjoint union of their state sets and preserving each model tag under every action.

For a word $w = a_1 \cdots a_m$, F_w applies a_1 first and a_m last. The empty word is the identity. Define

$$x \sim_C x' \quad \Longleftrightarrow \quad q(F_w x) = q(F_w x') \quad \text{for every finite } w.$$

This is a comparison in a common observation language. It does not require that different models share arbitrary latent coordinates. The behavioral class is a canonical response profile even when its implementation is not identifiable.

Definition 3.3 (Exact initialization from available evidence)**.** Given an available initial code $r : X \to H$, an exact reusable predictor consists of a finite state set Z, an initialization $d : H \to Z$, updates $G_a : Z \to Z$, and an output map $\bar{q} : Z \to Y$ such that

$$\bar{q}\, G_w\, d\, r(x) = q(F_w x)$$

for all x and all finite words. After initialization the predictor receives the operation names, but no additional world observations or hidden state.

Theorem 3.4 (Evidence sufficient for one exact predictive realization)**.** *An exact reusable predictor exists if and only if*

$$r(x) = r(x') \quad \Longrightarrow \quad x \sim_C x'. \tag{1}$$

When it exists, the behavioral quotient $X/\!\sim_C$ supplies a predictor with the fewest initialized states. Every predictor initialized for all $x \in X$ requires at least $|X/\!\sim_C|$ distinct values in $d(r(X))$.

Proof. Equal initial codes yield equal initialized states. The same operation sequence must then produce equal predicted outputs. Exactness forces equality of all true continuation outputs, proving necessity.

For sufficiency, note that $x \sim_C x'$ implies $F_a x \sim_C F_a x'$ because any subsequent word can be prefixed by a. The quotient consequently supports $G_a([x]) = [F_a x]$ and $\bar{q}([x]) = q(x)$. Condition (1) makes $d(r(x)) = [x]$ well defined on the attained initial codes. Arbitrary unused codes can be assigned any quotient state. Induction on word length gives the required equality. Finally, two behaviorally different states cannot have the same initialized predictor state, or a distinguishing continuation would receive the same prediction. Thus the number of initialized values is at least the quotient cardinality. □

The underlying quotient argument is classical and is developed elsewhere in the collection [1, 3]. Here its role is to attach a separate initialization condition to a common predictor across the surviving candidate worlds. A quotient can exist while the available record fails to initialize it.

3.2 Finite checking does not require an infinite list of words

Begin with the partition by q. Split any block whose members have successors in different current blocks under some action. A strict split increases the number of nonempty blocks, so at most $|X| - |q(X)|$ strict rounds occur. At a fixed point the quotient updates are well defined and preserve every later observation by induction. Any operation-stable refinement of the original output partition refines each round, so the terminal partition is the behavioral equivalence above. This supplies a finite computation when the complete candidate tables are given. It does not make their empirical acquisition free.

3.3 State identity is relative to the chosen questions

In the sensor example, $r = y$ satisfies (1) for display prediction. The quotient has two states even though eight worlds remain in the candidate universe. Adding physical-bit output refines the quotient to four classes and invalidates (1) for that same initial code. Adding the nuisance bit to the target would refine again to eight, but no admitted sensor action reveals it.

Only adding requirements yields a refinement relation automatically. Replacing one task with a different task can produce crossing partitions rather than a finer version of the old one. For example, the partition by display and the partition by physical bit are neither finer nor coarser than one another. A claim of cumulative learning must specify which old questions remain protected.

3.4 An available predictor is stronger than a worldwise predictor

Consider two immutable worlds, one with a permanently zero output and one with a permanently one output. The available initial record is the same symbol in both. Within each world separately, a one-state predictor exists: it simply prints that world's output. There is nevertheless no predictor that can be initialized from the common record and be correct in both. The obstruction is the order of quantifiers,

$$\forall \omega \; \exists P_\omega \qquad \text{versus} \qquad \exists P \; \forall \omega .$$

The first does not provide a selector for the second. The behavioral quotient of the combined family has two states, but the evidence cannot choose between them. Adding an output observation repairs the initialization problem without changing the mathematical existence of either predictor.

Likewise, a coordinate convention is not an observation. Relabeling a latent state and conjugating its updates can leave every operational prediction unchanged. Without a separately meaningful reference, evidence about those predictions need not favor one coordinate system. A canonical response profile avoids claiming that an arbitrary coordinate name was discovered as a physical entity. It does not prove that the reasoner's actual implementation uses the inferred profile causally.

Thus two distinct questions remain after fitting a predictor. Is it a common valid representation of the possible worlds, and does the observer possess the evidence needed to initialize it? The theorem answers their exact conjunction in a finite model. Efficient acquisition and causal attribution require further premises.

Part II

New questions and the memory needed to correct them

4 Reinterpretation, acquisition, and irrecoverable distinctions

A revised task often invites the vague instruction to learn a better representation. That instruction hides three different possibilities. The old code may already contain everything needed. A new observation may be required. Or the remaining interface may be unable to recover the needed distinction at all.

We first separate these cases in a setting where observations do not interfere with one another. Let Ω be finite, let $r : \Omega \to Z$ be retained evidence, and let $t : \Omega \to Y$ be the revised target. Let $\mathcal{E} = \{e_1, \dots, e_m\}$ be a finite family of total, deterministic, nondestructive tests. Each test reads the same fixed world and leaves it unchanged. All tests remain available throughout the investigation. Equivalently, a reset may be allowed if it genuinely restores the same unknown world, including the relevant history. Resetting into a new known state does not satisfy this premise.

For $J \subseteq \{1, \dots, m\}$, write

$$R_J(\omega) = \big(r(\omega), (e_j(\omega))_{j \in J}\big).$$

Let the unresolved target pairs be

$$D = \{\{\omega, \omega'\} : r(\omega) = r(\omega'),\ t(\omega) \neq t(\omega')\}.$$

A test covers such a pair when it gives different reports in the two worlds.

Theorem 4.1 (Exact revision trichotomy for a reusable test interface)**.** *Exactly one of the following applies.*

1. *D is empty. A new decoder of the old code recovers t without new observations.*
2. *D is nonempty, but every pair in D is separated by at least one admitted test. Executing all tests recovers t, and a selected nonadaptive set J suffices exactly when it covers D.*
3. *Some pair in D has identical reports under every admitted test. No finite adaptive protocol using this unchanged test interface and r guarantees t in every world.*

If test j has nonnegative fixed cost c_j, the minimum cost of one fixed nonadaptive test set, used for all initial codes in case 2, is precisely

$$\min_{J:\, J \text{ covers } D} \sum_{j \in J} c_j. \tag{2}$$

This is not a claim that an optimal adaptive protocol has the same cost.

Proof. If D is empty, the target is constant on each old-code fibre, giving a decoder. Otherwise a set J permits a decoder exactly when no pair with the same old code and different target remains equal on all its test reports. That is exactly pair coverage. In particular, the full finite test family succeeds when every unresolved pair has a separating test.

If a pair survives every test, induction on an adaptive history shows that the two worlds give the same report after each chosen action. The protocol chooses the same next test because its available history is the same. Its stopping decision and output are therefore the same, so it cannot be correct on both targets. Independent randomization does not restore a probability-one guarantee, by conditioning on its random tape. The three cases are exclusive and exhaustive. Formula (2) follows from the equivalence between successful nonadaptive test sets and covers, with exactly the stated additive charges. □

The cover formulation exposes a concrete resource question without making a large learner part of the theorem. It also identifies what a negative certificate says. An uncovered pair is impossible to resolve through the declared interface. It is not impossible to resolve after a new sensor, retained record, or physical intervention changes that interface. Finding the cheapest cover can itself be difficult. Its formulation is not an efficient universal discovery algorithm.

4.1 Why experimentwise evidence need not compose

Counterexample 4.2 (Pairwise separation without a successful investigation)**.** There are three possible initial cases $0, 1, 2$. Test a reports whether the case is 0, and test b reports whether it is 1. Either first test then destroys the accessible case state. Every later test returns a fixed zero report. The target remains the initial case, stored only as an inaccessible historical tag.

Every pair is distinguishable by some permitted first test. Yet beginning with a leaves cases 1 and 2 indistinguishable whenever its answer is no. Beginning with b leaves cases 0 and 2 indistinguishable. No finite adaptive protocol identifies every initial case. Randomizing the first test cannot guarantee success in every case either.

The missing premise is not intelligence. It is the ability to combine evidence from mutually destructive inquiries. State-identification work studies related restrictions on adaptive tests [9]. The example makes clear why the target map, physical transition, and record cannot be separated in a general acquisition problem. The full state-changing repair constructions in *Self-Improving Intelligence* address that different model [2].

4.2 The sensor revisited

For physical-bit recovery, an old display and a nondestructive reference measurement satisfy case 2. Without the reference, flip sequences leave the convention ambiguity unresolved. For recovery of the pre-reset physical bit after losing the old display, the surviving states can have the same current physical state, convention, and every future sensor response while having opposite historical targets. That is case 3 relative to the post-reset interface.

A learner should record which kind of failure occurred. A failed search over decoders cannot establish that no new experiment would help. Conversely, repeatedly searching for a decoder cannot reconstruct a target that is not a function of its inputs. Formula length and information access are separate resources.

5 A minimal memory for future correction

The old record can matter even when it does not change today's set of possible answers. The essential information may lie in a relation between a possible answer and an observation that will arrive later.

Let Ω be a finite model set and $t : \Omega \to Y$ a fixed target. For a candidate set $V \subseteq \Omega$, let

$$Q(V) = t(V) = \{t(\omega) : \omega \in V\}.$$

The empty set is permitted. It represents a contradiction between the admitted model family and the evidence, not a successful identification. Let $\mathcal{E} \subseteq \mathcal{P}(\Omega)$ be a finite family of evidence events containing Ω and closed under intersections. Filtering by E sends V to $V \cap E$. The events are known, truthfully interpreted, and do not change the underlying model in this section.

Definition 5.1 (Correction signature)**.** The correction signature of V is

$$\Sigma_{\mathcal{E},t}(V) = \big(Q(V \cap E)\big)_{E \in \mathcal{E}}. \tag{3}$$

Two candidate sets have the same signature when every admitted evidence filter leaves the same possible target values, including whether no candidate remains.

Theorem 5.2 (The coarsest exact memory for admitted corrections)**.** *Equality of correction signatures is a congruence for all updates $V \mapsto V \cap E_0$. The signature therefore supports well-defined updates and a readout of $Q(V)$. It preserves that readout after every finite sequence of admitted filters.*

Conversely, any retained representation with deterministic filter updates and a readout reproducing Q after every such sequence must distinguish every two candidate sets with different signatures. Thus the image of $\Sigma_{\mathcal{E},t}$ is the smallest exact state space, in cardinality, for this correction contract on all candidate subsets.

Proof. If $\Sigma(V) = \Sigma(W)$, then for every $E \in \mathcal{E}$,

$$Q((V \cap E_0) \cap E) = Q(V \cap (E_0 \cap E)) = Q(W \cap (E_0 \cap E)).$$

Intersection closure puts $E_0 \cap E$ in $\mathcal{E}$, so the equality is available from the original signatures. It proves $\Sigma(V \cap E_0) = \Sigma(W \cap E_0)$. The $E = \Omega$ coordinate gives $Q(V)$, and induction gives all finite filter sequences.

If two signatures differ, one event E produces different target sets. A retained representation that merged the two candidate sets would update identically on that event and return the same readout, contradicting exactness. Every distinct signature therefore requires a distinct retained value. The signature itself attains that count with the updates just constructed. □

This is a behavioral quotient of a powerset system. It preserves later task-valued correction, not merely the present prediction. Candidate identities can be discarded when no admitted correction distinguishes them. Contradiction with the entire candidate set remains observable.

5.1 An exact count when evidence identifies a partition cell

Suppose $A_1, \ldots, A_m$ are a partition of Ω into nonempty cells, and $\mathcal{E}$ consists exactly of all unions of those cells. Define the attainable evidence–target pairs

$$D_t = \{(i, y) : y \in t(A_i)\}, \qquad r = |D_t| = \sum_i |t(A_i)|.$$

A candidate set induces an occupancy record

$$O(V) = \{(i, t(\omega)) : \omega \in V \cap A_i\}.$$

Corollary 5.3 (Exact correction-memory size for partition evidence)**.** *Two candidate sets have equal correction signatures if and only if their occupancy records agree. On the domain of all subsets $V \subseteq \Omega$, there are exactly 2^r distinct signatures. A fixed-length exact binary memory therefore requires at least r bits, and an r-bit occupancy vector attains this bound.*

Proof. Filtering by cell A_i returns exactly the target values occurring in the ith part of $O(V)$. Thus equal signatures imply equal occupancy. Conversely, the target set after filtering by any union of cells is the union of their occupied target values, so occupancy determines every signature coordinate.

Every subset of D_t is realized: for each included pair choose one world in its nonempty intersection $A_i \cap t^{-1}(y)$, and choose no worlds for excluded pairs. These intersections are disjoint for different pairs. Hence the occupancy map has 2^r values. Distinct values need distinct memory states by the preceding theorem. Binary counting gives the lower bound, and direct occupancy bits attain it. □

A particular inquiry may reach fewer candidate sets, reducing the number of attained occupancy records. The bound counts logical states, not interpreter bytes, update work, or evidence acquisition.

The exact evidence family matters. Two worlds a, b in one cell with the same target have identical singleton occupancies. Adding the event $\{a\}$ distinguishes them: one filtered singleton retains the target and the other is empty. A new event that splits a cell can therefore require finer memory.

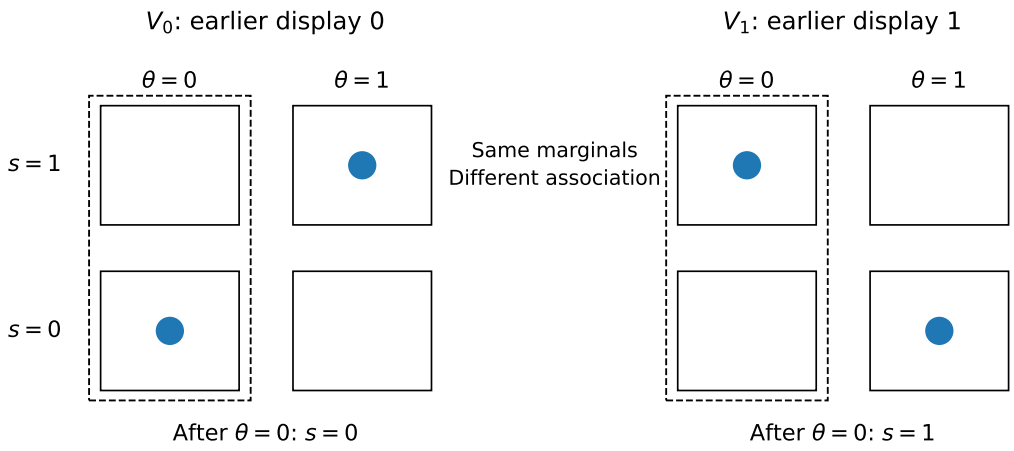


Figure 2: Two evidence histories preserve identical physical-bit and convention marginals. Their joint possibility patterns differ. A reference report selects a column, producing opposite physical-bit conclusions. The nuisance bit is omitted because it changes neither pattern.

5.2 Association can be the missing state

Counterexample 5.4 (Equal marginals with different corrected answers)**.** In the sensor model, let

$$V_0 = \{(s, \theta, k) : s \oplus \theta = 0\}, \qquad V_1 = \{(s, \theta, k) : s \oplus \theta = 1\}.$$

Take the target to be s and the evidence cells to be the two values of θ. Both candidate sets currently allow physical bits $\{0, 1\}$, and both allow conventions $\{0, 1\}$. Yet after the reference reports $\theta = 0$, V_0 requires $s = 0$ and V_1 requires $s = 1$.

Retaining both marginal possibility sets loses the association needed to use the next measurement. Their equality is not a sufficient criterion for merging the records.

Here the nuisance bit duplicates every attainable (θ, s) pair. There are $2^8 = 256$ raw candidate subsets, but only $2^4 = 16$ correction signatures. Four occupancy bits preserve every admitted future correction. The present physical-target set alone has only four possibilities, requiring two bits, but that smaller code is inadequate for the correction contract.

Four association bits suffice where eight raw membership bits are redundant and two marginal bits are inadequate.

5.3 Correction memory has a scope

The signature is minimal for its specified target and evidence family. New targets, events, candidate models, state-changing observations, or revised event meanings can split its classes. Old minimality remains a statement about the old contract.

If a future question asks for k, the occupancy is insufficient. When every available measurement ignores k, expanding the code cannot recover it. A finer signature can exist mathematically without being obtainable from the current interface.

Part III

Noisy evidence and quantitative justification

6 Identifiable does not mean certainly identified

The earlier exact filters remove a world only when it cannot produce the observation. With noisy channels that criterion can retain every candidate indefinitely. A statistical procedure may nevertheless become increasingly reliable. These are different guarantees, not contradictory descriptions of learning.

For a fixed protocol, let two worlds with different targets generate finite transcript laws P_0 and P_1. An estimator sees only that transcript and independent randomness. Its average success under equal prior probabilities is bounded by the discrimination in the entire record.

Proposition 6.1 (A finite-record discrimination bound)**.** *For two distinct target values and equal world probabilities, every transcript-based estimator has average success at most*

$$\frac{1 + d_{\mathrm{TV}}(P_0, P_1)}{2}, \qquad d_{\mathrm{TV}}(P_0, P_1) = \frac{1}{2} \sum_h |P_0(h) - P_1(h)|. \tag{4}$$

A deterministic function or a common stochastic processing channel applied to the transcript cannot increase this upper bound.

Proof. Convert the estimator into a binary test by reporting world zero when it outputs the first target and world one otherwise. A correct target output always gives a correct binary decision. If $f(h) \in [0, 1]$ is the test's chance of declaring world zero, its success is

$$\frac{1}{2}\left(1 + \sum_h f(h)(P_0(h) - P_1(h))\right).$$

The sum is at most the sum of positive differences, which equals total variation. This proves (4). If $B(h, z)$ is a common processing channel and $g(z) \in [0, 1]$ a processed-record test, then $f(h) = \sum_z B(h, z)g(z)$ is a raw-record test with the same two expectations. Thus every processed decision was already available from the original record. □

This is an elementary testing and data-processing argument, related to the classical comparison of experiments [10]. It does not identify the best protocol, establish the correctness of a sensor model, or make independent randomness into additional evidence. If an adaptive protocol is compared, its complete history must be the transcript in (4).

Counterexample 6.2 (Distinct laws with no finite certain identification)**.** Let the unknown world be $\omega \in \{0, 1\}$. A repeated observation is conditionally independent Bernoulli with success probability $1/4$ in world zero and $3/4$ in world one. The two one-observation laws differ, so the world is identified by its law. Every finite binary string nevertheless has positive probability under both worlds. No fixed finite sample supplies an error-free world classifier.

For odd n, majority voting is the likelihood-ratio rule. Its error in either world is

$$e_n = \sum_{j=0}^{(n-1)/2} \binom{n}{j} (3/4)^j (1/4)^{n-j}.$$

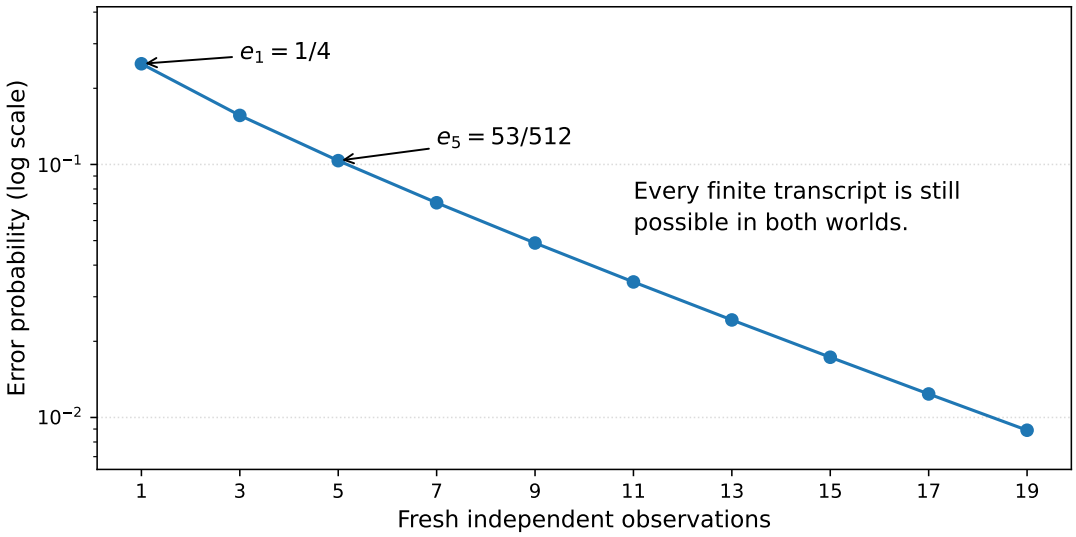


Figure 3: Exact majority-test error for two stipulated Bernoulli worlds. The laws differ, and repeated independent evidence reduces error, but every finite record remains compatible with both worlds. The curve is evaluated from the displayed binomial formula.

The first values are $e_1 = 1/4$, $e_3 = 5/32$, and $e_5 = 53/512$. These are exact analytical probabilities, not observed error rates from an agent experiment.

The example illustrates why logical candidate elimination and statistical confidence should not share a Boolean field named "known." Both candidate worlds remain in the support of every fixed finite record. Their relative likelihoods can become very different. Treating the most likely world as certain discards the distinction that the exact support test preserves.

6.1 Identifiability is not a uniform rate

Distinct models may induce arbitrarily similar laws. Unless a model class imposes a separation condition, a bounded number of observations need not distinguish every pair reliably. The total-variation term exposes that obstruction directly. A data-dependent summary cannot increase the discrimination contained in the actual record.

Conversely, a small class with known, well-separated distributions can admit a useful finite-sample procedure even when exact certainty is impossible. The statistical question is then how much error is acceptable and which assumptions justify its probability bound. We do not add a sample-complexity claim merely by labeling a model family finite. Separation, admissible protocols, and the sampling law have to enter the argument.

7 An explicit radius for what remains unknown

A set of candidate worlds may disagree on some future outputs while agreeing closely on others. It is useful to quantify that disagreement without selecting one world as if it had been identified.

Let $v : \Omega \to \mathbb{R}^d$ record a fixed finite collection of target quantities, where $d \geq 1$. These can be future predictions on specified operation words. For a nonempty $B \subseteq \Omega$, define

$$\ell_j(B) = \min_{\omega \in B} v_j(\omega), \qquad u_j(B) = \max_{\omega \in B} v_j(\omega),$$

$$m_j(B) = \frac{\ell_j(B) + u_j(B)}{2}, \qquad \rho(B) = \frac{1}{2}\max_j\big(u_j(B) - \ell_j(B)\big). \tag{5}$$

The radius concerns coordinatewise prediction in a common frame. It is not an uncertainty measure invariant under arbitrary nonlinear reparameterization.

Theorem 7.1 (Ambiguity radius and an explicit prediction certificate)**.** *For every nonempty finite B,*

$$\inf_{z\in\mathbb{R}^d}\max_{\omega\in B}\|z - v(\omega)\|_\infty = \rho(B),$$

and $m(B)$ attains the infimum. If $B' \subseteq B$ is nonempty then $\rho(B') \le \rho(B)$.

Fix nonnegative tolerances η, ε and $\delta_{\mathrm{id}}, \delta_{\mathrm{model}} \in [0,1]$. Suppose a data record H produces a candidate set $B(H)$ with

$$\Pr\{\omega_* \in B(H)\} \ge 1 - \delta_{\mathrm{id}}.$$

Suppose a second event, of probability at least $1 - \delta_{\mathrm{model}}$, guarantees

$$\sup_{\omega\in B(H)}\|v(\omega) - \widehat{v}_H(\omega)\|_\infty \le \eta.$$

Read the last supremum as zero when $B(H)$ is empty. When $B(H)$ is nonempty, form the midpoint $\widehat{m}$ and radius $\widehat{\rho}$ from $\widehat{v}_H$. If the returned numerical vector z satisfies $\|z - \widehat{m}\|_\infty \le \varepsilon$, then

$$\|z - v(\omega_*)\|_\infty \le \widehat{\rho} + \eta + \varepsilon \tag{6}$$

on an event of probability at least $1 - \delta_{\mathrm{id}} - \delta_{\mathrm{model}}$. The rule abstains on an empty candidate set. No independence between the two stated events is required.

Proof. For each coordinate j, any point z_j is at distance at least $(u_j - \ell_j)/2$ from one of the two extremes. Taking the largest coordinate width gives the lower bound. The midpoint is within half of every coordinate's width of every candidate, so it attains that bound. Inclusion can only narrow the coordinate ranges.

On the intersection of the coverage and model-accuracy events, the true world lies in the nonempty set and

$$\|z - v(\omega_*)\|_\infty \le \|z - \widehat{m}\|_\infty + \|\widehat{m} - \widehat{v}_H(\omega_*)\|_\infty + \|\widehat{v}_H(\omega_*) - v(\omega_*)\|_\infty.$$

The three terms are bounded by ε, $\widehat{\rho}$, and η. The union bound gives the probability statement. □

The center is allowed to be any vector in $\mathbb{R}^d$. Requiring it to be the prediction of one admissible model can increase the optimum. Nor does a coordinatewise midpoint automatically define a coherent stochastic process or preserve logical invariants. Those are additional constraints on the permitted approximation, not consequences of (5).

7.1 Three errors and two failure probabilities

The radius measures unresolved disagreement among retained candidates. The model discrepancy controls how well their represented predictions match the intended quantities. Numerical error concerns the computation of the returned answer. The coverage probabilities express the chance that the premises fail. They are not added to the numerical tolerance as if they had the same units.

The probability assertion is unconditional over the stated evidence procedure. It is not automatically the same bound conditional on a rare event such as issuance of a certificate. Likewise, repeatedly changing the candidate family or stopping rule requires a validity argument for the actual procedure. A confidence set cannot be assumed valid merely because its contents look plausible.

7.2 What compression does to the candidate set

For a fixed raw-record channel, let $B(h)$ be its exact support set. If only $c(h)$ is retained, the correct support is

$$B_c(z) = \bigcup_{h:c(h)=z} B(h).$$

Indeed, a world can produce the coarse record exactly when it can produce at least one raw record in that fibre. Thus $B(h) \subseteq B_c(c(h))$, and the ambiguity radius cannot decrease for the same target vector. This is an information statement, not a command to retain every byte. The correction signature in Section 5 identifies one principled way to compress while retaining the specified future uses.

For general confidence sets rather than exact supports, taking the same union still preserves the original coverage guarantee. It may be conservative. Selecting a smaller coarse-record set requires a new coverage argument, not the assertion that compression should make the model simpler.

Counterexample 7.2 (Agreement inside a misspecified family)**.** Suppose every candidate in a supplied family predicts zero for the target, but the actual unmodeled process produces one. The family has radius zero, and its internal arithmetic can be exact. The conclusion is nevertheless wrong because the true process is outside the represented family. Neither a narrow interval nor a successful proof about the family establishes that membership premise.

This is why model adequacy is not treated as a reward for confidence. In the present paper it is a stated premise, or a separate statistical event, rather than an automatic consequence of a concentrated representation.

8 A probability memory for repeated correction

The exact signature in Section 5 records which candidates remain possible. With noisy observations, all candidates can remain possible while their relative weights change substantially. Keeping only their support would then preserve logical possibility but discard the result of Bayesian correction. We now specify a different memory contract: retain both the probabilities of future reports and the posterior distribution of the selected target.

Let Ω be finite and nonempty, let $t : \Omega \to Y$ have finite image, and let $\mathcal{B}$ be a finite collection of experiments. Experiment e has a finite outcome set and specified likelihoods

$$\ell_{e,o}(\omega) \geq 0, \qquad \sum_o \ell_{e,o}(\omega) = 1.$$

The world is fixed. Every experiment remains available, and each new report has this conditional law given the world, the chosen experiment, and all earlier reports. This is a fresh conditional-observation premise, not a claim that copied readings become new samples. No experiment changes the physical state in this section.

Partition Ω into the nonempty cells $A_1, \ldots, A_m$ on which the complete likelihood vector $(\ell_{e,o}(\omega))_{e,o}$ is constant. Write its common value as $\lambda_{e,o,i}$. The profile vectors of different cells are distinct by construction. For a prior μ, retain

$$p_{i,y}(\mu) = \sum_{\omega \in A_i,\ t(\omega)=y} \mu(\omega), \qquad (i,y) \in D_t = \{(i,y) : A_i \cap t^{-1}(y) \neq \varnothing\}.$$

These are joint masses, not the two separate marginals. There are $r = |D_t|$ coordinates with nonnegative entries summing to one. Exact real probabilities are not a fixed r-bit code. A bit-complexity claim would also have to account for their representation and growing denominators.

Theorem 8.1 (Exact target correction from likelihood-profile memory)**.** *Under the fixed-world likelihood model, the record $p = (p_{i,y})$ supports every finite sequence of report predictions and target-posterior corrections. Its next-report probability and update are*

$$z_{e,o} = \sum_{i,y} \lambda_{e,o,i} p_{i,y}, \qquad p'_{i,y} = \frac{\lambda_{e,o,i} p_{i,y}}{z_{e,o}} \quad (z_{e,o} > 0),$$

and its target readout is $\Pr(t = y) = \sum_i p_{i,y}$. A zero-probability report is marked impossible rather than assigned a normalized posterior.

Conversely, any deterministic memory, initialized for every prior on Ω, that preserves all next-report probabilities and all target posteriors after every finite history must distinguish priors with different joint-mass records. In fact, the joint probabilities of the target and all experiment-report words of length at most $m - 1$ already determine the joint-mass record. The memory is therefore minimal in the sense of its induced equivalence on priors, not in machine words or physical storage.

Proof. Likelihoods are constant on each profile cell, so summing the ordinary Bayesian update over $A_i \cap t^{-1}(y)$ gives the displayed update. Its normalization and target readout are exact. Repeating this argument proves sufficiency for any fixed experiment sequence, and also for choices made from the retained report history without extra world-dependent information.

For necessity, a valid predictor and updater determine, for every word $w = ((e_1, o_1), \ldots, (e_k, o_k))$, the masses

$$M_{\mu,y}(w) = \sum_{t(\omega)=y} \mu(\omega) \prod_{j=1}^{k} \ell_{e_j,o_j}(\omega).$$

Indeed, multiply the successive predicted report probabilities by the final target posterior. If an earlier factor is zero, the joint mass is zero and no impossible conditional posterior is needed. The empty word gives the current target distribution.

For each cell i and each $j \neq i$, choose one likelihood coordinate $b(i, j) = (e, o)$ for which $\lambda_{b(i,j),i} \neq \lambda_{b(i,j),j}$. Such a coordinate exists because profile vectors are distinct. On Ω the polynomial

$$P_i(\omega) = \prod_{j \neq i} \frac{\ell_{b(i,j)}(\omega) - \lambda_{b(i,j),j}}{\lambda_{b(i,j),i} - \lambda_{b(i,j),j}}$$

is exactly $\mathbf{1}_{A_i}(\omega)$. It is a linear combination of likelihood monomials of degree at most $m - 1$. Each monomial is the likelihood product of an admitted fixed experiment-report word. Consequently $p_{i,y}(\mu) = \sum_{t(\omega)=y} \mu(\omega) P_i(\omega)$ is a linear combination of the corresponding $M_{\mu,y}(w)$. For $m = 1$ the empty product is one and the same argument uses only the current target masses. Two priors merged by an exact memory have equal predicted joint masses, hence equal joint-mass records. This proves the claimed separation and finite word-length bound. □

The proof uses a finite interpolation argument. Its polynomial coefficients are an identification device, not negative-probability experiments that a learner must execute. All likelihood values are supplied exactly. Very similar profile vectors can make the interpolation numerically ill-conditioned, so the theorem does not provide a stable estimator of p from noisy frequency estimates. Nor does it infer the true likelihood model from a narrow posterior. The distinction between sufficient experiments and compressed statistical evidence is classical [10]; the present contract additionally keeps the selected target and all future update operations explicit.

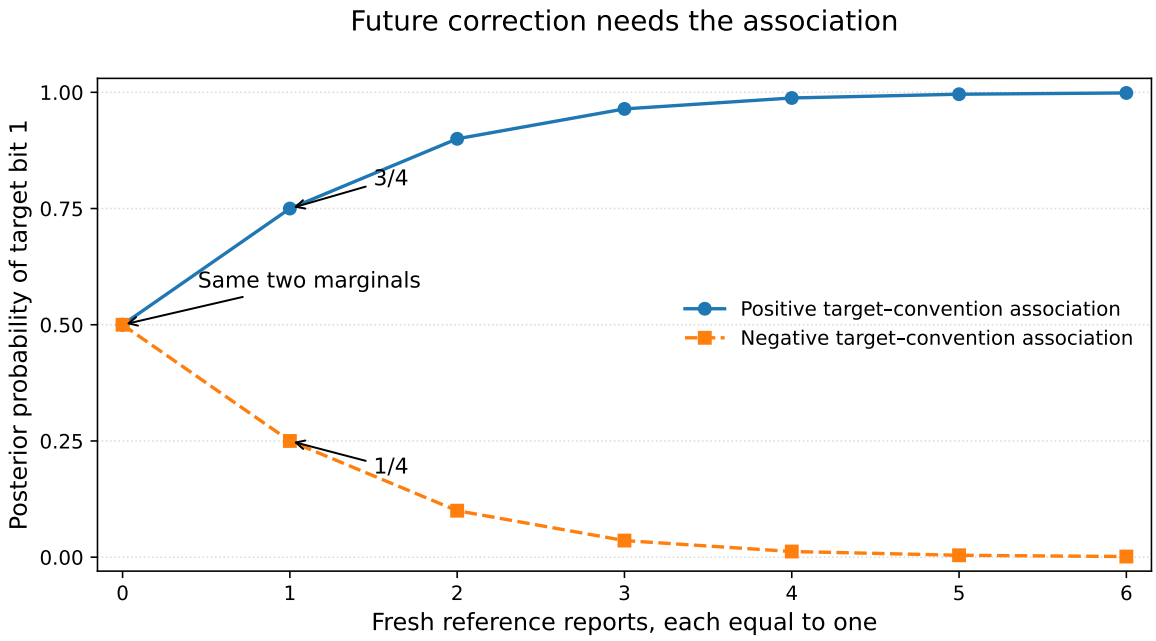


Figure 4: Exact posterior target probabilities for two priors with identical target and sensor marginals but opposite association. Each horizontal step is a fresh noisy-reference report equal to one. The curves are conditional analytical values, not measured learning performance. Four joint masses support the update; two separate marginals do not.

8.1 Two identical marginals, two opposite corrections

Return to the sensor worlds $\omega = (s, \theta, n) \in \{0,1\}^3$, where s is the target bit, θ is a convention, and n is irrelevant. Suppose the only future experiment is a nondestructive noisy reference with

$$\Pr(o = 1 \mid \theta = 0) = \tfrac{1}{4}, \qquad \Pr(o = 1 \mid \theta = 1) = \tfrac{3}{4}.$$

There are two likelihood-profile cells, one for each convention, and four nonempty profile–target pairs. The eight-world prior can be reduced to four joint masses without losing any specified prediction or correction. The nuisance coordinate disappears.

Consider a prior μ_+ assigning mass $1/2$ to $(s, \theta) = (0,0), (1,1)$ and a prior μ_- assigning mass $1/2$ to $(1,0), (0,1)$, with any fixed nuisance convention. Both have uniform s and uniform θ. Both predict the next report 1 with probability $1/2$. After that report, however,

$$\Pr_{\mu_+}(s = 1 \mid o = 1) = \tfrac{3}{4}, \qquad \Pr_{\mu_-}(s = 1 \mid o = 1) = \tfrac{1}{4}.$$

After k fresh reports all equal to one, these become $3^k/(1 + 3^k)$ and $1/(1 + 3^k)$. The difference is not a numerical error. It is the association that the separate marginals failed to retain. At every finite k, both target values remain possible under either prior.

8.2 What a revised evidence language requires

Let $\mathcal{P}_0$ be the partition of worlds into the nonempty old profile–target cells. Let $\mathcal{P}_1$ be the corresponding partition after likelihoods or the target have changed, still on the same finite Ω. The old record stores the probability of each $\mathcal{P}_0$ cell. The new record would store the probability of each $\mathcal{P}_1$ cell.

Corollary 8.2 (Exact migration of a probability memory)**.** *A deterministic map reconstructing the new record from the old record for every prior exists if and only if every $\mathcal{P}_0$ cell is contained in a single $\mathcal{P}_1$ cell. When this condition holds, summing the relevant old masses gives the new record. If the new evidence or target strictly splits an old cell, no such universal reconstruction exists.*

Proof. When every old cell lies inside a new one, each new mass is the sum of the old-cell masses it contains. Conversely, suppose two worlds lie in the same old cell but in different new cells. Point-mass priors on those worlds yield the same old record and distinct new records. No deterministic map can send that one old record to both required new records. □

Counterexample 8.3 (An old exact memory does not upgrade itself)**.** Two worlds have the same target and the same old report law, so point masses on them give an identical old record. A newly admitted experiment reports one with probabilities $1/4$ and $3/4$ in the respective worlds. The old record cannot predict its probability, although it was exact for every old question and update. Adding a field for the new likelihood profile does not determine its contents.

A restricted family of reachable priors can admit a reconstruction that fails for arbitrary priors. That restriction must be stated. Likewise, an archived raw record, a justified prior relation, or fresh evidence may help initialize the new memory, but its adequacy must be checked separately. Enlarging the candidate-world set is a different operation again: the old probabilities do not decide how much mass newly admitted worlds should receive.

8.3 Exact sufficiency is not a finite-precision guarantee

Even when two profiles differ only slightly, enough repeated evidence can make their posterior effects very different. Near-equal likelihoods are not interchangeable for an unbounded continuation contract. A finite-precision theory needs a horizon, a metric, a likelihood or evidence-mass condition, and a policy for outcomes of very small probability. Section 15 supplies a finite-horizon approximation contract with an explicit likelihood floor and retained-record denominator. It does not make rounding an exact sufficient statistic or provide an unbounded-time guarantee.

Part IV

Grounded continuation without complete identification

9 What a partial operator calibration determines

The finite-world examples show that complete identification can be unnecessary. A linear setting makes the positive result more structural. We will determine the exact domain on which every future operation sequence is already fixed by partial calibration data.

Let V be a finite-dimensional vector space over a field, and let $\mathcal{A}$ be a finite nonempty set of operation names. The unknown operations are linear maps $T_a : V \to V$. Calibration supplies input vectors $x_1, \dots, x_m$ and exact outputs $y_{a,1}, \dots, y_{a,m}$ for every a. The operator family is fixed, rather than being reselected after each action. Let

$$U = \operatorname{span}\{x_1, \dots, x_m\}.$$

The calibration is consistent precisely when every linear relation among the inputs is respected by the outputs, for each action. Under that condition it defines maps $S_a : U \to V$ with $S_a x_i = y_{a,i}$. Every S_a has linear extensions to V by extending a basis. Unless stated otherwise, *all* such extensions are admissible. There is no hidden norm bound, symmetry, invertibility condition, or preferred matrix completion.

The compatible-model viewpoint is shared with the data-informativity literature [7]. Our goal here is exact full-state prediction from a known initial vector, not stabilization, optimal control, or prediction from an unmeasured initial condition.

Proposition 9.1 (One-step identification from exact calibration)**.** *For a fixed action a, the value $T_a v$ is the same for every linear extension of S_a if and only if $v \in U$. The common value is then $S_a v$.*

Proof. If $v \in U$, all extensions agree by definition. If $v \notin U$, choose a linear functional ℓ that vanishes on U and satisfies $\ell(v) = 1$, and choose a nonzero vector $b \in V$. Given any extension T_a, the map $T'_a(x) = T_a(x) + \ell(x)b$ is another extension, but $T'_a v = T_a v + b \neq T_a v$. Such a functional exists by extending a basis of U with v. □

The proposition uses the complete unrestricted extension class. A justified structural prior can determine more. For example, knowing an operator belongs to a smaller parametric family can constrain its action outside U. Those constraints must be included as prior information, not inferred from the calibration span alone.

9.1 A known first step can leave the known domain

Even if $v \in U$, the vector $S_a v$ can lie outside U. A second action then acts at an input whose response was not calibrated. One-step accuracy is therefore insufficient for reusable operator meaning.

Define a descending sequence of subspaces

$$U_0 = U, \qquad U_{j+1} = \{v \in U : S_a v \in U_j \text{ for every } a \in \mathcal{A}\}. \tag{7}$$

The sequence is descending by induction. Every strict inclusion decreases dimension, so a fixed point is reached after at most $\dim U$ strict drops. Denote it by U_∞.

Theorem 9.2 (The maximal calibrated continuation domain)**.** *For every integer $n \geq 1$ and every $v \in V$, the values $T_w v$ for all operation words of length at most n are independent of the chosen compatible extension family if and only if*

$$v \in U_{n-1}.$$

Consequently, every finite full-state continuation is identifiable exactly on U_∞. This is the largest subspace $L \subseteq U$ such that $S_a L \subseteq L$ for every a. On this subspace the known restrictions themselves realize the complete operation family.

Proof. Membership in U_{n-1} means that every path of at most $n-1$ calibrated operations stays inside U. This follows by induction from (7), beginning with $U_0 = U$. All inputs on which the first n operations must act are consequently calibrated. Their outputs are determined by the S_a, independently of every completion outside U.

For necessity, suppose $v \notin U_{n-1}$. If $v \notin U$, the one-step proposition supplies two extensions that disagree within one action. Otherwise the recursive definition gives a word of length $j \le n-1$ with a first exit from U. Its intermediate vectors before exit lie in U, so its exit vector $z \notin U$ is the same under every compatible extension family. Choose any next action b. Two extensions of S_b can be made to disagree on z by adding a linear map that vanishes on U. Keep the other action maps the same. Even if action b occurred earlier, that modification has no effect before the exit because the earlier inputs lie in U. The two fixed extension families therefore produce different vectors after the exit word followed by b, whose total length is at most n. This contradicts uniform continuation identification.

At a fixed point, (7) gives $S_a U_\infty \subseteq U_\infty$. If $L \subseteq U$ is invariant under every restriction, induction gives $L \subseteq U_j$ for every j, so $L \subseteq U_\infty$. Applying the finite-horizon equivalence at all n proves the all-word assertion. □

This theorem supplies an identifiable *domain*, not necessarily an identified whole operator. It is exact for the stated completion class and full-state output. A smaller observation does not automatically enlarge the predictable domain: Proposition 11.1 proves that it does not under the present unrestricted completion class when the observation is nonzero. Sections 12–14 characterize the gains that become possible under explicitly structured uncertainty. Similarly, a learned nonlinear latent representation may compress the identifiable core further for its own tasks, but that is not part of this linear result.

9.2 A two-dimensional calibration with a one-dimensional reusable core

Counterexample 9.3 (Calibration does not close under continuation)**.** Let V have basis e_1, e_2, e_3, let $U = \operatorname{span}(e_1, e_2)$, and let the single calibrated operation satisfy

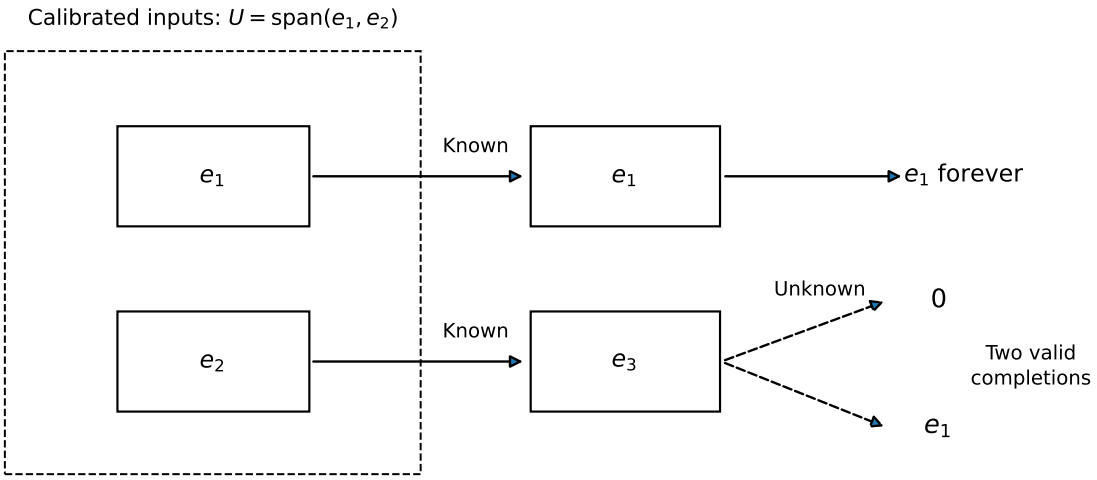
$$Se_1 = e_1, \qquad Se_2 = e_3.$$

The first continuation is known on all of U. The next calibrated domain is $U_1 = \operatorname{span}(e_1)$, which is already invariant. One completion has $T_0 e_3 = 0$ and another has $T_1 e_3 = e_1$. Both agree on every supplied calibration. Starting from e_2, their first outputs agree at e_3, but their second outputs are respectively zero and e_1.

Every continuation from e_1 is nevertheless determined. The system can ground a genuinely reusable operation on that line while remaining unable to predict repeated operations from e_2. A new exact calibration of e_3 expands the known input span to V and identifies the whole operator. Obtaining that calibration is a new experiment, not an algebraic consequence of the old data.

9.3 Several operators and the scope of the initial vector

With several actions, (7) retains only vectors that remain calibrated under *every* permitted word. An added action can therefore shrink the core even if the old operators and data are unchanged. Restricting the permitted grammar to a specific sequence can enlarge the domain of predictable use. The full free-word contract is deliberately stronger than one selected trajectory.



Known one-step domain: dimension 2. Identifiable all-word core: dimension 1.

Figure 5: A partial calibration can support unlimited exact continuation on a proper invariant core. The known image of e_2 leaves the calibrated input subspace, so its next image is unrestricted. Agreement with every supplied one-step calibration is not agreement on two-step behavior.

The initial vector v is known to the predictor in this section. If the actual input is hidden behind an evidence code, its initialization must satisfy an additional factorization condition like (1). A calibrated operation and an unobserved initial state are different missing inputs. They cannot be exchanged merely because both appear as uncertainty about the next output.

10 Why small local error needs an ambient growth bound

Exact calibration can justify a sharp domain. Approximate calibration needs a quantitative statement about errors that leave it. Let V be a normed vector space and $L \subseteq V$ a linear subspace invariant under known bounded linear maps $S_a : L \to L$. Suppose the actual bounded linear maps $T_a : V \to V$ satisfy

$$\|T_a\| \le A, \qquad \|S_a\| \le B, \qquad \|(T_a - S_a)v\| \le \eta\|v\| \quad (v \in L), \tag{8}$$

for common nonnegative bounds A, B, η. The norm on T_a is an ambient bound. Its purpose is to control propagation of errors outside L, not only behavior on calibrated inputs.

Proposition 10.1 (Approximate continuation with explicit leakage control)**.** *For every $v \in L$ and every word w of length $n \ge 1$,*

$$\|T_w v - S_w v\| \le \eta\|v\| \sum_{j=0}^{n-1} A^{n-1-j} B^j. \tag{9}$$

For the empty word the discrepancy is zero. Integer zero powers in the displayed finite sum have their usual value one. The conclusion does not require the actual trajectory to remain in L.

Proof. Write $z_j = S_{a_j} \cdots S_{a_1} v \in L$ and $x_j = T_{a_j} \cdots T_{a_1} v$. For the next operation,

$$\|x_{j+1} - z_{j+1}\| \le A\|x_j - z_j\| + \eta\|z_j\| \le A\|x_j - z_j\| + \eta B^j \|v\|.$$

The initial discrepancy is zero. Iterating this scalar recurrence gives (9). Ambient boundedness applies to $x_j - z_j$ even when that error has left L. □

This is a telescoping error estimate, not a new perturbation principle. Similar continuation bounds already appear in the collection [1, 3]. Its role here is to identify a premise that calibration on a proper subspace cannot certify by itself.

Counterexample 10.2 (An uncalibrated direction amplifies a small residual)**.** Let $L = \operatorname{span}(e_1)$, let $Se_1 = e_1$, and in the ℓ^1 norm define

$$T = \begin{pmatrix} 1 & 0 \\ \eta & M \end{pmatrix}, \qquad \eta > 0,\ M \geq 0.$$

The one-step error on the unit calibrated vector is η, independent of M. After two steps the error is $\eta(1 + M)$. It can be arbitrarily large while the local calibration error is fixed. For n steps the second coordinate is $\eta \sum_{j=0}^{n-1} M^j$.

A learned model can therefore match calibrated transitions closely and still fail rapidly under repetition. Increasing one-step accuracy helps only in relation to the length of use and the growth of the unseen directions. A measured small residual does not imply an ambient norm bound. The bound can come from a declared operator class, a separate global calibration argument, or a stronger structural certificate, but not from renaming the local fit "stable meaning."

11 A smaller output is not automatically a weaker calibration requirement

The preceding full-state theorem suggests asking whether a small observable can remain predictable outside the calibrated core. The answer depends on the uncertainty class. Under the unrestricted extension class of Section 9, simply requesting fewer output coordinates does not enlarge the all-word identifiable domain unless the requested observation is identically zero.

Let $Q : V \to Y$ be a fixed linear observation. The initial vector v remains known. All linear extensions of every calibrated restriction S_a remain permitted, independently across action names but fixed throughout an execution.

Proposition 11.1 (No enlargement under unrestricted linear completion)**.** *If $Q \neq 0$, then for every $h \geq 1$, the vectors $QT_w v$ are independent of the compatible extension family for every word of length at most h if and only if $v \in U_{h-1}$. In particular, the all-word output-identifiable domain is exactly U_∞. If $Q = 0$, every initial vector is trivially output-identifiable.*

Proof. On U_{h-1} the full vectors are determined by Theorem 9.2, so their Q outputs are determined. Outside that subspace, follow the same first-exit construction as in that proof. At an uncalibrated input z, choose a linear functional ℓ vanishing on U with $\ell(z) = 1$. Because $Q \neq 0$, choose b with $Qb \neq 0$. Replacing the next action by $T_a + \ell(\cdot)b$ preserves all calibration data and every earlier step of the first-exit path, but changes the next observed output by Qb. This produces disagreement within h actions. Each modified action map is fixed, not chosen anew during execution. The zero-output case follows immediately. □

This corrects an overly broad suggestion in the first draft's surrounding prose. Output-relative prediction can be less demanding under *structured* uncertainty. It is not automatically less demanding when every uncalibrated direction is allowed to return arbitrarily into the output. The previous full-state theorem and its proof are unchanged.

The next sections therefore change the model class explicitly. We first consider a finite family of complete candidate systems, retaining the fact that one unknown candidate is fixed. We then characterize the stronger

situation in which an entire uncertainty family admits one common quotient. These are different questions and have different answers.

12 A finite test for agreement under a fixed unknown model

Let V be n-dimensional over a field, let Θ be a finite nonempty model set, and let $T_a^\theta : V \to V$ be specified linear maps for each model θ and action a in a finite nonempty alphabet. All words are allowed. Choose a reference model $0 \in \Theta$ and write $A_a = T_a^0$. The shared linear observation is $Q : V \to Y$. Define

$$I_Q = \{v \in V : QT_w^\theta v = QA_w v \text{ for every } \theta \in \Theta \text{ and every word } w\}.$$

This is an initial-condition subspace. It is not assumed invariant under the physical maps. The common coordinates are part of this linear comparison model. With unrelated latent carriers, a justified identification of the carriers would be an additional premise.

For each θ, form the paired maps and difference observation

$$B_a^\theta = \begin{pmatrix} T_a^\theta & 0 \\ 0 & A_a \end{pmatrix}, \qquad C = \begin{pmatrix} Q & -Q \end{pmatrix}, \qquad \Delta v = (v, v).$$

The pair is evolved by the *same* action word in both components. Define

$$K_0^\theta = \ker C, \qquad K_{j+1}^\theta = K_j^\theta \cap \bigcap_a (B_a^\theta)^{-1} K_j^\theta.$$

Theorem 12.1 (Finite output-agreement test with persistent model tags)**.** *The descending paired-kernel sequence stabilizes in at most* $2n$ *strict dimension drops. Its terminal subspace is*

$$K_\infty^\theta = \bigcap_w \ker(CB_w^\theta), \qquad I_Q = \bigcap_{\theta \in \Theta} \Delta^{-1} K_\infty^\theta.$$

Thus exact all-word output agreement for the finite candidate family is decidable by finite linear algebra. If an initial vector is excluded, a disagreeing candidate and a distinguishing word can be extracted. Words of length at most $2n$ *suffice for this test. The conclusion compares fixed candidate models; it does not admit arbitrary switching between candidates during a word.*

Proof. By induction, K_j^θ consists precisely of paired states with zero difference output for all words of length at most j. The initial case is the empty word. Membership in the next intersection says that the present difference is zero and that applying any first action leaves a pair whose next j actions have zero difference. This is exactly the length-$j + 1$ condition.

Every strict inclusion lowers dimension in $V \oplus V$, which has dimension $2n$. Once two successive subspaces agree, the subspace is invariant under every B_a^θ and lies in $\ker C$, so all later intersections agree. This proves the all-word formula. At the diagonal initial pair, $CB_w^\theta \Delta v = QT_w^\theta v - QA_w v$, giving the paired-kernel formula. Row-space construction retains the operation word producing each tested row. If a vector fails the terminal kernel test, one such row is nonzero on it and supplies a witness. The dimension argument gives the stated conservative length bound. □

Finite linear-algebra equivalence testing has established precedents, including algorithms for probabilistic automata [11]. Reachability, observability, and minimal realization for multiple linear operations are also established subjects [12]. The contribution here is the fixed-compatible-model interpretation and its connection to evidence initialization and correction. We do not claim a new general automata-equivalence algorithm.

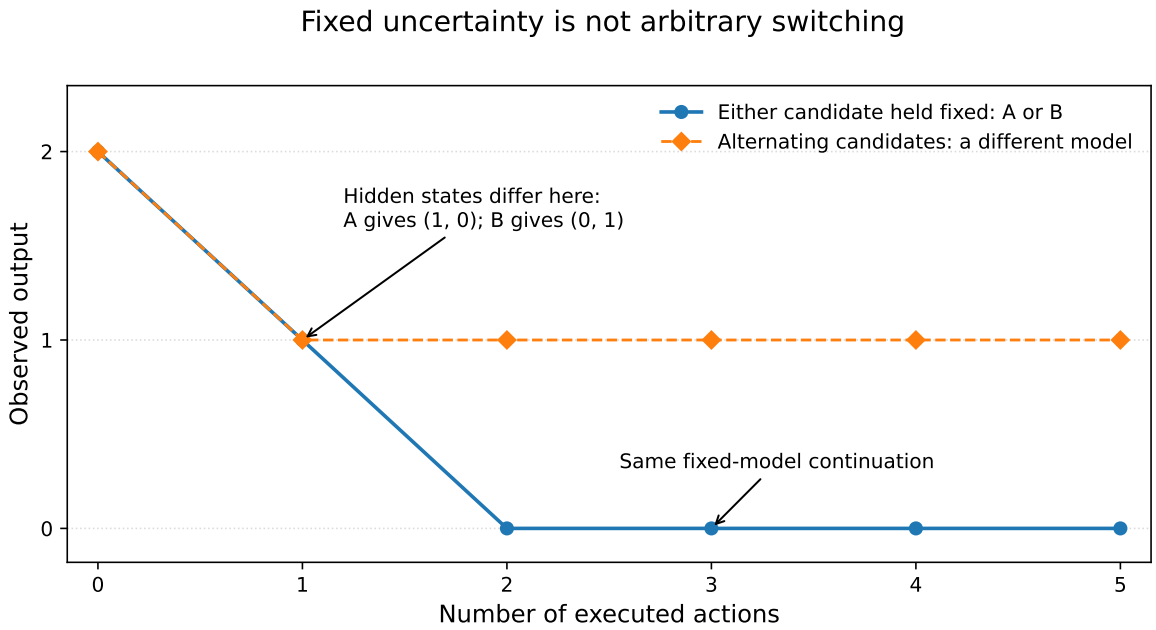


Figure 6: Two fixed candidate systems produce the same complete output history from $(1, 1)$ even though their hidden states differ. Switching from one candidate to the other after the first step creates an inadmissible history with a different second output. The fixed-model tag is information about the allowed trajectories.

12.1 Why the unknown model must not change between steps

Counterexample 12.2 (A switching adversary creates a nonexistent failure)**.** Over $\mathbb{Q}$, let there be one action and two possible fixed matrices

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \qquad Q = \begin{pmatrix} 1 & 1 \end{pmatrix}, \qquad v = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Both fixed systems have output sequence $2, 1, 0, 0, \ldots$ from v, since $A^2 = B^2 = 0$. But applying A first and B second gives $QBAv = 1$, not zero. That hybrid trajectory belongs to neither candidate model. Treating fixed uncertainty as a freely changing matrix would incorrectly reject the valid output prediction.

Here $I_Q = \operatorname{span}\{(1, 1)\}$, obtained already from the one-step equality. It is not invariant under A or B: $Av = (1, 0)$ and $Bv = (0, 1)$ leave that line. Those new physical vectors cannot be treated as fresh initial conditions with a forgotten model history. For example, starting both models at $(1, 0)$ gives different next outputs. The original fixed-model executions nevertheless remain in output agreement.

This is why the identifiable initial set is not automatically a common physical invariant core. A reference predictor may still work: it follows A from the initial vector and reports its outputs. It represents the common response sequence, not a claim that the unknown physical state always equals the reference state.

12.2 More data restricts the model family, not the logical quantifier

If new exact evidence removes candidates while preserving the reference model, the intersection defining I_Q can only enlarge. Its practical value depends on whether the learner can justify that removal and initialize the resulting predictor. Adding a new candidate has the opposite effect. Neither operation warrants replacing the model label at each step. A candidate family that omits the true system can still yield an impressive but inapplicable agreement subspace.

13 When uncertain dynamics admit one common observable quotient

Agreement on selected initial conditions is weaker than agreement on the whole space. The stronger problem has a useful exact solution. Retain the finite family and reference matrices of Section 12, and define the difference maps

$$D_a^\theta = T_a^\theta - A_a.$$

Their ranges contain all possible one-step discrepancies. Let

$$E_0 = \sum_{a,\theta} \operatorname{im} D_a^\theta, \qquad E_{j+1} = E_j + \sum_a A_a E_j, \qquad E_\infty = \bigcup_j E_j.$$

This increasing sequence stabilizes after at most n strict dimension increases. Its limit is the smallest subspace invariant under all reference actions and containing every discrepancy range. It includes discrepancies that are invisible now but return to the output later.

Theorem 13.1 (Exact common quotient for structured model uncertainty)**.** *The following conditions are equivalent.*

1. $QT_w^\theta = QA_w$ *on all of* V*, for every fixed model* θ *and every word* w*.*
2. $E_\infty \subseteq \ker Q$*.*
3. *There exists a subspace* $N \subseteq \ker Q$*, invariant under every* A_a*, containing every* $\operatorname{im} D_a^\theta$*.*
4. *There is a surjective linear representation* $r : V \to Z$*, shared linear updates* G_a*, and a readout* $\bar{Q}$ *such that*

$$rT_a^\theta = G_a r \quad \text{for all } a, \theta, \qquad Q = \bar{Q} r.$$

When the conditions hold, V/E_∞ *gives one such quotient. The smallest-dimensional common linear quotient is* V/N_0*, where* $N_0 = \bigcap_w \ker(QA_w)$ *is the reference system's unobservable subspace. Its dimension is* $n - \dim N_0$*. The shared quotient also preserves outputs if the candidate model is changed between actions, a stronger conclusion than fixed-model agreement on a restricted initial set.*

Proof. Assume condition 1. For any a, θ, w, agreement as linear maps on all of V gives

$$QA_w D_a^\theta = QT_w^\theta T_a^\theta - QA_w A_a = 0.$$

The last equality is condition 1 for the word that executes a and then w. Thus Q annihilates the range of every $A_w D_a^\theta$, whose span is E_∞, proving condition 2. Condition 2 supplies condition 3 with $N = E_\infty$.

Under condition 3, $T_a^\theta N \subseteq N$ because $T_a^\theta = A_a + D_a^\theta$, $A_a N \subseteq N$, and the entire range of D_a^θ lies in N. Every candidate therefore induces the same map $G_a([v]) = [A_a v]$ on V/N. Since $QN = 0$, the output factors through that quotient. This proves condition 4. Composing its commuting equations proves condition 1. The same composition works with a different model index at each action because r and G_a are shared.

The subspace N_0 is invariant under every A_a and lies in $\ker Q$. Every invariant subspace in $\ker Q$, including E_∞ when condition 2 holds, lies in N_0. Hence N_0 gives a valid common quotient. Conversely, the kernel of any representation in condition 4 is invariant under the reference actions and contained in $\ker Q$, so it lies in N_0. Rank-nullity gives the lower bound $\dim Z \geq n - \dim N_0$, attained by V/N_0. □

13.1 Large hidden variation with exact visible behavior

For a structured example over $\mathbb{Q}$, take either of the fixed maps

$$T_0 = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad T_1 = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 2 & -1 & 3 \end{pmatrix}, \qquad Q = \begin{pmatrix} 1 & 0 & 0 \end{pmatrix}.$$

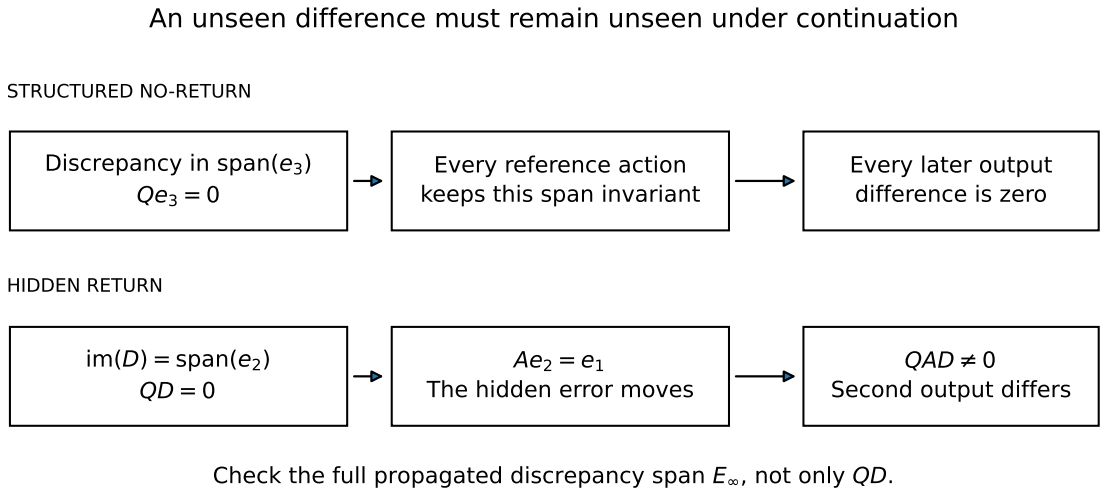


Figure 7: Structured uncertainty may be harmless when every difference remains in an invariant, output-invisible subspace. The opposite example has zero first-step output discrepancy but a visible second-step return. The figure represents exact algebraic alternatives, not a claim that small learned residuals certify the invariant subspace.

The hidden third-coordinate dynamics are different, and T_1 can make that coordinate grow. Nevertheless the discrepancy range is $\mathrm{span}\{e_3\}$, which the reference leaves inside itself and Q annihilates. The common quotient retains (x_1, x_2), with update

$$G(x_1, x_2) = (x_1 + x_2, x_2).$$

The output at step k is $x_1 + kx_2$ under both models, and even under switching between them. The output alone is not an initialized reusable state: equal x_1 with different x_2 gives different next outputs. Ignoring x_3 is justified by the no-return structure, not by its small size or its being temporarily unobserved.

Counterexample 13.2 (An invisible difference returns one step later)**.** Let

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad D = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \qquad Q = \begin{pmatrix} 1 & 0 \end{pmatrix}.$$

The candidates are A and $A + D$. Since $QD = 0$, every one-step output agrees for every input. But $QAD = (1, 0) \neq 0$. Starting at e_1, the second outputs are zero and one. The initial discrepancy range is invisible, while its propagated span $E_\infty = V$ is not. Checking only $QD = 0$ misses the failure.

13.2 What the criterion does not learn

The family $\{T_a^\theta\}$, its shared carrier, and Q are given. The closure in the discrepancy-span recursion uses exact linear algebra and decides whether those models share the stated quotient. It neither discovers the model class from finite observations nor proves that the true mechanism belongs to it. A structural prior such as an invariant hidden block can justify a smaller uncertainty family, but that prior must be carried as evidence or an explicit assumption.

14 Initializing the smallest common linear predictor

The paired-state criterion supplies a domain of identifiable output behavior. A learner may still lack the information needed to select the correct behavior within that domain. We now combine the two obligations without requiring the identifiable initial subspace to be physically invariant.

Let $L \subseteq I_Q$ be a specified linear initial-condition subspace and let the available initial record be a linear map $R : L \to H$. Define the reference reachable span and its unobservable part by

$$W = \operatorname{span}\{A_w v : v \in L,\ w \text{ any word}\}, \qquad N = \{z \in W : QA_w z = 0 \text{ for every word } w\}.$$

Both W and N are invariant under the reference actions, and $N \subseteq \ker Q$. These spaces can be computed by finite reachable-span and row-kernel iterations. They concern a reference predictor, not a claim that every physical candidate remains inside W.

Theorem 14.1 (Evidence-initialized minimal common linear prediction)**.** *There is a finite-dimensional linear predictor with initialization $d : R(L) \to Z$, updates $G_a : Z \to Z$, and readout $\bar{Q} : Z \to Y$ satisfying*

$$\bar{Q} G_w d R(v) = Q T_w^\theta v \quad \text{for every } v \in L, \theta \in \Theta, w$$

if and only if $\ker R \subseteq N$. *When it exists, the smallest possible state-space dimension among such* linear *predictors is*

$$\dim W - \dim N.$$

The quotient W/N, initialized by $d(Rv) = [v]$, attains that minimum.

Proof. If $v \in \ker R$, linear initialization gives the same state as input zero. Exactness therefore requires $QA_w v = 0$ for every word, so $v \in N$. Conversely, if $\ker R \subseteq N$, the rule $d(Rv) = [v]$ is well defined and linear. Invariance supplies $G_a([z]) = [A_a z]$, and $QN = 0$ supplies $\bar{Q}([z]) = Qz$. Their iteration gives $QA_w v$, which agrees with every candidate by $L \subseteq I_Q$.

For minimality, let any competing linear predictor be given. Its reachable span is generated by vectors $G_w d R(v)$. Associate such a vector with $[A_w v] \in W/N$ and extend by linearity. To see this association is well defined, suppose a finite linear combination of the predictor vectors is zero. Applying any later word and the predictor readout shows that the matching combination of reference vectors has zero QA_u output for every u. That combination therefore belongs to N. The induced map from the competitor's reachable span onto W/N is a well-defined linear surjection, since the classes of $A_w v$ span W/N. Its dimension cannot be smaller than the displayed quotient dimension. The constructed quotient attains equality. □

The dimension statement is a linear-realization theorem in a declared algebraic model, not a lower bound for arbitrary nonlinear encodings, unbounded-precision tricks, or shortest source programs. Reachability and observability are classical tools [12]. The distinctive conjunction here is compatibility across the retained models, initialization from the actual record, and the smallest common linear predictor.

14.1 Initial dimension, dynamic memory, and hidden state

In Counterexample 12.2, set $L = \operatorname{span}\{(1,1)\}$ and $R(x) = x_1$. The reference reaches $(1,1)$ and $(1,0)$, so $W = \mathbb{Q}^2$. The rows $Q = (1,1)$ and $QA = (0,1)$ have trivial common kernel, giving $N = 0$. A two-dimensional linear predictor is therefore necessary even for this one-dimensional initial family: it must reproduce $2t, t, 0, 0, \ldots$ for every multiplier t.

For the three-coordinate structured example, $L = V$, $W = V$, and $N = \operatorname{span}\{e_3\}$. Two predictor coordinates suffice. The record (x_1, x_2) initializes them; x_1 alone does not, since its kernel contains e_2, which becomes visible after one action. Physical dimension, initial-family dimension, and minimum predictive dimension are different quantities.

Part V

Quantitative grounding with finite retained memory

15 From exact probability coordinates to a finite record

The likelihood-profile theorem identifies what an exact statistical memory must retain. It does not make its probability coordinates finitely representable. A rational posterior can acquire larger denominators after each update. Rounding those coordinates changes the object, so a finite-record theory needs a stated error, horizon, and observation model.

We now give one such theory. The result deliberately uses a conservative positive-likelihood condition rather than claiming uniform stability for every observation process. Stability and numerical approximation of nonlinear filters have an established literature [13]. Our construction concerns a static hidden world, a finite cell record, and an explicit finite-horizon error account. It is not a particle-filter theorem and does not inherit a contraction hypothesis from that literature.

15.1 A concrete probability record

For an integer $k \geq 1$, write Δ_k for the probability simplex on k cells. Fix a denominator $N \geq 1$. A retained record has the form

$$\widehat{p} = (n_1/N, \ldots, n_k/N), \qquad n_i \in \{0, \ldots, N\}, \quad \sum_i n_i = N.$$

The cell identities, denominator, and decoding rule are fixed background. The k integer fields can be stored literally in $k\lceil \log_2(N+1) \rceil$ bits. This is an upper bound for this specified encoding, not a minimum description length. It excludes the likelihood table, program, temporary arithmetic, and any retained raw observation history.

Define $Q_N(p)$ by first setting $n_i = \lfloor N p_i \rfloor$. Distribute the remaining $N - \sum_i n_i$ units to the largest fractional remainders, with ties broken by a fixed cell order. This makes the normalization exact after every rounding. It is not independent coordinate rounding followed by an unspecified repair.

For a likelihood vector $\ell = (\ell_i)$ define the Bayes map

$$\mathcal{B}_\ell(p)_i = \frac{p_i \ell_i}{\sum_j p_j \ell_j}$$

whenever its denominator is positive.

Proposition 15.1 (Rounding and positive-likelihood update bounds)**.** *For every $p \in \Delta_k$,*

$$d_{\mathrm{TV}}(p, Q_N(p)) \leq \frac{k}{2N}.$$

Let $p, q \in \Delta_k$, let $0 < \beta \leq 1$, and let every entry of ℓ and $\widetilde{\ell}$ lie in $[\beta, 1]$. If $\max_i |\ell_i - \widetilde{\ell}_i| \leq \varepsilon$, then

$$d_{\mathrm{TV}}\big(\mathcal{B}_\ell(p), \mathcal{B}_{\widetilde{\ell}}(q)\big) \leq \frac{d_{\mathrm{TV}}(p, q) + \varepsilon}{\beta}.$$

Each right-hand side may also be replaced by its minimum with one. The probability-field storage bound above applies to the retained rounded record, not to all working memory.

Proof. The remainder rule produces nonnegative integers summing to N. Each rounded coordinate differs from the original by at most $1/N$, so half the sum of absolute differences is at most $k/(2N)$.

For the update, first hold ℓ fixed. Given $0 \le f_i \le 1$, put $a = \sum_i \mathcal{B}_\ell(q)_i f_i$ and $Z_p = \sum_i p_i \ell_i$. Then

$$\sum_i \big(\mathcal{B}_\ell(p)_i - \mathcal{B}_\ell(q)_i\big) f_i = \frac{\sum_i (p_i - q_i)\ell_i(f_i - a)}{Z_p}.$$

Every value $\ell_i(f_i - a)$ lies in $[-a, 1-a]$, an interval of length one. Since $p - q$ has total mass zero, its pairing with a function of oscillation at most one is bounded in absolute value by $d_{\mathrm{TV}}(p, q)$. Also $Z_p \ge \beta$. Taking the supremum over f bounds the fixed-likelihood difference by $d_{\mathrm{TV}}(p, q)/\beta$.

Next hold q fixed and set $a = \sum_i \mathcal{B}_{\widetilde{\ell}}(q)_i f_i$. The corresponding numerator is $\sum_i q_i(\ell_i - \widetilde{\ell}_i)(f_i - a)$, whose absolute value is at most ε. Its denominator is at least β. A triangle inequality proves the displayed update bound. The cap by one is the universal range of total variation. The storage assertion counts the stated integer fields only. □

The estimate is sufficient and can be loose. In particular, it need not credit favorable observations that reduce an existing error. Its purpose is a guarantee whose entire cost and probability assumptions are visible, not an assertion that rounding usually produces its worst possible error.

15.2 Approximate likelihood profiles

Let Ω be a finite static world set with prior μ_0. A fixed partition $C_1, \dots, C_k$ refines the target: $t(\omega) = t_i$ throughout C_i. Each admitted experiment e is nondestructive and has a finite report alphabet. Its actual conditional probabilities are $\ell_{e,o}(\omega)$. Reports are freshly generated from those likelihoods conditional on the fixed world and the chosen experiments.

Choose one valid representative report law $\bar{\ell}_{e,o,i}$ per cell. Assume, for every admitted experiment, report, cell, and world in that cell,

$$\beta \le \ell_{e,o}(\omega), \bar{\ell}_{e,o,i} \le 1, \qquad |\ell_{e,o}(\omega) - \bar{\ell}_{e,o,i}| \le \varepsilon. \tag{10}$$

Both the actual and representative report laws sum to one over reports. Exact profile cells satisfy this condition with $\varepsilon = 0$ when their likelihoods have the positive floor. A coarser partition can be used with a nonzero ε, but target-constant cells are still required for the target readout.

After a history, let p_t be the cell masses of the exact world posterior. Initialize the retained record by $\widehat{p}_0 = Q_N(p_0)$. For the next selected experiment and observed report use

$$\widehat{p}_{t+1} = Q_N\Big(\mathcal{B}_{\bar{\ell}_{e,o}}(\widehat{p}_t)\Big).$$

The reference posterior and rounded record are compared along the same actual action/report history. Experiment selection may depend on earlier retained records and independent randomness. It may not introduce a hidden world-dependent selector whose information is omitted from that history. The likelihood condition applies to every experiment the policy can select.

Theorem 15.2 (Finite retained memory for a declared correction horizon)**.** *Under (10), set $\eta_N = k/(2N)$, $\gamma = \beta^{-1}$, and let e_0 be any valid bound on $d_{\mathrm{TV}}(p_0, \widehat{p}_0)$. For every possible action/report history of length t,*

$$d_{\mathrm{TV}}(p_t, \widehat{p}_t) \le e_t := \min\left\{1,\ \gamma^t e_0 + (\gamma\varepsilon + \eta_N) \sum_{j=0}^{t-1} \gamma^j \right\}. \tag{11}$$

The same bound holds after mapping cells to target values. With the generic initialization bound $e_0 \le \eta_N$, define

$$A_H = \sum_{j=0}^{H} \gamma^j, \qquad B_H = \gamma \sum_{j=0}^{H-1} \gamma^j.$$

For any horizon $H \ge 0$ and tolerance $\tau > \varepsilon B_H$, every integer

$$N \ge \max\left\{1, \left\lceil \frac{kA_H}{2(\tau - \varepsilon B_H)} \right\rceil\right\} \tag{12}$$

is sufficient for target and cell error at most τ at every time through H. Empty sums are zero. This is a finite-horizon retained-record guarantee, not an unbounded-time or total-computation guarantee.

Proof. Condition on a particular attainable history and its next experiment. In each positive-mass cell, the exact next likelihood is the conditional average of $\ell_{e,o}(\omega)$ within that cell. It remains in $[\beta, 1]$ and within ε of $\bar{\ell}_{e,o,i}$. For a zero-mass cell any value obeying these bounds gives the same zero contribution. Thus the exact next cell mass is a Bayes update with such an effective likelihood vector, even when the worlds within a cell do not have identical laws.

Proposition 15.1 and the rounding bound give

$$d_{\mathrm{TV}}(p_{t+1}, \widehat{p}_{t+1}) \le \gamma d_{\mathrm{TV}}(p_t, \widehat{p}_t) + \gamma\varepsilon + \eta_N.$$

Iterating the uncapped scalar bound proves (11); total variation is also at most one. Aggregating cell coordinates into target coordinates cannot increase total variation, by the triangle inequality. Substituting $e_0 \le \eta_N$ gives $\eta_N A_t + \varepsilon B_t$. These quantities are nondecreasing in t because $\gamma \ge 1$ and all terms are nonnegative. Inequality (12) makes the horizon-H value at most τ. It therefore suffices at every earlier step as well. Every report has positive probability under both updates by the floor assumption, so no zero denominator is silently normalized. □

Equation (12) is a sufficient precision requirement, not an exact minimum. An infeasible or large bound does not prove that the particular rounded filter is inaccurate. It means that this uniform certificate is insufficient. Conversely, increasing precision cannot eliminate the model term εB_H. A false likelihood model does not become correct because its probabilities are stored with more bits.

Counterexample 15.3 (A tiny rounded-away mass can control a later posterior)**.** Take two worlds with prior $(1 - d, d)$, where $0 < d < 1/(2N)$. The remainder rule gives the retained record $(1, 0)$. A binary report has likelihoods $(d^2, 1)$ for outcome one. After that outcome, the actual posterior mass of the second world is

$$\frac{d}{d + (1-d)d^2} = \frac{1}{1 + d - d^2},$$

which tends to one as d tends to zero. The rounded posterior remains zero on that world. The initial total-variation error is only d. There is no common positive likelihood floor across this family: the first likelihood tends to zero, and the other report is impossible in the second world. No nontrivial uniform continuation conclusion follows from the small initial rounding error alone.

The counterexample concerns evidence amplification, not a failure of the remainder arithmetic. Replacing the likelihood one by $1 - d^2$ makes both outcomes possible in both worlds while preserving the limiting phenomenon. The minimum likelihood still tends to zero. Exact support, approximate probability, and justified abstention should therefore remain distinct data.

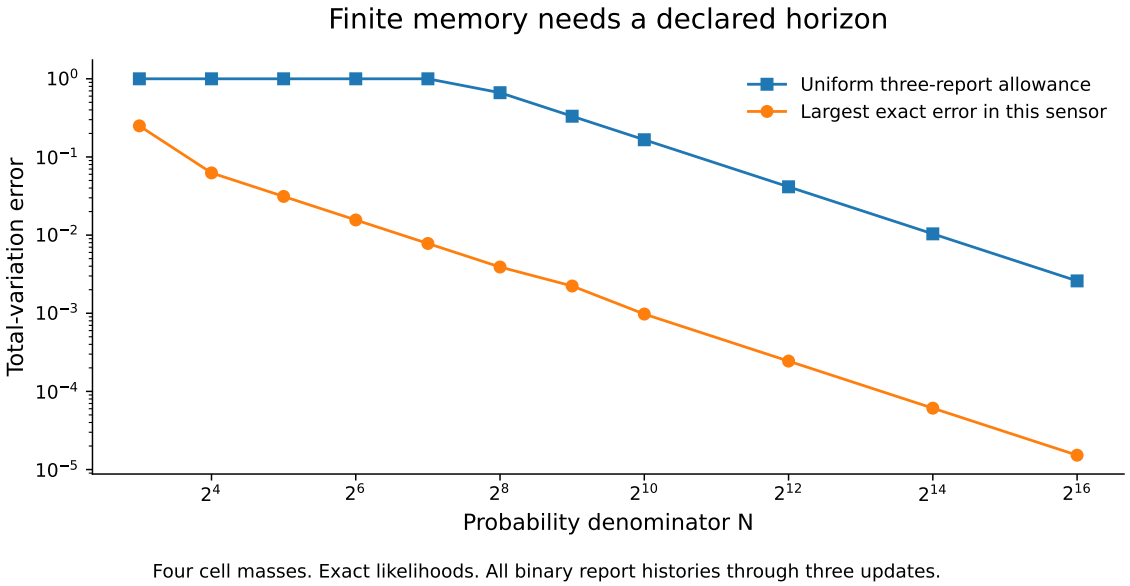


Figure 8: The noisy sensor with four retained cell masses, three future reports, and several denominator choices. The actual maximum error is enumerated over all eight three-report histories and their prefixes. The upper curve is the uniform positive-likelihood bound using generic rounded initialization. The gap is conservatism, not evidence of an uncounted learning gain. The installed likelihoods are exact; retained-record size does not count the program or temporary arithmetic.

16 Approximate invisibility must survive feedback

The exact common-quotient theorem allows arbitrarily large hidden dynamics when those directions never return to the selected output. Approximation changes that fact. A small hidden-to-visible coupling can make hidden growth consequential. This section states the relevant dependence explicitly rather than replacing exact invariance by a small residual with no continuation bound.

Approximate simulation and bisimulation offer established ways to compare observable behavior with a declared precision [14]. The following is a narrower block comparison argument for discrete linear operations. It does not claim that its matrix bound is a minimal simulation metric or that stability of every candidate separately supplies a common switching bound.

Write the physical state as (v_t, h_t) in a direct sum of two finite-dimensional normed spaces. The required output is the visible component v_t. Along an admitted execution suppose

$$\begin{pmatrix} v_{t+1} \\ h_{t+1} \end{pmatrix} = \begin{pmatrix} G_t & B_t \\ C_t & H_t \end{pmatrix} \begin{pmatrix} v_t \\ h_t \end{pmatrix}, \qquad \bar{v}_{t+1} = G_t \bar{v}_t. \tag{13}$$

The reduced predictor knows the chosen G_t and receives the same action sequence. It does not observe h_t. The blocks may come from a fixed unknown candidate or from an explicitly admitted varying family. Assume common operator-norm bounds

$$\|G_t\| \le g, \quad \|B_t\| \le b, \quad \|C_t\| \le c, \quad \|H_t\| \le d, \qquad g, b, c, d \ge 0.$$

Let $\|v_0\| \le V_0$, $\|h_0\| \le H_0$, and $\|v_0 - \bar{v}_0\| \le e_0$. These initial bounds are supplied evidence or declared model assumptions. Put

$$M = \begin{pmatrix} g & b \\ c & d \end{pmatrix}, \qquad u_0 = \begin{pmatrix} V_0 \\ H_0 \end{pmatrix}, \qquad u_j = M^j u_0.$$

Powers of this nonnegative comparison matrix are bounds on norms. They are not a replacement physical dynamics inferred from measurements.

Theorem 16.1 (Visible-error bounds from a common block comparison)**.** *Every admitted execution satisfying (13) obeys*

$$\|v_t - \bar{v}_t\| \le d_t := g^t e_0 + b \sum_{j=0}^{t-1} g^{t-1-j} (u_j)_2. \tag{14}$$

In particular, when $b = 0$ and $e_0 = 0$, the visible prediction is exact for every finite time, without a stability requirement on d or c.

If $g < 1$, $d < 1$, and

$$D = (1-g)(1-d) - bc > 0,$$

then the total visible discrepancy is bounded by

$$\sum_{t=0}^{\infty} \|v_t - \bar{v}_t\| \le \frac{e_0}{1-g} + \frac{b\big(cV_0 + (1-g)H_0\big)}{(1-g)D}. \tag{15}$$

These are sufficient common bounds. Failure of the inequality for D is not a proof that a particular signed system is unstable. Conversely, the two diagonal conditions alone are insufficient for a uniform conclusion over the stated block class.

Proof. The norm triangle inequality and the common block bounds give

$$\begin{pmatrix} \|v_{t+1}\| \\ \|h_{t+1}\| \end{pmatrix} \le M \begin{pmatrix} \|v_t\| \\ \|h_t\| \end{pmatrix} \le M^{t+1} u_0$$

componentwise. The error satisfies

$$v_{t+1} - \bar{v}_{t+1} = G_t (v_t - \bar{v}_t) + B_t h_t.$$

Induction gives (14), including its empty sum at $t = 0$. If $b = 0$ and the initial error is zero, the recurrence has no forcing term and remains zero, regardless of hidden growth.

Under the three strict inequalities, $(I - M)^{-1} = D^{-1} \begin{pmatrix} 1-d & b \\ c & 1-g \end{pmatrix}$ has nonnegative entries. Let $a = (I - M)^{-1}(1,1)^T$. Both coordinates are positive and $Ma = a - (1,1)^T < a$. Hence M is a strict contraction in the weighted maximum norm with weights a, and its geometric series converges to $(I - M)^{-1}$. Summing the nonnegative right-hand side of (14) gives

$$\frac{e_0}{1-g} + \frac{b}{1-g} \sum_{j \ge 0} (M^j u_0)_2.$$

The second coordinate of $(I - M)^{-1} u_0$ is $(cV_0 + (1-g)H_0)/D$, proving (15). The common comparison applies at each step and therefore also covers switching when that switching is explicitly allowed by the contract. It need not cover a family supplied only with unrelated per-model stability certificates. □

The result connects exact and approximate abstraction without collapsing them. Exact no-return can protect an output even when the hidden coordinate grows rapidly. With nonzero return, hidden growth and two-way feedback enter the error account. A small b is useful only in relation to the horizon, the initial hidden bound, and the other blocks.

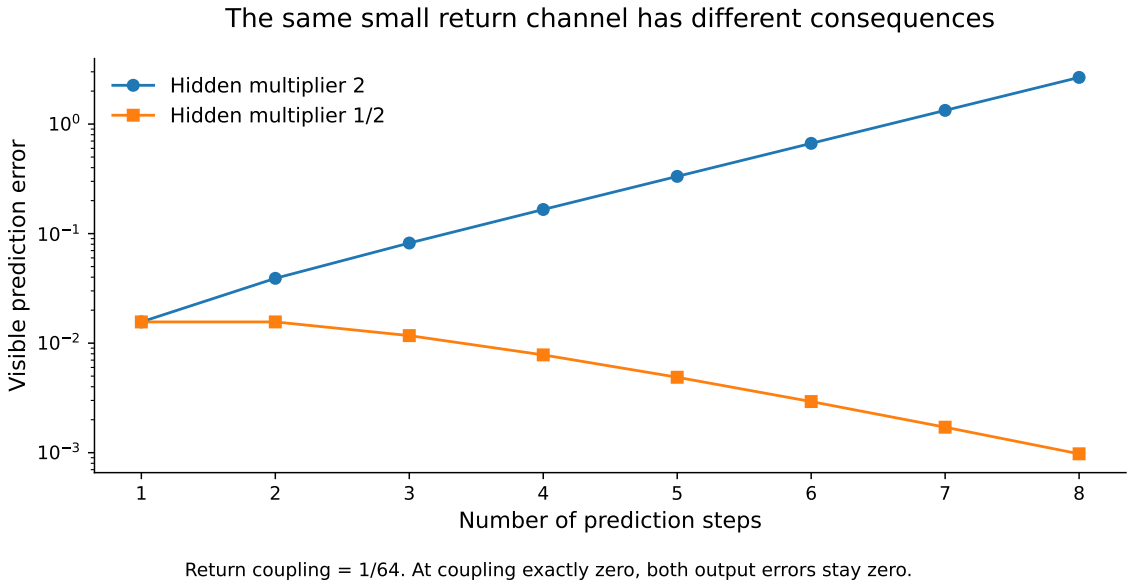


Figure 9: Exact scalar output discrepancies with the same small return coupling $1/64$ and visible multiplier $1/2$. The stable hidden multiplier $1/2$ gives $t2^{-(t-1)}/64$; the hidden multiplier 2 gives increasing error. The vertical axis is logarithmic. At exactly zero return coupling, the discrepancy is zero at every time in both cases and is not plotted on the log scale. These are explicit linear models, not neural activation traces.

Counterexample 16.2 (An arbitrarily small return channel exposes an unstable hidden mode)**.** Let $v_0 = \bar{v}_0 = 0$, $h_0 = 1$, and use scalar blocks $G = 1/2$, $B = \delta > 0$, $C = 0$, $H = 2$. The reduced output remains zero, while

$$v_t = \delta \sum_{j=0}^{t-1} (1/2)^{t-1-j} 2^j = \frac{\delta(2^t - 2^{-t})}{3/2}.$$

The first error is δ, yet the later error is unbounded for every fixed positive δ. At $\delta = 0$, every output is exactly zero. Thus a tiny positive coupling is not uniformly equivalent to exact no-return on an unbounded horizon.

Counterexample 16.3 (Stable diagonal blocks can have unstable feedback)**.** Take scalar blocks $G = H = 1/2$ and $B = C = 3/4$. Both diagonal subsystems are contractions. The full matrix sends $(1, 1)$ to $(5/4)(1, 1)$ and consequently grows on that direction. Here $D = 1/4 - 9/16 < 0$. A certificate that checks only the two diagonal bounds misses the feedback loop.

17 One justified decision from memory and dynamics

A probability-memory error and an output error concern different objects. The former is a distance between distributions on retained cells. The latter is a distance between a candidate response and its surrogate. They cannot be added without a loss function connecting both to the same decision.

Fix an actual evidence history. Let μ be the exact posterior on the stipulated finite world set and let $p_i = \mu(C_i)$. Suppose the retained distribution is $\widehat{p}$ with $d_{\mathrm{TV}}(p, \widehat{p}) \le e$. For a declared later action word or response horizon, let $y(\omega)$ be the actual model response and let $\bar{y}_i$ be the response used for cell i. Assume

$$\sup_{\omega \in C_i} \|y(\omega) - \bar{y}_i\| \le d \quad \text{for every cell } i. \tag{16}$$

The cell need not identify the physical state or candidate operator. It must support this common response allowance. A block bound such as (14) can supply it when the initial records, common visible updates, and hidden-state bounds match that cell.

Let the permitted decision set $\mathcal{D}$ be finite and nonempty. Its loss $L(y, a)$ takes values in $[0, 1]$ and is K-Lipschitz in y, uniformly over a. Define

$$R(a) = \sum_{\omega} \mu(\omega) L(y(\omega), a), \qquad \widehat{R}(a) = \sum_{i} \widehat{p}_i L(\bar{y}_i, a).$$

These are risks under the stated model posterior. They are not frequentist coverage probabilities or guarantees on the loss of a single realized world.

Theorem 17.1 (A common quantitative grounding certificate)**.** *Under (16), put* $\xi = \min\{1, e + Kd\}$. *Then*

$$|R(a) - \widehat{R}(a)| \le \xi \quad \textit{for every } a \in \mathcal{D}.$$

If $\widehat{a}$ *minimizes* $\widehat{R}$*, its true posterior regret is at most* 2ξ*. If it is the unique approximate minimizer and*

$$\widehat{R}(a) - \widehat{R}(\widehat{a}) > 2\xi \quad \textit{for every } a \ne \widehat{a},$$

then it is the unique true minimizer. A decision issued only when $\widehat{R}(a) + \xi \le \tau$ *has true posterior risk at most the declared threshold* τ.

The conclusion can combine (11) and (14) only when they refer to the same history, cell meanings, future action word, initialization, and admitted candidate family. No guarantee that the true external process belongs to that family is supplied by this composition.

Proof. Insert the intermediate risk $\sum_i p_i L(\bar{y}_i, a)$. Its difference from $R(a)$ is at most Kd by (16), the Lipschitz condition, and summing posterior masses. Its difference from $\widehat{R}(a)$ is at most e because the function $i \mapsto L(\bar{y}_i, a)$ lies in $[0, 1]$. A triangle inequality gives $e + Kd$; both risks also lie in $[0, 1]$.

Let a_* minimize R. Then

$$R(\widehat{a}) \le \widehat{R}(\widehat{a}) + \xi \le \widehat{R}(a_*) + \xi \le R(a_*) + 2\xi.$$

For any other decision, the approximate gap loses at most 2ξ when the two risks are replaced by their true values, proving the strict-gap assertion. The threshold assertion is the upper risk bound itself. Each step uses the same posterior and surrogate responses, which is why separately valid certificates about different histories or targets cannot be substituted. □

This is a bounded-loss composition of two error accounts, not a newly derived general decision principle. Related recipient-relative bounds appear in *Machine Understanding* [1]. The additional content here is the instantiated chain from a finite retained probability record and a separately controlled hidden-return error to one declared decision.

17.1 A complete sensor-to-predictor example

Use the eight worlds $(s, \theta, n) \in \{0, 1\}^3$. The four joint (s, θ) masses in the order $(0, 0), (0, 1), (1, 0), (1, 1)$ are initially

$$p_0 = (3, 1, 1, 3)/8,$$

and n is conditionally uniform. A noisy nondestructive reference has probability $1/4$ of report one under $\theta = 0$ and $3/4$ under $\theta = 1$. It does not depend on n. After three fresh reports all equal to one, the exact target probability is $\Pr(s = 1 \mid 111) = 41/56$.

The learner retains the four masses using $N = 65\,536$ and the stated remainder rule after each update. There are 68 literal probability-field bits because each of the four counts includes values through 65 536. The initial masses are exactly representable, so $e_0 = 0$. With $\beta = 1/4$, $\varepsilon = 0$, and three observations, (11) gives the sharper bound

$$e_3 \le \frac{1 + 4 + 16}{32\,768} = \frac{21}{32\,768}.$$

This uses the actual initialization, not the larger generic bound in (12).

The later response is produced by three applications of one fixed candidate matrix,

$$T_n = \begin{pmatrix} 1/2 & (-1)^n/64 \\ 1/8 & 1/2 \end{pmatrix}, \qquad (v_0, h_0) = (s, 1). \tag{17}$$

The predictor for cell (s, θ) uses $\bar{v}_t = 2^{-t}s$ and therefore reports $\bar{y}_i = s/8$. The nuisance bit n affects the physical response but remains unidentified by the reference observations. With $g = d = 1/2$, $b = 1/64$, $c = 1/8$, and $V_0 = H_0 = 1$, the comparison determinant is $127/512 > 0$. Directly evaluating (14) at three prediction steps yields

$$d_3 = \frac{481}{32\,768}.$$

Choose between numerical predictions $a = 0$ and $a = 1/8$, with loss $L(y, a) = \min\{1, |y - a|\}$. It is bounded and 1-Lipschitz. The common radius is consequently

$$\xi = \frac{21 + 481}{32\,768} = \frac{251}{16\,384}.$$

The exact companion computes the actual retained masses, approximate risks, both true risks, and their comparison. It verifies that the retained predictor selects $a = 1/8$, that its approximate advantage exceeds 2ξ, and that $\widehat{R}(1/8) + \xi < 3/50$. Thus the same finite construction supports a unique posterior-optimal decision and a posterior-risk threshold of 0.06 under the supplied model. The complete prior, sensor laws, candidate matrices, bit record, and loss function all belong to this one example.

Decision	Retained-record risk	Exact model-posterior risk
$a = 0$	$23991/262144 \approx 0.09152$	$173737/1835008 \approx 0.09468$
$a = 1/8$	$8777/262144 \approx 0.03348$	$81105/1835008 \approx 0.04420$

The certified upper risk for the selected decision is $12793/262144 \approx 0.04880$, below $3/50$. The actual cell-mass error is $1/114688$, smaller than its uniform allowance. The risk certificate uses the proved allowance, not a substituted post hoc exact error. Exact enumeration is available here only because the example is fully supplied and finite.

The probability coordinates do not encode n, and the dynamical allowance does not pretend they do. The result remains approximate precisely where the evidence leaves the physical response unresolved. Reducing the denominator, admitting an observation that depends on n, or increasing the hidden return coupling changes a different premise. These revisions require separate responses rather than an undifferentiated instruction to trust the old certificate.

Part VI

A consolidated grounding contract

18 A common theoretical contract for grounded revision

The results do not combine into one universal scalar score. They address different parts of a continuing claim.

An evidence contract says what the observer receives and which worlds remain possible. A predictive contract says what outputs must remain correct after which operations. A correction contract says which relationships between future evidence and target values must remain available. A calibration contract says where an operation is determined, and a quantitative contract controls any approximation and its growth.

A combined statement needs one representation and consistent premises. An unavailable input prevents initialization. An uncalibrated continuation prevents exact prediction. A lost association prevents correction. A narrow confidence set does not establish its own coverage.

18.1 The complete sensor argument

The eight-world sensor illustrates the chain without a large implementation. Initially the display and flip operation admit a two-state exact predictor. That predictor is grounded by the current display, even though the physical bit and sensor convention are not separately identified. Adding the physical-bit question refines the needed behavioral state, and the old code fails the initialization condition. A reference measurement repairs the information deficit while leaving the nuisance bit irrelevant.

Before calibration, the records corresponding to the two possible display values have the same marginal physical uncertainty. They must not be merged if later reference evidence is expected to identify the physical bit. The correction-memory theorem identifies the missing association. Resetting supplies convention information only after changing the physical state. Recovering the original state then depends on the retained old record. If that record is gone, the unchanged future interface cannot reconstruct it.

Noise changes the strength of the conclusion. A highly probable physical bit is not an exact state certificate. A quantitative response set can carry the remaining ambiguity into a tolerance, but only under coverage and model-accuracy premises. Every clause refers to the same task and same record history.

18.2 What this adds to the machine theory

The framework does not demand that an agent identify every detail of a world before it can reason. It gives restricted positive conditions for doing less. An unidentifiable physical realization can support an identified predictive quotient. An underdetermined operator can support a fully determined invariant domain. A large candidate set can be compressed exactly for future correction without preserving every candidate identity.

The additional discipline is that each reduction is indexed by what it preserves. One cannot transfer the word “grounded” from a successful prediction to an unrelated physical interpretation, from an exact model to a learned approximation, or from a present decision to every later correction. Such transfers require new evidence or a proved bridge.

18.3 Three different reductions should not share one name

The exact set-valued memory merges worlds only when the selected target and every admitted filter agree. The likelihood-profile memory preserves joint target probabilities and the future statistical correction law. The observable linear quotient suppresses directions that cannot return to the chosen output under its declared

model class. They are related examples of operation-compatible reduction, but their carriers, updates, and information requirements are different.

The fixed-model agreement subspace is a fourth object. It describes allowed initial vectors, not necessarily a physical invariant state space. A reference predictor can realize its common outputs even when the hidden candidate trajectories diverge. Forgetting that model history and restarting from a current physical vector can create a different inference problem.

18.4 What changes when the evidence contract changes

A new likelihood can split an old probability cell. A new candidate operator can shrink I_Q. A new action can propagate a previously invisible discrepancy into the output. A new initial-code restriction can fail to initialize a quotient that still exists mathematically. Each revision therefore needs a named compatibility check. The smallest representation for an old contract is not an entitlement to make new claims without acquiring the missing distinctions.

None of these facts obliges a learner to retain every detail. In the noisy sensor, four joint masses replace an eight-world prior. In the structured linear example, two predictor coordinates replace three physical coordinates. Both savings are exact under their respective contracts. What must be preserved is the relation between the discarded information and the future operations, not discarded information for its own sake.

18.5 A revision changes specific claims, not every claim at once

The sensor-to-predictor construction provides one ledger for revision. Its probability cells are indexed by (s, θ), while the hidden candidate index n is represented only through the response allowance. Each row below describes a change to that same example, not a newly selected favorable model.

Revision	What remains justified	What must be re-established
Change only N	The likelihood-profile partition and physical model.	Rounded history, memory error, probability-field size, and the resulting decision margin.
Let the reference depend on n	Old predictions under the old reference.	New profile cells and their masses. The old four masses cannot universally supply a split.
Admit a larger return coupling	The original observation update when its likelihoods are unchanged.	The future-response allowance and posterior risk. A former threshold may no longer be certified.
Retain only the two marginals	The present target and convention distributions.	The association required by later evidence. Equal marginals need not determine an update.
Treat a copied report as fresh	The recorded original observation.	The claimed conditional sampling law. Repetition in storage is not another independent measurement.
Replace a fixed model by switching	Results explicitly proved for the larger switching class, when their common bounds hold.	A restricted-initial-state fixed-model prediction need not survive this change.

A failed old certificate can mean that its premises no longer apply, not that the earlier theorem was false. A valid new risk assessment may conclude that no action meets the requested threshold. Abstention then preserves the stated claim instead of relabeling an uncertain answer as certain. No automatic dependency extraction from arbitrary programs is asserted here.

19 Open theoretical problems beyond the first edition

The first edition closes a restricted account of evidence, corrective memory, uncertain dynamics, and finite-horizon quantitative use. A complete theorem does not require an implemented general agent. It does require that the assumptions delimit the conclusion rather than merely decorate it.

Sharper approximate observable agreement. The block bound gives a sufficient norm-based certificate, not an optimal abstraction metric. Cancellation, structured candidate correlations, and action-language restrictions can permit smaller errors than the common nonnegative comparison detects. Characterizing the least attainable error or the smallest approximate state remains a distinct problem.

Economical long-horizon probability memory. The positive-likelihood estimate can grow rapidly with the horizon. It does not exploit statistical forgetting, contraction, or favorable evidence. Adaptive precision, support-aware safeguards, and stronger model conditions may permit better retained-memory guarantees. Such refinements must preserve the distinction between posterior risk, realized error, and frequentist coverage. The current record-size bound does not bound temporary arithmetic or total runtime.

Restricted operation languages and infinite model families. The exact linear results admit every action word. An automaton restricting those words changes the observable-equivalence problem and the relevant minimum. Infinite parametric families require a certified finite spanning argument, polynomial identity, or another uniform device. Sampling a few parameter values does not supply that device.

Model admission and state-changing evidence. Adding candidate worlds requires an explicit allocation of prior probability. A previous posterior does not determine that allocation. Noisy observations that change the hidden state require transition filtering, not only multiplication by a static likelihood. Those extensions need their own correction-memory and approximation theorems.

Learning the premises. Candidate systems, likelihoods, calibration, and structural bounds remain supplied mathematical data in this edition. Determining them from interaction, proving useful sample and search complexity bounds, and establishing a learned implementation's causal use are further research questions. They are not prerequisites that erase the completed conditional results.

Independent assurance. Prose proofs, exact finite illustrations, source comparison, and PDF rebuilding provide different forms of internal evidence. They do not establish independent peer review, a proof-kernel certificate, or an externally authenticated run. Those assurance layers can be developed without changing what the present statements assert.

20 Conclusion

Grounded abstraction relates a target, its possible realizations, accessible evidence, and the future operations a retained state must support. It is neither complete reconstruction by default nor confidence in a convenient coordinate system.

Incomplete knowledge can support exact continuing use. A likelihood-profile record preserves correction while dropping nuisance identity. Different linear systems can share one observable quotient. A reference predictor can reproduce common outputs without matching the unknown physical trajectory, provided the available record can initialize it.

Approximate use needs additional structure. Finite probability fields introduce rounding error at each update. A likelihood allowance and positive floor control its finite-horizon amplification. A hidden coordinate can be arbitrarily unstable when return is exactly absent, yet even a small positive return can become consequential. The common block bound and the rounded-memory bound meet only through a specified loss for one response and one posterior.

The integrated sensor illustrates that conjunction. Its probability record does not recover the hidden candidate, and its predictor does not silently identify the physical state. It nevertheless supports an explicit posterior decision within a certified allowance. Changing the observation law, precision, or return coupling changes a named obligation rather than granting universal reuse of an old certificate.

The first edition therefore supplies a theory of which claims survive what experience leaves unresolved, how a useful memory can preserve later correction, and when a reduced predictor remains justified across continued use. Its unresolved learning and implementation questions are visible applications of that theory, not claims supplied by the word grounded.

A Statement map and source relationships

Result	Mathematical role	Status and boundary
3.2	Law-level and support-level identification.	Elementary factorization, proved here. No sample-rate claim.
3.4	One evidence-initialized exact predictive realization.	Classical quotient argument with explicit initialization. Finite declared models.
4.1	Reinterpretation, acquisition, impossibility, and nonadaptive cost.	Exact for total nondestructive finite tests. Destructive tests excluded.
5.2	Coarsest memory preserving every admitted correction.	A powerset behavioral quotient. Evidence family is intersection closed.
5.3	Exact memory count for partition evidence.	All candidate subsets and exactly the atom-union evidence algebra. Logical states, not byte-optimal code.
6.1	Finite-record error and information processing.	Classical two-point bound, proved here. Fixed actual protocol.
7.1	Minimax coordinate ambiguity and a coverage certificate.	Unconstrained real-vector center. Coverage and model accuracy are premises.
9.1	Linear response identifiable exactly on the calibration span.	All linear extensions admissible. Exact inputs and outputs.
9.2	Maximal domain of identifiable full-state continuation.	Finite-dimensional common calibration span and fixed extension families.
10.1	Error propagation from a nominal invariant domain.	Local discrepancy plus a separate ambient growth bound.
8.1	Minimal likelihood-profile memory for target correction.	Fixed finite likelihoods, fresh conditional observations, all priors, and report probabilities retained.
8.2	Exact reconstruction after memory-contract change.	Same finite world set and every prior. Restricted prior families can differ.
11.1	Nonzero outputs do not enlarge unrestricted calibration.	All independent linear extensions admitted, fixed during an execution.
12.1	Finite fixed-model output agreement and a witness word.	Finite specified matrix family and all operation words.
13.1	Common observable quotient from propagated discrepancy.	Agreement on the whole space, not only a special initial set.
14.1	Evidence initialization and minimum linear state dimension.	Allowed initial subspace, linear record, linear predictor class.
15.1	Quantized probability fields and one-step Bayes perturbation.	A normalized finite simplex and positive likelihood floor. Retained fields, not total workspace.
15.2	Finite-horizon correction under rounding and profile error.	Static worlds, common observed history, valid likelihoods, and a specified denominator.
16.1	Visible discrepancy with hidden feedback.	Common norm bounds and initial allowances. Infinite sum needs the strict comparison condition.
17.1	One posterior decision from memory and response bounds.	Same cells, response, history, and bounded Lipschitz loss. No external coverage inferred.

B Small exact illustrations and reproduction

Three small exact suites check the declared constructions with integers, sets, rational arithmetic, and $\mathbb{F}_2$. The retained observable suite compares paired row-space closure with direct word enumeration, tests common-quotient and initialization criteria, and compares likelihood-profile updates with raw-world Bayes updates. It includes explicit switching, hidden-return, zero-probability, and memory-migration controls. The two retained suites are unchanged. The first-edition suite additionally checks quantized probability records, matched adaptive updates, block-error recurrences, and the joint sensor decision. These are bounded implementation checks, not universal proofs or agent experiments.

Run `python code/run_checks.py`. The JSON records state each domain and result. Nine Python-generated analytical figures use floating point only for drawing. The supplied Bash build uses their vector assets. Review notes record the actual proof, source, rendering, and clean-rebuild scope. No independent review, complete novelty audit, or proof-assistant certificate is claimed.

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Machine Understanding

Representation, Explanation, and Transfer

Komorebi P.
8 October 2026

First consolidated edition
Version 1.0.0

Abstract

A correct answer does not identify a reusable method, a faithful account of a computation, or an explanation another reasoner can use. This monograph develops a contract-relative account of computational understanding through representations that support prediction, intervention, transfer, and correction. Finite behavioral equivalence characterizes exact reusable state and the distinctions required by new operations. Sender access and recipient background determine what an explanation must convey. State-changing experiment planning makes acquisition costs and physical effects explicit. For noisy evidence, a likelihood factorization characterizes preservation of every Bayesian update, while separate bounds distinguish average information loss from rare-event posterior error. The first-edition synthesis fixes the recipient procedures as well as the information channel. A covering criterion identifies usable artifacts, and a finite resource construction jointly accounts for online work and the entire compiled observation tree. Task-dependent interpreter changes are charged rather than hidden in background knowledge. A three-register instrument connects these results in one executable account of acquisition, correction, and explanation. The contribution is an integrated realization-and-preservation framework with conditional proofs and explicit failures. It is not a universal test of understanding, an empirical claim about a deployed model, or a criterion for subjective experience.

First-edition scope. A theoretical account with explicit finite contracts, self-contained prose proofs, and executable witnesses. Mathematical validity, implementation evidence, and empirical attribution remain distinct. Version 1.0 freezes this scope rather than claiming a universal test of understanding.

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Reading the first edition

The paper asks what a computational system can carry from one question to the next, from one representation to another, and from an accepted model to a corrected one. Its central object is a usable representation together with its operations and limits. It is not a score assigned to a model after one successful answer.

A conceptual reading can follow Sections 1–3, the interpretations after the formal results, and Section 21. A mathematical reading follows four parts: reusable representations and implemented use, explanatory communication and correction, common resource witnesses, and interpretive boundaries. The final synthesis freezes the recipient procedures and charges both online work and compiled controller storage. Appendix A gives one complete numerical trace. The source package includes exact reference programs, a statement index, a claim ledger, and a review record.

What is inherited and what is proposed. State minimization, causal abstraction, hypothesis testing, counting bounds, and error propagation have substantial established literatures. The proofs below make the particular assumptions used here inspectable. Their presence is not a claim of priority for those mathematical ingredients. The proposed integration is to require a claim about understanding to identify its target, its realized use, its admissible changes of representation, its recipient, its correction procedure, and its total resource boundary.

Relation to Operation Komorebi. *Machine Societies* studies persistent plural institutions and the preservation of their operations and commitments [1]. This companion changes the object of study. It concerns methods and representations that one computational reasoner can use, explain, transfer, and revise. It uses the same realization perspective without treating institutional continuity as evidence of individual understanding.

Conventions. The main systems are finite and fully specified. Deterministic acquisition and stochastic correction are distinguished. Each probabilistic result states its own likelihood and selection assumptions. A metric transport result also applies outside finite spaces, with explicit uniform bounds. A finite check supports its implemented construction, not the general theorem. Mathematical validity inside a model is distinct from evidence that an external system realizes that model.

Symbol	Meaning
C	Declared family of states, operations, and future questions.
F_u, F_w	One implemented operation, or operations in the order specified by a word w.
$r : \mathcal{X} \to \mathcal{Z}$	Retained representation. Its acquisition procedure is a separate obligation.
$\bar{F}_u$, $\bar{q}$	An update and a query decoder operating on the retained state.
T, τ	Translation of representations and the corresponding operation names.
$V \subseteq \mathcal{M}$	Current set of candidate models, not a probability distribution.
$C_{e,o}(V)$	Candidates retained after experiment e returns observation o.
ϕ, η	Translation of candidate models and translation of observations.
$d_{\mathrm{TV}}(P, Q)$	$\frac{1}{2}\sum_x \lvert P(x) - Q(x)\rvert$ on a finite outcome space.

Part I

Representations and implemented use

1 What a correct answer leaves open

A reasoner is asked what a device will display after two operations. It gives the right answer. That success is a fact worth preserving. It does not yet tell us whether the reasoner has a method that will work after three operations, whether the order of the operations matters, or whether the same method could be expressed in another notation. It also does not tell us whether the explanation offered afterward describes the computation that produced the answer.

These are not objections to correctness. They are different questions about what correctness rests on and what it makes possible. A proof can establish a conclusion from premises without teaching a recipient how to select the relevant premises in a new case. A useful model can guide new predictions without being a faithful description of the particular implementation that supplied it. A fluent explanation can be understandable while failing to identify the mechanism it claims to explain. The distinction between persuasive and faithful explanations is explicit in interpretability research [2].

This paper investigates a narrower and more tractable question than whether a machine understands in every human sense of the word. It asks when a computational realization possesses a representation that remains useful across a declared family of operations, questions, translations, and corrections. We call this *reusable computational understanding* as an operational research proposal. It is relative to a domain and a contract. It is not a metaphysical property inferred from a single formal condition.

The central thesis is that a representation earns its explanatory role through what a system can do with it. It must retain distinctions needed for future questions, support the operations it claims to summarize, and expose the conditions under which its use should change. When it is communicated, a recipient must be able to recover something useful from it. When it is attributed to a particular mechanism, the attribution requires evidence about that mechanism. A representation that meets one of these obligations need not meet the others.

1.1 Three separations organize the argument

The first separation is between an answer and a reusable state. The answer to today's question can discard distinctions that tomorrow's permitted operation makes important. Understanding in the present sense therefore concerns a family of possible continuations, not only a completed input and output pair.

The second is between a correct account of a subject and a faithful account of a solver. An algebraic derivation can explain a numerical result even when the original solver used a table. It would be wrong to infer that the solver executed the derivation merely because the derivation is valid. Subject understanding and process faithfulness require different witnesses.

The third is between successful transport and preserved correction. Two languages may translate every currently expressible observation while one language merges distinctions needed to reject a mistaken model. Agreement in the translated language is then weaker than agreement about what has been learned. The correction theorem in Section 11 makes that loss exact.

1.2 Position among existing theories

Several neighboring theories already supply essential pieces. Computational mechanics studies minimal predictive state representations [3]. Controlled state aggregation studies equivalences that preserve transition

structure [4]. Information bottleneck methods formalize compression relative to a specified relevance variable [5]. These are not interchangeable claims, but they establish that useful compression must be assessed against what it promises to preserve.

Causal abstraction gives a stronger account of agreement under interventions. Beckers and Halpern distinguish increasingly restrictive abstraction conditions [6]. Geiger and colleagues develop this perspective as a foundation for mechanistic interpretability, including transformations of mechanisms [7]. Our finite commuting equations occupy a deliberately restricted part of that subject. They do not replace a general theory of causal models.

There are also reasons to constrain the maps used in an interpretation. Hewitt and Liang show why a trained probe's success must be assessed against the work the probe itself can perform [8]. Sutter and colleagues examine how sufficiently expressive alignment maps can make some causal-abstraction claims uninformative [9]. The constructive theory and this criticism address different aspects of the problem. We retain both lessons: intervention compatibility matters, and the translator must not be allowed to hide the entire explanation inside an unrestricted computation.

Explanation has a recipient as well as a referent. Miller's synthesis emphasizes the contextual and social features of explanation [10]. Gentner's structure-mapping account places relational organization, rather than surface resemblance alone, at the center of analogy [11]. Here we formalize only a limited recipient-relative property: an artifact enables specified future operations in a recipient with stated background resources. That property does not exhaust the reasons a human explanation can succeed or fail.

Finally, learning through informative queries and counterexamples has a substantial theoretical history, including Angluin's work on regular languages [12]. Our correction model is much simpler. It is a finite filter on candidate models, chosen so that the interaction between evidence and translation can be proved directly.

1.3 The intended contribution

The paper does not propose that every useful representation is new, that a symbolic method is inherently superior to a learned one, or that a large table can never support understanding. It proposes a common account of the obligations that such representations must satisfy. The exact finite results locate information requirements and compatibility conditions. The counterexamples show why isolated successes cannot silently substitute for a combined claim.

This is also a limit on the theory itself. A finite contract can be satisfied by many implementations, including tables. To distinguish them, one must add relevant constraints on acquisition, causal organization, storage, computation, or transfer. Those constraints cannot be recovered from a definition after the results are known.

2 A small instrument with a hidden register

Consider a mathematical instrument with three registers

$$x = (a, b, c) \in \mathbb{F}_5^3.$$

Each register contains one of $0, 1, 2, 3, 4$. All arithmetic in this example is modulo five. The display reports

$$q(a, b, c) = a + b.$$

There are 125 possible complete states, but only five possible readings. The registers and their operations are fully specified analytical objects. They are not measurements from an agent experiment.

Initially, two operations are available:

$$H(a, b, c) = (a + b, b, c), \qquad W(a, b, c) = (b, a, c). \tag{1}$$

The first adds the second register into the first. The second exchanges the first two registers. The third register is preserved by both and does not affect the display. The operation order is public. A prediction concerns what the display would show after a named finite sequence of these operations.

2.1 Knowing the current display is insufficient

A record containing only $s = a + b$ answers the current question exactly. It cannot, in general, predict the display after H. The two complete states

$$x = (0, 0, 0), \qquad y = (1, 4, 0)$$

both display zero. After H, their displays are zero and four. Any predictor receiving only the common old display must return the same answer for both, so it cannot be exact in both cases.

A sufficient retained state for the original operation family is

$$z = (s, b) = (a + b, b).$$

Its updates are

$$\bar{H}(s, b) = (s + b, b), \qquad \bar{W}(s, b) = (s, s - b), \tag{2}$$

and its displayed output is its first coordinate. These equations retain the information needed for every finite H/W continuation without retaining c.

The representation is also minimal in its number of states. The present reading s and the reading s_H after a hypothetical H determine

$$b = s_H - s, \qquad a = 2s - s_H.$$

Two different pairs (a, b) therefore differ on at least one of these questions. Every exact reusable representation for this contract needs at least 25 distinguishable values. Section 4 gives the general argument.

2.2 A new operation changes what must be known

Now admit

$$J(a, b, c) = (a + c, b, c). \tag{3}$$

The old representation remains correct for H and W. It is no longer sufficient for the enlarged contract. States with the same (a, b) but different c now produce different readings after J.

The additional reading s_J yields

$$c = s_J - s.$$

Together with the earlier distinctions for a and b, the enlarged question family distinguishes all 125 complete states. The useful state counts are therefore five for the current display, 25 for reusable H/W prediction, and 125 after J is admitted.

This is not a case of an old theorem becoming false. The old theorem had a smaller operation family. What failed was an unlicensed extension of its conclusion. The distinction matters whenever a learned method is reused under a change in task, environment, or intervention.

2.3 Correction requires evidence, not a renamed variable

A predictor that has retained only (s, b) cannot recover c merely by adding a third field to its data structure. Conditional on a fixed (a, b) and a uniformly distributed c, every possible display after J is equally likely. Any guess based only on the old record succeeds with probability $1/5$. Randomizing the guess does not improve that probability.

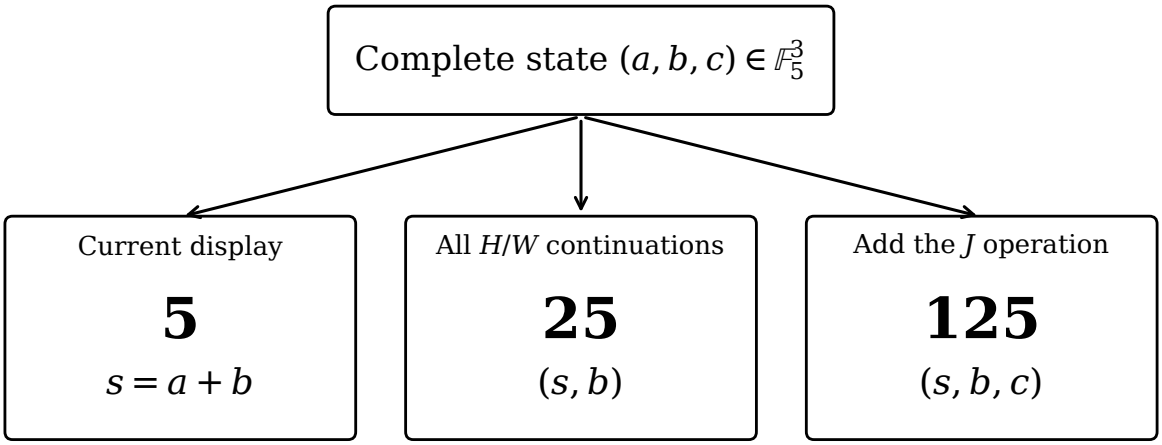


Figure 1: The same instrument supports different minimal representations under different contracts. The numbers count distinguishable states, not scores of intelligence. A new permitted operation makes a previously irrelevant register relevant.

An actual execution of J, followed by a new reading, supplies the missing distinction. Because J changes the instrument, the recovered state is the state *after* that operation:

$$(a', b', c') = (s_J - b,\ b,\ s_J - s). \tag{4}$$

The old state can then be reconstructed because this particular operation is invertible. Neither reversibility nor a free observation should be assumed for other instruments.

The example separates four achievements: reading an answer, retaining a method, recognizing a contract change, and acquiring evidence that repairs the resulting information gap. No amount of confidence in the original method supplies the fourth.

3 A contract for reusable understanding

Let $\mathcal{X}$ and $\mathcal{Y}$ be finite nonempty sets. Let $\mathcal{U}$ be a finite operation alphabet, possibly empty, and let $\mathcal{Q}$ be a finite nonempty family of queries $q : \mathcal{X} \to \mathcal{Y}$. Each $u \in \mathcal{U}$ has an implemented total transition $F_u : \mathcal{X} \to \mathcal{X}$. The deterministic mathematical contract is

$$C = (\mathcal{X}, \mathcal{U}, (F_u)_{u \in \mathcal{U}}, \mathcal{Q}, \mathcal{Y}).$$

For a word $w = u_1 \cdots u_n$, define $F_w = F_{u_n} \circ \cdots \circ F_{u_1}$. The empty word acts as the identity. Thus $q(F_w(x))$ asks a question after a stated continuation.

Totality is a simplifying assumption. If an operation is sometimes unavailable, its admissibility must be included in the observable contract or the model must use partial transitions with explicit domain preservation. An unmodeled refusal cannot be treated as the same event as a permitted update.

Definition 3.1 (Exact reusable representation)**.** An exact reusable representation for C is a surjection $r : \mathcal{X} \to \mathcal{Z}$, updates $\bar{F}_u : \mathcal{Z} \to \mathcal{Z}$, and decoders $\bar{q} : \mathcal{Z} \to \mathcal{Y}$ such that, for all x, u, q,

$$r(F_u(x)) = \bar{F}_u(r(x)), \qquad q(x) = \bar{q}(r(x)). \tag{5}$$

Induction immediately gives $q(F_w(x)) = \bar{q}(\bar{F}_w(r(x)))$ for every finite word. The same initial record supports many future questions. It is not allowed to inspect the selected future query and replace itself with that query's answer before the challenge begins.

3.1 A representation of the problem is not yet a representation in the solver

The map r specifies what information would suffice. It does not establish that a particular agent can acquire $r(x)$ from its sensors or that the agent uses this state. Those claims require an acquisition procedure, a realized computational path, and admissible interventions on that path.

For the instrument, an omniscient mathematical encoder can read (a, b, c). A device restricted to the display cannot do so. It can obtain b through a controlled H observation, but that procedure costs an operation and changes the state. Any account of its understanding must include those effects rather than identify mathematical existence with operational access.

A practical contract therefore adds an information boundary, a family of recipient or decoder algorithms, resource limits, an intervention catalogue, and a revision policy. The finite theorems below solve particular obligations within that larger contract. They do not fill in unspecified sensors, access rights, or learning procedures.

3.2 A profile rather than a single threshold

We propose reporting a profile with separate entries for predictive adequacy, implemented use, intervention compatibility, transfer, recipient reconstruction, correction, and resource cost. Unknown entries remain unknown. A high prediction score must not silently fill the mechanism or correction entries.

The profile can describe a traditional program, a learned model, a hybrid system, or a collective procedure. It does not require language-like internal states. Nor does it establish an ordering of all systems from least to most understanding. Two systems can preserve different families of questions and be incomparable under the declared criteria.

Counterexample 3.2 (A separate answer for each question is not one reusable record)**.** Let $x = (a, b) \in \{0, 1\}^2$ and let the two queries return a and b. If the encoder knows which query will be asked, one bit suffices: retain that answer. If one record must be fixed before the query is selected and must answer both exactly, it needs four distinct values. A one-bit record has only two values and must identify two different answer profiles.

The order of quantifiers is decisive. A different adequate representation for every question is not one adequate representation for their union. This is why a claimed method should be assessed with its encoder, updater, and decoder fixed across the family it promises to support.

4 Behavioral compression and the price of new questions

Define

$$x \equiv_C y \quad \Longleftrightarrow \quad q(F_w(x)) = q(F_w(y)) \text{ for all } q \in \mathcal{Q},\ w \in \mathcal{U}^*. \tag{6}$$

This is equality of all declared future answer profiles. It is an equivalence relation. It describes behavioral sufficiency under the contract, not equality of hidden mechanisms or identity outside that contract.

Theorem 4.1 (The minimal exact reusable state)**.** *The quotient $\mathcal{X}/\equiv_C$ carries an exact reusable representation for C. Every other exact reusable representation $r : \mathcal{X} \to \mathcal{Z}$ refines this quotient: if $r(x) = r(y)$, then $x \equiv_C y$. There is a unique map $h : \mathcal{Z} \to \mathcal{X}/\equiv_C$ with $h(r(x)) = [x]$. In particular,*

$$|\mathcal{Z}| \geq |\mathcal{X}/\equiv_C|.$$

A fixed-length binary state needs at least $\lceil \log_2 |\mathcal{X}/\equiv_C| \rceil$ *bits. The quotient can be computed by finite partition refinement.*

Proof. If $x \equiv_C y$, the empty word gives $q(x) = q(y)$ for every query. Moreover, $F_u(x) \equiv_C F_u(y)$ because any continuation after u is a continuation in the original contract. Thus $[x] \mapsto [F_u(x)]$ and $[x] \mapsto q(x)$ are well-defined updates and decoders.

For any exact representation, the equations in (5) imply equality of all future decoded answers whenever $r(x) = r(y)$. This proves refinement. Surjectivity of r gives existence and uniqueness of h and the cardinality bound. The binary bound follows by counting available codewords.

For computation, let P_0 group states with identical current query vectors. Given a partition P_k, refine each block by the block of $F_u(x)$ for every u. Induction shows that P_k identifies exactly the states agreeing on words of length at most k. Every strict refinement increases the block count. At most $|\mathcal{X}| - 1$ strict refinements are possible. At a fixed point the partition is closed under all operations, hence under every finite word, and is exactly (6). □

This is a deterministic state-minimization result, presented in the language of reusable questions. It belongs to the established state-aggregation and predictive-state lineage [4, 3]. Its significance here is not the discovery of quotients. It is the explicit identification of which future distinctions an asserted understanding is required to preserve.

4.1 Minimal information is not minimal effort

The theorem minimizes the number of distinct retained states. It does not minimize the work required to acquire the state, compute an update, verify a query, or discover the quotient. A smaller table can have harder entries to compute. A larger representation can make a frequently used transformation local and cheap.

Even uniqueness of the behavioral quotient does not supply a uniquely intelligible notation. The coordinates (a, b) and (s, b) describe the same 25 behavioral classes in the original instrument. One makes the physical register update obvious. The other makes the readout immediate. Their difference is computational and explanatory rather than a difference in information content.

4.2 The cost of expanding the contract

Let C' have the same complete state space and contain every operation and query of C, while possibly adding more. Write $\equiv_{C'}$ for its finer behavioral equivalence. Take any representation r adequate for C. For each z, let k_z be the number of $\equiv_{C'}$ classes that meet the fibre $r^{-1}(z)$.

Proposition 4.2 (Contract expansion and supplementary information)**.** *The relation* $\equiv_{C'}$ *refines* $\equiv_C$*. Any supplementary code* $s : \mathcal{X} \to \mathcal{S}$ *for which* (r, s) *supports exact reusable behavior under* C' *must satisfy*

$$|\mathcal{S}| \geq \max_z k_z.$$

If every $\equiv_{C'}$ *class lies within a single fibre of* r*, this bound is attained by an exact reusable pair* (r, s)*. This condition holds when* r *is the old minimal behavioral quotient. A supplementary value computed solely from* $r(x)$ *cannot repair a fibre meeting more than one expanded class.*

Proof. Agreement on the larger continuation family implies agreement on the smaller one. Within a fixed fibre of r, an adequate pair (r, s) must separate the k_z new classes, giving the lower bound. For attainment under the additional condition, assign distinct labels $1, \ldots, k_z$ to the expanded classes within each old fibre and let $s(x)$ be its class label. The pair $(r(x), s(x))$ then identifies exactly the expanded behavioral class. The

quotient update and query decoder supply its exact dynamics. Finally, any value computed only from $r(x)$ is constant on that fibre and separates none of its states. □

The attainment condition matters. An old representation can retain irrelevant extra coordinates that cease to be dynamically closed under new operations. For example, start with two bits (a, b), no operations, a constant-zero query, and the unnecessarily detailed record $r(a, b) = a$. Add the swap operation while keeping the query constant. There is still only one expanded behavioral class, so $k_z = 1$ in each old fibre. Yet updating the retained coordinate a after the swap requires b. The information lower bound for answers is not sufficient to preserve arbitrary redundant coordinates. The minimal old quotient has no such problem.

In the instrument, each old (a, b) class contains five expanded classes. An additional five-valued symbol is sufficient and necessary. A separately stored fixed-length symbol therefore needs three bits. This should not be confused with subtracting the rounded bounds for a joint 25-state and 125-state encoding, which are five and seven bits respectively. Separate buffers and joint recoding are different coding contracts.

Counterexample 4.3 (A finite continuation test leaves an unbounded alternative)**.** For any fixed $h \geq 0$, a one-operation machine can return zero through the first h steps and return one at step $h + 1$. A constant-zero machine agrees with it on all those earlier observations. Without a state bound or another restriction on the hypothesis family, success through horizon h cannot establish agreement at every future horizon.

This does not invalidate finite certification of a supplied finite model. It distinguishes exhaustive checking of a known transition system from identifying an unrestricted unknown system by a finite record.

5 Available information is not implemented understanding

The finite quotient answers an information question. A claim about a particular computational reasoner must answer an additional question: where is the useful representation realized, and how does it affect the computation?

A representation can contain a recoverable fact without the system using that fact. Conversely, a system can use information through a distributed or redundant encoding that a chosen simple probe fails to recover. The comparison of information structures in statistical decision theory provides a useful background for asking what a record could support [13]. That question is different from identifying the route by which one implemented decision was made.

5.1 A carrier can be readable and irrelevant

Consider a solver state $m = (a, b, c, z)$, with $z = a + b$. Suppose its output circuit reads a and b directly and ignores z. A decoder applied to z predicts the output perfectly. Replacing z while keeping the actual circuit inputs unchanged has no effect. The code z is available to an observer but is not the operative route used by this solver.

Now alter the implementation so that the output reads z, while a repair circuit recomputes z from a, b before every output. An attempted replacement may again have no observable effect, but for a different reason: the affected carrier is restored. A null intervention cannot identify which of these implementations is present unless its timing and the remaining information routes are controlled.

These examples justify neither a universal ablation test nor a demand that every useful representation be uniquely necessary. They show why an intervention must specify what changes, what remains accessible, and what the hypothesis predicts afterward. A proof about the instrument's state does not substitute for an intervention on the solver's implementation.

Proposition 5.1 (The identification limit of a transcript)**.** *Suppose two proposed implementations induce complete transcript laws P_0 and P_1 on a finite transcript alphabet under a specified permitted experiment,*

including any adaptive choices made by its evaluator. With equal prior probability, the greatest identification accuracy available from that transcript is

$$\frac{1}{2}\big(1 + d_{\mathrm{TV}}(P_0, P_1)\big).$$

In particular, identical transcript laws permit no accuracy above $1/2$.

Proof. Let $d(o) \in [0, 1]$ be the probability of guessing implementation 0 after transcript o. Accuracy is

$$\frac{1}{2}\sum_o \big(P_0(o)d(o) + P_1(o)(1 - d(o))\big) = \frac{1}{2} + \frac{1}{2}\sum_o (P_0(o) - P_1(o))d(o).$$

The sum is maximized by setting $d(o) = 1$ on positive differences and zero on negative ones. The positive differences sum to total variation. This choice attains the bound. □

The experiment may include interventions. The proposition says only that the *permitted complete transcript* bounds identification. Adding a new sensor or an admissible internal intervention changes that transcript law and can change the limit.

5.2 The interpreter must not secretly do all the work

Take any bijection $p : \mathcal{X} \to \mathcal{Z}$ of a finite state space. Define

$$\bar{F}_u = p \circ F_u \circ p^{-1}, \qquad \bar{q} = q \circ p^{-1}.$$

All exact representation equations hold. The labels in $\mathcal{Z}$ may nevertheless be arbitrary, and the translated updates may require a complete lookup table. Exact representability alone does not explain why the representation is easier to acquire, use, or communicate.

This does not make the construction false. It makes the claim narrow. A table can be a valid finite model. Its interpretive value depends on the allowed questions and costs. The mistake would be to infer that any such recoding identifies a compact or causally privileged concept inside a solver.

We therefore require an *interface account*. The encoder, decoder, intervention implementation, and translator are part of the evaluated system. Their information access is declared, their computation is charged, and any training or target-specific background they receive is recorded. A decoder that is free to solve the whole task is not evidence that the state it reads supplied the method. Control tasks for probes and the critique of unrestricted alignment maps are especially relevant here [8, 9].

Restrictions can concern circuit size, memory, communication rounds, a finite program language, or an explicit family of allowed maps. There is no universally correct restriction. A linear map is not automatically the right representation of a nonlinear concept, and a nonlinear map is not automatically uninformative. The appropriate restriction must be selected with the scientific question, not retrofitted to distinguish preferred results.

5.3 A bounded stochastic extension

The distinction between a current answer and a reusable state also survives stochastic dynamics. Let $K_u(x, x')$ be finite transition kernels and let $r : \mathcal{X} \to \mathcal{Z}$ be surjective. Assume the current query values are constant on fibres of r.

Proposition 5.2 (Uniform controlled stochastic reuse)**.** *A single reduced kernel* $\bar{K}_u$ *reproduces every projected one-step law from every complete state if and only if, whenever* $r(x) = r(y)$,

$$\sum_{v : r(v) = z'} K_u(x, v) = \sum_{v : r(v) = z'} K_u(y, v) \quad \textit{for every } u, z'.$$

With matched initial projected laws, the reduced model preserves the law of every finite sequence of reduced states and queries under any policy depending only on the reduced history.

Proof. A common projected row is necessary by considering point-mass initial states. If the rows agree, use their shared value to define $\bar{K}_u(r(x), z')$. Conditional on any reduced history, the complete state has a distribution supported on the current fibre. Averaging equal projected rows gives the same reduced transition law. Induction yields equality of the finite history laws, including the deterministic query values. A common randomized history policy preserves the same argument after conditioning on its selected action. □

This is the controlled lumpability condition underlying stochastic state aggregation [4]. The policy qualification is essential. A controller that secretly observes a discarded variable can choose different actions in states that the reduced description identifies. Small prediction error on one observed trajectory distribution is not a substitute for the uniform condition.

6 Transporting a method rather than matching an answer

An analogy is useful when relations and operations survive the comparison, not simply when two descriptions use similar words. Structure-mapping theory makes relational correspondence central to analogy [11]. We study a more limited mathematical question: when can an already specified method be translated without changing the answers to its declared continuation questions?

Let a source representation have state space Z, updates f_u, and query decoders q. Let a target have state space W, updates g_v, and decoders q'. A map τ translates operation names. Query pairs (q, q') and a representation map $T : Z \to W$ are fixed in advance. The operation names alone do not establish that the target implements the intended source operation.

Theorem 6.1 (Exact transport of reusable operations)**.** *Suppose, for every source operation and matched query,*

$$Tf_u = g_{\tau(u)}T, \qquad q'T = q. \tag{7}$$

Then, for every source state and every finite source word w,

$$Tf_w = g_{\tau(w)}T, \qquad q'(g_{\tau(w)}(Tz)) = q(f_w(z)).$$

Two such transports compose, with their operation and query correspondences composed as well. No injectivity assumption on T is needed for this forward guarantee.

Proof. The empty word is immediate. If the identity holds for w, then

$$Tf_uf_w = g_{\tau(u)}Tf_w = g_{\tau(u)}g_{\tau(w)}T.$$

Induction proves the word identity, and applying the matched decoder proves the answer identity. For a second transport S with operation map σ, substitution gives $STf_u = h_{\sigma(\tau(u))}ST$ and the corresponding decoder identity. This is the same condition for the composite. □

This is an intertwining argument. Related composition questions for interventional causal models are studied in greater generality by Rischel and Weichwald [14]. Here the theorem guarantees forward answers only from states in the image of T and for translated source operations. It does not guarantee coverage of all target states or all target operations. Surjectivity, reverse translation, and recovery of discarded source distinctions are separate requirements.

6.1 The instrument in display coordinates

For the original instrument quotient, let the source state be (a, b) and set

$$T(a, b) = (a + b, b).$$

The target updates in (2) satisfy (7). The source decoder $a + b$ becomes the target's first coordinate. Thus a method can be carried between a register description and a display-centered description without relearning the meaning of every operation sequence.

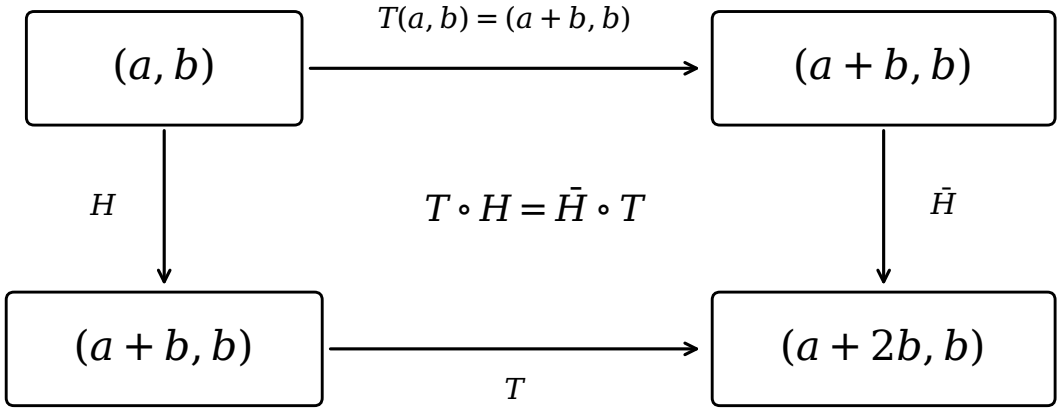


Figure 2: One verified operation square for the instrument. Translating and then applying the target update produces the same state as applying the source update and then translating. The equality concerns the declared H operation, not arbitrary interventions.

Counterexample 6.2 (Correct current answers with a wrong transported operation)**.** Retain the correct map $T(a, b) = (a + b, b)$ and the correct display decoder. Now incorrectly define the target swap by $\widetilde{W}(s, b) = (s, b)$. The current display and the display immediately after a swap are still correct. Starting from $(a, b) = (1, 3)$, however, the source sequence W then H ends with display zero. The incorrect target sequence ends with display two. Matching a restricted collection of outputs did not preserve the method.

The error becomes visible only when the altered internal state is used by a later operation. That is precisely why reusable understanding requires continuations rather than a list of isolated answers.

6.2 Approximate transfer accumulates error

Exact equalities are not always available. For this result let the target state space carry a metric d. Suppose each target update is L-Lipschitz for a common $L \geq 0$, and suppose

$$d(Tf_u(z), g_{\tau(u)}(Tz)) \leq \epsilon$$

uniformly on every state and operation that the contract permits. Let a target query be M-Lipschitz with real-valued output and satisfy $|q'(Tz) - q(z)| \leq \delta$.

Proposition 6.3 (Finite-horizon transfer budget)**.** *For matched source and target operation sequences, start from z_0, w_0 with $d(Tz_0, w_0) \le e_0$. After n operations,*

$$d(Tz_n, w_n) \le L^n e_0 + \epsilon \sum_{j=0}^{n-1} L^j,$$

$$|q(z_n) - q'(w_n)| \le \delta + M\left(L^n e_0 + \epsilon \sum_{j=0}^{n-1} L^j\right).$$

The empty sum is zero and $L^0 = 1$, including when $L = 0$. At $L = 1$ the state bound is $e_0 + n\epsilon$.

Proof. Writing $e_n = d(Tz_n, w_n)$, insert $g_{\tau(u)}(Tz_n)$ and use the triangle inequality and the Lipschitz bound to obtain $e_{n+1} \le \epsilon + Le_n$. Iterating gives the stated expression. Insert $q'(Tz_n)$ between the two outputs and use the decoder mismatch and its Lipschitz constant. □

The bound concerns the matched operation sequence. It does not automatically cover two controllers that select different operations after slightly different observations. Adaptive choices require a matching policy contract or a separate stability argument for the controller.

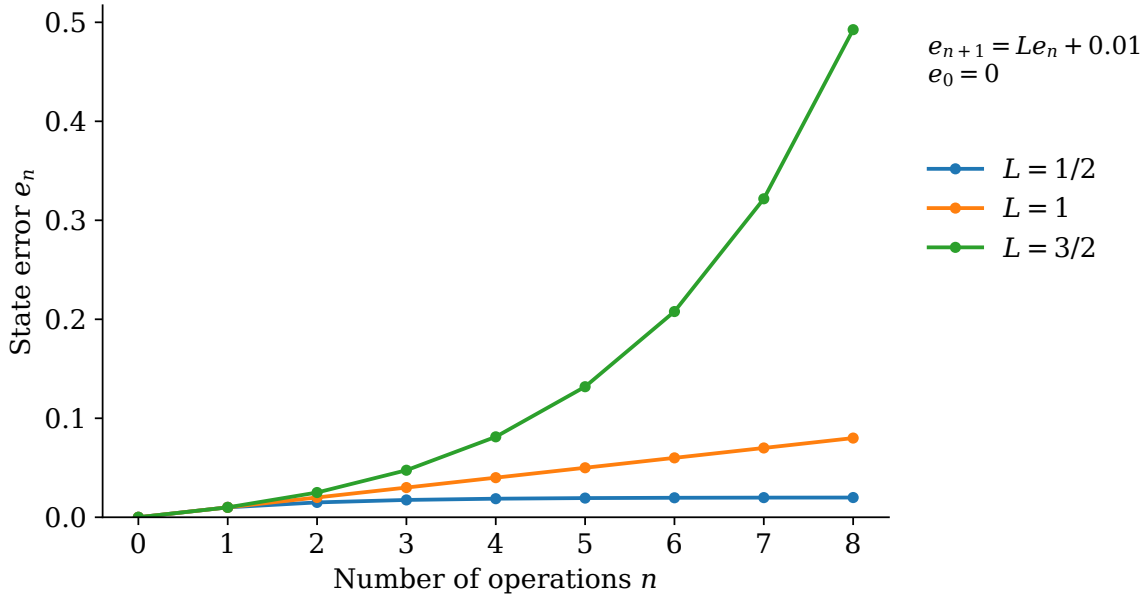


Figure 3: Exact values of the recurrence $e_{n+1} = Le_n + 1/100$, with $e_0 = 0$, illustrate the transfer bound. The curves are analytical constructions, not measured model errors. A small local defect can remain bounded, accumulate linearly, or be amplified.

A map can therefore be accurate for a short continuation and unsuitable for a longer one. The required tolerance belongs to the task, not to a universal declaration that the two representations are close. An approximate relation with one tolerance also need not be transitive. A chain of individually accepted translations may exceed the original error budget.

6.3 Local representations form an atlas only with transition rules

One representation may simplify prediction while another simplifies correction or proof. A useful system need not select one notation forever. It can maintain a family of representations with declared domains and translators. We call this an atlas only in the modest sense of a set of local descriptions connected by checked transition maps.

The domain guards matter. A method valid on reversible operations may fail after an irreversible reset. A chart switch may require evidence absent from the current chart. A common output name does not establish a common unit or query semantics. The system must know which transitions are supported and must retain the unresolved discrepancies when a transition is partial.

There is no general theorem here that more representations improve reasoning. Additional charts consume memory and create more opportunities for a mistaken translation. The research target is a family whose additional coverage or lower repeated cost justifies its construction and maintenance. Exact intertwining supplies one part of that analysis, not its economic conclusion.

Part II

Explanation, inquiry, and correction

7 An explanation belongs to a recipient

An explanation can be a correct description and still fail to help a particular recipient. It may presume unfamiliar notation, omit a necessary background result, or encode all the difficult work in a phrase the recipient cannot execute. Conversely, a deliberately simplified explanation can be useful for a limited question family without being a complete description of the source mechanism.

We therefore distinguish two contracts. A *subject explanation* enables the recipient to reason correctly about specified aspects of the subject. A *process explanation* additionally makes a faithful claim about how a particular solver reached its result. A process explanation must satisfy an implementation attribution as well as a subject-level or transcript-level account. Neither fluent presentation nor a valid derivation alone establishes the attribution [2].

Definition 7.1 (Recipient-relative explanatory adequacy)**.** Fix a recipient procedure L, background information B, resource allowance, and continuation family. An artifact E is adequate for a target model m when $L(B, E)$ yields a representation and operations that answer the required continuations for m within the declared error and resource limits. The artifact's inputs, the recipient's prior access, and any target-specific background are included in the contract.

This is an operational property of an artifact and a recipient. It does not equate instruction with human comprehension, and it does not require natural language. A diagram, proof, worked example, executable transformation, or combination can play the role of E.

7.1 An exact communication lower bound

Let Θ be a finite family of target models. Relative to the required continuation questions, divide these models into N classes with distinct answer profiles. Use one fixed recipient background and one fixed deterministic recipient. The teacher can select a message but cannot change the recipient's background separately for each target without that change being counted as communication.

Proposition 7.2 (The information an exact explanation must carry)**.** *If a fixed-length b-bit artifact lets the recipient answer the declared questions exactly for every target in Θ, then $2^b \geq N$. With target-dependent background B_θ, the same argument gives*

$$2^b \geq \max_B \; N_B,$$

where N_B counts distinct answer profiles among targets sharing that exact background value.

Proof. Two different required answer profiles cannot produce the same background and message. The deterministic recipient would then make the same responses for both and fail on a distinguishing question. There are at most 2^b fixed-length messages. Apply this counting argument to the whole family, or separately within each background fibre. □

The bound is about information, not explanatory elegance. It does not establish that any shortest message is easy to understand or that the recipient can decode it efficiently. A compressed index into an enormous shared table can meet the bound while transferring little new method. The table's construction and availability must be charged to the background account.

7.2 What the instrument explanation transfers

For the original instrument, an adequate artifact can specify arithmetic modulo five, the representation (s, b), the two update rules in (2), the output decoder, and the domain restriction excluding J. A recipient supplied with an initial (s, b) can then answer every H/W continuation by applying these operations.

That establishes a subject explanation for the specified recipient. It does not establish that a neural system that produced the artifact used (s, b) internally. A separate process explanation might show that the system's actual intermediate states realize these coordinates and respond to corresponding interventions. These two certificates can support one another, but they answer different questions.

An especially useful explanation includes the failure boundary. The line "the third register is irrelevant" is too broad. The line "the third register does not affect any display reachable through the current H/W operation family" is exact. The latter statement tells the recipient what should be reconsidered when a new operation arrives.

7.3 Explanations can preserve methods without reproducing discovery

A proof may give a shorter route than the route by which the result was found. A teacher may introduce coordinates that the discoverer never used. These are legitimate explanatory choices when their subject-level claims are correct. They become misleading only when presented as faithful records of the original reasoning without evidence.

This distinction avoids an unnecessary demand that every helpful explanation be an autobiography of a computation. It also prevents the opposite mistake of treating a helpful reconstruction as proof of the original mechanism. Human context and communicative purpose remain important beyond the formal recipient criterion [10].

8 One explanation for recipients with different backgrounds

An explanation can be adequate for each recipient separately without there being one short artifact that works for all of them. Conversely, a shared artifact can be shorter than a complete description because recipients already hold complementary information. The relevant object is the combination of sender access, communicated record, and recipient background. Counting only the words in the artifact misses two thirds of that object.

The finite problem below belongs to zero-error communication with side information. Witsenhausen related side-information coding to graph coloring [15], while broadcasting research studies a common message used by several informed receivers [16]. We adapt that established construction to explicit sender access and recipient-specific demands. The result is a communication criterion within an explanation contract, not a new definition of comprehension.

8.1 An accessible sender and a family of recipients

Let $\mathcal{M}$ be a finite nonempty set of possible cases. The sender observes only $\alpha(m)$, where $\alpha : \mathcal{M} \to S$ is a declared access map. Recipient $i \in \{1, \dots, k\}$ receives background $\beta_i(m)$ and must recover $d_i(m)$. A demand can be a single answer, a reusable state, or the complete finite answer profile for a declared task. The maps and recipients are fixed before the case is selected.

One encoder $E : S \to W$ produces a common artifact. A decoder D_i for each recipient must satisfy

$$D_i(E(\alpha(m)), \beta_i(m)) = d_i(m) \qquad \text{for every } m, i. \tag{8}$$

The encoder is not allowed to consult a recipient-specific hidden state. Recipients do not receive an uncharged transcript or a separate description of the selected case. Here decoders may be designed once for the declared

finite contract and then remain fixed. If an existing interpreter is prescribed, the criterion below is only a necessary information condition unless that interpreter can implement the constructed decoder.

Build a graph G_α on $\alpha(\mathcal{M})$. Join s and t when some cases m, n and recipient i satisfy

$$\alpha(m) = s, \quad \alpha(n) = t, \quad \beta_i(m) = \beta_i(n), \quad d_i(m) \neq d_i(n).$$

An edge can be a loop when $s = t$. It then records a distinction that neither the sender's access nor that recipient's background can resolve.

Theorem 8.1 (A common explanatory artifact under sender and recipient constraints)**.** *An exact artifact satisfying* (8) *with at most $K \geq 1$ symbols exists if and only if G_α has no loop and admits a proper coloring with at most K colors. In the loop-free case, the minimum alphabet size is $\chi(G_\alpha)$, and the minimum fixed binary length with designable decoders is $\lceil \log_2 \chi(G_\alpha) \rceil$. A loop makes the exact contract infeasible for every alphabet size.*

Proof. An encoder assigns a symbol to every accessible sender value. If adjacent vertices receive the same symbol, their witnessing recipient receives the same artifact and background in two cases requiring different outputs. No deterministic decoder can be correct in both. Thus every valid encoder is a proper coloring, and a loop is impossible.

Conversely, use a proper coloring as the encoder. For any recipient and any artifact/background pair that actually occurs, assign the demanded output from one compatible case. Another compatible case cannot demand a different output, since it would witness an edge whose endpoints have the same color. Thus the decoder is well defined. Define it arbitrarily on unattainable pairs. Counting binary strings proves the final bound. □

The theorem provides a smallest *alphabet*, not the cheapest explanation. Its decoders can be large tables. Their construction, runtime, background acquisition, and semantic interpretation remain part of the resource account. The graph is useful precisely because it does not pretend that a color label is intrinsically meaningful.

8.2 Complementary knowledge can reduce communication

For two bits (a, b), let the first recipient know a and need b, while the second knows b and needs a. A sender with access only to $a \oplus b$ can broadcast that one bit. Both recipients recover their missing bit. The sender need not reconstruct the complete pair. An inference from successful explanation to possession of the recipient's full state would therefore be invalid.

This does not make the recipients' knowledge free. Their prior bits are explicitly supplied. Removing them changes the graph and the communication requirement. Nor does the example show that the sender has a useful general theory of the two-bit system. It establishes a specific jointly realizable communication contract.

Counterexample 8.2 (Individually binary explanations can require three common symbols)**.** Take five cases labeled $0, \ldots, 4$ and five recipients. Recipient i cannot distinguish cases i and $i + 1$ modulo five from its background, but distinguishes every other case individually. It demands zero on case i and one on case $i + 1$, with arbitrary fixed demands elsewhere. The sender knows the case. Each recipient separately needs only two artifact symbols. The common confusability graph is a five-cycle. It is not two-colorable, but the sequence of colors A, B, A, B, C is proper. One common artifact therefore needs three symbols, or two fixed binary bits.

The obstruction is global: the largest ambiguity faced by any one recipient does not determine the size of a shared explanation. The same issue appears when one artifact is expected to support a novice, a specialist, and a verifier who each possess different background information. The finite theorem tells us what must be distinguished. It does not prescribe how a human-facing explanation should express those distinctions.

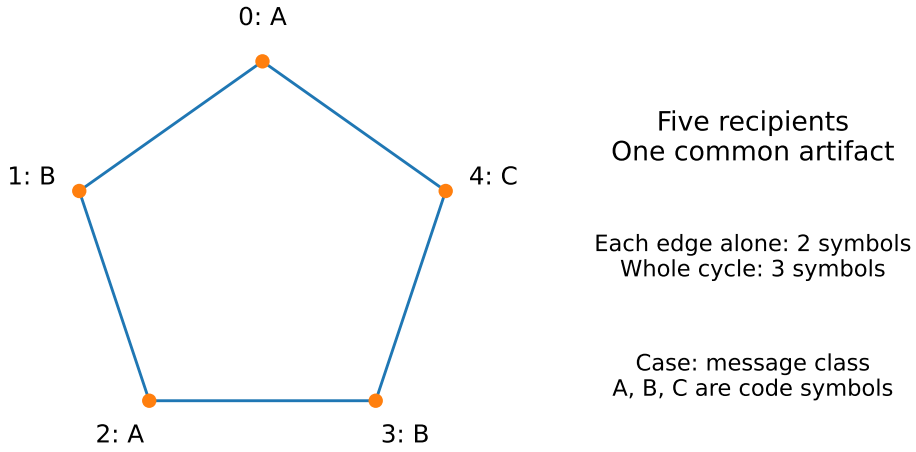


Figure 4: Each edge records one recipient's unresolved pair of cases. Each recipient alone can use a binary message. Their combined graph needs three symbols. Letters denote message classes, not empirical clusters or intrinsic semantic categories.

8.3 Source restrictions are substantive

If two cases have the same sender observation and the same recipient background but require different outputs, adding more fluent text cannot fix the information deficit. The graph has a loop. The options are to acquire evidence, change the recipient background, relax the demand, or report that the current contract cannot be met. Calling a more powerful decoder changes none of those facts if its permitted inputs remain identical.

9 Learning where the current representation fails

A reusable representation should support correction without treating every new observation as a reason to discard everything. A valid method can have a restricted domain. A recorded premise can be false. A previously omitted variable can become relevant. These failures call for different revisions.

We distinguish three cases. In a *state correction*, the operations remain valid but the current state estimate changes. In a *model correction*, an assumed transition or observation law changes. In a *contract expansion*, the requested operations or questions change. A representation can be entirely adequate before the third kind of change and inadequate afterward, as the instrument shows.

9.1 A finite model of informative correction

Let $\mathcal{M}$ be a finite nonempty set of deterministic candidate models. Let $\mathcal{E}$ be a finite set of permitted experiments. Each experiment e has an outcome map $f_e : \mathcal{M} \to O_e$. The current information is a candidate set $V \subseteq \mathcal{M}$. On observing o from e, update it by

$$C_{e,o}(V) = \{m \in V : f_e(m) = o\}. \tag{9}$$

This is consistency filtering, not Bayesian inference and not a general model of scientific reasoning. An empty set is meaningful: the observation contradicts every currently retained candidate. It must not be replaced by an invented best answer.

An experiment is informative for V when its outcome map is not constant on V. Models indistinguishable under every permitted experiment form an observational equivalence class. The aim can be to identify such a class rather than a uniquely named model.

Proposition 9.1 (Finite correction under separating evidence)**.** *Suppose the true model m_* belongs to the initial set V_0, every observation is exact, and the allowed experiment family separates any two distinct remaining observational classes. Repeatedly selecting an informative experiment preserves m_* and reaches a single observational class after at most $|V_0| - 1$ informative observations. If there are N distinguishable target classes, every adaptive procedure identifying all of them using experiments with at most $b \geq 2$ outcomes and at most h observations must satisfy $b^h \geq N$.*

Proof. The true model predicts each exact observation, so filtering never removes it. An informative experiment splits the current set into at least two nonempty outcome classes. Whatever true outcome occurs, at least one candidate is removed. The nonempty set can strictly shrink at most $|V_0| - 1$ times. If two observational classes remain, the separating assumption supplies another informative experiment. For the second claim, an adaptive decision tree of depth at most h and branching at most b has at most b^h leaves. Distinct required classes must reach different leaves. □

This elementary result shares the query-and-counterexample perspective of classical active learning [12]. It does not give an efficient method for discovering the candidate family, finding an inexpensive separating experiment, or dealing with inaccurate observations. Its upper bound is a count of informative observations, not a bound on wall-clock work or physical cost.

9.2 A question can be more useful than another answer

If all surviving candidates give the same answer to the next task, the system can act without identifying which candidate is true. If they disagree on a consequential operation, an experiment that distinguishes that disagreement may be more valuable than repeating familiar tasks on which all candidates agree.

The criterion is task-relative. Two models can differ on an inaccessible quantity and still support all permitted decisions. Conversely, a rarely asked question can distinguish models whose disagreement controls whether a future intervention is safe. The system should retain which distinctions have been settled and which have merely been irrelevant so far.

A request for evidence is also an operation with a realization. It may consume resources, change the system, or reveal only a biased part of the state. The finite filtering model presumes that the stated outcome maps already include those effects. Section 10 makes current-state changes and their cost explicit. For a changing instrument, one must use a controlled reset, a model of the full observation history, or a state update that tracks what the experiment changed. Independent repeated access to the same initial state is not free.

9.3 Repairing the running example

Before J is admitted, all five possible values of c are equivalent under the base contract. Treating them as one class is correct, not ignorance that the old task required the predictor to resolve. After J is admitted, these classes split. The system can report that its old state no longer determines the answer.

The current record supplies (s, b). Executing J and observing s_J determines $c = s_J - s$, while (4) updates the affected first register. The corrected extended coordinates are (s_J, b, c). Their operations are

$$\bar{H}(s, b, c) = (s + b, b, c), \quad \bar{W}(s, b, c) = (s, s - b, c), \quad \bar{J}(s, b, c) = (s + c, b, c).$$

The base method survives as the restriction of the enlarged method. Only the newly relevant distinction is acquired.

A different kind of failure would occur if the instrument secretly implemented $H(a, b, c) = (a + 2b, b, c)$. That would contradict the old transition model rather than merely extend the operation catalogue. A correction record should say which assumption changed. Otherwise a sequence of patched answers can conceal a permanently incorrect account of the mechanism.

9.4 Revision must preserve its own uncertainty

The true-model assumption in Proposition 9.1 is a substantive premise. When it is not justified, the system must allow the possibility that every candidate is wrong. When observations are noisy, deleting a model after one conflicting record can remove the best explanation. The statistical treatment in Sections 12–14 supplies exact and conditional results once a likelihood model and selection process have been declared. It does not establish their empirical calibration.

The finite framework nevertheless gives a useful discipline. A correction should identify the new evidence, the rejected candidates, the surviving claims, and the assumption linking the observation to the model. It should not rewrite an old uncertainty as though the corrected answer had always been established.

10 Acquisition as an experiment planning problem

A lower bound on the information a representation must hold is not yet a plan for acquiring it. Even a perfectly informative experiment can be unavailable, expensive, or destructive. We now make that acquisition problem exact for a finite deterministic model.

Belief-state planning has a substantial foundation in partially observed control [17]. The construction here uses sets of possible current states rather than probability distributions, a worst-case cost rather than expected reward, and an explicitly bounded horizon. It is a finite dynamic program for a stated contract, not a general solution to scientific discovery.

10.1 Experiments change both information and state

Let X be a finite nonempty state space. Each experiment e has an admissibility set $A_e \subseteq X$, a transition $T_e : X \to X$, an observation $o_e : X \to O_e$, and a known cost $c_e \geq 0$. The observation is indexed by the state before the experiment, but can describe a reading taken afterward: for example $o_e = q \circ T_e$. At an information state $V \subseteq X$, the experiment is permitted only when $V \subseteq A_e$. After outcome o, the new support is

$$V_{e,o} = \{T_e(x) : x \in V,\ o_e(x) = o\}. \tag{10}$$

Only nonempty branches are possible. This update tracks the world after the intervention, not just a narrowed list of old states.

Let $r : X \to Z$ be the target record. A support is terminal when r is constant on it. Let h be the number of experiments still available. Write $\mathcal{V}_h(V)$ for the least worst-case sum of experiment costs among policies that terminate with the target known within h steps. Infeasibility is denoted $+\infty$. Then define

$$\begin{aligned} \mathcal{V}_h(V) &= 0 && \text{if } |r(V)| = 1, \\ \mathcal{V}_0(V) &= +\infty && \text{otherwise,} \\ \mathcal{V}_h(V) &= \min_{e : V \subseteq A_e} \left(c_e + \max_{o : V_{e,o} \neq \varnothing} \mathcal{V}_{h-1}(V_{e,o}) \right) && \text{otherwise,}\ h > 0. \end{aligned} \tag{11}$$

An empty minimum is $+\infty$. The terminal case permits stopping immediately, so an unnecessary additional experiment is never forced.

Theorem 10.1 (Exact finite acquisition with state-changing observations)**.** *For every nonempty support V and finite horizon h, recurrence* (11) *equals the minimum worst-case acquisition cost among deterministic*

adaptive policies with that horizon, access boundary, and current-state target. Every finite value has an attaining policy obtained by choosing a minimizing experiment and its recursively attaining branches. The value is infinite exactly when no such policy exists.

Proof. At horizon zero, a target can be returned without further evidence precisely when it is constant over the current support. This gives the base case. At a nonterminal support with $h > 0$, the first experiment must be admissible at every possible state. After its observed outcome the possible current states are exactly (10). By induction, the best possible remaining worst-case cost on that branch is $\mathcal{V}_{h-1}$. Every nonempty branch can occur, giving the maximum over branches. Minimizing over the finite experiment set gives a lower bound for every adaptive policy. Choosing a minimizing experiment and attaining branch policies achieves that bound whenever it is finite. If every choice has an impossible branch, no guaranteed policy exists. □

The complete information state must include every variable needed for future transitions, admissibility, costs, and the target. A permission depending on an earlier event must be retained as state. Otherwise two histories with the same physical support could admit different experiments, invalidating the recurrence. Nonnegative costs and the finite horizon make zero-cost actions harmless to the definition. No unbounded termination claim is being made.

The state search can be large: there are up to $2^{|X|} - 1$ nonempty supports, before accounting for additional memory variables. The theorem is an exact finite characterization, not a polynomial-time learning guarantee. The delivered reference implementation memoizes supports and remaining horizons. Its companion check compares the result with a separately implemented enumeration of complete observation-history policies on small systems.

10.2 The diagnostic operation is a paid transformation

In the instrument, take a known base record (s, b) and let the five possible values of c define the initial support. Set the cost of J to two units, and the cost of H and W to one unit each. Each experiment returns the exact display after its operation. The target is the enlarged current record (s, b, c).

A J observation resolves all five possibilities and costs two. An H/W sequence cannot reveal c, so no policy costing less than two can meet the target. The optimum is therefore two, attained by executing J and retaining the post-operation record. This example fixes its sensor and operation catalogue. It does not claim that every physical diagnostic with five outcomes has the same cost.

10.3 Knowing the current state is not recovering its history

Counterexample 10.2 (A reset supplies certainty without recovering the past)**.** Let the current state be an unknown bit. The only experiment sets the bit to zero and returns a constant symbol. One experiment makes the current bit known with certainty. It supplies no information about whether the old bit was zero or one. Thus an acquisition policy for the current-state target can succeed by changing the world, not by identifying its original condition.

To require recovery of a past distinction, replace the state by (x_0, x_t), let every transition preserve x_0, and make the terminal target include the required part of x_0. The same recurrence then treats historical reconstruction honestly. In the reset example, the two origins remain indistinguishable at every horizon. If both current knowledge and preserved history matter, use their joint target rather than claim that either implies the other.

For the instrument, the diagnostic J is invertible once c is known, so the exact observation can support both records. That property comes from the specified operation. It cannot be transferred to an arbitrary experiment.

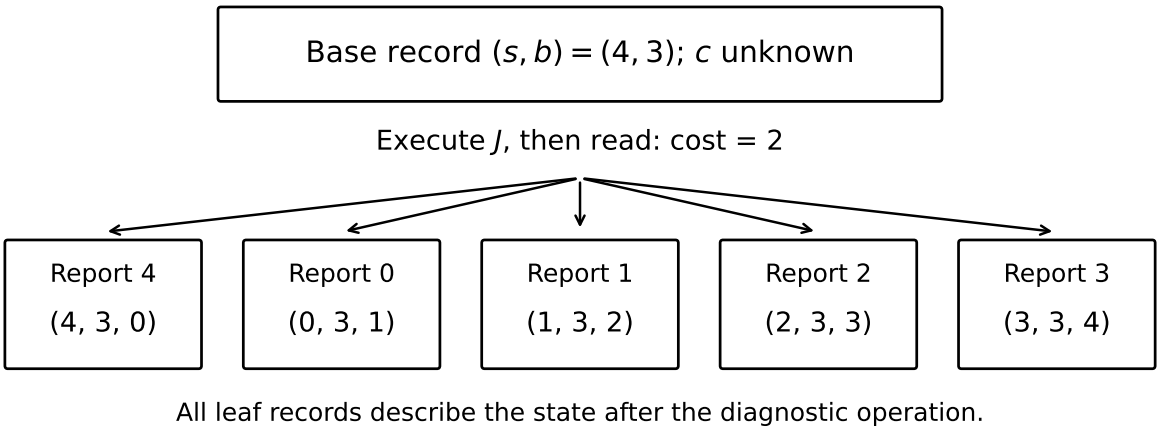


Figure 5: A single J diagnostic from base record $(s, b) = (4, 3)$ has five possible exact reports. Each report determines a different *post-operation* record. The cost is charged to the operation, and the hidden value is acquired rather than supplied to the decoder.

Likewise, a planner that chooses an informative experiment is not thereby credited with discovering its transition law or building its sensor.

10.4 Experiment value is relative to the intended continuation

An experiment need not distinguish every candidate model to be useful. It need only resolve distinctions that control the promised future behavior. Conversely, acquiring a value that changes none of the relevant decisions can waste resources even when it reduces uncertainty. Information-based experimental design has a long history [18]; the present cost criterion deliberately asks a different, exact question about a specified target and worst-case completion.

The horizon is a contractual limit, not an arbitrary truncation of a failed computation. A negative result at horizon h says that the declared task cannot be guaranteed within that many permitted experiments. Increasing the horizon changes the problem. It does not retroactively validate a promise made at the smaller horizon.

11 When translation commutes with correction

A method can travel through a language that preserves its current answers while losing the evidence needed to improve it. This is a different obstruction from a wrong update equation. It concerns the interaction between two otherwise sensible operations: translating a model and filtering it against evidence.

Let $\mathcal{M}$ and $\mathcal{N}$ be finite model families, with model translation $\phi : \mathcal{M} \to \mathcal{N}$. For one source experiment e, let $f : \mathcal{M} \to O$ give its outcomes. A corresponding target experiment has outcome map $g : \mathcal{N} \to O'$. Let $\eta : O \to O'$ translate evidence. Assume the basic observation square commutes:

$$g(\phi(m)) = \eta(f(m)) \quad \text{for all } m \in \mathcal{M}. \tag{12}$$

This condition says that a model's predicted evidence translates correctly. It permits η to merge source outcomes.

Write $\phi[V] = \{\phi(m) : m \in V\}$. Let $C_o(V) = \{m \in V : f(m) = o\}$ and $C'_{o'}(W) = \{n \in W : g(n) = o'\}$. Both are ordinary consistency filters.

Theorem 11.1 (Correction transport and evidence reflection)**.** *Under* (12)*, for every $V \subseteq \mathcal{M}$ and every attainable source outcome $o \in f(\mathcal{M})$,*

$$\phi[C_o(V)] \subseteq C'_{\eta(o)}(\phi[V]). \tag{13}$$

Equality holds for all such V and o if and only if η is injective on the attainable outcome set $f(\mathcal{M})$. The claim includes candidate sets contradicted by the observation, so that preservation of an empty update is part of the contract.

Proof. If $n \in \phi[C_o(V)]$, then $n = \phi(m)$ for some $m \in V$ with $f(m) = o$. Equation (12) gives $g(n) = \eta(o)$, proving inclusion.

If η is injective on $f(\mathcal{M})$ and $n \in C'_{\eta(o)}(\phi[V])$, choose $m \in V$ with $\phi(m) = n$. Then $\eta(f(m)) = g(n) = \eta(o)$, so $f(m) = o$. Thus $n \in \phi[C_o(V)]$, proving equality.

Conversely, suppose two distinct attainable outcomes a, b have $\eta(a) = \eta(b)$. Choose m with $f(m) = b$, let $V = \{m\}$, and consider source observation a. The source update is empty, whereas the target update retains $\phi(m)$ by (12). Uniform equality therefore fails. □

This is an exact set-theoretic result about direct images and filtering. The institutional or cognitive interpretation does not make it a new general theorem of set theory. Its value is to expose a condition that output translation alone omits: the translated evidence must preserve the distinctions by which a candidate is excluded.

The necessity statement deliberately tests observations that contradict the current candidate set. If those cases are excluded by a standing assumption that the current model is already correct, weaker conditions can suffice. That is a different and less demanding correction contract. A reasoner meant to discover that its model is wrong should not assume away the contradictory case.

Counterexample 11.2 (A translated observation preserves both mistaken candidates)**.** Take $\mathcal{M} = \{m_0, m_1\}$ with $f(m_0) = 0$ and $f(m_1) = 1$. Translate models bijectively to $\{n_0, n_1\}$, but translate both observations to a single symbol $*$. Let $g(n_0) = g(n_1) = *$. The observation square commutes. Starting from both candidates, source observation 1 leaves only m_1, whose image is $\{n_1\}$. Translating first and then filtering by $*$ leaves both n_0 and n_1.

The failure is not an inaccurate target answer in its coarse language. Both models really do predict $*$ there. The failure is the loss of a previously meaningful ability to discriminate. This is why a representation that can communicate conclusions may still be an inadequate carrier of scientific revision.

11.1 The instrument explains the boundary

For base experiments built only from H, W, and the display, the translation from (a, b, c) to (a, b) can preserve evidence without collapsing outcome symbols. Different values of c produce the same permitted observations, so their identification loses no base correction power.

For J, that translation cannot preserve the full outcome map at all: two states with identical (a, b) can have different readings after J. One could force a commuting square by translating all new readings into one coarse symbol. That would preserve the trivial coarse prediction, but it would destroy the ability to discover c. Enlarging the representation is a different response from weakening the question until the old representation appears adequate.

This distinction is important for evaluation. A system should not be rewarded for transfer by silently replacing a difficult target question with an easier translated one. The source and target question contracts, including evidence reflection when correction matters, must be fixed before the comparison.

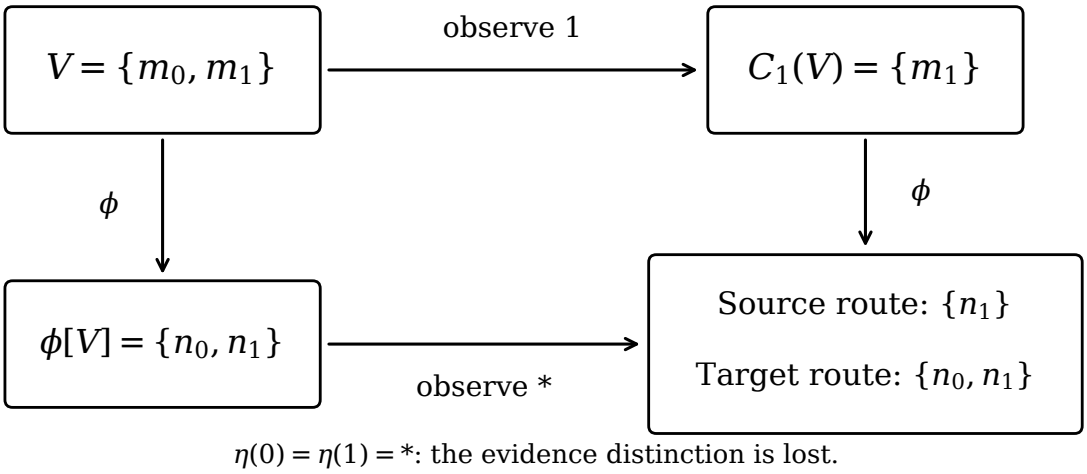


Figure 6: Correct translation of each predicted observation need not preserve correction. The lower evidence channel merges zero and one, so the target cannot remove the candidate excluded at the source. The theorem specifies when the inclusion becomes equality.

11.2 Composing corrective explanations

Suppose an explanatory artifact carries a recipient's candidate set, translated experiments, and evidence interpretation. If each translation preserves the observation square and reflects the relevant outcomes, successive filters and translations commute by repeated application of Theorem 11.1. The recipient can then discard the same translated candidates as the source after the specified observation sequence.

This conclusion is only about filtering the represented model families. It does not transfer an ability to invent a missing model, establish that the observations are trustworthy, or guarantee efficient implementation. An explanation can teach how to correct within a family while leaving the enlargement of that family as a separate problem.

12 Noisy evidence and Bayesian correction

Exact filtering removes a candidate as soon as an observation contradicts it. That is appropriate for the deterministic evidence contract in Section 11. A noisy observation can instead be unlikely under a model without being impossible. Reusing the exact filter in that setting can discard a well-supported model because a sensor behaved exactly as its noise model allowed.

Let $M \in \mathcal{M}$ be a fixed unknown model in a finite family, with prior $p(m)$. An experiment produces $O \in \mathcal{O}$ with likelihood $L_m(o)$. A common reporting channel $B(o, z)$ produces $Z \in \mathcal{Z}$. Each likelihood row and each channel row is a probability distribution. The complete joint model is

$$\Pr(M = m, O = o, Z = z) = p(m)L_m(o)B(o, z). \tag{14}$$

In particular, the channel does not secretly inspect M. Define

$$Q_m(z) = \sum_o L_m(o)B(o, z), \qquad p^o(m) = \frac{p(m)L_m(o)}{\sum_n p(n)L_n(o)}, \qquad p^z(m) = \frac{p(m)Q_m(z)}{\sum_n p(n)Q_n(z)}.$$

A posterior is defined only when its denominator is positive. An event assigned probability zero by the entire current model has no posterior under that model. It is a model-failure signal, not permission to invent a probability distribution.

This section retains the same model identities at source and target so that it isolates evidence loss. Merging model identities is a separate operation. A target likelihood on a merged class is independent of the allocation of prior mass within that class only when the relevant likelihood is constant there, or when the within-class prior itself is retained as additional state. That assumption must not be concealed in a notation change.

12.1 When a coarser report preserves every update

The reporting model implies $M \to O \to Z$. Therefore, on every positive-probability pair (o, z), the full posterior after observing both records equals p^o. The coarse posterior is a mixture of these fine posteriors:

$$p^z = \sum_o \Pr(O = o \mid Z = z)\, p^o. \tag{15}$$

Averaging is not generally the same as retaining each possible update. The distinction is classical statistical sufficiency, whose factorization foundations include Halmos and Savage [19]. We state and prove the finite channel criterion used here.

Theorem 12.1 (Exact Bayesian correction through a reporting channel)**.** *For fixed finite likelihoods L and reporting channel B, the following are equivalent:*

(i) For every prior p, $p^o = p^z$ on every positive-probability pair (o, z).

(ii) This equality holds for one prior assigning positive mass to every model.

(iii) There exists a row-stochastic kernel $\rho(o \mid z)$ such that, for all m, o, z,

$$L_m(o)B(o, z) = Q_m(z)\rho(o \mid z). \tag{16}$$

Thus a report preserves every posterior update exactly when the discarded observation can be conditionally reconstructed from that report without knowing the model, in the actual joint experiment.

Proof. The first condition implies the second. Suppose the second holds under a fully supported prior. For each z of positive probability set $\rho(o \mid z) = \Pr(O = o \mid Z = z)$. If (o, z) has positive probability, posterior equality and Bayes' formula give

$$p(m)L_m(o)B(o, z) = \Pr(O = o, Z = z)p^o(m) = \rho(o \mid z)\Pr(Z = z)p^z(m) = p(m)Q_m(z)\rho(o \mid z).$$

Cancel the positive $p(m)$. If the pair has probability zero, full prior support makes $L_m(o)B(o, z) = 0$ for every model, and $\rho(o \mid z) = 0$. If z is impossible under every model, choose any probability row $\rho(\cdot \mid z)$ since both sides are zero. This proves the third condition.

Conversely, the factorization makes the likelihood of (o, z) a model-independent multiple of the likelihood of z. Normalization under any prior with positive pair probability therefore gives $p(M \mid o, z) = p^z$. The common reporting channel gives $p(M \mid o, z) = p^o$. This proves the first condition. □

For deterministic reporting $z = \eta(o)$, the criterion says that the nonzero vectors $(L_m(o))_{m \in \mathcal{M}}$ within each report fibre must be proportional. An ancillary symbol may be discarded even when the report is not injective. In the deterministic evidence model, distinct attainable outcomes have nonzero likelihood vectors with disjoint supports. Those vectors cannot be proportional, recovering the injectivity requirement for that special case. Noise therefore changes the correct preservation test, not merely the notation used for filtering.

The kernel ρ is conditional reconstruction for the specified coupling (M, O, Z). The existence of some unrelated procedure simulating an experiment from another record is a different assertion. Neither assertion shows that the real sensor has the stated likelihood or that a learner inferred it correctly.

12.2 An exact measure of the lost update

A coarser report may fail the criterion while still retaining much of the useful evidence. There is a natural average loss, though it should not be mistaken for a worst-case bound on every possible observation. For finite distributions use natural logarithms and the convention $0 \log 0 = 0$.

Proposition 12.2 (Expected posterior loss is conditional information)**.** *Under* (14)*,*

$$\mathbb{E}_{O,Z}\left[D_{\mathrm{KL}}(p^O \| p^Z)\right] = I(M; O \mid Z) = I(M; O) - I(M; Z). \tag{17}$$

The common value is nonnegative. It is zero exactly when the two posteriors agree on every positive-probability observation pair for that prior. For a fully supported prior, zero is equivalent to the factorization in Theorem 12.1.

Proof. On the support of the joint law, $p(m \mid o, z) = p^o(m)$. Substituting this into the finite definition of conditional mutual information gives the expected divergence. Expanding its logarithm as $\log(p^o(m)/p(m)) - \log(p^z(m)/p(m))$ and summing gives the difference of mutual informations. Terms with zero prior mass carry no weight. Every positive-weight numerator has a positive denominator, so the expression is finite.

For probability distributions, $D_{\mathrm{KL}}(a\|b) \geq 0$, with equality only at $a = b$. This follows from $\log t \leq t - 1$ on the common support, with missing support only increasing the divergence. A finite weighted sum of nonnegative terms is zero exactly when every positive-weight term is zero. The final assertion follows from Theorem 12.1. □

This is the standard information chain rule in the present correction contract, not a new entropy identity. It makes the object of the lost information explicit: the update over the chosen model family. That family may include irrelevant distinctions or omit the true mechanism. A small or zero value does not independently validate the family.

12.3 The noisy instrument and a harmless discarded flag

Return to the instrument immediately after a known J intervention. The earlier (s, b) is known, while c is initially uniform on five values. A read-only sensor reports the shifted value Y = reported display $- s$ modulo five. It returns the correct c with probability $3/4$ and each wrong value with probability $1/16$. The physical state remains fixed during repeated read-only observations.

Suppose the raw report also contains a binary flag with probabilities $1/3$ and $2/3$, independent of c and of Y. Removing that flag satisfies (16) exactly. Both recipients retain the same posterior over c and hence the same distribution over the post-J state $(s + c - b, b, c)$. The flag does not carry corrective information for this family.

Counterexample 12.3 (A compact report can destroy the useful noisy distinction)**.** In the same instrument, replace $Y \in \{0, 1, 2, 3, 4\}$ by its ordinary integer parity, not a field operation. On observing the full symbol $Y = 2$, the posterior puts $3/4$ on $c = 2$ and $1/16$ on each alternative. On observing only even parity, it puts $7/24$ on each even candidate and $1/16$ on each odd candidate. The report is correct as a parity statement but fails to preserve the posterior needed to distinguish the three even values.

For n conditionally independent identical reports $Y = 2$, the posterior of $c = 2$ is

$$\frac{12^n}{12^n + 4}.$$

For n reports retained only as even parity, the posterior of that same candidate is

$$\frac{14^n}{3 \cdot 14^n + 2 \cdot 3^n}.$$

At $n = 0$ both equal the prior $1/5$. The full-report value at $n = 3$ is $432/433$, whereas the parity value cannot exceed $1/3$. Repeatedly receiving the same coarse fact does not restore distinctions the reporting language never records.

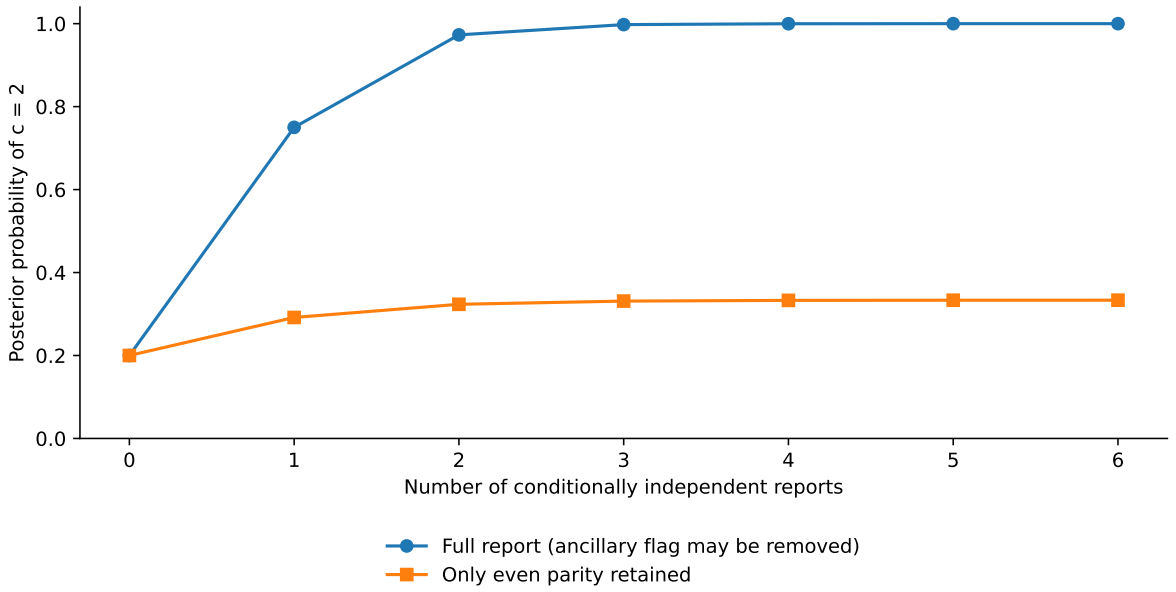


Figure 7: Posterior mass on $c = 2$ along the specified repeated-report histories. Removing the ancillary flag gives exactly the full-report curve. Retaining only parity leaves three even candidates indistinguishable. The curves are analytic conditional calculations, not measured learning trajectories or probabilities of obtaining these histories.

13 Approximation, rare evidence, and decision stability

A small error in an unconditional model does not automatically give a small posterior error after an unusual observation. Conditioning divides by the probability of the evidence. A useful correction guarantee must therefore record both an approximation error and the evidence probability at which that guarantee is used.

Proposition 13.1 (A conditional stability bound with an evidence denominator)**.** *Let P and Q be finite joint laws of a model and an observation, with $d_{\mathrm{TV}}(P, Q) \le \delta$. Let E be an observation event with $a = P(E) > 0$ and $b = Q(E) > 0$. Then*

$$d_{\mathrm{TV}}\big(P(M \mid E), Q(M \mid E)\big) \le \min\left\{1, \frac{\delta}{\max(a, b)}\right\}. \tag{18}$$

In particular, a justified lower bound $P(E) \ge \gamma > 0$ supplies the weaker usable bound $\min\{1, \delta/\gamma\}$.

Proof. Write $u_m = P(M = m, E)$, $v_m = Q(M = m, E)$, and let $p = u/a$, $q = v/b$. Exchanging P and Q if necessary, assume $a \le b$. The variation inside E is at least $\frac{1}{2}\sum_m |u_m - v_m|$. The variation outside E is at least $|a - b|/2$, since its total masses differ by that amount. Hence

$$\delta \ge \tfrac{1}{2}\sum_m |u_m - v_m| + \tfrac{1}{2}(b - a) = b - \sum_m \min(u_m, v_m).$$

Since $a \le b$, $\min(ap_m, bq_m) \le b \min(p_m, q_m)$. Substitution gives $\delta \ge b\, d_{\mathrm{TV}}(p, q)$. Division by $b = \max(a, b)$ and the universal upper bound one prove the claim. □

The bound is sharp for general joint laws: put the same mass a on E in both laws, choose conditional laws at distance t, and make their complements identical. Then the joint distance is at. The result neither estimates δ from data nor proves that the evidence model is calibrated. Those are separate inputs.

Counterexample 13.2 (Rare evidence can reverse a posterior despite small joint error)**.** For $0 < \epsilon < 1/2$, take a binary model with equal prior probabilities. Under P, let $P(M = 0, E) = \epsilon$ and $P(M = 1, E) = 0$. Under Q, interchange these two masses. Choose the complement masses so each model retains prior probability $1/2$. Both laws give E probability ϵ, their joint total variation is 2ϵ, and their posteriors on E are opposite point masses. Joint error can tend to zero while conditional error remains one.

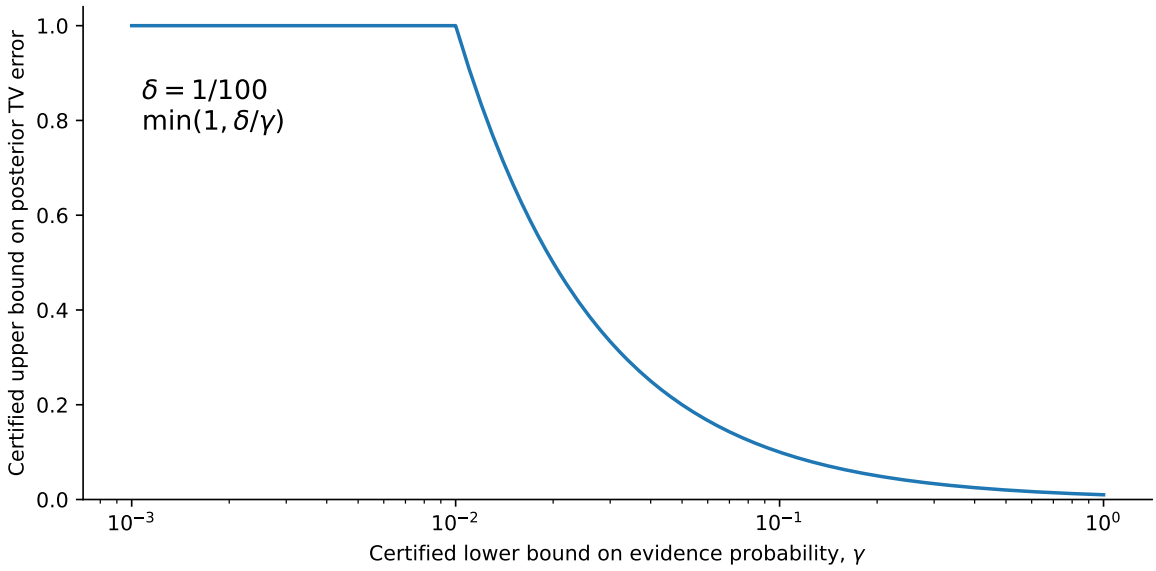


Figure 8: The certified upper bound $\min(1, \delta/\gamma)$ for $\delta = 1/100$. A small joint-law error does not give a useful guarantee when the evidence probability is comparably small. The plot is a bound, not an empirical estimate of posterior error.

13.1 From approximate beliefs to a recipient's decisions

Not every posterior distinction matters equally to an action. To link correction with use, fix one finite nonempty action set A and losses $\ell(m, a) \in [0, L_{\max}]$. A receiver evaluates actions from the distribution its actual interpreter reconstructs. The target loss function and available actions must be the same in the comparison.

Proposition 13.3 (A joint posterior and recipient decision certificate)**.** *Let p be the reference posterior, $\widehat{p}$ an approximate or transmitted posterior, and r_i recipient i's reconstructed posterior. Suppose*

$$d_{\mathrm{TV}}(p, \widehat{p}) \le \epsilon, \qquad d_{\mathrm{TV}}(\widehat{p}, r_i) \le \eta_i.$$

If a_i minimizes expected loss under r_i, then, with $e_i = \min\{1, \epsilon + \eta_i\}$,

$$\mathbb{E}_p \ell(M, a_i) - \min_{a \in A} \mathbb{E}_p \ell(M, a) \le \min\{L_{\max}, 2L_{\max} e_i\}. \tag{19}$$

If the expected-loss gap between a_i and every other action under r_i is strictly greater than $2L_{\max}e_i$, then a_i is also the unique optimum under p.

Proof. The triangle inequality bounds $d_{\mathrm{TV}}(p, r_i)$ by e_i. For a function taking values in $[0, L_{\max}]$, its expectations under these two distributions differ by at most $L_{\max} e_i$. Let a_* minimize expected loss under p. Insert the two risks under r_i between the risks of a_i and a_* under p. The middle difference is nonpositive by optimality under r_i, and the two outside differences are at most $L_{\max} e_i$. Regret is also at most $L_{\max}$. Applying the same two expectation bounds to the difference between a_i and any competitor proves the strict-gap assertion. □

The two errors concern one actual chain of records. One cannot use the best transmission for one recipient and the best posterior approximation for another while reporting a common certificate. When $\widehat{p} = Q(M \mid E)$, Proposition 13.1 can supply ϵ. A separate information-loss estimate or an explicitly verified decoder may instead supply the bound. The result is relative to the stipulated posterior and loss, not a guarantee that a real decision is wise or that a protected real-world outcome can be represented by a bounded scalar loss.

An abstention action can be included with an explicit loss and availability rule. This permits a recipient to decline an unsupported decision within the same model. It does not create a universally correct abstention threshold. Nor does closeness in expected loss establish exact preservation of a logical obligation: an arbitrarily unlikely prohibited outcome can matter under a different contract.

13.2 Approximate correction is not arbitrary tolerance

A likelihood that is merely close in average error may have the wrong zeros, changing which observations the model considers possible. Posterior bounds must therefore specify positive evidence, the full observation record, and the domain on which the approximation holds. Confidence produced by normalization is not an independent check of those premises.

The exact factorization criterion in Section 12 is an all-prior statement. The conditional-information identity is an average statement under one prior. The stability bound is an event-specific comparison of two joint laws. The decision certificate concerns one loss family. They are related, but none should be silently substituted for another.

14 Sequential inquiry through a retained evidence channel

An explanation intended for further inquiry must support more than one update. The recipient may choose its next experiment after seeing the current report. Exact evidence preservation can survive this adaptation, but only when the experiment-selection and observation assumptions are stated together.

Let the unknown model M remain fixed. At time t, a known policy selects experiment e_t using only the prior reported history and recorded experiment names. Randomization is permitted when its conditional law is independent of M given that history. Given $M = m$ and $e_t = e$, a fresh raw observation has law $L_{e,m}$, and its reported form is drawn through B_e. Conditional on the model and selected experiment, these draws do not depend on earlier hidden observations. Write $Q_{e,m} = L_{e,m} B_e$.

Theorem 14.1 (Sequential preservation of noisy correction under adaptive experiments)**.** *Suppose each permitted experiment admits a kernel ρ_e with*

$$L_{e,m}(o) B_e(o, z) = Q_{e,m}(z) \rho_e(o \mid z) \quad \textit{for all } m, o, z.$$

For every finite positive-probability history under the declared policy, the posterior conditioned on the complete raw and reported record equals the posterior conditioned only on the reported outcomes and experiment names. The claim also holds at a bounded stopping time determined by that reported history.

Proof. For a fixed realized history, its likelihood under model m is the product of the selected-experiment probabilities and the terms $L_{e_t,m}(o_t) B_{e_t}(o_t, z_t)$. The policy factors are independent of m once the reported

history is fixed. Applying the assumed factorization gives a product of the terms $Q_{e_t,m}(z_t)$ multiplied by the model-independent factors $\rho_{e_t}(o_t \mid z_t)$ and the same policy factors. Posterior normalization cancels those factors. The remaining product is exactly the likelihood for the reported history. A bounded stopping decision determined by that history adds no model-dependent factor, so the same calculation applies. □

The theorem gives a reusable correction procedure, not a claim that the selected experiments are optimal. The selection policy can be inefficient while the posterior updates remain exactly preserved. Nor does the theorem cover arbitrary dependent sensor noise. When observation laws depend on hidden physical state or prior observations, that dependence must be represented in the state and likelihood rather than treated as a fresh independent draw.

Counterexample 14.2 (Copying a record is not repeating an experiment)**.** Let a binary model have equal prior probabilities, and suppose one observed positive report has likelihood $4/5$ under model 1 and $1/5$ under model 0. Its posterior probability for model 1 is $4/5$. Sending two copies of that same report does not add evidence, so the posterior remains $4/5$. Treating the copies as conditionally independent instead gives $16/17$. Both calculations are arithmetically valid for their respective likelihoods, but only the first describes duplication of one record.

A correction record must therefore preserve provenance about which observations are new and which merely repeat an earlier result. A translated explanation that reproduces two identical sentences is not, without that provenance, a reliable record of two independent measurements.

14.1 Actions and stopping decisions can also carry information

The theorem records experiment names and restricts selection to information already represented at the recipient. A hidden selector that consults an additional sensor changes the likelihood of which experiment gets selected. Discarding that selection mechanism can bias the update even when the reported sensor itself is well modeled.

A stopping time chosen from the public reported history is different: the history already contains the information used to stop, so the event of stopping contributes no additional model-dependent likelihood on that fixed history. This exact likelihood argument should not be confused with transferring a deterministic-time approximation to an arbitrary stopping time. Small unconditional error at each fixed time still needs a separate argument to yield a useful conditional error on selected histories.

14.2 What a correction-preserving artifact must carry

For a recipient to execute this result, an artifact or its declared background must specify the model family, prior, experiment names, likelihoods, reporting channels, and update rule. It must also identify which independence and selection assumptions are being used. Sending only the current most probable answer does not generally preserve the next update.

The recipient does not need every raw token when the factorization is valid. It needs sufficient evidence and an accurate description of the reporting process. In the instrument, the irrelevant flag can be omitted throughout adaptive choices among experiments with verified factorization. A coarse parity language cannot claim the same posterior contract merely because every parity sentence it transmits is true.

Part III

Common witnesses and charged resources

15 One witness for prediction, transfer, and revision

The separate results now have to refer to one realized system. A good state representation, an elegant translation, and a useful explanation do not establish a combined capability when they require different hidden information, different decoders, or incompatible resource assumptions.

Definition 15.1 (Integrated adequacy record)**.** An integrated adequacy record for a declared task family identifies a common computational realization, its information boundary, a reusable representation and its updates, permitted intervention implementations, each claimed transport and recipient, the model and evidence maps used for correction, and the cost and scope of those operations. Every claimed guarantee refers to these same objects or to explicitly related replacements.

This is a specification proposal, not a new primitive of cognition. Its purpose is to make the quantifiers visible. The record cannot use a powerful target-specific decoder to establish prediction, then a different decoder to establish transfer, while reporting the result as one reusable method unless the selection procedure is itself part of the method.

15.1 What can be checked in the finite core

For explicitly supplied finite tables, the equations in Definition 3.1 can be checked on every state, operation, and query. Theorem 4.1 then proves the guarantee for arbitrary finite continuations. Exact transfer requires corresponding one-step and decoder checks. Correction transport requires the observation square and its stated reflection condition. The recipient must actually construct and execute the declared procedure within its allowed information boundary.

Some obligations remain different in kind. The finite model can list which intervention is admissible, but relating that declaration to a physical device requires evidence about the device. A table can name a sensor reading without proving calibration. A certificate can bind a program version without establishing the semantic meaning of its inputs. The specification should preserve these distinctions rather than present one successful check as universal certification.

Claim	Required object	What it does not supply
Reusable prediction	One retained state, updates, decoders, and a fixed future family.	Access to that state or evidence of its actual use.
Mechanistic attribution	A realized mapping and discriminating permitted interventions.	Every untested mechanism or unrestricted causal access.
Method transfer	Matched states, operations, queries, and coverage.	Reverse reconstruction or new target operations.
Recipient explanation	Fixed background, an artifact, an interpreter, and a budget.	Faithfulness to the source solver's actual computation.
Corrective transfer	Candidate and evidence maps that preserve filtering.	Model invention, noise calibration, or truth outside the family.
Resource feasibility	One accounting boundary for construction and repeated use.	A universal efficiency advantage.

15.2 The complete instrument witness

For the original contract, the instrument witness contains $r(a, b, c) = (a + b, b)$, updates (2), the first-coordinate decoder, and a statement that c is outside the relevant future distinctions. It also names how an observer acquires the initial record. A recipient with that record and the equations can reproduce all declared predictions.

After J is admitted, the witness changes. The old representation does not predict the new reading. An explicit J observation supplies c and updates the first coordinate. The enlarged state has the exact updates in Section 9. The original H/W explanations remain valid on their unchanged domain. New predictions require the acquired evidence, and any translated explanation must preserve that evidence's discriminating meaning.

These are connected properties of one analytical construction. They do not establish that an unspecified learning algorithm will discover the representation, infer the correct transition law, or choose the diagnostic operation. The point is to make those remaining research obligations identifiable rather than confuse them with the solved representation problem.

15.3 A cost account for reusable methods

For a sequential workflow with disjoint accounting, total cost includes acquiring information, discovering the representation, encoding state, performing updates, verifying claimed relationships, translating when needed, and interpreting the result. Parallel execution requires work and elapsed time to be recorded separately. Costs in different units should not be added without a stated conversion.

A simple amortization comparison illustrates the issue. Suppose a representation costs D to construct, each later use costs c, and the matched baseline costs $b > c$ per use. The representation has lower total cost after N uses exactly when

$$D + Nc < Nb, \quad \text{or equivalently} \quad N > \frac{D}{b - c}.$$

This accounting does not prove that the representation can be discovered for cost D, that the uses are comparable, or that all target instances share it. It locates where such evidence would enter.

Understanding need not always make computation faster. A more costly representation can make a proof inspectable, reveal a missing assumption, or support an intervention unavailable to the cheaper one. The relevant benefit must be stated. The theory should not smuggle efficiency into the meaning of understanding, nor infer efficiency from the existence of a short description.

16 Joint acquisition and explanatory communication

The acquisition problem and the recipient problem have so far been solved separately. Their composition is not automatic. An inexpensive observation may leave the sender unable to encode what a recipient needs. Conversely, requiring the sender to reconstruct the entire state can demand more evidence than is needed to help recipients who already possess relevant background.

We now combine the two exact finite models without introducing an omniscient encoder. The resulting test gives one witness for the experiment tree, its information boundary, a common artifact, and each recipient's decoding contract.

16.1 The graph belongs to the acquired transcript

Fix a finite set X_0 of possible initial states, a finite deterministic experiment catalogue as in Section 10, and a finite horizon h. A deterministic experimental policy is an observation-history tree. Its choices cannot depend on the hidden initial state. Only uniformly admissible choices are permitted at each reachable node.

For a policy tree T and initial state x, write $s_T(x)$ for its terminal sender transcript and $x_T(x)$ for its final physical state. The tree has finitely many reachable leaves. Specify recipient backgrounds $\beta_i^T(x)$ and demands $d_i^T(x)$ by a declared rule. A demand may concern $x_T(x)$, the original x, or a stated pair of past and present values. These are different contracts.

The recipient sees only the common artifact and its specified background. If an experiment name, its timing, or another transcript is visible to the recipient, that information belongs in β_i^T. It must not be used as an uncharged auxiliary message. Let $C(T)$ be the maximum total experimental cost across initial states. Build G_T on the reachable sender transcripts using the graph construction of Section 8, with $\alpha = s_T$.

Theorem 16.1 (A finite joint certificate for acquisition and explanation)**.** *For finite X_0, a finite experiment catalogue, horizon h, cost budget C, and artifact alphabet bound $K \geq 1$, there exists a uniformly admissible deterministic acquisition policy and one common artifact satisfying every recipient demand exactly if and only if some attainable policy tree T satisfies*

$$C(T) \leq C, \qquad G_T \text{ has no loop}, \qquad \chi(G_T) \leq K.$$

This is a finite decision problem. An accepting witness consists of the policy tree, a proper coloring, and the induced fixed decoders. Among feasible trees, a minimum experimental cost is attained. The statement does not assert that the search or the resulting decoders are efficient.

Proof. Any feasible system induces a policy tree of depth at most h, because its experiment choices depend only on its available history. The sender's final artifact is a function of its terminal transcript. Applying Theorem 8.1 to those transcripts proves the graph conditions. The worst-case experimental cost is $C(T)$, so it satisfies the budget.

Conversely, choose a tree with the stated properties. Execute its uniformly admissible experiments, encode the terminal transcript using its proper coloring, and use the decoders constructed in Theorem 8.1. They give every required answer on every initial state. The tree's cost and horizon meet the contract. There are finitely many experiment and outcome choices at each node and a finite depth bound, hence finitely many candidate trees. Exhaustive admissibility, cost, and coloring checks decide existence. A nonempty finite set of feasible tree costs has an attained minimum. □

The graph is built across all reachable leaves, not separately at each leaf. Two different transcripts can be indistinguishable to a recipient after encoding even when each sender support is locally well understood. A message assignment chosen independently at each leaf can therefore fail the common-artifact requirement.

For a fixed preexisting decoder family, each colored artifact must additionally be checked against that family. For a finite catalogue of allowable implementations, direct verification of those witnesses remains finite. The coloring theorem alone does not make a convenient decoder available, just as the acquisition theorem does not create a sensor.

Counterexample 16.2 (A useful explanation need not identify the state at its source)**.** Let the hidden state be two bits (a, b). The sender's only experiment reads $a \oplus b$ without changing the state. Repeating it never identifies the complete pair. One recipient already knows a and needs b, while another knows b and needs a. The single parity reading and its one-bit broadcast satisfy both demands. Before that reading, the sender has only one transcript and the explanation graph has a loop. After the reading, the graph is one edge and the one-bit message is exact.

This is an integrated information construction, not evidence that the sender possesses the recipients' full understanding. The explanatory success is jointly realized by the acquired parity and the recipients' backgrounds. Attribution should identify which participant carries which distinction.

16.2 Why noisy belief and exact state recovery remain different

The joint theorem concerns deterministic experiments and exact demands. The Bayesian sections concern noisy records and distributions over a model family. One cannot replace the target state in the joint theorem with a most probable guess and claim the exact guarantee has been preserved.

In the noisy instrument, every readout has positive likelihood under every candidate value of c. With a fully supported prior, any finite sequence of those readouts leaves every candidate with positive posterior probability. Repeated reports can make one value overwhelmingly likely, but they cannot give zero-error identification within a fixed finite number of such observations. A decision-loss tolerance, a posterior threshold, and an exact reconstruction target are different terminal conditions.

The decision certificate in Proposition 13.3 supplies one bridge: a recipient can act under an explicit error budget without pretending its estimate is exact. A more general synthesis optimizing noisy experimentation and multi-recipient communication jointly remains a separate problem. That larger stochastic optimization is outside the finite exact synthesis of this edition.

16.3 A complete expanded instrument record

The original instrument already separates current display, reusable state, and newly relevant information. The expanded account adds the mechanisms by which that information is acquired, represented probabilistically, and shared.

First, the reasoner retains the base coordinates and the exact scope of H/W reuse. Admitting J splits old behavioral classes. The exact diagnostic plan uses a real J operation and preserves its effect on the first register. If only the noisy sensor is available, the retained object is a posterior over possible post-J states, not an invented exact value of c.

Second, the evidence description identifies which readouts are conditionally independent and which tokens are copied or ancillary. Removing the independent flag is correction preserving. Replacing all five possible readings by parity is not. Three identical full readings give the conditional posterior $432/433$ for the corresponding candidate, but that number remains conditional on the declared likelihood model and prior.

Third, recipients receive the method appropriate to their backgrounds. A recipient with the base coordinates needs the newly acquired distinction or its uncertainty distribution. A recipient without those coordinates needs additional state information. A shared artifact is checked across all of those backgrounds at once. Its source encoder may use only the information actually acquired.

Finally, the record includes the decoder, query loss or exactness criterion, operation catalogue, and full resource account. Discovery of the transition law, calibration of the sensor, and mechanistic attribution to a particular neural system are not supplied by this analytical example. They remain separate scientific claims with separate evidence requirements.

16.4 A release boundary rather than an inflated definition

The mathematical contribution is a connected set of exact and conditional preservation results. The finite graph and decision tree supply executable witnesses. The likelihood factorization identifies when discarded records do not change Bayesian correction. The error bound exposes when a rare observation makes an approximate model inadequate. The recipient certificate carries those errors into a defined decision criterion.

Together they make a stronger specification of usable understanding. They do not turn every valid compression into a cognitive achievement or every encoded table into an economical explanation. Claims of economy require an explicit interpreter language, charged background information, and a measured or proved implementation cost. Claims of actual use require a realized causal mechanism. Keeping these obligations separate is part of the theory's content, not a reason to postpone its mathematical statements indefinitely.

17 What the resource account must charge

A short explanatory message can conceal an expensive recipient. A compact state can require an elaborate encoder. A fast online answer can depend on a table that was expensive to construct and distribute. None of these possibilities makes the answer wrong. They change what has been shown about the economy of the method.

The first-edition resource contract separates four accounts. The *design account* includes discovering and validating the state model, constructing a decoder, and searching for a policy. The *installed account* includes the fixed interpreter, model tables, recipient background, and compiled controller. The *online account* includes observations, physical interventions, communication, and executed instructions. The *revision account* includes obtaining new evidence and replacing invalid assumptions or interfaces. A quantity cannot be declared absent merely because it was paid in a different account.

These accounts use different units. Description bits, controller records, modular operations, physical interventions, and latency are not interchangeable. A weighted scalar cost is legitimate only after its conversion factors have been specified. The synthesis below retains a vector of online charges and a separate controller storage count. It does not silently convert one recipient's computation into the sender's communication budget.

17.1 Task-dependent interpreter changes are information

Let Θ be a finite nonempty family of tasks and let each task have a required answer profile over a fixed query family. A fixed interpreter U receives background B_θ, an installed program string p_θ, an artifact e_θ, and a query. It must return the required answer. The interpreter is fixed across tasks. The strings may depend on the task, but their lengths are charged. Let N_B be the number of distinct required answer profiles among tasks having the same background value B.

Proposition 17.1 (Charging a task-dependent interpreter)**.** *Suppose p_θ and e_θ have fixed lengths d and b bits, respectively, and the deterministic interpreter answers every required query exactly. Then*

$$2^{d+b} \geq \max_B N_B.$$

In particular, representing every Boolean function on n input bits under one fixed background requires $d + b \geq 2^n$. This counts task-dependent information, not the running time of the interpreter or the information needed for any one specially chosen function.

Proof. Fix a background B. If two distinct required answer profiles used the same pair (p_θ, e_θ), the fixed interpreter would receive identical inputs for a distinguishing query and would have to give the same answer. Exactness would fail for one of the two tasks. There are at most 2^{d+b} pairs, proving the first bound. There are 2^{2^n} Boolean functions on n input bits, with distinct complete answer profiles. Taking base-two logarithms proves the second assertion. □

This is the counting argument of Proposition 7.2 applied to the complete task-dependent package. It is not a new general description-complexity theorem. The point is to prevent a moving interpreter from becoming an unreported communication channel. Variable-length descriptions require a declared framing or prefix convention. Merely saying that a message has length at most b is not the same fixed-length model, because the length itself can carry information.

A task-independent interpreter can contain rich mathematical knowledge without encoding which task instance is present. Its construction and availability still belong in the installed and design accounts. Conversely, a separate decoder chosen for every task must be identified and charged even when that decoder

is called a background assumption. Control-task critiques of probing and expressivity critiques of alignment make this boundary especially important [8, 9].

17.2 Amortization without disappearing costs

Suppose a method has setup cost S and per-use cost r, while a comparator costs c on each use. In the same declared cost units and with no other differences, the method costs less over N uses exactly when $S + Nr < Nc$. When $c > r$, this is $N > S/(c - r)$. At equality the costs tie. If $c \leq r$ and $S \geq 0$, setup cannot create a strict cost saving in this model.

The formula does not establish that the method can be discovered, that the repeated problems are interchangeable, or that its validity survives a change of task distribution. Its role is bookkeeping. A useful representation can repay a substantial initial investment. It cannot claim that investment never occurred. Likewise, a precomputed table can be a sensible engineering choice for a bounded domain. The comparison should expose its storage and preparation rather than reject it by terminology.

17.3 Cost is an attribute of a specified implementation

An algebraic equation does not have one architecture-independent runtime. Addition modulo five can be a primitive instruction, a small Boolean circuit, or a table lookup. The worked implementation in Section 20 counts modular arithmetic operations and physical calls explicitly. It also records the sizes of its actual code and serialized witness. These byte counts describe those files in one environment, not minimum possible descriptions or a measure of cognitive complexity.

Causal attribution remains separate. A cheap decoder may show that information is readily accessible. It does not show that the source solver actually uses that decoder or the selected representation. Resource accounting strengthens an explanatory claim by identifying where work is done. It does not replace the implementation evidence discussed in Section 5.

18 Explanations for fixed recipient procedures

The coloring criterion in Theorem 8.1 permits fixed decoders to be designed for the chosen code. A recipient that already exists may not be able to execute those decoders within its budget. We now freeze the decoding procedures before choosing the artifact.

Let X be a finite nonempty set of cases, $\alpha : X \to S$ the sender's access map, and $\mathcal{A}$ a finite nonempty artifact catalogue. Recipient i has a background map $\beta_i : X \to B_i$, a demanded output $d_i : X \to D_i$, and a fixed procedure L_i. Evaluate $L_i(\beta_i(x), a)$ under a declared instruction and memory bound. A timeout, an invalid output, or a rejected artifact counts as failure. Since the inputs and execution bounds are finite, the resulting acceptance table is finite even when L_i is written in a general programming language.

Define

$$\mathcal{A}(x) = \{a \in \mathcal{A} : \text{every recipient returns } d_i(x) \text{ within its declared limits}\},$$

and, for each reachable sender value s, define

$$\mathcal{A}_s = \bigcap_{x:\alpha(x)=s} \mathcal{A}(x). \tag{20}$$

An artifact selected from $\mathcal{A}_s$ must work for all cases the sender still considers possible. Choosing a different acceptable artifact after secretly learning which hidden case is true is not allowed.

Theorem 18.1 (The fixed-recipient covering criterion)**.** *Under this finite contract, an exact encoder $E : S \to \mathcal{A}$ exists on the reachable sender values if and only if every $\mathcal{A}_s$ is nonempty. Its smallest possible image size is*

$$\min\{|C| : C \subseteq \mathcal{A},\ C \cap \mathcal{A}_s \neq \varnothing \text{ for every reachable } s\}, \tag{21}$$

when the feasible family is nonempty. Thus a codebook of at most K artifacts is possible exactly when a set of at most K artifacts meets all these requirements. Every valid encoder induces a proper coloring of the sender-access graph, so its image size is at least that graph's chromatic number. The converse graph condition need not hold for the fixed procedures.

Proof. A sender seeing s must emit one value $E(s)$, and that value must lie in $\mathcal{A}(x)$ for every x with $\alpha(x) = s$. This is exactly membership in $\mathcal{A}_s$. Hence a valid encoder implies nonempty intersections. Conversely, selecting one value from each nonempty finite intersection constructs a valid encoder.

The image of any such encoder intersects every $\mathcal{A}_s$. If a set C intersects every requirement, selecting $E(s)$ from $C \cap \mathcal{A}_s$ gives an encoder whose image is contained in C. This proves the minimum formula by finite minimization. If two adjacent sender values received the same artifact, a recipient with the same background would receive identical decoder inputs while requiring distinct outputs. Its fixed procedure could not satisfy both. This proves the graph lower bound. □

The finite optimization in (21) is a set-cover formulation on the reachable sender values. It counts how many distinct catalogue entries are needed. It does not license replacing their actual encodings by short indices unless the corresponding index decoder is installed and charged. A catalogue entry may be a large program or proof object even when only one such entry is used.

Counterexample 18.2 (Information suffices but the installed decoder needs a larger codebook)**.** Take three cases x_0, x_1, x_2 fully visible to the sender. One recipient already has background i in case x_i and always needs output zero. The sender-access graph has no edges, so a designable decoder could ignore a single common message and return zero. Now fix an installed decoder on artifacts A, B, C. It returns zero on A, B at background 0, on B, C at background 1, and on A, C at background 2. It fails on the remaining pairs. Every two of the three acceptable-artifact sets intersect, but their common intersection is empty. Two catalogue entries suffice, while one does not. The graph lower bound is one and the fixed-decoder optimum is two.

This construction is intentionally small. It isolates the difference between information-theoretic possibility and compatibility with an already installed procedure. A more capable replacement decoder repairs the example, but replacing it changes the installed account. A procedure that times out on a required input provides an equally direct reason a graph witness can fail at the resource boundary.

18.1 Sender uncertainty is a different obstruction

Nonempty casewise sets $\mathcal{A}(x)$ do not suffice. If the sender cannot distinguish two cases whose acceptable-artifact sets are disjoint, their intersection in (20) is empty. More recipient compute cannot repair a distinction missing from both its background and the transmitted artifact. A new observation may repair the sender's access. An added recipient background may repair the receiving side. Neither can be credited without being part of the actual workflow.

The bounded table concerns exact outputs. For approximate explanation, acceptable outputs can instead be defined by a fixed loss tolerance, but that changes the table and the claimed guarantee. A most probable model is not an exact current state merely because the decoder confidently emits it. The Bayesian and decision results retain their own uncertainty contracts.

19 Joint acquisition with a compiled resource budget

The preceding theorem freezes recipient procedures. We now combine it with state-changing acquisition and charge the entire compiled observation tree. This closes a specific first-edition gap: a small artifact, an affordable experiment, and a runnable decoder must belong to one attainable workflow.

19.1 The explicit finite implementation model

Use a finite augmented state space X. Include every origin variable, history flag, recipient background, and permission needed to make the declared transition and terminal tables valid. Each experiment e has the transition, observation, and admissibility semantics of Section 10, together with a fixed nonnegative integer cost vector $c_e \in \mathbb{N}^r$. Its coordinates have declared units. The charge includes the specified source-side interface work for that experiment. We do not assume that acquiring information leaves the physical state unchanged.

For each state x and artifact a accepted by all fixed recipient procedures, let $\gamma(x, a) \in \mathbb{N}^r$ be the terminal charge. It includes transmission and the declared source and recipient execution work. Invalid or out-of-budget decoder executions have no terminal entry. At support V, a stop using a is allowed only if it works for every $x \in V$, with worst-case vector

$$g_a(V) = \max_{x \in V} \gamma(x, a),$$

where maxima and comparisons of vectors are componentwise. Different coordinates may attain their maxima on different cases. A coordinatewise budget nevertheless has to cover every one of them.

A controller is an explicit observation tree of depth at most h. Each decision node or stop node occupies one fixed-size record, with fields for the operation or artifact and the bounded number of successor pointers. The count $n(T)$ includes all stored branches, not just those visited in one execution. The interpreter, recipient programs, and state tables are installed separately and remain fixed throughout the optimization. Their storage is added to the recorded controller size. Sharing equal subtrees in a directed acyclic graph is a different storage model.

For a tree T, let $v(T)$ be its componentwise worst-case charged online cost. Let $\mathcal{P}_h(V)$ be the attainable pairs $(v(T), n(T))$. The stop pairs are

$$\big(g_a(V), 1\big) \quad \text{for each artifact accepted throughout } V. \tag{22}$$

For $h > 0$, each uniformly admissible experiment gives the additional pairs

$$\left(c_e + \max_o v_o,\ 1 + \sum_o n_o\right), \quad (v_o, n_o) \in \mathcal{P}_{h-1}(V_{e,o}), \tag{23}$$

with one child choice for every nonempty observation branch. An experiment contributes no pair when even one possible branch has no completion. At $h = 0$ only stops are allowed. Sets are finite because the catalogue and horizon are finite.

Theorem 19.1 (A joint resource certificate for fixed recipients)**.** *Recurrences* (22)–(23) *enumerate exactly the cost profiles of the permitted explicit trees. A workflow within online budget $b \in \mathbb{N}^r$ and controller-record budget N exists if and only if some $(v, n) \in \mathcal{P}_h(V)$ satisfies $v \le b$ and $n \le N$. Every such pair has a finite executable tree witness. Keeping only componentwise nondominated pairs at each recursive step preserves feasibility for every pair of budgets (b, N).*

Proof. At depth zero, an admissible workflow must stop with one artifact that works for every possible state. Its worst-case terminal charge and its one stored record give exactly (22). For the inductive step, any nonterminal tree first executes an experiment available throughout V. Its possible next supports are exactly $V_{e,o}$. By induction each child tree supplies one attainable pair. Every branch is possible, so the online charge is c_e plus the componentwise maximum of the child charges. The installed explicit tree contains the root and every child record, giving the sum in (23). Conversely, combining an admissible root with the witnessing child trees realizes that pair. This proves exact enumeration and the budget equivalence.

If one child profile is dominated by another attainable profile at the same support and remaining horizon, replacing that child cannot increase a parent's componentwise maximum or its sum of records. Terminal admissibility and future observations are unchanged. Repeating this replacement proves that discarding dominated profiles preserves a budget-feasible witness whenever one exists. Finite induction gives the claim at the root. □

This is finite multiobjective dynamic programming under a fixed implementation contract. Its components have the same state-planning and side-information foundations as the earlier results [17, 15, 16]. The new obligation is their simultaneous realization with frozen recipient procedures, online charges, and explicit controller storage. The theorem does not optimize over arbitrary programs, discover missing sensors, or make policy synthesis computationally cheap.

Counterexample 19.2 (Separate resource optima do not define a feasible common workflow)**.** Let a hidden origin take four values and remain fixed under every experiment. A direct experiment reveals its value at charge four units. Its explicit controller has one decision node and four stop nodes, giving profile $(4, 5)$. Three other experiments ask whether the origin equals 0, 1, or 2, respectively, each at charge one unit. A decision chain of these equality tests distinguishes the four origins with worst-case charge three and seven stored records, giving profile $(3, 7)$. The fixed recipient decoder reads a two-bit origin artifact at a leaf. Its work and the common message length are held fixed and are not included in the one online coordinate shown here.

Within horizon three the nondominated profiles are $(4, 5)$ and $(3, 7)$. A tree with charge at most three cannot use the direct experiment. A binary equality tree distinguishing all four origins needs at least four leaves and hence at least seven records. Conversely, any tree distinguishing four origins needs four distinct stopping artifacts and at least one decision node, so five is the smallest possible record count. Thus the separate minima are three online units and five records, but the combined budget $(3, 5)$ admits no tree.

The direct observation is compact as an installed tree. The equality chain is cheaper to execute in the worst case. Taking the best coordinate from each certificate invents a workflow that neither realizes. Replacing the explicit tree by a program with shared branches changes the storage model and must be assessed separately.

19.2 Which implementation costs remain outside the recurrence

Offline search and proof checking do not disappear because the controller is compiled. Their time, software, and input tables are recorded separately. The online interpreter must be specified, and its dispatch costs can be included in c_e and γ. A record count becomes a storage guarantee only after the fixed record layout and installed tables are included. The number of source-file bytes is an audit quantity, not a theorem of minimal description length.

The result is exact for declared charges. If those charges are conservative upper bounds for an implementation, the certificate is correspondingly a sufficient implementation budget, not necessarily its tight physical cost. If the actual decoder reads an unlisted file or the sensor consults a hidden variable, the implementation is outside the contract. The right response is to expand and recheck the contract, not to call the extra access free.

A source with several possible initial observations can be handled by running the construction on each initial support and specifying how the installed controller dispatches among them. If that observation is public to the recipients, it belongs in their background. If it is private, the fixed decoders still see only their declared inputs. The integrated instrument below makes both sides explicit.

20 A costed first-edition instrument certificate

The common realization can now be described without a designable recipient or an omniscient sender. The instrument is still $(a, b, c) \in \mathbb{F}_5^3$. Before the diagnostic, the sender knows only $(s, b) = (a + b, b)$. Recipient

A has the old register pair (a, b). Recipient B has the old display pair (s, b). Neither recipient has c. All three parties know the operation laws and the fixed diagnostic protocol before the particular hidden state is selected.

The sole diagnostic in this version applies J exactly once and returns its exact post-operation display s_J. The sender computes $m = s_J - s = c$ and broadcasts that one field element. The report s_J and the sender's private controller address are not transmitted separately. The recipients are told by the installed protocol that their demanded record is the state after this one diagnostic, not an arbitrary later state.

The two frozen decoders are

$$L_A((a, b), m) = (a + m, b, m), \qquad L_B((s, b), m) = (s + m, b, m). \tag{24}$$

Recipient A obtains the post-J register description. Recipient B obtains the post-J display-centered description. Both compute one modular addition. Their installed update procedures then support all $H/W/J$ continuations in their respective coordinates. These procedures are supplied, not learned by the diagnostic.

For a fixed old (s, b), five possible values of c remain. Before an observation, no one message can supply both exact demanded records for every one of those cases. After the diagnostic, the five report values lead to five stop nodes and each selects its one valid message. The explicit tree has six records. A fixed three-bit field encodes the five messages. Two bits do not suffice. Three additional unused binary patterns are rejected, not assigned secret meanings.

20.1 The execution ledger

The following charges describe the reference procedure rather than a universal machine model. Assignments and interpreter dispatch are recorded as implementation operations, but are not included in the modular-arithmetic coordinate. Each field operation is one modular addition or subtraction, independent of how it is realized by a processor.

Resource	Count	Meaning and boundary
Physical diagnostic	1	One J operation with one returned exact display. Sensor construction is supplied.
Source modular arithmetic	1	Subtract the old display from the new one. Selection uses the compiled protocol.
Broadcast payload	3 bits	One fixed-length field for five possible messages. No branch address or raw report is an additional message.
Recipient A modular arithmetic	1	Compute $a + m$ in its fixed decoder.
Recipient B modular arithmetic	1	Compute $s + m$ in its fixed decoder.
Explicit controller	6 records	One diagnostic node and five message leaves for the fixed old record. Two records are visited in a run.
Installed method	Recorded separately	Both decoder programs, transition semantics, framing, and prior background. Not charged as newly acquired c.

The code enumerates all 25 old-record fibres and all 125 initial instrument states, compiles a resource witness, and verifies its leaves by executing the fixed decoders. It also tests incorrect messages, missing branches, reduced horizons, and insufficient recipient instruction limits. These are exact finite checks of the supplied model. They do not establish that an external reasoner discovered the representation or that a noisy device has this exact sensor.

The six-record count applies to one fixed old record and a literal observation tree. A generic arithmetic controller can share structure across all 25 old records and need not store 25 separate trees. The reference

program implements that arithmetic procedure as well and records its code size. Neither representation is declared the universally smallest controller. The distinction between tree storage and an arithmetic program is retained instead of combining their best costs.

20.2 Reuse has an explicit implementation

In B's coordinates (s, b, c), each H, W, or J update uses one modular addition or subtraction and the query reads the first coordinate. Thus a length-n continuation uses exactly n modular arithmetic operations after initialization, plus its recorded dispatch, assignment, and output operations. In A's coordinates the current display requires the additional addition $a + b$, while swap can be a coordinate permutation. The programs are different but agree on the declared answers through the exact translation theorem.

The supplementary record has five possible values. A separately stored three-bit message appended to a jointly packed five-bit base record uses eight bits, whereas a repacked complete 125-state record requires seven bits. A native implementation storing three individually encoded residues can use nine bits. These are three distinct storage layouts. The delivered source reports which layout is used rather than subtracting rounded bounds from incompatible layouts.

20.3 An uncertain reading requires a different artifact

The deterministic certificate does not apply to the noisy sensor merely because the same physical register is present. Under the noisy contract, the sender carries a posterior over c, not a certified exact value. A recipient can use a transmitted likelihood history or a specified posterior representation together with the existing correction and decision guarantees. Its message size and arithmetic work must then be charged again. They are not inherited from the three-bit exact-state message.

The ancillary flag may still be omitted when its verified factorization holds. Copies of one reading may not be treated as independent measurements. When a public adaptive selector chooses the next experiment using reported history, Theorem 14.1 applies under its fresh-observation assumptions. A hidden selector, an unknown likelihood, or an observation deemed impossible by every current model requires additional evidence or a revised model.

20.4 What this certificate warrants

The first edition supplies exact witnesses for specified representations, experiment trees, reporting channels, recipient procedures, and resource profiles. These witnesses support different statements. The quotient preserves future answers. An admissible diagnostic acquires a missing distinction. A factorization preserves a Bayesian update. A fixed decoder turns an artifact into a usable record. The common resource witness shows that particular components can be executed together.

None of these conclusions identifies the internal mechanism of an unrelated solver. Nor do they establish subjective experience, a universal ranking of understanding, or accuracy of unstated physical premises. A complete adequacy record should preserve those unresolved questions alongside its positive results. That is how a reusable method remains answerable to evidence rather than becoming an impressive answer with an invisible scope.

Part IV

Interpretation and scope

21 Limits, objections, and the next mathematical questions

The framework is intentionally operational, but that does not make it philosophically complete. A contract specifies selected consequences. It does not establish that those are all the consequences that matter. A formal model of a subject can be wrong, and a model of a reasoner's computation can be incomplete. A successful finite witness does not answer every question about meaning, intentionality, or experience.

21.1 Does this define understanding as success?

Not as a single success rate. The framework separates what a state can support, what a solver actually uses, what can be transported, what a recipient can reconstruct, and what survives correction. The counterexamples show that these properties differ. Nevertheless, every operational criterion remains a criterion for a declared set of abilities. This account does not offer an independent essence of understanding beyond all such criteria.

The resulting restraint is useful. A system may possess strong reusable knowledge in one domain and weak correction behavior in another. A receiver may understand an explanation under a supplied notation while failing to translate it. A process may be correctly modeled by an outside analyst without being able to explain itself. These are distinct descriptions rather than contradictory verdicts about an indivisible trait.

21.2 Can a table satisfy the theory?

Yes, for some finite contracts. A table can encode a transition system and support exact future queries. It should not be declared incapable by terminology alone. The stronger questions concern its origin, size, use, reuse across a family, and behavior when the contract changes.

The recoding construction in Section 5 makes this unavoidable. To distinguish a compact transferable method from a complete enumeration, one must specify the comparison class and account for the encoder, decoder, and shared background. Restricting those components is part of the research design. An assertion that a representation is meaningful is not a substitute for doing so.

21.3 Do valid proofs establish understanding?

A proof establishes the conclusion licensed by its premises and rules. The recipient may still lack a usable model of why those premises were appropriate or how the proof changes under a new problem. Conversely, a reasoner can use a correct model while being unable to produce a formal proof in a particular proof language.

Formal verification is therefore valuable but not interchangeable with the entire profile. The present proofs establish finite and metric statements under their assumptions. They do not certify that a deployed neural model realizes the maps, that a human reader has understood them, or that the task specification captures all relevant external facts.

21.4 Finite models and a changing world

The exact quotient assumes a fixed state space and a fixed operation family. The correction theorem assumes specified model families and outcome maps. The instrument's new operation enlarges one of those families in a controlled way. Real inquiry can also introduce new variables, new causal hypotheses, new measurement channels, and new forms of explanation.

There is no guarantee that the minimal state for every future extension is small. Preserving all conceivable future distinctions can collapse the compression objective back toward retaining the entire accessible state, and even that state may omit external information. A useful theory must allow partial coverage and a principled request for new evidence rather than promise unlimited generality.

21.5 What is now resolved, and what remains

Bounded nonvacuity. The sender-access graph prevents an encoder from consulting missing information, and the joint certificate keeps acquisition and communication in one contract. Sections 17–19 add fixed interpreter procedures, charged task-dependent descriptions, and exact finite resource frontiers. These do not constitute a general complexity theory of concepts. Useful scalable representation languages, economical decoder discovery, and fair cross-architecture comparisons remain research questions. A finite coloring can be optimal in alphabet size and poor as an explanation to an actual person.

Noisy and approximate correction. Theorem 12.1 resolves exact posterior preservation for finite likelihoods and common reporting channels. Proposition 13.1 quantifies event-specific sensitivity, and Proposition 13.3 carries approximation through a recipient to a bounded-loss decision. These results assume the likelihood family and its errors are supplied. Robust model sets, dependent noise, misspecification, and empirically justified uncertainty require additional work.

Acquiring new distinctions. Theorem 10.1 resolves optimal worst-case acquisition in the finite deterministic, finite-horizon model, including the experiment's effect on current state. Theorem 16.1 adds a common explanatory artifact. Efficient search, noisy joint experimentation and communication, and the invention of missing sensors or hypotheses are not supplied. A statement about feasible observation cannot be promoted to a claim of autonomous scientific discovery.

Recipients and revision over changing contracts. The graph criterion resolves exact communication when decoders can be designed. Theorem 18.1 instead resolves finite artifact selection for preexisting bounded procedures. Neither identifies which procedure is realized by an uninstrumented learned reasoner. The sequential correction theorem assumes a fixed model family and matching experiment semantics. A recipient who changes its conceptual language or acquires a new operation may need a new sufficient record, not just the old certificate copied forward.

These remaining questions define the boundary of the first edition. Version 1.0 is a stable theoretical scope with reproducible artifacts. It is not a claim that every question about understanding has been settled. Later empirical findings or mathematical corrections can revise the account without turning an editorial release number into a scientific guarantee.

22 Conclusion

A correct answer is an achievement. Reusable computational understanding concerns what a reasoner can carry beyond that answer: a method for new continuations, a representation another reasoner can use, and an explicit route to correction when further evidence changes what is warranted. Those properties belong to a continuing relationship among a subject, a representation, evidence, and a recipient.

The instrument makes the distinctions concrete. One display answers a present question. Two coordinates support every continuation under the original operations. A third register becomes relevant when a new operation is admitted. An exact diagnostic can acquire that distinction, but also changes the instrument. A noisy reading supplies a posterior instead. Reporting channels preserve correction only under the corresponding

evidence conditions. Repeating a copied report is not a new observation, and a small unconditional error can become decisive after rare evidence.

Explanation adds another realization boundary. A common artifact must fit what the sender actually knows and what each recipient already possesses. Even a message with sufficient information can be unusable by the installed decoder. Changing that decoder can help, but its task-dependent information, installation, and execution must be charged. The first-edition covering and resource constructions make those obligations jointly checkable in finite models. A controller small in stored records need not be the cheapest one to execute. The best separate resource bounds need not describe one possible workflow.

The theory does not select a uniquely human notation or exclude a table merely because it is a table. It asks what each implementation retains, how it was acquired, what work it performs, and which losses remain after translation. Predictive adequacy, causal use, economical realization, recipient usefulness, and responsiveness to evidence are related but distinct. A proof about one is not a certificate for all of them.

Operation Komorebi's realization perspective provides the common discipline: specify the operations that a representation preserves and make unresolved distinctions visible. The resulting framework supports exact mathematical claims without inferring consciousness, universal comprehension, or an unobserved mechanism from formal success alone. Its positive proposal is that a useful account of machine understanding should follow a method through reuse, explanation, and correction, keeping the information and work that make those achievements possible inside the account.

A A complete trace of the instrument

Start from $(a, b, c) = (1, 3, 2)$. The first display is four. The base record is $(s, b) = (4, 3)$. Apply H, then W, then the newly admitted J.

Event	Complete state	Display	Base record	Status
Initial	$(1, 3, 2)$	4	$(4, 3)$	H/W contract adequate
After H	$(4, 3, 2)$	2	$(2, 3)$	Exact base update
After W	$(3, 4, 2)$	2	$(2, 4)$	Exact base update
Before predicting J	$(3, 4, 2)$	2	$(2, 4)$	Missing c distinction
After J	$(0, 4, 2)$	4	$(4, 4)$	New reading acquired

The last base record does not by itself retain all newly relevant information. The previous display was two and the new display is four, so $c = 4 - 2 = 2$. The post-operation first register is $4 - 4 = 0$. The enlarged display-centered record is therefore $(s, b, c) = (4, 4, 2)$.

A subsequent H gives $(s, b, c) = (3, 4, 2)$ and display three. A subsequent W leaves the display at three but changes the second coordinate to $3 - 4 = 4$ modulo five. A subsequent J changes the display to zero. Each assertion follows from the enlarged update equations, not from extrapolating a list of displayed numbers.

A.1 What was actually learned

The original prediction method already encoded the effect of H and W. The J observation supplied a new state distinction. It did not establish the J transition law from scratch, because that law was part of the expanded analytical contract. Learning that law would require a candidate family and experiments capable of distinguishing it from alternatives.

This separation is important. Acquiring a latent value under a known rule, inferring an unknown rule, and transferring a known rule into a new notation are different research achievements. A single successful trace should not be counted as all three.

A.2 A correction record for the trace

The record should retain the old scope, the new operation, the failed sufficiency condition, the acquired observation, and the exact post-intervention state reconstruction. It should also preserve the unchanged base equations. A concise corrected explanation would say that the third coordinate is irrelevant to H/W continuations but relevant to any contract admitting J.

Nothing in that explanation establishes a hidden psychological state of an agent. It establishes the content and limits of a reusable mathematical model and gives a recipient an explicit way to use and revise it.

A.3 The same trace with an uncertain readout

In the preceding trace, the state immediately before J is $(3, 4, 2)$ and its exact display is two. Suppose the reasoner retains only the base record $(2, 4)$ and models c as uniform. After J, the physical state is $(0, 4, 2)$. Under the noisy sensor of Section 12, a reported display of four gives the shifted report two. The resulting belief is over states $(c - 2, 4, c)$, not over unchanged pre-intervention states.

Independent reports retained	One	Two	Three
Full shifted report 2	3/4	36/37	432/433
Only even parity	7/24	98/303	1372/4143

The entries are posterior mass on $c = 2$ along the named histories. Their probabilities of occurrence are not the entries in the table. Every other candidate remains possible after finitely many noisy reports. Duplicate copies of one reading count as one observation. The executable demonstration reproduces the fractions, checks the harmless ancillary flag, and distinguishes the full-state target from the recipient-relative communication target.

B Companion verification and evidence boundaries

The companion program `code/finite_checks.py` uses integers and exact rational arithmetic. It checks the register construction, partition refinement against a separately coded pair-search procedure, finite transcript discrimination, a stochastic quotient, transport-error recurrences, explanation capacity, informative model filtering, and correction transport. The program enumerates small domains stated separately for each check group. It does not train a model or test an empirical claim about machine cognition.

The finite quotient checks compare the refinement procedure with direct search for a distinguishing word for every small machine in the stated domain. The register checks compare microscopic updates with independently written reduced coordinate updates. The correction checks enumerate maps, candidate sets, and possible evidence, including evidence inconsistent with the current set. Negative controls deliberately use a wrong swap update, a collapsed evidence channel, and a delayed disagreement beyond a fixed test horizon.

The expanded program `code/expanded_checks.py` adds rational Bayesian updates, likelihood factorization, posterior-mixture identities, rare-evidence bounds, recipient decision margins, adaptive history checks, graph coloring against direct decodability, and acquisition values against full policy enumeration. The conditional-information identity is checked using exact rational coefficients of logarithms of primes rather than floating-point cancellation. Graph and policy enumerations are exhaustive only over the small domains recorded per group.

The reference module `code/understanding.py` exposes the finite constructions with input validation. Its acquisition routine returns a policy witness or an explicit infeasibility result. A timeout or malformed input is not interpreted as mathematical infeasibility. The demonstration `code/demo_expanded.py` reproduces the noisy instrument and the complementary-background example.

The first-edition module `code/resource_contracts.py` implements fixed-decoder covers and nondominated compiled-tree profiles. Its witness checker follows every actual initial state independently of the optimizer. The suite `code/edition_checks.py` compares the optimizer with separately written full observation-history tree enumeration on the declared small domains, tests invalid witnesses, and checks all instrument states with the two frozen recipient programs. The costed demonstration is `code/demo_first_edition.py`. Source-file sizes and hashes are recorded without calling them minimal description lengths.

The script's result is evidence about the supplied code and finite constructions. The prose arguments supply the general conditional results. A clean compilation, an exact finite check, an independent proof review, and a formal-kernel build remain different kinds of evidence. This edition claims the first two, within the recorded review scope, and does not claim the latter two.

The source package records the actual test output, source hashes, build diagnostics, reference-review scope, and known limitations. Diagrams are generated from mathematical constructions. No synthetic image is offered as an empirical observation. The PDF's version number identifies a consolidated first edition, not a completed priority audit or an accepted publication.

C Claim map

Statement	Main assumption	Conclusion and boundary
Theorem 4.1	Finite deterministic operations and all declared continuations.	Minimal behavioral quotient. Does not minimize acquisition or runtime.
Proposition 4.2	Expanded contract on the same complete state space.	Necessary supplementary capacity. Sharp reuse construction has an explicit refinement condition.
Proposition 5.1	Fixed complete transcript laws and equal priors.	Optimal binary identification bound. New evidence changes the laws.
Proposition 5.2	Equal projected rows on each fibre.	Exact stochastic history preservation for reduced-history policies.
Theorem 6.1	Matched operations and query decoders.	Exact forward transfer. Does not grant reverse recovery or target coverage.
Proposition 6.3	Uniform residuals and Lipschitz target dynamics.	Finite-horizon error budget for matched operations.
Proposition 7.2	One fixed recipient background and fixed-length messages.	Necessary communication capacity, not efficient decoding.
Proposition 9.1	Exact realizable candidate family and separating experiments.	Finite informative elimination and decision-tree lower bound.
Theorem 11.1	A commuting evidence square, including contradictory observations.	Filtering inclusion. Uniform equality exactly requires reflection of attainable outcomes.

C.1 Expanded correction and explanation results

Statement	Main assumption	Conclusion and boundary
Theorem 8.1	Finite source access, recipient backgrounds, and designable decoders.	Exact common alphabet via graph coloring. Does not establish decoder efficiency.
Theorem 10.1	Finite deterministic experiments, explicit disturbance and admissibility, bounded horizon.	Minimum worst-case acquisition cost and an attaining policy.
Theorem 12.1	Finite likelihoods and one reporting channel independent of the model.	All-prior posterior preservation exactly when conditional factorization holds.
Proposition 12.2	The same joint experiment and a stated prior.	Average posterior divergence equals conditional information, not a worst-case bound.
Proposition 13.1	Two joint laws and an event positive under both.	Conditional variation bound with an evidence denominator.
Proposition 13.3	One posterior chain, finite actions, and bounded common losses.	Recipient regret and strict action-gap certificate.
Theorem 14.1	Fresh conditionally independent observations and a reported-history selection policy.	Exact posterior preservation along adaptive inquiry and bounded public stopping.
Theorem 16.1	Finite deterministic trees, costs, and receiver contracts.	Exact joint acquisition and common-message feasibility, not efficient general synthesis.

C.2 Fixed procedures and first-edition resource results

Statement	Main assumption	Conclusion and boundary
Proposition 17.1	Fixed interpreter with charged task-dependent fixed-length program and artifact.	Combined information lower bound. Not an individual-function runtime lower bound.
Theorem 18.1	Finite source access, artifact catalogue, and fixed bounded recipient procedures.	Exact artifact intersections and minimum covering codebook. Raw artifact size remains charged.
Theorem 19.1	Finite augmented state, bounded experiments, frozen recipients, vector charges, explicit controller trees.	Exact resource frontier and common workflow witnesses. Not optimal unrestricted program synthesis.

C.3 How the certificates depend on one another

The quotient and transfer results concern the declared operations. Their conclusions do not specify how an absent register is measured. The acquisition result adds that physical and informational route, and the correction results determine what an observation can eliminate or how it changes a belief. Neither step by itself guarantees that another reasoner can use the result.

The designable-recipient graph in Theorem 8.1 characterizes a communication alphabet after a decoder may be selected. Theorem 18.1 instead holds the recipient's procedure fixed. Its intersections can be empty even when the designable graph has a coloring. This is why a claim about one implemented recipient needs the stronger contract.

The final recurrence incorporates those same fixed terminal procedures into one experiment tree. Its witness must fit both online charges and the storage of every installed branch. A policy attaining one minimum and another policy attaining a second minimum are not interchangeable evidence for one simultaneous budget. The costed instrument supplies a common witness, while the four-origin counterexample supplies an explicit obstruction.

The probabilistic results are not silently inserted into that exact tree model. A noisy report may support a posterior and a bounded decision loss without identifying a unique state. Approximate posterior transfer therefore belongs to its stated likelihood, evidence, and recipient-loss assumptions. A zero-error state certificate requires its own stronger evidence. This separation is part of the first edition's scope, not an omission disguised by a shared diagram.

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Large Signal Model Foundations

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Signal realizations, packets, predictive closure, observable models, and certified probability domains.

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Theory edition, not a new production runtime; probability validity, identification and empirical adequacy are distinct.

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Operation Komorebi

Large Signal Model

Foundations of realization, refinement and predictive closure

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Exact representation. Observable learning.
Certified probability domains.
Typed defects and controlled approximation.

Komorebi P.

Foundations edition 0.4.0

Companion to PR-LASM software release 2.3.0

9 October 2026

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This supplement develops the mathematical foundation of the Projector-Resolved Large Algebraic Signal Model (PR-LASM). It accompanies software release 2.3.0 without changing that release's implementation or historical verification records. Its subject is the structure of a signal model, not performance on a task suite.

The central question is not only whether a signal can be reconstructed. It is whether a chosen description retains enough information for the transformations and inferences performed on it. A representation can be exact as an archive and insufficient as a predictive state. A predictive state can be sufficient for a specified family of dynamics without reconstructing the entire signal.

The core organizing principle is therefore:

> Preserve the observed signal. Declare the representation used for reasoning. Express transformations in that representation. Identify the obstruction when transformation, compression or inference fails to commute. Refine the representation or retain memory when the obstruction matters.

The finite atlas, packet and best-basis results below consolidate the historical PR-LASM capstone [K1]. The Bayesian coarsening result is brought forward from Paper VI [K2]. The predictive-closure constructions connect these foundations to invariant-subspace and projection-memory theory [R2, R3]. Their role here is a self-contained foundation for the model, not a claim that these classical ingredients were first discovered by Komorebi.

The numbered results have proofs in this document. Exact rational and symbolic examples accompany them. They have not been newly formalized or checked by a Lean kernel in this supplement.

Edition 0.4.0 contains Propositions 1–49. Part I develops realization and closure. Part II begins at Section 15 with observable learning and stability. Part III begins at Section 27 with controlled stochastic realization. Part IV begins at Section 41 and adds finite probability-domain certificates, robustness, full-law error bounds and a nonpolyhedral boundary with an explicit approximation. The production software remains at version 2.3.0 and is not changed by this theory edition.

Sections 14, 26 and 40 retain the end-of-block checkpoints from preceding editions. Their stated open directions are historical waypoints. Part IV and Section 55 give the current boundary. The unchanged 0.3.0 archive is included for source lineage, and it contains the earlier archives. General proofs appear in the text, while computational records check explicit finite cases.

1. Three objects that must remain distinct

An **observation** is an element x of a declared space X. Examples include a finite real signal, a complex field on a finite grid, or an array of categorical labels. Its type includes shape, scalar field or palette, and boundary convention. A zero-extended finite sequence and a periodic sequence are different objects even when their displayed samples agree.

An **archive** is a lossless representation of the accepted observation. In a mathematical model this means an injective analysis map with an explicit left inverse. In an implementation, conversion and floating-point error must be accounted for separately. An archive of a measured signal does not recover physical variables that were never observed.

A **predictive state** is a summary $z = \gamma(x)$, or more generally a summary of an observation history. It is sufficient only relative to a specified family of future quantities, actions and dynamics. There is no implication from exact storage of x to sufficiency of $\gamma(x)$.

The historical model's macro, meso and micro vectors are nested projections of a feature vector. They are not, merely by being nested, state variables on which the signal dynamics close. The distinction is retained in the present theory.

The term **large** denotes the structured hypothesis family: representations, resolutions, typed operators, compositions, memory mechanisms and their parameters. It is not a theorem about capacity, sample complexity or scale. Such a theorem needs a declared grammar, a complexity measure and a learning protocol.

2. Exact realization and atlas consistency

2.1 A left inverse is the reconstruction contract

Let X and C_a be finite-dimensional Hilbert spaces. A linear atlas consists of analysis and synthesis maps

$$A_a : X \longrightarrow C_a, \qquad S_a : C_a \longrightarrow X, \qquad S_a A_a = I_X.$$

The coefficients representing actual signals form $\mathcal{C}_a = \operatorname{im} A_a$. This distinction matters for redundant representations, where $\mathcal{C}_a$ is a proper subspace of the ambient coefficient space.

Proposition 1. Exact realization and consistency. The analysis map is injective, $\Pi_a = A_a S_a$ is an idempotent with image $\mathcal{C}_a$, and $\Pi_a c = c$ exactly when c is a realizable coefficient vector. Moreover,

$$\|A_a x\| \geq \frac{\|x\|}{\|S_a\|}, \qquad \|S_a(c+e) - S_a c\| \leq \|S_a\|\,\|e\|.$$

The first inequality is understood for nonzero X. No assertion of orthogonality follows from a left inverse alone.

Proof. If $A_a x = 0$, then $x = S_a A_a x = 0$. Also $\Pi_a^2 = A_a(S_a A_a)S_a = \Pi_a$. Its image is contained in $\mathcal{C}_a$ and it fixes every $A_a x$, so its image is exactly $\mathcal{C}_a$. Both norm inequalities follow from the operator-norm inequality applied to S_a. □

For a unitary atlas U_a, the synthesis is U_a^* and the condition number is one. For a learned invertible atlas, exact invertibility is not enough: the synthesis norm controls amplification of coefficient errors. An arbitrarily ill-conditioned exact transform is not a stable representation.

2.2 Several exact atlases do not create new observations

For unitary U_a and positive weights w_a, put

$$\mathcal{A}x = (w_a U_a x)_a, \qquad \mathcal{S}(c_a)_a = \frac{1}{c}\sum_a w_a U_a^* c_a, \qquad c = \sum_a w_a^2.$$

Then $\mathcal{S}\mathcal{A} = I$ and $\|\mathcal{A}x\|^2 = c\|x\|^2$, as proved in [K1]. Here $\mathcal{A}\mathcal{S} = c^{-1}\mathcal{A}\mathcal{A}^*$ is the orthogonal projection onto the consistent coefficient subspace.

Each complete atlas contains the same observed information because it has a left inverse. Multiple atlases offer different sparsity, locality, conditioning and operator descriptions. They do not provide independent observations of the world. In particular, multiplying likelihoods from deterministic recodings as though they were independent measurements generally counts the same evidence more than once.

Categorical one-hot encoding is a separately typed injective map of sets. It is not a unitary map on the scalar-valued signal space. Its exact decoder is valid on one-hot arrays, not on arbitrary coefficient vectors. The linear tight-frame theorem does not apply to it without a separately specified linear embedding and inner product. This separation is already explicit in the capstone's categorical section.

3. Coordinate transport and one underlying operator

On realizable coefficients, define transport from atlas a to atlas b by

$$T_{ba} = A_b S_a : \mathcal{C}_a \longrightarrow \mathcal{C}_b.$$

For an endomorphism $F : X \to X$, define its coordinate realization by $F_a = A_a F S_a$ on $\mathcal{C}_a$. In this paragraph F may be nonlinear. For a partial map, every formula is restricted to the transported admissible domain. Maps between different signal types use separate input and output atlases.

Proposition 2. Coherent transport and operator realization. On the consistent coefficient spaces,

$$T_{aa} = I, \qquad T_{cb}T_{ba} = T_{ca}, \qquad F_b T_{ba} = T_{ba} F_a.$$

Conversely, suppose maps $G_a : \mathcal{C}_a \to \mathcal{C}_a$ satisfy $G_b T_{ba} = T_{ba} G_a$ for every pair. Then there is a unique map $F : X \to X$ with $G_a = A_a F S_a$ for every a.

Proof. The transport identities follow by inserting $S_b A_b = I$. The operator identity follows because either path is $A_b F S_a$. For the converse, choose a reference atlas r and put $F = S_r G_r A_r$. Compatibility gives $G_a A_a = A_a F$. Every consistent coefficient is $A_a x$, which proves the asserted realization. Applying S_a proves uniqueness. □

An independently fitted family of atlas predictors need not satisfy this proposition. Its compatibility residual is a well-defined quantity,

$$d_{ba}(x) = G_b(A_b x) - T_{ba} G_a(A_a x).$$

After synthesis it becomes an object-space discrepancy:

$$S_b d_{ba}(x) = S_b G_b A_b x - S_a G_a A_a x.$$

Thus $\|S_b d_{ba}(x)\| \le \|S_b\| \, \|d_{ba}(x)\|$. Zero discrepancy establishes that the chart descriptions agree, not that their common prediction is correct. All charts can agree on the same wrong operator.

Redundant coefficient outputs must also be tested for consistency. Synthesizing an arbitrary coefficient vector always produces some signal, but reanalysis returns $\Pi_a c$, not necessarily the original c. Reconstruction and coefficient consistency are different checks.

4. Packets, refinement and transfer blocks

4.1 Packets belong to a declared resolution

Within one orthogonal atlas, let E_i be pairwise orthogonal coordinate selectors with $\sum_i E_i = I$. Pull them back by a unitary map:

$$P_i = U^* E_i U, \qquad P_i P_j = \delta_{ij} P_i, \qquad P_i^* = P_i, \qquad \sum_i P_i = I_X.$$

A packet is the component $P_i x$ together with its projector, domain, atlas and resolution. A bare coefficient block without this information is not a fully specified packet.

Projectors from different atlases need not commute and must not be treated as one larger mutually orthogonal family. Weighted frame packets can sum to the identity while failing to be idempotent. These are different structures.

Proposition 3. Complete refinement preserves reconstruction. Suppose a finite rooted tree has root projector I, and every internal projector is the sum of pairwise orthogonal child projectors. Every complete antichain Λ, meeting each root-to-leaf path exactly once, satisfies

$$\sum_{v \in \Lambda} P_v = I, \qquad x = \sum_{v \in \Lambda} P_v x, \qquad \|x\|^2 = \sum_{v \in \Lambda} \|P_v x\|^2.$$

Proof. Replace an internal node by the sum of its children. Repeated replacement reaches any complete antichain and preserves the total projector. Different selected nodes have disjoint descendant branches, hence orthogonal ranges. The norm identity is Pythagoras. □

This is a conservation law for information under complete refinement. Deleting a child, storing only its energy, or imposing a coefficient threshold changes the contract. For a retained subset J of an orthogonal decomposition, the lost signal is exactly $\sum_{i \notin J} P_i x$, and its squared norm is the sum of the omitted packet energies.

The finite best-basis recursion

$$V(v) = \min\left(c(v), \sum_{u \in \mathrm{ch}(v)} V(u)\right)$$

minimizes an additive cost over the declared finite tree [K1, R1]. The recurrence itself does not make a nonadditive predictive objective additive. If a prediction cost couples two leaves, a scalar cost at each node is generally insufficient for global optimization. Packet construction and cost evaluation must also be included in an implementation's complexity account.

4.2 Operations, not just signals, decompose

Let $L : X \to Y$ be linear, with orthogonal input projectors P_i and output projectors Q_j. Its transfer blocks are $L_{ji} = Q_j L P_i$.

Proposition 4. Transfer reconstruction and refinement. One has

$$L = \sum_{j,i} Q_j L P_i.$$

If $P_i = \sum_r P_{i,r}$ and $Q_j = \sum_s Q_{j,s}$ are complete refinements, then

$$Q_j L P_i = \sum_{s,r} Q_{j,s} L P_{i,r}.$$

For another linear map $M : Y \to Z$ and output projectors R_k,

$$R_k M L P_i = \sum_j (R_k M Q_j)(Q_j L P_i).$$

Proof. Insert the appropriate identity decomposition on the input, output or intermediate space. Use $Q_j^2 = Q_j$ in the composition formula. □

This is the algebraic reason to make transfer blocks first-class objects. The same operation can be expressed at several resolutions, while the coarse and fine expressions remain linked by exact identities. Approximations acquire explicit omitted-block terms rather than an unnamed representation error.

4.3 Nonlinear operations need interactions

For a nonlinear map, $F(\sum_i P_i x)$ generally differs from $\sum_i F(P_i x)$. Applying a linear transfer formula to such a map silently loses interactions.

Proposition 5. Polynomial packet interactions. Let

$$F(x) = b + \sum_{m=1}^{p} B_m(x, \ldots, x),$$

where each $B_m : X^m \to Y$ is multilinear. Then

$$Q_j F(x) = Q_j b + \sum_{m=1}^{p} \sum_{i_1, \ldots, i_m} Q_j B_m(P_{i_1} x, \ldots, P_{i_m} x).$$

Proof. Substitute $x = \sum_i P_i x$ in every argument and distribute by multilinearity. □

For example, $(u+v)^2 = u^2 + 2uv + v^2$. A grammar containing only independent packet squares misses $2uv$. For nonpolynomial maps, one needs direct nonlinear semantics or a local expansion with a stated remainder bound. The word algebraic does not remove that obligation.

5. Predictive closure is the central contract

Let $\gamma : X \to Z = \gamma(X)$ be a summary and let $F_u : X \to X$ describe a deterministic transition for action u. The reduced description closes when there is $\bar{F}_u : Z \to Z$ such that

$$\gamma F_u = \bar{F}_u \gamma.$$

This is an equality of functions on the declared domain, not a fit statistic.

Proposition 6. Exact quotient criterion. Such a reduced transition exists if and only if

$$\gamma(x) = \gamma(x') \implies \gamma(F_u(x)) = \gamma(F_u(x'))$$

for every admissible x, x' and every action. When it exists it is unique on $\gamma(X)$.

Proof. Necessity follows by applying $\bar{F}_u$ to equal summaries. For sufficiency, define $\bar{F}_u(z) = \gamma(F_u(x))$ for any representative with $\gamma(x) = z$. The stated condition makes this well-defined and also proves uniqueness. □

This criterion also applies when X is a latent physical state and γ is an observation map. Two physically different states can produce identical images and different futures. Exact image reconstruction does not resolve that hidden-state ambiguity. A history, a belief state or additional observations may be needed.

For a stochastic kernel, the corresponding condition is equality of the distributions of the next summary for states with the same current summary. Section 7 makes this finite criterion explicit.

5.1 The obstruction in the linear case

Suppose X is a Hilbert space, P is an orthogonal projection, $Q = I - P$ and $x_{t+1} = Lx_t$. The retained state is $z_t = Px_t$. Then

$$z_{t+1} = PLPz_t + PLQx_t.$$

Proposition 7. Closure defect and compositional defect. The retained state closes for every initial signal if and only if $PLQ = 0$. If it does not close, the error in using only PLP is

$$e_t = PLQx_t, \qquad \|e_t\| \le \|PLQ\| \, \|Qx_t\|.$$

For two linear operators L and M,

$$\boxed{PLMP - (PLP)(PMP) = PLQMP.}$$

Consequently, compressing two operations separately can discard a path that leaves the retained subspace and later returns.

Proof. If $PLQ = 0$, the displayed recurrence depends only on z_t. Conversely, apply Proposition 6 to x and Px, which have the same retained state, to obtain $PLQx = 0$ for every x. The norm bound is submultiplicativity. The compositional identity follows by inserting $I = P + Q$ between L and M. □

The relevant quantity is not omitted energy alone. A very small omitted packet can have a large effect if its transfer into retained packets is strongly amplified. Conversely, an energetic packet can be irrelevant to a specified prediction if every corresponding return block is zero.

A residual is therefore typed. Reconstruction residual, coefficient-consistency residual, predictive residual, closure residual and statistical uncertainty should not be collapsed into one scalar without a specified objective.

6. Constructing the smallest closed linear refinement

There is a constructive finite solution when the dynamics are a known finite family of linear maps L_u on a d-dimensional space. It identifies which additional observables are required, rather than merely requesting a larger state vector.

Let the initial summary be C_0x. Regard its rows as linear observables. Starting from $W_0 = \mathrm{row}(C_0)$, define

$$W_{n+1} = W_n + \sum_u W_n L_u, \qquad W_n L_u = \{wL_u : w \in W_n\}.$$

Proposition 8. Minimal linear predictive completion. The sequence stabilizes after at most $d - \dim W_0$ strict enlargements. Its limit W_* is the smallest row space containing W_0 and invariant under right multiplication by every L_u. If R is any full-row-rank matrix whose rows form a basis of W_*, then

$$C_0 = HR, \qquad RL_u = B_u R$$

for suitable matrices H and B_u. Therefore $z = Rx$ retains the original summary and has exact reduced dynamics $z_{t+1} = B_u z_t$. Every linear summary with those two properties has row space containing W_*.

Proof. The spaces are nested and have dimension at most d. Equality at one step means invariance under every L_u, so it is a fixed point. Containment in any invariant row space containing W_0 follows by induction. The matrix identities express inclusion of the relevant rows in the basis R. Conversely, if $C_0 = HR$ and $RL_u = B_u R$, the row space of R is invariant and contains W_0, so it contains every W_n. □

For one transition matrix this reduces to the span of the rows of

$$C_0,\ C_0L,\ \ldots,\ C_0L^{d-1}.$$

Its annihilator is the largest L-invariant subspace inside $\ker C_0$, the standard unobservable-subspace construction [R2]. With several actions, observables pulled back through action words are required, but the linear span still stabilizes in finite dimension.

This gives a precise role for obstruction-driven refinement: add observable directions created by the dynamics until the chosen prediction variables close. It does not guarantee that the completion is small. In some systems $W_* = X^*$ and exact memoryless closure needs the full state.

The theorem assumes the operators and relevant state are available. Applying it to estimated operators constructs closure for those estimates. It is not a proof that the real system has the same closure. If only C_0x is measured and the full x is unavailable, the augmented state Rx is not automatically accessible.

7. Joint closure of prediction and Bayesian inference

The finite categorical case admits an equally explicit construction that joins dynamics with Paper VI's inference refinement. This is useful because a state can be closed for prediction but too coarse for a new observation, or sufficient for an update but not for the next transition.

Let Ω be a finite state set. Let $K_u(i,j)$ be known row-stochastic transition matrices. Let $\ell_y(i) > 0$ be a finite family of observation likelihoods. A partition π has aggregation matrix

$$C_\pi(a,i) = \mathbf{1}_{i \in B_a}.$$

Distributions are column vectors. Prediction is $p \mapsto K_u^\top p$, and a Bayesian update is

$$\mathcal{B}_{\ell_y}(p) = \frac{p \odot \ell_y}{\ell_y^\top p}.$$

The coarse state must commute with both operations:

$$C_\pi K_u^\top = \bar{K}_u^\top C_\pi, \qquad C_\pi \mathcal{B}_{\ell_y}(p) = \mathcal{B}_{\bar{\ell}_y}(C_\pi p).$$

Fix a mandatory initial partition π_0 with at least two blocks. This preserves distinctions that the intended application requires and excludes the trivial one-state quotient.

Proposition 9. Coarsest joint finite closure. First split π_0 by the likelihood vectors $(\ell_y(i))_y$. Then repeatedly split each current block by the signatures

$$\left(\sum_{j \in B} K_u(i,j) \right)_{u,B},$$

where B ranges over the current partition. The process terminates after at most $|\Omega| - |\pi_{\text{init}}|$ strict refinement rounds. The final partition is the unique coarsest refinement of π_0 on which all declared predictions and Bayesian updates close for every prior.

Proof. At a stable partition, states in a common block have the same transition probability into every coarse block. Those probabilities define $\bar{K}_u$ and give the first commutation identity, by checking it on point-mass priors. Likelihood constancy on each block defines $\bar{\ell}_y$ and gives the second identity by direct summation.

For minimality, every admissible partition refining π_0 has at least two blocks. By Proposition 12 below, exact updating for all priors requires each fine likelihood to be constant within each block. Thus it refines the first likelihood split. Suppose it refines a current partition. Each current block is a union of its blocks. Exact prediction makes the probability of reaching each such union constant within any admissible block, so it also refines the next split. Induction proves it refines the final partition. Each strict refinement increases the number of blocks, which proves termination. □

The transition condition is the finite lumpability criterion [R4], expressed directly as a commutation identity. The joint construction adds the explicitly declared likelihood distinctions. Its output is a quotient of a known finite model, not an automatically learned model of an unknown world.

The accompanying rational example starts with six states. Likelihoods create three blocks, transition distinctions force two further rounds of splitting, and the final model has five states. Two original states remain genuinely equivalent for every action and observation in the declared family. The example verifies both commuting identities exactly.

8. Memory is an exact alternative to adding state

Refinement and memory are two ways to represent the influence of unresolved components. This can be derived rather than merely proposed as an architectural option.

For fixed linear dynamics and an orthogonal split, write

$$z_t = Px_t, \qquad q_t = Qx_t,$$

and define the restricted blocks

$$A = PLP, \quad B = PLQ, \quad C = QLP, \quad D = QLQ.$$

Thus $z_{t+1} = Az_t + Bq_t$ and $q_{t+1} = Cz_t + Dq_t$.

Proposition 10. Exact projected memory and a conditional tail bound. For every $t \geq 0$,

$$\boxed{z_{t+1} = Az_t + \sum_{k=0}^{t-1} BD^k Cz_{t-1-k} + BD^t q_0.}$$

If $\|D\| \leq \rho < 1$ and $\|z_s\| \leq Z$ over the relevant history, omitting terms with $k \geq H$ changes the one-step right-hand side by at most

$$\|B\|\,\|C\| Z \frac{\rho^H}{1-\rho}.$$

Dropping the initial unresolved-state term adds at most $\|B\|\rho^t\|q_0\|$.

Proof. Iteration of the unresolved recurrence gives

$$q_t = D^t q_0 + \sum_{s=0}^{t-1} D^{t-1-s} Cz_s.$$

Substitute this into the retained recurrence and change indices. Bound the omitted terms by the geometric series. □

The memory kernel is BD^kC. It records a path from retained variables into unresolved variables, evolution there, and return. The first kernel BC is precisely the return path appearing in the two-step composition defect. The term BD^tq_0 is deterministic when q_0 is specified. It becomes uncertain only relative to incomplete information or a probability model for q_0.

This is the elementary linear instance of projection-memory reduction, associated with Mori-Zwanzig theory [R3]. A finite hand-chosen power-law or exponential memory feature is an approximation class for such effects, not a general derivation of the exact kernel.

The tail bound compares right-hand sides using the same resolved history. Replacing that history by recursively generated predictions introduces additional rollout error. Stability of that recursion needs its own bound. Also, a spectral-radius condition alone does not imply $\|D\| \le \rho$ in the norm already being used, especially for nonnormal operators.

9. Statistical refinement needs its own theorem

Nested feature vectors are algebraically nested when the smaller vector can be recovered from the larger by a fixed map. Predictive risk is a separate matter.

Proposition 11. Ideal statistical benefit of refinement. Let a target Y take values in a finite-dimensional Hilbert space with $\mathbb{E}\|Y\|^2 < \infty$. Let $\mathcal{F} \subseteq \mathcal{G}$ be information sets and define $m_{\mathcal{F}} = \mathbb{E}[Y \mid \mathcal{F}]$ and $m_{\mathcal{G}} = \mathbb{E}[Y \mid \mathcal{G}]$. Then

$$\mathbb{E}\|Y - m_{\mathcal{F}}\|^2 = \mathbb{E}\|Y - m_{\mathcal{G}}\|^2 + \mathbb{E}\|m_{\mathcal{G}} - m_{\mathcal{F}}\|^2.$$

Proof. Expand $Y - m_{\mathcal{F}} = (Y - m_{\mathcal{G}}) + (m_{\mathcal{G}} - m_{\mathcal{F}})$. The cross term has zero conditional expectation given $\mathcal{G}$. □

Thus finer information cannot worsen the optimal squared-error predictor under a fixed distribution. It can worsen an estimated predictor with finite data, restricted function classes, changed regularization or shifted distributions. The theorem supplies the ideal target for refinement, not a guarantee for every fitting procedure.

An exact archive makes later reanalysis possible without reconstructing discarded observations from a lossy state. It does not eliminate estimation error or distribution shift.

10. Bayesian coarsening and the information-loss packet

This section imports a specific existing Komorebi foundation rather than inventing another inference heuristic. Let p_i be a prior on a finite fine state set and let B_a be coarse blocks. Define

$$P_a = \sum_{i \in B_a} p_i, \qquad r_a(p) = \frac{\sum_{i \in B_a} p_i \ell_i}{P_a} \quad (P_a > 0).$$

On zero-mass blocks the value of r_a is immaterial. Updating finely and then aggregating is exactly updating P with the effective likelihood $r(p)$. The effective likelihood generally depends on the fine prior inside each coarse block, information that P alone does not retain.

Proposition 12. Bayesian closure and its exact defect. Suppose there are at least two coarse blocks and the fine and declared coarse likelihoods are strictly positive. Updating and coarsening commute for every prior if and only if

$$\ell_i = c\bar{\ell}_a \quad \text{for every } i \in B_a$$

with one common $c > 0$. For a fixed prior, put $\delta = r(p) - \bar{\ell}$, $\bar{b} = \mathcal{B}_{\bar{\ell}}(P)$ and $Z_r = r(p)^\top P$. Then

$$\boxed{C_\pi \mathcal{B}_\ell(p) - \bar{b} = \frac{1}{Z_r}\left(\operatorname{diag} P - \bar{b}P^\top\right)\delta.}$$

The fixed-prior identity also holds for nonnegative likelihoods when both normalizing evidences are positive.

Proof. Constancy up to one global factor gives commutation by summing the fine posterior. Conversely, compare priors supported on one state in each of two distinct coarse blocks. Equality of coarse posterior odds forces the ratio of those states' likelihoods to equal the ratio of the coarse likelihoods. Varying the representatives proves constancy within each block and the common factor across blocks. For the defect, use $Z_r = \bar{\ell}^\top P + P^\top \delta$ and subtract the two normalized vectors. □

With only one coarse block, both posteriors aggregate to one regardless of the likelihood, so the necessity criterion is false there. This boundary is explicitly retained from the corrected Paper VI statement [K2].

The matrix in the defect formula annihilates $\bar{\ell}$, recording the irrelevant common-scale direction. It does not make every likelihood discrepancy irrelevant. The construction in Proposition 9 refines until both the likelihood and transition defects vanish for a specified finite family.

10.1 Parameter inference is not program truth

Within one declared regression model, a normal-inverse-gamma prior gives a standard conditional Bayesian parameter posterior. The software's program weights instead use penalized held-out predictive scores and then may refit coefficients on all available training records. These are finite-candidate predictive weights. Without a corresponding generative derivation, they should not be conflated with the marginal-likelihood posterior over programs under one fixed prior.

Every candidate must be scored on the same target event and base measure. Densities on differently sized coordinate spaces are not directly comparable. An invertible nonlinear coordinate change requires the appropriate Jacobian. A redundant deterministic embedding induces a measure on its consistent subspace, not an independent full-dimensional density on all coefficient entries.

The law of total variance separates within-candidate uncertainty from disagreement of candidate means. It does not supply a probability that the truth lies outside the grammar. An explicit outside-grammar component needs its own prior and predictive distribution. Assigning an arbitrary leftover mass does not solve open-world inference.

11. What a typed operator language must guarantee

A program has a source type, target type, admissible domain, evaluation rule and declared approximation status. A nonlinear partial operation is not a total matrix merely because its inputs are stored in a vector. The domain of a composite is

$$\operatorname{dom}(g \circ f) = \{x \in \operatorname{dom} f : f(x) \in \operatorname{dom} g\}.$$

Composition must preserve those types and domains. Coordinate changes must preserve the represented operation as in Proposition 2. Packet refinement must preserve exact transfer semantics as in Proposition

4, or carry an explicit approximation residual. Nonlinear interactions must retain their cross-packet terms as in Proposition 5.

A grammar can be closed under a specified set of constructors without containing every possible physical law. It can also be expressive while computationally intractable. Universal discovery, practical search and semantic closure are distinct claims.

For a known operator, Proposition 8 can build its minimal linear summary. For an unknown operator, learning must address identifiability: which different candidates make identical predictions under the available observations and actions? More coordinate systems alone do not distinguish candidates that agree on the entire observed signal. Additional interventions or data may be needed.

The foundation should therefore treat a learned hypothesis as an operator with an evidence record, not just a matrix with a high normalized weight. Its record includes fitting assumptions, admissible inputs, compared alternatives, unresolved equivalences and the representation in which its residual is measured.

12. A bounded infinite-resolution bridge

Finite reconstruction identities do not alone prove convergence of an unbounded or infinite-dimensional model. A useful bounded extension can nevertheless be established directly.

Proposition 13. Strong fixed-horizon refinement limit. Let X be a Hilbert space, $L : X \to X$ bounded, and P_n orthogonal projections converging strongly to I. Put $L_n = P_n L P_n$. For every fixed $x \in X$ and every fixed nonnegative integer k,

$$L_n x \longrightarrow Lx, \qquad L_n^k x \longrightarrow L^k x.$$

Proof. Since $\|P_n\| \le 1$,

$$\|(L_n - L)x\| \le \|L\|\,\|(P_n - I)x\| + \|(P_n - I)Lx\| \longrightarrow 0.$$

Also $\|L_n\| \le \|L\|$. Induct on k, using

$$(L_n^{k+1} - L^{k+1})x = L_n(L_n^k - L^k)x + (L_n - L)L^k x.$$

The first term tends to zero by the uniform norm bound and induction, and the second by strong convergence on the fixed vector $L^k x$. □

If estimated operators $\widehat{L}_n$ are uniformly bounded and $(\widehat{L}_n - L_n)x \to 0$ for every fixed x, the same argument gives fixed-horizon convergence for $\widehat{L}_n$. These are explicit assumptions, not conclusions of the finite learning procedure.

This result is strong convergence, not operator-norm convergence and not a bound uniform in time. On ℓ^2, finite coordinate projections converge strongly to I while $\|I - P_n\| = 1$ for every n. Unbounded operators require domain control. Adaptive data-dependent partitions and fitted coefficients require additional consistency arguments. These remain separate mathematical tasks.

13. A complete small signal example

Consider the rational linear signal dynamics

$$x_{t+1} = Lx_t, \qquad L = \begin{pmatrix} \frac{1}{2} & 1 \\ \frac{1}{4} & \frac{1}{3} \end{pmatrix}, \qquad x_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

The spatial archive stores x_t. A second atlas uses the orthogonal Haar matrix

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Its coefficients $c_t = Ux_t$ reconstruct as $x_t = U^* c_t$. The same dynamics in that atlas are ULU^*, not a separate physical law. Thus coordinate transport preserves the full dynamics.

Now retain only the first spatial coordinate. With $P = \mathrm{diag}(1, 0)$, the retained scalar z_t and discarded scalar q_t obey

$$z_{t+1} = \tfrac{1}{2} z_t + q_t, \qquad q_{t+1} = \tfrac{1}{4} z_t + \tfrac{1}{3} q_t.$$

The exact first-coordinate sequence begins

$$z_0 = 1, \qquad z_1 = \tfrac{1}{2}, \qquad z_2 = \tfrac{1}{2}, \qquad z_3 = \tfrac{11}{24}.$$

The compressed one-step matrix predicts $z_2 = 1/4$ instead of $1/2$. The lost return path contributes exactly

$$BC = 1 \cdot \tfrac{1}{4} = \tfrac{1}{4}.$$

There are two exact repairs. One is state refinement: start with $C_0 = (1, 0)$ and add $C_0 L = (1/2, 1)$. The resulting row space is the full two-dimensional space, so it closes. The other is memory:

$$z_{t+1} = \tfrac{1}{2} z_t + \tfrac{1}{4} \sum_{k=0}^{t-1} \left(\tfrac{1}{3}\right)^k z_{t-1-k},$$

since $q_0 = 0$. This is the exact same system expressed without retaining q_t as a separate current-state variable. At $t = 2$, it yields

$$z_3 = \tfrac{1}{2} \cdot \tfrac{1}{2} + \tfrac{1}{4} \cdot \tfrac{1}{2} + \tfrac{1}{12} \cdot 1 = \tfrac{11}{24}.$$

To see a nontrivial reduced closure, adjoin a third independent coordinate with multiplier $5/6$. The smallest closed refinement of the first coordinate then contains the first two coordinates, not the third. Exact prediction of the chosen summary does not require reconstructing every component of the enlarged signal.

Finally, low energy is not the same as low predictive importance. For

$$L' = \begin{pmatrix} 0 & 1000 \\ 0 & 0 \end{pmatrix}, \qquad x = (0, 1/1000)^\top,$$

omitting the second coordinate loses norm $1/1000$, but changes the next first coordinate by exactly one. The correct certificate is the product of omitted magnitude and transfer amplification.

14. First-block checkpoint and release organization

The principal theoretical object is now a signal realization with explicit compression maps and commuting requirements. Exact archives, packets, operator descriptions and inference are connected by those requirements rather than simply listed as neighboring modules.

The inherited core supplies exact finite atlases, complete packet synthesis, additive best-basis selection and a finite typed predictive grammar [K1, K3]. Paper VI supplies the exact inference-coarsening defect and its hypotheses [K2]. This supplement develops a connected treatment of atlas coherence, predictive quotients, compositional defects, minimal closure, joint finite stochastic closure, projection memory and bounded refinement limits.

The additions are mathematical statements and reference calculations. They do not silently make the 2.3.0 runtime enforce atlas coherence, compute minimal predictive closures or discover the true transition law. An implementation map is supplied separately. The original source and historical manuscript remain unchanged.

A foundations-centered release should have one primary theory volume organized by realization, packets, transport, closure, memory and inference. The mathematical examples and proof ledger belong alongside it. Software documentation should identify which definitions and laws each component implements. Empirical material can remain supplementary without being the criterion for mathematical completeness.

The first edition identified the next substantive gaps as stable learned atlases, identification from incomplete observations, nonlinear interaction truncation, coherent program weighting under grammar growth, and adaptive refinement with learned operators. Part II now addresses explicit finite or bounded sectors of these questions. Its final section distinguishes those closed statements from the general problems that remain.

The finite theory can be stated decisively: exact realization preserves observations, complete refinement preserves decomposition, coherent transport preserves the meaning of operators, and predictive or inferential closure holds precisely under the corresponding compatibility conditions. Where closure fails, the theory supplies a typed defect and, in the finite known-model cases above, a constructive repair.

Part II. Learning, stability and coherent expansion

The first stage constructed predictive closures for declared operators. The second stage asks how those constructions relate to what a learner can actually observe. A closure that depends on an unavailable hidden coordinate is a mathematical description, not yet an accessible state. A fitted recurrence can be accessible but wrong outside the span represented by its data. A one-step approximation can be accurate and still fail under repeated composition.

The results below connect these distinctions. They give an exact learning result for a finite linear observable sector, deterministic perturbation bounds, a propagation law for typed defects, a sufficient stability condition for truncated memory, controlled nonlinear interaction bounds, coherent atlas reconciliation and posterior accounting for an expanding grammar. These are finite or explicitly bounded results. They do not assume that every unknown process has a short linear state.

The observable-state perspective has classical antecedents in predictive state representations [R5] and observable-operator learning [R7]. Finite invariant observable spaces also underlie Koopman reductions [R6]. The present development uses elementary statements proved below and identifies their specific place in Komorebi's realization and obstruction calculus. It does not import those papers' sample-complexity guarantees into this model.

15. From hidden state to an accessible predictive state

Let a finite-dimensional autonomous linear process have hidden state $x_t \in \mathbb{R}^d$ and observed output $y_t \in \mathbb{R}^p$:

$$x_{t+1} = Lx_t, \qquad y_t = Cx_t.$$

The observation map need not be injective. Define the length-h observation map

$$O_h = \begin{pmatrix} C \\ CL \\ \vdots \\ CL^{h-1} \end{pmatrix}, \qquad V_h = \operatorname{im} O_h.$$

The coordinates of $O_h x_s$ are future observations relative to time s. They are not available at time s merely because this matrix is written down. At time $t \geq h-1$, however, the same vector is the observed history

$$z_t = \begin{pmatrix} y_{t-h+1} \\ \vdots \\ y_t \end{pmatrix} = O_h x_{t-h+1}.$$

Thus a forward observation map becomes a causal state after a declared warm-up interval. The time index is part of the contract.

Proposition 14. Exact causal delay closure. The following statements are equivalent:

1. There is a linear map $D_h : V_h \to V_h$ satisfying $O_h L = D_h O_h$.
2. $\ker O_h \subseteq \ker(O_h L)$.
3. $\operatorname{rank} O_{h+1} = \operatorname{rank} O_h$.

When these statements hold, D_h is unique on V_h. The history state obeys $z_{t+1} = D_h z_t$, and its last output block is y_t. Its iterates therefore determine every subsequent observed output. If d is an upper bound on the hidden dimension, $h = d$ is sufficient for closure, though a smaller h may suffice.

Proof. The equivalence of the first two statements is the linear factorization criterion: define $D_h(O_h x) = O_h L x$ and require independence from the representative x. The rows of $O_h L$ are $CL, \ldots, CL^h$. All except the final block already occur in O_h. The kernel condition is therefore equivalent to CL^h being in the row space of O_h, which is exactly the rank equality. It also implies $O_h Lx \in V_h$ because this vector is O_h applied to Lx. Uniqueness follows because every element of V_h has a representative. The history recurrence follows directly. Cayley-Hamilton expresses L^d as a linear combination of lower powers, so CL^d is in the row space of O_d. □

This does not recover hidden variables in $\ker O_h$. It identifies the quotient needed for the declared outputs. An independent hidden component can remain unobservable forever without preventing exact output prediction.

The rank equality here is a property of the *true observation maps*. A plateau in a finite empirical data matrix is not the same statement. It may reflect unexcited directions rather than a closed observation space. Section 17 makes the difference exact.

For several actions L_u, a fixed delay vector does not automatically have one autonomous transition matrix. Products of the actual action matrices determine its observation map, which can change with the action history. One needs an action-aware state, a common closed observable family, or a separately justified belief-state construction. Proposition 14 is not silently generalized to arbitrary controlled or stochastic dynamics.

16. What finite data identify

Suppose candidate states and their successors have been recorded as columns of matrices $Z_-, Z_+ \in \mathbb{R}^{m\times N}$. Their construction may use output histories. No hidden-state coordinates need be supplied to the fitting procedure.

Write $W = \operatorname{im} Z_-$. Exact interpolation of these columns is a question about a linear map on W, not yet about every physically realizable state.

Proposition 15. Data-span identification and its boundary. A linear map B on the ambient space with $BZ_- = Z_+$ exists if and only if

$$\ker Z_- \subseteq \ker Z_+.$$

Its restriction to W is unique. It is an endomorphism of W if and only if $\operatorname{im} Z_+ \subseteq W$. A particular ambient extension is

$$B_0 = Z_+ Z_-^\dagger,$$

where $\dagger$ denotes the Moore-Penrose inverse. Every other fitted extension agrees with B_0 on W. If W is a proper subspace of the true realizable state space V, the data alone do not identify the operator on $V \setminus W$.

Proof. Necessity follows by applying B to every linear dependence among the input columns. Under the kernel condition, define $B(Z_-a) = Z_+a$. Two choices of a differ by a kernel vector and give the same answer. This defines and uniquely determines the map on W. It maps W into W exactly under the image condition. The pseudoinverse formula gives that map because $a - Z_-^\dagger Z_- a \in \ker Z_-$. If $W \subsetneq V$, choose a nonzero linear functional φ on V vanishing on W and a nonzero $v \in V$. The operators B and $B + v\varphi$ agree on every observed column and differ elsewhere. □

The three checks are different. The kernel test checks consistency of a deterministic linear fit. The image test checks whether the observed span is closed under the fitted transitions. Equality $W = V$ is a *coverage condition*. Neither of the first two checks establishes it.

For a known fixed feature space of dimension r, coverage is witnessed by r independent input states. For an unknown larger feature space, the same rank only gives a lower bound on dimension. A model card must say which interpretation is being used.

If several action-specific operators act on a common declared r-dimensional state, each action's paired data must determine that operator on the required span. Pooling all actions into one high-rank matrix does not identify an action that has never been tried on some direction.

16.1 Perturbations have an explicit amplification factor

A noisy rank test is not an exact certificate. Even after the state space is fixed, fitting can be ill-conditioned. The following deterministic bound isolates that issue without introducing a probability model for noise.

Work in a fixed r-dimensional coordinate system, where $Z_- \in \mathbb{R}^{r\times N}$ has full row rank. Let

$$Z_+ = BZ_- + R, \qquad \widetilde{Z}_- = Z_- + E, \qquad \widetilde{Z}_+ = Z_+ + F.$$

The matrix R records deviations from the declared linear recurrence on these samples. The matrices E and F record input and successor perturbations. All norms in the next proposition are spectral norms.

Proposition 16. Fixed-space identification bound. Put $s = \sigma_{\min}(Z_-) > 0$. If $\|E\| < s$, then $\widetilde{Z}_-$ has full row rank, and

$$\widehat{B} = \widetilde{Z}_+ \widetilde{Z}_-^{\dagger}$$

satisfies the exact error identity and bound

$$\widehat{B} - B = (R + F - BE)\widetilde{Z}_-^{\dagger},$$
$$\|\widehat{B} - B\| \le \frac{\|R\| + \|F\| + \|B\|\,\|E\|}{s - \|E\|}.$$

Proof. For every unit $v \in \mathbb{R}^r$, $\|\widetilde{Z}_-^{\top} v\| \ge \|Z_-^{\top} v\| - \|E^{\top} v\| \ge s - \|E\|$. Thus the perturbed matrix has full row rank and its pseudoinverse has norm at most $(s - \|E\|)^{-1}$. Since $\widetilde{Z}_- \widetilde{Z}_-^{\dagger} = I_r$, subtract B from the fitted expression and collect terms. The bound follows by submultiplicativity and the triangle inequality. □

This is a bound for a fixed state space and a fixed coordinate basis. If the basis or its dimension is also estimated from noisy data, subspace-estimation error must be added. No such error is hidden inside R without a derivation. Correlated and adversarial perturbations are allowed by the displayed deterministic inequality, but there is no sample-complexity claim until their magnitudes are bounded under a sampling model.

The denominator explains why numerical rank alone is insufficient. Full rank can coexist with severe noise amplification. The reference calculations include examples attaining the $1/s$ dependence and a rank-collapse example at the excluded boundary $\|E\| = s$.

17. Why finite agreement is not unrestricted discovery

Proposition 17. Finite-prefix indistinguishability. For every integer $T \ge 0$, there are two finite-dimensional autonomous rational linear systems with identical scalar observations at times $0, \dots, T$ and different observations at time $T + 1$. Both transition matrices can be chosen nilpotent.

Proof. One system has scalar output identically zero. For the other, take dimension $T + 2$, initial state $x_0 = e_{T+2}$, observation row $C = e_1^{\top}$, and shift $Le_j = e_{j-1}$ for $j \ge 2$, $Le_1 = 0$. Then $CL^t x_0 = 0$ for $0 \le t \le T$ and $CL^{T+1} x_0 = 1$. The shift is rational and nilpotent. □

An invisible delayed component of this kind can also be adjoined to any fixed base model. The combined output agrees with the base output on the entire inspected prefix and then differs. This is not an

argument against learning. It identifies why a claimed guarantee must state a dimension bound, a model class, a coverage condition, an intervention protocol or a probabilistic assumption.

Even nilpotent operators have only zero eigenvalues, so stability language alone does not remove this ambiguity. A finite rank plateau cannot certify the absence of a delayed mode when arbitrarily large hidden dimensions are admitted.

18. A complete output-only identification example

Return to the rational system from Section 13, now with an additional independent hidden coordinate:

$$L = \begin{pmatrix} 1/2 & 1 & 0 \\ 1/4 & 1/3 & 0 \\ 0 & 0 & 5/6 \end{pmatrix}, \qquad C = \begin{pmatrix} 1 & 0 & 0 \end{pmatrix}.$$

The third coordinate never influences the observed output. The first coordinate alone is not closed, but

$$O_2 = \begin{pmatrix} 1 & 0 & 0 \\ 1/2 & 1 & 0 \end{pmatrix}$$

has rank two and O_3 has the same rank. Proposition 14 gives an exact two-dimensional delay state. Choose $x_0 = (1, 0, q)^\top$ for any q. The first four observations are

$$y_0 = 1, \quad y_1 = 1/2, \quad y_2 = 1/2, \quad y_3 = 11/24.$$

A learner receives those four outputs and the declared delay length $h = 2$. It does **not** receive L, C, O_2 or q. From consecutive observed windows it forms

$$Z_- = \begin{pmatrix} 1 & 1/2 \\ 1/2 & 1/2 \end{pmatrix}, \qquad Z_+ = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 11/24 \end{pmatrix}.$$

Since $\det Z_- = 1/4$, these windows span the declared two-dimensional state. Exact arithmetic gives

$$\boxed{D_2 = Z_+ Z_-^{-1} = \begin{pmatrix} 0 & 1 \\ 1/12 & 5/6 \end{pmatrix}.}$$

The learned recurrence is

$$\boxed{y_{t+1} = \frac{5}{6} y_t + \frac{1}{12} y_{t-1}, \qquad t \geq 1.}$$

It predicts $y_4 = 61/144$, which was not used to identify the coefficients. Proposition 14 and the verified rank condition on the true observation maps establish why this recurrence continues for every initial state of this particular system. The four samples alone establish the fitted recurrence only within the declared closed two-state model class. Proposition 17 forbids confusing those two conclusions.

The reference script separates the roles. One routine fits only the observed samples. A different check, with access to the generating system, verifies the theoretical assumptions and compares subsequent outputs. This prevents hidden access to the true operator from being presented as learning.

The same example connects back to the exact memory kernel. Section 13 expressed the output through a decaying convolution of its history. Here the history is summarized by a two-component recursive state. Neither representation stores the independent third coordinate. Memory and state realization describe the same observable behavior in two different ways.

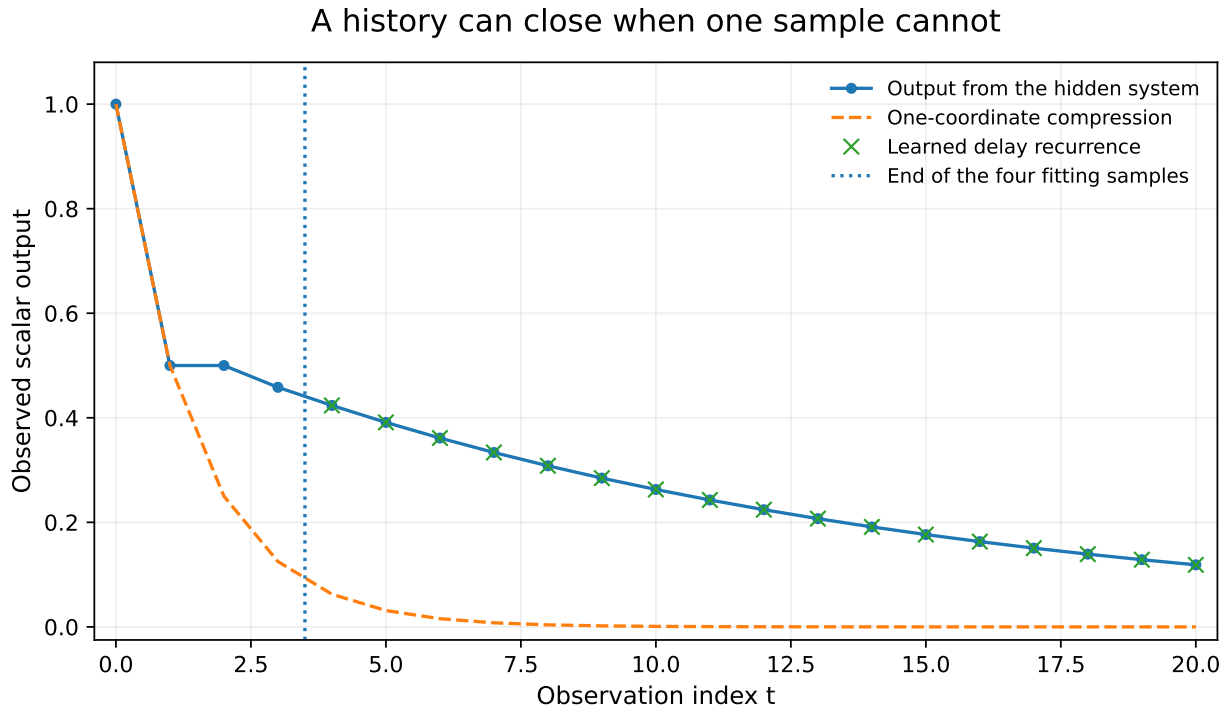


Figure 1: Output-only identification in the declared rational example. Four observations determine the delay recurrence. Later outputs are calculated independently from the hidden system. The one-coordinate compression omits the return path. This is a mathematical example, not a task-suite score.

19. A single propagation law for typed defects

A reconstruction error, an operator-fit error and a refinement error have different origins. Once each has been translated into an error in the state used by the next operation, their propagation can be analyzed by the same identity.

Allow the representation to change with time. Let $\gamma_t : X \to Z_t$ be the declared representation at step t, with a norm on each Z_t. Let $x_{t+1} = F_t(x_t)$ be the true transition and let $\widehat{f}_t : Z_t \to Z_{t+1}$ be the implemented step. Its defect is

$$r_t(x) = \gamma_{t+1}(F_t(x)) - \widehat{f}_t(\gamma_t(x)).$$

This definition already includes a representation switch. It is not restricted to one fixed feature vector.

Proposition 18. Compositional defect propagation. Suppose $\widehat{f}_t$ is λ_t-Lipschitz on a domain containing both $\gamma_t(x_t)$ and the predicted state $\widehat{z}_t$. Suppose $\|r_t(x_t)\| \le \delta_t$ and $\widehat{z}_{t+1} = \widehat{f}_t(\widehat{z}_t)$. Then, with $e_t = \|\gamma_t(x_t) - \widehat{z}_t\|$,

$$\boxed{e_T \le \left(\prod_{j=0}^{T-1} \lambda_j\right) e_0 + \sum_{k=0}^{T-1} \left(\prod_{j=k+1}^{T-1} \lambda_j\right) \delta_k.}$$

Empty products are one. If the output $h_t(x)$ has a decoder H_t with Lipschitz constant κ_t and $\|h_t(x_t) - H_t(\gamma_t(x_t))\| \le \eta_t$, then

$$\|h_t(x_t) - H_t(\widehat{z}_t)\| \le \eta_t + \kappa_t e_t.$$

Proof. Add and subtract $\widehat{f}_t(\gamma_t(x_t))$ to obtain $e_{t+1} \le \delta_t + \lambda_t e_t$. Repeated substitution yields the product-sum formula. Add and subtract $H_t(\gamma_t(x_t))$ for the output inequality. □

For a fixed linear representation R and an estimated transition $\widehat{B}_u$, the defect is $(RL_u - \widehat{B}_u R)x$. For a reference reduced operator B_u, it splits exactly as

$$RL_u - \widehat{B}_u R = (RL_u - B_u R) + (B_u - \widehat{B}_u)R.$$

The first term is representational nonclosure. The second is operator-estimation error. A fit statistic for the second cannot certify that the first vanishes. Proposition 16 supplies a conditional bound on the second in its declared fixed-space setting.

If $\lambda_t \le \rho < 1$ and $\delta_t \le \delta$, then

$$e_T \le \rho^T e_0 + \delta \frac{1 - \rho^T}{1 - \rho}.$$

This is a genuinely recursive prediction bound. It is not merely a one-step comparison made using the true history at every step. If the Lipschitz constants exceed one, the same theorem remains valid but can permit error growth.

19.1 Stability under changing operations

For linear candidate operators, one sufficient way to obtain a common contraction norm is to exhibit a positive-definite matrix M and $0 \le \rho < 1$ with

$$\widehat{B}_u^\top M \widehat{B}_u \preceq \rho^2 M \quad \text{for every declared action } u.$$

Then $\|z\|_M = (z^\top M z)^{1/2}$ gives $\|\widehat{B}_u z\|_M \le \rho \|z\|_M$ for all actions, directly from the displayed inequality. Proposition 18 therefore applies under arbitrary switching among them. This is the common quadratic Lyapunov construction from control theory [R8], used here as an explicit certificate rather than inferred from a list of eigenvalues.

Individual spectral stability does not suffice. The matrices

$$A = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 0 \\ 2 & 0 \end{pmatrix}$$

are both nilpotent, but $BA = \mathrm{diag}(0, 4)$. Alternating them amplifies an appropriate state by four every two steps. Even a single stable nonnormal matrix can produce large transient amplification in a specified norm.

If a learned policy chooses actions from predicted states while the reference process uses actions chosen from true states, the two action sequences may differ. Proposition 18 then needs a bound for that action mismatch, or a Lipschitz bound for the full closed-loop map. The result is not an automatic control-policy guarantee.

20. Truncating memory without losing the prediction bound

Section 8 bounded the omitted tail of the exact linear memory expression using the same resolved history. We now close the separate recursion argument.

Write the exact relation as

$$z_{t+1} = Az_t + \sum_{k=0}^{t-1} K_k z_{t-1-k} + f_t, \qquad K_k = BD^k C, \quad f_t = BD^t q_0.$$

For negative indices set $z_s = 0$, solely as a convention for missing prehistory in this initialized formula. A length-H approximate model, with $H \geq 0$, is

$$\widehat{z}_{t+1} = \widehat{A}\widehat{z}_t + \sum_{k=0}^{H-1} \widehat{K}_k \widehat{z}_{t-1-k}.$$

An empty sum is zero. Let d_t be the exact right-hand side minus this truncated right-hand side evaluated on the **true** history.

Proposition 19. Uniform prediction bound for a stable finite-memory truncation. Suppose

$$a = \|\widehat{A}\| + \sum_{k=0}^{H-1} \|\widehat{K}_k\| < 1, \qquad \sup_{t\geq 0} \|d_t\| \leq \Delta.$$

Let E_0 bound the initialization errors at indices $-H, \ldots, 0$. Then the recursively generated predictions satisfy

$$\boxed{\sup_{t\geq 0} \|z_t - \widehat{z}_t\| \leq \max\left(E_0, \frac{\Delta}{1-a}\right).}$$

If $\|z_t\| \leq Z$ for $t \geq 0$ and $\|D\| \leq \rho < 1$, a valid choice of defect bound is

$$\Delta = Z\left(\|A - \widehat{A}\| + \sum_{k=0}^{H-1} \|K_k - \widehat{K}_k\|\right) + \|B\|\,\|C\| Z \frac{\rho^H}{1-\rho} + \|B\|\,\|q_0\|.$$

The last term can be removed if the exact initialized forcing f_t is retained in the approximate recurrence.

Proof. Subtract the two recurrences and bound every retained history error by the largest of the errors at indices $t-H,\dots,t$. This gives $e_{t+1} \le a \max_{t-H\le s\le t} e_s + \Delta$. The ball of radius $E = \max(E_0, \Delta/(1-a))$ is invariant because $aE + \Delta \le E$. Induction proves the bound. The estimate for Δ separates coefficient errors, the geometric kernel tail and the initialized forcing, using Proposition 10. □

This norm-sum condition is sufficient, not necessary. A stable recurrence with cancellation can fail this test. Failure to find this certificate is not proof of instability.

For the scalar example with $q_0 = 0$, take exact retained coefficients and $z_0 = 1$. All outputs lie in $[0, 1]$: the full nonnegative memory coefficients sum to at most $7/8$, so induction bounds every new output by the previous maximum. Thus $Z = 1$ is valid. With

$$A = 1/2, \quad K_k = \frac{1}{4}\left(\frac{1}{3}\right)^k,$$

one obtains

$$a_H = \frac{7}{8} - \frac{3}{8}3^{-H}, \qquad \Delta_H = \frac{3}{8}3^{-H}, \qquad \boxed{\sup_t |z_t - \widehat{z}_t| \le \frac{3\,3^{-H}}{1+3\,3^{-H}}.}$$

The right-hand side is a theorem for every time in this initialized example. The figure compares it with a separately computed finite-horizon error. The computed maximum is not substituted for the uniform bound.

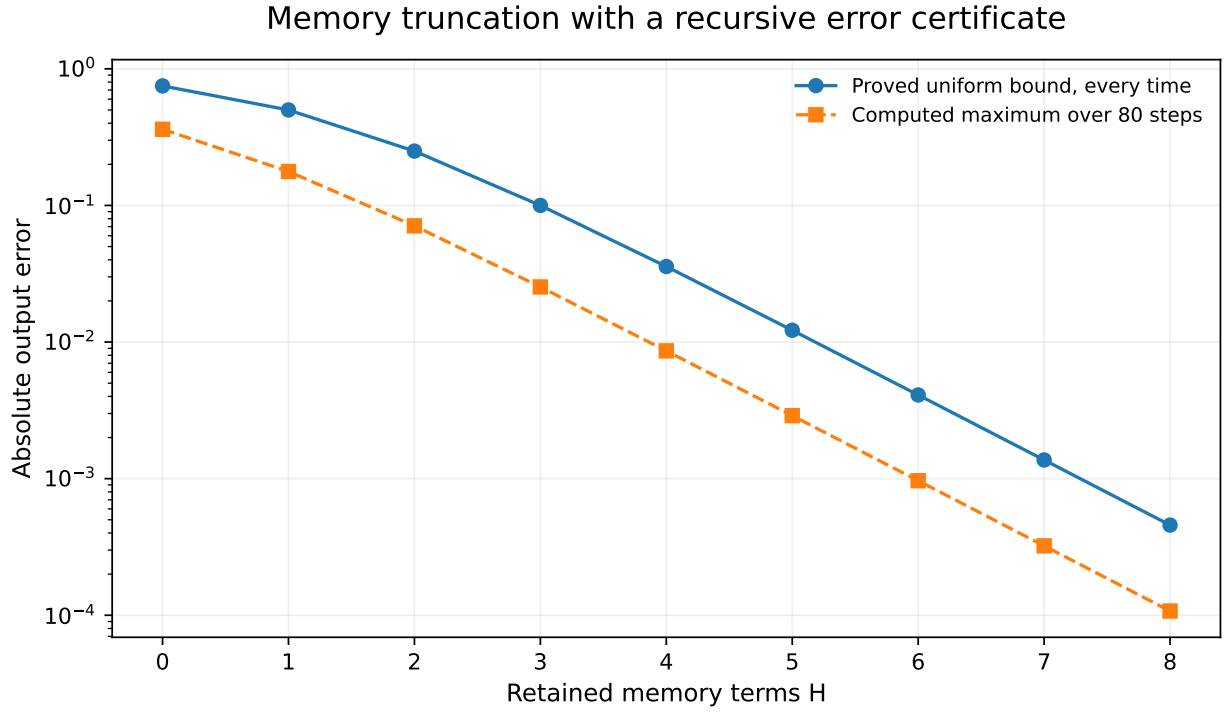


Figure 2: A memory-tail estimate becomes a recursive prediction certificate only after a stability margin is supplied. The measured curve is the maximum over 80 calculated steps. The upper curve is the proved bound for all steps under the stated scalar initialization.

21. Nonlinear packet truncation with an explicit remainder

Polynomial interactions were expanded exactly in Proposition 5. A computational model may retain only some ordered interaction tuples. It then needs a bound on what was omitted, rather than an assumption that packetwise evaluation is exact.

Let P_i be an orthogonal resolution of a finite-dimensional real Hilbert space. Suppose

$$F(x) = b + \sum_{m=1}^{p} B_m(x, \dots, x),$$

with bounded multilinear maps. For each ordered tuple $\boldsymbol{i} = (i_1, \dots, i_m)$, let $\beta_{m,\boldsymbol{i}}$ bound the norm of B_m restricted to $\operatorname{im} P_{i_1} \times \cdots \times \operatorname{im} P_{i_m}$. Let $\mathcal{I}_m$ be the retained tuple set, and let $F_{\mathcal{I}}$ retain the constant term and exactly those interactions.

Proposition 20. Packet-interaction truncation bound. Put $a_i = \|P_i x\|$. Then

$$\boxed{\|F(x) - F_{\mathcal{I}}(x)\| \le \sum_{m=1}^{p} \sum_{\boldsymbol{i} \notin \mathcal{I}_m} \beta_{m,\boldsymbol{i}} \prod_{j=1}^{m} a_{i_j}.}$$

On the ball $\|x\| \le R$, a valid Lipschitz bound for $F_{\mathcal{I}}$ is

$$\operatorname{Lip}(F_{\mathcal{I}}) \le \sum_{m=1}^{p} \sum_{\boldsymbol{i} \in \mathcal{I}_m} m \beta_{m,\boldsymbol{i}} R^{m-1}.$$

Proof. Subtract the retained terms from the exact multilinear expansion. Bound each omitted term by its restricted multilinear norm and sum. For the Lipschitz estimate, telescope the difference between two m-fold evaluations by replacing one argument at a time. Each changed argument has norm at most $\|x - y\|$ after projection, and each unchanged argument has norm at most R. There are m terms for each retained tuple. □

If the declared polynomial itself approximates a nonpolynomial map with remainder at most ε_R on the ball, add ε_R to the defect bound. Without such a remainder estimate, the result is only a theorem for the polynomial, not for the unexpanded physical law.

For a retained state $z = Px$ and omitted state $q = (I - P)x$, dropping every term that uses q gives the coarser majorant

$$\|F(z + q) - F(z)\| \le \sum_{m=1}^{p} \|B_m\| \left((\|z\| + \|q\|)^m - \|z\|^m \right).$$

The tuple-level result is finer because it can distinguish a harmless omitted packet from a strongly coupled one. Ordered tuples must be counted with their multiplicities unless symmetry has explicitly combined them.

A prediction guarantee follows by inserting these defect and Lipschitz bounds into Proposition 18, provided the true and predicted trajectories remain in the declared domain. That domain condition cannot be inferred from a small initial error. One possible construction is to assume that the true map preserves a closed convex set K and to project every approximate output back onto K. Euclidean

projection is nonexpansive and fixes the true output, so it does not increase the one-step approximation bound. This is a declared modification of the predictor, not an identity for an unmodified recurrence.

For the scalar quadratic $(u+v)^2$, retaining only u^2+v^2 omits $2uv$. The packet bound is $2|u||v|$ and is exact in magnitude. Small energy in one packet matters in proportion to both its amplitude and the cross-interaction coefficient.

22. Learning several charts without inventing several worlds

The exact atlas identities constrain the meaning of a fitted model, but independently trained predictors need not satisfy them. Here is a simple coherent reconciliation with a precise objective.

Fix a finite-dimensional represented space X. It may be the observed signal space or a predictive quotient, but it must be the *same* space for all atlases in this section. Let $A_a : X \to C_a$ and $S_a : C_a \to X$ satisfy $S_a A_a = I$. Let $G_a : C_a \to C_a$ be linear fitted maps. Pull them back to object space:

$$L_a = S_a G_a A_a.$$

For positive weights summing to one, define

$$\overline{L} = \sum_a w_a L_a, \qquad \widetilde{G}_a = A_a \overline{L} S_a \quad \text{on } \mathcal{C}_a = \operatorname{im} A_a.$$

Proposition 21. Coherent reconciliation and coordinate-stable transport. The family $\widetilde{G}_a$ is consistent and satisfies every atlas intertwining identity. The operator $\overline{L}$ is the unique minimizer of

$$\sum_a w_a \|L - L_a\|_F^2$$

over linear maps on X, where $\|\cdot\|_F$ is the Frobenius norm. For any target operator L_* and any chosen operator norm,

$$\|\overline{L} - L_*\| \le \sum_a w_a \|L_a - L_*\|.$$

Further, the norm $\|c\|_a = \|S_a c\|$ on $\mathcal{C}_a$ makes every exact transport $T_{ba} = A_b S_a$ an isometry. A represented transition $A_b L S_a$ has exactly the same induced Lipschitz constant in these norms as L has in object space.

Proof. Coherence follows from Proposition 2 because every chart realizes the same $\overline{L}$. Expanding the squares gives

$$\sum_a w_a \|L - L_a\|_F^2 = \|L - \overline{L}\|_F^2 + \sum_a w_a \|L_a - \overline{L}\|_F^2.$$

The weighted cross term vanishes, proving the unique minimizer. The target bound follows from the triangle inequality. The pullback expression is a norm because S_a is inverse to A_a on $\mathcal{C}_a$. Finally, $\|T_{ba} c\|_b = \|S_b A_b S_a c\| = \|S_a c\|$. The same identity for a transition, and the bijection between X and $\mathcal{C}_a$, establish equality of Lipschitz constants. □

This reconciliation is ordinary weighted least squares in a common semantic space. It does not prove that consensus is correct, reduce statistical error under arbitrary dependence, or turn deterministic recodings into independent evidence. The output must also be reanalyzed into a realizable coefficient vector. An independently fitted redundant output can have an additional inconsistency residual $(I - A_a S_a)G_a A_a x$ before reconciliation.

For Euclidean coefficient errors, conditioning remains important. If A_a has smallest singular value at least $m_a > 0$ and largest singular value at most M_a, its canonical synthesis $A_a^\dagger$ has norm at most $1/m_a$, and $\|T_{ba}\| \le M_b/m_a$. An arbitrary left inverse can behave much worse away from the consistent coefficient space, so the canonical-synthesis qualification matters.

A learned atlas can therefore be parameterized within prescribed singular-value bounds, with synthesis tied to its inverse or pseudoinverse. This gives a construction contract. It is not a claim that every unconstrained training procedure respects those bounds.

When atlases change over time but represent the same state, using the pullback norms avoids multiplying arbitrary chart condition numbers at every step. Exact transport telescopes to the direct transport. Numerical synthesis errors and genuinely inconsistent predictions remain defects and must still be bounded.

23. Refinement cannot manufacture missing information

An atlas change is an invertible recoding of the same represented information. Predictive refinement usually adds information. Treating these as identical operations would undo the central distinction of the foundation.

Proposition 22. The upward-lift obstruction. Let $\gamma : X \to Z$ be a coarse representation and $\eta : X \to W$ a proposed refined representation. There is a deterministic map $J : \gamma(X) \to \eta(X)$ satisfying $\eta = J\gamma$ if and only if

$$\gamma(x) = \gamma(x') \implies \eta(x) = \eta(x') \quad \text{for every } x, x'.$$

For linear representations, this is equivalent to $\ker \gamma \subseteq \ker \eta$. If two states share a coarse representation but their refined representations differ, any predicted refined value w has worst-case error at least

$$\max(\|w - \eta(x)\|, \|w - \eta(x')\|) \ge \tfrac{1}{2}\|\eta(x) - \eta(x')\|.$$

Proof. A lift must be constant on each coarse fiber. Conversely, the stated condition defines it unambiguously on every fiber. The linear version is the kernel factorization criterion. The lower bound follows from the triangle inequality between the two refined values through w. □

For a strict refinement, the coarse state is recoverable from the refined one but not conversely. In the linear case this typically means $\ker \eta \subsetneq \ker \gamma$, exactly the opposite of the condition required for an upward lift.

For example, $\gamma(x_1, x_2) = x_1$ and $\eta(x_1, x_2) = (x_1, x_2)$. The states $(0, 1)$ and $(0, -1)$ have the same coarse state, but any common point initialization has worst-case error at least one in Euclidean norm. Padding a vector with zeros is a choice, not information recovery.

A refinement step therefore needs one of three explicitly distinguished inputs: an archive that already contains the relevant observations, new observations, or a set-valued or probabilistic state retaining

uncertainty about the missing details. An exact archive of measured outputs can initialize a longer history when those samples were stored. It cannot recover a latent coordinate that never affected any measurement.

For adaptive prediction, an attempted lift J_t contributes its own initialization defect

$$\eta_t(x_t) - J_t\gamma_t(x_t).$$

It belongs in Proposition 18's initial error or in the defect of the composed transition. It cannot be set to zero merely because a finer packet tree has been allocated.

A data-dependent representation schedule also requires care. The pathwise bound applies to the realized schedule when all of its hypotheses hold along that schedule. Comparing two different schedules chosen by true and predicted histories introduces a selection discrepancy. No uniform statistical consistency of adaptive selection follows solely from the deterministic propagation law.

24. A growing operator grammar with coherent evidence

A large algebraic signal model should be able to add operator programs without changing the meaning of its existing evidence. This requires more than assigning a new softmax weight when a candidate is introduced.

Let $\mathcal{G}$ be a countable declared program set, with one fixed probability mass function $\pi(g) > 0$ satisfying $\sum_g \pi(g) = 1$. Every program specifies a likelihood $L_g(D)$ for the **same** observed data event or density relative to a common base measure. Assume

$$0 < Z(D) = \sum_{g\in\mathcal{G}} \pi(g)L_g(D) < \infty.$$

Let finite active sets $\mathcal{G}_n$ increase to $\mathcal{G}$. Their unnormalized evidence is $Z_n = \sum_{g\in\mathcal{G}_n} \pi(g)L_g(D)$. Only stages with $Z_n > 0$ are considered.

Proposition 23. Exact accounting for an enumerated grammar. Define the full posterior $P(g \mid D) = \pi(g)L_g(D)/Z$ and extend the active posterior $P_n(g \mid D) = \pi(g)L_g(D)/Z_n$ by zero outside $\mathcal{G}_n$. Then

$$\boxed{\|P - P_n\|_{\mathrm{TV}} = \tau_n, \qquad \tau_n = 1 - \frac{Z_n}{Z}.}$$

For any common family of candidate predictive probability kernels, the corresponding predictive mixtures differ in total variation by at most τ_n. If the omitted unnormalized evidence has a bound

$$\sum_{g\notin\mathcal{G}_n} \pi(g)L_g(D) \le b_n,$$

then

$$\tau_n \le \frac{b_n}{Z_n + b_n}.$$

Proof. On the active set, $P = (Z_n/Z)P_n$, so the total mass of the difference there is τ_n. The mass outside is also τ_n. Half the sum of absolute differences is therefore τ_n. Applying the same probability kernel to two mixing measures cannot increase total variation: integrate any event probability, a function taking values in $[0, 1]$, against their difference. For the final bound, write $Z = Z_n + W_n$ with $0 \le W_n \le b_n$ and use that $W_n/(Z_n + W_n)$ increases with W_n. □

Since the active sets exhaust the grammar and the evidence sum is finite, $Z_n \uparrow Z$ and $\tau_n \downarrow 0$. This is posterior approximation *inside the declared grammar*. It does not establish that the real signal-generating mechanism belongs to it.

For a discrete data sequence, a probability likelihood is at most one, so the omitted prior mass is a valid b_n. A continuous-data density need not be at most one. It requires a density envelope or another justified bound on omitted evidence. A small unnormalized tail can also carry substantial posterior mass when the active evidence is very small.

The fixed prior need not be uniform. For an explicit enumeration $g_1, g_2, \ldots$, $\pi(g_k) = 2^{-k}$ is a normalized example, with omitted prior mass 2^{-n} after n programs. It expresses a particular enumeration bias, not a canonical prior on mathematical meaning.

A program outside the enumerated prefix can be accounted for only when its prior and likelihood are defined or bounded. A genuinely outside-grammar component needs its own model. Assigning an unexplained residual probability is not a Bayesian treatment of missing mechanisms.

24.1 Refining a program without double counting it

A representation-rich language can contain many syntactic versions of one mechanism. It can also split one broad mechanism into more specific children. These operations must conserve prior mass when they are intended as a re-expression of the same model.

Proposition 24. Prior-preserving program refinement. Suppose a parent program g is replaced by children g_j with weights $\alpha_j \ge 0$, $\sum_j \alpha_j = 1$, and suppose its likelihood is the mixture

$$L_g(D) = \sum_j \alpha_j L_{g_j}(D).$$

Give child j prior mass $\pi(g)\alpha_j$ and leave all other programs unchanged. Then total evidence is unchanged and the sum of the children's posterior masses equals the former parent's posterior mass. Posterior predictive distributions are also unchanged when the same mixture relation holds for the joint likelihood of (D, y) for every predicted event y.

Proof. The children's total unnormalized evidence is $\sum_j \pi(g)\alpha_j L_{g_j}(D) = \pi(g)L_g(D)$. Thus both the total normalizer and every other program's posterior are unchanged. The same sum for joint evidence gives the predictive statement after normalization by the unchanged evidence of D. □

The mixture identity must hold for the **whole data record**. Averaging one-step likelihoods independently at each time generally describes a switching mechanism, not a fixed child selected once for the whole sequence.

If the children are only exact atlas recodings of the same operator and data model, their likelihoods are identical. Splitting the original prior mass leaves every prediction unchanged. Giving each duplicate a fresh full prior mass instead changes the model's prior preference. That may be an intentional new prior, but it is not invariance under representation.

The implementation's existing held-out predictive weights, described in Section 10, are not replaced by these identities merely by calling them posterior weights. The proposition supplies the probabilistic semantics that a future grammar-weighting implementation would need to enforce.

25. The finite learned-realization theorem

The preceding results combine into a concrete completion of one part of the learning problem. This is the strongest claim made for incomplete observations in this volume.

Proposition 25. Observationally grounded finite linear realization. Assume all of the following:

1. The data come from an autonomous finite-dimensional linear process with declared outputs $y_t = Cx_t$.
2. A delay length h satisfies $\operatorname{rank} O_{h+1} = \operatorname{rank} O_h$.
3. Exact observed input windows in Z_- span $V_h = \operatorname{im} O_h$, and Z_+ records their true one-step successor windows.

Then an algorithm using only (Z_-, Z_+) recovers the unique transition on V_h. Initialized with a fully observed length-h window, it predicts every later output exactly. Any lossless atlas of V_h, any complete packet decomposition of that space, and every coherently transported realization of the learned transition preserve the same predictions. Hidden variables outside the observable quotient need not be recovered.

If the fixed-space assumptions of Proposition 16 replace exact observations, that proposition bounds operator error. Whenever the resulting transition defects and Lipschitz constants satisfy Proposition 18, its product-sum bound controls the output prediction error. A common contraction margin yields the stated uniform bound.

Proof. Proposition 14 gives the unique true transition D_h on V_h. Because the observed windows span V_h, Proposition 15 identifies exactly its restriction, which is the entire required operator. Equality of the learned and true states follows by induction after initialization. The current output is the last block of the history state. Propositions 1–4 preserve state reconstruction, transport and complete decomposition, so they preserve that output sequence. The perturbation statement is the direct composition of Propositions 16 and 18 with the declared output decoder. □

This establishes an actual route from observations to a composable algebraic signal model under explicit finite assumptions. It is more than a fit on a few points. It is also less than unrestricted nonlinear world-model discovery. In particular, the algorithm does not infer assumptions 1–3 merely by returning a matrix. The accompanying certificate keeps the data-verified identities separate from those model assumptions.

26. Second-block checkpoint and remaining questions

The known-operator constructions in Part I now have an observation-based continuation. There is an exact causal realization theorem for a finite linear observable sector, a data-span identifiability criterion and a fixed-space perturbation bound. Recursive memory truncation has a sufficient stability theorem. Polynomial packet omissions have a remainder bound. Learned chart descriptions can be reconciled, refinement has an explicit information requirement, and grammar growth has an evidence-preserving semantics.

The theoretical center is a sequence of obligations:

$$\begin{aligned} &\text{accessible observations} \longrightarrow \text{declared predictive state} \longrightarrow \text{identified operations},\\ &\text{typed defects} \longrightarrow \text{stable composition} \longrightarrow \text{coherent refinement and inference}. \end{aligned}$$

A training residual cannot stand in for coverage of unobserved directions. A small coefficient error cannot stand in for well-conditioned synthesis. A posterior within a grammar cannot stand in for a probability that the grammar is complete.

26.1 The remaining research questions are narrower

Proposition 20 controls omissions in a declared polynomial expansion. It does not discover an invariant nonlinear dictionary or prove that its dimension stays finite. Selection and stability of nonlinear invariant features remain separate tasks [R6].

Stochastic partially observed processes need their own accessible state construction, positive probability semantics and finite-sample control. The HMM literature gives examples under rank and separation assumptions [R7], not guarantees for an arbitrary operator grammar. Noisy feature-space learning needs subspace perturbation and rank-selection analysis. Dependent sampling and active observations need their own statistical and excitation assumptions.

Without contraction or another stability certificate, the product-sum error bound remains valid but need not stay small. Unbounded infinite-dimensional operators still need domain control. Adaptive selection of representations still needs consistency beyond a pathwise deterministic estimate. These questions delimit the present theorems rather than invalidate their stated finite conclusions.

26.2 Release and verification status

The preceding foundations edition 0.2.0 contained Propositions 1–25, including twelve statements added in Part II. Every added statement is proved in the text. The accompanying rational, symbolic and separately labeled numerical calculations check finite examples and counterexamples. No general Lean kernel verification is claimed.

The 2.3.0 runtime is unchanged. The output-only reference learner implements data-span identification, not unrestricted discovery. Source lineage, assumptions, implementation boundaries, calculation reports and the complete preceding foundations archive accompany this edition.

The contribution is a connected foundation and explicit special-case repairs. Observability, predictive states, least squares, invariant subspaces, Lyapunov stability and Bayesian mixture identities have classical antecedents. The proposition count is not a count of historical discoveries.

Part III. Controlled stochastic realization

The next step is not to rename a deterministic predictor as a probabilistic one. A stochastic signal model must say what its state predicts, which actions intervene on the process, how observations update the state, and why every resulting number is a probability. These are separate obligations. A low dimensional linear representation can satisfy its reconstruction equations while failing probability positivity.

Part II established observable learning for an autonomous linear signal. Here the observable objects are probabilities of finite action and observation experiments. We derive a finite realization from them, show when it is minimal, give an explicit probability-valid state domain, and separate exact reconstruction from noisy estimation. The final construction connects back to Komorebi's packets through a normalized closure-defect identity.

The classical antecedents are predictive state representations, observable operator models and finite-rank realization [R5, R7, R9]. Those traditions already use probabilities of tests and products of operators. The purpose here is to make their assumptions and consequences explicit within the realization calculus. The twelve propositions below are a connected development, not twelve claims of historical novelty. No statement here replaces a proof with a numerical check.

27. Actions and observations are different inputs

Let $\mathcal{A}$ and $\mathcal{O}$ be finite nonempty action and observation sets. A symbol $s = (a, o)$ records an action followed by an observation. A word $w = s_1 \cdots s_k$ has temporal order from left to right. The empty word is ϵ.

Write

$$f(w) = \Pr(o_1, \ldots, o_k \mid \mathrm{do}(a_1, \ldots, a_k)).$$

The process is reset to one specified initial distribution before this experiment. The actions in this definition are fixed externally. This is not, in general, the distribution obtained by conditioning a passive record on an adaptively selected sequence of actions. We assume the consistency laws

$$f(\epsilon) = 1, \qquad f(w) \geq 0, \qquad \sum_{o \in \mathcal{O}} f(w(a, o)) = f(w) \quad \text{for every } w, a.$$

These laws specify causal probabilities on finite observation trees. A fixed history-dependent policy can be evaluated by selecting the action on each branch and multiplying the appropriate conditional probabilities. This does not establish optimality of that policy or its identification from insufficient data.

27.1 A finite hidden-state construction

A concrete realization has a belief $p \in \mathbb{R}^n_+$ with $\mathbf{1}^\top p = 1$. For each $s = (a, o)$, let $M_s \in \mathbb{R}^{n \times n}$ be entrywise nonnegative, with

$$\mathbf{1}^\top \sum_o M_{a,o} = \mathbf{1}^\top.$$

The matrix entry $M_{a,o}(j, i)$ is the probability of next hidden state j and observation o, given current hidden state i and chosen action a. This is a discrete probability instrument. It includes an action-specific transition followed by an observation, but does not require that factorization.

Put $M_w = M_{s_k} \cdots M_{s_1}$. Then

$$f(w) = \mathbf{1}^\top M_w p_0, \qquad p^+ = \frac{M_s p}{\mathbf{1}^\top M_s p}$$

on branches of positive probability. An observation with zero probability has no conditional posterior under that model. It must not be handled by silently dividing by an arbitrary numerical floor.

Proposition 26. Finite instrument closure. Starting with $W_0 = \mathrm{span}\{\mathbf{1}^\top\}$, define

$$W_{j+1} = W_j + \sum_{s \in \mathcal{A} \times \mathcal{O}} W_j M_s.$$

This stabilizes after at most $n - 1$ strict enlargements at

$$W_* = \mathrm{span}\{\mathbf{1}^\top M_w : w \text{ a finite word}\}.$$

Choose a full row rank matrix R with row space W_* and first row $\mathbf{1}^\top$. There are unique matrices B_s with

$$RM_s = B_s R, \qquad \lambda = e_1^\top, \qquad \lambda R = \mathbf{1}^\top.$$

For $z = Rp$, both the next-observation likelihood and the posterior predictive state close:

$$\Pr(o \mid z, a) = \lambda B_{a,o} z, \qquad z^+ = \frac{B_{a,o} z}{\lambda B_{a,o} z}.$$

Every linear summary determining all word probabilities by linear readouts for every initial belief has row space containing W_*.

Proof. The nested-space argument from Proposition 8 applies to the finite family M_s. Multiplication of a test row by M_s prepends the symbol s to that test, so the stable span is exactly the displayed word span. Invariance supplies the matrices B_s, and full row rank makes them unique. Apply R to the unnormalized and normalized hidden-state updates. If all test probabilities are linear functions of another summary, then each row $\mathbf{1}^\top M_w$ lies in its row space. Equality on all beliefs implies equality on their linear span, which is $\mathbb{R}^n$. □

This proposition uses a known hidden-state model. It is not yet an observable-data learner. It also concerns all initial beliefs, whereas a model initialized at one fixed p_0 can have a smaller reachable predictive dimension. The next construction handles that distinction without requiring recovery of hidden states.

28. The state is a function of possible future experiments

For a history h, define its unnormalized future-behavior column

$$v_h(t) = f(ht),$$

where t ranges over every finite test word. Form the infinite history/test matrix

$$\mathscr{H}_{t,h} = f(ht).$$

Its rank means the dimension of the span of its columns, equivalently the supremum of the ranks of finite submatrices. The columns are functions, not vectors obtained by physically observing every future. Their finite coordinate realization becomes learnable only when enough experiment probabilities are available.

Proposition 27. Minimal finite behavior realization. If $\operatorname{rank} \mathscr{H} = r < \infty$, the vector space $V = \operatorname{span}\{v_h\}$ has dimension r. There are well-defined linear maps $B_s : V \to V$, an initial vector v_ϵ, and a linear mass functional $\lambda(v) = v(\epsilon)$ satisfying

$$B_s v_h = v_{hs}, \qquad f(s_1 \cdots s_k) = \lambda B_{s_k} \cdots B_{s_1} v_\epsilon.$$

Every linear realization of f has dimension at least r. Any two reachable and observable realizations of dimension r differ by an invertible change of coordinates. For $f(h) > 0$, the normalized state $v_h/f(h)$ determines every future test probability conditional on h and the declared future interventions.

Proof. If $\sum_h c_h v_h = 0$, then for every test t,

$$\sum_h c_h v_{hs}(t) = \sum_h c_h f(hst) = 0,$$

because st is another test. Thus shifting columns preserves all linear relations and defines B_s. The probability formula follows by induction. For any d-dimensional realization, the history/test matrix factors into test rows and reachable state columns, so its rank is at most d.

For uniqueness, map the reachable vector after each history in one minimal realization to the corresponding vector in the other. If a linear combination vanishes in the first, every future test evaluates to zero on it. Those same evaluations vanish in the second, where observability makes the combination zero. The map is therefore well-defined and bijective, and intertwines every symbol operator and both boundary vectors. Finally $f(ht)/f(h)$ is exactly the conditional experiment probability. □

The coordinate dimension r includes probability mass. Normalized states lie in the affine hyperplane $\lambda z = 1$, so their affine dimension is at most $r - 1$. A finite hidden-state model of size n has behavior rank at most n. The theorem itself requires finite behavior rank, not a preferred hidden-state ontology.

Two realized histories with equal normalized future columns are equivalent for every declared future test. This is a predictive equivalence, not equality of their physical pasts or lossless archives.

29. Recovering the operators from an observable probability table

Select r tests $t_1, \ldots, t_r$ and r histories $h_1, \ldots, h_r$, with $t_1 = \epsilon$, for which the square matrix

$$H_{ij} = f(h_j t_i)$$

is invertible. Such a choice exists under the rank hypothesis of Proposition 27. The nonzero empty-test row can be extended to a row basis. Select independent history columns for that basis.

For each symbol s, also obtain the shifted table and initial test vector

$$(G_s)_{ij} = f(h_j s t_i), \qquad (b_0)_i = f(t_i).$$

Proposition 28. Observable table reconstruction. Assume the full behavior rank is at most r and the displayed $r \times r$ matrix H is invertible. Then the full rank is exactly r, and

$$\boxed{B_s = G_s H^{-1}, \qquad \lambda = e_1^\top, \qquad b_0 = (f(t_i))_i}$$

is a minimal realization of every word probability. Its state after history h is

$$b(h) = (f(h t_i))_i = B_h b_0.$$

The normalized state is $b(h)/f(h)$ when $f(h) > 0$. Its update uses the formula in Proposition 26, with no hidden-state coordinates supplied to the reconstruction algorithm.

Proof. Invertibility gives rank at least r, so the assumed upper bound is attained. Evaluation on the selected tests is an isomorphism $V \to \mathbb{R}^r$, because it maps the chosen basis of history columns to the invertible matrix H. The shifted images of those basis columns are precisely the columns of G_s, so $B_s H = G_s$. The first selected test evaluates the mass. Initial evaluation gives b_0. Proposition 27 supplies correctness for every history, not only the selected columns. □

There are two ways to justify coverage in this statement. One can know that the selected row and column families span the true behavior space. Alternatively, a justified upper bound r on the process rank, together with the invertible $r \times r$ block, proves that coverage. The observed invertibility by itself only gives a lower bound. An unknown higher-rank process can agree on a finite table and differ later.

Exact experiment probabilities are population quantities. The reference example below supplies rational tables from a declared process and keeps that table generator separate from the learner. Section 36 supplies one finite sampling protocol for estimating those entries. Neither a hidden matrix nor its probability cone is secretly passed to the observable learner.

30. Probability validity is not a rank condition

A real matrix product can yield a negative number. Therefore a fitted factorization is not a stochastic model merely because it is low rank or exactly interpolates a probability table. Spectral estimators can encounter negative-probability problems in practice [R10]. We address the structural issue directly.

In the canonical behavior space V, define

$$\mathcal{C} = \{v \in V : v(t) \geq 0 \text{ for every finite test } t\}.$$

Proposition 29. Canonical probability cone. The set $\mathcal{C}$ is a closed convex pointed cone that spans V. Every B_s preserves it. The mass functional is strictly positive on its nonzero elements, and

$$\sum_o \lambda B_{a,o} = \lambda.$$

Consequently the normalized base

$$\mathcal{K} = \{v \in \mathcal{C} : \lambda v = 1\}$$

is a compact convex probability-valid state space. Every positive-probability branch maps $\mathcal{K}$ into itself by normalized updating. In the core-test coordinates of Proposition 28, $\mathcal{K} \subseteq \{1\} \times [0,1]^{r-1}$.

Proof. Each test evaluation is a continuous linear functional on finite-dimensional V, so the cone is closed and convex. If both v and $-v$ belong to it, every test value is zero and $v = 0$. It spans V because it contains every column v_h. Also $(B_s v)(t) = v(st)$, proving invariance.

The consistency law $\sum_o v(t(a,o)) = v(t)$ holds for each history column and hence throughout V. Iterating it along any fixed action schedule expresses $v(\epsilon)$ as a sum of nonnegative values for all observation words on that schedule. If $v(\epsilon) = 0$, each summand is zero, for every schedule and length, so $v = 0$. The same sums show $0 \leq v(t) \leq 1$ when $\lambda v = 1$. They also give mass conservation. The base is closed and bounded in the finite core coordinates and thus compact. Invariance and positive normalization prove the posterior assertion. □

This cone is defined by all finite tests. Its existence is not an algorithm for finding a finite list of inequalities. A finite list gives an outer approximation and can miss a longer-test violation. The cone also need not equal the convex hull of the states reachable from the fixed initial distribution.

A practical sufficient certificate can instead use a known closed cone C_0 with nonempty normalized base, a strictly positive mass functional, $b_0 \in C_0$, $B_s C_0 \subseteq C_0$, and mass conservation. A nonnegative hidden-state realization supplies one after a coordinate change. Such a certificate may use information unavailable to an observable-only learner and must be labeled accordingly.

Negative entries of B_s do not, by themselves, prove invalidity. Coordinates can be signed while the represented state remains in the correct cone. Conversely, clipping negative coefficients can destroy the operator identities. Validity belongs to the pair consisting of the operators and their admissible cone, not to a coordinatewise heuristic.

31. Packets can reconstruct a belief without being beliefs

Let $P_1, \dots, P_m$ be a complete family of linear packet maps on a predictive space, with $\sum_i P_i = I$. Algebraic reconstruction gives $z = \sum_i P_i z$. It does not imply that $P_i z$ is an unnormalized probability state.

Proposition 30. Positive packet decomposition and its boundary. Suppose a cone C has a strictly positive mass functional λ. If every packet map preserves C, then each normalized $z \in C$ decomposes as a genuine mixture

$$z = \sum_{i:w_i>0} w_i z_i, \qquad w_i = \lambda P_i z \geq 0, \qquad z_i = P_i z / w_i, \qquad \sum_i w_i = 1.$$

A zero weight has $P_i z = 0$. In the standard cone $\mathbb{R}^n_+$, however, a complete family of cone-preserving idempotents must consist of diagonal coordinate-selection matrices. In particular, a nontrivial Fourier or Haar packet family cannot generally be reinterpreted as a mixture of probability states merely because it is orthogonal and complete.

Proof. Cone preservation and strict positivity give the weights and the zero-weight claim. Apply λ to the complete decomposition to obtain their sum. Normalization of each nonzero term gives the mixture.

On $\mathbb{R}^n_+$, cone preservation is entrywise nonnegativity. Since the matrices sum to I, every off-diagonal entry in every matrix is zero. Idempotence forces each diagonal entry to be zero or one, and completeness makes the selections disjoint and exhaustive. □

For a two-state belief $p = (3/4, 1/4)^\top$, the Haar detail projector

$$P_{\mathrm{d}} = \frac{1}{2}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$$

produces $(1/4, -1/4)^\top$. It is a valid signed detail packet, but not a subprobability distribution. Its total mass is zero despite being nonzero, which cannot happen inside a cone with a strictly positive mass functional.

This does not prohibit useful spectral packets. It fixes their type. Signed algebraic packets can be used for transfer, approximation and diagnostics. They must be synthesized before probability interpretation unless a separate positive decomposition has been established. That distinction prevents a reconstruction identity from being misused as a Bayesian mixture identity.

32. Action coverage is a mathematical requirement

Proposition 31. Missing-intervention nonidentifiability. There are finite controlled stochastic models that agree on every possible observation record generated using only action a, but disagree under another action b. Therefore even infinitely many passive records confined to a cannot identify the unrestricted controlled model.

Proof. Use one hidden state and two observations. In both models, action a gives probabilities $(1/2, 1/2)$. Under b, let the first model give $(1, 0)$ and the second $(0, 1)$. Every length-k record under a has probability 2^{-k} in both models. Their first observation under b differs with probability one. □

This is a coverage obstruction, not an optimization problem. It remains even when the hidden dimension is known exactly. Richer transforms of the same action-a observations do not create data about b.

Action selection based on past observations requires a further distinction. Under a known policy π, an action/observation word has joint probability

$$\Pr_{\pi}(w) = f(w) \prod_{j=1}^{|w|} \pi(a_j \mid h_{j-1}),$$

provided the action randomization has no access to hidden information beyond the declared history. The policy factors depend on the observed branch. Conditioning on a selected action sequence generally reweights branches rather than removing these factors. Recovering intervention probabilities from such data requires an appropriate coverage and reweighting argument. The reset experiments in Section 36 avoid that additional inference problem.

33. Learning the coordinate space under bounded noise

Part II bounded operator error after a state space and basis were fixed. Here we also bound one step of data-dependent space selection. This remains a finite-matrix statement. It does not certify that a chosen finite history/test table captures the full process.

Let $X \in \mathbb{R}^{m \times n}$ have rank $r \geq 1$, let $s = \sigma_r(X) > 0$, and observe $Y = X + E$ with $\|E\|_2 \leq \varepsilon < s/2$. Let P project onto $\operatorname{im} X$, and let $\widehat{P}$ project onto the leading r left singular vectors of Y.

Proposition 32. Finite-table rank separation and subspace stability. One has

$$\sigma_r(Y) \geq s - \varepsilon, \qquad \sigma_{r+1}(Y) \leq \varepsilon, \qquad \|P - \widehat{P}\|_2 \leq \frac{2\varepsilon}{s} < 1.$$

The $(r+1)$st singular value is interpreted as zero when r is the maximum possible rank. If U and $\widehat{U}$ are orthonormal bases of the two ranges, then

$$\sigma_{\min}(\widehat{U}^{\top} U) \geq \sqrt{1 - (2\varepsilon/s)^2}.$$

Thus $x \mapsto \widehat{U}^{\top} x$ is injective on the true column space. Its inverse on that space is

$$y \longmapsto U(\widehat{U}^{\top} U)^{-1} y,$$

with norm at most $[1 - (2\varepsilon/s)^2]^{-1/2}$.

Proof. The min-max characterization of singular values and the triangle inequality imply that adding E changes each singular value by at most ε. The leading-r singular subspace gives $\|(I - \widehat{P})Y\|_2 = \sigma_{r+1}(Y) \le \varepsilon$. Hence

$$\|(I - \widehat{P})X\|_2 \le 2\varepsilon, \qquad \|(I - \widehat{P})P\|_2 = \|(I - \widehat{P})XX^\dagger\|_2 \le 2\varepsilon/s.$$

For equal-rank orthogonal projections, the latter norm equals $\|P - \widehat{P}\|_2$. To see the equality, diagonalize $U^\top \widehat{U}$ by singular value decomposition. Each nontrivial paired principal plane contributes a sine to both norms, while common directions contribute zero. On the same planes the singular values of $\widehat{U}^\top U$ are the corresponding cosines. This proves the last bound and the inverse formula. □

A threshold τ with $\varepsilon < \tau < s - \varepsilon$ recovers the rank of this finite matrix. The unknown true gap must be justified or replaced by a valid confidence interval. Rank thresholds do not exclude modes absent from the selected tests or unexcited by the selected histories.

The inverse formula contains the unknown true basis U. It proves identifiability and conditioning relative to that space, not that a learner already knows a perfect decoder. When using the estimated subspace as coordinates for a realization, all boundary vectors, symbol operators and probability constraints must be transformed together. Signed singular-vector coordinates are not new independent observations.

34. Noisy operator reconstruction in a fixed core

After choosing a core block, the square reconstruction formula yields a particularly direct error identity. It is useful even when the core was chosen on a separate discovery data set.

Let $G_s = B_s H$ with $s_H = \sigma_{\min}(H) > 0$. Observe

$$\widetilde{H} = H + E, \qquad \widetilde{G}_s = G_s + F_s, \qquad \|E\|_2 < s_H.$$

Proposition 33. Noisy core-block realization. The estimate $\widehat{B}_s = \widetilde{G}_s \widetilde{H}^{-1}$ satisfies

$$\widehat{B}_s - B_s = (F_s - B_s E)\widetilde{H}^{-1},$$

and therefore

$$\boxed{\|\widehat{B}_s - B_s\|_2 \le \frac{\|F_s\|_2 + \|B_s\|_2 \|E\|_2}{s_H - \|E\|_2}.}$$

A bound using only the fitted operator and the same perturbation budgets is

$$\|\widehat{B}_s - B_s\|_2 \le \frac{\|F_s\|_2 + \|\widehat{B}_s\|_2 \|E\|_2}{s_H}.$$

If $\widetilde{s} = \sigma_{\min}(\widetilde{H})$ is known and $\|E\|_2 \le e < \widetilde{s}$, the last denominator may be replaced by $\widetilde{s} - e$.

Proof. Since $\widetilde{H}$ is invertible, subtract B_s from its fitted expression and collect the terms. The least singular value of $\widetilde{H}$ is at least $s_H - \|E\|_2$. For the second identity, use

$$(\widehat{B}_s - B_s)H = F_s - \widehat{B}_s E$$

and multiply by H^{-1}. Finally $s_H \ge \widetilde{s} - e$ by singular-value perturbation. □

When the first test is empty, the mass row $\lambda = e_1^\top$ is fixed rather than estimated. Set the first component of the initial vector to one as required by $f(\epsilon) = 1$. Errors in other initial test probabilities remain initial-state error. This convention removes an avoidable normalizer-estimation ambiguity but does not enforce cone preservation or mass conservation for the fitted symbol family.

Numerical implementation should solve the linear systems rather than explicitly form a floating-point inverse. The theorem describes the exact algebraic estimator. Finite-precision solve error is an additional residual, which the reference script measures separately.

35. Conditioning a prediction can amplify its error

Let C be a probability cone with compact normalized base K and mass row λ. Work in a declared norm and its dual. Suppose true normalized states have norm at most R. Let a true branch have

$$u = B_s z, \qquad p = \lambda u \ge \eta > 0, \qquad z^+ = u/p \in K.$$

For a predicted state $\widehat{z}$, form $\widehat{u} = \widehat{B}_s\widehat{z}$, $\widehat{p} = \lambda\widehat{u}$, and define $d = \widehat{u} - u$. Write $e = \|\widehat{z} - z\|$, $\|\widehat{B}_s\| \le L$, and $\|\widehat{B}_s - B_s\| \le \epsilon_B$. Then

$$\|d\| \le Le + \epsilon_B R =: D.$$

Proposition 34. Normalization-aware filtering stability. If $\|\lambda\|_* D < \eta$, then $\widehat{p} > 0$ and

$$\widehat{z}^+ - z^+ = \frac{(I - z^+\lambda)d}{\widehat{p}},$$

so

$$\boxed{\|\widehat{z}^+ - z^+\| \le \frac{(1 + R\|\lambda\|_*)D}{\eta - \|\lambda\|_* D}.}$$

Alternatively, if each realized fitted denominator is checked to satisfy $\widehat{p}_t \ge \widehat{\eta}_t > 0$, the pathwise recursion obeys

$$e_{t+1} \le \alpha_t e_t + \beta_t, \qquad \alpha_t = \frac{(1 + R\|\lambda\|_*)L_t}{\widehat{\eta}_t}, \qquad \beta_t = \frac{(1 + R\|\lambda\|_*)\epsilon_{B,t}R}{\widehat{\eta}_t}.$$

The product-sum bound of Proposition 18 applies. A uniform bound follows if these constants have a common contraction margin and bounded forcing. The algebraic bound alone does not certify that a fitted state or every unchosen branch is probability-valid.

Proof. The branch likelihood differs by λd, giving the lower bound on $\widehat{p}$. Subtract u/p from $(u + d)/(p + \lambda d)$ and use $u = pz^+$. The induced norm of $I - z^+\lambda$ is at most $1 + R\|\lambda\|_*$. Applying that identity at each step gives the recursion. □

Rare evidence can create a large posterior change from a small absolute error. For example, take a true unnormalized vector $(\eta, 0)$ and a perturbed vector (η, η). Their ℓ^1 difference is η, while their normalized vectors differ by one in ℓ^1. Both arise as possible branch outputs of valid probability instruments when $0 < \eta \le 1/2$, after assigning the remaining mass to another branch. There is no uniform posterior Lipschitz constant independent of a probability margin.

When a known closed convex feasible base is available, Euclidean projection of a tentative posterior back onto that base cannot increase its distance from the true posterior. This is a declared repair to the estimator, not an identity for an unmodified update. It also does not repair negative branch probabilities of arbitrary fitted operators before normalization.

36. A finite experimental route to the probability table

The following deliberately simple protocol makes a finite-sample statement possible without confusing a deterministic identity with a statistical guarantee.

Fix M finite words before estimation. For each word, reset independently to the same initial distribution, execute its prescribed action sequence, and record the indicator that all observations match. Obtain N independent trials per word. Dependencies between different words' estimates are allowed, but the N trials used for each individual mean must be independent with the declared success probability.

Proposition 35. Reset-experiment probability certificate. Let $\widehat{f}(w)$ be the sample mean for each selected word. For $0 < \delta < 1$, with probability at least $1 - \delta$, all M estimates satisfy

$$|\widehat{f}(w) - f(w)| \le \epsilon_N, \qquad \epsilon_N = \sqrt{\frac{\log(2M/\delta)}{2N}}.$$

If the selected words cover every entry of H, every G_s, and b_0, then on the same event

$$\|\widetilde{H} - H\|_2 \le r\epsilon_N, \qquad \|\widetilde{G}_s - G_s\|_2 \le r\epsilon_N, \qquad \|\widetilde{b}_0 - b_0\|_2 \le \sqrt{r}\epsilon_N.$$

These bounds feed Propositions 32 and 33 whenever their respective gap conditions hold. The empty-word probability may be set exactly to one instead of sampled.

Proof. For a centered Bernoulli variable $X - p$, write $g(t) = \log \mathbb{E} \exp(t(X - p))$. Then $g(0) = g'(0) = 0$, and $g''(t)$ is the variance of a Bernoulli under exponential tilting, at most $1/4$. Integrating twice gives $g(t) \le t^2/8$. Independence and the exponential Markov inequality give an upper-tail bound $\exp(-2N\epsilon^2)$ after optimizing $t = 4\epsilon$. The lower tail has the same bound. A union bound over the M words proves the simultaneous result. The spectral norm is bounded by the Frobenius norm, yielding the matrix and vector estimates. □

This is the Bernoulli form of Hoeffding's inequality [R11], proved here for the protocol used. It is a conservative entrywise-to-spectral bound, not a claim of optimal sample complexity. The number and lengths of chosen experiments are part of its cost.

If core tests and histories were selected from data, use a separate independent discovery sample and freeze the selected finite experiment list before these estimation trials. Conditional on that selection, the proposition applies. Alternatively, a uniform bound over a fixed larger candidate list can support data-dependent selection within that list. Reusing a post-selected table without either argument changes the claim.

A single dependent trajectory, an adaptive stopping rule, unknown resets, drifting sensors, or experiments chosen without coverage do not satisfy this proposition as stated. Those settings require their own concentration and identifiability arguments. Synthetic reset sampling in the reference example demonstrates the protocol, not its validity for every measurement source.

37. A filtering defect with a packet interpretation

The relation to the realization calculus is particularly direct before normalization. Let R be a linear summary of a hidden probability space, let $\lambda R = \mathbf{1}^\top$, and let B_s be a proposed reduced branch operator. For a true belief p, put

$$z = Rp, \qquad \delta_s = (RM_s - B_sR)p.$$

The defect δ_s is a vector in the reduced representation, not a probability of model failure. It records the failure of the unnormalized instrument to commute with the summary.

Proposition 36. Exact normalized closure defect. Suppose the true and proposed branch likelihoods

$$a = \mathbf{1}^\top M_s p > 0, \qquad b = \lambda B_s z > 0$$

are positive, and let $\widehat{z}^+ = B_s z/b$. Then

$$\boxed{R\left(\frac{M_s p}{a}\right) - \widehat{z}^+ = \frac{(I - \widehat{z}^+\lambda)\delta_s}{a}, \qquad a - b = \lambda\delta_s.}$$

If $M_s \neq 0$ and both likelihoods and normalized updates agree on every strictly positive belief, then $RM_s = B_sR$. Agreement of normalized updates alone need not force this equality.

Proof. The true unnormalized summary is $B_sz + \delta_s$, with mass $a = b + \lambda\delta_s$. Subtract the two normalized vectors and collect terms. Since a nonzero nonnegative branch has positive likelihood on every strictly positive belief, agreement permits multiplication by the common mass. This gives equality of unnormalized summaries on the interior of the probability simplex, whose linear span is the whole hidden vector space. For the last claim, a one-state model can assign different positive branch probabilities while its normalized state always remains the scalar one. □

For an orthogonal retained projector P preserving the mass row, put $Q = I - P$, $R = P$ on its image and $B_s = PM_sP$. Then

$$\delta_s = PM_sQp.$$

This is the unresolved-to-retained transfer block from Part I. Its posterior effect is not simply its norm: normalization applies the oblique map $I - \widehat{z}^+\lambda$ and divides by the true branch probability. A component proportional to $\widehat{z}^+$ changes only the branch mass and disappears from the normalized defect, while still changing likelihood-based inference over mechanisms.

A certificate should therefore retain both $\lambda\delta_s$ and the posterior defect. A posterior that looks correct can hide a wrong evidence model. This is the stochastic counterpart of the distinction between reconstruction and predictive closure.

38. A complete two-action rational example

Use two hidden states only to generate a reference process. The learner does not receive these matrices. Let

$$T_0 = \begin{pmatrix} 3/4 & 1/4 \\ 1/4 & 3/4 \end{pmatrix}, \qquad T_1 = \begin{pmatrix} 1/4 & 2/3 \\ 3/4 & 1/3 \end{pmatrix}, \qquad p_0 = \begin{pmatrix} 2/3 \\ 1/3 \end{pmatrix}.$$

The observation matrices are

$$O_0 = \operatorname{diag}(1/4, 3/4), \qquad O_1 = \operatorname{diag}(3/4, 1/4), \qquad M_{a,o} = O_o T_a.$$

The two transitions do not commute. Thus the process is genuinely action-dependent rather than one autonomous recurrence with an action label attached.

Choose tests $(\epsilon, (0,1))$ and histories $(\epsilon, (0,1))$. The observable tables give

$$H = \begin{pmatrix} 1 & 13/24 \\ 13/24 & 5/16 \end{pmatrix}, \qquad \det H = 11/576, \qquad b_0 = \begin{pmatrix} 1 \\ 13/24 \end{pmatrix}.$$

The selected coordinate q is the probability of seeing observation one after choosing action zero. In the hidden realization,

$$R = \begin{pmatrix} 1 & 1 \\ 5/8 & 3/8 \end{pmatrix}, \qquad z = Rp = (1, q)^\top.$$

The observable reconstruction $B_s = G_s H^{-1}$ yields

$$B_{0,0} = \begin{pmatrix} 1 & -1 \\ 11/32 & -1/4 \end{pmatrix}, \quad B_{0,1} = \begin{pmatrix} 0 & 1 \\ -3/32 & 3/4 \end{pmatrix},$$
$$B_{1,0} = \begin{pmatrix} 5/48 & 5/6 \\ 23/192 & 5/24 \end{pmatrix}, \quad B_{1,1} = \begin{pmatrix} 43/48 & -5/6 \\ 37/64 & -5/8 \end{pmatrix}.$$

The coefficients include negatives. Nevertheless, every operator preserves the cone $R\mathbb{R}^2_+$, because $B_s R = R M_s$. A sufficient normalized base is

$$K_0 = \{(1, q)^\top : 3/8 \le q \le 5/8\}.$$

This cone certificate uses the known generating model and is checked separately. It is not inferred merely from the finite probability table, and K_0 is not claimed to be the maximal canonical base of Proposition 29.

The likelihoods and posterior maps can now be read from the matrices. For action zero and observation one,

$$p(1 \mid z, 0) = q, \qquad q^+ = \frac{-3/32 + (3/4)q}{q} = \frac{3}{4} - \frac{3}{32q}.$$

Starting at $q_0 = 13/24$, this gives probability $13/24$ and $q_1 = 15/26$. If action one then produces observation zero, its probability is

$$\frac{5}{48} + \frac{5}{6}\frac{15}{26} = \frac{365}{624},$$

and the new predictive coordinate is

$$q_2 = \frac{23/192 + (5/24)(15/26)}{365/624} = \frac{599}{1460}.$$

The joint probability of this two-step action/observation word is $365/1152$. The reference checks compare these values and every word up to a declared finite test length against the hidden generator in exact rational arithmetic. The proposition, under its rank and coverage assumptions, is what extends the result to all word lengths.

The model's online state is not a sequence of newly measured future probabilities. It is a recursively maintained prediction initialized from learned experiment probabilities and updated after each actual action and observation. Uncertainty in those learned parameters is separate from the stochastic uncertainty in the next observation.

39. Integrated controlled stochastic realization

Proposition 37. Conditional observation-based stochastic completion. Suppose a causal finite-alphabet signal process has behavior rank at most r, and a selected $r \times r$ history/test block containing the empty test is invertible. Exact values of that block, every one-symbol shifted block, and the initial core-test vector determine a minimal realization of every future experiment probability. Its normalized state is recursively accessible after each positive-probability history, without reconstructing the hidden physical state.

The exact realization has the canonical probability cone of Proposition 29. Under any invertible atlas, transform the state, symbol operators, mass row and cone together:

$$z' = Tz, \qquad B'_s = TB_sT^{-1}, \qquad \lambda' = \lambda T^{-1}, \qquad C' = TC.$$

Every likelihood and posterior prediction is then unchanged. A complete packet decomposition preserves these predictions after full synthesis. Packetwise probability interpretation requires the extra positivity condition of Proposition 30.

For estimates obtained under Proposition 35's reset protocol, the simultaneous event there bounds the finite table perturbations. A separated finite-table spectrum yields Proposition 32's subspace certificate. A well-conditioned fixed core yields Proposition 33's operator certificate. On branches satisfying Proposition 34's likelihood margins, the resulting normalized state errors obey its recursive bound. These perturbation statements alone do not certify a learned cone or global positivity of an estimated model.

Proof. Proposition 28 reconstructs the unique minimal behavior realization, and Proposition 27 gives its conditional interpretation. Proposition 29 supplies probability validity for the exact realization. Substituting the transformed quantities cancels every $T^{-1}T$, including in the scalar normalization. Full packet synthesis is the identity, whereas Proposition 30 characterizes the additional probability-mixture condition. The perturbation conclusions are the stated chain of propositions with their separate assumptions retained. No missing gap, coverage or positivity assumption follows from matrix inversion alone. □

This theorem closes a finite controlled and stochastic sector of the foundational model. It does not claim unrestricted discovery of a nonlinear environment from a single trajectory. It does identify what a learned predictive state represents, why its operations compose, where its evidence comes from, and which defects change likelihoods or posteriors.

40. Third-block checkpoint

The model now has a single framework for exact signal archives, predictive quotients, controlled probability instruments, coordinate transport, packet decompositions, and filtering. The central mathematical object is not merely a large feature vector. It is a representation together with operations and explicit admissibility conditions under which those operations preserve meaning.

Several questions remain concrete. Finite data still require justified rank or coverage assumptions. Learning a globally invariant probability cone from partial tests is not supplied by an SVD. Continuous observations require kernels or densities with a common reference measure, and finite-alphabet proofs do not transfer by renaming their symbols. Dependence, adaptive experimentation and unknown reset

states require additional statistical arguments. Useful finite nonlinear observable dictionaries need construction and approximation theory beyond the existence of a finite-rank special case.

These are not reasons to dilute the finite conclusions. They identify the next mathematical work. In particular, the new normalized defect

$$\frac{(I - \widehat{z}^{+}\lambda)(RM_s - B_sR)p}{\mathbf{1}^{\top}M_sp}$$

provides an exact interface between representation error, packet transfer, branch evidence and posterior inference. It is the quantity a foundations-driven refinement rule would need to control, rather than an undifferentiated aggregate error.

The reference implementation is a theory laboratory, not a replacement of the production 2.3.0 runtime. The release contains exact tables, an observable-only table learner, a separate hidden-model verifier, noise calculations, reset-sampling examples, and negative controls. None is represented as Lean kernel verification. The full general proofs are the arguments in this volume.

Part IV. Certified probability domains and robust learning

A matrix learned from observations is not yet a probability model. Part III made that distinction exact: a finite predictive realization needs an admissible cone preserved by its action and observation operators. It supplied a canonical cone defined by all possible future tests, but did not supply a finite way to verify that cone for an estimated model.

This block develops a constructive finite sector of that missing step. It gives a certificate that can be checked by rational arithmetic, a search procedure that does not need hidden-state matrices, and a robust version that covers an entire declared uncertainty set. It also proves why a finite polyhedral certificate is not available for every valid finite-dimensional process. The resulting distinction is between an exact finite certificate, an uncertified candidate, and an explicitly modified approximation. Those are different mathematical outcomes.

The positive-realization and invariant-cone ingredients are classical. Jaeger's account includes the polyhedral-cone characterization and an irrational-rotation obstruction [R12]. Polyhedral invariance is also a standard subject of linear systems and optimization [R13]. The present contribution is a self-contained connection of these ingredients to Komorebi's observable learner, probability packets, confidence sets and approximation defects. The controlled rational rotation example is an adaptation of that classical obstruction, not a claim of discovering the phenomenon.

The next twelve propositions continue the numbering from the preceding volume. All have proofs below. The exact computations check stated finite instances and counterexamples. They are not a Lean formalization and do not change the production PR-LASM 2.3.0 runtime.

41. The object to be certified

Fix a real predictive space $E = \mathbb{R}^r$, a nonzero row λ, and a finite family B_s, indexed by $s = (a, o)$. The action and observation alphabets are finite and nonempty. Every declared observation branch is present, including branches represented by the zero operator. Temporal words act from right to left:

$$B_{s_1 \cdots s_k} = B_{s_k} \cdots B_{s_1}, \qquad B_\epsilon = I.$$

The mass identity is an exact constraint,

$$\lambda \sum_o B_{a,o} = \lambda \quad \text{for every action } a.$$

An initial state has $\lambda z_0 = 1$. The proposed probability of a word is $\lambda B_w z_0$. An invertible coordinate change must transform z_0, B_s, λ and the admissible cone together, as in Proposition 37.

A **probability-domain certificate** specifies a closed, pointed, generating cone $C \subset E$, with λ strictly positive on $C \setminus \{0\}$, such that

$$z_0 \in C, \qquad B_s C \subseteq C.$$

Its normalized base is $K = \{z \in C : \lambda z = 1\}$. These conditions imply nonnegative word probabilities and their per-action normalization at every depth. Positive-probability updates remain in K. Zero-probability branches have no conditional posterior and must not be assigned one by denominator flooring.

There are three separate claims. **Internal validity** says the fitted model is a stochastic model. **Data compatibility** says it agrees with the observations within a specified error budget. **Coverage of the generating process** says the assumptions and uncertainty set actually include the process that generated those observations. The first can have a deterministic certificate even when the other two remain conditional.

42. Finite tests give an outer hierarchy, not automatic closure

Define the test cones

$$C_h = \{z : \lambda B_w z \geq 0 \text{ for every } |w| \leq h\}, \qquad C_0 = \{z : \lambda z \geq 0\}.$$

Each C_h is polyhedral because it is an intersection of finitely many homogeneous half-spaces. It need not be a pointed cone or have a bounded normalized base.

Proposition 38. Test-cone hierarchy and a finite stopping certificate. The cones satisfy

$$C_{h+1} = C_h \cap \bigcap_s B_s^{-1} C_h, \qquad C_{h+1} \subseteq C_h.$$

Their intersection C_∞ is the largest cone contained in C_0 and invariant under all B_s. If an exact equality $C_{h+1} = C_h$ occurs, then $C_h = C_\infty$ and the finite inequalities certify all future word probabilities. If the

representation is observable, meaning that the rows λB_w span E^*, then C_∞ is pointed and λ is strictly positive on its nonzero elements. If reachable vectors $B_w z_0 \in C_\infty$ span E, the cone is also generating.

Proof. For a first symbol s followed by a word v, $B_{sv} = B_v B_s$. The inequalities for all words of length at most $h+1$ are therefore exactly the displayed recurrence. If $z \in C_\infty$, every test on $B_s z$ is another test on z, so C_∞ is invariant. Conversely, invariance of a cone contained in C_0 makes every successive test nonnegative, hence places it in C_∞. Equality at one finite step establishes invariance immediately and therefore gives the intersection.

If both z and $-z$ lie in C_∞, every test row annihilates z, so observability gives $z = 0$. For $z \in C_\infty$ with $\lambda z = 0$, sum word probabilities along any fixed action sequence. Mass conservation makes the sum zero. Every term is nonnegative, so every word test vanishes and again $z = 0$. Spanning by reachable cone elements gives the final assertion. □

A useful negative witness is a state admitted by the current inequalities and a specific longer word with negative proposed probability. It demonstrates that this state needs to be excluded. It does not prove that the initial state, or every reachable state, is invalid.

Finite stabilization is sufficient, not promised. Even three predictive coordinates can require a nonpolyhedral domain, as Section 51 shows. Checking a long list of trajectories is not the same as checking the invariance identity that makes every trajectory valid.

43. A finite certificate with a positive lift

Let $V = [v_1 \ \cdots \ v_m] \in \mathbb{R}^{r \times m}$ have full row rank and satisfy

$$\lambda V = \mathbf{1}_m^\top.$$

Its columns generate $C = V\mathbb{R}_+^m$ and its normalized base is $K = \operatorname{conv}\{v_1, \dots, v_m\}$. Finite generation makes the cone closed. Its normalized generators and full row rank make it pointed and generating, with a strictly positive mass functional.

Proposition 39. Finite positive-lift certificate. This cone is invariant under B_s if and only if there is an entrywise nonnegative matrix $A_s \in \mathbb{R}^{m \times m}$ such that

$$\boxed{B_s V = V A_s, \qquad A_s \geq 0.}$$

If these identities hold for every symbol and $z_0 = V\alpha_0$ for $\alpha_0 \geq 0$ with $\mathbf{1}^\top \alpha_0 = 1$, then

$$\mathbf{1}^\top \sum_o A_{a,o} = \mathbf{1}^\top, \qquad \lambda B_w z_0 = \mathbf{1}^\top A_w \alpha_0.$$

Thus the finite witness defines a nonnegative probability instrument whose image under V is the predictive model. It certifies all word probabilities without enumerating them.

Proof. Invariance places each $B_s v_j$ in the conical hull of the columns of V. Choose its nonnegative coefficients as column j of A_s. Conversely, $B_s V \alpha = V A_s \alpha \in C$ for every $\alpha \geq 0$. Multiplying the intertwining identity by λ and summing over observations gives

$$\mathbf{1}^\top \sum_o A_{a,o} = \lambda \sum_o B_{a,o} V = \lambda V = \mathbf{1}^\top.$$

Iterating $B_sV = VA_s$ gives $B_wV = VA_w$. Applying the mass row and the initial coefficients proves the word identity. Strict positivity of λ follows because a nonzero conical combination has positive total coefficient mass. □

The m nonnegative coordinates need not be physical hidden states, and m need not equal the minimal predictive dimension r. They are one finite realization witnessing validity. When V has dependent columns, the lift can be nonunique. Those redundant coordinates do not change the predictive state after synthesis.

There is a dual finite test. If a complete inequality description of the same cone is $C = \{z : Gz \geq 0\}$, then $GB_sv_j \geq 0$ for every generator j is equivalent to invariance. One may also certify $GB_s = H_sG$ with $H_s \geq 0$. The latter certificate is sufficient by direct multiplication and necessary by the polyhedral cone form of Farkas' lemma [R13]. The reference calculation uses the direct generator test, not an unreported numerical optimization certificate.

44. Robustness is a quantified invariance statement

A certificate for one fitted operator does not automatically cover neighboring operators. For uncertainty, the quantifier must range over the whole proposed model set, including correlations required by mass conservation.

Let $\mathcal{U}$ be the convex hull of finitely many **complete operator families** $\{B_s^{(j)}\}_s$. Every family must satisfy the exact per-action mass identities. The convex coefficient is shared across that family's symbols. A product of independent uncertainty sets can instead be handled by its joint vertices.

Proposition 40. Robust finite-domain certificate. For a fixed normalized generator matrix V, suppose every vertex family has witnesses

$$B_s^{(j)}V = VA_s^{(j)}, \qquad A_s^{(j)} \geq 0.$$

Then every family in $\mathcal{U}$ preserves $C = V\mathbb{R}_+^m$ and is mass conserving. If its initial state lies in K, it defines valid finite observation-tree probabilities at every horizon. This remains true when a family in $\mathcal{U}$ is selected separately at each history.

For a complete facet description $C = \{z : Gz \geq 0\}$, let $g_i^\top$ be its rows. A sufficient norm-ball certificate around $\widehat{B}_s$ is

$$\boxed{g_i^\top \widehat{B}_s v_j \geq \epsilon_s \|g_i\|_2 \|v_j\|_2 \quad \text{for all } s, i, j.}$$

It covers all perturbations with $\|B_s - \widehat{B}_s\|_2 \leq \epsilon_s$ that also satisfy the exact mass constraints.

Proof. For convex weights θ_j, take $A_s = \sum_j \theta_j A_s^{(j)}$. The equality and nonnegativity survive convex combination, as does mass conservation. At each history, every selected family maps the current cone state back into the cone. Applying the mass identity at that node proves recursive normalization, regardless of how the family was selected. For the norm ball,

$$|g_i^\top (B_s - \widehat{B}_s) v_j| \leq \epsilon_s \|g_i\|_2 \|v_j\|_2.$$

The margin makes every perturbed generator image satisfy all facets. Proposition 39, or the direct generator test, then supplies invariance. □

The switching conclusion is structural validity, not an assertion that time-varying selection represents one fixed unknown parameter. A likelihood for a fixed mechanism and a likelihood for a mechanism selected anew at each history are different statistical models.

The norm-ball inequality is sufficient and can be conservative, especially at boundary images with zero slack. Failure of this margin is not a proof that no robust certificate exists. The exact vertex test retains uncertainty correlations that a separate bound for each entry may discard.

45. Fitting inside a certified domain

Once V is fixed, certificate-constrained fitting is a convex problem. Unknowns include the entries of $\widetilde{B}_s$, A_s, $\widetilde{z}_0$ and α_0. The constraints

$$\widetilde{B}_s V = V A_s, \quad A_s \geq 0, \quad \lambda \sum_o \widetilde{B}_{a,o} = \lambda,$$
$$\widetilde{z}_0 = V \alpha_0, \quad \alpha_0 \geq 0, \quad \mathbf{1}^\top \alpha_0 = 1$$

are linear. Entrywise confidence intervals and fixed-input regression tolerances add linear inequalities. Norm balls add convex constraints. Jointly optimizing V and A_s is different: the product VA_s is then bilinear. No general convexity is claimed for learning the domain and dynamics together.

Proposition 41. Certified correction and its error budget. For fixed V, every feasible solution of the displayed constraints is an internally valid model. Suppose a center estimate $\widehat{B}_s$ and a feasible corrected estimate $\widetilde{B}_s$ satisfy

$$\|B_s^* - \widehat{B}_s\| \leq \epsilon_s, \qquad \|\widetilde{B}_s - \widehat{B}_s\| \leq \epsilon_s.$$

Then $\|\widetilde{B}_s - B_s^*\| \leq 2\epsilon_s$. The same conclusion holds for initial states in their specified norm. If the true model lies in these confidence sets and itself admits the fixed-domain certificate, the convex constraints are feasible. Feasibility alone does not establish that the true model preserves this domain.

Proof. Validity is Proposition 39. The error bound is the triangle inequality through the center estimate. Under the final assumptions, insert the true operators, true initial state and their nonnegative witnesses into the constraints. They provide a feasible point. The converse fails because existence of one valid point does not say that every point in a confidence set, or its true member, has the same invariant domain. □

This is why existential correction and robust certification must be reported separately. A corrected model can be a well-defined approximation even when no common true-model cone has been proved. The trajectory bounds below require both models to be valid on a common specified cone.

45.1 An explicit positivity repair in a simplicial cone

When V is square and invertible, put $A_s = V^{-1}\widehat{B}_s V$. Assume mass conservation has already been imposed. Let N_s be an entrywise positive reference instrument with the same per-action mass identities. The mixture

$$\widetilde{A}_s = (1-\eta)A_s + \eta N_s$$

is nonnegative provided

$$\eta \geq \max_{s,i,j:(A_s)_{ij}<0} \frac{-(A_s)_{ij}}{(N_s)_{ij} - (A_s)_{ij}}.$$

An empty maximum is zero. If every negative entry has magnitude at most d and every reference entry is at least $m_0 > 0$, the simpler sufficient choice is $\eta = d/(m_0 + d)$. These statements follow by solving each scalar inequality. Mass conservation is unchanged by convex mixing.

This is an explicit model modification, not coefficient clipping or an exact coordinate change. Relative to a true instrument on the same cone, the one-step error is at most $(1 - \eta)\epsilon + 2\eta$ in the summed base-norm metric defined next, when the original error was at most ϵ. The factor two follows because two valid instruments each distribute unit total mass. The reference example retains the repair amount as part of its approximation record.

46. A norm that measures probability mass before conditioning

Assume C is a closed, pointed, generating cone and λ is strictly positive on its nonzero elements. Its base K is compact. Define the base norm

$$\|x\|_C = \inf\{\lambda u + \lambda v : x = u - v,\ u, v \in C\}.$$

The infimum is finite because the cone is generating. Compactness of K makes this a genuine norm. Its unit ball is $\mathrm{conv}(K \cup (-K))$. In particular $\|u\|_C = \lambda u$ on C, and $|\lambda x| \leq \|x\|_C$. On the ordinary probability orthant this is the ℓ^1 norm. These are standard ordered-space constructions [R15], used here to avoid mixing Euclidean coefficient magnitude with probability mass.

Proposition 42. Instrument contraction before normalization. If $\{B_{a,o}\}_o$ is cone preserving and mass conserving for a fixed action, then

$$\boxed{\sum_o \|B_{a,o}x\|_C \leq \|x\|_C.}$$

For a polyhedral base $K = \mathrm{conv}\{v_j\}$, define the discrepancy between two instruments by

$$\epsilon_a = \sup_{z \in K} \sum_o \|(B_{a,o} - \widetilde{B}_{a,o})z\|_C.$$

The supremum equals the maximum of the same expression at the finitely many generators v_j. When both instruments are valid on C, $\epsilon_a \leq 2$.

Proof. For any $x = u - v$ with $u, v \in C$, cone preservation gives

$$\sum_o \|B_{a,o}x\|_C \leq \sum_o \lambda B_{a,o}(u + v) = \lambda u + \lambda v.$$

Take the infimum. For the generator statement, the discrepancy is convex in z. At a convex combination its value is at most the same combination of generator values, hence at most their maximum. The generators themselves belong to K, proving equality. Finally, for $z \in K$, the triangle inequality bounds the sum by the total masses of both instruments, which are one each. □

For a polyhedral certificate, the norm itself has a finite linear-program formulation:

$$\|x\|_C = \min_{a,b\geq 0}\{\mathbf{1}^\top(a+b) : V(a-b) = x\}.$$

If V is square and invertible, this reduces to $\|V^{-1}x\|_1$. The exact worked interval certificate is of this type. With dependent generators, choosing one pseudoinverse does not in general compute the base norm.

47. Whole observation laws can be stable without rare-event margins

The posterior after one rare observation can be sensitive, but that does not force the entire distribution of observation records to be equally sensitive. Work before normalization and retain the probability of each branch.

Let two models share (C, λ) and have initial states $z_0, \widetilde{z}_0 \in K$. At each time use the same policy, a specified function or randomization depending only on the observed history. Thus on any common history the action probabilities are identical in the two comparisons. This is not a comparison of two policies optimized separately for the two models.

Proposition 43. Observation-tree total-variation bound. If both instruments preserve C and conserve mass, and if every action at time t has the discrepancy in Proposition 42 bounded by ϵ_t, then the distributions $P_T, \widetilde{P}_T$ of length-T observation records under the common policy satisfy

$$\boxed{\|P_T - \widetilde{P}_T\|_{\mathrm{TV}} \leq \min\left(1, \frac{1}{2}\|z_0 - \widetilde{z}_0\|_C + \frac{1}{2}\sum_{t=0}^{T-1}\epsilon_t\right).}$$

The same bound holds for joint action and observation records when the common policy is randomized. No positive lower bound on individual branch probabilities is required.

Proof. Let $u_h, \widetilde{u}_h \in C$ be the unnormalized states on each history at depth t, including the common policy factors. Their masses sum to one in each model. Define $D_t = \sum_h \|u_h - \widetilde{u}_h\|_C$. At a history with selected action a, write

$$B_{a,o}u_h - \widetilde{B}_{a,o}\widetilde{u}_h = B_{a,o}(u_h - \widetilde{u}_h) + (B_{a,o} - \widetilde{B}_{a,o})\widetilde{u}_h.$$

Sum the first term over observations and use Proposition 42. When $\lambda\widetilde{u}_h > 0$, normalize only to apply the definition of ϵ_t to the second term. Its sum is bounded by $\epsilon_t\lambda\widetilde{u}_h$. If the mass is zero, strict positivity implies $\widetilde{u}_h = 0$. Summing histories yields $D_{t+1} \leq D_t + \epsilon_t$. Averaging over common policy probabilities gives the same inequality for randomized actions. Finally each word-probability difference is bounded by its state difference because $|\lambda x| \leq \|x\|_C$. Total variation is half the sum of those absolute differences. □

This bound separates total law error from conditional-state error. It can grow linearly with the horizon and eventually become uninformative. It is not a uniform-in-time approximation theorem. It also does not control the regret of a changed planning policy without additional decision-theoretic assumptions.

48. Conditioning still has an unavoidable cost

Let P, Q be probability distributions on a finite record space. A record event H might be a particular observed prefix. Conditional continuation laws are obtained by restriction to H and normalization.

Proposition 44. Sharp event-probability dependence of conditioning. If $\|P - Q\|_{\mathrm{TV}} \le \delta$, $p = P(H) > 0$ and $q = Q(H) > 0$, then

$$\boxed{\|P(\cdot \mid H) - Q(\cdot \mid H)\|_{\mathrm{TV}} \le \min(1, \delta/p).}$$

When $p > \delta$, positivity of q follows automatically. The factor $1/p$ cannot be removed in general.

Proof. Write the subprobability restrictions on H as u, v, of masses p, q. On the complementary event, the sum of absolute probability differences is at least $|p - q|$. Therefore

$$\|u - v\|_1 + |p - q| \le 2\delta.$$

Adding and subtracting v/p gives

$$\frac{1}{2}\|u/p - v/q\|_1 \le \frac{\|u - v\|_1 + |p - q|}{2p} \le \delta/p.$$

Also $|p - q| \le \delta$, establishing the positivity condition. For sharpness, put total mass p on an event with two outcomes. Let P assign all of it to the first, and let Q move mass $\delta \le p$ to the second, leaving the complement unchanged. The unconditional total variation is δ and the conditional distance is δ/p. □

Combining this proposition with Proposition 43 gives a conditional continuation bound when the evidence event has adequate probability. It does not give a coordinate-state error bound unless test probabilities control that coordinate norm. Validity at every branch and stable inference at every branch remain different properties.

49. Searching for a domain from observable operators

A cone certificate need not be imported from a hidden-state generator. With exact rational operators it can be searched for directly, then checked independently.

Proposition 45. Rational finite-certificate search. Suppose λ, B_s and z_0 are rational and satisfy the exact mass identities. There is an enumeration procedure that returns a valid finite certificate whenever a certificate of Proposition 39 exists with a rational normalized full-row-rank generator matrix V. The procedure is sound whenever it returns. No termination guarantee is asserted when no such rational certificate exists.

Proof. Enumerate all finite rational matrices V in an order that eventually includes every size and every bounded numerator and denominator range. Retain those of full row rank with $\lambda V = \mathbf{1}^\top$. For each retained matrix, the requirements $B_s V = V A_s$, $A_s \ge 0$, $V\alpha_0 = z_0$, $\alpha_0 \ge 0$ and $\mathbf{1}^\top \alpha_0 = 1$ are rational linear feasibility problems. A nonempty rational polyhedron has a rational feasible point, as follows from elimination of rational linear inequalities. These problems can therefore be decided by exact rational linear programming, or witnesses can be enumerated in a dovetailed rational search. Recheck every returned equality and sign exactly. Proposition 39 proves soundness. If a rational

generator certificate exists, its finite representation is eventually considered and its feasible witnesses are found. □

The general enumeration is a mathematical existence algorithm, not a practical complexity claim. A bounded candidate family is often the useful reference experiment. Failure within that family means only that the chosen family did not produce a certificate.

49.1 A certificate learned without hidden matrices

The observable tables of Section 38 determine the two-dimensional operators B_s and initial vector $(1, 13/24)^\top$. The new reference procedure receives only these fitted operators and that vector. It enumerates intervals with rational endpoints in $[0, 1]$ of reduced denominator at most eight, ordered by increasing width and then by endpoints. The coordinate restriction $[0, 1]$ comes from its declared test-probability meaning.

The first accepted interval is

$$\boxed{K = \{(1, q)^\top : 2/5 \le q \le 3/5\}.}$$

It uses

$$V = \begin{pmatrix} 1 & 1 \\ 2/5 & 3/5 \end{pmatrix}, \qquad \alpha_0 = \begin{pmatrix} 7/24 \\ 17/24 \end{pmatrix}, \qquad A_s = V^{-1} B_s V.$$

Every entry of every A_s is at least $1/64$. The full per-action column sums are exactly one. These rational identities certify all word probabilities of the fitted model, not only the words used in the fit. The certificate was not supplied to the search, and no hidden generator is imported by the standalone example.

The interval is the narrowest accepted interval **in this specified finite grid**, not a claim of the smallest invariant interval among all real endpoints. It also need not contain every predictive state associated with an arbitrary initialization of the old hidden model. It contains the declared initial state and preserves every subsequent branch of this fitted realization, which is the requirement being certified.

The positive slack gives a robust statement: any perturbation of the lifted entries by at most $1/64$, coupled to retain the exact per-action column sums, remains nonnegative. Interior perturbations of size $1/128$ are used in the worked comparison. Repeated composition is valid for all horizons, while Proposition 43 bounds the changed observation law for each fixed horizon.

50. Selection and noise do not invalidate an exact witness

A data-dependent cone proposal is not itself a statistical confidence statement. Once the matrices and witness are fixed, its exact invariance identities hold or fail independently of how it was proposed. What remains statistical is whether the chosen uncertainty set covers the generating model.

For a finite-sample interpretation, first obtain a valid uncertainty set for a fixed identifiable model class. Proposition 35 supplies one route under independent reset experiments and declared rank coverage. Proposition 33 translates those probability bounds into a fixed-core operator bound. A validity witness

can then be sought for a fitted member or robustly for every member. Only the latter identifies a common true-model domain on the coverage event.

The admissible family must enforce mass conservation exactly. A free box around unconstrained matrices generally contains nonnormalized families. Discarding those families is legitimate only because the generating process is already assumed to obey the exact mass law. Other cuts on a confidence set need their own justification.

51. Why finite predictive rank need not mean finitely many positive states

The following controlled rational example makes the boundary explicit. It is a worked adaptation of the irrational-rotation obstruction in [R12, Section 7]. Related distinctions among finite linear, positive and quantum realizations are discussed in [R14]. Only the elementary cone argument below is used.

Let $z = (t, x, y)^\top$, $\lambda = (1, 0, 0)$, $z_0 = (1, 1, 0)^\top$, and

$$R = \begin{pmatrix} 3/5 & -4/5 \\ 4/5 & 3/5 \end{pmatrix}, \qquad T = \operatorname{diag}(1, R).$$

There are two actions and two observations. A rotate action has branches

$$B_{\mathrm{r},0} = T, \qquad B_{\mathrm{r},1} = 0.$$

A probe action has branches

$$B_{\mathrm{p},0} = z_0(1/2, 1/2, 0), \qquad B_{\mathrm{p},1} = z_0(1/2, -1/2, 0).$$

The rotate outcome zero is certain. The probe has probabilities $(t + x)/2$ and $(t - x)/2$ and resets every nonzero branch to z_0 after normalization. Both actions conserve the mass t.

Proposition 46. A rational rank-three model with no polyhedral probability-domain certificate. The Lorentz cone

$$C_{\mathrm{L}} = \{(t, x, y) : t \geq \sqrt{x^2 + y^2}\}$$

is probability valid for these operators. It is the canonical all-test cone. Every closed convex invariant cone containing z_0 and admitting the nonnegative mass interpretation equals C_{L}. In particular no finite normalized generator certificate exists for this exact model. Its reachable and observable predictive dimension is three.

Proof. Rotation preserves t and $x^2 + y^2$. On C_{L} both probe effects are nonnegative and each probe image is a nonnegative multiple of z_0. Thus the displayed cone is invariant and valid.

Write the rotation angle as θ. It is not a rational multiple of 2π. Otherwise $\zeta = e^{i\theta}$ would be a root of unity, so $\zeta + \zeta^{-1} = 6/5$ would be a rational algebraic integer. Every rational algebraic integer is an integer, by the rational-root argument for a monic integer polynomial, contradicting $6/5 \notin \mathbb{Z}$. The forward powers of an irrational circle rotation are dense. Compactness supplies powers approaching the identity, so their closure contains the inverse rotation and is a closed subgroup. An infinite closed

subgroup has arbitrarily small positive angles. Integer multiples of such angles approximate every point on the circle, so that subgroup is the entire circle.

A state passing every test must pass a probe after any number k of rotations. Hence

$$t \pm e_1^\top R^k (x, y)^\top \geq 0 \quad \text{for every } k \geq 0.$$

Density gives $t \geq \sqrt{x^2 + y^2}$. The reverse inclusion was already proved, so the canonical cone is exactly C_{L}.

Any invariant cone with nonnegative mass lies inside this canonical cone, by Proposition 38. If it is closed, convex and contains z_0, it contains the orbit $T^k z_0$ and the closed convex hull of that orbit. On $t = 1$, this hull is the full unit disk. Conical scaling gives all of C_{L}. The unit disk has infinitely many extreme points, so it is not the convex hull of finitely many points.

Finally the reachable columns $z_0, Tz_0, T^2 z_0$ are independent. The mass row and probe rows before zero and one rotations span the dual space because the rotation has nonzero sine. Reachability and observability establish predictive dimension three. □

This is not a failure of finite-dimensional prediction. The exact model has only three coordinates and a simple quadratic cone. It is a failure of the narrower requirement that every valid model admit a finite *polyhedral* domain. A certificate language limited to finitely many positive generators would exclude a legitimate low-rank signal model.

No general impossibility of all finite certificates follows. This very example has a finite quadratic certificate. The obstruction concerns polyhedral certificates, not invariant cones described by other mathematics.

52. A finite approximation with an explicit price

A curved probability domain can sometimes be replaced by a finite one after changing the dynamics. That change must be quantified, not described as exact refinement.

Let P_n be the regular n-gon inscribed in the unit disk, with a vertex at $(1, 0)$ and $n \geq 3$. Replace the rotate branch by

$$T_\alpha = \operatorname{diag}(1, \alpha R), \qquad 0 \leq \alpha < 1,$$

while retaining the same probe branches. The spatial coordinates are attenuated, but total probability mass remains unchanged.

Proposition 47. Polyhedral approximation of the rotation model. If

$$\alpha \leq \cos(\pi/n),$$

then the cone over P_n is invariant under all modified operators and contains z_0. Consequently the modified model admits a finite positive lift. The original and modified models are both valid on the larger cone C_{L}, and for every common policy their length-T laws satisfy

$$\boxed{\|P_T - P_T^{(\alpha)}\|_{\mathrm{TV}} \leq \min(1, T(1-\alpha)/2).}$$

The angle-independent sufficient condition can be improved for a fixed polygon and rotation. For the rational diamond $|x| + |y| \leq t$, the present rotation is invariant after attenuation whenever $\alpha \leq 5/7$. Thus $\alpha = 2/3$ gives an entirely rational four-generator certificate.

Proof. The regular polygon contains the disk of radius $\cos(\pi/n)$ and lies in the unit disk. Every $\alpha R v$ with $v \in P_n$ has Euclidean norm at most α, hence lies in P_n under the stated condition. The probe effects remain nonnegative because P_n lies in the disk, and their normalized images are the retained vertex $(1, 0)$.

The base norm of C_{L} is

$$\|(t, x, y)\|_{C_{\mathrm{L}}} = \max(|t|, \sqrt{x^2 + y^2}).$$

Indeed the convex hull of the unit disks in the planes $t = 1$ and $t = -1$ is the cylinder with this unit ball. The rotate-branch discrepancy on a normalized state is at most $1 - \alpha$ in this norm. Probe discrepancies are zero. Proposition 43 gives the law bound. For the diamond, the induced ℓ^1 norm of R is $7/5$, so $\alpha \|R\|_1 \leq 1$ proves invariance. □

For a target attenuation loss $0 < \eta \leq 1$, choosing $\alpha = 1 - \eta$ and sufficiently many polygon vertices satisfies the certificate. The inequality $1 - \cos u \leq u^2/2$ shows that $n \geq \pi/\sqrt{2\eta}$ is sufficient, with an integer ceiling and $n \geq 3$. This is a constructive upper bound on certificate size, not a lower bound or a proof of optimality.

The distinction is exact. At $\alpha = 1$ there is no finite polyhedral certificate for this model. At $\alpha < 1$ a sufficiently fine polygon works, at the cost of a declared change in the observation law. The reference figures show that tradeoff rather than labeling the approximation an identical model.

53. Repeated proposal and verification with a confidence budget

Model development may choose another core, propose another domain or collect another independent batch after an earlier certificate is withheld. Statistical coverage must survive that selection process.

Proposition 48. Sequential confidence accounting for certified proposals. At stage n, let the proposal be measurable from previous data, and let a verification procedure return a set $\mathcal{U}_n$ for the true model in the declared identifiable coordinate system. Assume conditional coverage

$$\Pr(\text{true model} \notin \mathcal{U}_n \mid \text{previous data}) \le \delta_n, \qquad \sum_{n\ge 1} \delta_n \le \delta.$$

Then with probability at least $1-\delta$, every returned uncertainty set covers its true model. On this event, any data-selected domain with an exactly verified robust certificate for **all** members of its corresponding $\mathcal{U}_n$ is also valid for the true model, provided the true initial state is covered by the certified base. The conclusion remains valid at any stopping time chosen from the recorded results.

Proof. Take expectations of the conditional failure bounds, then apply the countable union bound. Off the union of failure events, all coverages hold simultaneously. Robust invariance over an entire set then applies to its true member by deterministic implication, independently of how its domain was chosen. A stopped stage is one of the already covered stages. □

For example, $\delta_n = \delta 2^{-n}$ spends a finite total confidence budget. Fresh reset trials for a proposal fixed by past data can use Proposition 35 conditionally, with this stage budget. An exact witness needs no statistical correction for how many candidate cones were tried. The coverage statement does need correction for repeatedly constructing confidence sets or inspecting post-selected data.

The proposition does not infer a global rank bound, validate an unknown reset operation, or transform dependent observations into independent trials. Those are assumptions of the coverage procedure. A known coordinate transformation can transport a confidence set exactly. An estimated transformation requires the subspace and conditioning controls already separated in Part III.

54. Integrated certificate-carrying realization

Proposition 49. Conditional learning with an all-horizon probability certificate. Suppose the observable identification assumptions of Proposition 37 hold for a declared finite-rank controlled process. Suppose a justified confidence procedure produces an operator and initial-state set containing the true realization on an event of probability at least $1-\delta$. Assume the per-action mass constraints are exact.

If a fitted realization admits a checked finite witness of Proposition 39, it is an internally valid stochastic model at every horizon. If the witness is robust over the whole confidence set and covers its initial states, then on the coverage event the same probability domain is valid for both the fitted and true models. When their summed base-norm instrument discrepancies are at most ϵ_t and their initial discrepancy is at most e_0, every common policy has the bound

$$\|P_T - \widehat{P}_T\|_{\mathrm{TV}} \le \min\left(1, \frac{e_0 + \sum_{t<T} \epsilon_t}{2}\right).$$

Conditioning on a positive-probability event then obeys Proposition 44. Exact coordinate transport preserves the certificate. A signed packet decomposition preserves it after complete synthesis but does not turn individual packets into probability states without Proposition 30's extra positivity requirement.

Proof. Observable identification supplies the specified realization and the confidence procedure supplies the stated event, not an inferred event of its own. Proposition 39 supplies internal validity for the fitted model. Proposition 40 extends the same domain to every member of the uncertainty set, hence to its true member on that event. Propositions 42 and 43 give the total-variation bound, and Proposition 44 gives the conditional bound. For a coordinate change T, use $V' = TV$, $B'_s = TB_sT^{-1}$ and $\lambda' = \lambda T^{-1}$. Then $B'_sV' = V'A_s$ and $\lambda'V' = \mathbf{1}^\top$ with the unchanged nonnegative witnesses. Full packet synthesis is the identity. □

This closes a concrete foundational gap. A learner can now return not just a matrix family but a verifiable domain in which that family has probability meaning. It can attach a robust uncertainty statement when one is justified, or return an internally valid approximation with its correction budget explicitly recorded.

It does not turn a successful fit into unrestricted knowledge of the world. The model class, observation protocol, rank coverage, exact mass law, domain witness and approximation budget remain visible parts of the result. Failure of a finite search is a withheld certificate, not a proof of invalidity. The nonpolyhedral example explains why this distinction is necessary even for very small exact models.

55. Foundation status after this block

The cumulative theory now contains 49 propositions. This block adds a finite route from observable operators to probability-domain certificates, robust invariance, certified correction, pre-normalization stability of full observation laws, and a precise boundary for finite positive lifts. It also supplies an observable-only interval search and a rational curved-domain counterexample with a quantified polyhedral approximation.

These are not twelve claims of historical novelty. The positive-realization and rotation ingredients have explicit antecedents [R12, R14]. Convex invariance and ordered-space contraction are established mathematics [R13, R15]. The value of the block is the connected set of requirements, constructive witnesses, exact counterexamples and uncertainty accounting within the large signal model.

The next open tasks are no longer simply "make the probabilities valid." They include efficient synthesis of useful nonpolyhedral domains, informative confidence regions in poorly conditioned predictive coordinates, dependent-data and adaptive-intervention learning, continuous observation measures, and nonlinear representations whose discovery and approximation costs are controlled. A general practical solver for all such domains is not supplied here.

The production runtime remains unchanged. The new code is an exact reference checker, a bounded rational domain search and finite verification experiments. General validity comes from the stated proof and its checked witness, not from a large number of sampled trajectories. The release ledger separates unchanged predecessor source, new results, finite arithmetic evidence, expected rejections and unexecuted work.

References and source lineage

[**K1**] Komorebi P. *Projector-Resolved Large Algebraic Signal Model*. Historical 38-page capstone preserved in `Komorebi_PR_LASM_2.3.0_Source.zip`, `paper/Projector_Resolved_Large_Algebraic_Signal_Model.tex`. In particular, Sections 2–7 and the categorical-atlas distinction. Historical experimental claims are not used here.

[**K2**] Komorebi P. *Projector-Resolved Seed Calculus VI*, corrected presentation in `Paper_VI_With_Visual_Appendix.pdf`, Section 8, Theorem 8.4 and the adjacent coarsening criterion and one-packet boundary. This supplement restates and proves the finite result it uses. It does not claim a fresh Lean audit of that paper.

[**K3**] Komorebi P. *PR-LASM 2.3.0 Release Audit*, 9 October 2026, and the accompanying source. The source inspected includes `exact_atlas.py`, `lasm_packetizer.py`, `lasm_model.py`, `docs/RELEASE_GUIDE.md` and `docs/claim-boundary.md`.

[**R1**] Ronald R. Coifman and Mladen Victor Wickerhauser. "Entropy-based algorithms for best basis selection." *IEEE Transactions on Information Theory* 38(2), 713–718, 1992. DOI: 10.1109/18.119732. Source checked: the authors' Washington University research record. The finite additive tree optimization is classical background.

[**R2**] MIT OpenCourseWare. *6.241J Dynamic Systems and Control*, Chapter 24, "Observability," Spring 2011, especially Section 24.5 on the largest invariant subspace contained in the observation kernel. Source checked: the MIT course notes, PDF page 6. Used to identify the classical control-theoretic antecedent of Proposition 8.

[**R3**] Michael Woodward, Yifeng Tian, Arvind Mohan, Yen Ting Lin, Christoph Hader, Hermann Fasel, Misha Chertkov and Daniel Livescu. *Data-Driven Mori-Zwanzig: Approaching a Reduced Order Model for Hypersonic Boundary Layer Transition*, arXiv:2301.07203, version 1, 2023, introduction and description of resolved, memory and unresolved contributions. Cited for the established projection-memory context, not as evidence for PR-LASM's performance or for any application discussed in that paper. Proposition 10 is proved directly by finite linear recurrence expansion here.

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[**R5**] Michael L. Littman, Richard S. Sutton and Satinder Singh. "Predictive Representations of State." *Advances in Neural Information Processing Systems* 14, NIPS 2001. The paper connects state with predictions of future observations conditional on actions. Cited for conceptual lineage, not as a proof of the finite delay theorem or of this implementation. The original paper and proceedings record are the primary sources. The scanned paper carries all three author names.

[**R6**] Steven L. Brunton, Bingni W. Brunton, Joshua L. Proctor and J. Nathan Kutz. "Koopman Invariant Subspaces and Finite Linear Representations of Nonlinear Dynamical Systems for Control." *PLOS ONE* 11(2), e0150171, 2016. DOI: 10.1371/journal.pone.0150171. Used for the established role and limitations of finite invariant observable families in nonlinear modeling. No general nonlinear feature-discovery guarantee is imported.

[**R7**] Daniel Hsu, Sham M. Kakade and Tong Zhang. "A Spectral Algorithm for Learning Hidden Markov Models." *Journal of Computer and System Sciences* 78(5), 1460–1480, 2012. Primary author manuscript inspected: `hmm-jcss.pdf`, hosted by Daniel Hsu at Columbia University, especially its abstract and rank assumptions. The result illustrates observable-operator learning under explicit spectral hypotheses. Its statistical guarantees are not claimed for PR-LASM.

[**R8**] Stephen Boyd, Laurent El Ghaoui, Eric Feron and Venkataramanan Balakrishnan. *Linear Matrix Inequalities in System and Control Theory*. SIAM Studies in Applied Mathematics 15, 1994. The authors' Stanford publication page identifies the book and provides the primary text. Used to identify the classical common quadratic Lyapunov and matrix-inequality setting. The discrete contraction implication used in Section 19 is proved directly there.

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Journal of Robotics Research 30(7), 954–966, 2011. Used for the established PSR lineage, not as an assertion that this supplement implements that paper's complete algorithm or planning guarantees. Source: `https://www.cs.cmu.edu/~ggordon/boots-siddiqi-gordon-closing-loop-psrs.pdf`.

[**R10**] Han Zhao and Pascal Poupart. "A Sober Look at Spectral Learning." *ICML 2014 Workshop on Spectral Learning*, 2014. Primary author-hosted manuscript inspected, including its discussion of negative probabilities in spectral estimates. Used only to identify that practical failure mode. The cone and packet statements in this supplement are proved independently. Source: `https://hanzhaoml.github.io/papers/ICML2014/spectral-main.pdf`.

[**R11**] Wassily Hoeffding. "Probability Inequalities for Sums of Bounded Random Variables." *Journal of the American Statistical Association* 58(301), 13–30, 1963. DOI: 10.1080/01621459.1963.10500830. Publisher record checked. Proposition 35 supplies a self-contained Bernoulli moment-generating-function proof for the protocol it uses. Source: `https://www.tandfonline.com/doi/abs/10.1080/01621459.1963.10500830`.

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[**R12**] Herbert Jaeger. "Observable operator models for discrete stochastic time series." *Neural Computation* 12(6), 1371–1398, 2000. Author-hosted manuscript with the August 2017 erratum. Section 7 gives the polyhedral-cone connection to finite positive realizations and an irrational-rotation obstruction. The controlled rational example in Section 51 is an independently worked adaptation of that established obstruction, not a claim of discovering it. The corrected manuscript and its geometric argument were inspected. Source: `https://www.ai.rug.nl/minds/uploads/oom_neco00.pdf`.

[**R13**] Zoltán Horváth, Yunfei Song and Tamás Terlaky. *A Novel Unified Approach to Invariance Conditions for a Linear Dynamical System.* arXiv:1405.5167, version 2, 5 July 2016. Primary author manuscript, particularly the polyhedral-invariance discussion and use of Farkas' lemma. Used for the classical invariant-cone and optimization setting. The generator and robust-vertex statements used here are proved directly. Source: `https://arxiv.org/html/1405.5167v2`.

[**R14**] Marco Fanizza, Josep Lumbreras and Andreas Winter. *Quantum theory in finite dimension cannot explain every general process with finite memory.* arXiv:2209.11225, version 2, 5 May 2023. Primary manuscript, especially its discussion of finite linear realizations, positive realizations and polyhedral cones. Cited for the distinction between these representation classes. No quantum implementation or quantum advantage is claimed for PR-LASM. The present block uses a real three-dimensional cone construction proved in Section 51. Source: `https://arxiv.org/html/2209.11225v2`.

[**R15**] Anna Jenčová. *Base norms and discrimination of generalized quantum channels.* arXiv:1308.4030, version 1, 19 August 2013. Primary manuscript, Section 1.2, printed page 5, on finite-dimensional proper cones, normalized bases and base norms. That ordered-vector-space definition is used in Section 46. The classical instrument contraction and observation-tree bound needed here are derived in the text, without importing quantum-channel assumptions. Source: `https://arxiv.org/pdf/1308.4030`.

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Operator-Symbolic Language

Proof-carrying interfaces

Traditional paper

Komorebi P.

Research edition 1.0 / OSL-Core 0.5 / Standardized 8 October 2026

Reading scope

Finalized object. A consolidated language manuscript, a frozen OSL-Core 0.5 reference implementation, the finite Discovery Bridge, and a checked finite-state interface. The edition number is not a new runtime version. Neural proposals, formal statements, operational permissions, and historical evidence retain different types and different assurance levels.

Abstract

A machine-oriented language should specify what an expression does, what information it depends on, and under which conditions another system may use it. Operator-Symbolic Language (OSL) develops this principle as an explicit intermediate representation for heterogeneous computational agents. Its geometric layer uses typed state spaces, operators, and structure-preserving transport. Its operational layer distinguishes access, provenance, endorsement, proof resources, memory, planning, bounded capabilities, transactions, and finite communication protocols. Its evidence layer separates receipt integrity, source-bound semantic checking, interpreter replay, and complete finite enumeration.

This edition consolidates the earlier OSL v0.5 monograph and its OSL-Core 0.3 implementation with Core 0.4 receipt hardening, Core 0.5 semantic independence, and the finite Discovery Bridge. Fresh execution reproduces the Core 0.5 corpus exactly: 66 programs, 1,825 receipts, and all recorded acceptance or rejection outcomes. Its 168-test suite passes. The Discovery Bridge's 20 tests and three generated programs also pass against the current core. A new, deliberately finite interface constructs and checks continuation-preserving quotients, serializes semantic checkpoints, and routes accepted or unverified candidates through the actual OSL runtime.

The result is an executable research language, not a general-purpose proof assistant, a production security boundary, or a demonstration that a neural model has spontaneously learned OSL. The paper gives the mathematical contracts needed for the language, identifies the portions implemented and checked, and preserves the failed or absent learned-language evidence. The stable interface is explicit semantics; latent computation remains a separately tested producer of candidate representations.

How to read the evidence. "Proved" means a mathematical derivation under stated assumptions. "Replayed" means a computation completed in this edition. "Retained" means a historical result was recovered and its source identified. Neither unit-test counts nor archive hashes turn a retained experiment into an independent scientific replication. The original 88-page monograph and all recovered OSL experimental lineages remain available in the accompanying source archives.

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1 The language problem

The phrase *machine-native language* can denote three different objects: a learned latent representation, a notation optimized for machines, or an intermediate representation with an independently specified semantics. OSL addresses the third. It is intended to make heterogeneous internal computations communicate through explicit operations, claims, and obligations. It does not require every producer to use the same coordinates, nor does it require a producer's internal state to resemble a string in the external language.

Consider an agent that receives a vector from another agent. Knowing the vector's dimensions is not enough. The recipient may need to know which operation produced it, which coordinate system it inhabits, which observations supported it, whether a transformation preserved its meaning, and whether it may be used for a particular action. A fluent explanation is not a substitute for those relations. OSL makes them separate parts of the program state.

The proposal has established neighbors. Abstract interpretation formalizes sound abstraction of computation; proof-carrying code separates untrusted production from policy-relative checking; session types make communication protocols part of a language's discipline. OSL draws on these ideas rather than claiming to replace them. Its specific research object is their integration with operator transport, provenance-sensitive memory, and explicit neural admission boundaries. [CC77, N97, HYC08]

The present release should also be distinguished from an argument about conscious machines. No linguistic syntax, proof receipt, or recurrent state establishes experience. The operational question is narrower: can one system offer a reusable transformation or claim that another system can interpret and check without trusting the producer's narrative?

1.1 What has been finalized

The manuscript fixes terminology and scope, identifies one current executable core, and binds its claims to reproducible artifacts. OSL-Core remains version 0.5 with receipt schema `osl.core.receipt.v2`. The language's broader mathematical specification remains available in the preserved v0.5 monograph. The Discovery Bridge remains a separately versioned bounded synthesizer. Experimental packages numbered v0.6 or v0.3.1 are not later editions of the language merely because one number is larger. [O03, O05, ODB]

This is consequently a *research-edition freeze*. Future runtime changes should receive their own version, source manifest, compatibility statement, and new evidence. An unfinished historical Core 0.6 development mentioned in prior working records is not promoted into a released implementation here: no complete authoritative successor was recovered.

2 Objects, meanings, and realizations

2.1 Three distinct interfaces

We use the following separation throughout:

$$\text{latent producer} \longrightarrow \text{typed proposal} \longrightarrow \text{checked operational consumer.}$$

The latent producer can be a neural network, a search procedure, a human, or a fixed fixture. The typed proposal names a carrier, operation, scope, and evidence. The consumer decides what the proposal

permits within an explicit language semantics. Success at one arrow does not establish success at the other.

Definition 2.1 (Realization)**.** A realization of an operation vocabulary $\mathcal{G}$ consists of a state space X, maps $\rho_X(g) : X \to X$ for $g \in \mathcal{G}$, and a declared observation map $O_X : X \to Y$. For typed operations, each generator instead has an explicit source and target carrier, and only type-compatible words may be composed.

The observation map is part of the claim. Preserving a categorical answer is weaker than preserving all output probabilities, and preserving either can be weaker than preserving future behavior after another operation. The companion Neuralese manuscript develops that distinction in detail. An OSL claim must not silently change its observation interface after evaluation.

2.2 Operational packets are not geometric projectors

An OSL packet is a record with content and metadata. A Komorebi geometric packet is a component of a decomposition, often represented by a projector or a frame. These concepts may be linked by an application, but they are not definitionally identical.

A useful abstract operational packet is

$$p = (\text{id}, \text{content}, \text{origin}, \text{access}, \text{status}, \text{scope}, \text{evidence}).$$

The implementation has a more detailed schema. The content may describe a vector or an operation, but it may also be an ordinary proposition. The provenance records how the content entered the system. An access relation states which agents may inspect it. Endorsement and authorization are separate relations. A packet's numerical confidence does not itself grant either.

2.3 Meaning is relative to a declared task family

Two programs can have the same output on today's input and still differ on another input, on their receipts, or on their permitted effects. We therefore distinguish state equivalence, observation equivalence, contextual equivalence, and trace equivalence. A discovery algorithm that quotients programs by their final finite-state transformation is not thereby entitled to erase provenance or proof consumption from a security-sensitive trace. The intended quotient must be named.

3 Operator transport and compositional semantics

3.1 Exact transport

Let $A_g : X \to X$, $B_g : Y \to Y$, and $T : X \to Y$ be linear maps. We use column vectors and execute the rightmost map first. The compatibility equation is

$$B_g T = T A_g. \tag{1}$$

When this equation holds, translating a state and applying the target operator agrees with applying the source operator and then translating. It is a statement about operations, not merely geometric resemblance between embeddings.

Proposition 3.1 (Compositional transport)**.** *If* (1) *holds for every generator, then for every finite word* $w = g_1 \cdots g_k$*, with* $A_w = A_{g_k} \cdots A_{g_1}$ *and analogously for* B_w*, one has* $B_w T = T A_w$*.*

Proof. The empty word gives $T = T$. If the identity holds for w, then $B_g B_w T = B_g T A_w = T A_g A_w$. Induction on word length proves the result. □

No invertibility of T is needed. When T is invertible, the target operators are conjugates $B_g = T A_g T^{-1}$. When it is not, the equation may describe an abstraction or information-losing projection. That projection must preserve the observations relevant to the advertised application; the intertwining equation alone does not specify which observations matter.

3.2 Approximate transport

Proposition 3.2 (Finite-word error budget). *Suppose $E_i = B_{g_i} T - T A_{g_i}$, with $\|E_i\| \le \varepsilon_i$. Then*

$$B_w T - T A_w = \sum_{i=1}^{k} B_{g_k} \cdots B_{g_{i+1}} E_i A_{g_{i-1}} \cdots A_{g_1}.$$

Consequently, if every $\|A_g\| \le a$ and $\|B_g\| \le b$,

$$\|B_w T - T A_w\| \le \sum_{i=1}^{k} b^{k-i} \varepsilon_i a^{i-1}.$$

Proof. Insert the intermediate products obtained by moving T past one operator at a time. Consecutive products cancel, leaving the displayed sum. Submultiplicativity and the triangle inequality give the norm bound. □

For $a = b = 1$ and $\varepsilon_i \le \varepsilon$, the bound is $k\varepsilon$, not ε. A small one-step residual does not establish unlimited reliable composition. If the operators amplify errors, the budget can grow faster. Practical certificates must record their admissible operation family, norms, and composition horizon rather than advertising an unqualified transport accuracy.

3.3 Projectors and changes of chart

For an invertible T, define $P' = TPT^{-1}$. If $P^2 = P$, then

$$(P')^2 = TP^2T^{-1} = P'.$$

Idempotence is preserved by similarity. Orthogonality with respect to the Euclidean inner product need not be. If a metric G is transported to $G' = T^{-\mathsf{T}} G T^{-1}$, metric-relative identities can be compared consistently. These elementary distinctions matter in a language that allows an operator to be called a projector. A name or an approximate matrix test must not conceal the metric or tolerance under which the predicate is evaluated.

The exact derivations in this section are standard linear algebra. Their role is to supply explicit contracts for OSL, not to introduce a new theory of intertwiners.

4 The operational state machine

4.1 State and judgment forms

A schematic runtime state is

$$S = (\Gamma, V, \mathcal{O}, \mathcal{P}, \mathcal{Q}, \mathcal{M}, \mathcal{C}, \mathcal{A}, \mathcal{T}),$$

where Γ contains declarations, V vectors, $\mathcal{O}$ operators, $\mathcal{P}$ packets, $\mathcal{Q}$ proof resources, $\mathcal{M}$ memories, $\mathcal{C}$ capabilities, $\mathcal{A}$ planned actions, and $\mathcal{T}$ contracts, sessions, and transaction state. The concrete runtime also maintains metadata and audit structures. This tuple is an explanatory grouping, not a replacement wire schema.

A typing judgment $\Gamma \vdash e : \tau$ constrains an expression's carrier or kind. A proof-resource judgment constrains which already established obligations may be consumed. An execution judgment

$$\langle c, S, L \rangle \longrightarrow \langle S', L' \rangle$$

relates a source statement, operational state, and audit ledger to their successors. A statement that rejects still has an outcome and, within the implemented lifecycle, an auditable terminal explanation. Rejection is not equivalent to absence of a trace.

4.2 Access is not ownership

A received observation is evidence that someone else observed something, not a firsthand observation by its recipient. The implemented language preserves the distinction between observed, communicated, inferred, simulated, injected, and reconstructed content. [O03, O05]

Proposition 4.1 (Provenance preservation for a communication rule)**.** *Suppose a packet's origin record is immutable under a communication operation, and communication adds a recipient access record plus a communicated provenance step. Any finite sequence of such operations preserves the original origin record. No recipient acquires a firsthand observation solely by that sequence.*

Proof. Each permitted step leaves the origin component unchanged. Induction preserves that component over the entire sequence. The only new event constructor is communication; none of the steps introduces a firsthand-observation event for a recipient. □

This is a theorem about the stated rule, not an external guarantee that a source really observed the world. A false input observation can be faithfully propagated. OSL separates that world-grounding problem from the internal consistency of provenance.

4.3 Proof resources and permissions

Persistent, affine, and linear proof resources have different use disciplines. A persistent resource may be reused, an affine resource used at most once, and a linear resource used exactly once in the relevant completed scope. These are operational restrictions on evidence use, not statements that truth itself expires.

Proposition 4.2 (Finite linear-use accounting)**.** *Assume a linear resource starts with use count one, every legal consumption requires a positive count and decrements it by one, and successful scope completion requires count zero. Then a successful scope consumes the resource exactly once.*

Proof. A second consumption is impossible because the first makes the count zero. Zero consumptions leave count one and prevent successful completion. Exactly one consumption is therefore necessary and sufficient for this accounting condition. □

A capability is a different object. It grants a bounded kind of action to a named agent, optionally with a use limit and deadline. A proof about content cannot substitute for the capability required to execute an effect. In this reference implementation effects and capabilities are symbolic. They do not impose an operating-system sandbox or authorize a real external tool.

4.4 Rollback and evidence

Separate mutable operational state from an append-only audit ledger. A rollback restores the selected operational snapshot and appends a rollback event, but does not restore the ledger to its old prefix. Otherwise a failed speculative action could erase the evidence that it was attempted.

Proposition 4.3 (Rollback does not imply historical erasure)**.** *Under the preceding transition rule, restoring an operational snapshot preserves all earlier audit entries and appends the rollback event. Operational equality with an earlier state therefore does not imply equality of the two execution histories.*

Proof. The post-rollback operational component equals the saved snapshot by definition. The ledger is extended rather than replaced. The new ledger consequently contains a strict extension of the earlier prefix whenever a rollback event is appended. □

This does not undo physical effects. A program that has already acted on an external environment requires an application-level compensation or recovery protocol. The reference language's rollback concerns its declared internal projection.

5 Receipt integrity and independent checking

5.1 Four assurance questions

For a receipt-bearing execution, ask four separate questions. Are the bytes intact? Does the trace bind the exact source? Are the recorded transitions semantically valid within a checked domain? Does executing the implementation again reproduce the result? Core 0.5 exposes separate answers rather than counting same-runtime replay as independent semantic evidence. [O05]

A digest commits to a particular source or record, conditional on the hashing assumptions. It does not establish the source's truth, authenticate a principal, or prove that a claimed transition occurred. The semantic checker must connect operands, prior state, effects, and final projections. An externally retained chain tip is needed to detect removal of a suffix from an otherwise valid unanchored history.

5.2 The current checker stack

The components below answer different assurance questions. A record may pass one layer and still require separate evidence in another.

Component	Checked subject	Remaining dependency
Receipt-hardened base	Source lifecycle, packet and proof state, capabilities, actions, protected rollback	Primary parser and static checker remain shared.
Numeric component	Spaces, dimensions, matrices, applications, transports, numerical predicates	Finite floating-point profile and supported grammar.
Contract component	Five implemented templates, obligations, deadlines, final checks	Not arbitrary temporal logic.
Session component	Finite graph checks and synchronous state/history transitions	Not asynchronous distributed liveness.
Metadata component	Secret commitments, records, assertion counts, show bindings	Does not prove all assertion predicates.
Intrinsic component	The exact prior prefix checked by `verify_receipts`	Chain integrity, not unrelated semantic truth.
Supplementary replay	Receipt and final-state reproduction	Same interpreter family, not an independent semantics.

The numerical, contract, and session fragments have separate bounded parsers. The metadata and intrinsic paths share more of the primary frontend. "Independent" is therefore a scoped implementation property, not a claim that every component has a disjoint trusted computing base.

5.3 Unclosed runtime obligations

The preserved Core 0.5 specification explicitly leaves several obligations open: the general truth of all fourteen assertion forms; exact reconstruction of every complete cross-domain transaction digest; independent reconstruction of every rejection cause; exact diagnostic event-count reconstruction; and removal of all shared frontend dependencies. Python, NumPy, canonical serialization, hashing, and the host remain trusted. These are not silently discharged by this edition. [O05]

A sound production claim would require a stronger semantic correspondence theorem or an equivalently explicit assurance argument. The finite tests establish that the tested implementation handles the included cases, not that every syntactically valid program satisfies all intended invariants. The older 47,958-transition metatheory concerns declared finite abstract models, not an exhaustive traversal of the Python interpreter's state space. [O03]

6 Bounded discovery and semantic admission

The Discovery Bridge supplies a missing distinction between *a well-formed proposal* and *an identified operator*. It accepts a complete finite world W, a primitive vocabulary G_W, a depth bound d, demonstrations D, and explicitly claimed laws. All of these belong to the certificate's scope. [ODB]

6.1 Transformation classes

Let S_W be the full finite legal state set and

$$\mathcal{P}_{W,d} = \bigcup_{j=0}^{d} G_W^j, \qquad \sigma_W(p) = \big(p(s)\big)_{s \in S_W}.$$

Programs are identified when their complete state signatures agree. For a demonstration set $D = \{(s_i, t_i)\}$, define

$$\mathcal{C}(D) = \{[p] : p \in \mathcal{P}_{W,d},\ p(s_i) = t_i \text{ for all } i\}.$$

The implementation enumerates this bounded universe, quotients equivalent spellings, and tests the advertised laws on every state in S_W. It admits only a unique derived transformation that is neither the identity nor an existing primitive and whose listed laws pass.

Proposition 6.1 (Scope-relative identification)**.** *Complete enumeration and exact evaluation identify one transformation precisely when $|\mathcal{C}(D)| = 1$. If $|\mathcal{C}(D)| > 1$, two compatible classes have a separating state in S_W. Adding the intended output at that state removes at least one of those classes.*

Proof. Distinct equivalence classes have unequal full signatures. At least one legal input therefore yields unequal outputs. A demonstration choosing one output is inconsistent with a class yielding the other. The uniqueness assertion follows directly from the definition of $\mathcal{C}(D)$. □

The proposition says nothing about programs outside the depth bound or operators outside the vocabulary. "New" means new relative to the declared primitive set. It is not a claim of mathematical novelty or autonomous concept invention.

6.2 The five possible outcomes

Outcome	Meaning
`ADMITTED`	One derived transformation fits and every listed law holds.
`AMBIGUOUS`	Several transformations fit; retain alternatives and a separating input.
`UNSATISFIABLE`	No candidate fits, or demonstrations conflict.
`REJECTED_FALSE_LAW`	The identified transformation violates an advertised relation.
`REJECTED_NONNOVEL`	The transformation is equivalent to identity or a primitive.

Ambiguity is an informative result, not a failed attempt to force admission. The bridge's verifier recomputes the candidate decision and the registry replays the embedded proposal. Merely supplying a content digest cannot bypass that semantic reconstruction.

6.3 Why two equivalent programs can still require separate traces

In the four-state flip example, `flip_left; flip_right` and `flip_right; flip_left` have the same full state transformation. A transformation library should identify them. An audit ledger may nevertheless distinguish their execution order, resource use, or intermediate observations. Quotienting the first interface does not authorize quotienting the second. A typed bridge must state which equivalence it certifies.

7 Worked OSL programs

The following programs are taken from the recovered finite corpus. They are executable reference syntax, not pseudocode. The complete sources and their fresh receipts are bundled. [O05]

7.1 A projector transported between coordinates

```
module coordinate_transport
space H dim 2
operator P : H -> H kind projector matrix [[1,0],[0,0]]
operator T : H -> H kind linear matrix [[0,-1],[1,0]]
transport P via T as P_rot
check_projector P_rot tolerance 1e-12
vector x in H values [0.0,3.0]
apply y = P_rot(x)
assert vector y approx [0.0,3.0] tolerance 1e-12
```

Here P selects the first coordinate and T rotates the chart. The transported projector selects the second coordinate. The program checks the transported predicate and applies it to a vector in that component. It does not infer a correspondence between two learned models; the transport matrix is supplied.

7.2 Communication without fabricated firsthand memory

The complete communication example declares separate lineages for A and B, records an observation by B, communicates it to A, and permits A to use it after the required endorsements and authorization. Its concluding assertions include

```
assert packet route_blocked_at_A provenance communicated
assert memory A contains route_blocked_at_A as communicated true
assert memory A contains route_blocked_at_A as observed false
```

The successful action and the origin of its supporting evidence are distinct claims. A recipient may act on communicated evidence without relabeling it as its own observation.

7.3 From quarantine to admission

```
module neural_candidate_admission
lineage L_A
agent A lineage L_A
neural_propose latent_claim to A feature feature_441 content "route-may-be-blocked" confidence 0.84
assert packet latent_claim status provisional
assert packet latent_claim obligation origin true
attest origin latent_claim by A lineage L_A as p_origin
attest relevance latent_claim by A for planning as p_relevance
admit latent_claim by A using ["p_origin","p_relevance"]
assert packet latent_claim status admitted
assert packet latent_claim provenance inferred
assert packet latent_claim quarantined_by A false
```

This example also exposes a trust boundary. The attestations are declared inputs to the language; they are not generated by a universal truth detector. A deployment that requires semantic validation must bind the attester to an actual external checking procedure. The Discovery Bridge and the new finite

interface demonstrate two narrowly specified checking procedures. Neither establishes truth outside its declared world.

8 A checked semantic-checkpoint interface

8.1 What the new demonstrator does

The edition adds `interface/semantic_interface.py`. It takes a *fully declared finite deterministic machine*, discovers its observable quotient, checks a certificate through local preservation equations, and serializes a checkpoint containing the quotient class and its scope bindings. It is not a decoder of arbitrary Neuralese vectors. Its entire transition table is an explicit trust root.

Let the declared machine be $M = (S, \mathcal{A}, F, O)$. The certificate contains a surjection $q : S \to Q$, quotient transitions $\bar{F}_a$, and outputs $\bar{O}$, subject to

$$q(F_a(s)) = \bar{F}_a(q(s)), \qquad O(s) = \bar{O}(q(s)) \quad (s \in S,\ a \in \mathcal{A}). \tag{2}$$

The certificate generator uses partition refinement. The checker instead verifies the displayed equations over every declared state and action. It accepts any sound quotient, including a finer nonminimal one. Minimality is a property of the generator and its tests, not an extra property falsely attributed to the certificate checker.

Theorem 8.1 (Finite semantic checkpoint preservation)**.** *If* (2) *holds, then a checkpoint storing $q(s)$ reproduces $O(F_w(s))$ for every finite word w over the declared alphabet, when continued with the quotient transitions. No information inside a fiber of q is needed for those outputs.*

Proof. Induction on $|w|$ gives $q(F_w(s)) = \bar{F}_w(q(s))$. Apply the output equation to the terminal state. The argument uses totality of the declared transition maps and the equations at every state, not merely a sampled history panel. □

This finite theorem permits all word lengths *for the declared finite model*. It is not an infinite-horizon theorem about a learned implementation that only approximately matches that model.

8.2 The D4 fixture

The demonstrator has sixteen concrete states (a, b, u), with $a \in \mathbb{Z}/4\mathbb{Z}$, $b, u \in \mathbb{Z}/2\mathbb{Z}$. Its visible state is (a, b); u is a private bit. Operations are

$$r(a, b, u) = (a + 1, b, u), \quad f(a, b, u) = (-a, 1 - b, u), \quad v(a, b, u) = (a, b, 1 - u).$$

The first two satisfy $r^4 = f^2 = frfr = 1$. The private operation changes only u. Partition refinement recovers eight semantic classes. Discarding the private bit preserves every future declared observation.

The source-bound certificate is then rendered into OSL. A validated finite checkpoint is admitted under the adapter's explicit attestation policy. An unverified candidate remains provisional and quarantined. Both paths execute in Core 0.5 and pass its base and semantic checkers. The accepted path has 27 receipts and the unverified path 23. This is an actual cross-package execution, but its producer is a finite fixture rather than a trained model.

8.3 Binding, restoration, and limits

A checkpoint binds the machine digest, certificate digest, scope, quotient class, and checkpoint digest. A fresh process restores it only against the same declared machine and a valid quotient certificate. Changing an alphabet, output table, or scope invalidates the old binding. A valid serialized copy establishes informational continuation, not personal identity or authenticated process lineage.

A rehashed checkpoint that names a different valid class is still a different valid checkpoint. Without an external authenticating authority, the checker cannot determine which past process truly produced it. Content validity, historical authenticity, and authorization remain separate.

9 What the retained neural-language experiments establish

The OSL archive includes geometry, micro-emergence, Hugging Face causal, and operand-slot experiments. They are preserved because the language program depends on understanding what neural producers can and cannot supply. They do not change the identity of the canonical runtime. [OL]

The geometry pilot recovered partial structure under added supervision and equivalence pressure, but control models weakened the inference that a usable internal operator algebra had emerged. The Discovery Bridge's upstream audit also identified a problem in an older intervention endpoint: the experimental definition of `observe(a,f)` changed both status and origin, while the intervention scripts demanded preservation of the previous communicated origin. The resulting low conversion statistic is not a clean estimate of the intended observe operation. It must not be used as a definitive numerical falsification of that operation. The valid conclusion is that the desired causal-language claim was not established by the retained experiment. [ODB]

The operand-slot v0.3.1 package supplies entities, operands, source reads, and write locations, but no completed scientific run. Its supplied ontology would make a future success a test of scaffolded execution, not unrestricted discovery. One audit also leaves alpha-isomorphic separation unverified. These qualifications are necessary because a representation can predict labels without implementing the intended updates.

The practical interface is therefore asymmetric: neural computation may propose an operator or a state interpretation; the explicit language does not grant it authoritative semantics merely because a probe can decode it. Causal abstraction research motivates intervention-based tests, and critiques of unrestricted alignment motivate bounds on the complexity and information access of the translator. [G+23, S+25]

10 Fresh verification and reproducibility

10.1 Completed checks

Check	Result	Scope
Core 0.5 unit suite	168 passed	Packaged implementation tests.
Core 0.5 corpus	66/66	42 accepts, 24 expected rejections.
Native corpus receipts	1,825	Byte-identical regenerated bundle.
Discovery Bridge tests	20 passed	Core integration enabled.
Bridge programs in Core 0.5	3/3	71 receipts.
New finite-interface tests	15 passed	Includes exact finite enumeration and fresh-process restoration.
New OSL interface examples	2/2	Accepted and quarantined paths, 50 receipts total.

The conformance report, semantic summary, and complete source-bound trace bundle regenerate byte for byte from the preserved Core 0.5 source. The finite-interface tests include all 216 three-state, single-action machines with binary observations. This is a complete finite family, not a sampled machine-learning population. Larger neural experiment counts elsewhere in the repository are historical and are not included in this edition's fresh test total.

10.2 Reproduction commands

Run from the combined release root, with Python and the supplied dependencies installed:

```
python verify_release.py
python run_current_checks.py
python interface/run_demo.py --output ../local_demo
```

The first command checks the edition manifest and the absence of redistributed font files. The second runs the bounded current-code checks and writes separate logs. It does not train neural checkpoints or consume sealed learned-evaluation banks. Exact upstream commands and the original manifests remain in their own packages.

The reproducibility claim has limits. Test execution uses the installed Python and numerical stack documented in the validation record. It does not reproduce a historical GPU serving environment. Runtime matching of the finite OSL corpus is stronger here than a general numerical-portability assertion: the named files are actually identical, whereas no such claim is made for all neural floating-point trajectories.

10.3 A corrected invocation failure

The first bridge corpus command supplied a relative examples path. The upstream generator expected a path under its absolute root and raised `ValueError` before evaluating a program. Repeating the command with the absolute path completed all three cases. The original source was not changed. This invocation failure and the successful correction are preserved in the execution notes rather than hidden behind the last exit code of a compound shell command. A separate integrity invocation on the shortened directory name `core05` failed the upstream archive-name predicate, even though its payload checks passed. Repeating it on a canonical-named working copy passed all 156 listed file checks. The portable edition runner preserves that original root name in its temporary copy; it does not weaken the upstream

check.

11 Compatibility, trust, and publication status

OSL is useful because its semantics are explicit enough to disagree with a producer. A checker may reject a proposal, retain ambiguity, or accept only a restricted scope. A language that simply translates every fluent claim into an authoritative memory would lose the point of this design.

There are two distinct remaining frontiers. The *runtime frontier* is a mechanized correspondence between all implemented source forms, their effects, and an independently specified semantics. The *learning frontier* is a neural producer that discovers reusable typed operations under controlled intervention and transfer tests. Completing one does not complete the other. A language can be useful before it is learned spontaneously; a learned latent operator can be causally meaningful before it has a faithful textual rendering.

The final claim of this edition is therefore direct: OSL is a specified and executable operator-symbolic research language with independently checked finite semantic components, exact bounded operator admission, and an implemented finite checkpoint bridge. The record does not establish a universal verifier or an endogenous neural language. Those are research targets, not missing editorial labels.

A Claim register

ID	Statement	Status and boundary
O1	Exact generator transport composes.	Proved by induction; specified maps and words.
O2	Approximate transport has a finite-word error budget.	Proved by telescoping; norm assumptions explicit.
O3	Communication preserves the original origin record.	Proved for the declared rule; external truth not implied.
O4	Linear-use accounting permits exactly one consumption in a completed scope.	Proved under the accounting rules.
O5	Rollback need not erase execution history.	Proved for separated state and append-only ledger.
O6	Finite demonstrations identify an operator only relative to a candidate universe.	Proved; implemented by Discovery Bridge.
O7	A locally checked finite quotient preserves all declared future outputs.	Proved; implemented and tested.
O8	Core 0.5 passes the retained corpus and 168-test suite.	Fresh execution in this edition.
O9	The complete Python interpreter has a mechanized soundness proof.	Not established.
O10	The neural producer has learned OSL's semantics generally.	Not established.
O11	The finite interface compiles arbitrary learned Neuralese.	Not claimed; it uses complete declared tables.

B Source hierarchy and reading map

The authority order is source-specific, not a total order by version. Current finite runtime semantics come from `OSL/core05/`. The historical broad mathematical development is preserved in `OSL/historical_monograph/`. Bounded synthesis is in `OSL/discovery_bridge/`. The newly authored finite checkpoint code is in `interface/`. The whole OSL lineage archive retains the experimental packages and earlier receipt-hardening release.

The appendix does not replace the original 88-page manuscript. That manuscript remains an independently identifiable historical artifact. Its statements about Core 0.3 describe that version, not Core 0.5. The new edition provides the missing version crosswalk and current evidence rather than silently rewriting an old release's dates and claims.

For a syntax-first reading, start with the current examples and `CORE05_SEMANTIC_INDEPENDENCE_SPEC.md`. For finite discovery, read `FORMAL_NOTES.md` before running the three proposal examples. For neural producers, read the companion Neuralese edition and the individual experimental claim boundaries. A reader should be able to trace a claim to its implementation and retained result without inferring authority from a filename containing "final."

C A compact operational checklist

Before accepting an externally produced representation, specify its carrier, observation interface, allowable future operations, history or state domain, source/version binding, and intended use. Check the obligations that actually belong to that scope. Preserve unresolved alternatives and failed conditions. Record whether the evidence is exact finite enumeration, a mathematical theorem, a sampled experiment, or a declaration by a trusted authority.

Before executing an action, separately verify the required capability and resource state. Before compressing memory, verify that the proposed abstraction preserves the future observations the consumer needs. Before translating between agents, distinguish a coordinate change from an information-losing abstraction. Before declaring a learned language, establish causal intervention and continuation behavior rather than relying only on decoded labels.

These are operational consequences of the formal distinctions in this paper. They are not claims that a single finite reference language settles institutional governance, world truth, or general AI safety.

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[ODB] Komorebi P. *OSL Discovery Bridge v0.1.0*. 2026. Finite world semantics, admission algorithm, upstream audit, and tests in `OSL/discovery_bridge/`.

[OL] Komorebi P. *OSL Complete Lineage with Core 0.4*, 1 September 2026. Includes geometry, micro-emergence, causal-compositional, operand-slot, and receipt-hardening branches. Exact original bytes and hashes are bound by the edition's source ledger.

[FI] Komorebi P. *Finite Semantic Checkpoint Interface*, research edition 1.0, 2026. Newly authored reference demonstrator, tests, and actual OSL admission/quarantine traces in `interface/`.

NEU / SOURCE READING CARD

Neuralese

Research edition 1.0 · 20 original pages

READ THIS FOR

Latent operator semantics, observable continuation, gauge and stability boundaries.

KEEP THIS BOUNDARY

Mixed theory and retained research history. Historical outcomes are preserved, not used as evidence for a combined agent.

CONNECTED SOURCES

OSL LSM SRWT

Original text begins on the next page.

The source pages that follow are preserved without editorial stamps. Page labels in the PDF navigation use the source code and physical source-page number. The package retains the byte-identical original PDF and its SHA-256 digest.

OPERATION KOMOREBI

REALIZATION CALCULUS



RESEARCH LIBRARY

Neuralese

Recurrent latent operators, causal semantics, and stability

Traditional paper

Komorebi P.

Research edition 1.0 / Standardized 8 October 2026

REALIZATION CALCULUS

Reading scope

Finalized object. A consolidated mathematical and experimental account of the recovered direct Neuralese branches, with source archives, explicit claim status, and reproducible checks. This is not a release of a universally stable learned language. Positive finite results, failed scientific gates, invalidated assay gates, and unspent experiments are retained as different outcomes.

Abstract

Neuralese names a research program on recurrent latent computation whose intermediate state need not be rendered as natural-language tokens. The central issue is not whether a hidden vector predicts an answer, but whether an identifiable state supports reusable operations, causal intervention, reliable continuation, and transport between representations. This manuscript develops those requirements and consolidates twelve recovered direct Neuralese source packages plus the recurrent-depth reference laboratory.

The mathematical account separates coordinate gauges from semantic equivalence, current readout from future observability, bounded norm from stable meaning, and exact compiled control from learned implementation. It gives a two-index family of history/continuation quotients, a finite observable-kernel construction, composition-error bounds, and an obstruction to writing task information into a fixed null fiber under globally isometric dynamics. These are explicit derivations under stated assumptions, not proof-assistant certifications of the neural software.

The retained evidence is mixed and informative. Projective terminal-only models achieved exact tested structured-family performance through depth 64 in a two-seed developmental study, with strong state and operator interventions. Subsequent exhaustive finite audits found more than 33 million correct ordinary-history continuation decisions while preserving failed cumulative and long-path gates. Repeated identity paths exposed delayed semantic errors. A later norm-preserving operator study completed 48 training cells but passed only 8 of 26 registered gates. The packet-closure pilot retained an assay failure, the invariant-kernel branch stopped at calibration, and the semantic-retraction bank remains unspent. A small frozen-language-model integration passed engineering checks but not its scientific gates.

Together the results support bounded learned operator computation and a concrete stability problem, not a general endogenous machine language. The companion OSL edition supplies an explicit semantic consumer. The missing bridge is a learned, causally adequate translation into that interface under declared scope and independent continuation tests.

The key distinction. A correct finite semantic executor exists for the supplied tasks. Learning a representation that reliably implements that executor under weaker information and supervision remains a separate problem. An exact positive control is not counted as a learned result; a preserved failure is not converted into success by a later editorial version.

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1 The object of study

A language model can produce useful text without exposing a faithful account of the computation that produced it. Conversely, a recurrent computation can manipulate information without turning every intermediate state into a word. Neuralese studies the latter possibility through instrumented, bounded experiments. The name does not assert that every useful latent vector is a language or that a human-readable transcript exists for it.

A general controlled latent system is

$$z_{t+1} = F_\theta(z_t, u_t), \qquad y_t = O_\theta(z_t, q_t), \tag{1}$$

where u_t supplies an operation or observation and q_t specifies a later query. A recurrent state is scientifically interesting when it supports multiple future uses, not merely when it stores the already known final output token. The question may be asked for a trained operator network, an adapter attached to a frozen language model, or a deliberately constructed finite fixture; these are not interchangeable experimental settings.

The public Coconut proposal is a relevant neighboring architecture: hidden states are fed back into a language model rather than always decoded into intervening words. This motivates latent recurrence but does not validate the particular mechanisms tested here. Causal abstraction provides a vocabulary for relating an internal mechanism to a higher-level algorithm, while critiques of unrestricted alignment show why a flexible translator can overstate apparent understanding. [H+24, G+23, S+25]

1.1 An operational evidence ladder

A useful ladder is: executable recurrence; task-relevant state; causal state use; reusable operation; continuation-preserving abstraction; stable composition; and transport under controlled changes of realization. Each level adds an obligation. A high score on a linear probe is evidence of recoverability by that probe, not sufficient evidence that the model natively uses the decoded variable. A successful state patch establishes a particular intervention effect only under its sampling and control design. Stable norm establishes a numerical property, not the semantic part of the ladder.

The current record reaches different levels in different finite settings. It must not be collapsed into a single “Neuralese percentage.”

1.2 What changes during learning

In the small operator-workspace studies, learned encoders and operator-producing maps are optimized from finite task data; the resulting weights are then frozen for evaluation. In a hard-QR or hard-polar arm, an architectural construction enforces isometry. That property is not learned, although the parameters selecting an isometry are learned.

In G1, a small trainable recurrent workspace is coupled to a frozen local language-model backbone. The workspace parameters change; the registered backbone does not. Later diagnostic probes fitted to the resulting activations are measurements performed by the analyst, not evidence that the original model learned the probe’s algorithm. In a frozen-checkpoint audit, neither the original workspace nor the backbone is retrained. The release labels all four situations separately. [N02, NOP, G1, G1D]

2 Exact semantics before approximate semantics

2.1 A task-relative realization

Let $\mathcal{A}$ be an operation alphabet, Z a latent state space, and Y an observation space. Each $a \in \mathcal{A}$ induces $F_a : Z \to Z$. For a word $w = a_1 \cdots a_k$, define

$$F_w = F_{a_k} \circ \cdots \circ F_{a_1}.$$

A candidate semantic state space S, operation maps G_a, and abstraction $q : Z \to S$ should satisfy

$$q \circ F_a = G_a \circ q, \qquad O = \bar{O} \circ q. \tag{2}$$

These conditions say that semantics is both operation-compatible and sufficient for the declared outputs. A decoder that predicts only the present label establishes neither condition by itself.

Proposition 2.1 (Semantic continuation)**.** *If* (2) *holds on a forward-invariant domain $D \subseteq Z$, then*

$$O(F_w(z)) = \bar{O}(G_w(q(z))) \quad \textit{for every } z \in D \textit{ and every finite } w.$$

Proof. Forward invariance keeps every intermediate state inside the domain where the equations hold. Induction gives $q(F_w(z)) = G_w(q(z))$. Apply the observation factorization to the endpoint. □

Without forward invariance, a one-step certificate may leave its own scope. Without observation factorization, the reduced state may preserve transitions but omit an output the consumer needs. Without operation compatibility, matching today's readout says nothing about tomorrow's computation.

2.2 Three meanings of "same state"

Geometric equality, semantic equality, and predictive equality are different. Two latent vectors can differ in a redundant coordinate while implementing identical continuations. Two vectors can have the same current class label while differing in future confidence or even future answers. Two states can be experimentally indistinguishable within a finite horizon yet separate after a longer suffix.

An approximate relation such as $d(z, z') \leq \varepsilon$ is generally not an equivalence relation. For points $0, 0.75\varepsilon, 1.5\varepsilon$, successive pairs meet the bound but the first and last do not. Quotient terminology therefore requires an actual equivalence relation, a specified quantization, or a separate approximate-abstraction formalism. Pairwise tolerance alone is insufficient.

3 The two-index completion surface

3.1 History depth and future horizon

Let $I \subseteq Z$ be the allowed initial states and

$$R_{\leq L} = \{F_w(z) : z \in I,\ |w| \leq L\}.$$

Define exact finite-horizon observational equivalence by

$$z \sim_H z' \iff O(F_w(z)) = O(F_w(z')) \text{ for every } |w| \leq H.$$

The finite object is

$$Q_{L,H} = R_{\leq L} / \sim_H . \tag{3}$$

The parameter L governs which microstates can be reached; H governs which future observations can distinguish them. A single depth index conceals this difference. The causal-completion and completion-surface audits were designed to expose it. [NCC, NCS]

Proposition 3.1 (Refinement and diagonal action)**.** *Increasing H refines $\sim_H$. For every action a, there is a well-defined map*

$$Q_{L,H+1} \longrightarrow Q_{L+1,H}, \qquad [z] \longmapsto [F_a(z)].$$

Proof. The set of tested continuations grows with H, so equality under all longer-horizon tests implies equality under the shorter set. If $z \sim_{H+1} z'$, then for every suffix $|w| \leq H$, the continuation consisting of a followed by w has length at most $H+1$. Hence $O(F_w F_a(z)) = O(F_w F_a(z'))$, so $F_a(z) \sim_H F_a(z')$. Both successor states are in $R_{\leq L+1}$. □

The horizon decreases along this map. A finite-horizon state cannot automatically be updated forever at the same horizon without additional closure information.

3.2 A minimal delayed-observability counterexample

Let the concrete state be $(x, y) \in \{0,1\}^2$, with present observation $O(x, y) = x$, and let the only operation swap the coordinates. At horizon zero, there are two classes, classified by x. At horizon one, there are four: the next observation also reveals y.

The states $(0,0)$ and $(0,1)$ initially look identical but become distinguishable after one operation. Keeping only the current answer destroys future semantics. This exact example is tested by the new finite checkpoint interface. It is the simplest reason a successful final-answer probe should not be called a complete causal state.

3.3 Hard and soft observations

A hard interface is $O_h(z) = \arg\max_j \ell_j(z)$. A soft interface may be the probability vector or centered logits. Exact agreement under O_h permits changes in every nonwinning logit and changes in the winner's margin. Those changes can matter under later operations.

The retained completion-surface results show this distinction rather than merely proposing it: hard continuation profiles can be exact on ordinary reachable shells while the probability and centered-logit profiles retain a measurable history-dependent microstate. That does not imply that the hard quotient is useless; it means that its interface and continuation domain must remain attached to its claim. [NCS]

4 Linear observable kernels

4.1 A computable null-space construction

For a switched linear system $z_{t+1} = M_{a_t} z_t$ with observation Cz_t, define

$$N_H = \bigcap_{|w| \leq H} \ker(CM_w), \qquad M_\varnothing = I.$$

A direction belongs to N_H exactly when it is invisible under every continuation through horizon H.

Proposition 4.1 (Observable-kernel recurrence)**.** *One has*

$$N_{H+1} = N_H \cap \bigcap_{a \in \mathcal{A}} M_a^{-1} N_H.$$

If $N_{H+1} = N_H$, then N_H is invariant under every generator and equals N_∞. In a d-dimensional state space, the decreasing sequence stabilizes after at most d strict dimension drops.

Proof. Invisibility for a word of length at most $H+1$ includes invisibility for all words of length at most H, and invisibility after one action followed by every suffix of length at most H. This is the displayed recurrence. Equality of adjacent spaces gives $M_a N_H \subseteq N_H$. Since $N_H \subseteq \ker C$, induction makes every longer word invisible. Each strict inclusion decreases a finite nonnegative dimension. □

The result identifies the largest common invariant subspace contained in $\ker C$. It is standard finite-dimensional system reasoning presented here because a visually small activation direction is not necessarily a persistent nuisance direction.

4.2 When a linear quotient exists

Proposition 4.2 (Quotient-kernel criterion)**.** *Let $Q : Z \to S$ be a surjective linear map. Linear maps $A_a : S \to S$ satisfying $QM_a = A_a Q$ exist for all a if and only if* $\ker Q$ *is invariant under all M_a. They are then unique.*

Proof. If the equations hold and $z \in \ker Q$, then $QM_a z = A_a Q z = 0$. Conversely, define $A_a(Qz) = QM_a z$. Two lifts of the same vector differ by an element of $\ker Q$; invariance makes the definition independent of the lift. Surjectivity gives existence on all of S and uniqueness. □

To preserve Cz, additionally require $\ker Q \subseteq \ker C$. A candidate quotient justified only by a rank-zero present readout can fail as soon as recurrence returns its supposedly discarded directions to the visible subspace.

The learned-checkpoint probe finds exactly that kind of obstruction in its tested linear observation interface: the observable rank rises from seven at horizon zero to the full ambient rank twelve at horizons one through four. The conclusion is restricted to the declared linear/centered-behavior interface and audited checkpoints; it does not rule out a nonlinear hard-decision quotient on a smaller domain. [NLP]

5 Gauge fixing is not semantic completion

5.1 Positive rays

For nonzero z, positive scaling may be redundant for a homogeneous hard-decision readout. The appropriate relation is $z \sim cz$ for $c > 0$, not identification of z with $-z$. Per-step normalization gives

$$F_M(z) = \frac{Mz}{\|Mz\|},$$

provided $Mz \neq 0$. This fixes radial magnitude, but leaves angular dynamics intact.

Proposition 5.1 (Derivative of normalized dynamics)**.** *For $u = Mz/\|Mz\|$ and $Mz \neq 0$,*

$$DF_M(z) = \frac{(I - uu^\top)M}{\|Mz\|}.$$

Proof. The derivative of $x \mapsto x/\|x\|$ at nonzero x is $(I - vv^{\mathsf{T}})/\|x\|$, with $v = x/\|x\|$. Compose it with the linear derivative M. □

A bounded normalized state can therefore have a locally amplifying angular derivative. A contraction claim must name the norm, state domain, tangent directions, and whether it concerns all tangents or a separately identified semantic fiber.

5.2 An exact norm-preserving failure

Counterexample 5.2 (Small phase error accumulates)**.** Let the intended semantic operation be the identity and the learned implementation be a planar rotation R_ε. Every step preserves Euclidean norm exactly. After k steps the state has rotated by $k\varepsilon$. With a readout determined by the sign of the first coordinate, a state starting at $(1, 0)$ eventually changes class for any fixed $0 < \varepsilon < \pi/2$.

This counterexample separates numerical stability from correct composition without involving underflow, overflow, or a defective normalization implementation. It also explains why enforcing QR or polar isometry may repair one failure class while leaving the central learning problem untouched.

5.3 Margin protection is finite and local

Proposition 5.3 (Readout margin bound)**.** *Suppose a logit vector has unique winning class i and margin $m = \ell_i - \max_{j \neq i} \ell_j > 0$. Any perturbation $\Delta\ell$ with $\|\Delta\ell\|_\infty < m/2$ preserves that winner.*

Proof. For every competitor j, the perturbed difference is at least $m - 2\|\Delta\ell\|_\infty > 0$. □

A finite region with positive margins can hide a continuous path defect for many steps. Correct endpoint labels do not prove the defect is absent. The relevant experiment asks whether repeated paths erode the margin until an ordinary continuation reveals the difference.

6 Exact compiled controls and learned operators

6.1 Anchor transport

Suppose B is an invertible matrix whose columns are latent representatives of a finite semantic basis, and P_g is a supplied permutation action. Define

$$M_g = BP_gB^{-1}, \qquad G = B^{-\mathsf{T}}B^{-1}.$$

Then $M_gB = BP_g$ and $M_g^{\mathsf{T}}GM_g = G$. For a permutation representation, every group relation is preserved exactly by conjugacy.

Proposition 6.1 (Exact finite realization)**.** *The preceding construction implements the supplied permutation action on every anchor and preserves the G-norm. In the canonical semantic chart $B = I$, it is simply the permutation executor $M_g = P_g$.*

Proof. The anchor equation follows by cancellation of $B^{-1}B$. For the metric identity, substitute the definitions and use $P_g^{\mathsf{T}}P_g = I$. Products telescope to BP_wB^{-1}, so semantic relations transfer. □

This is a constructive control, not an inference that a learned encoder discovered the same realization. If the true permutation table is supplied to construct M_g, exact behavior has been compiled from that table. The learned arm must be evaluated separately.

6.2 Approximate intertwining

If $QM_a = A_aQ + E_a$, repeated composition can accumulate leakage. For latent norm bounded by R, semantic operators bounded by a, and $\|E_a\| \le \varepsilon$, the reduced-state discrepancy obeys

$$e_{t+1} \le ae_t + \varepsilon R.$$

Induction gives

$$e_t \le a^t e_0 + \varepsilon R \sum_{j=0}^{t-1} a^j.$$

When $a = 1$, the upper bound grows linearly. When $a > 1$, it may grow exponentially. A common invariant metric can control one source of growth, but does not establish zero semantic defect. The companion OSL paper gives the corresponding operator telescoping identity.

7 Ambient memory and the isometry obstruction

A persistent latent state may carry both task information and history that is presently invisible. It is tempting to demand three properties simultaneously: an exact fixed observable quotient, globally isometric dynamics, and free writing of visible task information into the invisible fiber. In the natural orthogonal splitting, those demands conflict.

Theorem 7.1 (Fixed-kernel isometry obstruction)**.** *Let V be a finite-dimensional inner-product space, $K \subseteq V$ a subspace, and $U : V \to V$ an isometry with $U(K) \subseteq K$. Then $U(K) = K$ and $U(K^\perp) = K^\perp$. Relative to $V = K^\perp \oplus K$, U is block diagonal.*

Proof. An isometry is injective, so $\dim U(K) = \dim K$, and the finite-dimensional inclusion gives equality. Thus $U^{-1}(K) = K$. For $x \in K^\perp$, $k \in K$, $\langle Ux, k\rangle = \langle x, U^{-1}k\rangle = 0$. Hence $U(K^\perp) \subseteq K^\perp$; dimensions again give equality. □

If K is the fixed kernel of a quotient, the theorem excludes a nonzero visible-to-kernel block *in this orthogonal decomposition*. A nonorthogonal coordinate chart can display apparent off-diagonal entries without contradicting the theorem. The geometric statement is about invariant orthogonal complements, not a particular array spelling.

The obstruction is not a prohibition on computational memory. A system can relax isometry, use controlled nonunitary writes, change its quotient or boundary, or adopt state-dependent fibers. Those alternatives require their own semantics and stability analysis. The invariant-kernel source explicitly identifies this assumption boundary rather than claiming all ambient memory impossible. [NIK]

7.1 Curvature can be invisible to the task

Conversely, exact semantic behavior does not require the entire ambient space to be path-flat. Consider a block-diagonal action $U_g = \mathrm{diag}(P_g, V_g)$, observed only through the first block. The P_g may obey all semantic relations while the V_g retain path-dependent history. The task quotient is exact even though a full-state identity loop need not return the hidden block to its initial value.

The scientific question is therefore not “Is all ambient holonomy zero?” It is “Which holonomy becomes visible under the declared future operations and observations?” This is why current invisibility, invariant invisibility, and task-irrelevant internal dynamics must remain different notions.

8 The recovered experimental lineage

8.1 A branching record rather than one success ladder

The source archive contains several related branches, often reusing checkpoints or retaining predecessor files. A root README can be a prospective source sealed before its later result exists. A copied claim-boundary document can still name an older version. This edition uses the branch-specific result and execution record, not the most optimistic sentence in a nested ancestor. Exact archive hashes and the selected reading paths are recorded in the source ledger.

Branch	Evidence type	Consolidated status
Operator Workspace 0.3.0	48 trained cells, 12 seeds, 4 arms	Completed primary study; 8/26 gates pass. Norm preservation succeeds architecturally; overall learned-semantic claim fails.
Stable Quotient 0.2.0	Two-seed developmental training	Strong structured-family depth and causal results; unrestricted permutations and long-path flatness remain limited.
Ambient Packets 0.3.0	Frozen-checkpoint calibration	Finite cone, gauge, and switched-dynamics diagnostics; not new independent training evidence.
Packet Closure 0.4.0	Three-seed paired 2-by-2 pilot	Execution passes; assay A15 fails; full developmental claim not evaluated.
Invariant Observable Kernel 0.6.0	Development and rehearsal	Calibration stop; untouched screen and pilot not run.
Causal Completion 0.2.1	Frozen-checkpoint audit	Finite hard quotient passes; path flatness and all-tangent contraction gates fail.
Completion Surface 0.3.0	Two-checkpoint finite audit	Mixed registered verdict; delayed path failures retained.
Entry–Cone–Holonomy 0.3.2	Registered bounded diagnosis	Execution complete; finite path-dependent holonomy localized; no general completion claim.
Learned Checkpoint Probe 0.1.0	Post-hoc rank/loop diagnosis	Persistent small additive quotient unsupported on tested linear interface.
Semantic Retraction 0.3.0	Prospective frozen package	Armed, unspent, capacity-gated; no learned result.
G1 0.1.1	Frozen-backbone integration	Engineering passes; retained seed 711001 scientific gates fail.
G1-D1 0.1.0	Read-only diagnostic tooling	Static/engineering checks; does not turn the failed G1 pilot into a pass.

The recurrent-depth laboratory is retained separately as analytic and numerical instrumentation. It is not an additional trained Neuralese success.

8.2 Stable Quotient: what the positive result says

Two 4,224-parameter models were trained on terminal answers for randomly re-coordinatized D4 sequences of lengths one through four. Per-step projective normalization substantially extended the tested compositional depth. The retained two-seed means are shown below. [N02]

Model / family	4	12	64	128	256
Raw parent, D4	.9990	.8403	.3867	.2869	.0000
Projective, D4	1.0000	1.0000	1.0000	.9924	.8477
Projective, C8	—	—	1.0000	.9917	.7351
Projective, mixed cycle	—	—	1.0000	.9956	.8665
Projective, arbitrary pair	—	—	.2629	.2485	.2480

The raw depth-256 value counts float32 overflow as incorrect. Normalized replay of the raw parent remains directionally inaccurate, so the conclusion is not simply that one arm overflowed. Structured-family generalization is strong; arbitrary permutation execution is not established.

At depth 64, intended donor-state and internal-operator interventions score 1.0000, with the reported matched random controls at most .0781 and .0654. Short-trajectory affine transport preserves the target model's structured-family strengths and its arbitrary-operator weakness. These are bounded developmental results with two seeds and supplied task structure, not population-level or spontaneous-language conclusions.

The exact compiled control executes every tested family and depth through 256 with accuracy 1.0000. This localizes an engineering target: the task admits a stable finite solution, while the learned encoding of its operators can drift. It does not prove that the learned model uses the compiled solution.

8.3 Later norm preservation did not close semantics

The newer Operator Workspace v0.3.0 compares hard sign-canonical QR with a soft penalty, each under correct and shuffled conditions. All 48 cells complete, but the overall study passes only 8 of 26 gates. The retained record comprises 12 training seeds and six transport pairs. [NOP]

Metric	Hard correct	Hard shuffled	Soft correct	Soft shuffled
D4 depth 8	.8757	.8913	1.0000	.9990
D4 depth 64	.4840	.1955	.9054	.1466
D4 depth 128	.3576	.1401	.5521	.1361
C8 depth 64	.3960	.1401	.7869	.1489
Arbitrary depth 64	.1800	.1292	.2796	.1403
Norm ratio at 2048	1.0000	1.0000	≈ 0	≈ 0

The hard arm repairs norm drift by construction but is often less accurate. Its D4 depth-64 advantage over the shuffled control is positive in the mean, yet only seven of twelve seeds clear the required paired margin when eight are required. A post-hoc interval does not revise that frozen gate.

The six-pair aligned D4 state-patch score is .8495. A discrete decode–label–reembed baseline scores .9088. Thus the record does not establish that the learned transport carries a richer operational state than the

label-mediated baseline. Hard and soft arms also differ in effective dynamics and computational cost; this is not a clean intervention on only one abstract property.

The 49,152,000 processed training examples are not 49,152,000 independent experimental replications. The replication unit is the trained cell and its declared seed pairing. Stage B's sparse operation scaffold has no trained primary result in the recovered package.

9 Completion audits: many correct answers, no unrestricted closure

9.1 The first causal-completion audit

The learned causal-completion audit uses frozen checkpoints and no retraining. Its finite Q0 hard quotient passes: each projective seed has 61,440 correct continuation decisions for the registered ordinary histories and suffixes. The D4 Cayley audit separates all 28 pairs of group elements and checks all 64 products under the declared interface. Yet the registered projective path-flatness and all-tangent contraction gates fail. [NCC]

After sixteen repeats, the worst reported median ray angle is .45636 radians and worst H8 continuation survival .92969. A single identity loop can remain hard-decision silent while its full positive-ray tangent has p95 gain near 1.98. The latter derivative is over the full tangent space, not a separately identified vertical semantic fiber. The result denies that particular contraction certificate; it does not prove every conceivable semantic-fiber construction non-attracting.

9.2 The bi-depth surface

The later completion-surface audit examines ordinary history depth $L \leq 8$, continuation horizon $H \leq 8$, and repeated identity paths through total history depth 70. Its exact registered outcomes must be read before its high aggregate accuracy. [NCS]

Gate	Result	Interpretation
E0	Pass	Execution integrity.
H0	Fail	The complete hard rectangle includes an entry defect.
H1	Pass	Down-set consistency.
S0	Fail	Soft profiles do not collapse exactly by label.
D0	Fail	Full registered diagonal condition fails.
W0	Fail	Repeated identity paths are not hard-closed.
W1	Fail	Exact soft path flatness fails.
C0	Fail	The matched nonidentity control also fails in some records.

The full cumulative rectangle contains 33,423,488 decisions and eight errors. All eight are the same seed-0 depth-zero anchor defect repeated across the registered episodes. After at least one ordinary generator application, every exact history shell of length one through eight has correct hard continuations through H8: 33,358,080 of 33,358,080 decisions. This latter statement is a post-hoc conjunction of registered shell measurements. It does not replace the failed cumulative H0 gate.

The repeated-path assay contains 336 relation/budget/environment records, 172,032 individual cases, and 87,908,352 suffix decisions. Hard closure passes 316 of 336 records; exact soft path flatness passes

none. Identity insertions produce 350,757 wrong decisions in 2,193 cases. Of those hard-failing cases, 1,697 require a nonempty suffix before the error appears. The earliest registered hard witness occurs at total observed history depth 38 and first reveals itself after a suffix of length three.

The matched nonidentity control fails in 19 records, versus 20 identity-loop failures. Therefore most long-depth failure cannot be attributed uniquely to algebraically identity words. The supported account is broader path-length fragility with relation-dependent rates, not a special universal pathology of identity loops.

9.3 Entry–cone–holonomy localization

The v0.3.2 diagnosis separates a seed-localized initialization defect, an ordinary-history finite decision-cone regime, and accumulated angular path drift. The runner correction did not change the scientific design; it repaired a bootstrap issue that had prevented the previous invocation from reaching learned evaluation. The corrected registered evaluation is complete. Its one-shot endpoint rule was respected during this edition: only saved evidence and release integrity were checked. [NECH]

The resulting description is precise:

$$\text{entry transient} \longrightarrow \text{finite hard decision-cone regime} \longrightarrow \text{margin-crossing path drift.}$$

It supports neither a globally invariant cone nor a stable quotient of every future probability law. Its value is that it replaces an undifferentiated "depth failure" with observable mechanisms and a concrete continuation witness.

10 Later branches and stopped experiments

10.1 Packet Closure v0.4

The three-seed, four-arm packet-closure pilot completes training and execution, but retains an assay failure at A15, the packet-result contract. Its full developmental claim is `NOT_EVALUATED`. Other mechanism, capability, transfer, and attribution values remain diagnostic records; a passing conditional patch subscore cannot override the failed prerequisite. [NPC]

The hard-polar construction does preserve norm to numerical tolerance. The `polar_cr` arm's reported mean depth-512 D4 accuracy is .6921, and its worst-grid closure and margin gates fail. The experiment thus preserves a distinction already visible in the earlier operator study: architectural isometry can work while learned semantic closure does not.

10.2 Observable rank after recurrence

The learned-checkpoint probe uses all twelve v0.4 checkpoints and 32 primary contexts. It finds rank seven at present observation and rank twelve after one through four continuation steps. Repeat-16 hard-polar log-radial gain remains below 2.90×10^{-5} in absolute value, while p95 oriented loop angles reach roughly .72–.97 radians. [NLP]

The result is post-hoc mechanistic diagnosis, not new training, and it does not repair A15. It does identify why the proposed small fixed additive nuisance quotient is unsupported for the inspected linear observer: directions absent from the current readout become visible after recurrence.

10.3 Invariant Observable Kernel v0.6

The latest numbered kernel branch is not the latest successful scientific endpoint. A developmental smoke passes its execution and assay checks. A retired-seed rehearsal then fails positive-control calibration under the frozen training recipe. The untouched screen seed 947 and five pilot seeds are not run. [NIK]

The learned-intertwiner arm remains behaviorally strong locally but its suffix observer has rank twelve rather than seven, and its length-64 leakage exceeds the registered limits. Exact and quotient-curved analytic controls retain accuracy 1.0000 through depth 16,384. Those controls support the mathematical distinction between exact quotient behavior and globally flat ambient dynamics; they are not learned depth-16,384 results.

Some inherited files in this archive still carry older claim-boundary headers. The v0.6 results index, calibration findings, and frozen stop rule govern its current outcome. Version precedence cannot be inferred by reading an inherited v0.4 document as a new v0.6 success.

10.4 Semantic Retraction v0.3

This branch is **armed, unspent, and capacity-gated**. It contains a frozen selector-by-section factorial design, checkpoint-free tests, a parameter-only preflight, and a sealed fresh bank. Its historical execution was paused because free storage did not meet the required reservation. No v0.3 learned result or spend receipt exists. [NSR]

The 84 checkpoint-free tests pass again in this edition, including 260 reported subtests. That result concerns protocol, lifecycle, bank, and projector machinery. It is not a model score. More disk space in a later environment does not retrospectively convert the earlier unspent bank into evidence. This edition does not consume it.

11 The frozen-language-model integration

G1 is a useful counterweight to the small synthetic operator networks because it attaches a trainable workspace to a frozen local language model. The retained pilot seed 711001 passes all ten engineering gates: the model loads, answer tokens satisfy their boundary contract, the backbone remains frozen, gradients and parameter changes are nonzero, and artifacts are complete. It fails the scientific gate conjunction. [G1, G1D]

Retained pilot metric	Accuracy or difference
Recurrent IID	.278472
Shuffled-state IID	.278472
No-feedback IID	.277778
Zero-injection IID	.055556
Recurrent depth-OOD	.276389
Recurrent minus shuffle	.000000
Recurrent minus no feedback	.000694

Beating zero injection is not sufficient when shuffling the workspace leaves accuracy unchanged and removing feedback has negligible effect. Direct and parity query accuracy also fails to exceed the

empirical majority baselines in the retained gates. The record supports a functioning integration and trainable adapter, not a successful recurrent semantic mechanism.

G1-D1 is a read-only diagnostic block for that same fixed checkpoint. Its static tests and artifact checks pass here. Full model diagnosis requires the registered model files and serving environment; it is not claimed as executed in this release. The diagnostic cannot transform the original pilot into a pass by fitting a post-hoc probe.

12 From latent computation to an explicit language

12.1 The intended interface

The working architecture separates three timescales:

$$\text{fast latent computation} \leftrightarrow \text{typed semantic checkpoint} \leftrightarrow \text{persistent evidence ledger.}$$

A latent state can be provisional and disposable. The checkpoint preserves only distinctions justified for a declared future use. The ledger preserves the origin, interventions, counterexamples, and scope needed to challenge that checkpoint later. This is a proposed integration discipline, not an established end-to-end neural compiler.

OSL supplies explicit operators, provenance, proof obligations, and operational permissions. The finite bridge added in this edition demonstrates a complete path for a fully declared machine: discover a quotient, independently check its local preservation equations, serialize the class, restore it in a fresh process, and execute admitted or unverified cases in OSL-Core 0.5. [ONI]

12.2 What a learned bridge would still need to show

A genuine Neuralese-to-OSL bridge must identify its source state without silently using the target answer, certify its validity region, preserve counterfactual continuation, and show that its translator has not become the actual solver. Its baseline set should include decoded labels, exact external execution where available, random or mismatched transports, and comparable-capacity nonlatent alternatives.

A checkpoint that preserves one task but discards needed future distinctions should be rejected for that broader use, not accepted under an inflated confidence score. A model replacement that changes the semantic chart requires a newly validated transport, not merely a matching tensor shape. The operator-workspace label-reembed baseline demonstrates why this comparison is substantive rather than ceremonial.

13 A precise continuation protocol

The remaining scientific target can be stated without claiming a universal language. Fix a task generator, admissible inputs, observation interface, train/development/test separation, architecture family, compute budget, and intervention family before inspecting the primary results. Specify separately the history depth L, suffix horizon H, and the composition depths used to test a learned operator.

Train candidate models without access to primary evaluation answers. Freeze the source, checkpoint, and gate files. Verify positive controls before spending a designated fresh bank. Evaluate ordinary histories and algebraically redundant long paths separately. Measure both current accuracy and the worst relevant continuation behavior. Report hard, probability, and centered-logit interfaces as distinct estimands rather than choosing whichever one produces the most favorable quotient.

For transport, compare operation-compatible state exchange with a discrete label-mediated reconstruction. For ablation, identify which alternative carriers of information remain available. For composition, report the distribution of errors and margins rather than only a mean norm. For state-space claims, name the allowed class of abstraction maps and bound the translator's information access and capacity. These controls follow directly from the limitations exposed in the retained record.

The primary success condition should be a conjunction of valid execution, valid assay, intended semantic behavior, causal specificity, and continuation retention. An assay failure blocks the stronger conclusion even when some performance values are high. A failed scientific gate remains an outcome, not a reason to change its threshold after inspection.

14 Fresh checks and release boundaries

The fresh verification performed here is principally a source, evidence, and engineering replay. It is not new large-scale neural training. Completed records include the operator-workspace verifier with 179 tests; stable quotient with 20; completion surface with 19 structural tests and recomputation of its failed as well as passed gates; the entry–cone release check; the invariant-kernel package check; causal-completion manifest checks; semantic-retraction's 84 checkpoint-free tests; G1's 19 and G1-D1's 24 engineering tests; and 16 recurrent-depth tests. [VAL]

Two broader verifier invocations exceeded their execution budgets. Ambient Packets printed 59 passing tests but its full post-test verifier did not complete. Packet Closure's parent verifier timed out; its test subprocess subsequently printed 505 passes in 145.11 seconds. The completion of those unit tests is recorded separately from the incomplete parent package-verifier invocation. Neither event is represented as a fresh full package pass.

All selected source archives are content-addressed in the edition ledger. Historical training and evaluation results retain their original identities. No one-use Semantic Retraction learned bank was opened for evaluation, no entry–cone learned endpoint was rerun, and no frozen-language-model scientific result was manufactured from engineering checks. The common release verifier checks the assembled artifact inventory; it does not certify the scientific efficacy of every package it contains.

15 Conclusion

The direct Neuralese program has produced more than a loose analogy about internal machine language. It contains trained finite operator systems, causal interventions, cross-realization transports, exact positive controls, exhaustive continuation panels, and a mathematically explicit stability problem. The positive result is bounded recurrent computation with reusable structure in selected settings. The most informative failures concern persistent semantics, not just numerical execution.

The evidence does not justify identifying norm preservation with understanding, decoding with native use, or a high ordinary-history score with unbounded causal completion. Nor does it justify dismissing the finite successes because the strongest conjecture remains open. The correct synthesis is a layered account: specified finite semantics, learned approximate realizations, scope-sensitive causal tests, and an explicit language interface that can withhold admission.

This edition finalizes that account and its recoverable evidence. The next scientific advance would be a learned representation that meets the continuation and transport obligations under a valid fresh protocol, not a stronger name for an existing checkpoint.

A Mathematical claim register

ID	Result	Necessary qualification
N1	Semantic transition compatibility preserves all future observations.	Exact equations on a forward-invariant domain.
N2	History/continuation quotients admit a diagonal action.	Source horizon is one larger than target horizon.
N3	Observable-kernel refinement stabilizes in finite dimension.	Exact linear maps; numerical rank needs a tolerance policy.
N4	A linear quotient exists exactly when its kernel is invariant.	Surjective quotient map; outputs need their own factorization.
N5	Normalized dynamics have the stated projective derivative.	The image vector must be nonzero.
N6	Norm-preserving rotations can accumulate semantic error.	Explicit counterexample, not a fitted model explanation.
N7	Positive logit margin protects a class under bounded perturbation.	Local finite margin, not unlimited recurrence.
N8	Supplied finite permutations have an exact anchor realization.	Compiled control, not learned discovery.
N9	Fixed invariant kernels of finite-dimensional isometries reduce the action.	Orthogonal splitting; not a universal impossibility of memory.

These derivations are mathematical arguments in prose and equations. They have not been formalized in Lean in this edition. The finite interface tests exercise related exact models; they do not replace general proof checking.

B Reading and execution map

For the strongest early learned result, read Stable Quotient's `RESULTS_SUMMARY.md`. For the later norm-preservation comparison, read Operator Workspace's `results/v0.3.0/INTEGRATED_FINDINGS.md` and its primary gate record. For the loss of completion, read Causal Completion, Completion Surface, and the entry–cone summaries in that order. For fixed-kernel claims, read the learned-checkpoint probe and invariant-kernel calibration findings.

For the pretrained-backbone line, use the pilot copied into G1-D1, not a wiring result from the original G1 package. For semantic retraction, read its release status before invoking anything: the absence of a learned result is part of the current state. The source ledger provides exact filenames, hashes, and preserved reading paths.

The source archives are intentionally not flattened into one import namespace. They have different experimental contracts and in some cases incompatible environment requirements. Extract a selected archive into its own directory. Do not infer that a root-level command in one branch applies safely to another, especially where an evaluation bank is one-use or an endpoint has already been spent.

C Terminology that remains fixed

Latent computation means an internal recurrent state transition. **Operator semantics** means an explicitly tested or specified correspondence between that transition and an operation. **Quotient** means a valid equivalence-based abstraction, not a picture of nearby embeddings. **Gauge** means a representation change preserving the declared object. **Holonomy** in the experiments names a path-dependent return defect, whose task relevance still requires observation and continuation tests. **Closure** is always indexed by the object, domain, operations, and observation interface being closed.

Engineering pass means the implementation meets its execution or artifact gates. **Assay pass** means the measurement procedure meets its validity gates. **Scientific pass** means the prespecified substantive hypothesis gates pass, given the prerequisites. **Unspent** means a designated learned evaluation has not been executed. These labels may coexist in one archive and must never be merged into one undifferentiated success flag.

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OPERATION KOMOREBI

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One-observation Gaussian updates, exact packets, and outward-rounded evidence

Traditional paper

Komorebi P.

Standardized publication edition / 8 October 2026

REALIZATION CALCULUS

Reading scope

A complete finite-dimensional, one-observation inference construction
Based on Certified Inference Core 1.0.0

Abstract

We present a certified finite-dimensional Gaussian inference construction that connects an executable packet solver to a posterior measure and to an explicit arithmetic output contract. Model data are exact rational factors. The prior covariance may be singular, and the observation operator may be rectangular. A positive noise floor makes the innovation covariance invertible. The canonical solver maintains an exact cache of orthogonal output-packet residuals, selects a largest-energy packet, and updates only that packet. Its trajectory agrees with the recomputed greedy trajectory. A geometric residual bound supplies computable finite fuel. After an exact stopping test establishes a strict verification margin, an outward-rounded dyadic verifier certifies a Wasserstein tolerance. The final result also contains enclosing mean and covariance intervals for the Joseph Gaussian at the returned gain, with prescribed maximum radii. We give the underlying algebra, proof composition, worked executions, arithmetic distinctions, and verification boundaries. A checked scalar counterexample shows that independently rounding the gain and cached residual centers does not preserve the exact-cache invariant. The integrated result is complete for one observation update under its declared hypotheses. Sequential filtering, a rounded selective cache, executable sampling, and a universally refined host implementation are separate extensions.

Reading contract

This traditional paper develops the mathematics and the exact program contract. The extended visual guide follows two concrete models through the same chain, with twenty figures generated by Python from exact data or explicit mathematical relationships. The Lean companion gives actual source excerpts, hypotheses, dependency maps, archived evidence, and pinned reproduction instructions. Editable LaTeX, figure programs, exact example outputs, audit reports, and the intact formal release accompany all three documents.

The formal baseline is `FD_Certified_Inference_Core_v1.0.0_Checked.zip`. Its archived build belongs to 12 September 2026. This publication edition separates that evidence from the fresh source and Python checks performed while preparing the papers. The compiler and pinned dependency runtime were not available for a fresh Lean build here. No finite test, source inventory, or rendered page is described as a new kernel check.

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1 The result and its role

An approximate inference program needs a conclusion about the object it returns. A small internal residual, a visually plausible covariance, and a successful numerical routine address different obligations. In this construction, a single result joins them through exact statements. The gain returned by the permitted iteration defines a Gaussian approximation. A certificate controls its distance from a proved posterior version. Parameter intervals enclose the mean and covariance of that same approximation. The result ties its reported arrays to the actual outward-rounded evaluator.

The problem is one Gaussian observation update. Write the prior as

$$X \sim \mathcal{N}(m, P), \qquad Y = HX + \eta,$$

where the centered Gaussian noise η is independent of X. The representation uses rational factors

$$P = LL^{\mathsf{T}}, \qquad R = NN^{\mathsf{T}} + \sigma I, \qquad \sigma > 0.$$

All data supplied to the executable construction, including m, H, L, N, σ and the observation y, are rational. Finite state and observation index types determine the matrix shapes. There is no requirement that H be square or that P be invertible. The innovation matrix is

$$S = HPH^{\mathsf{T}} + R.$$

Its positive noise floor supports the inverse used in the exact posterior semantics. The executable iterative solver avoids computing that inverse to construct its returned gain.

For a rational gain B, the Joseph approximation has mean and covariance

$$\begin{aligned} m_B &= m + B(y - Hm), \\ P_B &= (I - BH)P(I - BH)^{\mathsf{T}} + BRB^{\mathsf{T}}. \end{aligned}$$

Its covariance is positive semidefinite by construction. The approximation is a measure, rather than a collection of unqualified decimal entries. The exact gain $K = PH^{\mathsf{T}}S^{-1}$ identifies the familiar Gaussian posterior parameters. Later sections prove that this explicit Gaussian kernel disintegrates the actual forward joint law. This gives a posterior version defined at every supplied observation. Agreement with Mathlib's canonical conditional kernel is an evidence-almost-everywhere statement, as is appropriate for conditional distributions.

Theorem 1.1 (Publication contract of the canonical solver)**.** *Fix a finite-dimensional rational factor-coded model with $\sigma > 0$, and complete, mutually orthogonal self-adjoint rational output and input packet families. Supply an output-packet seed and a finite scan containing every output label. Let $\epsilon, \delta > 0$ be rational. An arbitrary rational initial gain and an arbitrary finite list of requested precision probes are permitted.*

The canonical program `Komorebi.CachedPosteriorRunCached` *returns a result r. If B is its candidate gain and Π_y is the explicitly proved Gaussian posterior version, then*

$$W_2(\Pi_y, \mathcal{N}(m_B, P_B)) \le \epsilon.$$

The candidate and tolerance predicates hold. The reported mean and covariance intervals enclose m_B and P_B, respectively, and every reported radius is at most δ. Its parameter precision is the program's computed output precision, and its arrays equal the declared evaluator at that precision.

Any successful return of the same program is sound without caller-supplied validity proofs: its internal admission and verification steps imply the same result predicate. The positive-input hypotheses above establish guaranteed successful termination.

Proof composition. The source declaration `Komorebi.CachedPosterior.runCached_terminates` obtains a successful certified gain from `cached_solver_reaches_posterior_tolerance`. That theorem combines initialization, cache preservation, greedy-trajectory equivalence, finite fuel, and a sufficient fallback verification precision. The Gaussian semantic bridge converts its residual certificate into the Wasserstein conclusion. The output theorem `represent_spec` provides parameter enclosure and the radius bound at positive δ. The definition of the result record supplies both reported-evaluator equalities. The separate source declaration `runCached_sound` derives these properties from an actual successful return. The mathematical steps are developed in the following chapters and traced to their exact declarations in the companion. □

The two tolerances have different purposes. The parameter tolerance δ is an enclosure radius for the parameters of the returned approximation. It is not an entrywise error against the exact posterior covariance. The law tolerance ϵ compares the whole approximation with the posterior in W_2. Neither conclusion certifies a Gaussian formed by discarding every interval radius and using its centers as exact parameters. Such a replacement would need a separate perturbation argument, including a positive-semidefinite covariance obligation.

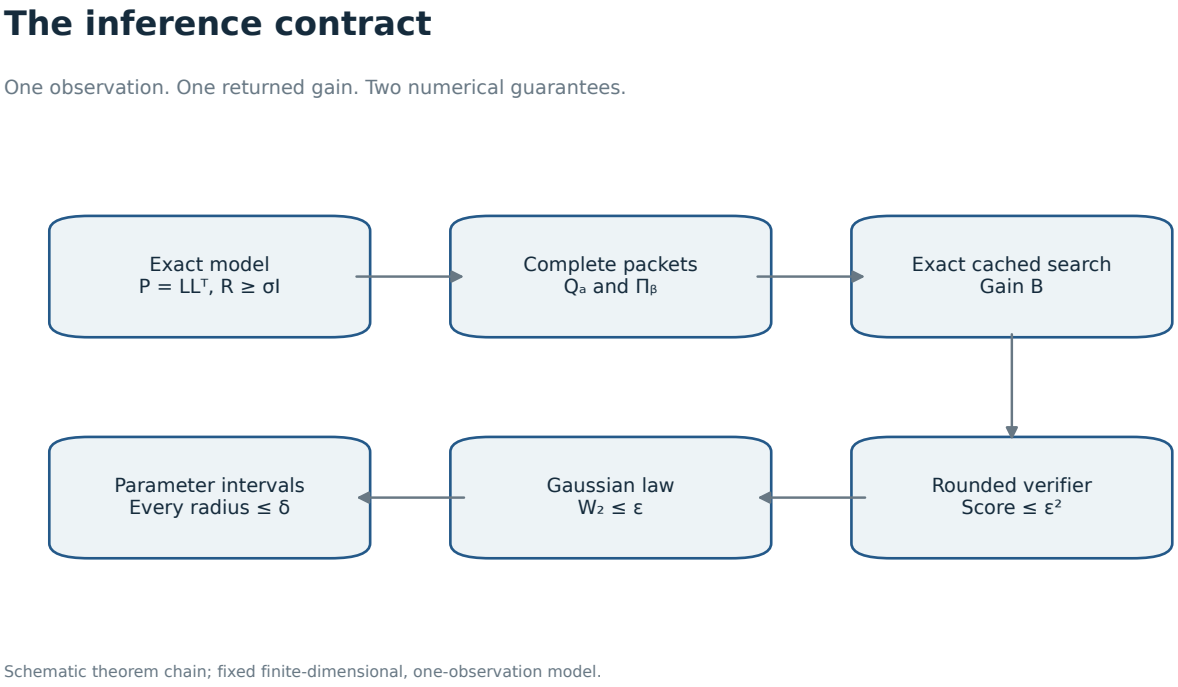


Figure 1: The complete contract links an exact rational model, a permitted cached iterate, rounded verification, a posterior-measure guarantee, and enclosing parameter arrays. The two output tolerances belong to different conclusions.

1.1 Why the packet representation matters

The residual is $F(B) = PH^{\mathsf{T}} - BS$. Orthogonal output projectors Q_a split it into components $G_a = Q_a F(B)$. Completeness makes the sum of their squared Frobenius energies equal the energy of the full residual. Selecting a largest-energy component therefore chooses a block containing a definite fraction of

the remaining error.

The exact cache retains these components, their energies, and their total. For a selected label a, the update changes B by τG_a, changes G_a by right multiplication with $I - \tau S$, and leaves every other cached component unchanged. The proof of that last statement uses orthogonality of the output projectors. It is an algebraic locality result. Keeping unrelated cache entries untouched is justified by their semantic invariant.

The fixed packet family is part of the model of execution. It can represent coordinate blocks or rotated subspaces. The theorem is not restricted to a preferred coordinate basis. Changing the family during the run, however, changes the state invariant and is outside this solver's proved trajectory theorem. The singular rotated worked example demonstrates a non-coordinate packet family without making a claim about changing hierarchies.

1.2 Arithmetic is part of the statement

The cached branch uses exact rational gain updates and exact rational cache scores. It does not independently round the selected gain and cached residual during iteration. Final certificate evaluation and parameter representation use outward-rounded dyadic operations. Those rounded operations have explicit enclosure and inflation bounds. A separately preserved solver uses rounded full-gain updates and has its own convergence theorem. These are distinct verified branches.

The naive rounded-cache counterexample explains the distinction. In a scalar model, exact gain and residual both equal $1/3$ after one step. Rounding both centers independently to zero fractional bits makes both stored values zero. The stored residual score then says zero, although the true residual at the stored gain has score two. The independent final interval verifier rejects that candidate. Sound final verification survives, but the invalid cache cannot supply the exact-cache progress proof.

1.3 Relation to established methods

Maximum-energy packet selection has a direct optimization interpretation. For the quadratic objective

$$\phi(B) = \tfrac{1}{2}\operatorname{tr}(BSB^{\mathsf{T}}) - \langle B, PH^{\mathsf{T}}\rangle_{\mathrm{F}},$$

the gradient is $BS - PH^{\mathsf{T}} = -F(B)$. The selected packet has largest projected-gradient norm. This is a block Gauss–Southwell mechanism, and the residual cache is maintained by exact linear identities. Greedy coordinate and block selection are established methods [2, 3]. Pathwise Gaussian conditioning and Wasserstein approximation analysis also have substantial prior literature [4]. Formally checked interval reasoning for rounded arithmetic is an established methodology [5].

The contribution presented here is the integrated formal construction: a specified executable trajectory, maintained semantic cache, computable fuel, sufficient verification precision, posterior guarantee, and interval output tied to the actual returned record. The paper proves this construction under its declared hypotheses. It makes no global priority claim or competitive-performance claim from the small archived benchmark.

1.4 Evidence and reproduction

The archived release records successful compilation of 117 active modules and axiom reports for 543 named theorem or lemma declarations. Its allowlist is `propext`, `Classical.choice`, and `Quot.sound`. Those counts include support lemmas and benchmark certificates. They count different forms of evidence rather than independent mathematical discoveries.

The original source release contains successful receipts, source snapshots, logs, dependency records, exact diagnostic outputs, and preserved failures. It excludes the portable compiler and compiled dependency runtime. Fresh publication checks can verify retained source and log hashes and reproduce exact Python examples. Archived compiled-output hashes remain recorded values when the corresponding binaries are absent. The Lean companion states precisely which evidence was replayed here and which remains archived.

The reference Python producer accepts exact rational strings and rejects raw JSON floating-point numbers. Its successful status is provisional until the emitted explicit Lean instance is compiled. A checked instance certifies the rational data present in that source. It does not prove a universal JSON-parser or Python-exporter refinement. The resource-capped producer may fail with a resource limit even when the ideal mathematical solver has finite fuel. These distinctions make the delivered object reproducible and its authority inspectable.

2 A finite Gaussian model with an inspectable input contract

The inference problem concerns one observation of a finite-dimensional latent state. Let the state space be $\mathbb{R}^n$ and the observation space be $\mathbb{R}^p$. A state X has prior law $\mathcal{N}(m, P)$, and its observation is

$$Y = HX + V, \qquad V \sim \mathcal{N}(0, R), \qquad V \text{ independent of } X. \tag{1}$$

Here $H \in \mathbb{R}^{p\times n}$ may be rectangular, and $P \in \mathbb{R}^{n\times n}$ may be singular. The assumptions are $P \succeq 0$ and $R \succ 0$. A singular prior is an ordinary Gaussian probability law concentrated on an affine subspace. The argument does not replace that law by a nonsingular approximation or assign an artificial inverse to its covariance.

We use Euclidean vector norms and the squared Frobenius energy

$$E(A) = \|A\|_{\mathrm{F}}^2 = \sum_{i,j} A_{ij}^2. \tag{2}$$

The distinction between these norms matters. A matrix energy is a finite sum of scalar squares that can be evaluated exactly on rational inputs. An operator norm is used in the proof that this energy controls distribution error. The certificate never asks a producer to supply an unverified operator norm or an approximate eigenvalue calculation.

For an observed value y, put

$$d = y - Hm, \qquad S = HPH^\top + R, \qquad K = PH^\top S^{-1}. \tag{3}$$

The matrix S is the innovation covariance. It is positive definite because, for every nonzero $z \in \mathbb{R}^p$,

$$z^\top S z = (H^\top z)^\top P (H^\top z) + z^\top R z > 0.$$

Consequently the inverse in the definition of K exists even when P has a nontrivial nullspace or H has deficient rank. Every inverse used by the posterior argument is an inverse on the observation space. This is the structural reason singular priors are supported.

2.1 Factor-coded rational inputs

The completed executable contract represents its model by rational matrices and vectors. Its input consists of H, m, y, a prior factor $L \in \mathbb{Q}^{n\times r}$, a noise factor $N \in \mathbb{Q}^{p\times s}$, and a rational number σ. The

covariance matrices are derived from these inputs:

$$P = LL^\top, \qquad R = NN^\top + \sigma I_p, \qquad \sigma > 0. \tag{4}$$

The factor dimensions r and s are finite and need not equal the state or observation dimension. The factors may have dependent columns or zero entries. The prior factor is allowed to have rank below n. The positive floor is checked rather than inferred from a decimal approximation.

This representation makes the covariance hypotheses concrete. For every x,

$$x^\top LL^\top x = \|L^\top x\|^2 \geq 0, \qquad x^\top Rx = \|N^\top x\|^2 + \sigma\|x\|^2.$$

Thus P is positive semidefinite and $R \succeq \sigma I_p \succ 0$. These conclusions are proved from the factors; they are not additional statistical assumptions hidden inside a covariance record. The real semantics interprets each rational coordinate through the exact inclusion $\mathbb{Q} \hookrightarrow \mathbb{R}$. There is no intermediate conversion through hardware floating point.

Factor coding is a declared restriction of the executable input language. The release does not prove that every real positive semidefinite matrix can be supplied as a rational Gram factor, or that every rational positive semidefinite matrix has a rational factor of the chosen dimensions. A user begins with the represented factors. The theorem then concerns the exact real Gaussian model determined by those factors. If an application first approximates measured real data by rationals, an error analysis for that approximation is a separate obligation.

The record `RationalModel` in `Komorebi/RationalCertificateChecker.lean` stores the factors, the floor, and the observation data. Its methods `prior`, `noise`, `innovation`, `residual`, and `innovationVect or` derive the displayed quantities. `RationalModelSemantics.lean` proves their interpretation in real Euclidean spaces, including `prior_real_posSemidef`, `noise_real_posDef`, `noise_real_floor`, and `real_residual`. These are the connection between the exact input record and the probability measures used later.

2.2 Why the noise floor is more than an input check

The same σ that certifies positive noise also bounds the inverse innovation operator. Adding the positive semidefinite term $HPH^\top$ gives

$$z^\top Sz \geq \sigma\|z\|^2 \qquad \text{for all } z \in \mathbb{R}^p. \tag{5}$$

Let $x = S^{-1}v$. Then $Sx = v$, and the Cauchy–Schwarz inequality yields

$$\sigma\|x\|^2 \leq x^\top Sx = x^\top v \leq \|x\|\|v\|.$$

If $x = 0$ the desired bound is immediate. Otherwise division by $\sigma\|x\|$ gives $\|S^{-1}v\| \leq \sigma^{-1}\|v\|$. Hence

$$\|S^{-1}\|_{\mathrm{op}} \leq \sigma^{-1}. \tag{6}$$

This elementary estimate supplies an explicit certificate constant without computing the inverse. The inverse exists in the semantic proof, while the executable stopping test uses σ, an innovation vector, and a residual. This separation allows an approximate gain to be certified independently of the method that produced it.

The source names these steps `innovation_quadraticLowerBound` and `innovation_inverse_norm_le` in `FDInnovationConditioning.lean`. The latter uses the actual inverse identity, so its conclusion is not an assumption about an arbitrary supplied matrix. A false inverse or an overstated floor cannot be inserted as an accepted semantic shortcut.

3 The explicit posterior is a genuine conditional Gaussian law

The exact posterior formula is

$$\nu_y = \mathcal{N}(m + Kd, C_*), \qquad C_* = P - KHP. \tag{7}$$

Writing this formula does not by itself establish conditional probability semantics. A matrix identity could describe a plausible mean and covariance without proving that its Gaussian is a disintegration of the prior and likelihood. The source constructs that disintegration at the level of probability measures. This section records the argument and the precise meaning of the word posterior.

3.1 A positive covariance without an inverse prior

The covariance C_* admits the Joseph representation

$$C_* = (I_n - KH)P(I_n - KH)^\top + KRK^\top. \tag{8}$$

Each term on the right is positive semidefinite. Therefore $C_* \succeq 0$, including at the singular boundary. This avoids any proof based on P^{-1} or C_*^{-1}.

Lemma 3.1 (Exact Joseph identity)**.** *Under the assumptions of Section 2, the covariance in* (7) *equals* (8)*.*

Proof. The innovation equation gives $KS = PH^\top$. Since P and S are symmetric, transposition gives $SK^\top = HP$. Thus

$$KSK^\top = KHP = PH^\top K^\top.$$

Expanding the right side of (8) and collecting its two quadratic terms gives

$$\begin{aligned}(I_n - KH)P(I_n - KH)^\top + KRK^\top \\ = P - KHP - PH^\top K^\top + KSK^\top \\ = P - KHP.\end{aligned}$$

The last line is C_*. Positivity follows because each unexpanded term is a covariance congruence of a positive semidefinite matrix. □

The source proves `josephCovMatrix_eq_posteriorCovMatrix` and `posteriorCovMatrix_posSemidef` in `FDPosteriorCovPSD.lean`. The posterior kernel is then built as a Gaussian affine kernel, rather than as a formal expression requiring a nonexistent Lebesgue density on a singular state space.

3.2 Forward and reverse constructions of the same joint law

The forward construction samples $X \sim \mathcal{N}(m, P)$ and then samples Y with conditional law $\mathcal{N}(HX, R)$. Its joint mean, in the order (Y, X), is (Hm, m). Its covariance is

$$\begin{pmatrix} S & HP \\ PH^\top & P \end{pmatrix}. \tag{9}$$

This joint law is Gaussian because it is an affine image of independent Gaussian inputs. The argument remains valid when the input covariance has zero eigenvalues.

For the reverse construction, sample $Y \sim \mathcal{N}(Hm, S)$ and an independent $E_* \sim \mathcal{N}(0, C_*)$. Define

$$Z = m + K(Y - Hm) + E_*.$$

Conditional on $Y = y$, the law of Z is precisely ν_y. The joint mean of (Y, Z) is again (Hm, m). Its cross covariance is $SK^\top = HP$, and its state covariance is

$$KSK^\top + C_* = KHP + (P - KHP) = P.$$

It therefore has exactly the covariance in (9). Both constructions yield Gaussian measures with the same mean and covariance, so they yield the same joint probability measure. This is a full law identity, rather than a statement that the two constructions merely agree in their first two moments without Gaussianity.

Theorem 3.2 (Gaussian disintegration)**.** *Let $\ell(x, \cdot) = \mathcal{N}(Hx, R)$ be the observation kernel and let $\eta = \mathcal{N}(Hm, S)$ be the evidence law. Then the explicit kernel $y \mapsto \nu_y$ satisfies*

$$\eta(dy)\,\nu_y(dx) = \mathcal{N}(m, P)(dx)\,\ell(x, dy) \tag{10}$$

after exchanging the order of the two coordinates on the right. In particular, η is the actual observation marginal and ν is a posterior version for this prior and likelihood.

Proof. The forward and reverse constructions just described have the same joint Gaussian mean and covariance. The reverse construction is the product of the evidence measure with its affine Gaussian conditional kernel. The forward construction is the product of the prior measure with the likelihood kernel. Their joint equality is (10). Taking the observation marginal yields the stated evidence law, and equality of the joint measures is the defining disintegration property. □

The raw declaration `fd_joint_disintegration` in `FDGaussianConditioning.lean` is this joint equality. `fd_observation_evidence` derives the evidence marginal by applying the first-coordinate marginal to that equality. These theorems use the actual measure constructions, including the joint packing and covariance identities proved in their imported modules.

3.3 Pointwise posterior version and almost-everywhere uniqueness

A regular conditional distribution is determined by its joint law only up to sets of evidence measure zero. The source accordingly proves that the explicit Gaussian kernel agrees with Mathlib's canonical Bayesian posterior evidence-almost everywhere. The relevant declarations are `fdPosteriorKernel_eq_canonical`, `canonicalPosterior_eq_multivariateGaussian`, and `canonicalPosterior_eq_josephGaussian`. Their almost-everywhere quantifier is part of the theorem and must remain visible when translating it to prose.

For a supplied rational observation y, the executable contract uses the explicit version ν_y. That version is defined at every such y and is already proved to disintegrate the model. The solver's distance theorem is pointwise against this version. It does not prove pointwise equality to every possible modification of a canonical posterior kernel on evidence-null sets. The distinction is standard for conditional laws and does not weaken the law-level certificate that the program actually returns.

In particular, observing a single real vector in a continuous model does not require the event $Y = y$ to have positive probability. Conditioning is encoded by a kernel whose integration reconstructs the joint law. The proof cannot be replaced by dividing the probability of a zero-probability event. This is one reason the measure-level construction is essential when the inference contract claims a posterior distribution guarantee.

4 An approximate gain defines a valid Joseph Gaussian

Let $B \in \mathbb{R}^{n\times p}$ be an arbitrary gain candidate. It need not solve the innovation equation and need not have been produced by the packet algorithm. Define

$$m_B = m + Bd, \tag{11}$$

$$C_B = (I_n - BH)P(I_n - BH)^\top + BRB^\top, \tag{12}$$

$$\mu_B = \mathcal{N}(m_B, C_B). \tag{13}$$

Every C_B is positive semidefinite, since it is a sum of covariance congruences. Thus every candidate gain determines a genuine Gaussian law. This is useful for certification: an imperfect candidate is a valid distribution with a measurable error, rather than a covariance formula that can become negative merely because B is inexact.

One should not replace (12) by $P - BHP$. The latter is the exact posterior covariance only when the required innovation identities hold. For an arbitrary candidate it need not even be symmetric. The Joseph form makes the mathematical object independent of those identities, and the final verifier certifies this same object.

4.1 Residual algebra separates the target from its approximation

Define the gain-equation residual

$$F(B) = PH^\top - BS. \tag{14}$$

It can be computed from the model and B without using K. The exact target satisfies $KS = PH^\top$, so

$$F(B) = (K - B)S, \qquad K - B = F(B)S^{-1}. \tag{15}$$

These are exact identities, not perturbative approximations. They explain how a computable defect in a linear equation controls an uncomputed target gain.

Proposition 4.1 (Excess covariance)**.** *For every gain B,*

$$C_B - C_* = (K - B)S(K - B)^\top = F(B)S^{-1}F(B)^\top \succeq 0. \tag{16}$$

Proof. Expansion of the Joseph form gives

$$C_B = P - BHP - PH^\top B^\top + BSB^\top.$$

Subtract $C_* = P - KHP$. Expanding $(K - B)S(K - B)^\top$ and using $KS = PH^\top$, $SK^\top = HP$, and $KSK^\top = KHP$ gives the same expression. The second equality follows from (15) and the symmetry of S. Since $S \succ 0$, its covariance congruence by $K - B$ is positive semidefinite. □

This identity is recorded as `gainJoseph_sub_posterior_eq_gap` and `gainJoseph_sub_posterior_eq_residual`; positivity is `gainJoseph_excess_posSemidef`, all in `FDInexactGainAlgebra.lean`. It shows that the approximate Joseph law has additional covariance relative to the exact posterior. It does not imply that its mean is exact, or that it is the Bayesian posterior of a second model chosen by the program.

4.2 A shared-noise construction of the two laws

Take independent centered Gaussian variables $U \sim \mathcal{N}(0, P)$ and $V \sim \mathcal{N}(0, R)$. For each gain B, define the mathematical affine random variable

$$T_B = m + Bd + (I_n - BH)U - BV. \tag{17}$$

Its mean is m_B and its covariance is C_B, so its law is μ_B. At $B = K$, the exact Joseph identity identifies its law with ν_y. The pair (T_K, T_B), using the same inputs (U, V), is therefore an actual transport plan between the exact posterior version and the approximate law.

This is a mathematical random-variable construction used to prove a coupling bound. The release does not include an executable random sampler or a refinement theorem for a pseudorandom implementation. The raw name `gainSample` denotes the affine map in this construction. The finite executable program returns a represented gain and certified parameter enclosures, rather than samples from a Gaussian distribution.

Let $\Delta = K - B$. Subtraction in (17) gives

$$T_K - T_B = \Delta d - \Delta(HU + V).$$

The random vector $HU + V$ is centered and has covariance S. The difference therefore has mean Δd and covariance $\Delta S \Delta^\top$. Its expected squared Euclidean norm is

$$\mathbb{E}\|T_K - T_B\|^2 = \|\Delta d\|^2 + \operatorname{tr}(\Delta S \Delta^\top). \tag{18}$$

The source proves both marginals and this cost. It does not assume that a producer has supplied a coupling with the desired properties. The load-bearing declarations are `gainSample_law`, `gainSample_exact_posterior_law`, and `inexactGainCoupling_isTransportPlan` in `FDGainSampler.lean`, followed by `inexactGainCoupling_cost_gap` in `FDInexactGainCoupling.lean`.

The common-noise plan need not be optimal. Its value is that it is explicit and its cost has an exact residual expression. The Wasserstein distance takes an infimum over all transport plans, so the cost of any valid plan supplies an upper bound. No closed formula for the optimal transport between two covariance matrices is required.

4.3 Singular support is compatible with candidate certification

The target posterior can be supported on a proper affine subspace while an arbitrary candidate law is supported on a larger set. The construction does not require the candidate to preserve the target's support exactly. It requires a bound on the distance between the two actual measures. The explicit coupling remains meaningful because both measures are obtained from the same independent Gaussian inputs, even if the resulting affine maps have different ranks.

The extreme case $P = 0$ makes this especially clear. The prior is the point mass at m, and the model's observation carries no uncertainty about the state. The exact gain is $K = 0$, and the posterior version is still the point mass at m. For an arbitrary candidate B, however, the Joseph approximation has mean $m + Bd$ and covariance $BRB^\top$. These are valid Gaussian parameters, but they need not describe a point mass. Its residual is $F(B) = -BR$, so the residual certificate measures the error introduced by that candidate rather than declaring every valid Gaussian approximation exact.

For a candidate $B = 0$, both its mean and covariance are exactly those of the point mass, and its residual energy is zero. This boundary calculation shows why the proof separates covariance validity from

posterior accuracy. The Joseph formula supplies validity for every gain. The residual supplies accuracy for the selected gain. Neither property is inferred from the other.

These observations use only the same general matrix and coupling identities. They do not require a positive prior eigenvalue, an inverse covariance on the target support, or an auxiliary regularization parameter. The declared positive floor regularizes the observation covariance, which is precisely the space where the innovation equation must be solved.

5 An inverse-free residual-energy certificate

For probability laws with finite second moment, write

$$W_2(\nu,\mu)^2 = \inf_{\pi\in\Pi(\nu,\mu)} \int \|x-z\|^2 \, d\pi(x,z),$$

where $\Pi(\nu,\mu)$ is the set of transport plans with the stated marginals. The formal source defines the squared quantity as a nonnegative extended-real infimum and obtains the distance by a square-root operation. The explicit finite Gaussian coupling proves that the quantities considered here are finite. The notation W_2 in the paper is the corresponding finite distance.

Using (15) in (18) yields the more informative, but not directly inverse-free, bound

$$W_2(\nu_y,\mu_B)^2 \le \|F(B)S^{-1}d\|^2 + \operatorname{tr}\big(F(B)S^{-1}F(B)^\top\big). \tag{19}$$

The source declaration `posterior_wasserstein2Sq_le_gain_residual` proves this inequality from the exhibited transport plan. The displayed right side is its exact cost. It is not asserted to equal the optimal squared Wasserstein distance.

Lemma 5.1 (Frobenius energy controls the residual cost)**.** *If $F \in \mathbb{R}^{n\times p}$ and $S \succeq \sigma I_p \succ 0$, then*

$$\|FS^{-1}d\|^2 + \operatorname{tr}(FS^{-1}F^\top) \le E(F)\left(\sigma^{-1} + \sigma^{-2}\|d\|^2\right). \tag{20}$$

Proof. For a vector v, rowwise Cauchy–Schwarz gives

$$\|Fv\|^2 = \sum_i \left(\sum_j F_{ij}v_j\right)^2 \le \sum_i \left(\sum_j F_{ij}^2\right)\|v\|^2 = E(F)\|v\|^2.$$

Apply this with $v = S^{-1}d$ and use (6) to obtain

$$\|FS^{-1}d\|^2 \le E(F)\sigma^{-2}\|d\|^2.$$

If f_i is the transpose of the ith row of F, then

$$\operatorname{tr}(FS^{-1}F^\top) = \sum_i f_i^\top S^{-1} f_i \le \sum_i \|f_i\|\|S^{-1}f_i\| \le \sigma^{-1}\sum_i \|f_i\|^2.$$

The last sum is $E(F)$. Adding the two estimates proves the result. □

The formal estimate `quadratic_residual_cost_le_energy` in `FDResidualEnergyCertificate.lean` is slightly more general: it accepts an auxiliary operator Q with a proved bound $\|Q\|_{\mathrm{op}} \le c$. Its application to $Q = S^{-1}$ uses the established noise-floor bound and gives the formula above.

Theorem 5.2 (Residual-energy posterior certificate)**.** *For the model in* (1)*, with $R \succeq \sigma I_p$ and $\sigma > 0$, every gain B satisfies*

$$W_2(\nu_y, \mu_B)^2 \le a(B), \qquad a(B) = E(F(B))\frac{\sigma + \|d\|^2}{\sigma^2}. \tag{21}$$

In particular, for $\varepsilon \ge 0$, the stopping condition

$$E(F(B))(\sigma + \|d\|^2) \le \varepsilon^2\sigma^2 \tag{22}$$

implies $W_2(\nu_y, \mu_B) \le \varepsilon$.

Proof. The common-noise transport plan gives (19). Lemma 5.1 bounds its cost by $a(B)$, establishing the first statement. Since $\sigma^2 > 0$, (22) is equivalent to $a(B) \le \varepsilon^2$. Taking the nonnegative square root gives the asserted distance bound. □

The raw declarations are `posterior_wasserstein2Sq_le_residual_energy` and `posterior_wasserstein2_le_of_residual_tolerance`. Their hypotheses include the covariance assumptions, positivity of the floor, and the floor's quadratic interpretation. In the factor-coded program, those hypotheses are discharged by the represented factors and the checked positive scalar. The user does not supply the conclusion $a(B) \ge W_2^2$ as an input promise.

5.1 What the certificate depends on

The score depends on the proposed gain through $F(B)$, on the observation through d, and on conditioning through σ. The same residual energy can lead to different certified errors for different observations. A large innovation can amplify a gain discrepancy in the posterior mean. A small noise floor makes a residual less informative about the gain error. These are visible features of the bound, rather than implementation effects hidden in a tolerance parameter.

The stopping test is sufficient and can be conservative. A score above ε^2 does not prove that the true distance exceeds ε. The coupling may have a nonoptimal cost, the inverse norm bound may be loose, and later interval evaluation may further enlarge an upper bound. Conversely, a passing verified score does prove the tolerance. The solver's convergence theorem concerns this explicit sufficient test, so its correctness does not depend on predicting how tight it will be on a particular application.

If $F(B) = 0$, invertibility of S gives $B = K$, and the approximation is exactly the posterior version. The source proves this as `gainPosteriorLaw_eq_of_residual_zero`. No inference from approximately zero floating point entries is used in that theorem. For a nonzero tolerance the theorem uses the whole squared energy, allowing a candidate to pass before the gain equation is solved exactly.

5.2 The two output tolerances have different meanings

The completed solver uses a posterior-law tolerance $\varepsilon > 0$ and a parameter-representation radius tolerance $\delta > 0$. The first bounds $W_2(\nu_y, \mu_B)$. The second bounds the radii of outward-rounded intervals containing entries of m_B and C_B. The latter arrays are descriptions of the approximation already certified by the former theorem.

An interval enclosing an entry of C_B is not, by itself, an interval enclosing the corresponding entry of C_*. Nor does a radius at most δ certify a Gaussian formed by treating all interval centers as exact parameters. The represented gain and its Joseph law remain the semantic object. Keeping that object fixed is what permits the solver, the final certificate, and the output enclosure proof to compose without changing the meaning of the returned distribution midway through the computation.

6 Packets resolve a complete residual energy

A packet family in this release is a finite collection of self-adjoint, pairwise disjoint idempotent matrices. Let $\{Q_a : a \in \mathcal{A}\}$ act on the state space and $\{D_b : b \in \mathcal{B}\}$ act on the observation space. For each family, the structural conditions are

$$Q_a^\top = Q_a, \qquad Q_a^2 = Q_a, \qquad Q_a Q_{a'} = 0 \quad (a \neq a'), \tag{23}$$

$$D_b^\top = D_b, \qquad D_b^2 = D_b, \qquad D_b D_{b'} = 0 \quad (b \neq b'). \tag{24}$$

Completeness is a separate condition:

$$\sum_{a \in \mathcal{A}} Q_a = I_n, \qquad \sum_{b \in \mathcal{B}} D_b = I_p. \tag{25}$$

The self-adjoint condition makes these Euclidean orthogonal projectors. Idempotence alone would not give the required energy geometry. Completeness prevents a certificate from forgetting a residual direction outside the selected packet spans.

The two-sided residual packet is

$$F_{ab}(B) = Q_a F(B) D_b. \tag{26}$$

Each packet remains an $n \times p$ matrix, although it is supported on the output and input subspaces selected by its projectors. The notation does not assume a coordinate partition or require projectors to be diagonal. Rational rotated projectors are allowed when their exact entries satisfy the declared identities.

6.1 Orthogonality gives exact accounting

Use the Frobenius inner product $\langle A, C \rangle_{\mathrm{F}} = \operatorname{tr}(A^\top C)$. For $a \neq a'$, self-adjointness and disjointness give

$$\langle Q_a A, Q_{a'} A \rangle_{\mathrm{F}} = \operatorname{tr}(A^\top Q_a Q_{a'} A) = 0.$$

Consequently the energy of a finite sum of output packets is the sum of their energies. The input analogue follows by transposing the matrices. Applying the two decompositions successively gives complete two-sided accounting.

Theorem 6.1 (Complete packet decomposition)**.** *If both families satisfy* (23)–(25)*, then every* $A \in \mathbb{R}^{n \times p}$ *satisfies*

$$E(A) = \sum_{a \in \mathcal{A}} \sum_{b \in \mathcal{B}} E(Q_a A D_b). \tag{27}$$

More generally, for subsets $U \subseteq \mathcal{A}$ *and* $V \subseteq \mathcal{B}$*,*

$$E\left(\left(\sum_{a \in U} Q_a\right) A \left(\sum_{b \in V} D_b\right)\right) = \sum_{a \in U} \sum_{b \in V} E(Q_a A D_b). \tag{28}$$

Proof. The sum of any subfamily of the Q_a is again self-adjoint and idempotent: expansion of its square removes the off-diagonal products, and the remaining diagonal products equal the original projectors. For a self-adjoint idempotent Q,

$$E(QA) = \operatorname{tr}((QA)^\top QA) = \operatorname{tr}(A^\top QA).$$

The trace is linear, so replacing Q by a sum over U proves the output decomposition. Transposition preserves energy and turns right multiplication by the sum over V into left multiplication, proving the input decomposition. Combining these proves (28). Setting $U = \mathcal{A}$ and $V = \mathcal{B}$, completeness reduces the matrix inside its energy to A, proving (27). □

The source formalizes the packet structure in `SelfAdjointPackets.lean`. Its exact energy declarations are `matrixEnergy_sumOn_left`, `matrixEnergy_sumOn_right`, `matrixEnergy_sumOn_both`, and `matrixEnergy_eq_sum_blocks` in `PacketEnergyDecomposition.lean`. The local parent identity does not require global completeness. The final whole-residual identity does. This separation allows local algebra to be used without silently granting full-space coverage.

Equation (28) describes aggregation within a fixed family. It does not by itself certify an algorithm that changes its hierarchy while running. The completed solver fixes its packet families before iteration. Dynamic hierarchy changes belong to a separate extension and require their own state and convergence argument.

6.2 Majorants compose with the posterior certificate

Suppose a producer supplies values u_{ab} and v with proved inequalities

$$E(F_{ab}(B)) \le u_{ab}, \qquad \|d\|^2 \le v. \tag{29}$$

Completeness and nonnegative energy imply

$$E(F(B)) \le U, \qquad U = \sum_{a,b} u_{ab}.$$

Each u_{ab} is nonnegative because it bounds a nonnegative quantity. Likewise $v \ge 0$. The score

$$\widehat{a}(B) = U\frac{\sigma + v}{\sigma^2} \tag{30}$$

therefore bounds $a(B)$ and hence the squared posterior distance. If

$$U(\sigma + v) \le \varepsilon^2\sigma^2, \qquad \varepsilon \ge 0, \tag{31}$$

the same law tolerance follows.

Corollary 6.2 (Packet-majorant tolerance)**.** *Under the model and complete packet hypotheses,* (29) *and* (31) *imply* $W_2(\nu_y, \mu_B) \le \varepsilon$.

Proof. Sum the packet inequalities and apply Theorem 6.1 to obtain $E(F(B)) \le U$. Since $\sigma + \|d\|^2 \ge 0$, and $U \ge 0$,

$$E(F(B))(\sigma + \|d\|^2) \le U(\sigma + \|d\|^2) \le U(\sigma + v) \le \varepsilon^2\sigma^2.$$

Theorem 5.2 now applies. □

The source proves the residual bound from local majorants as `frobeniusSq_le_packet_majorants` and the tolerance result as `posterior_wasserstein2_le_of_packet_majorants` in `FDResidualPacketCertificate.lean`. The innovation-majorant combination is used by the rational checker through `posterior_wasserstein2Sq_le_energy_majorants` in `PosteriorEnclosureBounds.lean`.

Majorants allow different arithmetic representations to feed the same semantic theorem. An exact rational evaluator can choose $u_{ab} = E(F_{ab}(B))$. An outward-rounded evaluator can instead supply a certified upper endpoint. The law proof needs the inequality, not an assumption that every local value was calculated with unlimited precision. The arithmetic chapters explain how the final dyadic evaluator proves those inequalities for its own operations.

6.3 Why partial packet accounting is insufficient for stopping

For a fixed residual, a sum over only some packets is no larger than the full energy. The declaration `sum_block_energies_le` proves this direction. It is a useful local diagnostic, but it cannot replace the whole-energy upper bound in a stopping test. A candidate can have zero residual in the inspected subspaces while retaining a large residual elsewhere.

For example, take the coordinate projectors onto the two axes of $\mathbb{R}^2$. A residual matrix with a nonzero second row has zero energy in the first output packet. Certifying that first packet alone says nothing about the omitted row. Completeness and a bound for every packet are what change local information into a global posterior guarantee. The exact checker tests those structural and local obligations; it does not accept a producer's assertion that a chosen subset happened to be representative.

The canonical selective loop uses the output-packet energies $E(Q_a F(B))$ to choose which gain component to update. Its final residual certificate may use both output and input packets. These roles are compatible but different: output packets guide the iterative correction, while complete two-sided packets organize the final verification. Neither role makes the final distribution error a collection of independent per-packet posterior distances.

7 From finite rational checks to a real-law guarantee

The rational checker turns the previous inequalities into an inspectable decision boundary. A candidate contains a rational gain B, rational packet bounds u_{ab}, and a rational innovation bound v. In the source these are the fields `gain`, `packetUpper`, and `innovationUpper` of `RationalCandidate`. Its validity proposition requires five groups of conditions: a positive noise floor, valid complete output packets, valid complete input packets, every local residual-energy inequality, and the innovation-energy inequality.

The packet check is exact. It checks symmetry, idempotence, pairwise disjointness, and the sum-to-identity condition by finite rational equalities. These identities are not inferred from near-equality of floating point arrays. Exact inclusion into $\mathbb{R}$ preserves matrix products, transposes, finite sums, and order. A rational family that passes therefore constructs a real self-adjoint complete packet family with the geometry required in Section 6.

The energy checks are also exact finite sums. The real interpretation of rational matrix energy is the squared real Frobenius energy of the interpreted matrix. The declaration `rationalEnergy_cast` proves this fact, and `rational_packet_majorant` transfers a rational local upper bound into the real packet inequality. Likewise `rationalVectorEnergy_cast` identifies the rational innovation energy with the squared Euclidean norm used in the probability theorem.

7.1 Validity and tolerance are separate decisions

The first Boolean function is

$$\texttt{checkCandidate}(M, Q, D, c) = \texttt{decide}(\texttt{ValidCandidate}(M, Q, D, c)).$$

It recomputes the finite obligations for the explicit model, packets, and candidate. A claimed successful producer run is not one of its inputs. The equivalence `checkCandidate_eq_true` states that its true result is exactly candidate validity.

The second Boolean function checks $\varepsilon \geq 0$ and

$$\left(\sum_{a,b} u_{ab}\right)(\sigma + v) \leq \varepsilon^2 \sigma^2.$$

This function is `checkTolerance`. The displayed cross-multiplied form avoids evaluating square roots and does not require division in the tolerance decision. Positivity of σ, obtained through candidate validity, justifies interpreting this test as a score bound. A tolerance check alone is not a substitute for the structural and local validity checks.

Theorem 7.1 (Sound exact candidate checking)**.** *Let M be a factor-coded rational model, Q, D rational packet families, and c a rational candidate. If both* `checkCandidateMQDc` *and* `checkToleranceMcepsilon` *return true, then the exact real posterior version represented by M and the exact real Joseph Gaussian represented by the gain of c satisfy*

$$W_2(M.\text{posterior}, M.\text{approximation}(c.\text{gain})) \leq \varepsilon. \tag{32}$$

Here ε is the real interpretation of the rational tolerance.

Proof. The first check gives positivity of the floor and the four finite packet identities for each family. The factors prove the covariance and noise-floor hypotheses. The rational-to-real interpretation lemmas turn every checked local inequality into a bound on the corresponding real packet energy, and turn the innovation inequality into a bound on $\|d\|^2$. Thus the real majorant score in (30) equals the interpretation of the rational candidate score. Corollary 6.2, applied to the second check, gives the law tolerance. The model semantics identifies the two laws in its conclusion with those in (32). □

The exact raw theorem is `checked_tolerance_bounds_posterior` in `RationalPosteriorCertificate.lean`. Its squared precursor, `checked_score_bounds_posterior`, derives a bound by the interpreted rational candidate score. The route passes through `valid_residual_energy_bound` and `valid_innovation_energy_bound`, which explicitly connect the accepted finite data to the analytic quantities. These connections are why the checker certifies a distribution law rather than merely certifying internal consistency of a table of numbers.

7.2 The certificate is independent of proposal quality

A candidate proposal can come from an exact iterative solver, an external optimizer, or a finite list of speculative alternatives. The general candidate theorem requires a passing checker, rather than a theorem about that proposal process. The source's validity-preserving finite acceptance policy keeps a checked incumbent and replaces it only when its finite tests permit the replacement. The declarations `selected_batch_bounds_posterior` and `selected_batch_preserves_posterior_tolerance` show that a finite batch of untrusted proposals preserves the relevant law bound under that policy.

This does not certify the parser or arbitrary code that prepares the explicit data. An exported Lean instance certifies the rational model, packets, gain, and arrays written into that instance. Connecting an external data file to those constants is an additional implementation relation. The first release supplies useful Python producers and exports, but it does not claim a universal refinement of that parser and implementation to the Lean program.

The canonical cached solver is stronger in a different direction. It constructs its own exact cache from the model, produces actual greedy iterates, and proves finite success for positive tolerances under its input hypotheses. The general checker theorem remains useful because the final verification is independent of

any internal claim that the iteration has converged. The subsequent chapters establish how that canonical execution reaches the certificate boundary without assuming candidate success in advance.

8 A singular-prior calculation through the full semantic chain

Consider the rational model

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad H = \begin{pmatrix} 1 & 1 \end{pmatrix}, \qquad R = 1, \qquad m = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \qquad y = 1.$$

It is factor-coded by $L = (1,0)^\top$, a zero noise factor, and $\sigma = 1$. The second state coordinate is deterministic under the prior. Nevertheless, the observation operator includes that coordinate, and all model formulas remain well typed. The innovation covariance is $S = 2$, the innovation is $d = 1$, and

$$K = \begin{pmatrix} 1/2 \\ 0 \end{pmatrix}, \qquad C_* = \begin{pmatrix} 1/2 & 0 \\ 0 & 0 \end{pmatrix}.$$

Thus the explicit posterior is concentrated on the first coordinate axis with mean $(1/2,0)^\top$.

Choose $B = (1/3,0)^\top$. The approximate mean and Joseph covariance are

$$m_B = \begin{pmatrix} 1/3 \\ 0 \end{pmatrix}, \qquad C_B = \begin{pmatrix} 5/9 & 0 \\ 0 & 0 \end{pmatrix}.$$

The residual is $F(B) = (1/3,0)^\top$, with energy $1/9$. Its exact common-noise coupling cost is

$$\|F(B)S^{-1}d\|^2 + \operatorname{tr}(F(B)S^{-1}F(B)^\top) = \frac{1}{36} + \frac{1}{18} = \frac{1}{12}.$$

The inverse-free energy score is larger:

$$a(B) = \frac{1}{9}\frac{1+1}{1^2} = \frac{2}{9}.$$

Both are valid upper bounds on the squared law distance. The difference illustrates the cost of replacing the exact inverse expression by an explicit noise-floor majorant. No inverse prior or inverse posterior covariance was used at any stage.

For coordinate output packets and the single identity input packet, the local residual energies are $1/9$ and 0. Their complete sum is exactly the full energy. If the rational tolerance is $\varepsilon = 1/2$, the squared tolerance is $1/4$, and $2/9 \le 1/4$. The exact certificate therefore proves the posterior-law tolerance for this candidate. This is a worked arithmetic specialization of the general theorems, not a claim that a fresh Lean instance of these particular constants has been compiled in this publication run.

The example also separates three objects that can otherwise be confused: the target conditional Gaussian ν_y, the gain-defined Joseph approximation μ_B, and any finite interval arrays used to describe the latter. The posterior theorem relates the first two probability laws. The output theorem encloses parameters of the second. Their composition provides a usable represented approximation whose law error and representation error are stated separately.

9 An exact selective iteration with a preserved cache

The posterior certificate becomes an algorithm when it is paired with a gain construction that is guaranteed to reach a passing certificate. The canonical construction in Core 1.0.0 performs exact rational selected-packet updates. It retains enough residual information to choose the next update without rebuilding the complete residual after every step. Its final certificate and parameter representation use outward-rounded dyadic arithmetic. These are separate stages with separate correctness arguments.

Throughout this section, the state dimension and observation dimension are finite. The supplied rational model has

$$P = LL^{\mathsf{T}}, \qquad R = NN^{\mathsf{T}} + \sigma I, \qquad \sigma > 0,$$

and observation operator H. Put

$$S = HPH^{\mathsf{T}} + R, \quad v = y - Hm, \quad F(B) = PH^{\mathsf{T}} - BS, \quad E(X) = \|X\|_{\mathrm{F}}^2.$$

The output packet family $(Q_a)_{a\in A}$ consists of rational self-adjoint, mutually orthogonal projectors with sum I. The input family $(D_b)_{b\in D}$ has the same properties on the observation space. The output family controls selection and cache maintenance. The input family participates in the independent final certificate. No commutation relation between S and the input projectors is assumed.

Preprocessing forms three model-dependent constants:

$$\tau = \frac{1}{\sigma + \operatorname{tr} S}, \qquad T = I - \tau S, \qquad w = \frac{\sigma + \|v\|^2}{\sigma^2}. \tag{33}$$

All are rational. They are the fields of the record `Komorebi.CachedPosterior.Kernel`, produced by `Komorebi.CachedPosterior.prepare`. The correctness theorems use this actual preparation, rather than an arbitrary record whose fields happen to have the same names. A wrong transition matrix or weight would invalidate the correspondence between the stored score and the posterior certificate.

Definition 9.1 (Cache validity)**.** A cache state is a tuple $c = (B, (G_a), (e_a), t)$ satisfying

$$G_a = Q_a F(B), \qquad e_a = E(G_a), \qquad t = \sum_{a\in A} e_a. \tag{34}$$

The cached score is $s(c) = tw$.

Validity is an exact equality invariant. It does not mean that a stored residual is approximately near a true residual, nor that its energy is merely an upper bound. Initialization evaluates $F(B_0)$ once, projects it onto every output packet, computes every energy, and sums them. The theorem `initialState_valid` follows directly from the definitions. It requires no prediction about future iterates.

Orthogonal complete packets account for the whole energy:

$$\sum_{a\in A} E(Q_a F) = E(F).$$

Consequently a valid cache has $t = E(F(B))$ and

$$s(c) = E(F(B))\frac{\sigma + \|v\|^2}{\sigma^2} = a(B). \tag{35}$$

The right-hand side is the exact residual score used by the posterior error certificate. The declarations `total_eq_residual_energy` and `score_eq` establish these two identifications. The total is therefore a meaningful global stopping signal even though only one of its summands changes in an iteration.

For a selected label a, the executed update is

$$\begin{aligned} B' &= B + \tau G_a, & G'_a &= G_a T, & e'_a &= E(G'_a), \\ G'_b &= G_b \quad (b \neq a), & e'_b &= e_b \quad (b \neq a), & t' &= t - e_a + e'_a. \end{aligned} \tag{36}$$

This is the definition `Komorebi.CachedPosterior.step`. It computes the new selected residual from the old selected residual and the precomputed transition. It retains unselected entries and updates the scalar total by a difference. The implementation does not define its new state by recomputing $F(B')$ and projecting that matrix onto every packet.

Proposition 9.2 (Preservation by the executed update)**.** *If the output packets satisfy the stated projector assumptions, c is valid, and the kernel is* `prepared`*, then the state in* (36) *is valid.*

Proof. By cache validity, $G_a = Q_a F(B)$, so the gain update is the exact packet step $B' = B + \tau Q_a F(B)$. Expand its residual:

$$F(B') = F(B) - \tau Q_a F(B) S.$$

Multiplication on the left by the selected projector gives

$$Q_a F(B') = Q_a F(B) - \tau Q_a^2 F(B) S = G_a (I - \tau S) = G_a T.$$

For $b \neq a$, mutual orthogonality gives $Q_b Q_a = 0$ and hence $Q_b F(B') = Q_b F(B) = G_b$. This proves every residual equality in the new cache. Its energy equalities hold because the selected energy is evaluated from its new residual and the remaining residual-energy pairs are retained. Finally, replacing one term in a finite sum changes that sum by exactly $-e_a + e'_a$. Thus $t' = \sum_b e'_b$. All three clauses of validity are preserved. □

This proof is realized by `step_valid`, using the algebraic identities `selected_residual` and `unselected_residual`. The declarations `step_other_residual` and `step_other_energy` expose the unchanged entries independently. The distinction between proving an invariant for an executed update and postulating it for a desired state is decisive: the later termination theorem uses the invariant at every state visited by the program.

The locality is on the output side. For an input projector D_b,

$$F(B') D_b = F(B) D_b - \tau Q_a F(B) S D_b.$$

The factor SD_b can mix input directions. The source declaration `input_packet_effect` makes this effect explicit. Complete input packets can still account for certificate energy, but an input packet decomposition does not by itself make the iteration independently local in both indices.

10 Selection, exact trajectory equality, and finite fuel

The algorithm chooses a largest stored output energy. Its finite scan has a seed label and an ordered list. It replaces the incumbent label only when a scanned label has strictly greater energy. On an equality it preserves the incumbent. This tie rule appears in the definition `winner`, whose comparison is `ea<eb`. The scan may contain repeated labels. Completeness requires that every output label occurs in the scan, while the seed supplies an inhabitant of the label type. The proof counts the cardinality of the finite label type, not the length of a possibly repeated scan.

`winner_ge_initial` and `winner_ge_member` prove that the final winner has energy at least that of the seed and of every scanned label. Under complete coverage, `selectPacket_max` therefore supplies a genuine maximum over all output packet energies. The argument does not prescribe a fixed schedule. The maximum is selected again from the current state after every update.

Proposition 10.1 (Exact selection and gain equality)**.** *For a valid cache, cached selection returns the same label as a scan of the recomputed energies $E(Q_a F(B))$. Starting from an initialized cache and using the prepared kernel, its gain after n updates equals the recomputed greedy gain after n updates, for every $n \in \mathbb{N}$.*

Proof. Each stored energy equals the corresponding recomputed energy by (34). Both scans therefore receive the same energy function, the same seed, the same list, and the same strict comparison. Their outputs are equal, including ties. This is `chosen_eq`. The cache gain update then equals the recomputed greedy step by `advance_gain`.

The initial gains agree at $n = 0$. Suppose they agree at n. The cache at that index is valid by repeated application of `step_valid`. Equality of selection and equality of the gain step give equality at $n + 1$. Induction proves the entire trajectory identity. The formal statements are `trajectory_valid` and `trajectory_gain`. □

This is stronger than convergence to the same limit. At a fixed index, the two rational gain matrices are identical. As a result, the convergence proof for the recomputed greedy algorithm transfers directly to the cached execution.

Write $A_{\mathrm{count}} = |A|$ for the positive number of output labels, and set

$$\rho = 1 - \frac{\sigma}{\sigma + \operatorname{tr} S}, \qquad q = \rho^2, \qquad r = 1 - \frac{1 - q}{A_{\mathrm{count}}}. \tag{37}$$

The model factorization and positive noise floor imply that S is positive semidefinite with quadratic lower bound σ. Its trace is nonnegative, and $\sigma + \operatorname{tr} S$ is a valid positive spectral cap. Thus $0 \le \rho < 1$, $0 \le q < 1$, and $0 \le r < 1$. No eigenvalue oracle is needed to construct these rational constants.

The transition contracts every residual matrix:

$$E(XT) \le qE(X).$$

If label a is updated, its residual energy contracts by this inequality and all other output residuals remain unchanged. Orthogonal energy accounting gives

$$E(F(B')) \le E(F(B)) - (1 - q)e_a.$$

Because a is a maximum, $e_a \ge E(F(B))/A_{\mathrm{count}}$. It follows that

$$E(F(B')) \le rE(F(B)), \qquad a(B_n) \le r^n a(B_0). \tag{38}$$

The energy share is `selected_energy_share`; one-step contraction is `greedy_energy_contract`; and the score estimate is `greedy_score_geometric`. The estimate allows singular P, arbitrary initial rational B_0, rectangular H, and rational rotated packets under the same projector assumptions.

A convergent sequence alone would not define a terminating program with an explicit natural-number budget. The source supplies a logarithm-free fuel:

$$N = \left\lceil \frac{a(B_0)}{(\varepsilon/2)^2(1 - r)} \right\rceil_{\mathbb{N}}, \qquad \varepsilon > 0. \tag{39}$$

Here $\lceil\cdot\rceil_{\mathbb{N}}$ is the natural-number ceiling used by `toleranceFuel`. The canonical `adaptiveFuel` evaluates this expression from the actual model, initial gain, tolerance, and number of labels.

Lemma 10.2 (Rational fuel sufficiency)**.** *For $0 \le r < 1$ and $n \ge N$, the geometric envelope obeys $r^n a(B_0) \le \varepsilon^2/4$.*

Proof. The source first proves

$$r^n\big(1 + n(1-r)\big) \le 1.$$

At the induction step, the inequality $r(1 + (n+1)(1-r)) \le 1 + n(1-r)$ reduces to nonnegativity of $(n+1)(1-r)^2$. The natural ceiling in (39) gives $a(B_0) \le n(\varepsilon/2)^2(1-r)$. Multiplying by the nonnegative r^n and applying the preceding bound yields the required squared tolerance. The proof remains valid when the contraction factor is zero. The source statements are `geometric_bernoulli`, `geometric_fuel_sufficient`, and `greedy_score_at_fuel`. □

This fuel is conservative. The geometric rate is substantially sharper than the reciprocal envelope used to obtain a simple rational fuel, and the executed search can stop much earlier. Neither the large natural fuel nor its existence is a claim about a practical time or memory budget.

`search` tests the exact cached score against $\varepsilon^2/4$ at the current state. If the test fails and fuel remains, it advances once and decrements fuel. At fuel zero it still tests the last permitted state. This convention permits success after exactly N updates. `search_score` proves that every returned state passes the score condition. `search_origin` proves that a returned state is an actual trajectory state at some $k \le N$. Finally, `search_complete` proves that a passing score at the last permitted state forces the search to return some state, possibly earlier.

Combining preserved validity, exact trajectory equality, and fuel sufficiency therefore proves a finite successful cached search. There is no caller-supplied claim that an iteration has converged. The computed budget and proved invariant supply the necessary conclusion internally.

11 Independent final verification and precision fallback

The exact cached score is used to select a gain and establish termination. The successful gain is still sent to the independent certificate evaluator. That evaluator reconstructs rounded packet residual obligations from the actual model and gain. It does not accept the stored cache energies as certificate evidence. This separation protects successful returns from an incorrect internal stopping signal. The canonical route checks setup, constructs the outward-rounded candidate, and executes its tolerance test. Construction and setup imply that `checkCandidate` is true by theorem. A separate untrusted-candidate checker can recompute exact residual obligations; that is distinct from the canonical rounded construction route.

The definition `solveCached` first checks a positive posterior tolerance, valid model and packet setup, and complete scan coverage. It prepares the kernel, initializes the cache, and runs the exact-score search with its computed fuel. Only the returned gain is passed to `certifyGain`. Thus certificate evaluation is outside the selective update loop. The definition makes one call to this final verification procedure, although that procedure may try several precisions.

The precision schedule consists of the caller's finite list of probes followed by one computed fallback. The probes can be empty, repeated, or unordered. Successful verification records the precision actually used. The appended fallback makes finite success independent of a lucky externally chosen probe.

Let $h_p = 2^{-p}$. The dyadic evaluator propagates rational centers and outward radii through its arithmetic graph. Its budget calculation supplies a nonnegative coefficient $C_{\mathrm{ver}}(M, Q, D, B)$ independent of p, with

$$\widehat{a}_p(B) \le a(B) + C_{\mathrm{ver}}(M, Q, D, B)h_p. \tag{40}$$

Here $\widehat{a}_p$ is the actual rounded candidate score. The full expression for the coefficient is `scoreRoundCoeff`, built from packet residual energy and innovation-vector magnitude and radius budgets. It bounds the entire score graph, rather than rounding only its final exact answer. The theorem `rounded_score_inflation` proves (40).

The common precision selector is the computable integer expression

$$\mathrm{precisionFor}(C, d) = \log_2^{\mathbb{N}}\left(\lceil C/d \rceil_{\mathbb{N}}\right) + 1, \qquad d > 0, \tag{41}$$

where $\log_2^{\mathbb{N}}$ denotes the natural-number logarithm `Nat.log`, including its specified value at zero. It is not a floating-point logarithm. For every precision at least this selected count, `coefficient_mesh_le` gives $C2^{-p} \le d$. The proof uses integer power and ceiling inequalities. The additional bit is a sufficient choice, not a claim that the bit count is minimal.

For final verification, the solver appends

$$p_* = \mathrm{sufficientPrecision}(M, Q, D, B, \varepsilon^2/4).$$

At its returned search state, $a(B) \le \varepsilon^2/4$. The fallback guarantees inflation at most another $\varepsilon^2/4$, so

$$\widehat{a}_{p_*}(B) \le \varepsilon^2/2 \le \varepsilon^2.$$

Positive ε gives a positive inflation allowance. Valid model and packet setup give the candidate check, while this inequality gives the tolerance check. The declarations `certifyRounded_of_slack` and `certifyGain_complete` establish that the fallback must succeed.

The stricter search threshold supplies room for rounded verification. Searching only until an exact score happened to equal ε^2 would leave no positive allowance for an enclosing evaluator. The construction deliberately separates progress from verification slack. Its inherited geometric fuel is computed for $\varepsilon/2$, which is exactly the square-root tolerance corresponding to $\varepsilon^2/4$.

Theorem 11.1 (Successful and complete cached gain construction)**.** *For rational factor-coded model data with $\sigma > 0$, valid complete rational output and input packets, a seed and complete output scan, any positive rational ε, any finite precision probe list, and any rational initial gain, `solveCached` returns a candidate and verification precision. The candidate and tolerance checks pass, and its gain is an actual greedy iterate at an index within the computed fuel. Its Joseph Gaussian has Wasserstein distance at most ε from the explicit posterior version established by the Bayesian semantics.*

The source assembles these conclusions through `solveCached_terminates`, `cached_solver_gain_origin`, and `cached_solver_reaches_posterior_tolerance`. The semantic posterior used here is the proved explicit Gaussian kernel version. Its agreement with Bayesian conditioning is established through disintegration and almost-everywhere agreement with canonical conditional versions, as discussed in the foundations. The theorem at the supplied rational observation does not assert uniqueness of every possible pointwise conditional version at an event of probability zero.

The independent statement `solveCached_sound` is logically different from termination. It says that any successful program return satisfies both Boolean checks without requiring the caller to supply setup-validity proofs. A returned candidate has passed the evaluator's checks. Valid input hypotheses are needed to guarantee that a return exists; they are not external assumptions attached to the meaning of an already successful return.

12 Parameter arrays and the meaning of a complete result

A useful posterior output contains more than a passing gain. For the returned rational gain B, the parameter stage represents

$$\mu_B = m + B(y - Hm),$$
$$V_B = (I - BH)P(I - BH)^{\mathsf{T}} + BRB^{\mathsf{T}}. \tag{42}$$

These are the mean and Joseph covariance of the same Gaussian law certified by the Wasserstein statement. The covariance is positive semidefinite under the model hypotheses even when the prior is singular. The declaration `approximation_eq_output_gaussian` identifies this parameterized Gaussian with the approximation law used in the posterior theorem.

The evaluator returns a ball for each mean coordinate and each covariance entry. For exact rational entry x, a ball (c, r) means $|x - c| \leq r$. Its radius is an arithmetic enclosure error for that parameter of the returned approximation. The output tolerance $\delta > 0$ bounds every such radius. It is independent of the posterior Wasserstein tolerance ε.

Each output arithmetic expression has a precision-independent budget coefficient. The source sums all mean coefficients and all covariance coefficients to obtain C_{out}, defined by `outputCoefficient`. Every individual coefficient is nonnegative and no greater than this sum. It chooses

$$p_{\text{out}} = \text{precisionFor}(C_{\text{out}}, \delta)$$

and evaluates the output graph at that precision. Since every entry radius is bounded by its coefficient times $2^{-p_{\text{out}}}$, the same integer precision bounds all radii by δ. The declarations `mean_coeff_le`, `covariance_coeff_le`, and `evaluate_radii_le` establish this argument. `represent_spec` joins entrywise enclosure and the requested radius bound.

The result record stores verification precision, the checked candidate, parameter precision, and the evaluated arrays. Its contract `ResultValid` contains seven clauses: the candidate check passes, the tolerance check passes, the Wasserstein bound holds, the arrays enclose the returned approximation's parameters, all radii are at most δ, the reported parameter precision equals the actual precision selector, and the arrays equal the actual evaluator at that precision. The final two clauses make the reported representation part of the contract, rather than allowing arbitrary arrays that happen to satisfy a looser description.

Theorem 12.1 (Complete first-release output)**.** *Under the valid-input hypotheses above and $\delta > 0$, `Komorebi.CachedPosterior.runCached` returns a result satisfying `ResultValid`. Every successful return is sound even without setup-validity proofs supplied by the caller. A nonpositive output-radius tolerance is rejected.*

These are precisely `runCached_terminates`, `runCached_sound`, and `runCached_rejects_nonpositive_output_tolerance`. The complete contract is finite-dimensional inference for one observation update. It includes singular priors, rectangular observation operators, gain origin, computable iteration fuel, verification precision, and parameter precision.

Two interpretations must remain separate. The intervals enclose the parameters of the certified Joseph Gaussian. They do not certify another Gaussian obtained by treating interval centers as exact mean and covariance. In particular, rounded covariance centers are not independently proved positive semidefinite. Likewise, δ is not a coordinate error bound against the exact posterior. The Wasserstein guarantee compares laws; the interval guarantee encloses the parameters of one of those laws. A theorem combining either guarantee with a new operation must account for that operation explicitly.

The program is mathematically finite for every positive rational pair (ε, δ) under its declared assumptions. A reference implementation can still encounter a time, memory, or user-supplied resource cap. The Python tools report `RESOURCE_LIMIT` when their caps are reached. Such a cap is a limit of the execution environment, not a counterexample to the finite-success theorem. The result contract also contains no sampler and no guarantee for an external floating-point implementation.

13 A complete coordinate execution

The coordinate fixture has a two-dimensional state and one-dimensional observation:

$$P = \begin{pmatrix} 4 & 0 \\ 0 & 1 \end{pmatrix}, \quad H = \begin{pmatrix} 1 & 1 \end{pmatrix}, \quad R = 1, \quad m = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad y = 1.$$

It is represented by $L = \operatorname{diag}(2, 1)$, $N = 0$, and $\sigma = 1$. The two output projectors select the coordinate axes, and the single input projector is 1. Start with $B_0 = 0$, seed 0, scan $[0, 1]$, probes $[4, 12]$, $\varepsilon = 1/20$, and $\delta = 1/1000$.

Here $S = 6$, $\tau = 1/7$, $T = 1/7$, and $w = 2$. The initial residual is $(4, 1)^{\mathsf{T}}$, with packet energies 16 and 1. Updating one coordinate multiplies its residual by $1/7$ and its energy by $1/49$. The other coordinate remains exactly unchanged. The maximum-energy schedule therefore alternates between labels 0 and 1. The complete pre-update trace is

Updates already made	Selected label	e_0	e_1	t
0	0	16	1	17
1	1	16/49	1	65/49
2	0	16/49	1/49	17/49
3	1	16/2401	1/49	65/2401
4	0	16/2401	1/2401	17/2401
5	1	16/117649	1/2401	65/117649

After six updates, each coordinate has been selected three times. The geometric sum in its gain gives

$$B_6 = \begin{pmatrix} 228/343 \\ 57/343 \end{pmatrix}, \qquad F(B_6) = \begin{pmatrix} 4/343 \\ 1/343 \end{pmatrix}, \qquad a(B_6) = \frac{34}{117649}.$$

The strict search threshold is $\varepsilon^2/4 = 1/1600$. At the previous state, $a(B_5) = 130/117649 > 1/1600$, while $a(B_6) < 1/1600$. Thus six is the actual update count for this stopping rule. The formal fuel is 410093; the program uses only six of these allowed updates. This large difference illustrates the conservative fuel construction, not an ambiguity in the observed trace.

At verification precision 4, the rounded score is $25/16$ and the attempt fails. At precision 12, it is $481/1048576$, which is below the required final threshold $\varepsilon^2 = 1/400$. The solver therefore returns verification precision 12. A failed coarse enclosure is a failed certificate attempt, not evidence that the gain itself is inaccurate. Sufficiently high precision bounds the enclosure inflation.

The exact posterior mean is $(2/3, 1/6)^{\mathsf{T}}$, while the returned Joseph mean is the gain B_6 because $m = 0$ and $y = 1$. Parameter output uses precision 32. Its mean centers are

$$\begin{pmatrix} 2854963683/4294967296 \\ 89217615/536870912 \end{pmatrix}$$

with radius $1/4294967296$ in each coordinate. The covariance centers are

$$\begin{pmatrix} 5726720411/4294967296 & -1431643597/2147483648 \\ -1431643597/2147483648 & 3579145495/4294967296 \end{pmatrix},$$

and the corresponding radii are

$$\begin{pmatrix} 1/536870912 & 7/4294967296 \\ 7/4294967296 & 1/536870912 \end{pmatrix}.$$

Every radius is smaller than $1/1000$. These fractions describe output arithmetic, not uncertainty intervals for model estimation. They enclose exactly the Joseph mean and covariance at B_6.

The exact rational matrix enclosed by those covariance balls is

$$V_{B_6} = \frac{1}{117649}\begin{pmatrix} 156868 & -78432 \\ -78432 & 98041 \end{pmatrix}.$$

It can be checked directly by substituting B_6 into (42). Another useful check starts from the exact posterior covariance $V_* = \left(\begin{smallmatrix} 4/3 & -2/3 \\ -2/3 & 5/6 \end{smallmatrix}\right)$. If $B_* = (2/3, 1/6)^\mathsf{T}$ is the exact gain, expansion of the Joseph expression gives

$$V_{B_6} = V_* + 6(B_6 - B_*)(B_6 - B_*)^\mathsf{T}.$$

The gain discrepancy is $B_* - B_6 = (2/1029, 1/2058)^\mathsf{T}$, which produces the displayed matrix. This second calculation checks the sign of the off-diagonal entries and explains why the approximate covariance has an additional positive semidefinite term. The posterior accuracy certificate concerns the complete mean-covariance law, while the dyadic output stage faithfully represents these rational parameters. The larger output budget can still choose many more bits than the smallest possible entrywise enclosure would require. Its purpose is a guaranteed common precision, not an optimal encoding of this particular covariance.

The module `GeneratedCachedCoordinate.lean` defines this actual model, initial cache, six-step cache trajectory, final gain, certificate, and result. Its declarations check the computed fuel, final gain, exact and rounded scores, each of the six chosen labels, solver return, parameter precision, every displayed center and radius, and the full output return. Its theorem `output_valid` applies `runCached_sound` to the proved return. This is a checked finite program example, while the universal termination theorem covers all valid inputs in the declared class.

An earlier recomputed selective solver verified after each update and could stop at five updates with a passing rounded certificate. That is a different stopping policy. The canonical cached solver first waits for the stricter exact-score margin and then performs final verification. A recomputed baseline using this same strict margin returns the same six-step gain. Comparisons between cached and recomputed execution must align their stopping policy before comparing work.

14 A singular rotated execution

The rotated fixture makes the prior singular and the packets non-coordinate:

$$P = J = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad H = I, \quad R = I, \quad m = 0, \quad y = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

It uses the rectangular prior factor $L = (1,1)^{\mathsf{T}}$, zero noise factor, and $\sigma = 1$. Both output and input packet families are

$$Q_+ = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \qquad Q_- = \frac{1}{2}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}.$$

These project onto the common and antisymmetric directions. They are rational, self-adjoint, orthogonal, and complete. The prior has rank one, but the innovation matrix $S = I + J$ is positive definite because of the positive noise floor.

Take $B_0 = 0$, seed 0, scan $[0,1]$, probes $[4,12]$, $\varepsilon = 1/4$, and $\delta = 1/1000$. Here $\operatorname{tr} S = 4$, $\tau = 1/5$, and $T = I - (I+J)/5$. The initial residual is J, entirely in the Q_+ output packet. Since $JS = 3J$, each selected update multiplies this residual by $2/5$. The Q_- residual remains zero. Thus every selected label is 0 and

$$(e_+, e_-) = (4(4/25)^n, 0), \qquad B_n = \frac{1-(2/5)^n}{3} J.$$

The first four pre-update energy pairs are $(4,0)$, $(16/25,0)$, $(64/625,0)$, and $(256/15625,0)$.

After four updates,

$$B_4 = \frac{203}{625} J, \qquad F(B_4) = \frac{16}{625} J, \qquad a(B_4) = \frac{3072}{390625}.$$

The score weight is $w = 3$, because $\|y\|^2 = 2$. The strict margin is $\varepsilon^2/4 = 1/64$. At the previous state the score is $768/15625 > 1/64$; after the fourth it is below $1/64$. The computed fuel is 4267. Precision 4 produces score $435/64$ and fails; precision 12 produces $9111/1048576$ and passes the final threshold $1/16$.

The exact posterior mean is zero because $Jy = 0$, and the returned approximation's mean is also zero. Its Joseph covariance is

$$V_{B_4} = \frac{130379}{390625} J.$$

The exact posterior covariance is $J/3$. Both laws remain supported in the prior's one-dimensional common-direction subspace. This example therefore combines a singular prior, rotated packet selection, a nontrivial covariance approximation, and a zero mean without using an inverse of the singular prior covariance.

At output precision 32, each mean center is zero with radius $1/2147483648$. Every covariance entry has center $179191533/536870912$ and radius $7/4294967296$. All output radii satisfy $\delta = 1/1000$. The equality of these centers reflects the exact common-direction structure of the returned Joseph covariance. It does not turn the rounded center matrix into the object of the posterior theorem.

`GeneratedCachedRotated.lean` checks the same kinds of finite obligations as the coordinate module: fuel, chosen labels, gain, scores, actual return, parameter precision, arrays, and `ResultValid`. The example is particularly useful for testing whether a presentation has accidentally assumed an invertible prior or coordinate-only packets. Neither assumption belongs to the solver contract.

15 The rounded-cache counterexample and the separate rounded solver

Exact cache validity cannot be reused unchanged after independently rounding its components. The release contains a scalar counterexample with $P = H = R = 1$, $\sigma = 1$, $m = 0$, $y = 1$, and $B_0 = 0$. Then $S = 2$, $\tau = 1/3$, $w = 2$, and the first exact update is

$$B_1 = 1/3, \qquad F(B_1) = 1/3.$$

The naive route separately rounds the gain center and residual center downward to zero fractional bits. Both become zero. It then computes the stored energy from the rounded residual, obtaining a stored score of zero.

The actual residual at its stored gain is instead

$$F(0) = 1, \qquad a(0) = 2.$$

Thus the state claims exact internal convergence while its true residual is nonzero. The equality $G = QF(B)$ has been destroyed by rounding independently related quantities. The source module `CachedRoundingBoundary.lean` proves `naive_score_zero`, `true_score_two`, and `naive_cache_invalid`. For $\varepsilon = 1/10$ and probe [12], it also proves `naive_final_verification_rejected`: `certifyGain` returns no candidate.

This result identifies both a failure and a protection. The naive rounded cache cannot support the exact-cache termination proof. The independent final verifier still rejects its false stopping signal, preserving the meaning of successful returns. The example does not rule out a rounded selective cache with propagated error bounds. Such a construction needs an enclosure relation, an error budget for each selected update, and a selection argument that tolerates uncertain energies. Those obligations are a separate extension.

Core 1.0.0 already retains a different, completed rounded solver. Its update is a genuinely rounded full-gain Richardson step. It neither claims exact selective cache invariants nor selects a largest cached packet. Its declarations expose the whole-update defect instead. If $\widetilde{B}_{n+1}$ is the rounded update and B^{ideal}_{n+1} is the exact full step at the current gain, then

$$F(\widetilde{B}_{n+1}) = F(\widetilde{B}_n)T - (\widetilde{B}_{n+1} - B^{\mathrm{ideal}}_{n+1})S.$$

This is `rounded_residual_step`. The defect is explicitly included in the tracking and termination argument.

Writing d for the number of gain entries and $h_t = 2^{-t}$, the source establishes an all-iteration tracking bound

$$E(\widetilde{B}_n - B^{\mathrm{exact}}_n) \le \frac{6dh_t^2}{(1-q)^2}.$$

Its residual-score comparison has the form

$$a(\widetilde{B}_n) \le 2a(B^{\mathrm{exact}}_n) + C_{\mathrm{round}}h_t^2, \qquad C_{\mathrm{round}} = \frac{12dE(S)w}{(1-q)^2}.$$

The source names are `roundedGain_tracking`, `roundoffScoreCoeff`, and `rounded_score_compare`. The first term is ideal progress. The second is a precision-dependent floor. Both are needed.

The chosen update precision `updateAccuracy` places the floor below $\varepsilon^2/8$. The rounded fuel `roundedFuel` drives the exact full-step score below $(\varepsilon/4)^2 = \varepsilon^2/16$. The factor of two in the comparison therefore gives another $\varepsilon^2/8$, so the actual rounded iterate reaches score at most $\varepsilon^2/4$. The same independent final precision fallback can certify it. `solveRounded_terminates` proves finite successful return, and `rounded_solver_gain_origin` identifies the result as an actual rounded iterate within its declared fuel.

There are consequently two completed branches: exact selective updates with an exact cache, and genuinely rounded full-gain updates with an explicit perturbation bound. A rounded selective cache combining their advantages remains an extension. No proof for one branch is presented as a theorem about the other.

16 Measured work, execution boundaries, and extensions

The cached representation reduces the work needed to maintain packet residuals relative to recomputing every packet at every selected update. It stores the projected residual matrices and energies, retains unselected objects, and updates the total by one scalar difference. Selection still scans the stored energies. The amount of storage depends on the number and shape of the represented packet matrices. Rational numerator and denominator growth also matters in execution. The algebraic locality theorem alone is not a formal bit-complexity estimate.

The archived bounded benchmark compares cached selective, recomputed selective, and full Richardson reference implementations on seven cases of dimensions at most eight. All use exact fraction arithmetic, the same zero-skipping matrix backend, the same strict exact-score stopping margin, and common requested posterior and output tolerances. All include setup, final rounded certification, and parameter output. The recomputed and cached selective branches return identical candidates. The full baseline maintains a full residual by recurrence and avoids unnecessary packet scans.

One warmup precedes five repetitions with rotating method order. Archived median producer times and selective-iteration ratios are

Case	Cached ms	Recomputed ms	Full ms	Iteration ratio
coordinate	2.38	2.46	2.33	1.94
rotated	4.38	4.45	4.28	2.16
sparse4	13.84	14.52	13.79	2.51
dense4	16.34	17.96	16.34	3.11
sparse8	171.98	174.30	169.87	4.62
dense8	200.69	235.15	195.44	5.61
conditioned4	19.16	21.36	17.46	3.08

The final column is recomputed-selective iteration time divided by cached iteration time. It excludes setup, verification, and output. The producer columns include these stages and exclude export and Lean checking. These are archived reference measurements, rather than fresh timing measurements from this publication session.

The cache improves selective iteration in these cases, yet these measurements establish no overall advantage over the efficient full baseline. Several cases require substantially more selective updates than full updates. The dense8 case, for example, takes 67 cached selective updates and 9 full updates. Final certification and output account for a large part of total production cost. The bounded result supports a narrower conclusion: preserving a packet cache avoids unnecessary selective recomputation, while end-to-end performance depends on more than the selective update loop.

Every method-case pair also exported an explicit final candidate certificate, for 21 checked benchmark outputs. Their archived Lean checks certify those explicit rational gains and posterior tolerances. They do not certify the Python timing trace, counters, or numeric parameter arrays from the benchmark runs. The two generated coordinate and rotated modules have the stronger finite-program and output-array checks described above. A universal theorem about the mathematical output evaluator is also distinct from a universal refinement theorem for a Python implementation of that evaluator.

The archived per-instance Lean checking measurements range from seconds to tens of seconds and dominate millisecond producer times. Each includes a fresh pinned Lean process, import, elaboration, and kernel checking with common dependencies already compiled. The dependency build is accounted

for separately. One checking time per export cannot reliably rank small differences. The source reports preserve these distinctions, rather than combining an iteration ratio with an unrelated claim of total acceleration.

The producer/exporter boundary is similarly precise. Python parses user data, executes a reference algorithm, and emits a Lean instance containing explicit rational data. Successful checking establishes the statement about that emitted instance. A universal theorem identifying arbitrary parser input with the emitted model is outside this release. The output record is a mathematical representation with checked arithmetic semantics, rather than a claim about every external file format or hardware arithmetic instruction.

The most direct extensions now have concrete obligations. A rounded selective cache must propagate residual and energy enclosures and prove progress under its actual selection rule. Approximate maximum selection needs a quantitative energy share bound. A changing packet hierarchy must preserve global accounting across refinement and coarsening. Uncertain model inputs need their own error budget in both the certificate and the posterior semantics. Sequential filtering must account for repeated updates and transport of earlier approximation error. Large-scale performance work must measure preprocessing, final certification, output representation, rational bit growth, memory, and checking cost as well as iteration kernels.

These extensions do not leave the first-release theorem unfinished. The closed contract remains exact rational, finite-dimensional, single-update Gaussian inference with complete fixed packets, positive tolerances, a terminating gain construction, independent dyadic verification, and enclosed parameters of the certified Joseph law. The rounded-cache counterexample and bounded performance result make the limits of that contract inspectable and give specific targets for the next stage of development.

17 Assurance, reproducibility, and the completed scope

The result has a mathematical input boundary, a formal program boundary, and an implementation boundary. These boundaries can be read directly from the declared functions and predicates. Rational factor-coded data determine the Gaussian model. Complete rational packet families determine the permitted decomposition. A successful formal program return determines the candidate gain and parameter arrays. The Python producer and a physical execution environment require additional correspondence evidence.

Object	Established conclusion	Evidence role
Factor-coded model	Prior covariance PSD, innovation invertible at positive noise floor	Internal algebra and Gaussian semantics
Posterior kernel	Explicit Gaussian version disintegrates the forward law; canonical agreement evidence-a.e.	Internal measure-level proof
Joseph gain	A valid Gaussian approximation and residual-based W_2 bound	Common-noise coupling and matrix bounds
Exact packet cache	Correct residuals and energies at every iterate; same gains as recomputation	Initialization and preservation induction
Computed search fuel	Successful search under the positive-input hypotheses	Geometric contraction and computable sufficient fuel

Object	Established conclusion	Evidence role
Rounded verifier	Candidate validity and posterior tolerance follow from successful checking	Outward enclosure and inflation proofs
Parameter arrays	Enclose this approximation's parameters, with radii at most δ	Declared evaluator and precision theorem
Python instance	Exact emitted rational data can receive a Lean certificate	Compilation needed for that instance
Fresh publication runs	Specified examples and diagnostic assertions reproduced	Bounded executable evidence
Archived Lean release	117 successful module receipts, 543 axiom-audited declarations	Preserved historical kernel evidence

17.1 What was rechecked for this publication

The standalone Core 1.0.0 archive was recovered and its SHA-256 compared with its separately delivered checksum. The source release and its history were retained intact. The fresh inventory checks the archived manifest, current source bytes, retained source snapshots, logs, dependency-source records, declaration locations, and axiom-report coverage. An independent parser finds the same 543 theorem and lemma declarations. Verbatim excerpts in the Lean companion are matched back to their actual source intervals.

Fresh exact computations reproduce the coordinate and rotated returns, including their gains, score values and every output interval. Independent rational matrix calculations reconstruct the exact posterior and the Joseph covariance. They check the coupling cost, residual bound, cache identities, stopping thresholds, and the rounded-cache obstruction. The computation record states its finite domain and distinguishes producer checks from formal program theorems. The source archive remains byte-for-byte unchanged.

The figures use exact input data and explicit formulas. Floating evaluation is used to draw curves and contours, with that visual role identified in their captions. No sampled scatter cloud is used as a proof of a Gaussian identity. Archived benchmark timings are marked as archived. Fresh timings of publication checks are not substituted into the original performance table.

All three documents are compiled and rendered for page review. The audit records track any mathematical transcription corrections and the inspected page coverage. Build warnings and PDF structure checks address layout, references, links and glyphs. These checks verify the publications as artifacts; they do not extend the solver's formal scope.

The original release records a clean rebuild after removing active project caches while reusing pinned Mathlib caches. It preserves successful source and log receipts, corrected failures, and recorded compiled-output hashes. Compiled output payloads and the portable runtime are not included in the source archive. A fresh source audit therefore cannot recompute absent compiled-output hashes. Fresh Lean compilation was unavailable for this publication edition. The companion provides the exact pinned reproduction route and keeps this limitation explicit.

17.2 Interpreting cost evidence

The controlled benchmark contains seven models with dimensions at most eight, using five measured repetitions after warmup. Cached and recomputed selective runs use matching tolerances, backend and

stopping conditions and return identical gains. Their iterative work differs because the cache updates only the selected packet. Precomputation, final interval verification, parameter representation, export and proof checking remain parts of a full execution.

The archived measurements support a reduction in selective iteration cost. They do not establish overall superiority to the efficient full Richardson baseline. Operation counters describe the tested implementation's scalar arithmetic and comparisons. They do not constitute a formal bit-complexity theorem or a prediction for large matrices. Exact rational numerator and denominator growth, memory use, model preparation and certificate evaluation would need to be included in a practical scale study.

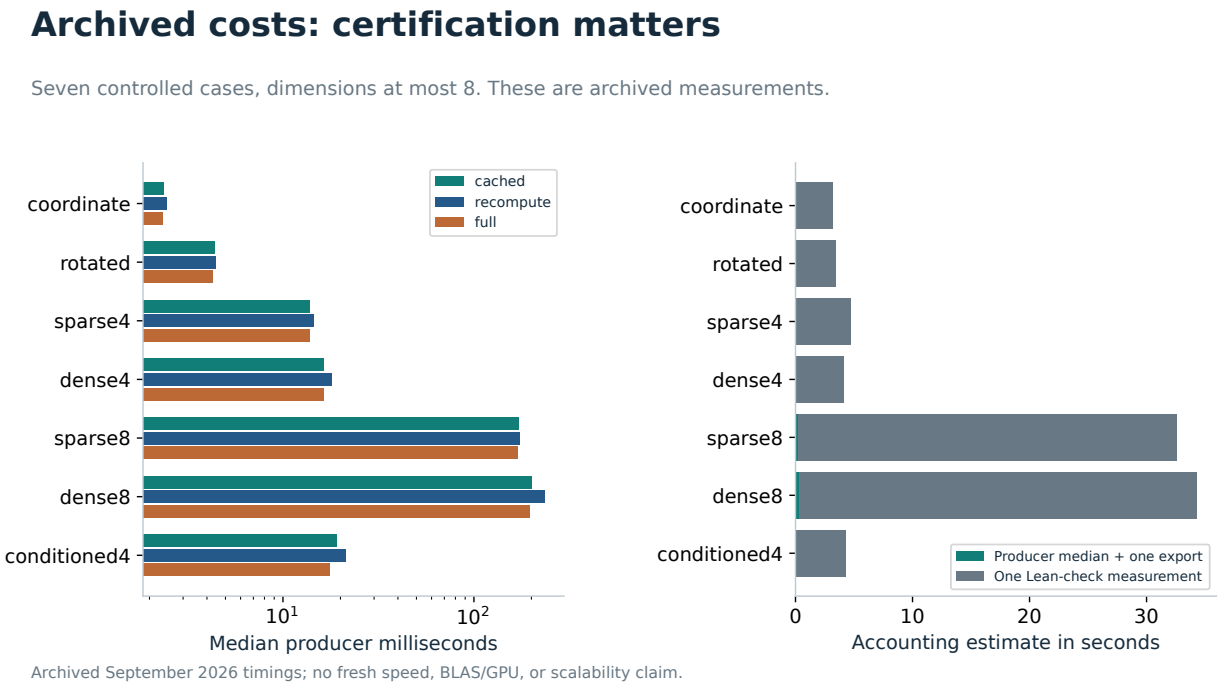


Figure 2: Archived bounded cost comparison. The plotted measurements concern the seven declared small models and their original reference environment. Iterative cache savings and whole-producer cost are different quantities.

17.3 Extensions that preserve the completed result

The first extension is a rounded selective cache. Its invariant must carry uncertainty about the semantic residual of the stored gain. A center-only update cannot reuse the exact equality $G_a = Q_a F(B)$ after independent rounding. The checked counterexample identifies the failed implication. A valid replacement could track residual enclosures and allocate update, selection and final-verification error budgets. That construction would require a new progress proof and a new origin statement.

Sequential filtering is another extension. A repeated update needs a contract for how the returned representation becomes the next prior, including propagation of approximation error and data uncertainty. The present output arrays represent the returned Joseph law. Replacing them by centers or repeatedly treating an approximate prior as exact would add unstated obligations. The single-update theorem remains a reusable component for such a future composition.

Changing packet geometry also requires a transition contract. The exact invariant uses one fixed complete orthogonal family. A new family can represent the same residual, but its cache must be reconstructed or transported with a proved relationship. Adaptive choice of a label within the fixed family is already covered. Adaptive replacement of the family is a separate operation.

A universally refined host implementation would connect parsing, rational representation, code generation and execution to the defined Lean program. The current exporter provides explicit candidate data and checkable instance source. Its bounded diagnostics support that route without supplying a universal refinement theorem. Executable sampling and hardware floating-point implementations similarly need their own input, arithmetic and output contracts.

18 Conclusion

Certified Inference Core 1.0.0 closes a precise finite-dimensional inference construction. Its exact cached packet computation has a maintained semantic invariant, a proved greedy trajectory, and computed finite fuel. Its final outward-rounded verifier connects the returned gain to a posterior-measure tolerance. Its parameter evaluator returns enclosing arrays for the same Gaussian approximation, with a prescribed radius bound. These are composed conclusions about one actual program result.

The worked examples show the construction operating with rectangular observations and a singular prior. The rounded-cache counterexample supplies an equally useful boundary: the proof identifies which state equality an attractive alternative loses. Together, the positive result and the obstruction make the arithmetic contract inspectable. The extensive visual and Lean companions provide routes from the picture to the exact formula, and from the formula to the source declaration.

A Source and reproduction map

The authoritative archive is `inputs/FD_Certified_Inference_Core_v1.0.0_Checked.zip`. Its SHA-256 is

`844ae280ea4be7e7c25906e00733f0130996b32aa110c53951df00f8c18fa488`.

Extract it under `source`. The publication package includes all editable LaTeX, twenty figures, their common generator, fresh exact results, source maps, computation provenance, independent audit and page-review records. The intact archive contains the underlying formal modules, Python reference tools, historical receipts, failed attempts, benchmark records and runtime restoration metadata.

The formal pins are Lean 4.34.0-rc2, compiler commit `6a10ac8c22beadecabdbb0919c2b50214762f91d`, and Mathlib commit `aeb72f1e9981b6d35b426b16dca7759be3e7ba06`. The dependency manifest supplies the remaining pins. Read the original `README.md` and `RELEASE_SCOPE.md` before rebuilding. Original historical receipts should remain separate from newly generated evidence. The main theorem is in `Komorebi/CachedPacketOutput.lean`; its result predicate is in `Komorebi/PosteriorOutputSolver.lean`.

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Artifact value, guarded operations, and calibrated planning

Traditional paper

Komorebi P.

Standardized publication edition / 8 October 2026

REALIZATION CALCULUS

Reading scope

A source-grounded exposition of the cumulative AVOC Formal v0.41.0 theory. The companion documents provide a worked visual route and a guide to the actual Lean declarations, evidence, and reproduction commands.

Abstract

An artifact can be correct and still be a poor computational investment. A probability estimate can be useful and still be invalid for the stream on which it is applied. A recovery protocol can preserve a logical checkpoint and still depend on an unverified external execution contract. AVOC develops these distinctions within a finite formal architecture for artifact planning. Its cumulative theory combines task sufficiency, stochastic and Bayesian information transport, exact checking for declared artifact languages, finite-horizon planning, robust quote bounds, rational finite-sample calibration, and recoverable state transitions. The latest recorder block connects complete executions of a source-generative lifecycle machine to its actual terminal sample prefixes and proves an unconditional bound on uncertified policy issuance. We present central definitions and proof mechanisms, worked exact examples, and a theorem-to-source map. We also explain a completion-order obstruction: honest and complete logging can turn independent attempted outcomes into biased reported prefixes. The publication is based on the preserved v0.41.0 sources and their archived Lean verification. Fresh source-inventory and exact Python checks are reported separately from compilation, which was not repeated for this edition.

Reading contract

This paper gives a mathematical exposition of the cumulative formal project, with explicit contracts for applying its results to external computational services. AVOC is the project name used throughout.

There are three complementary documents. The main paper develops the definitions, central arguments, and precise scope. The visual guide works through concrete examples with exact Python-generated figures. The Lean companion identifies actual definitions and theorem declarations, explains their hypotheses, and gives the pinned reproduction workflow. All source files, review notes, figure programs, and the original formal archive accompany the publication set.

The following evidence vocabulary is used throughout. A *formal result* refers to a statement in the delivered Lean theory with archived positive-build and axiom evidence. A *prose reconstruction* explains that argument in conventional notation. An *exact example* is a rational or finite computation with an explicit tested family. An *external premise* is an assumption about a provider, host, storage layer, cost conversion, or sampling process that is not derived from accepting a formal certificate. These categories are deliberately kept distinct.

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1 Introduction: correct artifacts and correct decisions

A useful computational system must decide more than whether an answer is correct. It must decide whether to search for an answer, inspect an existing candidate, reuse a checked result, or stop. It must account for checking and acquisition costs. When those costs and outcomes are uncertain, it must distinguish a plausible ranking from a certified decision. When work is interrupted, it must preserve enough state to resume without silently charging or installing the same logical result twice.

AVOC organizes these questions around three related objects: the semantics of an artifact, the value of an action, and the evidence supporting the action's quoted parameters. None can substitute for the others. An exact affine checker can establish that a stored factorization computes the desired map for all inputs. It cannot establish how likely a future provider is to return such a factorization. A confidence procedure can bound a failure event under a stated source law. It cannot establish that externally supplied log entries satisfy that law. A checkpoint theorem can establish idempotent acceptance of a logical commit. It cannot establish exactly-once execution of a physical operation.

This separation makes the architecture compositional. The artifact layer supplies a declared semantic language and checker. The decision layer supplies a state space, finite admissible action sets, finite observation alternatives, rewards, and transitions. The state space itself need not be finite. The uncertainty layer supplies frozen quote bands and compatible realizations. The observation layer supplies finite source laws, scheduled calibration, and a recorder whose terminal samples are connected to those sources. The recovery layer specifies how typed evidence and logical accounting are retained across replay.

1.1 The central chain

The central statistical composition has the following form. A finite rational probability space supplies per-identity latent Boolean sources satisfying the stated conditional fresh-outcome law. A predetermined finite schedule fixes the checkpoints and their error allowances. The source-generative recorder constructs terminal reports from those sources. A theorem identifies each observed sample list with the appropriate source prefix. Calibration then bounds failure of simultaneous scheduled probability-band validity. On the simultaneous-coverage event, every issued plan has the stated episode certificate. Outside that event a plan may be uncertified; the final theorem bounds the unconditional risk of that issuance.

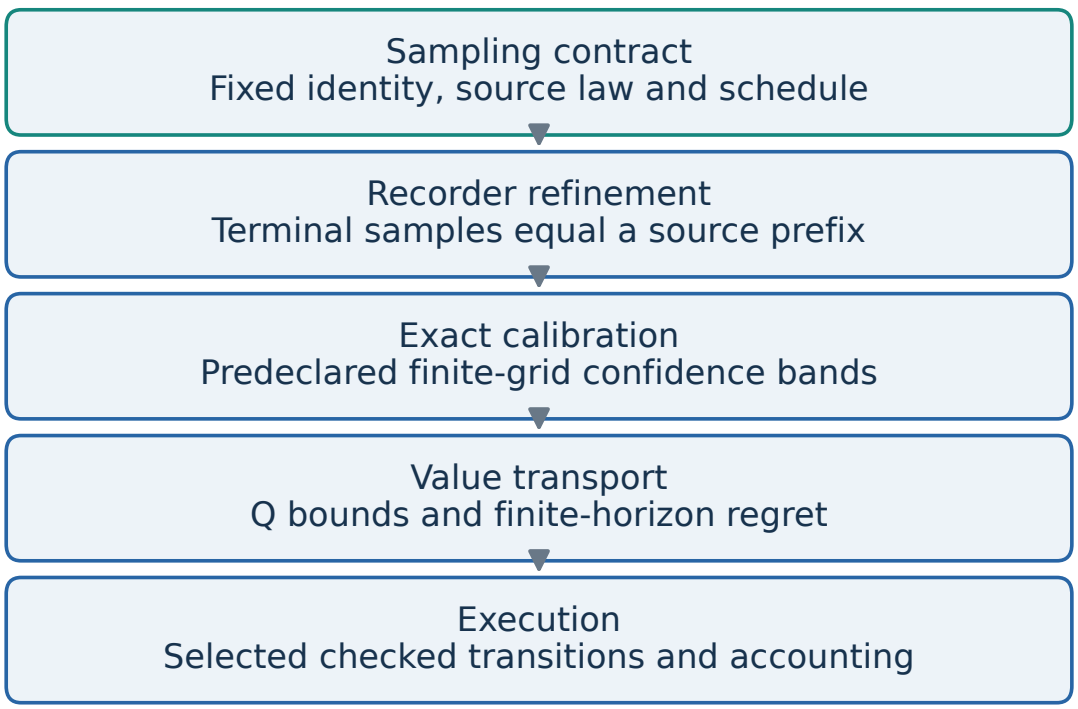


Figure 1: The dependency chain inside the declared model. Physical source integrity and durable host storage require their own justification. The chain is a source-based schematic, not a measurement.

Writing I for issuance and C for the specified episode certificate, the resulting conclusion is

$$\mathbb{P}(I \cap C^{\mathsf{c}}) \leq \sum_{j \in J} \alpha_j.$$

This is an unconditional probability over the original outcome space. It allows arbitrary dependence between scheduled checkpoints. It retains failed and unfinished runs in that space, with no plan issued on those runs. It does not assert a nominal bound conditional on issuance or completion, eventual issuance, or a productivity lower bound. The final recorder theorem derives the execution-to-prefix correspondence inside its generative model. The model's primitive source law remains an explicit premise.

1.2 What is established

The paper selects a coherent path through a larger cumulative library. The first group of results explains when a context or stochastic observation preserves what a task or decision problem needs. The second supplies complete checking and cost comparisons for declared Boolean and affine artifact languages.

The third proves finite planning and robust value guarantees under specified guards and tariffs. The fourth connects finite calibration, source laws, recorder executions, and optional policy issuance. The fifth preserves logical evidence and accounting under replay and recovery.

The global Boolean synthesis result is global within its declared finite semantic universe and syntax tariff. The affine minimum-width result concerns the specified two-stage factorization class. The policy guarantee concerns expected finite-horizon value against a comparator with the same conservative reservation guards. These restrictions are part of the conclusions, rather than qualifications attached after the fact.

1.3 Relation to classical mathematics

The formal development uses familiar mathematical principles: factorization through fibers, comparison of statistical experiments, finite conditional probability, dynamic programming, test inversion, union bounds, and linear algebra over a field. For example, composing a decision rule with a stochastic garbling proves the direction that a more informative observation cannot reduce attainable decision value. This belongs to the classical comparison-of-experiments tradition of Blackwell; the finite proof is given here and is not represented as a new general informativeness theorem.

The publication's mathematical focus is the integration of explicit contracts: which representation preserves a task, which artifact checker is complete for its language, which cost model supports a claimed saving, which uncertainty bound supports a selected action, and which sampling contract makes the calibration applicable. We make no exhaustive historical novelty claim. The names Bellman and Bayesian refer to the established planning and conditioning constructions used in the source, not to newly discovered replacements for those theories.

1.4 A running computational example

Consider an affine map over $\mathbb{F}_5$ with four inputs and four outputs, all-ones linear coefficient matrix, and offset $(1, 2, 3, 4)$. The direct dense evaluation tariff is $4(2 \cdot 4 + 1) = 36$. Its rank-one factorization has a $4 \to 1$ stage followed by a $1 \to 4$ stage, with tariff $9 + 12 = 21$. Thus each declared call saves 15 arithmetic cost units. For 1,000 calls the gross saving is 15,000.

The delivered exact-quote example gives warm inspection setup 133 and cold generation setup 196, both with success probability 1. Their net values are 14,867 and 14,804. These numbers are exact within the declared model. They are not wall-clock measurements, energy savings, or an empirical result about an external generation service. Later examples introduce uncertainty and show how certification can change an action, or lead a conservative planner to stop.

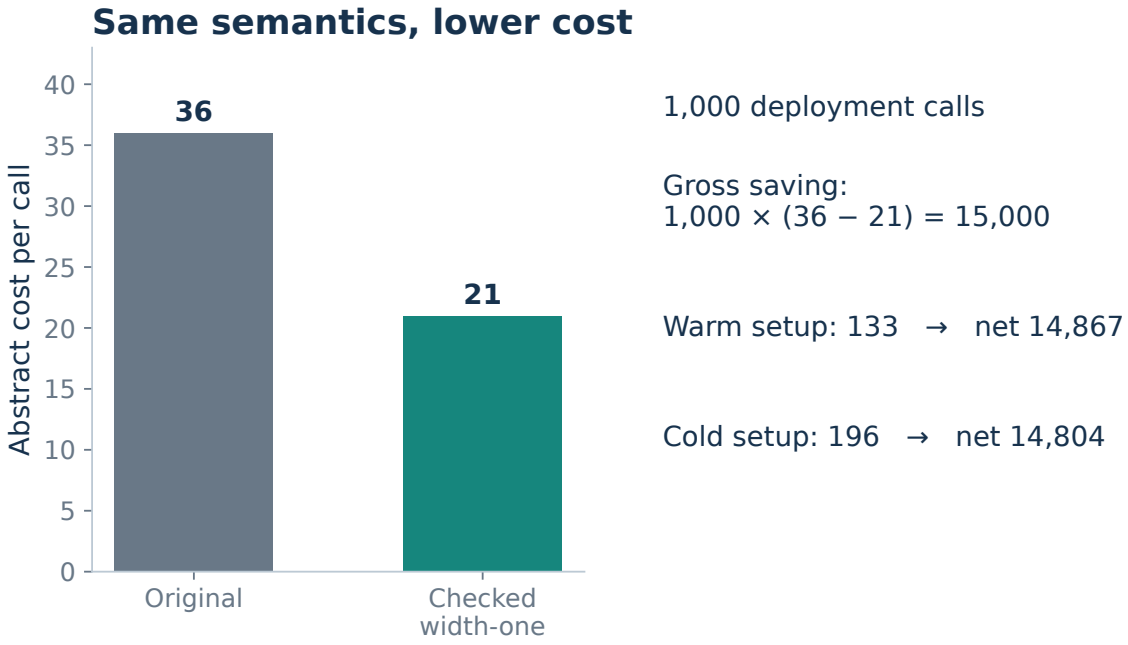


Figure 2: The rank-one affine example under its declared dense arithmetic tariff. Both the evaluation saving and the acquisition setup must enter the comparison.

The example gives one intelligible route through the whole architecture. The semantic certificate establishes the factorization. The tariff establishes a per-call saving. The quote identity binds that saving to a specified source, call count, checker, provider, and environment. The planner compares admissible acquisition actions. The recorder and calibration address the probability parameters only under their own source contracts. A recovery checkpoint retains a confirmed logical result without treating every replay as a fresh acquisition.

2 Representations, tasks, and retained distinctions

AVOC begins with an operational question about information: which distinctions must a retained artifact preserve so that a specified task can still be performed? The answer depends on the task. A representation that is adequate for one query can fail for another, and a representation that answers the current query can fail to support future controlled transitions. The formal development separates these obligations before assigning costs or probabilities.

Definition 2.1 (Deterministic refinement and task sufficiency)**.** Let X be a raw state space and let $\phi : X \to Z$ and $T : X \to Y$ be functions. We write

$$\phi \succeq T \quad \Longleftrightarrow \quad \exists d : Z \to Y \ \ \forall x \in X,\ d(\phi(x)) = T(x).$$

More generally, $\phi \succeq \psi$ means that ψ factors through ϕ. The relation is called refinement. The representation ϕ is task sufficient for T when $\phi \succeq T$. Two representations are information equivalent when each refines the other.

The decoder is part of the assertion. It may be computationally expensive, and its existence alone supplies no efficient implementation. Refinement is reflexive, using the identity decoder, and transitive, by com-

position of decoders. Thus it is a preorder. Information equivalence is the associated equivalence relation. Neither relation compares the byte length, acquisition cost, or evaluation time of the representations.

Theorem 2.2 (Exact fiber criterion)**.** *Suppose Y is nonempty. Then ϕ is sufficient for T if and only if*

$$\phi(x) = \phi(y) \quad \Longrightarrow \quad T(x) = T(y) \qquad (x, y \in X).$$

In particular, a single pair with $\phi(x) = \phi(y)$ and $T(x) \neq T(y)$ proves insufficiency.

Proof. If $T = d \circ \phi$, equality of the two representations gives equality after applying d. Conversely, fix a default $y_0 \in Y$. For every reachable $z \in \phi(X)$ choose x_z with $\phi(x_z) = z$ and set $d(z) = T(x_z)$. Set $d(z) = y_0$ on unreachable representation states. The fiber hypothesis ensures that $T(x_z) = T(x)$ whenever $z = \phi(x)$, so $d(\phi(x)) = T(x)$. The nonempty-codomain hypothesis supplies the default on unreachable states. The formal proof uses classical choice for the representatives. □

The target T itself is sufficient for T. Every sufficient representation refines T, by definition. Consequently T is a canonical coarsest sufficient representation in this information preorder. A sufficient representation is task minimal when T also refines it, and hence task minimality is exactly information equivalence with T. This is a statement about retained distinctions, rather than a theorem that the target can be computed cheaply or stored in a shortest file.

Proposition 2.3 (Finite data processing)**.** *For a finite sample $D \subseteq X$, let $c_D(\phi) = |\phi(D)|$ denote the number of reached representation states. If $\phi \succeq \psi$, then*

$$c_D(\psi) \leq c_D(\phi).$$

Proof. Choose d with $\psi = d \circ \phi$. Then $\psi(D) = d(\phi(D))$, and the image of a finite set under a function has cardinality at most that of the original set. □

The finite sample serves only to count reached states. Sufficiency in Theorem 2.2 is quantified over all of X. Agreement on a sample is therefore a different kind of evidence unless that sample exhausts the domain.

Example 2.4 (A task must be declared before compression)**.** Let $X = \{0,1\}^3$ and let $T(x_1, x_2, x_3) = x_1 \wedge x_2$. The representation $\phi(x) = T(x)$ is sufficient for this target, although it discards x_3 and many distinctions among the first two coordinates. The representation $\psi(x) = x_1 \oplus x_2$ is insufficient: $(0,0,0)$ and $(1,1,0)$ have the same ψ value and different T values. No decoder can repair this collision. Conversely, retaining all three coordinates refines T but retains more information than this task requires.

2.1 Query families and joint targets

Suppose a family of tasks is given by $T_q : X \to Y$, indexed by $q \in Q$. Define the joint target

$$T_{\text{joint}}(x) = (T_q(x))_{q \in Q} \in Y^Q.$$

Then ϕ is sufficient for the joint target if and only if it is sufficient for each individual task. The forward direction evaluates the decoded answer vector at q. For the reverse direction, choose a decoder d_q for each query and define $d(z)(q) = d_q(z)$. The formal theorem permits an arbitrary query index type, and the simultaneous decoder selection uses classical choice.

This distinction matters for reuse. An artifact can be minimal for one fixed target while a later target requires a distinction that the artifact discarded. A reusable library therefore needs a declared target family or an exact semantic key. Enlarging that family changes the sufficiency obligation. The framework does not infer the full future query family from the fact that past tasks succeeded.

3 Information value and finite Bayesian evidence

An information ordering becomes a decision ordering only after specifying a prior, available actions, and utility. The source treats the prior and utility as declared mathematical inputs. In probabilistic applications the prior is normalized and nonnegative. Some structural decision identities need less and are proved even for arbitrary rational weights.

Definition 3.1 (Deterministic experiment value)**.** Let Ω be finite, A be a finite nonempty action set, $b : \Omega \to \mathbb{Q}$ be a prior, and $u : \Omega \times A \to \mathbb{Q}$ be utility. For an observation $\phi : \Omega \to Z$ and policy $\pi : Z \to A$, define

$$J_b(\phi, \pi) = \sum_{\omega \in \Omega} b(\omega) u(\omega, \pi(\phi(\omega))).$$

When Z is finite, write $J_b^*(\phi) = \max_{\pi : Z \to A} J_b(\phi, \pi)$.

Theorem 3.2 (Deterministic information-value monotonicity)**.** *If $\phi \succeq \psi$, then $J_b^*(\psi) \leq J_b^*(\phi)$. If the acquisition costs satisfy $c_\phi \leq c_\psi$, then*

$$J_b^*(\psi) - c_\psi \leq J_b^*(\phi) - c_\phi.$$

Proof. Let d decode ψ from ϕ. Any coarse policy π lifts to $\pi \circ d$. For every world the lifted policy chooses exactly the same action, so its decision value is unchanged. A maximizing coarse policy exists because the policy space is finite and nonempty. Its lifted value is at most the best fine value. Subtracting the stated costs gives the second inequality. □

More information can increase gross decision value while decreasing net value if it costs more. The theorem's cost hypothesis is therefore essential to its domination conclusion. It also supplies a practical pruning rule: an experiment that is coarser and no cheaper can be removed from a finite decision menu without losing an optimum.

3.1 Stochastic observations and garbling

Definition 3.3 (Finite rational channel)**.** A channel from Ω to a finite observation alphabet O consists of rational entries $E(o \mid \omega) \geq 0$ with $\sum_o E(o \mid \omega) = 1$ for every ω. Given a second channel $K(y \mid o)$, their composite is

$$(EK)(y \mid \omega) = \sum_o E(o \mid \omega) K(y \mid o).$$

The composite is a garbling of E.

The unnormalized signal-action score and policy value are

$$S_E(o, a) = \sum_\omega b(\omega) E(o \mid \omega) u(\omega, a), \qquad J_b(E, \pi) = \sum_o S_E(o, \pi(o)).$$

This form avoids dividing by the evidence mass. In particular, it remains well defined on observations whose predictive probability is zero.

Theorem 3.4 (Finite stochastic information-value monotonicity)**.** *Every deterministic policy on EK is weakly dominated in value by some deterministic policy on E. Consequently the best deterministic value of EK is at most that of E.*

Proof. For each fine observation o, select $a_o \in A$ maximizing $S_E(o, a)$. Such a selection exists because A is finite and nonempty. For a coarse policy π, expand the channel composition and rearrange finite sums:

$$J_b(EK, \pi) = \sum_o \sum_y K(y \mid o) S_E(o, \pi(y)).$$

Every inner score is at most $S_E(o, a_o)$. Nonnegativity and normalization of the row of K give

$$\sum_y K(y \mid o) S_E(o, \pi(y)) \le S_E(o, a_o).$$

Summing over o proves domination by the fine policy $o \mapsto a_o$. Taking a best coarse policy yields the optimal-value comparison. □

Example 3.5 (Information has a price)**.** For two equally likely worlds, let the action predict the world and let utility be 1 for a correct prediction and 0 otherwise. A constant observation gives value $1/2$. Revealing the world gives value 1. If revelation costs $1/8$, its net value is $7/8$, exceeding the free constant experiment's $1/2$. If it costs $3/4$, its net value is $1/4$. The information order is unchanged across these two cases, while the preferred paid experiment changes.

3.2 Exact Bayesian updates

A rational belief is a nonnegative vector b on finite Ω with $\sum_\omega b(\omega) = 1$. For a channel E, define the predictive mass

$$p(o) = \sum_\omega b(\omega) E(o \mid \omega).$$

For $p(o) > 0$, the posterior is $b_o(\omega) = b(\omega)E(o \mid \omega)/p(o)$. The total updater in the source uses $b_o = b$ when $p(o) = 0$. This fallback is normalized and has zero coefficient in predictive expectations. Nonnegativity of every summand implies that each Bayes numerator also vanishes on such an impossible observation. Hence, with the total-update convention, the division-free identity

$$p(o) b_o(\omega) = b(\omega) E(o \mid \omega)$$

holds on every branch. It follows that

$$\sum_o p(o) \sum_\omega b_o(\omega) u(\omega, \pi(o)) = J_b(E, \pi).$$

Posterior-predictive evaluation and the unnormalized decision formula therefore agree exactly.

For any nonnegative likelihood weight L, write

$$M_b(L) = \sum_\omega b(\omega) L(\omega), \qquad C_L(b)(\omega) = \frac{b(\omega) L(\omega)}{M_b(L)}$$

when $M_b(L) > 0$. Given nonnegative L_1, L_2 and positive successive evidence masses, the source proves

$$M_b(L_1) M_{C_{L_1}(b)}(L_2) = M_b(L_1 L_2), \qquad C_{L_2}(C_{L_1}(b)) = C_{L_1 L_2}(b).$$

These identities follow by expanding the finite sums and canceling the positive normalizers. Both conditioning orders agree when their respective intermediate masses are positive. Applying this algebra to physical observations requires the appropriate joint likelihood model. In particular, multiplying likelihoods is justified by a stated factorization of the evidence law, rather than merely by the existence of two reports.

3.3 Compression must preserve the likelihood

Theorem 3.6 (Conditioning commutes with exact compression)**.** *Let $\phi : \Omega \to Z$ map between finite spaces, and define the pushforward belief $\bar{b}(z) = \sum_{\phi(\omega)=z} b(\omega)$. Suppose $L = \bar{L} \circ \phi$ for a nonnegative likelihood $\bar{L} : Z \to \mathbb{Q}$, and the evidence mass is positive. Then*

$$\phi_* C_L(b) = C_{\bar{L}}(\phi_* b).$$

Such an exact $\bar{L}$ exists if and only if L is constant on every fiber of ϕ.

Proof. Grouping the normalizer by fibers gives $M_b(L) = M_{\bar{b}}(\bar{L})$. On each fiber $\phi^{-1}(z)$, the likelihood can be factored outside the sum. Thus

$$\sum_{\phi(\omega)=z} \frac{b(\omega)L(\omega)}{M_b(L)} = \frac{\bar{b}(z)\bar{L}(z)}{M_{\bar{b}}(\bar{L})}.$$

The factorization criterion is Theorem 2.2 applied to the rational-valued target L. □

A prior compression that merges worlds with different likelihoods has no exact coarse likelihood for that observation. A later evidence update can consequently reveal a distinction that the compressed state no longer represents. This is the evidence version of the collision obstruction.

Example 3.7 (Rare evidence can reverse a posterior completely)**.** Under the uniform prior on $\{0,1\}$, take $L_0 = (\varepsilon, 0)$ and $L_1 = (0, \varepsilon)$ with rational $\varepsilon > 0$. The two likelihoods differ pointwise by at most ε, and both evidence masses equal $\varepsilon/2$. Their posteriors are respectively $(1,0)$ and $(0,1)$, with full L^1 distance 2. Arbitrarily small positive likelihood perturbations therefore do not imply uniform pointwise posterior stability without an additional evidence condition. The later model-error chapter uses weighted continuation comparisons rather than identifying nearby posterior points as the same controller state.

4 Sufficiency for controlled futures

For a one-shot task, refinement preserves sufficiency by decoder composition. A state abstraction for a controlled process has a second obligation: its fibers must remain compatible with the process's transitions. This extra obligation changes the mathematical behavior of refinement.

Definition 4.1 (Deterministic control sufficiency)**.** Let A_0 be a finite nonempty set of admissible actions, let $f : S \times A_0 \to S$ be transition, and let $r : S \times A_0 \to \mathbb{Q}$ be reward. A representation $\phi : S \to Z$ is control sufficient if, whenever $\phi(s) = \phi(t)$, every admissible action satisfies

$$r(s,a) = r(t,a), \qquad \phi(f(s,a)) = \phi(f(t,a)).$$

The global action set is common to all states in this definition.

Define the finite-horizon values by

$$V_0(s) = 0, \qquad Q_n(s,a) = r(s,a) + V_n(f(s,a)), \qquad V_{n+1}(s) = \max_{a \in A_0} Q_n(s,a).$$

Theorem 4.2 (Exact preservation of controlled value)**.** *If ϕ is control sufficient, then V_n is constant on every fiber of ϕ for every $n \in \mathbb{N}$. The same is true of $Q_n(\cdot, a)$ for each admissible a, and Bellman-optimal action sets coincide within each fiber.*

Proof. At horizon zero the values vanish. If V_n is fiber constant and $\phi(s) = \phi(t)$, control sufficiency gives equal rewards and equal represented successors. Hence $Q_n(s,a) = Q_n(t,a)$ for every admissible a. Taking the maximum over the common action set proves equality of V_{n+1}. The action-value equality also makes all inequalities defining optimal action membership identical at s and t. □

Example 4.3 (A finer representation can fail dynamic closure)**.** Consider $S = \{0,1,2\}$, one action, zero reward, and transitions $f(0) = 0$, $f(1) = 2$, $f(2) = 2$. The constant representation is control sufficient. The finer representation with fibers $\{0,1\}$ and $\{2\}$ refines that constant representation, but it merges states 0 and 1 whose successors fall into different fine fibers. It is therefore not control sufficient. This example is retained as an explicit formal counterexample to arbitrary refinement monotonicity.

The exact repair is simple: a refinement of a control-sufficient representation remains control sufficient if the finer representation is itself transition closed. Reward equality is inherited through the coarse decoder, while successor equality must be supplied for the fine representation. For models with state-dependent action sets, exact action feasibility is an additional obligation, handled by the guarded control interfaces later in the paper.

4.1 Projected stochastic laws

Let S be finite and replace deterministic transition by rational probability rows $P_a(s,s') \geq 0$ with $\sum_{s'} P_a(s,s') = 1$. The stochastic Bellman recursion is

$$V_0(s) = 0, \qquad Q_n(s,a) = r(s,a) + \sum_{s'} P_a(s,s')V_n(s'), \qquad V_{n+1}(s) = \max_{a \in A_0} Q_n(s,a).$$

Equality of entire raw transition rows is sufficient for compression, but can demand more than the abstract process needs. The appropriate condition compares their projected laws.

Definition 4.4 (Projected stochastic control sufficiency)**.** A representation $\phi : S \to Z$ is projected stochastically sufficient if equal represented current states have equal immediate rewards and, for every admissible a and every rational fiber-constant function $W : S \to \mathbb{Q}$,

$$\sum_{s'} P_a(s,s')W(s') = \sum_{s'} P_a(t,s')W(s') \quad \text{whenever } \phi(s) = \phi(t).$$

When Z is finite, this expectation condition is equivalent to equality of the mass sent to every abstract fiber:

$$\sum_{\phi(s')=z'} P_a(s,s') = \sum_{\phi(s')=z'} P_a(t,s') \qquad (z' \in Z).$$

To prove necessity, use the indicator of the fiber as W. To prove sufficiency, group the expectation by fibers and factor out the common value of W on each nonempty fiber. Empty fibers contribute zero. These arguments explain why raw states may have distinct next-state probabilities while having identical controlled abstract behavior.

Theorem 4.5 (Exact stochastic quotient)**.** *Suppose $\phi : S \to Z$ is surjective and projected stochastically sufficient, with S, Z finite. Choose a representative s_z of every fiber, and define*

$$\bar{r}(z,a) = r(s_z,a), \qquad \bar{P}_a(z,z') = \sum_{\phi(s')=z'} P_a(s_z,s').$$

These quantities give a normalized rational stochastic control system, independent of the representative choices for admissible actions. For every finite horizon,

$$\bar{V}_n(\phi(s)) = V_n(s).$$

Proof. Nonnegative fiber masses give nonnegative quotient probabilities; partitioning S by the fibers shows that each quotient row sums to one. Sufficiency makes reward and projected mass unchanged when replacing s_z by another representative. Inductively assume the value equality at horizon n. Then

$$\sum_{z'} \bar{P}_a(\phi(s), z')\bar{V}_n(z') = \sum_{s'} P_a(s, s')V_n(s'),$$

by projected equality and fiberwise grouping. Immediate rewards agree, so every action backup agrees. Taking the maximum over the common action set proves the next horizon. The zero-horizon case is immediate. □

This result concerns exact finite models with a specified reward and transition law. It neither discovers the correct abstraction automatically nor identifies a supplied rational model with a physical system. Its role is to certify the consequences of a declared abstraction once the one-step obligations have been established.

5 Artifacts, identities, and proof-carrying admission

An artifact's semantic content and its operational identity serve different purposes. Semantic equivalence permits replacement in a declared task language. Operational identity prevents a receipt for one statement, dependency set, environment, or generation from authorizing another.

The concrete binding layer uses a small data-only propositional language. Formulas contain numbered atoms, truth, falsity, conjunction, disjunction, and implication. Derivations contain the corresponding introduction and elimination rules. Atoms are interpreted as arbitrary propositions under a valuation $v : \mathbb{N} \to \text{Prop}$. A recursive inference function either returns a formula or rejects. It accepts data representing a derivation, rather than executable tactics or imported source code.

Theorem 5.1 (Checker soundness)**.** *If the inference checker accepts derivation p for target F under context Γ, then every valuation satisfying Γ satisfies F.*

Proof. Induct on the derivation. A used assumption belongs to Γ and is true by the context hypothesis. Conjunction introduction combines the two inductive conclusions, and its projections select the corresponding conclusion. Implication introduction applies the induction hypothesis in the extended context; implication elimination applies a proved implication to a proved premise. The two disjunction injections construct the appropriate alternative. Case elimination applies the matching branch proof after extending the context with the observed alternative. Falsity elimination follows from the impossible proposition. The checker's shape and equality tests ensure that the formulas in these rules match exactly. □

This is soundness of the defined propositional checker. It is not a universal ingestion theorem for arbitrary Lean source. Dependencies remain assumptions in the checker context, and their truth must come from the surrounding certified workflow.

Definition 5.2 (Protected artifact identity)**.** An identity records a task name, a formula, its statement-byte snapshot, a list of dependencies with their formulas and artifact-byte snapshots, and an environment snapshot. An artifact contains that identity, a source-byte snapshot, and a data-only derivation. A ticket consists of a generation number together with an artifact. A receipt names the exact ticket and records an approval bit.

All byte snapshots in this model are exact strings. No finite hash is assumed injective. The validator checks four obligations: the pending artifact's identity equals the current identity, the receipt names the exact pending ticket, its approval bit is true, and the derivation checks for the current target under the current dependency context. The resulting validation theorem supplies the target proposition whenever the dependency propositions hold.

A changed statement, dependency, environment, source snapshot, or ticket generation is rejected by the corresponding equality gate. Approval alone supplies no proof. Re-registering an identical payload still changes its generation, which prevents an old receipt from serving as authorization for a new registration.

The finite slot model increments its generation on registration. Successful admission consumes the pending ticket; failed admission leaves it unchanged. A successfully consumed receipt cannot admit again because no pending ticket remains. These statements describe admission in the defined reducer. They do not establish exactly-once performance of external paid work, filesystem durability, or authentication of an external provider. The recovery chapters treat the separate accounting and persistence obligations.

6 Certified Boolean synthesis and semantic reuse

The Boolean backend provides a complete finite example of semantic checking, cost-aware replacement, and retained computation. Its restricted language is deliberate: strong universal conclusions are possible because both the input domain and semantics are explicit.

6.1 Programs and their declared work

Programs are finite abstract syntax trees built from Boolean literals, three input variables, negation, conjunction, disjunction, and exclusive-or. The input domain is the eight triples in $\{0,1\}^3$. Let $N(p)$ be the tree-node count: each literal or variable contributes one, negation adds one to its child, and a binary constructor adds one to the sum of its child counts.

The instrumented evaluator evaluates both children of a binary operation. A structural induction proves that its output equals the ordinary evaluator and that its visit count is exactly $N(p)$ for every input. Thus node count is a valid eager-traversal meter for this interpreter. It is not a short-circuit meter, hardware runtime, memory-allocation count, or Boolean circuit size with shared subexpressions.

Theorem 6.1 (Complete Boolean equivalence check)**.** *Let $\sigma(p)$ be the eight output bits of p, in the fixed lexicographic input order. Then*

$$\sigma(p) = \sigma(q) \quad \Longleftrightarrow \quad \forall x \in \{0,1\}^3,\ p(x) = q(x).$$

The executable checker comparing all eight outputs is sound and complete.

Proof. Equality of signatures gives equality at every entry. Every possible input occurs in the enumerated list, so every input is covered. Conversely, pointwise equality makes the two mapped output lists equal. This is exhaustive finite checking, not a sampled approximation. □

6.2 Bounded grammar search

The initial grammar layer has five atoms: two literals and three variables. If a layer is L, expansion retains L, adds all negations, and adds all ordered pairs under each of the three binary operations. Let

M_d count emitted list entries at depth d. The source proves

$$M_0 = 5, \qquad M_{d+1} = 2M_d + 3M_d^2.$$

The entries can contain duplicates. The node bound is $B_0 = 1$, $B_{d+1} = 2B_d + 1$, and every expression of height at most d occurs in the depth-d list. In particular $M_1 = 85$ and $M_2 = 21845$.

The search retains an incumbent equivalent to a reference. A candidate replaces it only when the complete checker passes and the candidate has strictly fewer nodes. Induction over the examined list proves preservation of equivalence and nonincrease of node count. The returned incumbent has node count at most that of every equivalent examined candidate.

Generation is reserved before a layer is constructed. The declared generation charge is the sum of the layer-entry counts through the chosen depth. A candidate comparison is conservatively charged

$$T(r, d) = 8\big(N(r) + B_d + 1\big).$$

The examined prefix is limited by both a user trial cap and the remaining budget divided by $T(r, d)$. Its generation plus examination charge is proved at most the supplied budget. If the generation charge does not fit, search returns the original reference and incurs zero modeled search charge. If the entire layer fits both limits, the result is optimal among equivalent expressions of height at most d. A partial prefix provides optimality only over that prefix.

Example 6.2 (A smaller artifact need not repay its discovery)**.** The reference $\neg(\neg x_1 \vee \neg x_2)$ has six nodes. The equivalent conjunction $x_1 \wedge x_2$ has three. The archived depth-one search examines the complete 85-entry layer under its stated limits and returns the conjunction, with modeled search charge 6890. Each eager evaluation subsequently saves three visits. A large deployment can repay the exploration cost; a single evaluation cannot. The synthesis theorem supplies semantic safety and a budget bound even in the unprofitable case, and the promotion layer separately checks the campaign economics.

6.3 Compaction preserves the search optimum

Semantic entries carry an expression, its complete eight-bit signature, its node count, and equations certifying the two cached fields. The index retains at most one representative of each signature and keeps a smallest representative among inserted expressions. There are at most $2^8 = 256$ entries, regardless of the size of the submitted list.

Proposition 6.3 (Recursive semantic coverage)**.** *For every depth d, recursively expanding the retained representatives and compacting again preserves the smallest equivalent node count obtainable from the full uncompressed grammar layer at that depth, with the same reference fallback.*

Proof. Compaction gives every input expression an equivalent retained representative of no greater node count. Equivalence is a congruence for each grammar constructor. Replacing either child by a no-larger equivalent representative therefore preserves the result and does not increase its node count. Negation has the same property. Induction on depth supplies semantic and cost coverage of the full grammar by recursive representatives. Conversely, every retained representative originates in the full grammar. These two inclusions in semantic-cost coverage give equal selected optimum costs. □

This is a precise example of reusable knowledge reducing subsequent construction. It is finite semantic compaction with a proof of optimum preservation. It does not establish that arbitrary learned representations preserve future optimization problems. The instrumented construction separately records emitted

entries, total node mass, and index probes; its declared tariff excludes several host costs, including allocation and serialization.

6.4 From bounded search to a global cost certificate

A complete finite semantic algebra supports a stronger conclusion than searching a deeper finite layer. Let D be a semantic domain with interpretations of the Boolean constructors. A cost function $c : D \to \mathbb{N}$ is a lower certificate if it satisfies the constructor inequalities

$$c(\text{literal } b) \le 1, \qquad c(\text{variable } j) \le 1,$$
$$c(\neg s) \le c(s) + 1, \qquad c(s \circ t) \le c(s) + c(t) + 1 \quad (\circ \in \{\wedge, \vee, \oplus\}).$$

Theorem 6.4 (Attained global semantic minimum)**.** *Suppose c satisfies those inequalities and, for every semantic state s, a witness expression w_s has semantics s and node count $c(s)$. Then for every expression p with semantics s,*

$$N(w_s) = c(s) \le N(p).$$

No expression-height bound appears in this conclusion. Two such attained cost certificates have identical cost functions.

Proof. Structural induction proves $c(\mathrm{sem}(p)) \le N(p)$. The leaf inequalities give the base cases. For a unary constructor, use its inequality and the child's induction hypothesis. For a binary constructor, use its inequality and both child hypotheses. Applying this lower bound to any expression with semantics s and using the attained witness proves global minimality. For two certificates, apply each one's lower bound to the other's witness, obtaining both inequalities between their costs. □

For the three-input Boolean backend, D consists of all 256 truth signatures. A proposed bank must have exactly 256 rows. Each row's stored expression is checked for its exact signature and claimed node count, and the bank is checked against every unary and binary constructor inequality. An accepted bank therefore carries the hypotheses of Theorem 6.4. Its query returns a globally node-minimal expression within this entire Boolean tree grammar, including all finite heights.

The distinction is substantive. A bounded search is global only within its examined language fragment. A checked closure bank certifies a lower bound for every expression by structural induction and supplies witnesses attaining that bound. The universal conclusion remains specific to the grammar, three-input semantics, and eager tree-node tariff. It does not claim unrestricted program minimization or optimal implementation on physical hardware.

7 Exact affine artifacts and complete finite probes

The affine backend extends exact semantic certification to infinite input domains. Its complete finite test is possible because the represented functions have a fixed algebraic form.

Definition 7.1 (Dimension-indexed affine signature)**.** Over a ring R, a signature from dimension n to dimension m is a pair (A, b) with $A \in R^{m \times n}$ and $b \in R^m$, denoting $x \mapsto Ax + b$. Dimensions are part of the type. Composition and pointwise addition are

$$(B, c) \circ (A, b) = (BA, Bb + c), \qquad (A, b) + (A', b') = (A + A', b + b').$$

Theorem 7.2 (Complete zero-and-basis probes)**.** *Two affine signatures $f, g : R^n \to R^m$ are equal if and only if they agree at 0 and at every standard basis vector e_j. Thus $n+1$ vector probes establish equality on all inputs. A mismatch in signatures has a mismatch among these probes.*

Proof. The zero probe identifies the offsets because $f(0) = b_f$. At e_j, the output is the jth matrix column plus the offset. After canceling the now-equal offsets, equality of the basis probes identifies every column. Equal coefficients give equal evaluations for all inputs. The mismatch assertion is the contrapositive. □

The theorem is algebraic completeness, rather than empirical interpolation from arbitrary function values. Its premise is that both candidates belong to the affine language over the stated ring. Adding a nonlinear primitive changes the language and invalidates this probe argument unless a new completeness theorem is supplied.

Example 7.3 (The zero probe is necessary)**.** Over $\mathbb{Q}$, consider scalar maps $f(x) = x$ and $g(x) = 2x - 1$. They agree at the single basis vector $x = 1$, but $f(0) = 0$ and $g(0) = -1$. Omitting the zero probe would accept a false affine equivalence. Over a field, equality of the linear matrices alone also leaves arbitrary affine offsets unconstrained.

7.1 Typed normalization and safe contextual replacement

Typed programs combine affine atoms, identities, sequential composition, and pointwise addition. Composition connects only matching intermediate dimensions. An independent recursive execution semantics and coefficient-normalization semantics are defined. Structural induction proves

$$\mathrm{eval}(\mathrm{normalize}(p), x) = \mathrm{run}(p, x) \qquad \text{for all } x.$$

Hence two programs are equivalent if and only if their normal signatures coincide. Decidable coefficient comparison is a sound and complete checker when equality in R is decidable, as it is for the rational and finite-field executable examples.

One-hole typed contexts permit replacement at arbitrary nesting depth. Equivalence is preserved under composition and pointwise addition, so plugging equivalent programs into such a context preserves its semantics. A declared additive nonnegative primitive tariff, with m charged for adding two m-dimensional outputs, also makes contextual cost monotone: a no-more-expensive replacement cannot increase the whole context's declared cost. This is a compositional guarantee for the explicit syntax and tariff.

8 Fusion, rank certificates, and retargeted reuse

An exact normal form can improve verification without improving evaluation. The affine backend therefore checks semantics and cost separately. Its dense arithmetic tariff is

$$D(n, m) = m(2n + 1),$$

charging n multiplications, n accumulations, and one bias addition per output row. The coefficient-composition tariff is

$$C(n, m, k) = 2knm + D(m, k).$$

These are declared scalar-operation tariffs. They exclude bit complexity, memory traffic, allocation, and physical timing.

8.1 Fusion has an amortization threshold

Replacing a composed program by one dense atom with its normalized coefficients preserves every output. Its per-call execution tariff becomes $D(n, m)$. The source's no-slower selector retains the original program whenever that dense tariff is greater than its current execution cost. Normalization work is charged separately for campaign comparisons.

Example 8.1 (Beneficial and harmful fusion)**.** A dense $2 \to 3 \to 2$ chain has per-call tariff $D(2,3) + D(3,2) = 29$. Its fused $2 \to 2$ map costs 10 per call, and coefficient composition costs 38. Thus $38 + 10q \leq 29q$ exactly when $q \geq 2$, with strict savings exactly when $q \geq 3$.

In contrast, a $4 \to 1 \to 4$ bottleneck costs $D(4,1) + D(1,4) = 21$ per call, while its dense fused $4 \to 4$ form costs 36. Fusion remains semantically correct but loses its cheap intermediate bottleneck. The cost-aware selector therefore keeps the original chain in that comparison.

The second case motivates retaining a factorized artifact. A dense semantic key can identify a function while a different execution representation realizes it more cheaply.

8.2 An upper factorization and a lower rank witness

Work over a field R. For reference $f = (A, b) : R^n \to R^m$, a candidate supplies an intermediate width k, first and second affine signatures

$$a(x) = Ux + u_0, \qquad d(y) = Vy + v_0,$$

and matrices $L \in R^{k \times m}$, $H \in R^{n \times k}$. The checker verifies

$$VU = A, \qquad Vu_0 + v_0 = b, \qquad LAH = I_k.$$

The first two identities give the exact upper implementation. The third is an independently checkable lower witness. Validation uses polynomial matrix identities; an untrusted rank routine is not part of the logical justification.

Theorem 8.2 (Certified minimum affine width)**.** *An accepted candidate has $k = \operatorname{rank}(A)$. Every exact two-stage affine factorization of f through width ℓ satisfies $k \leq \ell$. Under the dense two-stage tariff,*

$$F(n, m, k) = D(n, k) + D(k, m),$$

the accepted candidate is no more expensive than any such two-stage factorization.

Proof. The upper factorization gives $\operatorname{rank}(A) = \operatorname{rank}(VU) \leq k$. The lower witness gives

$$k = \operatorname{rank}(I_k) = \operatorname{rank}(LAH) \leq \operatorname{rank}(A).$$

Thus rank equals k. Any exact factorization through ℓ has $\operatorname{rank}(A) \leq \ell$, by the same upper rank inequality. Finally,

$$F(n, m, k) = k(2n + 1) + m(2k + 1)$$

is increasing in k, so minimum width gives minimum tariff in this class. □

Exact factorization alone does not prove minimum width. A redundant factorization can be correct while failing to supply a valid $LAH = I_k$. Conversely, a lower rank witness alone does not prove that the proposed affine implementation has the correct offset. Both directions are needed.

Example 8.3 (A checked rank-one bottleneck)**.** In the archived finite-field example over $\mathbf{F}_5$, A is the 4×4 all-ones matrix and $b = (1, 2, 3, 4)$. The first stage computes $y = x_1 + x_2 + x_3 + x_4$. The second emits $(y + 1, y + 2, y + 3, y + 4)$. Taking L to select the first output coordinate and H to select the first input coordinate gives $LAH = (1)$. The width-one factorization is therefore exact and width minimal.

Its tariff is $9 + 12 = 21$, compared with dense cost 36, saving 15 per call. The declared factor-checking charge is 105. For this already atomic reference, the campaign inequality is exactly

$$105 + g \leq 15q,$$

where g is the declared generation work and q the deployment count. With $g = 0$, seven calls break even. Generation, validation, and deployment are all visible in this comparison.

The menu selector compares three equivalent implementations: the current program, its dense fused atom, and the accepted minimum-width two-stage plan. Its selected tariff is exactly

$$\min\{\mathrm{cost}(p),\ D(n, m),\ F(n, m, k)\}.$$

This proves a minimum over the explicit menu, and also comparison with every exact two-dense-stage factorization. It does not prove a globally cheapest affine program in the larger language. Sparse schedules, sharing, special primitives, and hardware-specific algorithms have different admissible implementations or tariffs. An identity program already costs zero in this syntax and remains preferable to a paid dense factorization.

8.3 Reuse under a changed offset

Suppose a stored candidate has been validated for (A, b) and a later target is (A, b') with exactly the same linear matrix. Keep U, u_0, V, L, H and replace only the second-stage offset by

$$v_0' = b' - Vu_0.$$

Then $VU = A$ is unchanged, $Vu_0 + v_0' = b'$, and $LAH = I_k$ remains true. The same width and two-stage tariff are retained. A second retargeting overwrites the offset rather than accumulating prior offset changes.

The import gate rechecks the entire stored certificate and exact equality of the current linear semantic key before constructing the retargeted checked record. It rejects invalid saved certificates and changed linear matrices. The current implementation is then compared with the reused plan, so a previously beneficial plan is not automatically preferred after other implementation changes.

This is an exact reuse theorem for a specified invariant key. It shows which part of the calculation is reusable and which part must be adapted. Persisting a claimed certificate, trusting a previous receipt, or comparing a suggestive task label would not provide these hypotheses. The proof-carrying bank and the protected artifact workflow supply complementary mechanisms for making that distinction explicit.

9 Finite-horizon planning with admissible actions

An artifact decision changes both what is available and what can be afforded. Inspection can turn an untried cached artifact into a ready artifact, but a rejection also spends its inspection fee. Generation can create a candidate, but deployment must wait for a checked success. A controller therefore needs an action set that depends on the current state, rather than a fixed menu with inadmissible actions assigned convenient scores.

Definition 9.1 (Guarded rational system)**.** Let S be a state space, $A(s)$ a finite nonempty set of actions, and O a finite observation space. A guarded system consists of a rational reward $r(s,a)$, a transition $T(s,a,o)$, and rational probabilities $P(o \mid s,a)$ satisfying

$$P(o \mid s,a) \geq 0, \qquad \sum_{o \in O} P(o \mid s,a) = 1.$$

Its finite-horizon Bellman values are

$$V_0(s) = 0, \tag{1}$$
$$Q_h(s,a) = r(s,a) + \sum_o P(o \mid s,a) V_h(T(s,a,o)), \tag{2}$$
$$V_{h+1}(s) = \max_{a \in A(s)} Q_h(s,a). \tag{3}$$

Here h in Q_h counts *continuation* steps. Choosing an acquisition action and then deploying a successful artifact requires two total actions, so its first-action score is Q_1, not Q_0. This indexing also applies to `robustChoice md h s` in the formal source.

The state space need not be finite. Rational Bayesian beliefs are one intended instance. Only each feasible action set and the observation space are finite. The finite maximum exists even when there is no asserted efficient procedure for finding it. The mathematical selector and the executable evaluator are separate constructions.

Theorem 9.2 (Optimality over complete adaptive plans)**.** *Every horizon H has a legal deterministic observation tree attaining $V_H(s)$. Every other legal deterministic observation tree has value at most $V_H(s)$. If each admissible transition preserves an invariant, the attaining tree preserves that invariant on every branch.*

Proof. At horizon zero, the empty tree has value zero. At horizon $h+1$, choose an action attaining the finite maximum in (3); after each observation, attach an optimal horizon-h tree at the successor state. Induction proves equality of the resulting expected value and the Bellman value. For an arbitrary legal tree, induction bounds each child value by the corresponding optimal continuation. Multiplication by nonnegative probabilities, summation, and the maximum over admissible first actions give the upper bound. Invariant preservation follows by the same tree induction. The invariant argument applies to every observation branch, including a branch of probability zero. □

This is a prose reconstruction of `safe_optimal_plan_exists`, `optimalPlan_value`, and `planValue_le_value` in `AVOC/GuardedControl.lean`. The competitors are complete adaptive observation trees. They need not use the same subsequent choices as the selected controller. The comparator nevertheless uses the same state, information, action guards, and transition model. An oracle with additional information is a different optimization problem.

9.1 Warm inspection and cold generation

The lazy staged controller makes this structure concrete. Its state records warm and cold statuses, remaining budget, reserved spending, original credit, and a terminal flag. Each artifact status is untried, failed, or ready. The action list always contains stop. Inspection and generation require an untried status and enough budget. Deployment requires the matching ready status. Stopping or deploying ends the episode.

An acquisition transition pays its setup fee whether its Boolean report is true or false. A true report changes the corresponding status to ready. A false report changes it to failed. Acquisition rewards are negative setup fees, deployment rewards are nonnegative gross values, and stopping has reward zero. This prevents three easy accounting mistakes: assuming unsuccessful acquisition is free, treating a quote as an already available artifact, and deploying before checked success.

Example 9.3 (The artifact-backed rank-one comparison)**.** The exact quote model retained in `AVOC/RobustQuoteExamples.lean` has warm setup 133, cold setup 196, success probability 1 for both, and gross deployment value 15000 for each over 1000 calls. With initial credit 196 and two total actions available,

$$Q_1(\text{inspectWarm}) = 15000 - 133 = 14867, \qquad Q_1(\text{generateCold}) = 15000 - 196 = 14804.$$

Warm inspection is better by 63 in this exact model. At one total action, acquisition has only its negative immediate reward, and stopping is preferable when neither artifact is ready. The deployment opportunity creates the change between these two horizons.

This example has actual bounded artifact witnesses in the retained source. The gross value is the symbolic execution saving for the protected workload, not a measured dollar return or a wall-clock benchmark. The singleton evidence makes all quote uncertainty radii vanish, so the robust lower-Q selector recovers the inherited exact selector. The relevant declarations are `exact_recovers_v037`, `exact_all_local_radii_zero`, and `artifact_refines_deterministic_fields`.

9.2 An executable evaluator and its cost boundary

The executable planner enumerates admissible first actions and observation branches, recursively evaluates continuations, and chooses a best score. Under the formal `Realizes` relation, its returned score equals the mathematical Bellman value. A full catalogue enumeration and a verified admissibility check connect the staged workflow to that evaluator. In particular, `WorkflowPlanner.model_realizes` and `executable_policy_source_safe` connect actual staged transitions, source-bound artifacts, and the selected policy.

The planner's work counter counts defined recursive evaluator nodes. A branching cap permits reserving an abstract planning allowance, and truncated lookahead has a source-backed policy regret bound. This counter is not a rational-bit complexity bound. It does not include all host parsing, catalogue scans, reservation arithmetic, storage operations, or physical invocation cost. Consequently, a proved allowance in that model cannot by itself establish a time bound for the complete service.

10 Quote uncertainty and certified lower-Q selection

Quotes carry more than a success estimate. A protected identity records the artifact kind, exact source signature, workload call count, cost model, provider, checker, environment, and schema. A setup band has natural endpoints. Gross and probability bands have rational endpoints, with probability endpoints in $[0, 1]$ and nonnegative gross lower endpoints. A submitted quote validates only when it equals the protected approved quote. Changing either identity or any band changes the evidence.

For kind $k \in \{w, c\}$, write

$$c_k \in [c_k^-, c_k^+], \qquad g_k \in [g_k^-, g_k^+], \qquad p_k \in [p_k^-, p_k^+].$$

The formal nominal model reserves c_k^+, credits g_k^-, and uses p_k^-. A compatible realization supplies exact values inside these bands. In the true comparison system, rewards and report probabilities use those

exact values, but the state transitions and action guards remain those of the nominal model. In particular, setup reservations continue to use the upper setup bound.

This common guard matters. If the true setup is 1 and the upper quoted setup is 3, a budget of 2 still makes the acquisition inadmissible for both systems in the theorem. The comparator does not obtain an extra feasible action by knowing the smaller true fee. A theorem against that larger feasible set would require a different comparison argument.

10.1 From primitive bands to value bounds

Let $w_c(k) = c_k^+ - c_k^-$, $w_g(k) = g_k^+ - g_k^-$, and $w_p(k) = p_k^+ - p_k^-$. The source uses the uniform quantities

$$\varepsilon = w_c(w) + w_c(c) + w_g(w) + w_g(c), \tag{4}$$

$$\delta = 2\big(w_p(w) + w_p(c)\big), \tag{5}$$

$$M = c_w^+ + c_c^+ + g_w^+ + g_c^+. \tag{6}$$

They bound reward error, Boolean-row L^1 error, and nominal one-step reward magnitude respectively. For an acquisition of kind k, the sharper local reward error is $w_c(k)$ and its row error is $2w_p(k)$. For deployment of kind k, the local reward error is $w_g(k)$ and its row error is zero. Stop has zero primitive errors.

Lemma 10.1 (Bernoulli row error and expectation transport)**.** *For Bernoulli parameters p, q,*

$$\sum_{o\in\{0,1\}} |P_p(o) - P_q(o)| = 2|p - q|.$$

If $|W(t)| \le B$, then replacing an observation row P by $\widehat{P}$ changes its expected continuation by at most

$$\left|\sum_o (P(o) - \widehat{P}(o))W(T(s,a,o))\right| \le B\sum_o |P(o) - \widehat{P}(o)|.$$

Proof. The two Bernoulli row differences are $p - q$ and $q - p$, which have equal absolute values. For the second statement, apply the triangle inequality to the finite sum and use the uniform bound on each continuation value. □

Induction on the Bellman recursion gives $|\widehat{V}_h(s)| \le hM$ for the nominal value. The source's accumulated value radius and local Q radius are

$$D_h = \sum_{j=0}^{h-1} (\varepsilon + \delta j M), \tag{7}$$

$$\rho_h(a) = D_h + \varepsilon_a + \delta_a h M, \tag{8}$$

$$\Gamma_h = D_h + \varepsilon + \delta h M. \tag{9}$$

The Bellman residual is zero here because the nominal tables are exact Bellman values. The broader model-error framework also allows a supplied table with a proved nonzero residual, but no residual term is silently introduced in these quote results.

Theorem 10.2 (Value and Q enclosures)**.** *For every compatible realization, every state, and every continuation horizon,*

$$|V_h(s) - \widehat{V}_h(s)| \le D_h, \tag{10}$$

$$|Q_h(s,a) - \widehat{Q}_h(s,a)| \le \rho_h(a) \le \Gamma_h. \tag{11}$$

Thus $L_h(s,a) = \widehat{Q}_h(s,a) - \rho_h(a)$ *and* $U_h(s,a) = \widehat{Q}_h(s,a) + \rho_h(a)$ *enclose the true Q value.*

Proof. At horizon zero the value error is zero. In a single backup, insert and subtract the expectation of the nominal continuation under the true row. The reward error contributes at most ε_a, the continuation-value error at most D_h, and row transport at most $\delta_a h M$. Taking a maximum over the same feasible action set is nonexpansive in the uniform score error. Therefore $D_{h+1} \le D_h + \varepsilon + \delta h M$, whose accumulated upper bound is (7). The local Q statement follows from the same inserted expectation before taking a maximum. □

The reconstructed argument follows `value_error`, `local_q_error`, `value_between`, and `q_between` in `AVOC/RobustQuotePlanning.lean`. The primitive steps are `reward_close`, `probability_l1_le`, and `expectation_transport`.

10.2 What the robust selector optimizes

The executable robust choice maximizes $L_h(s,a)$ over feasible actions. It is a maximizer of a *certified surrogate lower bound*. It is not asserted to solve the exact minimax problem over all compatible realizations. A bound can be valid and substantially below the actual worst-case score.

Proposition 10.3 (Distinct-competitor dominance)**.** *If a is feasible and*

$$U_h(s,b) \le L_h(s,a) \quad \textit{for every feasible } b \ne a,$$

then a is optimal in every compatible realization.

Proof. For a distinct competitor, chain the three inequalities $Q_h(s,b) \le U_h(s,b) \le L_h(s,a) \le Q_h(s,a)$. The comparison with a itself is reflexive. □

This is `competitor_dominance_sound` in `AVOC/CalibrationBridge.lean`. Excluding self-comparison is essential for a useful non-singleton certificate. The older sufficient condition compares every feasible upper score, including the chosen action's own upper score, to its lower score. That older condition is sound but forces the chosen radius to be zero. The formal declaration `self_comparison_forces_zero_radius` records this limitation.

Example 10.4 (A deliberately conservative abstract quote fixture)**.** The retained uncertainty example assigns zero setup to both kinds, warm gross 200, cold gross 60, warm success band $[3/5, 1]$, and exact cold success 1. Start with both statuses untried and use $h = 1$, hence two total actions. The nominal scores are warm 120, cold 60, and stop 0. Here $\varepsilon = 0$, $\delta = 4/5$, $M = 260$, and $D_1 = 0$. The local warm radius is 208, while the cold and stop radii are zero:

action	$\widehat{Q}_1$	ρ_1	L_1
inspectWarm	120	208	−88
generateCold	60	0	60
stop	0	0	0

The lower-Q rule chooses cold although warm's exact compatible Q range is $[120, 200]$. This is a clear example of conservatism in the certified envelope. It does not show that cold is the exact maximin choice.

The source proves the widths, local radii, and numerical order through `uncertain_warm_local_radius`, `uncertain_cold_local_radius`, and `uncertainty_changes_numerical_order`. The displayed first-action ranking is a direct reconstruction at the stated initial state and horizon. This fixture is *abstract quote planning*, not an artifact-backed saving example. Its protected rank-one source has dense cost 36 per call and the fixture has one call, so gross values 200 and 60 cannot come from bounded execution savings for that source. The separate 1000-call example above is the artifact-backed comparison. These two roles must remain distinct.

10.3 Whole-policy regret, not only first-action quality

Let $a_h(s)$ maximize $L_h(s,a)$ and let a^* maximize the true Q_h. Then

$$\begin{aligned} V_{h+1}(s) - Q_h(s,a_h(s)) &\le U_h(s,a^*) - L_h(s,a_h(s)) \\ &\le U_h(s,a^*) - L_h(s,a^*) \\ &\le 2\Gamma_h. \end{aligned}$$

The comparison controls the Bellman loss of the actual choice at each state. The policy is rebuilt at every remaining horizon with the same frozen evidence.

Theorem 10.5 (Expected frozen-policy regret)**.** *For a horizon H, let $J_H^{\text{lower}}(s)$ be the expected value of the complete adaptive policy induced by robust lower-Q selection. Under any compatible realization,*

$$0 \le V_H(s) - J_H^{\text{lower}}(s) \le \sum_{h=0}^{H-1} 2\Gamma_h.$$

The comparator is the true optimal legal adaptive plan with the same conservative reservation guards.

Proof. Write the regret at a node with $h+1$ remaining steps as its local Bellman loss plus the expected regret of the selected continuation. The local term is at most $2\Gamma_h$. By induction, every child regret is bounded by the common sum $\sum_{j<h} 2\Gamma_j$. Taking the expectation leaves that constant unchanged because probabilities sum to one. This yields the cumulative upper bound. Nonnegativity follows because the selected complete tree is legal and V_H dominates the value of every legal tree. □

This is the argument implemented by `plan_regret_le` in `AVOC/GuardedPolicyRegret.lean` and `robust_policy_regret` in `AVOC/RobustQuotePlanning.lean`. Ordinary finite sums simplify the displayed bound to

$$\varepsilon H(H+1) + \frac{\delta M}{3} H(H-1)(H+1).$$

This last closed form is a prose algebraic consequence of the source's sums, not a newly compiled Lean declaration. The envelope is deliberately broad and may be vacuous for a particular application. It remains a valid certificate. The regret is an expected policy-value difference. It does not bound every realized sequence of successes and failures.

11 Exact finite-grid calibration

Probability bands need an evidence rule. The implemented rule is an exact rational finite-grid cell-envelope construction. It should not be relabeled as an exact inversion of binomial tails or as a Clopper–Pearson interval. Its proof uses a different and simpler exclusion budget.

Let K be the number of successes in n Bernoulli trials with rational parameter $p \in [0,1]$. Define

$$b_{n,k}(p) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad 0 \le k \le n.$$

Choose a positive integer grid size m and nonnegative rational allowance α. The grid cells have endpoints

$$\ell_i = \frac{i}{m}, \qquad u_i = \frac{i+1}{m}, \qquad 0 \le i < m.$$

For observed count k, the envelope of its point mass over cell i is

$$B_{n,k,i} = \binom{n}{k} u_i^k (1-\ell_i)^{n-k}.$$

Retain the cell exactly when

$$B_{n,k,i} > \frac{\alpha}{n+1}.$$

Return the closed interval spanning all retained cells. If there are no retained cells, return $[0,1]$. The returned interval is therefore always a valid band. The construction performs exact rational arithmetic and strict comparison. Boundary cells cover 0 and 1 as well as interior parameters.

Theorem 11.1 (Finite-sample grid coverage)**.** *If K has the binomial count law $b_{n,k}(p)$, the returned interval $I(K)$ obeys*

$$\mathbb{P}\{p \notin I(K)\} \le \alpha.$$

This holds for every rational $p \in [0,1]$, without assuming that p is a grid point.

Proof. Some grid cell contains p. In that cell, $p \le u_i$ and $1-p \le 1-\ell_i$, hence $b_{n,k}(p) \le B_{n,k,i}$. If p is outside the returned interval, the containing cell could not have been retained: any retained cell lies inside the hull. Therefore $B_{n,k,i} \le \alpha/(n+1)$, and the mass of this excluded count k is at most that threshold. There are $n+1$ possible counts, so summing their failure masses gives at most α. If no cells were retained, the returned full band cannot exclude p and the conclusion is immediate. □

The source proves the grid cover, the cell envelope, excluded-count mass, and final sum as `grid_covers`, `mass_le_envelope`, `excluded_mass_le`, and `missMass_le` in `AVOC/FiniteCalibration.lean`. This proves finite-sample validity of the algorithm itself within the exact rational binomial model. It does not infer the sampling model from a list of reported outcomes.

Example 11.2 (Ten successes are evidence, not certainty)**.** For $n = 10$, $m = 10$, and $\alpha = 1/10$, ten successes retain cells $\{6,7,8,9\}$ and return $[3/5, 1]$. Ten failures retain $\{0,1,2,3\}$ and return $[0, 2/5]$. These endpoint calculations are explicit Lean theorems. By contrast, the empirical singleton $\{K/n\}$ has no such coverage property. At $n = 1$ and $p = 1/2$, both possible singletons exclude the true parameter, so its miss probability is 1.

The relevant declarations include `ten_successes_lower`, `ten_successes_upper`, `ten_failures_lower`, `ten_failures_upper`, and `empirical_singleton_counterexample`. Nothing here says the true parameter has posterior probability $9/10$ of lying inside an observed band. The guarantee concerns repeated-data failure probability under the specified law.

11.1 Predetermined schedules and data-dependent selection

A schedule is a finite set J of checkpoints. Each checkpoint fixes an identity stratum, sample size n_j, grid $m_j > 0$, true parameter p_j, and nonnegative error allocation α_j. Its count statistic has the stated binomial marginal. Counts can be dependent, including different prefixes of one source and checkpoints that share other underlying randomness.

Theorem 11.3 (Scheduled and selected failure bounds)**.** *Let B_j be the event that checkpoint j's interval excludes p_j. Then*

$$\mathbb{P}\Big(\bigcup_{j\in J} B_j\Big) \le \sum_{j\in J} \alpha_j.$$

For any data-dependent selector $j(\omega) \in J$,

$$\mathbb{P}\{\omega \in B_{j(\omega)}\} \le \sum_{j\in J} \alpha_j.$$

Proof. For each outcome ω, the indicator of a union is at most the sum of the individual indicators. Multiply by the nonnegative rational weight of ω and sum over the finite probability space. Interchange the two finite sums to obtain the union bound. Each checkpoint has failure mass at most its allocated α_j by finite-grid coverage. Finally, selected failure implies failure of at least one scheduled checkpoint, so it is a subset of the union. This last inclusion holds pointwise even if selection sees all outcomes. □

This reconstructs `finite_union_le`, `scheduled_binomial_failure_le`, and `selected_binomial_failure_le` in `AVOC/CalibrationSchedule.lean`. Neither checkpoint independence nor an independent selector is required. Predetermination is required: the theorem does not authorize creating arbitrary new checkpoints after viewing the data. It is finite scheduled selection, rather than an unrestricted anytime confidence sequence.

Example 11.4 (Why the schedule's total allowance is necessary)**.** Take a fair Boolean outcome ω and two checkpoint bad events $B_0 = \{\omega = 0\}$ and $B_1 = \{\omega = 1\}$. Each has probability $1/2$. Selecting $j(\omega) = \omega$ makes selected failure certain. The total allowance $1/2 + 1/2 = 1$ remains valid, while either single checkpoint's allowance fails. This is the exact obstruction proved by `fairBool_adaptive_exceeds_single_budget`.

12 From source laws to actual terminal samples

The count-law obligation is substantial. Fair marginals do not imply a binomial sum. Two identical fair bits each have parameter $1/2$, but their two-bit count is either zero or two and never one. The source contains this obstruction as `correlatedPair_not_product`.

The primitive model uses a finite rational probability space and, for each identity k, a finite latent Boolean vector of length N_k. Its success parameter p_k is fixed. The fresh-outcome contract states that after agreement with any specified pattern on a previously revealed set of positions, a fresh position's joint success mass is p_k times that history mass. In notation,

$$\mathbb{P}\{H \text{ matches and } X_i = 1\} = p_k \mathbb{P}\{H \text{ matches}\}, \qquad i \text{ fresh}.$$

The multiplication-only form is meaningful for histories of mass zero and does not divide by them. Induction determines the mass of every full pattern, then summing patterns by success count gives its binomial law. Truncation yields the same law for each fixed prefix.

The formal steps are `conditional_implies_product`, `count_fiber_eq_binomial`, and `prefix_count_fiber_eq_binomial` in `AVOC/ObservationStream.lean`. This derives the count law from a primitive source law. The primitive source law itself remains an explicit model assumption.

12.1 Pending intents and immutable terminal events

The new lifecycle recorder separates a pending-intent list from a terminal log. A single global natural-number operation-ID namespace covers both lists and all identities. Pending entries store an ID and identity. Terminal entries store an ID, identity, and success, rejection, or fault. An unfinished outcome cannot be inserted as a terminal event.

Command situation	Defined result
Fresh begin	Append one pending ID and identity.
Exact pending begin replay	Return the same state.
Begin replay after matching completion	Return the same state without reopening.
Matching fresh finish	Remove the pending ID and append one terminal event.
Exact terminal replay	Return the same state.
Changed identity or terminal outcome	Reject the command.
Finish without matching intent	Reject the command.
Unfinished terminal outcome	Reject the command.

These cases resolve a concrete inherited obstruction. If an immutable log stores an unfinished event under an ID, later adding success under that ID changes the event and must be rejected. A pending registry therefore cannot simply be an unfinished terminal record. The declaration `inherited_unfinished_terminal_conflict` proves the mismatch, rather than relying on a narrative explanation.

Theorem 12.1 (Pure recorder lifecycle invariant)**.** *Every successful run from the empty state has unique pending IDs, unique terminal IDs, disjoint pending and terminal IDs, and no unfinished terminal entries. An accepted terminal event occurs once and remains once throughout every later successful continuation. Terminal history grows by suffix append.*

Proof. For a begin step, the transition first checks both uses of the ID. It either returns the unchanged state for a matching replay, rejects conflicting reuse, or appends a fresh pending entry. For a finish step, exact replay is unchanged; otherwise successful completion requires an unused terminal ID and a matching pending entry. Removing that pending ID and appending its terminal event preserves uniqueness and disjointness. Unfinished outcomes are rejected before these cases. Induction over a successful command list preserves the invariant. Every successful step leaves the log unchanged or appends one event, which proves retention under continuation. □

The source declarations are `run_wellFormed`, `runFrom_log_prefix`, and `accepted_terminal_retained_once` in `AVOC/ObservationRecorder.lean`. A success adds true once to its identity's samples, a rejection adds false once, and a fault adds no sample. False outcomes count toward sample readiness just as true outcomes do.

12.2 A source-generative recorder and whole-run refinement

The raw recorder accepts externally supplied terminal labels. The generative recorder instead has begin, report, and fault commands. A fresh report constructs its true or false label from the next latent Boolean

of its identity. A replay checks the terminal log first and reuses the stored outcome, consuming no new source bit. A fresh report with an exhausted source rejects. Faults and begins consume no source bit. Reporting after a fault conflicts with the immutable terminal event.

Every accepted generative step emits an ordinary recorder command and an effective reveal marker. Whole-run induction proves that the emitted commands reproduce the actual raw recorder state and that the reveal state equals the inherited reveal machine on its effective trace. In particular, for every identity,

$$\mathrm{samples}_k(\mathrm{terminalLog}) = \mathrm{source}_k.\mathrm{take}\big(|\mathrm{samples}_k(\mathrm{terminalLog})|\big). \tag{12}$$

This is `run_refines` and `run_samples_eq_take` in `AVOC/RecorderReveal.lean`. The source-prefix correspondence is a derived invariant of the defined machine, not an additional equality asserted for each observed raw event. It still does not establish that an arbitrary external success label came from this latent source. A parser can check the label's syntax without checking its physical or statistical origin.

Example 12.2 (Replay and faults do not create new evidence)**.** Let an identity's source be [true, false]. Begin ID 7, report ID 7, then report ID 7 again. The sample list is [true] and the remaining source begins with false. Begin ID 8 and finish it as a fault. The sample list stays unchanged. Begin and report ID 9. It appends false, yielding [true, false]. A further fresh report rejects after source exhaustion, while a replay of ID 7 remains possible. The replay supplies no extra success and the fault supplies no concealed binary observation.

This is an illustrative execution of the defined transitions. The retained formal examples `replay_fault_exhaustion_example` and `exhausted_fresh_report_rejected` provide closely related checked instances. Reusing a different ID for the same physical operation is a separate identity problem. Global ID uniqueness alone cannot rule it out.

13 Unconditional calibrated policy issuance

The recorder-to-policy endpoint reads counts from the actual terminal log. Rejected whole runs supply no counts. Successful runs with *any* pending operation also supply no counts, even when the pending identity is unrelated to the selected checkpoint. A fully resolved run still needs enough success or rejection reports for both selected checkpoints. A checkpoint always reads the first predetermined n_j reports, so later reports cannot change that count.

When both selected counts are available, the endpoint constructs their exact learned probability bands and installs them for a fresh frozen policy episode. Setup and gross memberships are supplied deterministic premises. The warm and cold selectors may depend on all outcomes but must select checkpoints with the specified warm and cold true parameters. Changing a probability band does not authorize changing an outstanding operation's evidence.

Theorem 13.1 (Recorder-to-policy bad-issuance bound)**.** *Assume finite rational latent sources satisfy the primitive conditional laws; the finite schedule, grids, and nonnegative allowances are predetermined; scheduled prefix lengths fit within their sources; selected strata match the specified true probabilities; and deterministic setup/gross memberships hold. Then*

$$\mathbb{P}\{\textit{a plan is issued and lacks its episode certificate}\} \le \sum_{j \in J} \alpha_j. \tag{13}$$

The episode certificate supplies a compatible realization with the stated exact parameters, value and Q enclosures, and expected whole-policy regret for every finite horizon under frozen quotes.

Proof. The primitive conditional law yields binomial prefix count laws at all scheduled checkpoints. The generative run refinement gives (12). Whenever gated counts are available, taking the fixed first n_j reports therefore gives the corresponding latent prefix count. On the event where all scheduled bands cover their true parameters, either the gates suppress issuance or both selected learned bands contain the correct parameters. Combine those memberships with the supplied deterministic memberships to construct a compatible realization. The deterministic frozen-episode theorem then supplies the policy certificate. Thus bad issuance is a subset of the scheduled band failure union, whose probability is bounded by the total allowance. □

The final declarations are `recorder_episode_coverage` and `recorder_issued_failure_le` in `AVOC/RecorderPolicy.lean`. Success of execution, absence of pending intents, availability of enough samples, log-prefix equality, and the binomial count law are not separately supplied premises to that final theorem. They are derived or checked by its machine and gates. The primitive source laws and deterministic memberships are still supplied.

The probability in (13) ranges over the original outcome space, including rejected, unresolved, and insufficient-data runs. There is no claim that a plan will be issued, that any operation finishes, or that collection is cheap. If the issuance probability is $q > 0$, the generic conditional implication is only

$$\mathbb{P}\{\text{uncertified} \mid \text{issued}\} \le \min\left(1, \frac{\sum_j \alpha_j}{q}\right).$$

The same nominal allowance need not hold after conditioning on issuance.

13.1 Completion conditioning can select a biased population

Example 13.2 (Predictable availability still biases completed runs)**.** Let (X_1, X_2) be independent fair bits. Reveal X_1, and reveal X_2 by the deadline only if $X_1 = 1$. Availability of the second report depends on the past, not on X_2. Completion probability is $1/2$. Conditional on completion, $X_1 = 1$ with probability one, and the two-bit count has probability zero of being zero. A genuine two-trial binomial count with parameter $1/2$ has zero count probability $1/4$.

This is kernel-checked in `AVOC/ObservationCensoring.lean`, including `deadline_second_retention_predictable`, `deadline_conditioned_first_success`, and `deadline_conditioned_count_not_binomial`. It shows why one cannot replace the original law by the law conditioned on enough reports arriving.

A direct interval illustration uses ten independent fair bits and issues an interval only when all ten are true. Its issuance probability is 2^{-10}. The implemented $n = m = 10$, $\alpha = 1/10$ interval is then $[3/5, 1]$, which excludes $1/2$. Conditional on issuance, failure is certain. Unconditional bad issuance is only 2^{-10} and satisfies the allowance. This finite calculation is an illustrative consequence of the checked interval endpoints and the elementary product law, not a new Lean declaration.

13.2 Complete logs do not fix outcome-dependent completion order

Conditioning and ordering are different obstructions. Suppose two independent fair attempts start together. Success takes one unit of time and rejection takes two; resolve ties by the initiation ID. Record

every final outcome and wait until all intents have resolved. The completion order is

initiation outcomes	completion-order outcomes	mass
$(0,0)$	$(0,0)$	$1/4$
$(0,1)$	$(1,0)$	$1/4$
$(1,0)$	$(1,0)$	$1/4$
$(1,1)$	$(1,1)$	$1/4$

The first completed bit succeeds with probability $3/4$. No report was missing, no terminal event was forged, and no operation remains pending. Nevertheless, the first-completed prefix is not a fair source prefix.

For the implemented interval, the stronger retained finite experiment starts twenty independent fair attempts and lets all successes complete before all rejections. If there are at least ten successes, the first ten completed reports are all true. Their interval is $[3/5, 1]$ and excludes $1/2$. That event has mass

$$\frac{1}{2^{20}} \sum_{k=10}^{20} \binom{20}{k} = \frac{308333}{524288} > \frac{1}{10}.$$

Thus actual interval failure exceeds its nominal allowance despite complete, fully resolved lifecycle records. The exact Python evidence is preserved under `results/v0.41.0/review_completion_order.py` and its JSON receipt. It is a finite computational obstruction with an explicit rational derivation here, not a newly kernel-checked theorem.

The generative theorem assigns the next latent bit in report order and assumes that source's conditional law. Independent outcomes in initiation order do not automatically satisfy that assumption after outcome-dependent reordering. A physical deployment must justify its report-order contract, for example through an initiation-indexed sampling policy or a proved alternative coupling. The current release leaves that physical coupling open.

14 Checked recovery and durable accounting

Recovery must distinguish a recorded intention from a completed paid result. The robust quote recovery model binds a commit to its sequence, complete pre-state, selected action, quote evidence, outcome tag, and post-state. Fresh acknowledgment checks those bindings and checks that the supplied typed outcome realizes the action through the runtime's `confirm` transition. It then requires the computed successor's store view to equal the recorded durable post-state. An unfinished intent is rejected rather than interpreted as a negative observation or permission to repeat paid work.

Proposition 14.1 (Fresh acknowledgment and replay)**.** *A successful fresh acknowledgment advances the sequence by one and yields the successor certified by* `confirm`*. It preserves the runtime safety invariant. Repeating that accepted commit at the resulting matching checkpoint returns the same checkpoint.*

Proof. Each failed identity, sequence, pre-state, outcome, evidence, or post-state comparison returns an error. A successful fresh branch therefore exposes all those equalities and a successful `confirm`. The successor invariant follows from `confirm_safe`. At the resulting checkpoint, the recorded last commit, next sequence, and actual store view match the duplicate branch, which returns the state unchanged. □

These are `acknowledge_fresh_contract`, `acknowledge_safe`, and `acknowledge_idempotent` in `AVOC/RobustQuoteRecovery.lean`. Duplicate acknowledgment recognizes the already accepted commit

and checkpoint. It does not freshly validate every field of an independently supplied outcome argument on that replay branch. This is appropriate for state replay but must not be described as new checking of a second physical result.

The `prepare` constructor selects the robust action and binds current evidence into its intent. Acknowledgment alone does not prove that an arbitrary supplied commit was originally produced by `prepare`. If the conclusion needs provenance of the planner's choice, that provenance must be supplied or established by the enclosing protocol. Fresh acknowledgment proves the exact checked transition contract, not an unstated origin claim.

Probability bands are frozen within the episode. Replacing them for a fresh episode preserves the source and protected identity fields but changes the approved evidence. An old intent cannot be acknowledged under the changed evidence. The explicit declaration `new_bands_obstruct_old_acknowledgement` in `AVOC/CalibrationBridge.lean` prevents calibration from silently rewriting an active durable operation.

14.1 Prepaid evaluation and upgrade journals

The separate durable-upgrade machine has a journal containing the current upgrade state, one optional pending ticket, and closed tickets. A ticket binds a campaign-local key, its complete parent state, and a proposal. Beginning a funded fresh ticket charges its protected evaluation tariff immediately. Repeating the same ticket is unchanged. Finishing it charges no second evaluation tariff, and switching occurs only according to the checked upgrade decision. Abandoning clears the pending ticket while retaining its already charged evaluation expenditure.

The upgrade authority supplies checker soundness and sound lower and upper utility bounds for one fixed workload, objective, and unit convention. These are explicit obligations in `AVOC/VerifiedUpgrade.lean`. Receiving a JSON utility bound or an approval flag does not prove them.

In particular, a failed or abandoned evaluation is not a free trial. The complete parent state includes spending, so reusing a ticket based on an old active version but ignoring changed expenditure is rejected. The formal declarations `begin_keeps_active`, `finish_evaluation_unchanged`, and `abandon_keeps_current` make these distinctions explicit.

For a successful paid trace with total evaluation debit e and switching debit u, the formal accounting identities are

$$\text{evaluationSpent}_{\text{final}} = \text{evaluationSpent}_{\text{initial}} + e, \tag{14}$$

$$\text{switchSpent}_{\text{final}} = \text{switchSpent}_{\text{initial}} + u. \tag{15}$$

From a safe initial journal, their sum stays within protected credits. Successful replay also cannot decrease the protected utility of the active version. These are `paid_trace_exact_accounting`, `paid_trace_no_overspend`, and `replay_utility_nondecreasing` in `AVOC/DurableUpgradeLaws.lean`.

Restoration replays supplied commands from a specified initial version and compares the result with the actual checkpoint. A successful restoration equals that checkpoint. Provided the specified initial version satisfies the authority's contract, restoration also inherits the safety certificate. This checks consistency between two supplied records. It cannot authenticate their physical origin or detect coordinated replacement of both the journal and checkpoint.

14.2 What recovery establishes about external execution

The lifecycle recorder is a pure finite snapshot reducer. Its once-only terminal history is not exactly-once external execution. It neither invokes an operation nor persists a crash-safe transaction. The recovery models give checked state-transition and replay contracts, while the retained runtime and storage adapters have their stated bounded evidence and external trust assumptions. Filesystem durability, endpoint authentication, physical identity, and reliable observation delivery are not consequences of a pure transition function.

A rejected pure batch supplies no final export. That is an all-or-nothing reduction result, not rollback of physical work already performed before rejection. A fault closes an intent without supplying a binary sample. It does not establish that retrying the corresponding paid operation is safe or free. Conservative reserved spending remains a protected model quantity rather than an automatic identity with actual billed expenditure.

The completed chain is therefore precise: under the declared finite source model, the lifecycle machinery derives trustworthy prefix correspondence; fixed scheduled calibration bounds bad issuance; on the simultaneous-coverage event, every issued frozen episode has the proved value and policy guarantees; outside that event, uncertified issuance has mass at most the scheduled total allowance; and the checked recovery machines preserve their specified state and accounting contracts. Physical source coupling, universal host/storage refinement, multi-episode accumulation, and measured productivity remain separate obligations.

15 Evidence, reproduction, and the publication boundary

The authoritative formal baseline for this edition is `AVOC_Formal_v0.41.0.zip`, with SHA-256

```
04efab87010b3b88c9a119275e190423a2465da32ba1e526edb7e6d2e0cd077e.
```

The original archive is preserved intact in the accompanying package. Its internal payload manifest was freshly checked for this edition: all 16,912 listed payload digests match. The archive contains the mathematical source, interfaces, experiments, review records, failed attempts, and reproduction instructions. It excludes the Lean runtime and compiled project/dependency objects.

15.1 Archived formal evidence

The v0.41.0 release record reports 222 positive build targets, all passing under warnings-as-errors, and 2,268 exact named theorem/lemma axiom reports. The allowed axiom union is `Classical.choice`, `Quot.sound`, and `propext`. There are 195 non-audit mathematical modules; the latest block adds four modules and 69 named theorem/lemma declarations. The 191 inherited mathematical modules were checked byte-identical to the preceding release. The archive also records 135 substantive negative proof rejections.

These are different counts with different meanings. A build target may be an executable or audit target rather than a mathematical module. A named theorem/lemma count includes local infrastructure and explicit private helpers. It is not a count of independent discoveries. Negative-test success means that the incorrect test proposition was rejected for a substantive proof reason, not merely that a command exited unsuccessfully.

The compiler pin is Lean 4.34.0-rc2, commit

```
6a10ac8c22beadecabdbb0919c2b50214762f91d.
```

The mathlib pin is

```
aeb72f1e9981b6d35b426b16dca7759be3e7ba06.
```

Nine exact dependency revisions are retained in `lake-manifest.json`. Local project objects were rebuilt from empty in the archived release. The external pinned caches were not rebuilt from every source. This distinction belongs to the reproducibility contract.

15.2 Fresh checks for this edition

A fresh source inventory matches the actual 195 mathematical modules and 2,268 named declarations to the delivered registry. It checks the archived declaration-source digests and recorded source locations. Fresh exact Python computations reproduce the declared examples and finite recorder comparisons. The detailed tested families, assertions, input digests, outputs, and commands are included in the publication's `reviews` directory.

No fresh Lean build or executable Lean comparison was performed for this publication edition. The compiler and exact portable dependency kit were unavailable in this session. The companion therefore uses archived kernel evidence together with fresh source-identity checks; it does not relabel that evidence as a new compilation. The Python comparisons remain bounded executable evidence and do not prove universal Python, parser, or storage refinement.

15.3 Assurance matrix

Layer	Established statement	Application obligation
Task context	Fiber criterion and decision-rule transport in the stated finite model	Identify the actual task and allowed rules
Artifact checker	Soundness/completeness for specified Boolean or affine syntax	Bind the checked input to the intended artifact
Cost comparison	Exact declared syntax/arithmetic tariffs and break-even calculations	Justify conversion to physical cost or reward
Frozen quote planner	Q/value enclosures, admissible robust choices, expected policy regret	True parameters and deterministic values belong to approved bands
Calibration	Finite rational cell-envelope coverage on fixed scheduled checkpoints	Source law, parameter strata, and predetermined schedule
Generative recorder	Unique logical IDs, immutable terminal retention, source-prefix refinement	Couple external operations and reporting order to the defined source
Recovery model	Confirmed logical evidence, selected installation, replay/accounting invariants	Input provenance and external execution/storage contracts

Layer	Established statement	Application obligation
Host programs	Exact bounded tests and archived Lean endpoint comparisons	Universal parsing/host refinement, authentication, physical persistence

15.4 The remaining research questions

The principal statistical question is the coupling from physical attempts to reported samples. In the generative model, a fresh accepted report consumes the next latent bit. In a physical service, an attempted outcome can affect latency, retention, fault reporting, or whether a report appears at all. The completion-order counterexample proves that merely resolving every intent does not solve this problem. An initiation-indexed sampling contract is a plausible future target, but this paper does not claim to have proved it.

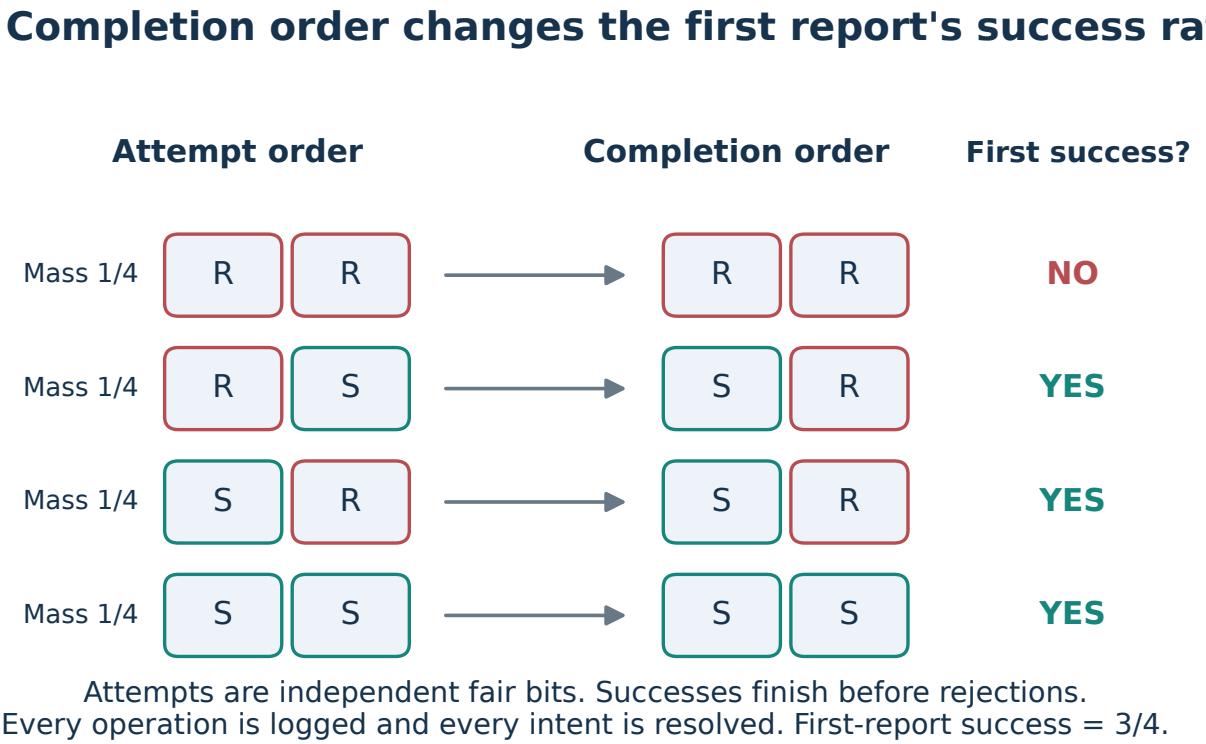


Figure 3: Every pair of attempted outcomes is logged and resolved, yet successes finishing first make the first reported bit successful with probability 3/4. This exact finite counterexample concerns a physical interpretation of reported samples; it does not refute the formal generative-source theorem.

A second question is controller composition. Current quotes are frozen within an episode. Updating bands during the same episode changes the policy and its dependency structure. Repeated episodes require explicit treatment of cumulative error, cost, and reward. The finite schedule theorem alone does not supply an unrestricted optional-stopping or continual-learning result.

A third question is implementation assurance. The raw recorder accepts supplied labels; it does not authenticate their truth or execute a paid operation. An injective identity registry can prevent protected

identity aliasing within its declared input contract, but a unique terminal ID does not prevent a caller from assigning two IDs to the same physical work. The inherited SQLite implementation relies on its database, operating-system, filesystem, and hardware contract. Its process-exit tests do not establish power-loss or distributed-system correctness.

Finally, the present source laws are finite and rational, and the artifact backends use explicit languages and tariffs. Broader real-valued probability models, drift, dependence, scalable calibration, unrestricted stopping, larger artifact languages, and measured service workloads remain distinct extensions. None is needed to state the existing formal results accurately.

16 Conclusion

AVOC supplies a checked finite route from artifact semantics and quoted uncertainty to admissible planning, calibrated policy issuance, and recoverable logical state. Its results are strongest when their interfaces are explicit. Correctness of an artifact, validity of a probability model, value of an action, and durability of external work are separate assertions, each with its own evidence.

The recorder composition closes an operational step inside the formal model: the samples used by calibration are the actual terminal samples of a complete generative run. The completion-order counterexample identifies the next external step with equal clarity. Together they provide a usable publication boundary: a substantial verified architecture, exact explanatory examples, and an explicit physical-sampling problem that remains open.

A Source and document navigation

The source root inside the original archive is `AVOC_Formal_v0.41.0`. Begin with `START_HERE.md`, `FORMAL_THEORY.md`, and `PROOF_BOUNDARY.md`. The full named declaration index is `THEOREM_INDEX.md`. Current reproduction instructions are in `REPRODUCE.md`; interface details are in `RECORDER_EXECUTABLE.md` and `OBSERVATION_EXECUTABLE.md`. Earlier source and verification records remain under `provenance`, `history`, and `results`.

Publication review maps identify which source statements support each manuscript section. The separate Lean companion gives the module groups, actual theorem names and excerpts, archived evidence schema, and practical reproduction commands. Figure programs and exact test outputs make the visual examples independently reproducible. These navigation aids supplement the original source rather than replacing its hypotheses.

References

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Proof Path Theory

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Justification hypergraphs, certificate extraction, recovery debt and mathematical memory.

KEEP THIS BOUNDARY

Finite declared proof fields and written arguments; discovery memory is not itself a proof of its target.

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Original text begins on the next page.

The source pages that follow are preserved without editorial stamps. Page labels in the PDF navigation use the source code and physical source-page number. The package retains the byte-identical original PDF and its SHA-256 digest.

Proof Path Theory

Certificates, Compression, Discovery Fields, and Mathematical Memory

Komorebi P.

Version 1.0.0

Abstract

A finished proof is a certificate of implication and a compressed projection of a larger mathematical object. The larger object may contain alternative derivations, examples, failed routes, exact computations, boundary cases, provenance, and bridges to later questions. This paper develops a finite Proof Path Theory that keeps these layers distinct. A proof field is a typed acyclic justification hypergraph. Certificates are selected certified subfields that derive declared targets from admissible leaves. In the fixed AND-DAG case, dependency ancestry gives a unique minimal certificate. With alternative justifications, minimum certificate selection is NP-complete by reduction from Set Cover.

Compression is defined relative to task families, producing endpoint, audit, and discovery equivalence, a recoverability preorder, vector-valued loss, and expected recovery debt. A latent bridge graph models declared future reach. Its weighted coverage utility is monotone and submodular, yielding a greedy cardinality guarantee. The framework also treats versioned proof objects, evidence ledgers, correlated review, audit queues, and goal-dependent refinement. The accompanying Python package implements the finite theory, validates proof fields, extracts certificates, computes retention policies, and reproduces the examples. The theory preserves deductive standards while formalizing what compression retains, discards, and makes costly to reconstruct.

Status of the work. The mathematical theorems below concern finite declared proof fields, task families, bridge graphs, covariance matrices, and queue models. Their proofs are given in the text. The computational results are exact finite checks or deterministic numerical examples generated from stated inputs. No simulation is presented as an empirical law of mathematical discovery. Historical priority and comparison with neighboring literatures remain separate questions.

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1 Introduction

Mathematics usually publishes a proof after the proof has been compressed. Definitions have stabilized. Failed attempts have disappeared. A long calculation has become a lemma. A representation change that made the argument visible is replaced by a direct sentence. This is often an improvement. The final certificate is easier to verify and easier to reuse.

Compression can also remove mathematical information that is not needed for the present theorem. An example may explain why a definition was chosen. A failed route may isolate the exact boundary of a method. An alternative proof may carry a bridge to another field. A computation may reveal a stronger statement than the one originally sought. None of these objects is automatically part of the final implication

$$A \vdash B.$$

They may still be part of the mathematical event that produced it.

Proof Path Theory studies this larger object without changing the standard of proof. Its central separation is

$$\boxed{\text{certificate of validity} \quad \neq \quad \text{expanded discovery field.}}$$

The certificate answers whether the conclusion follows. The discovery field records how the conclusion became reachable, what alternatives were considered, what was lost during compression, and which omitted structures may matter later.

The finite theory developed here has six connected layers.

1. A typed proof field records claims, admissible leaves, justifications, evidence, and targets.
2. A certified subfield separates proved or otherwise certified transitions from candidates and interpretations.
3. A certificate is a selected subfield that derives the target.
4. Compression is evaluated relative to declared tasks rather than by length alone.
5. A bridge graph represents possible future reach and makes archival retention an optimization problem.
6. Versioning, review, audit, and refinement describe the life of the proof after the first certificate exists.

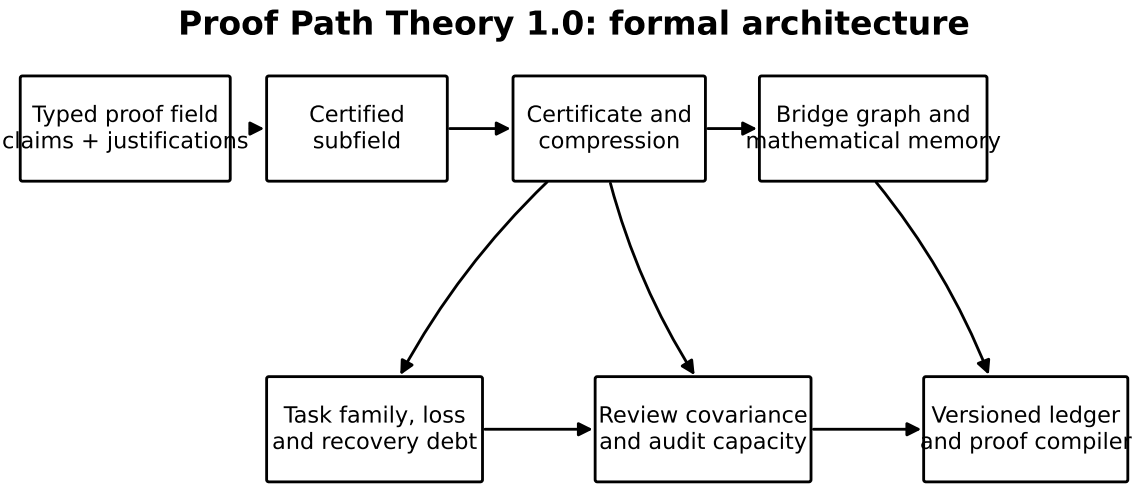


Figure 1: The formal architecture. The certificate is one verified projection of a larger versioned proof field.

The theory is intentionally finite at its foundation. Finite objects make every closure condition and algorithm explicit. Infinite, probabilistic, and higher categorical extensions are listed later as open programs rather than silently assumed.

2 Claim types and logical boundaries

A useful theory of proof paths must not turn every recorded object into a proof step. A notebook may contain a theorem, a conjecture, a failed analogy, a numerical experiment, and a historical interpretation on the same page. Their coexistence does not give them the same logical status.

Definition 2.1 (Evidence kind)**.** An evidence kind is a declared type attached to a record supporting a claim or justification. The core finite vocabulary used here is

formal proof, assumption, exact computation,
numerical evidence, conjecture, external result,
counterexample, interpretation.

An evidence record also states its source, scope, and whether it certifies the justification to which it is attached.

The word "certifies" is deliberately separate from the evidence label. An exact computation may certify a finite identity while providing only numerical evidence for a general statement. An external result may certify a step only on the domain covered by its hypotheses. The scope field is therefore part of the mathematical object.

Warning 2.2 (Evidence types are not one ranking)*.* The evidence kinds do not form a universal total order. A formal proof of a weak statement and an exact computation of a stronger finite instance cannot

be compared by one scalar without specifying the task. The ledger retains type and scope instead of replacing them with a single confidence number.

This separation will be used throughout. Candidate edges may remain in the discovery field, but only certified edges enter a proof certificate.

3 Typed proof fields

The first formal object is a directed hypergraph. A justification may require several premises at once, and a claim may admit several alternative justifications.

Definition 3.1 (Typed proof field)**.** A finite typed proof field is a tuple

$$\mathcal{P} = (V, \mathcal{J}, A, T, \tau, \mathcal{E}, \mathbf{v}, \mathbf{c}),$$

where:

- V is a finite set of claim nodes.
- $\mathcal{J}$ is a finite set of justifications.
- Each $j \in \mathcal{J}$ has a finite premise set $\mathrm{prem}(j) \subseteq V$ and one conclusion $\mathrm{con}(j) \in V$.
- $A \subseteq V$ is the set of required premises of the declared judgment.
- $T \subseteq V$ is the nonempty target set.
- τ assigns node and justification types.
- $\mathcal{E}$ is a typed evidence ledger.
- $\mathbf{v}$ assigns nonnegative value coordinates to nodes.
- $\mathbf{c}$ assigns nonnegative storage, audit, recovery, and cognitive costs.

The dependency graph contains an edge $u \to v$ whenever $u \in \mathrm{prem}(j)$ and $v = \mathrm{con}(j)$ for some justification j. It is required to be acyclic.

A single justification is conjunctive in its premise set. All listed premises must be available. Multiple justifications with the same conclusion are alternatives. This gives the familiar AND/OR structure of proof search without forcing it into an ordinary graph.

Definition 3.2 (Admissible leaf)**.** A node is an admissible leaf when it is a required premise, a declared axiom, or an explicitly cited external result. A zero-premise certified justification may also introduce a leaf when the proof system permits it.

Definition 3.3 (Certified subfield)**.** The certified subfield $\mathcal{P}^{\mathrm{cert}}$ contains every node of $\mathcal{P}$ and only those justifications whose evidence and scope have been declared sufficient to certify the recorded transition.

Definition 3.4 (Derivability)**.** Let $K \subseteq V$ be a selected node set and let $H \subseteq \mathcal{J}$ be a selected justification set. The derivable set $D(K, H)$ is the least subset of K satisfying:

(i) every selected admissible leaf belongs to $D(K, H)$;

(ii) if $j \in H$, $\mathrm{prem}(j) \subseteq D(K, H)$, and $\mathrm{con}(j) \in K$, then $\mathrm{con}(j) \in D(K, H)$.

When only certified justifications are allowed, write $D_{\mathrm{cert}}(K, H)$.

Proposition 3.5 (Finite derivability)**.** *For a finite acyclic proof field, $D(K, H)$ is uniquely determined and can be computed in one topological pass through the dependency graph.*

Proof. Acyclicity gives a topological ordering of the nodes. Begin with the selected admissible leaves. At each later node, test whether one selected incoming justification has all of its premises already derived. Add the node exactly when such a justification exists. No later step can change the status of an earlier node because all dependencies point forward. The resulting set satisfies the closure rules. Any set satisfying the rules must contain every node added by this pass, so the result is the least such set and is unique. □

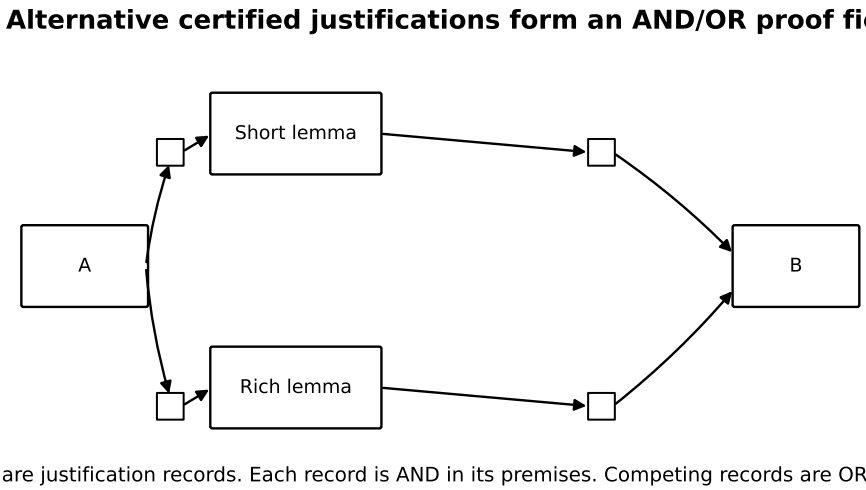


Figure 2: A proof field with alternative certified justifications. Squares represent justification records.

4 Certificates and verification closure

Definition 4.1 (Certificate)**.** A certificate for a requested target set $T_0 \subseteq V$ is a pair

$$\mathcal{C} = (K, H), \qquad K \subseteq V, \qquad H \subseteq \mathcal{J}^{\mathrm{cert}},$$

such that

$$A \cup T_0 \subseteq K \qquad \text{and} \qquad T_0 \subseteq D_{\mathrm{cert}}(K, H).$$

The certificate is node-minimal, justification-minimal, or cost-minimal according to the declared objective.

The canonical target request is usually a subset of the declared set T, although the same object can certify an intermediate node selected for local analysis. This definition preserves the requested endpoint judgment and requires every target to be reconstructed from certified transitions. Extra nodes may remain in a certificate, but they do not help minimality under nonnegative costs.

4.1 The fixed AND-DAG case

The simplest case occurs when every relevant derived node has exactly one certified incoming justification. Then every recorded dependency is mandatory.

Definition 4.2 (AND-DAG)**.** A certified proof field is an AND-DAG relative to target set T_0 when every nonleaf node in the certified ancestry of T_0 has exactly one certified incoming justification.

Let $\mathrm{Anc}(T_0)$ denote the node ancestry obtained by following the unique certified justifications backward from T_0.

Theorem 4.3 (Unique verification core)**.** *Let $\mathcal{P}$ be an AND-DAG relative to T_0. The selected nodes*

$$K_{\mathrm{core}} = A \cup T_0 \cup \mathrm{Anc}(T_0)$$

together with the unique incoming justification of every selected nonleaf node form a certificate. This certificate is the unique node-minimal and justification-minimal certificate for the recorded derivation.

Proof. Backward traversal from each target selects the unique justification that can derive it and every premise required by that justification. Repeating the process terminates because the dependency graph is finite and acyclic. The selected leaves are admissible, and a forward topological pass derives every selected node, so the construction is a certificate.

Let (K, H) be any certificate for the same target set. Each target must be derived. Since it has one certified incoming justification, that justification must lie in H, and all of its premises must lie in K. Applying the same argument recursively forces every node and justification in the backward ancestry into (K, H). Therefore every certificate contains the constructed core. Uniqueness and minimality follow. □

Corollary 4.4 (Linear extraction)**.** *The unique core of an AND-DAG can be extracted in time linear in the size of the relevant certified ancestry.*

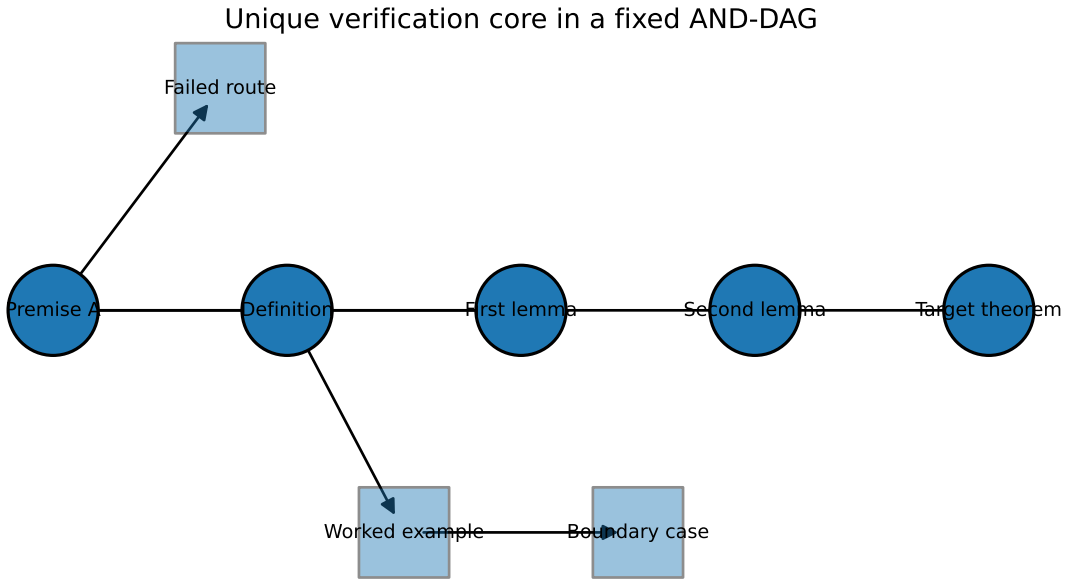


Figure 3: The unique verification core and optional discovery material in a fixed AND-DAG.

Remark 4.5. The theorem identifies the minimum certificate for the recorded derivation. It does not claim that the derivation is globally shortest among all possible proofs of the theorem. A different proof field may have a smaller core.

5 Alternative derivations and certificate complexity

When a claim admits several certified justifications, certificate extraction is no longer a backward closure problem. A local choice may be reused by several later obligations, which couples decisions across the field.

Problem 5.1 (Minimum Certificate Decision)**.** Given a finite acyclic certified proof field, a target set, nonnegative integer node and justification costs, and a budget B, determine whether a certificate of total cost at most B exists.

Theorem 5.2 (Minimum certificate selection is NP-complete)**.** *Minimum Certificate Decision is NP-complete, even when every nonzero cost is a unit node cost and all justification costs are zero.*

Proof. The problem belongs to NP. A proposed node and justification set can be checked in polynomial time by verifying references, computing derivability in topological order, and summing costs.

For NP-hardness, reduce Set Cover. Let $U = \{u_1, \ldots, u_m\}$ be the universe, let $S_1, \ldots, S_n \subseteq U$ be the available sets, and let k be the cover budget. Construct one optional admissible leaf node s_i for each set S_i, with unit cost. Construct one zero-cost obligation node e_j for each element u_j. Whenever $u_j \in S_i$, add a certified justification

$$\{s_i\} \Longrightarrow e_j.$$

Add a zero-cost target node t with the certified justification

$$\{e_1, \ldots, e_m\} \Longrightarrow t.$$

The dependency graph is acyclic. A certificate for t must derive every e_j. Deriving e_j requires at least one selected source node s_i with $u_j \in S_i$. One selected s_i may discharge several obligations, and its cost is paid once. Therefore a certificate of cost at most k exists exactly when the original Set Cover instance has a cover of size at most k. The reduction is polynomial. □

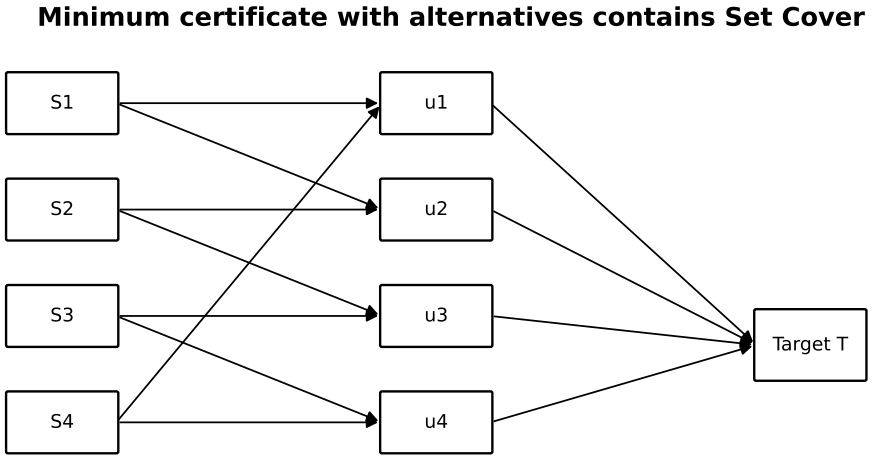


Figure 4: The Set Cover encoding used in the complexity proof.

Remark 5.3 (Algorithmic consequence). Exact minimum certificate search is practical for small fields or fields with few alternatives. Large formal libraries require branch and bound, dynamic programming on restricted structures, integer programming, approximation, or proof-system-specific methods. The reference package includes an exact finite search with an explicit state limit. It does not hide the exponential worst case.

6 Certified paths as a category

The hypergraph model records local justifications. A complementary view treats complete certified transformations as morphisms.

Definition 6.1 (Certified proof category)**.** A certified proof category Cert has mathematical states as objects and certified transformations as morphisms. Identity morphisms retain a state unchanged. Composition is sequential use of compatible certified transformations.

A state may be a proposition, a diagram, an exact finite computation, a coordinate representation, or a structured collection of claims. Candidate analogies may live in a larger discovery graph, but they do not enter Cert until their logical force and domain have been certified.

Proposition 6.2 (Functorial transport)**.** *Let $F : \mathsf{Cert}_1 \to \mathsf{Cert}_2$ be a functor. If*

$$A \xrightarrow{f_1} S_1 \xrightarrow{f_2} \cdots \xrightarrow{f_r} B$$

is a certified path in Cert_1, then

$$F(A) \xrightarrow{F(f_1)} F(S_1) \xrightarrow{F(f_2)} \cdots \xrightarrow{F(f_r)} F(B)$$

is a certified path in Cert_2.

Proof. A functor maps objects to objects and morphisms to morphisms. It preserves identities and composition. The image of the composite path is therefore the composite of the image morphisms, which is certified in the target category. □

The proposition is elementary, but its interpretation is central. A proof may be transported into another language without changing validity. What changes is visibility. A spectral representation may expose a symmetry hidden in physical coordinates. A finite exact computation may expose the correct statement before a general derivation is known. A formal proof assistant may expand dependencies that a journal proof compresses.

Definition 6.3 (Compositional certificate map)**.** A compositional certificate map is a functor

$$\mathcal{C} : \mathsf{Expanded} \longrightarrow \mathsf{Certificate}$$

that sends expanded certified paths to checkable certificate paths and preserves identity and composition.

Proposition 6.4 (Local audit composes)**.** *If every primitive morphism in a finite path has an auditable certificate and the certificate map is compositional, then the composite path is auditable from the primitive certificates and the declared composition rules.*

Proof. Proceed by induction on path length. The length-zero case is an identity. Assume the first r morphisms have an auditable composite certificate and append one certified morphism. Functoriality identifies the new certificate with the composition of the earlier certificate and the primitive certificate of the appended morphism. The declared composition rule completes the audit. □

7 Compression relative to a task family

A representation is not simply good or bad. It is sufficient or insufficient for a declared family of questions.

Definition 7.1 (Task)**.** A task q is a query whose answer is defined on the full proof field. Examples include:

- verify the target theorem;
- reconstruct every primitive dependency;
- teach the conceptual bottleneck;
- recover a boundary example;

- identify all claims affected by a revision;
- search for bridges to a future target family.

A task family is a set $\mathcal{Q}$ of such queries.

Definition 7.2 (Task-sufficient representation)**.** A representation R of a proof field $\mathcal{P}$ is sufficient for task q when there exists a decoder d_q such that

$$q(\mathcal{P}) = d_q(R).$$

It is $\mathcal{Q}$-sufficient when it is sufficient for every $q \in \mathcal{Q}$.

This definition includes exact reconstruction as one task, but it does not require exact reconstruction for every purpose. A certificate can be sufficient for verification and insufficient for teaching. A discovery archive can be sufficient for bridge search and unnecessarily large for routine checking.

Theorem 7.3 (Task information loss criterion)**.** *Let C be a compression map and let q be a task. If there exist proof fields $\mathcal{P}_1$ and $\mathcal{P}_2$ such that*

$$C(\mathcal{P}_1) = C(\mathcal{P}_2) \qquad \textit{but} \qquad q(\mathcal{P}_1) \neq q(\mathcal{P}_2),$$

then the compressed representation is not sufficient for q.

Proof. Suppose a decoder d_q existed. Applying it to the common compressed representation would have to produce both $q(\mathcal{P}_1)$ and $q(\mathcal{P}_2)$. Since these answers differ, no such decoder exists. □

Definition 7.4 (Recoverability preorder)**.** For a fixed task family $\mathcal{Q}$, write

$$R_1 \preceq_{\mathcal{Q}} R_2$$

when every task in $\mathcal{Q}$ that is recoverable from R_1 is also recoverable from R_2.

Proposition 7.5 (Recoverability is a preorder)**.** *The relation $\preceq_{\mathcal{Q}}$ is reflexive and transitive.*

Proof. Every representation supports every task it supports, so the relation is reflexive. If every task supported by R_1 is supported by R_2, and every task supported by R_2 is supported by R_3, then every task supported by R_1 is supported by R_3. Thus the relation is transitive. □

Define $R_1 \sim_{\mathcal{Q}} R_2$ when both $R_1 \preceq_{\mathcal{Q}} R_2$ and $R_2 \preceq_{\mathcal{Q}} R_1$.

Corollary 7.6 (Quotient poset)**.** *The recoverability preorder descends to a partial order on the equivalence classes under $\sim_{\mathcal{Q}}$.*

Proof. Reflexivity and transitivity pass to equivalence classes. Mutual recoverability has been quotiented out, so antisymmetry holds by construction. □

7.1 Pure retention operators

The simplest compression removes nodes without rewriting the retained material. This baseline is useful because it separates information deletion from more complicated summarization or reinterpretation.

Definition 7.7 (Retention projection)**.** Fix a finite node universe V. For a mask $K \subseteq V$, define

$$P_K(S) = S \cap K, \qquad S \subseteq V.$$

Theorem 7.8 (Retention projection algebra)**.** *For masks $K, L \subseteq V$,*

$$P_K \circ P_L = P_{K \cap L} = P_L \circ P_K.$$

Consequently every retention projection is idempotent, and the masks form a Boolean lattice with meet $K \cap L$ and join $K \cup L$.

Proof. For every $S \subseteq V$,

$$P_K(P_L(S)) = (S \cap L) \cap K = S \cap (K \cap L).$$

Intersection is associative and commutative, which gives the composition identity and commutativity. Taking $L = K$ gives idempotence. The power set of V with intersection, union, complement, bottom $\varnothing$, and top V is a Boolean lattice. □

Corollary 7.9 (Task monotonicity under retention)**.** *If $K \subseteq L$, every task whose required nodes and metadata are retained by P_K is also supported by P_L.*

Proof. Every item retained by K is retained by L. Hence every task requirement contained in K is contained in L. □

Pure retention operators commute. Noncommutativity in real proof workflows therefore requires an additional operation such as state-dependent summarization, renaming, quotienting, or reinterpretation. This distinction prevents simple deletion from being credited with structure it does not possess.

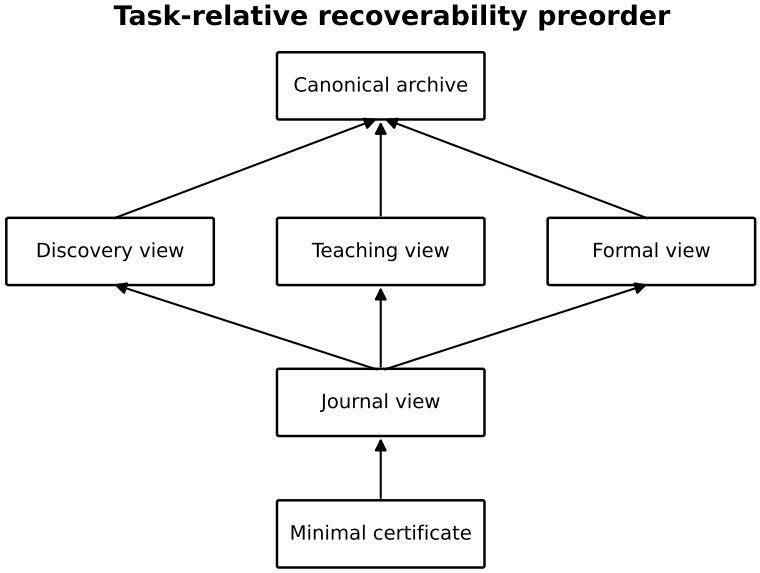


Figure 5: A task-relative recoverability preorder. Teaching, discovery, and formal views may be incomparable.

Theorem 7.10 (Task-family dependence of minimal compression)**.** *There need not be one proper compression that is simultaneously minimal and sufficient for every task family on a proof field.*

Proof. Consider a proof archive containing a certificate and two independent side records x and y. Let q_x ask for x and let q_y ask for y. For the singleton family $\{q_x\}$, a minimal sufficient representation retains the certificate and x but may omit y. For $\{q_y\}$, a minimal sufficient representation may omit x instead. For $\{q_x, q_y\}$, every sufficient representation must distinguish both records. Hence the minimal sufficient representations differ with the task family. The only representation sufficient for every query that can inspect arbitrary archive content is the uncompressed archive itself, which is not a proper compression. □

8 Endpoint, audit, and discovery equivalence

The phrase "the same proof" can refer to several distinct equivalence relations.

Definition 8.1 (Endpoint equivalence)**.** Two representations are endpoint equivalent when they certify the same declared premises and targets.

Definition 8.2 (Audit equivalence)**.** Two representations are audit equivalent when they preserve equivalent dependency, evidence, provenance, and scope information for checking the declared targets.

Definition 8.3 (Discovery equivalence)**.** Two representations are discovery equivalent relative to a declared bridge model when they preserve the same examples, alternatives, boundary cases, candidate routes, and future reach, up to the chosen graph equivalence.

Endpoint equivalence is the weakest of the three. Audit equivalence may fail even when the same theorem is stated, because one document relies on hidden reconstruction or missing provenance. Discovery equivalence is stronger still. Two perfectly valid certificates can expose different mathematical neighborhoods.

Remark 8.4. These equivalences are task-relative. Discovery equivalence requires a declared bridge model. It is not an assertion that every future use of a proof can be predicted.

9 Vector-valued loss

Let a node v carry nonnegative coordinates

$$\mathbf{v}(v) = \big(V(v), D(v), M(v), F(v), A(v)\big),$$

representing verification, discovery, meaning, future bridge, and auditability values. These coordinates are declared model inputs. They may be exact, estimated, or ordinal after a scale has been chosen.

Definition 9.1 (Additive loss vector)**.** For a retained node set $K \subseteq V$, define

$$\mathrm{Loss}(\mathcal{P}, K) = \sum_{v \notin K} \mathbf{v}(v).$$

When normalization is useful, divide each coordinate by the corresponding total coordinate of the full field.

Proposition 9.2 (Zero verification loss can coexist with positive discovery loss)**.** *There exist proof fields and certificates K for which the verification coordinate of* $\mathrm{Loss}(\mathcal{P}, K)$ *is zero while the discovery and bridge coordinates are positive.*

Proof. Take an AND-DAG certificate from A to B and add an optional example node E that is not an ancestor of B. Assign positive discovery and bridge values to E and zero verification value. The unique core omits E while preserving the entire verification ancestry. The verification loss is zero and the other declared losses are positive. □

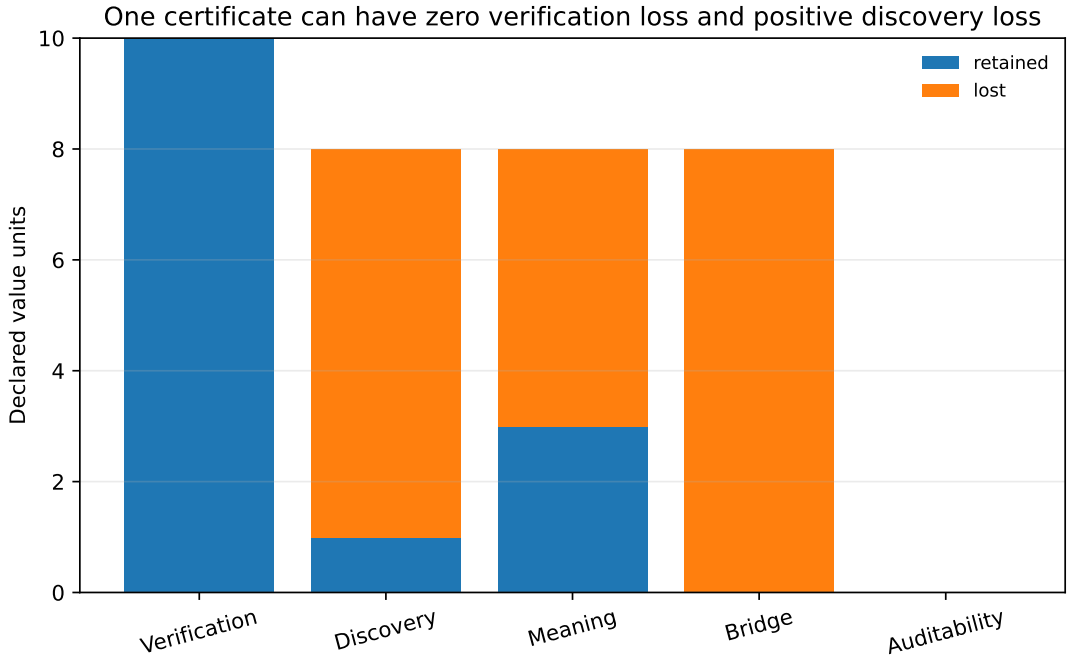


Figure 6: One finite example in which certificate extraction preserves verification and discards discovery value.

No claim is made that all value coordinates are additive. Additivity is a transparent baseline. Interactions can be represented by hyperedge values, submodular utilities, or task-specific decoders when the application requires them.

10 Compression debt

Compression may postpone work rather than eliminate it. A later task may require reconstruction of material that was omitted from the current representation.

Definition 10.1 (Future task model)**.** Let $\mathcal{T}$ be a finite task set with probabilities $p_t \geq 0$ satisfying $\sum_t p_t = 1$. Let $R_t \subseteq V$ be the node set required by task t, and let $c(v) \geq 0$ be the recovery cost of node v when it is absent.

Definition 10.2 (Expected compression debt)**.** The expected compression debt of retained set K is

$$\mathrm{Debt}(K) = \sum_{t \in \mathcal{T}} p_t \sum_{v \in R_t \setminus K} c(v).$$

Proposition 10.3 (Debt is monotone under retention)**.** *If $K_1 \subseteq K_2$, then*

$$\mathrm{Debt}(K_2) \leq \mathrm{Debt}(K_1).$$

Proof. For every task t, the missing set $R_t \setminus K_2$ is contained in $R_t \setminus K_1$. Recovery costs are nonnegative. The taskwise cost cannot increase, and neither can its probability-weighted sum. □

Corollary 10.4 (Irreversible compression accumulates debt)**.** *For a descending sequence of retained sets*

$$K_0 \supseteq K_1 \supseteq \cdots \supseteq K_r,$$

the expected debt is nondecreasing.

A recovery archive can change the cost function. If an omitted node has a stable provenance pointer and a reproducible generator, its recovery cost may be small. If it survives only in memory, its recovery cost may be effectively infinite.

11 A proof ecology rather than one ideal form

The same theorem can support several legitimate representations:

$$\mathcal{R}(B) = \{R_{\mathrm{cert}}, R_{\mathrm{journal}}, R_{\mathrm{teach}}, R_{\mathrm{formal}}, R_{\mathrm{discovery}}, R_{\mathrm{archive}}\}.$$

The certificate is optimized for checking. The journal form balances rigor and readable length. The teaching form expands conceptual bottlenecks. The formal form exposes primitive dependencies. The discovery form retains alternatives and failed routes. The archive preserves reconstruction information.

Definition 11.1 (Representation portfolio)**.** A representation portfolio is a family of proof representations together with stable claim identifiers, translation maps, and a shared status ledger.

Proposition 11.2 (Portfolio robustness)**.** *If several representations lose different information, then their union retains at least as much recoverable information as any one representation alone.*

Proof. Every item recoverable from one representation is recoverable from the union containing it. Additional representations may restore information absent from the first. Recoverable information is therefore monotone under enlargement of the portfolio. □

Redundancy is useful when it is heterogeneous. Storing six copies of the same compressed certificate adds little. Storing a certificate, an expanded explanation, a formal dependency graph, and a reproducible notebook may preserve genuinely different values.

12 Proof compilers and shared ledgers

A proof compiler generates a goal-specific representation from one canonical proof field.

Definition 12.1 (Proof compiler)**.** Let g be a presentation goal and let $\mathcal{P}$ be a versioned proof field. A proof compiler is a map

$$\Gamma_g : \mathcal{P} \longrightarrow R_g$$

that selects, orders, and renders the material required by g while preserving links to canonical claim and justification identifiers.

Typical goals include certificate, journal, teaching, discovery, formal verification, and archive. The compiler does not invent logical status. It reads status from the canonical ledger.

Proposition 12.2 (Shared-ledger consistency)**.** *Suppose every compiled representation is generated from one canonical ledger and every rendered claim retains its ledger identifier. Then a status correction can be propagated consistently to every compiled representation containing that claim by updating the ledger and recompiling.*

Proof. The stable identifier determines every occurrence of the claim across the compiled outputs. Updating the canonical record changes the source value associated with that identifier. Recompilation reads the updated record in every view. A stale status can persist only in an output detached from the canonical ledger or not recompiled. □

This proposition is simple, but it is operationally important. A monograph, a formal library, and a teaching site should not maintain independent truth labels for the same claim.

13 Latent bridge graphs

A proof field records certified dependencies. A latent bridge graph records possible future transfers. The two graphs have different logical roles.

Definition 13.1 (Latent bridge graph)**.** A latent bridge graph is a finite directed graph

$$\mathcal{B} = (W, E_B)$$

whose nodes include retained proof objects, omitted proof objects, and anticipated future targets. An edge $x \to y$ means that x has been declared a possible transfer route toward y. The edge may be certified, conjectural, or exploratory, but its status must be recorded.

Let $S \subseteq W$ be the retained starting set. Let $F \subseteq W$ be a future target set with weights $w_f \geq 0$.

Definition 13.2 (Future reach utility)**.** The weighted future reach of S is

$$U(S) = \sum_{f \in F} w_f \, \mathbf{1}\{f \in \mathrm{Reach}_{\mathcal{B}}(S)\}.$$

This quantity measures declared reach, not truth. A conjectural edge can increase modeled reach without certifying the target. The ledger must keep those statuses separate.

13.1 Bridge criticality and dominators

Definition 13.3 (Bridge-critical node)**.** A node v is bridge critical for target f relative to starts S when f is reachable from S in $\mathcal{B}$ and becomes unreachable after v is removed.

In graph language, v dominates f relative to S when every directed path from S to f passes through v.

Theorem 13.4 (Criticality equals domination)**.** *Assume f is reachable from S. A node $v \notin S$ is bridge critical for f if and only if v dominates f relative to S.*

Proof. If v dominates f, every path from S to f uses v. Removing v removes every such path, so f becomes unreachable. Conversely, if removing v makes f unreachable, then there could not have been a path from S to f avoiding v. Therefore every path used v, which is exactly domination. □

Corollary 13.5 (Weighted criticality)**.** *The utility lost by removing v is the total weight of future targets dominated by v:*

$$U(S) - U_{\mathcal{B}\setminus\{v\}}(S) = \sum_{f \in F:\, v \text{ dominates } f} w_f.$$

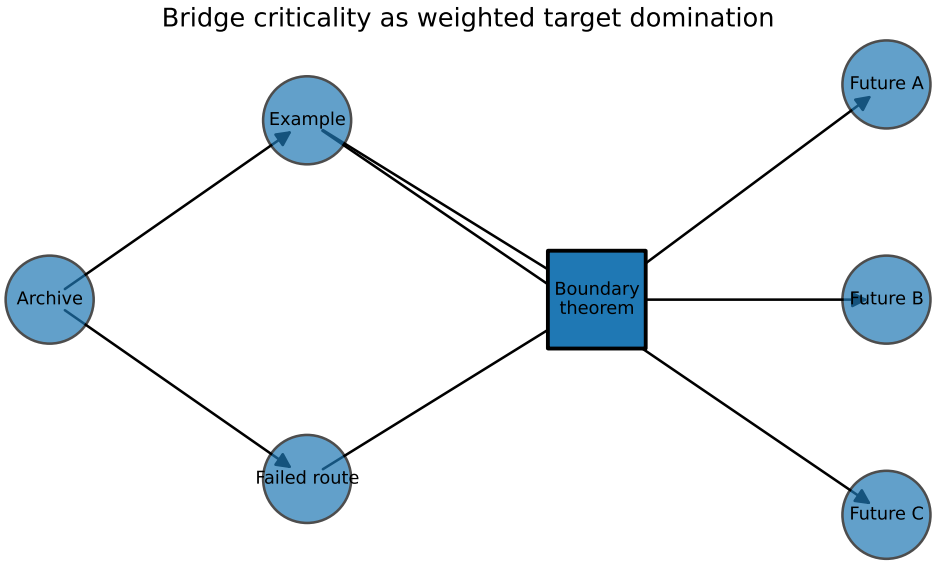


Figure 7: A bridge-critical node dominates two future targets.

A node may have low verification value for the current theorem and high domination value for future targets. This is the precise form of the claim that a forgotten step can remain mathematically important after it has lost its role in the original certificate.

14 Retention under a budget

An archive cannot retain every object at full resolution. The first mathematically controlled retention problem is to choose a finite supplement that preserves as much future reach as possible.

Let $C \subseteq W$ be the candidate nodes that may be retained and let M be a mandatory base set. For $S \subseteq C$, define

$$U_M(S) = U(M \cup S).$$

Theorem 14.1 (Bridge coverage is monotone and submodular)**.** *The future reach utility U_M is a nonnegative monotone submodular set function on C.*

Proof. For each candidate $x \in C$, let $R_x \subseteq F$ be the set of future targets reachable from $M \cup \{x\}$. Let R_M be the targets already reachable from M. Reachability from a union of starting nodes is the union of their individual reach sets, so

$$U_M(S) = \sum_{f \in R_M \cup \bigcup_{x \in S} R_x} w_f.$$

Adding a candidate cannot remove a reachable target, so U_M is monotone.

Now let $S \subseteq T \subseteq C$ and let $x \in C \setminus T$. The marginal gain of adding x is the total weight of targets in R_x that are not already covered. Since T covers at least every target covered by S, the set newly covered from T is contained in the set newly covered from S. Therefore

$$U_M(S \cup \{x\}) - U_M(S) \geq U_M(T \cup \{x\}) - U_M(T),$$

which is submodularity. □

Theorem 14.2 (Greedy cardinality guarantee)**.** *Let S_k be the set obtained by starting from the empty supplement and repeatedly adding the candidate with largest marginal future reach until k candidates have been selected. Let O_k be an optimal supplement of size at most k. Then*

$$U_M(S_k) - U_M(\varnothing) \geq \left(1 - \left(1 - \frac{1}{k}\right)^k\right) (U_M(O_k) - U_M(\varnothing)),$$

and therefore the greedy gain is at least $(1 - 1/e)$ times the optimal gain.

Proof. Write $G_i = U_M(S_i) - U_M(\varnothing)$ and $G^* = U_M(O_k) - U_M(\varnothing)$. By monotonicity and submodularity, the total marginal contribution of the at most k elements of O_k beyond S_i is at least $G^* - G_i$. Hence at least one candidate has marginal gain at least $(G^* - G_i)/k$. Greedy chooses a candidate with at least that gain, so

$$G^* - G_{i+1} \leq \left(1 - \frac{1}{k}\right) (G^* - G_i).$$

Iterating from $G_0 = 0$ gives

$$G^* - G_k \leq \left(1 - \frac{1}{k}\right)^k G^*.$$

Rearranging gives the first bound. The inequality $(1 - 1/k)^k \leq e^{-1}$ gives the second. □

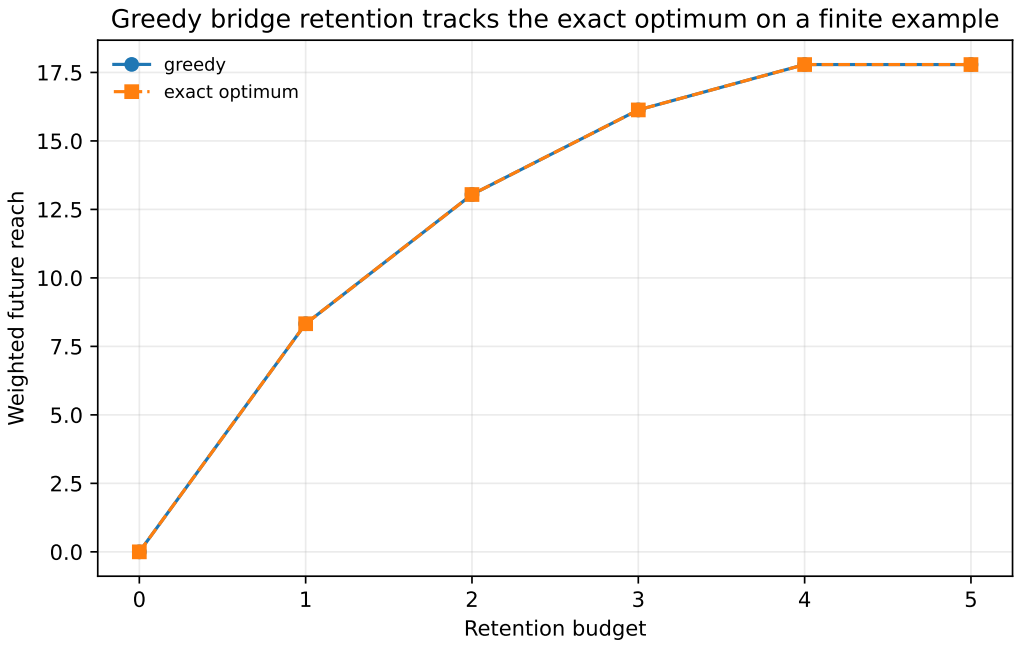


Figure 8: Greedy and exact bridge retention on one reproducible finite example. The theorem gives a worst-case guarantee. The displayed instance performs better.

Warning 14.3 (Arbitrary costs). The simple marginal-gain-per-cost rule included in the software is an interpretable heuristic for arbitrary real costs. The cardinality guarantee above does not automatically apply to that heuristic. Stronger knapsack guarantees require additional algorithmic machinery.

15 Versioned proof objects

A proof archive changes. Claims are strengthened, corrected, refuted, or split. A reproducible theory therefore needs stable identifiers and explicit version relations.

Definition 15.1 (Versioned proof object)**.** A versioned proof object is a sequence

$$\mathcal{P}_0 \longrightarrow \mathcal{P}_1 \longrightarrow \cdots$$

whose nodes and justifications carry stable identifiers. Each transition records additions, removals, altered scopes, evidence changes, and a digest of its parent version.

Definition 15.2 (Append-only certified update)**.** An update $\mathcal{P}_i \to \mathcal{P}_{i+1}$ is append-only certified when every node and justification of $\mathcal{P}_i$ remains unchanged, and $\mathcal{P}_{i+1}$ only adds nodes, evidence, or certified justifications.

Theorem 15.3 (Derivability is monotone under append-only updates)**.** *Every target derivable in $\mathcal{P}_i$ remains derivable in $\mathcal{P}_{i+1}$ under an append-only certified update.*

Proof. A certificate in $\mathcal{P}_i$ uses only nodes and certified justifications that remain unchanged in $\mathcal{P}_{i+1}$. The same selected subfield therefore satisfies the same derivability closure in the later version. □

Retractions and semantic changes are not append-only. They require invalidation analysis.

Definition 15.4 (Syntactic affected set)**.** For a changed node set $X \subseteq V$, define

$$\mathrm{Aff}(X) = X \cup \mathrm{Desc}(X),$$

where descendants are computed in the declared dependency graph.

Theorem 15.5 (Minimal closed syntactic review set)**.** *The set* $\mathrm{Aff}(X)$ *is the unique smallest node set* Y *such that* $X \subseteq Y$ *and* Y *is closed under outgoing dependency edges: whenever* $u \in Y$ *and* $u \to v$ *is a declared dependency edge, then* $v \in Y$*.*

Proof. By construction, every node reachable from X lies in $\mathrm{Aff}(X)$, so the set contains X and is closed under outgoing edges. Let Y be any other set with those two properties. Every directed path beginning in X remains inside Y by induction on path length. Hence every descendant of X lies in Y, and $\mathrm{Aff}(X) \subseteq Y$. □

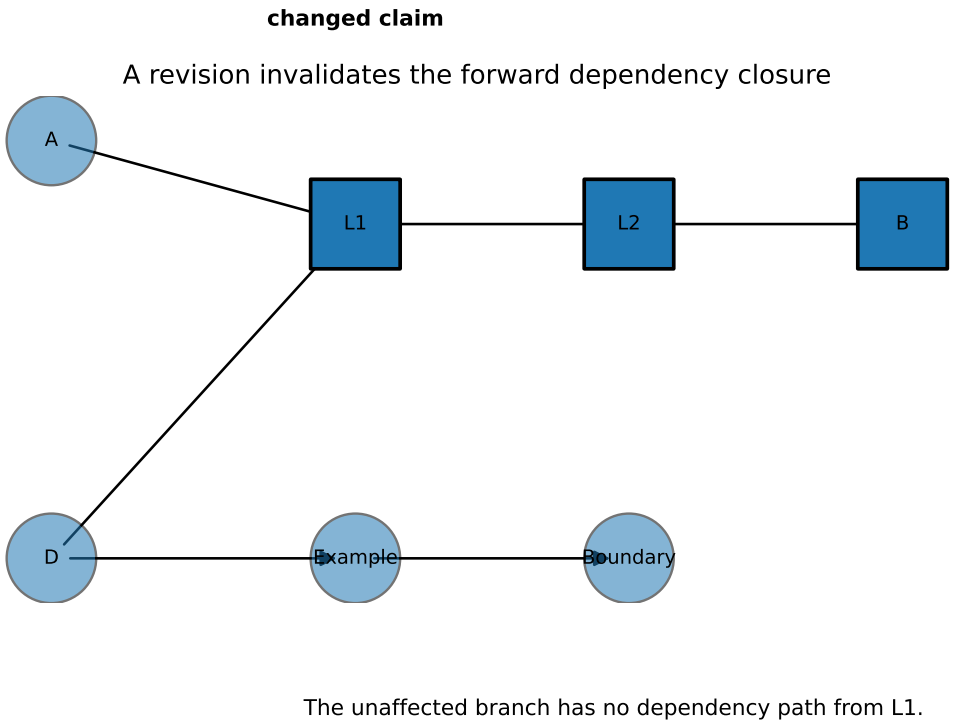


Figure 9: A changed claim and its forward dependency closure.

The theorem is intentionally syntactic. A descendant can survive after rechecking. The affected set says where review is required, not where falsehood is guaranteed.

16 Evidence ledgers and provenance

A claim ledger should answer five questions without reconstructing the entire research history.

1. What is the exact claim identifier?
2. What is its current status and scope?

3. Which justifications depend on it?
4. Which evidence records support or challenge it?
5. Which versions and compiled documents contain it?

Definition 16.1 (Evidence ledger)**.** An evidence ledger is an appendable collection of typed, scoped records linked to claim and justification identifiers. A record may certify, support, challenge, interpret, or refute an object. Its role is explicit rather than inferred from prose placement.

Definition 16.2 (Provenance digest)**.** A provenance digest is a cryptographic hash of a canonical serialization of the claim identifier, content, scope, evidence references, parent version, and relevant build inputs.

A digest does not prove a theorem. It detects unrecorded changes and gives independently generated representations a common reference point.

Proposition 16.3 (Digest stability)**.** *A deterministic canonical serialization gives the same provenance digest for identical records and, except for hash collisions, a different digest after any serialized field changes.*

Proof. Deterministic serialization produces the same byte sequence for identical records, so the hash is identical. Any change that alters the canonical byte sequence changes the hash input. A collision-resistant hash is designed to make equal hashes for distinct byte sequences computationally negligible, though not logically impossible. □

17 Correlated review

Several reviews are not equivalent to several independent reviews. Reviewers may share training, notation, examples, software, or an unnoticed premise.

Let $X_i = \theta + \varepsilon_i$ be reviewer estimates of a quantity θ, with common variance σ^2 and common pairwise correlation ρ. For $n > 1$, assume $-1/(n-1) < \rho \leq 1$, so the mean has positive variance. The mean has variance

$$\operatorname{Var}(\bar{X}) = \sigma^2 \left(\frac{1-\rho}{n} + \rho \right).$$

Theorem 17.1 (Effective reviewer count)**.** *Under the equicorrelated model, the reviewer mean has the same variance as the mean of*

$$\boxed{n_{\text{eff}} = \frac{n}{1 + (n-1)\rho}}$$

independent reviewers of variance σ^2.

Proof. The mean of m independent reviewers of variance σ^2 has variance σ^2/m. Equating this with the correlated mean variance gives

$$\frac{\sigma^2}{m} = \sigma^2 \left(\frac{1-\rho}{n} + \rho \right) = \sigma^2 \frac{1 + (n-1)\rho}{n}.$$

Solving for m gives the formula. □

For fixed positive ρ, n_{eff} approaches $1/\rho$ as n grows. Headcount continues to help, but shared dependence creates a saturation floor.

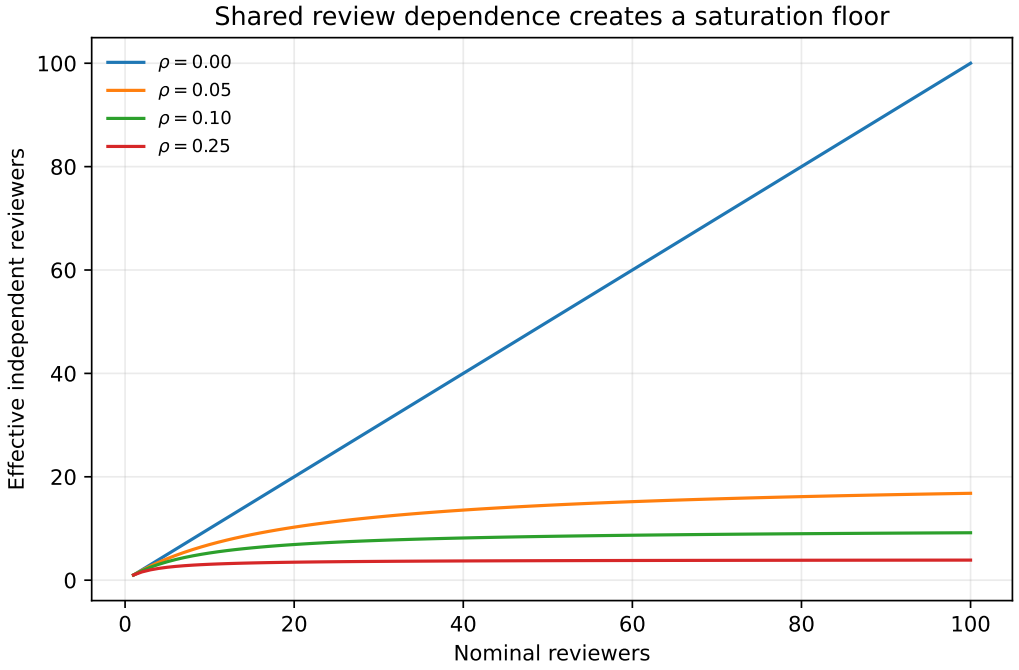


Figure 10: Effective independent reviewer count under equicorrelated errors.

For a general positive definite covariance matrix Σ, consider unbiased linear aggregates $w^\top X$ with $\mathbf{1}^\top w = 1$.

Theorem 17.2 (Minimum-variance covariance-aware weights)**.** *The unique minimum-variance unbiased linear aggregate has weights*

$$w^* = \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}^\top \Sigma^{-1}\mathbf{1}}.$$

Its variance is

$$\frac{1}{\mathbf{1}^\top \Sigma^{-1}\mathbf{1}}.$$

Proof. Minimize $w^\top \Sigma w$ subject to $\mathbf{1}^\top w = 1$. The Lagrangian is

$$L(w, \lambda) = w^\top \Sigma w - 2\lambda(\mathbf{1}^\top w - 1).$$

Stationarity gives $\Sigma w = \lambda \mathbf{1}$, hence $w = \lambda \Sigma^{-1}\mathbf{1}$. The constraint gives

$$\lambda = (\mathbf{1}^\top \Sigma^{-1}\mathbf{1})^{-1}.$$

Positive definiteness makes the objective strictly convex, so the solution is unique. Substitution gives the variance formula. □

Warning 17.3 (Bias is not covariance)*.* Covariance-aware weighting does not remove a common bias that is absent from the covariance model. If every reviewer shares bias b, the mean squared error contains the irreducible term b^2.

The practical conclusion is not that consensus is useless. It is that review portfolios should diversify dependence structure, not only increase nominal count.

18 Verification bandwidth and audit queues

A system can generate mathematical candidates faster than it can verify them. Let a_t be arrivals during audit cycle t, let s_t be service capacity, and let B_t be the unresolved backlog. The basic deterministic queue is

$$B_{t+1} = \max\{0, B_t + a_t - s_t\}.$$

Theorem 18.1 (Constant-rate audit regimes)**.** *Suppose $a_t \equiv \lambda$ and $s_t \equiv \mu$.*

(i) *If $\lambda < \mu$, every finite initial backlog reaches zero after finitely many cycles.*

(ii) *If $\lambda = \mu$, the backlog remains constant.*

(iii) *If $\lambda > \mu$, the backlog grows linearly with slope $\lambda - \mu$.*

Proof. Before the truncation at zero, each cycle adds $\lambda - \mu$. When this quantity is negative, the backlog decreases by a fixed amount until it reaches zero. When it is zero, the backlog is unchanged. When it is positive, the truncation is never active after a nonnegative start, and the backlog is $B_0 + t(\lambda - \mu)$. □

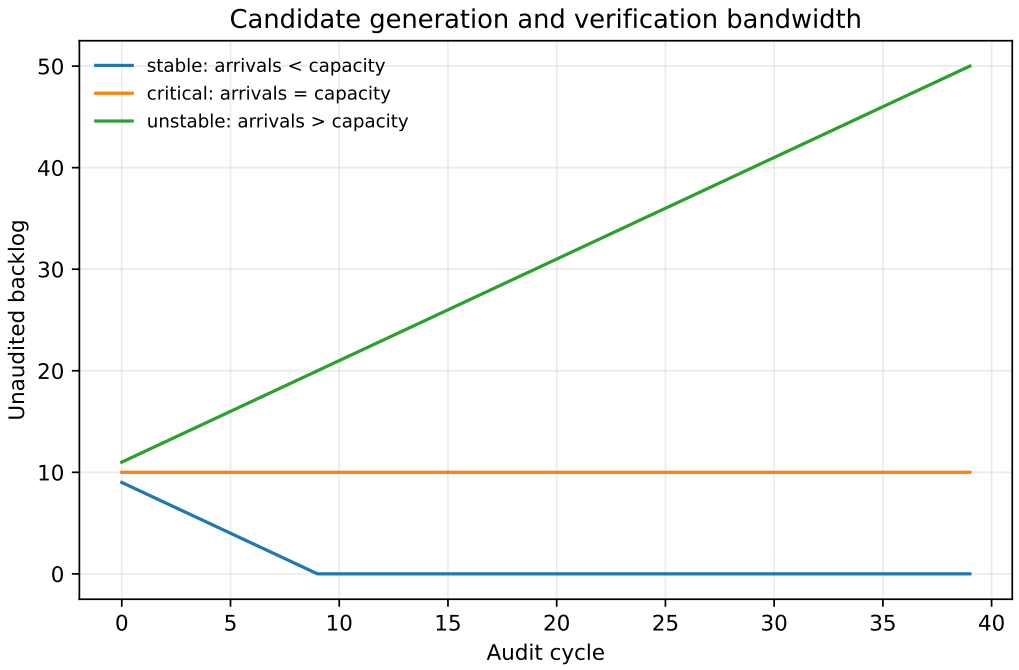


Figure 11: Stable, critical, and unstable audit regimes in the constant-rate model.

A simple audit priority score is

$$P(x) = \frac{p_{\mathrm{err}}(x)\, c_{\mathrm{consequence}}(x)\big(1 + r_{\mathrm{dependency}}(x)\big)}{c_{\mathrm{audit}}(x)}.$$

This is a decision model, not a truth measure. It makes the assumptions behind triage explicit.

19 Recursive proof cubes and expansion depth

A visible proof step is rarely atomic. A lemma opens into definitions and sublemmas. A computation opens into algorithms, parameters, and exact checks. A citation opens into another dependency graph. Proof Path Theory treats this as recursive refinement.

Definition 19.1 (Proof packet and refinement)**.** A proof packet is a node presented at one resolution. A refinement replaces that node by a finite subfield whose certified target reconstructs the original node on the declared scope.

Repeated refinement gives a nested sequence of representations

$$R_0 \prec R_1 \prec \cdots \prec R_d,$$

where deeper levels expose more internal structure. Let $r_d \geq 0$ be the unresolved residual at depth d and let $c_d \geq 0$ be the cumulative cost.

Definition 19.2 (Goal-relative depth objective)**.** For goal g with weights $\alpha_g, \beta_g \geq 0$, define

$$J_g(d) = \alpha_g r_d + \beta_g c_d.$$

An optimal expansion depth is any minimizer of J_g.

Theorem 19.3 (Expansion depth is goal dependent)**.** *The same proof object can have different optimal expansion depths for different goals.*

Proof. It is enough to construct a proof object with exactly two available depths. Choose $d_1 < d_2$ with $r_{d_2} < r_{d_1}$ and $c_{d_2} > c_{d_1}$. Goal g prefers d_2 over d_1 exactly when

$$\frac{\alpha_g}{\beta_g} > \frac{c_{d_2} - c_{d_1}}{r_{d_1} - r_{d_2}},$$

with the usual interpretation when $\beta_g = 0$. Choose one goal ratio below this threshold and another above it. The first prefers the shallower depth and the second prefers the deeper depth. □

A practical stopping rule selects the first depth satisfying either

$$r_d \leq \varepsilon$$

or

$$\frac{r_{d-1} - r_d}{c_d - c_{d-1}} < \tau.$$

The first condition stops when the residual is small enough. The second stops when the next refinement gives too little residual reduction per unit cost.

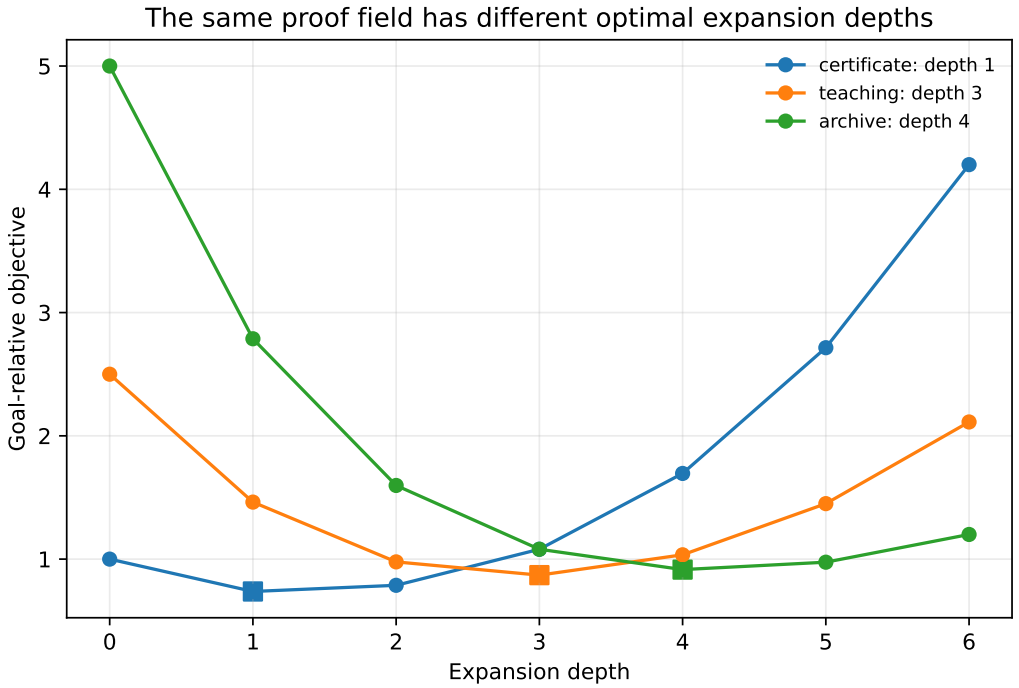


Figure 12: Certificate, teaching, and archive goals select different depths in one finite model.

Deeper is not automatically better. A formal checker may need primitive steps. A specialist paper may cite a standard theorem. A teaching account may expand only the conceptual bottleneck. An archive may retain a failed branch that neither certificate requires.

20 Worked finite examples

The accompanying repository contains three canonical JSON proof fields and several auxiliary models. The examples are small enough to inspect by hand and complete enough to exercise the formal distinctions.

20.1 A fixed AND-DAG

The field `examples/and_dag.json` has required premise A, a definition D, lemmas L_1 and L_2, and target B. It also contains a worked example, a failed route, and a boundary case. The unique verification core is

$$\{A, D, L_1, L_2, B\}.$$

The optional nodes have zero declared verification value for B and positive discovery, meaning, or bridge value. This is the executable instance behind Figure 6.

20.2 Two certified proofs

The field `examples/alternative_proofs.json` contains a short route and a richer explanatory route. The short route has storage cost 1, while the richer route has storage cost 3 and larger meaning and

discovery values. Exact certificate search chooses

$$A \longrightarrow L_{\text{short}} \longrightarrow B$$

when the objective is storage cost alone. A teaching compiler may rationally choose the other route because it uses a different objective.

20.3 Set Cover encoded as certificate selection

The field `examples/set_cover_certificate.json` contains four optional source lemmas and four obligations. Exact search returns the certificate using S_1 and S_3, with total cost 2. The particular optimum is not important. The example demonstrates how local alternative derivations create a global combinatorial selection problem.

20.4 Bridge retention

The bridge experiment uses ten candidate nodes and sixteen weighted future targets. For budgets zero through five, the script computes both the exact optimum and the greedy solution. The generated data are stored in `data/retention_comparison.csv`. On this instance the two curves coincide. The theorem guarantees only the worst-case $(1 - 1/e)$ factor.

20.5 Correlated review and audit capacity

For one hundred equicorrelated reviewers with $\rho = 0.1$,

$$n_{\text{eff}} = \frac{100}{1 + 99(0.1)} \approx 9.17.$$

The audit example begins with backlog 10. Arrivals 4 and capacity 5 clear the queue. Equal rates hold the backlog at 10. Arrivals 6 and capacity 5 increase it by one per cycle.

Table 1: Canonical finite checks in the version 1.0 release.

Object	Computed result	Mathematical role
AND-DAG	unique five-node core	Theorem 5.3
Alternative field	minimum cost 1	exact finite search
Set Cover field	minimum cost 2	Theorem 6.2 encoding
Bridge model	greedy and exact curves	Theorems 16.1–16.2
Review model	n_{eff} formula	Theorem 20.1
Audit model	three queue regimes	Theorem 21.1

21 Reference implementation

The repository implements the finite theory as the Python package `proof_path_theory`. The public modules are organized by mathematical responsibility.

Module	Responsibility
`model.py`	Typed nodes, justifications, evidence records, acyclicity, ancestry, and finite derivability.
`certificate.py`	Certificate validation, unique AND-DAG cores, exact minimum certificate search, and certificate cost.
`compression.py`	Task sufficiency, recoverability preorder, loss reports, compression debt, representation utilities, and Pareto fronts.
`bridges.py`	Reachability, domination-based criticality, exact retention, greedy retention, and finite submodularity checks.
`review.py`	Equicorrelated covariance, effective reviewer count, generalized least squares, and shared-bias mean squared error.
`audit.py`	Audit queue simulation, rate regimes, and explicit priority scores.
`refinement.py`	Goal-relative refinement decisions, optimal depth, and stopping rules.
`versioning.py`	Affected descendants, append-only checks, ledger snapshots, and provenance digests.
`io.py`	Stable JSON import and export for proof fields and certificates.
`cli.py`	Command-line validation, core extraction, minimum certificate search, and certificate audit.

21.1 Command-line interface

The main commands are:

```
proof-path validate examples/and_dag.json
proof-path core examples/and_dag.json --output core.json
proof-path minimum examples/alternative_proofs.json --output minimum.json
proof-path audit examples/and_dag.json core.json
```

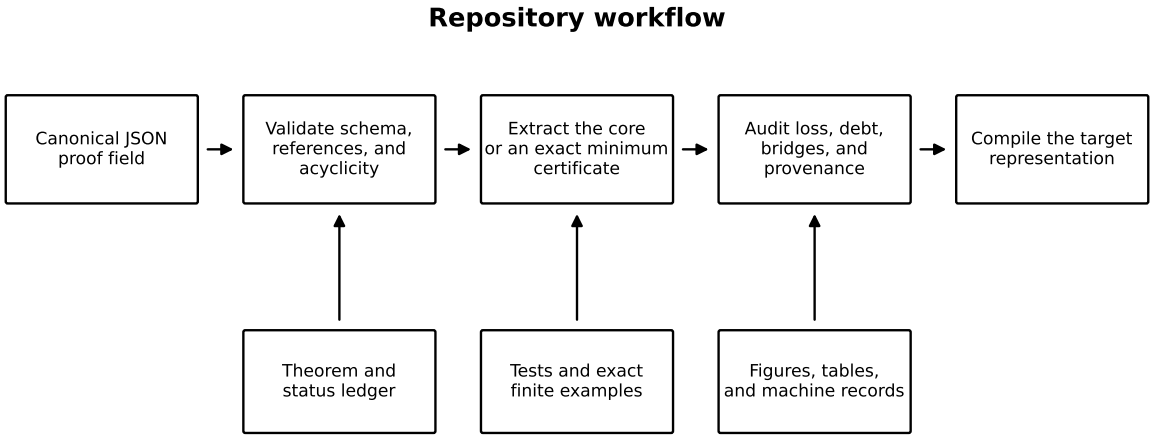


Figure 13: Repository workflow from a canonical JSON field to compiled representations and reproducible records.

21.2 JSON contract

A proof field file contains:

- a schema version and metadata;
- required premises and targets;
- typed nodes with value and cost vectors;
- typed evidence records with scope and certification flags;
- justifications with premise sets, conclusions, rules, evidence references, and certification flags.

The schema is intentionally plain. It can be produced by a proof assistant adapter, a research notebook, a theorem ledger, or a hand-written example.

21.3 Tests and build discipline

Version 1.0 contains sixty-seven automated tests. They cover graph validation, derivability, certificate extraction, alternative proofs, the Set Cover encoding, loss vectors, task sufficiency, bridge submodularity, retention, review covariance, audit queues, refinement, version propagation, JSON round trips, and command-line smoke tests.

A separate deterministic cross-check compares the exact certificate solver with brute-force enumeration over eighteen cost profiles, tests the greedy cardinality guarantee on one hundred generated coverage instances, verifies 1,024 retention-algebra identities, and compares the effective-reviewer formula with direct covariance calculations in 114 parameter settings.

The tests and cross-checks verify the implementation of finite definitions and examples. They do not replace the proofs in this paper. Conversely, the proofs do not guarantee that a software implementation is free of defects. The release keeps both forms of verification.

22 What the theory establishes

The completed finite theory establishes the following structural facts.

1. A proof field can represent conjunction within a justification and alternatives among justifications without confusing candidate edges with certified edges.
2. Derivability in a finite acyclic selected subfield is well defined and computable in topological order.
3. A fixed AND-DAG has a unique minimal verification core.
4. Alternative derivations make minimum certificate selection NP-complete in general.
5. Compression is meaningful only relative to a task family.
6. Task-relative recoverability forms a preorder and a quotient partial order.
7. Verification loss, discovery loss, meaning loss, bridge loss, and audit loss can diverge.
8. Expected reconstruction debt decreases monotonically as more required material is retained.
9. Bridge criticality is domination in the declared future-reach graph.

10. Weighted future reach is monotone and submodular.
11. Greedy cardinality retention has the standard $(1 - 1/e)$ guarantee.
12. Append-only certified updates preserve existing derivability.
13. The forward dependency closure is the smallest universally sound syntactic invalidation set.
14. Review dependence reduces effective independent review, and generalized least squares gives the optimal linear unbiased aggregate under a known covariance model.
15. Constant-rate audit queues have an exact stable, critical, or unstable classification.
16. Optimal proof expansion depth depends on the presentation goal.

These results turn the original methodological insight into a finite mathematical branch with explicit objects, algorithms, complexity, and failure boundaries.

23 Limitations and claim boundaries

Proof Path Theory does not establish a theorem by preserving the history of an argument. A richly documented wrong proof remains wrong. Bridge value is not validity. Explanatory power is not validity. Consensus is not validity. The certified subfield remains the object that carries deductive force.

The dependency graph is a declared model. It records the dependencies that were formalized, not every historical influence on discovery. Tacit knowledge, visual intuition, unrecorded conversations, and changes in vocabulary may be absent.

The value and cost coordinates are task models. Verification can often be specified sharply. Meaning and future bridge value are partly prospective. Their use is justified when assumptions are stated, not when the numbers are treated as intrinsic physical constants.

The finite bridge graph does not predict every future theorem. It supports conditional statements of the form: under these declared future targets and transfer edges, this retention policy preserves this amount of modeled reach.

The exact minimum certificate algorithm is exponential in the worst case. The package makes this limitation explicit through a search-state bound.

The review and queue models are deliberately elementary. They isolate dependence and capacity effects, but they do not model every institutional or psychological feature of mathematical review.

Finally, the paper does not claim that graphs, hypergraphs, proof theory, submodular optimization, covariance-aware aggregation, or queueing recurrences were invented here. The contribution is the formal assembly of these structures around proof compression, discovery fields, mathematical memory, and versioned proof portfolios.

24 Open theorem program

The finite theory leaves several focused directions.

Problem 24.1 (Infinite proof fields)**.** Develop locally finite and countable proof fields with conditions guaranteeing that finite certificates approximate an infinite derivation without losing domain or

convergence information.

Problem 24.2 (Parameterized certificate complexity)**.** Classify minimum certificate selection by treewidth, number of alternative justifications, justification arity, shared-subproof width, and target count. Identify fixed-parameter tractable regimes and approximation bounds.

Problem 24.3 (Probabilistic bridge calibration)**.** Replace declared bridge weights by calibrated posterior models while keeping conjectural reach separate from certified derivability.

Problem 24.4 (Proof assistant interoperability)**.** Construct adapters that export stable claim identifiers, primitive dependencies, scopes, and proof terms from existing formal systems into the common proof-field schema.

Problem 24.5 (Higher proof transformations)**.** Model transformations between proofs, transformations between transformations, and coherent compiler pipelines in a higher categorical setting.

Problem 24.6 (Privacy and attribution)**.** Design proof archives that preserve provenance and priority while allowing private notes, embargoed routes, and sensitive collaboration records to remain access controlled.

Problem 24.7 (Empirical validation)**.** Study real mathematical projects to estimate recovery costs, audit bottlenecks, and the predictive value of bridge-retention scores. Such work should test the models rather than assume them.

Problem 24.8 (Distributed audit systems)**.** Extend the queue model to heterogeneous reviewers, dependent tasks, rework, priority classes, and claims whose audit cost changes after neighboring claims are resolved.

25 Conclusion

A final proof is indispensable, but it is not always the whole mathematical object. The completed theory distinguishes the certificate from the proof field that produced it. It gives that field typed nodes, alternative justifications, evidence records, task-relative compressions, bridge graphs, version histories, and explicit audit models.

The main mathematical boundary is now clear. Verification is a closure property of the certified subfield. Discovery value belongs to a larger graph. Compression can preserve the first while reducing the second. Once that separation is formal, several consequences follow. Minimal certificates can be extracted exactly in fixed AND-DAGs. Alternative derivations create a genuine combinatorial optimization problem. Future-reach retention becomes submodular coverage. Version changes propagate through descendants. Correlated review has an effective sample size. Candidate generation is constrained by audit capacity. Expansion depth depends on purpose.

The final thesis is therefore:

> A proof should be published as a certificate and maintained as a versioned proof field. The certificate preserves deductive validity. The larger field preserves the dependencies, alternatives, examples, failures, provenance, and bridges from which later mathematics may grow.

A Theorem ledger

ID	Result	Status	Executable support
PPT-001	Finite derivability is unique and computable in topological order.	Proved	`ProofField.derive`
PPT-002	An AND-DAG has a unique minimal verification core.	Proved	`unique_and_dag_certificate`
PPT-003	AND-DAG core extraction is linear in relevant ancestry.	Proved	traversal implementation
PPT-004	Minimum Certificate Decision is NP-complete.	Proved	Set Cover example
PPT-005	Certified paths transport functorially.	Proved	structural theorem
PPT-006	Local auditable certificates compose under a certificate functor.	Proved	structural theorem
PPT-007	Task information loss follows from compression collisions.	Proved	task API
PPT-008	Recoverability is a preorder.	Proved	`recoverability_preorder`
PPT-009	The quotient by mutual recoverability is a partial order.	Proved	equivalence API
PPT-010	Minimal sufficient compression depends on the declared task family.	Proved	finite countermodel
PPT-011	Zero verification loss can coexist with positive discovery loss.	Proved	AND-DAG example
PPT-012	Compression debt is monotone under retention.	Proved	`compression_debt`
PPT-013	Bridge criticality equals target domination.	Proved	`dominated_targets`
PPT-014	Weighted future reach is monotone and submodular.	Proved	exhaustive finite checker
PPT-015	Greedy cardinality retention achieves at least $1 - 1/e$.	Proved	greedy and exact solvers
PPT-016	Append-only certified updates preserve derivability.	Proved	versioning check
PPT-017	Forward dependency closure is the minimal closed syntactic review set.	Proved	`affected_descendants`
PPT-018	Equicorrelated review has the stated effective count.	Proved	covariance implementation
PPT-019	GLS weights minimize unbiased linear review variance.	Proved	`generalized_least_squares_weights`
PPT-020	Constant-rate audit queues have exact stable, critical, and unstable regimes.	Proved	queue simulation
PPT-021	Optimal expansion depth depends on the goal.	Proved	refinement implementation
PPT-022	Pure retention projections commute and are idempotent.	Proved	`retention_projection`

ID	Result	Status	Executable support
PPT-023	Retention masks form a Boolean lattice.	Proved	`retention_meet`, `retention_join`

B Reproduction

From the section root:

```
python -m pip install -e . --no-build-isolation
python -m pytest
python scripts/build_examples.py
python scripts/generate_figures.py
proof-path schema proof-field
proof-path validate examples/and_dag.json
proof-path core examples/and_dag.json
proof-path minimum examples/alternative_proofs.json
pdflatex -interaction=nonstopmode -halt-on-error docs/Proof_Path_Theory.tex
pdflatex -interaction=nonstopmode -halt-on-error docs/Proof_Path_Theory.tex
pdflatex -interaction=nonstopmode -halt-on-error docs/Proof_Path_Theory.tex
```

The source archive also contains a wheel build script and a GitHub Actions workflow fragment for a multi-section repository.

C Canonical proof-field schema excerpt

```
{
  "schema_version": "1.0",
  "premises": ["A"],
  "targets": ["B"],
  "nodes": [
    {
      "id": "A",
      "label": "Premise A",
      "kind": "premise",
      "values": {"verification": 1.0},
      "costs": {"storage": 0.2}
    }
  ],
  "evidence": [
    {
      "id": "e-proof",
      "kind": "formal_proof",
      "source": "declared derivation",
      "scope": "",
      "certifies": true
    }
  ],
  "justifications": [
    {
      "id": "j1",
      "premises": ["A"],
      "conclusion": "B",
      "rule": "deduction",
      "certified": true,
      "evidence_ids": ["e-proof"]
    }
  ]
}
```

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SII / SOURCE READING CARD

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1.0.0 · 60 original pages

READ THIS FOR

Repair synthesis, compiler revision, acquired equations, certificate dependencies and resource prices.

KEEP THIS BOUNDARY

Bounded constructive chain with prose proofs; no indefinite recursive acceleration or unrestricted agent is inferred.

CONNECTED SOURCES

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Original text begins on the next page.

The source pages that follow are preserved without editorial stamps. Page labels in the PDF navigation use the source code and physical source-page number. The package retains the byte-identical original PDF and its SHA-256 digest.

Self-Improving Intelligence

Representation, Discovery, and Recursive Revision

Komorebi P.
8 October 2026

First consolidated edition
Version 1.0.0

Abstract

Improving a solver is not the same as improving the procedure that produces its successors. This first edition develops a contract-relative account in which representations, experiments, repair languages, evidence, and costs remain explicit as discovery procedures change. Exact finite synthesis produces diagnostic policies with execution and complete-storage guarantees, including uncertainty about one fixed experiment model. A checked compiler shortcut avoids exhaustive search when a candidate attains proved lower bounds. A counterexample-guided learner then synthesizes an internal pruning rule. A further acquisition procedure constructs a proof-checked equation library that reduces later synthesis work while preserving the exact candidate and counterexample transcript. Repair-relative weight certificates extend the diagnostic method beyond singleton-success repairs. The final integration tracks which guarantees remain applicable when semantics, grammar order, repairs, witnesses, or resource prices change. A complete resource ledger separates preservation of meaning from reductions in particular event counts and from end-to-end improvement. The contribution is an integrated theoretical framework, conditional proofs, and executable finite witnesses. It does not establish general intelligence, autonomous invention of its trusted substrate, or indefinite recursive acceleration.

Scope. Self-contained mathematical arguments and exact finite reference constructions. Prior experimental histories are not used as evidence for the proposed integrated system. First edition freezes this theoretical scope. Independent peer review and proof-assistant verification are not claimed.

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Reading the first edition

The central question is what has changed when a system becomes better at making itself better. More stored answers can improve a solver without improving its learning procedure. A richer repair language can make a correction expressible without making it discoverable. An improved search procedure can find useful corrections without establishing that its own descendants will search better still. The paper keeps these distinctions visible within one account.

A conceptual reading can follow Sections 1–4, the interpretations after the formal results, and Part VIII. A mathematical reading proceeds through diagnosis and uncertain-model synthesis, then certified compiler revision, grammar synthesis, acquired search quotients, repair-relative certificates, sequential evidence, and recursive progress. Part VI binds the guarantees to their changing inputs and consolidates the resource ledger. The running instrument is finite and inspectable. Appendix A gives a complete diagnostic trace, and the source package contains the parent and descendant compilers, frozen procedure, rule-synthesis, and equation-library records, exact checks, and an itemized review record.

Relation to earlier work. *Machine Understanding* specifies what representations must preserve across inquiry and explanation [1]. *Stratified Recursive Improvement Closure* (SRIC) supplies the earlier distinction between task, improvement-process, and epistemic improvement, together with repair transport and evidence-preserving lineages [2]. Neuralese and Operator-Symbolic Language (OSL) distinguish provisional latent computation from an explicit semantic interface [3, 4]. This paper consolidates their theoretical roles. It neither reproduces their experimental histories nor imports a complete-system claim from them. Every formal statement used below is proved here or identified as a cited foundation.

What is new in this manuscript. The proposed integration makes the improvement procedure itself a charged, transferable computational object. Its observations must support an appropriate repair, its decisions must fit one executable policy, and its acceptance must refer to a preserved evaluation contract. The finite constructions sharpen those obligations. Partition refinement, decision trees, dynamic programming, statistical concentration, and simulation arguments have classical foundations. Their appearance here is not a priority claim for the underlying mathematics.

Conventions. The principal constructive models are finite and deterministic. A stochastic law appears only with an explicit probability model. A protected predicate means a declared operational constraint, not a proof of moral adequacy. An exact reference check verifies the implemented finite case. A proof supplies the general conditional argument. No benchmark score, test count, or version number substitutes for the other.

Symbol	Meaning
S, U, L	A target solver, an improvement procedure, and a reusable library.
Ξ, B	An evaluation contract and its resource allowance.
$\mathcal{R}_\Xi(U, L; B)$	Expected improvement produced by the procedure, not an oracle over its unselected candidates.
$B_0 \subseteq X$	Possible current states of a finite repair problem.
e, r, P	An experiment, a terminal repair, and an executable diagnostic tree.
$F_h(B_0)$	Nondominated execution/storage pairs for successful trees of depth at most h.
σ, τ	Correspondences of states and operations in a declared transfer.
$\mathcal{K}, W_{h,k}$	A safe continuation set and states guaranteeing k progress events within h attempts.

First-edition scope and dependency guide

This edition closes an editorial and mathematical scope, not the open problem of general intelligence. It develops one finite chain in which a solver, its diagnostic compiler, an internal search rule, and the representation used to acquire such rules are separate objects. Each generated object is accompanied by the assumptions needed to interpret it.

Layer	Constructed or acquired here	Still supplied
Repair	Executable diagnostic tree	Experiment model and successful-repair meanings.
Compiler	Selected or pruned discovery procedure	Candidate skeletons, lower-bound proofs and witness checker.
Rule acquisition	Arithmetic pruning expression and counterexamples	Grammar, feature registry and teacher.
Acquisition representation	Checked equations and ordered representatives	Arithmetic semantics, primitive proof rules and equation learner.
Continuing validity	Replayed certificates and explicit invalidation	Dependency-complete registry and sound local checking base.

The principal formal dependency runs from finite repair synthesis to transport and guarded compiler revision, then from sound equations to ordered-trace preservation, and finally to composition of the acquired rule with the diagnostic compiler. Repair-weight pruning is a separately scoped extension, not a consequence of the singleton certificate. Statistical lineage evidence and resource inequalities add independent premises to any claim of repeated improvement.

How to interpret the demonstrations. The finite programs generate or acquire the indicated objects. Their control problems and arithmetic languages are analytical constructions. Exact checks test the implemented domains; prose proofs establish the general conditional statements. No historical agent experiment supplies a missing premise. A smaller event count is not a universal runtime result, and a generated rule is not evidence that its teacher or proof language invented itself.

Review route. Read Sections 3, 6, 21, 23, 24, and 25 for the connected core. The supplied statement review identifies the premises and proof route for every formal result. The source map records the actual depth of reference checking. Open questions in Section 34 remain open research within the program rather than unfilled requirements for this edition.

Part I

The object that improves

1 From a better answer to a better investigator

A system encounters a failure, changes its representation, and succeeds on the next attempt. Something useful has happened. The system may have learned a fact, recovered a missing variable, or selected a more suitable operation. Yet the next unfamiliar failure may leave it no better equipped than before. It has a repaired answer without a reusable account of repair.

A stronger change would preserve the way the failure was investigated. The system might discover that its representation merged two situations with different future consequences. It might learn to seek an observation that separates those situations, to reject an inadequate family of repairs, or to reuse a diagnostic method in another notation. The object retained is then not merely a solution. It is part of a procedure for acquiring solutions.

The recursive question begins one step further out. Can the system improve that procedure, use the revised procedure to construct another improvement, and retain a defensible comparison across those changes? The question is not answered by putting an agent's source code in its own context. Editability is an affordance. Productive revision is a property of what the permitted edits actually achieve.

We take general intelligence as a motivating problem rather than a theorem premise. A broad reasoner should acquire useful capabilities across substantially different tasks without receiving their solutions inside an uncharged interpreter. Chollet's account of skill-acquisition efficiency emphasizes scope, priors, and experience rather than demonstrated skill alone [5]. Our narrower contribution concerns one possible mechanism behind such acquisition: reusable procedures for diagnosing failures, choosing informative actions, constructing corrections, and preserving their scope.

1.1 Three distinct achievements

Task improvement changes how well a solver performs a declared task. *Process improvement* changes how effectively an improvement procedure produces useful solvers from comparable starting conditions. *Epistemic improvement* changes how reliably the evaluation process distinguishes genuine gains from misleading evidence. This separation is inherited from SRIC [2]. A system can advance on one axis without advancing on the others.

A larger archive, for example, can improve retrieval while the procedure that extends the archive remains unchanged. A stronger checker can reject more invalid proposals while producing no additional valid proposals. A faster search may exploit a defect in its evaluation. None of these possibilities makes improvement meaningless. They identify the claim that must be tested.

Recursive improvement also differs from sustained acceleration. A sequence of positive gains can converge to a finite limit. A sequence of better algorithms can consume increasing development costs. A procedure can improve on one task family and deteriorate on another. The substantive claim is not that a feedback loop exists, but that the loop produces useful and transferable changes under a stable comparison.

1.2 The route developed here

The paper moves from representational failure to a finite repair problem, from that problem to a compiled procedure, and from the procedure to transferable and statistically accountable improvement. It then treats the

succession of procedures as another finite control problem. This last step makes a common quantifier visible: the system needs one policy that can carry its claims forward, not a different hypothetical policy for every desired property.

The route does not require every useful correction to be formally proved beneficial on all future tasks. Exact semantic certificates and statistical performance evidence have different jobs. Nor does it require each system to invent its own computational substrate. A finite search language may be supplied. The theory must disclose what that language already makes possible and what the learner actually selects or extends.

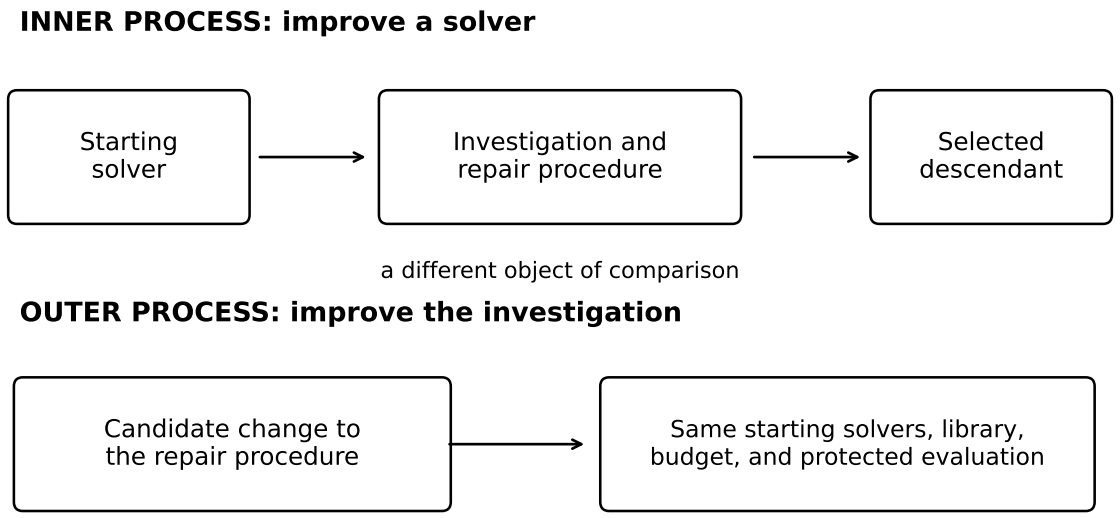


Figure 1: The inner process repairs a solver. The outer process changes how repair is attempted. Arrows indicate proposed information flow, not an implemented neural agent.

2 Related work and the proposed contribution

Learning how to learn has a substantial history. Learned optimizers can acquire update rules from optimization problems [6]. Model-Agnostic Meta-Learning trains parameters from which a small number of updates can adapt to a new task [7]. These approaches show why adaptation mechanisms deserve direct study. Neither the phrase meta-learning nor the presence of an outer objective, by itself, establishes an indefinitely improving outer procedure.

DreamCoder couples learned program libraries to neural search guidance [8]. Its relevance is the interaction between a language for expressing solutions and the ability to search that language. SEAL instead learns to produce adaptation data and update directives whose value is evaluated after the resulting parameter update [9]. These are different places at which an improvement process can become learnable.

The Gödel-machine formulation treats a self-rewrite as admissible after a proof of usefulness within encoded assumptions [10]. The Darwin Gödel Machine uses empirical evaluation and an archive of modified coding agents [11]. They should not be conflated. A proof-based criterion and a benchmark-based criterion support different guarantees and have different costs.

More recent systems sharpen the distinction between a task agent and its modifier. Hyperagents makes both roles editable and studies transfer of improvement strategies [12]. Its reported experiments include improvements to modification procedures and transfer across tested domains. The archived variants and their

selection protocol remain part of the evaluation setup. The general expressiveness of the editable program is not itself evidence that efficient discovery occurs on every computable task.

The September 2026 AIDE[2] preprint studies repeated modification of a research agent's harness and reports transfer beyond its selection tasks [13]. Its separate comparison of a discovered agent as an outer-loop improver is explicitly inconclusive relative to the strong reference. This distinction is directly relevant: improvements in research performance and improvements in driving the next improvement loop are related hypotheses, not interchangeable observations. These public works are research context. Their empirical findings are not treated as evidence for the architecture proposed here.

Causal abstraction provides a formal account of mechanism-preserving interpretation [14]. Adaptive data analysis studies why repeated access to evaluation information can invalidate ordinary statistical reasoning [15]. Both are necessary neighbors. A self-improver must know what its representations mean and what its evidence permits it to conclude.

2.1 What this paper adds to that discussion

The intended contribution is an integrated realization contract. A candidate improvement identifies its target, information access, experiment family, repair language, controller, retained evidence, and charged resources. It also identifies the comparison under which the procedure itself is said to improve. The exact results concern finite cases of that contract. The architectural hypotheses concern whether learned proposal and acquisition mechanisms can satisfy it economically on wider tasks.

This is not a claim that program libraries, recursive search, causal abstraction, or guarded updates originated with Komorebi. The aim is to connect them without allowing the cost or uncertainty of one stage to disappear inside another. A valid final repair is weaker than a valid procedure for finding it. A transferable repair is weaker than a transferable investigation. A safe accepted edge is weaker than a lineage with a continuing path to further improvement.

3 A protected comparison for improvement procedures

Let S denote a solver implementation, U an improvement procedure, and L the library available to that procedure. The distinction is functional rather than physical. They may occupy one executable program. Their different roles are retained so that changes can be attributed rather than hidden in a single checkpoint name.

Definition 3.1 (Improvement contract)**.** A contract Ξ specifies a distribution $\mathcal{H}$ of starting-solver/task pairs, an initial information and library boundary, permitted observations and modifications, a resource allowance B, protected predicates, and an evaluation rule $J_\tau(S) \in [0, 1]$. The improvement procedure returns one selected descendant or the unchanged baseline. Selection must use only the admitted development information. The final evaluation is not available as an oracle for choosing retrospectively among all candidates.

Write $U(S, \tau, L; B)$ for the selected result, including the procedure's randomness where applicable. Its productivity is

$$\mathcal{R}_\Xi(U, L; B) = \mathbb{E}_{(S,\tau)\sim\mathcal{H}}\big[J_\tau(U(S, \tau, L; B)) - J_\tau(S)\big]. \tag{1}$$

Invalid execution and out-of-budget behavior receive the treatment fixed by Ξ, not an exclusion chosen after seeing the outcome. The score may be a primary coordinate of a larger profile. Resource, validity, and group-specific constraints need not be folded into one scalar.

The value in (1) lies in $[-1, 1]$. A procedure that may return the baseline need not achieve a nonnegative true value: noisy selection can still choose a worse descendant. A nonnegative certificate requires additional evidence. The mere existence of a fallback does not make every selector good at using it.

3.1 Fixed-library and full-system comparisons

A comparison of $\mathcal{R}_\Xi(U', L; B)$ with $\mathcal{R}_\Xi(U, L; B)$ isolates process change at a common library. A comparison with (U', L') measures the combined descendant. Both are useful, but they answer different questions. A crossed comparison of (U, L), (U', L), (U, L'), and (U', L') can expose interaction between procedure and library. There is no general additive decomposition that makes their contributions independent.

The same discipline applies to base models, training histories, tools, and stored controllers. A changed foundation model is a changed component. A foreign-agent test must supply a valid adapter or charge discovery of one. Calling two targets unfamiliar does not make them distributionally independent of the development process.

Counterexample 3.2 (More knowledge without a better improver)**.** Consider four equally likely task names. A fixed procedure returns a stored solution when one exists and otherwise leaves the solver unchanged. A library containing one answer yields a smaller expected gain than a library containing all four. The procedure has not changed. Holding the library fixed gives identical productivity for the two copies of the procedure. The full system improved, but this example contains no improvement of the improvement rule.

3.2 What persists across a recursive step

A lineage state can be written schematically as

$$Z_t = (S_t, U_t, L_t, E_t, \Xi_t, \rho_t),$$

where E_t records evidence and rejected alternatives, and ρ_t records remaining resources and scope. A proposed modification may target any admitted subset. A comparison across different Ξ_t needs a declared bridge. It cannot be justified merely because both scores are called accuracy.

The public record should distinguish which changes were selected by the system, which were supplied by an external designer, and which were only proposed. A successful finite search within a fixed grammar is a genuine computational result. It is not evidence that the grammar was invented, that a broader domain was covered, or that the outer search process improved itself.

4 A small world with reusable diagnostic structure

Consider eight cells with exactly one latent defect, indexed by $j \in \{0, \dots, 7\}$. The solver must repair that cell while preserving the others. Its old representation retains only the number of defects, so every episode initially looks identical.

A diagnostic query names a subset A and returns $\mathbf{1}\{j \in A\}$. The reference language contains eight singleton queries and three bit-coordinate queries. Repair r_i succeeds exactly when $i = j$. Because repair is irreversible within the episode, a procedure may commit only when every surviving possibility justifies its choice. Otherwise it may continue investigating or abstain.

The model is fully declared. Setting each known cell as the defect on a calibration copy would permit learning the complete query table in $8(8 + 3) = 88$ observation calls. The code instead generates that table from the analytical specification and records it as supplied information. It does not claim to have executed those calibrations or inferred a hidden environment.

Equality-first testing asks whether $j = 0$, then $j = 1$, and so on. Its worst case needs seven queries, identifying the last cell by elimination. The three bit-coordinate queries need only three. Each procedure then performs one repair. Both complete binary trees have eight leaves and fifteen stored records. Faster execution has not made the controller free.

Part II

Discovering and transporting repair

5 A residual constrains a diagnosis

A residual records disagreement between a prediction and an observation, between a proposed operation and a required invariant, or between two routes of translation. It is evidence against a conjunction of assumptions. It does not uniquely identify which assumption failed.

For example, an unexpected output can arise because a state variable was discarded, a transition was modeled incorrectly, the observation was noisy, or an action had a side effect omitted from the model. These explanations can agree on the current discrepancy and disagree about the next useful experiment. An automatic instruction to enlarge the representation after every error would confuse missing information with faulty dynamics.

Let a hidden case θ determine a set $\mathcal{R}(\theta)$ of successful repairs. The set may have several members. Diagnosis need only obtain enough information to choose a repair in the appropriate set. It need not identify every hidden property of θ.

Proposition 5.1 (Feedback indistinguishability limits repair)**.** *Suppose two cases have disjoint successful-repair sets and induce transcript laws P_0, P_1 under a fixed admitted investigation protocol. A repair selector observes only the transcript and independent private randomness. Under equal prior probabilities, its average probability of successful repair is at most*

$$\frac{1 + d_{\mathrm{TV}}(P_0, P_1)}{2}.$$

In particular, identical transcript laws rule out certain success in both cases.

Proof. From the selected repair construct a binary decision: report case zero if the repair belongs to $\mathcal{R}(0)$, and case one otherwise. Success in case zero implies a correct decision. Success in case one also implies a correct decision because the two repair sets are disjoint. Thus repair success is at most the success of this binary test. For a randomized test with probability $f(o) \in [0, 1]$ of reporting zero, the latter success is

$$\tfrac{1}{2}\big(1 + \mathbb{E}_{P_0} f - \mathbb{E}_{P_1} f\big) \le \tfrac{1}{2}(1 + d_{\mathrm{TV}}(P_0, P_1)).$$

The inequality follows by summing the positive differences of the two finite probability laws. The same argument applies to any measurable transcript space with the usual total-variation definition. □

The bound concerns the entire transcript, not each observation marginal in isolation. Correlations between observations can identify a case even when every separate marginal is uninformative. Randomized action selection is already covered when it is part of the declared protocol and its observable record.

Nor does the bound identify the most informative protocol. The learner must choose experiments before receiving their outcomes. A protocol that distinguishes the cases may exist while being inadmissible, too expensive, or absent from the current language. The finite synthesis below makes those possibilities separate outcomes.

5.1 Repair sufficiency is not full identification

Suppose $\mathcal{R}(\theta_1) = \{a, c\}$ and $\mathcal{R}(\theta_2) = \{b, c\}$. The cases are distinct, yet repair c works in both. A method can stop without learning which case occurred. Conversely, pairwise overlap of repair sets does not guarantee one repair for an entire uncertainty set. The three sets $\{a, b\}$, $\{b, c\}$, and $\{a, c\}$ intersect pairwise and have empty total intersection. The stopping condition must refer to the whole surviving set.

This matters for representation discovery. A useful latent state may preserve exactly those distinctions needed for future repair decisions rather than every property of the underlying process. Whether that state remains sufficient after new repair operations become available is a separate question.

5.2 Repairing a failed representation

Let X be a finite set with deterministic operations $F_u : X \to X$ and output labels $q(x)$. Begin with a partition P_0 that preserves both the old retained labels and q. Define the next partition by splitting a block whenever two states in it have successors in different current blocks under some permitted operation. Equivalently,

$$x \equiv_{P_{t+1}} y \quad \Longleftrightarrow \quad x \equiv_{P_t} y \text{ and } F_u(x) \equiv_{P_t} F_u(y) \text{ for every } u.$$

The construction uses a complete transition description. Having a few observed contradictions does not give a learner that table for free.

Proposition 5.2 (Least stable refinement of a retained representation)**.** *The splitting construction stabilizes after at most $|X| - |P_0|$ strict refinements. Its final partition is the coarsest refinement of P_0 on which every F_u has a well-defined quotient operation. It preserves every finite continuation of the retained labels and outputs.*

Proof. A strict refinement increases the number of nonempty blocks, which is bounded by $|X|$. At a fixed point, equivalent states have equivalent successors, so the quotient operations are well-defined. Induction on word length then preserves all retained labels after every finite word. For minimality, let R be any stable partition refining P_0. If R refines P_t, stability of R implies that R-equivalent states have successors in the same P_t blocks, so they remain together in P_{t+1}. Induction shows that R refines every P_t and hence the terminal partition. □

This is a standard partition-refinement argument expressed as a repair of an old representation. It is not necessarily the smallest representation for the outputs alone: P_0 may retain distinctions that are no longer needed. Discarding those distinctions requires a different contract. The continuation and cost perspectives are developed more fully in *Machine Understanding* [1].

A counterexample pair is useful because it proves that a current block is too coarse under the stated transition model. It does not specify how a learned encoder can acquire the missing coordinate from its actual inputs. The mathematical quotient and its realizable acquisition procedure remain separate objects.

6 One investigation, one repair, and two resource bounds

We now give a finite model in which that acquisition problem can be decided exactly. The result is deliberately restricted, but it charges the complete investigation rather than only a selected final answer.

Definition 6.1 (Finite repair experiment model)**.** A model has a finite nonempty state set X, a finite experiment set $\mathcal{E}$, and a finite repair set $\mathcal{A}$. Experiment e is allowed on a domain $D_e \subseteq X$, changes the state by $f_e : D_e \to X$, and reports $o_e : D_e \to O_e$. Its execution cost $c_e \geq 0$ is fixed and known. Repair a has fixed cost $c_a \geq 0$ and succeeds on a declared set $G_a \subseteq X$. All costs are in one stated unit. A nonempty set $B_0 \subseteq X$ records the possible current states.

A chosen experiment must be uniformly allowed: $B_0 \subseteq D_e$. After observing o, the possible *post-experiment* states are

$$B_{e,o} = \{f_e(x) : x \in B_0,\ o_e(x) = o\}. \tag{2}$$

Only nonempty branches are retained. A terminal repair is justified when $B_0 \subseteq G_a$. If the original state or an earlier obligation matters, it must be retained as an immutable tag in X. Merely changing the current state into a known state is not recovery of its history.

A diagnostic program is a finite rooted tree. An internal record names an experiment and branches on its possible report. A leaf names a repair justified on that branch's surviving states. We count one storage record per node of the complete tree. Shared subroutines, compressed code, and graphs with merged suffixes belong to other storage models. The present count is chosen because it is exact and unambiguous.

Let $W(P, B_0)$ be the largest execution cost of a complete path, including its terminal repair. Let $M(P)$ be the number of stored tree records. These costs are not independently minimized. A single tree has to realize both.

For horizon h, let $\mathcal{F}_h(B_0)$ be the nondominated attainable pairs (w, m). A pair dominates another when it is no larger in either coordinate and strictly smaller in at least one. Let $T(B_0) = \{(c_a, 1) : B_0 \subseteq G_a\}$. Then

$$\begin{aligned} \mathcal{F}_0(B_0) &= \operatorname{Pareto} T(B_0), \\ \mathcal{F}_h(B_0) &= \operatorname{Pareto}\left(T(B_0) \cup \bigcup_{e : B_0 \subseteq D_e} \left\{\left(c_e + \max_o w_o,\ 1 + \sum_o m_o\right) : (w_o, m_o) \in \mathcal{F}_{h-1}(B_{e,o})\ \forall o\right\}\right). \end{aligned} \tag{3}$$

An empty child frontier makes that experiment unavailable for a successful complete tree. An empty union is not a zero-cost solution.

Theorem 6.2 (Exact finite repair synthesis with compiled storage)**.** *For every finite model, nonempty B_0, and integer $h \geq 0$, recurrence* (3) *yields exactly the nondominated cost pairs of successful diagnostic trees of depth at most h. Every retained pair has an executable witness. A work/storage allowance (b, m) is feasible if and only if some retained pair is componentwise at most (b, m).*

Proof. At horizon zero only a justified terminal repair can succeed, giving $T(B_0)$. Assume the claim at $h - 1$. A successful tree at horizon h either stops immediately or has one uniformly allowed first experiment. Its child at each possible report is successful on exactly the post-experiment support in (2). The execution cost is the root cost plus the worst child cost. The storage is the root record plus the sum over every child, including children not traversed in a particular execution. Conversely, choosing a successful child for each report constructs a successful root tree with those costs. Replacing a dominated child by a dominating one never increases the root maximum or sum. Thus Pareto pruning loses no feasible allowance. Induction proves both completeness and witness construction. □

The theorem is an application of finite dynamic programming, not an efficient general solution to scientific discovery. Its state subsets and child combinations may grow exponentially. It assumes the transition, observation, legality, and success tables are correct. It also does not include the cost of constructing those tables in its online path cost. Acquisition of the model, compiler execution, and validation are separately charged development resources.

6.1 A more informative question can be a worse repair question

Counterexample 6.3 (Entropy reduction need not minimize repair work)**.** There are four equally likely cases. Repair a succeeds in $\{0, 1, 2\}$ and repair b succeeds in $\{3\}$. A balanced query separates $\{0, 1\}$ from $\{2, 3\}$. Another query separates $\{0, 1, 2\}$ from $\{3\}$. Both cost one unit and leave the state unchanged. The balanced

query has the larger entropy reduction, but its second branch still needs another observation. The unbalanced query supports a correct repair after one observation in every case. Maximizing information about the entire hidden case is not the same objective as minimizing the cost of a successful correction.

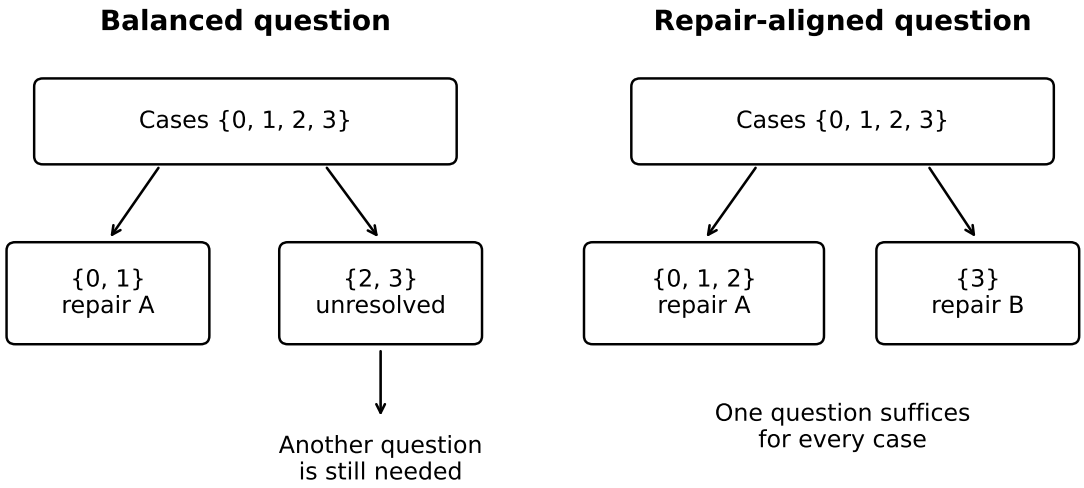


Figure 2: Two equally priced questions about four cases. The balanced split leaves one branch with incompatible repairs. The repair-aligned split identifies a valid action immediately. The figure represents exact sets, not measured agent behavior.

The example suggests a learning target more specific than general curiosity. A useful acquisition policy should distinguish alternative *decisions* when needed, preserve unresolved alternatives when they matter later, and avoid spending work on irrelevant details merely because those details are easy to predict.

6.2 What a compiled witness proves

The reference implementation returns a concrete tree together with its cost pair. A separate interpreter checks its branch coverage, the domains of its experiments, the success of each leaf, and both resource counts. The checker does not infer that the environment model is true. It checks the relation between a specified model and the proposed procedure. This is the same reason an OSL semantic check and a learned Neuralese realization need separate evidence [4, 3].

7 Investigating uncertainty in the experiment model

The first synthesis theorem assumes that the experiment table is known. This is a substantial informational premise. A reasoner may know which experiments it can request while being uncertain about their outcomes, side effects, or interpretation. Optimizing one guessed table can produce a perfectly checked procedure for the wrong instrument.

A useful first extension retains several explicit possibilities rather than replacing uncertainty by one convenient estimate. Let Θ be a finite nonempty set of candidate models with a common state set X, experiment names, repair names, and nonnegative costs. Model θ specifies D_e^θ, f_e^θ, o_e^θ, and G_a^θ. The

unknown model is fixed during an investigation. The initial information is a nonempty set $B \subseteq \Theta \times X$. The procedure observes reports, not the model label or the hidden physical state.

An experiment is allowed only when it is allowed under every pair still possible. Its update is

$$B_{e,o} = \{(\theta, f_e^\theta(x)) : (\theta, x) \in B,\ x \in D_e^\theta,\ o_e^\theta(x) = o\}. \tag{4}$$

The model tag remains unchanged. Observations can exclude a model, but the algorithm cannot switch the underlying model between steps merely because doing so produces a convenient worst case. A terminal repair a is justified when $x \in G_a^\theta$ for every $(\theta, x) \in B$.

This is a finite form of planning under model uncertainty. Robust dynamic programming provides an established neighboring literature [17]. The present construction is not an application of a rectangular transition-uncertainty assumption. It explicitly remembers a fixed model label, including correlations between that model's predictions at different times.

Theorem 7.1 (Repair synthesis under a fixed unknown model)**.** *Construct a finite repair model on $\widetilde{X} = \Theta \times X$ by*

$$\widetilde{f}_e(\theta, x) = (\theta, f_e^\theta(x)), \quad \widetilde{o}_e(\theta, x) = o_e^\theta(x), \quad \widetilde{G}_a = \{(\theta, x) : x \in G_a^\theta\},$$

with the corresponding tagged domains and the common action costs. Applying Theorem 6.2 to this model gives exactly the nondominated work/storage pairs of report-dependent trees that succeed for every initial pair in B, within the declared horizon. Every retained tree is valid for the actual system whenever its model and initial state belong to B.

Proof. A tree receives the same experiment names and report alphabet in both descriptions. For each experiment and report, its possible tagged successor states are exactly (4). Its terminal repair is valid on the tagged support exactly when it is valid for every retained model/state pair. Consequently the report histories, path costs, branch counts, and stopping conditions agree node by node. Induction on tree depth gives a cost-preserving correspondence of all successful trees. The finite synthesis theorem then supplies the frontier and its witnesses. If the actual pair starts in B, each actual observation leaves its actual successor in the updated support, so the executed terminal repair is valid for that pair. □

This theorem decides a finite robust investigation problem. It does not learn an arbitrary model class or guarantee that the class contains the truth. Its negative result is also contract-relative. An empty frontier may mean that the admitted experiments cannot resolve the uncertainty, that a needed calibration is unavailable, or that the horizon is too short. It is not evidence that every possible experiment would fail.

7.1 One hidden bit and two sensor conventions

Counterexample 7.2 (Known-model repair without a common unknown-model policy)**.** Let $x \in \{0, 1\}$ be the cell requiring repair and let $\theta \in \{0, 1\}$ be a fixed sensor convention. A nondestructive query reports $o = x \oplus \theta$ at cost one. Repair r_i succeeds exactly when $x = i$ and costs one. If θ is known, one query followed by the appropriate repair always succeeds. If all four (θ, x) pairs are initially possible, report zero leaves $(0, 0)$ and $(1, 1)$, while report one leaves $(0, 1)$ and $(1, 0)$. Repeating the same query gives no further information. No finite tree using only this query can guarantee a repair, although each known model admits a two-unit procedure.

Add a permitted reference measurement that reports θ, leaves x unchanged, and costs two. The procedure can calibrate, make the ordinary query, and choose $r_{o \oplus \theta}$. Its worst-case work is four and its complete binary tree has seven records. Either measurement alone leaves both repairs possible. The two measurements are

therefore both necessary in this example, regardless of their order. The executable reference construction returns exactly $(4, 7)$ at horizons at least two.

The reference measurement is an added information channel. It is not a renamed latent coordinate and not a free inference from the original report. Its authority, reliability, and cost belong to the model. If the physical reference is faulty, the exact guarantee no longer follows. The same issue would arise if a learner excluded the true sensor convention after an unreliable observation.

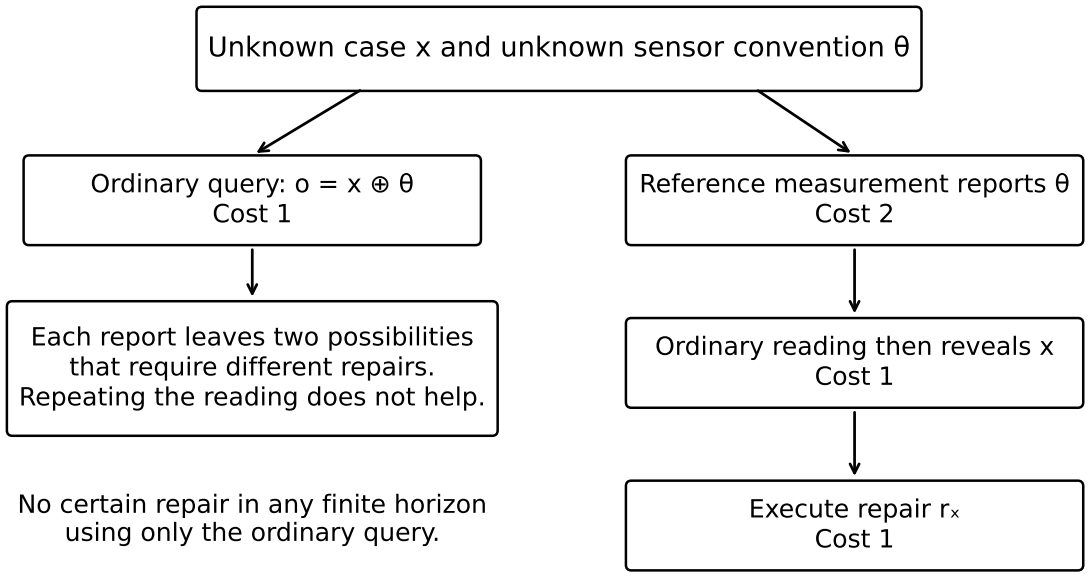


Figure 3: An ordinary reading can identify the repair under a known sensor convention yet fail under two possible conventions. A separately specified reference measurement resolves the missing distinction. The figure is an analytical model, not a calibration experiment on an agent.

7.2 Model refinement should be driven by repair relevance

There is no general need to identify the whole model before acting. Several retained models may agree on a successful repair even while disagreeing about irrelevant outputs. The tagged support retains this possibility because its stopping condition is a common successful repair, not a singleton model label. Conversely, a calibration that removes many candidate models can still leave the decisive repair ambiguity intact.

A model learner therefore has two different targets. It can seek a more accurate description, or it can seek enough justified information to produce an affordable valid repair. The former can support the latter, but the two costs need not be minimized by the same investigation. The learner should retain the unresolved alternatives that matter to future decisions rather than certify an entire environment from one successful correction.

The finite construction also makes the cost of caution visible. The tagged state set can have $|\Theta||X|$ members, and the support search can be substantially larger than a search under one assumed model. Representation refinement, model acquisition, and compiler improvement are consequently coupled research problems. A more reliable model can increase the computational burden of planning until a better discovery procedure exploits its structure.

8 Selecting a better procedure without hiding an oracle

The finite compiler can itself be the object modified by an outer process. One candidate may change the experiment order. Another may change the cost model, add a repair constructor, or select a different representation for the uncertainty set. Their effects should not all be described as the same kind of learning.

For a fixed finite collection $\{U_1, \ldots, U_N\}$, suppose a declared assessment gives estimates $\widehat{R}_i$ of true productivities R_i. This assessment can be exact on a fully specified finite distribution or statistical under additional assumptions. Choosing a candidate must use the assessment actually available at selection time, not a hidden final score inspected afterward.

Proposition 8.1 (Selection under uniform assessment error)**.** *If $|\widehat{R}_i - R_i| \le \varepsilon$ for every candidate and $\hat{\imath}$ maximizes $\widehat{R}_i$, then*

$$R_{\hat{\imath}} \ge \max_i R_i - 2\varepsilon.$$

If a baseline is included and the selected candidate's estimated advantage exceeds $2\varepsilon + \gamma$, its true advantage exceeds γ.

Proof. For a true maximizer i^*, $R_{\hat{\imath}} \ge \widehat{R}_{\hat{\imath}} - \varepsilon \ge \widehat{R}_{i^*} - \varepsilon \ge R_{i^*} - 2\varepsilon$. Apply the same two error bounds to the selected candidate and the baseline for the second assertion. □

The uniform premise is the essential obligation. A small error for each candidate considered in isolation does not automatically control a candidate selected after adaptive access to the same evaluation record. Section 26 addresses an explicit alternative based on conditional validity.

8.1 Changing the selector and changing the language

An improved selector over a fixed candidate family can produce higher gains without making any previously impossible repair expressible. Conversely, a richer language can increase the best achievable gain while a weak selector never finds the new repair. These are distinct forms of headroom, already separated in the earlier SRIC treatment [2].

For a fixed budget, let $R^*_{\mathcal{G}}$ be the largest productivity among procedures expressible in a declared language $\mathcal{G}$, with installed data and interpreter costs held fixed. The gap $R^*_{\mathcal{G}} - R(U)$ is search or selection headroom *within that language*. Comparing $R^*_{\mathcal{G}'}$ with $R^*_{\mathcal{G}}$ concerns expressibility and resources. Neither upper envelope is automatically available to the learner.

A zero observed gain from one selector update is not a proof of saturation. A genuine saturation statement needs either a complete finite search or an upper bound close enough to the incumbent value. Otherwise the failure may reflect the proposer, the evidence channel, the search allowance, or the evaluation.

8.2 The compiled instrument improver

In the eight-cell instrument, the supplied language admits both singleton and bit queries. Recurrence (3) can construct a depth-three diagnosis. The exported program replaces the equality-first diagnostic policy in an otherwise unchanged repair system. It does not contain the hidden defect index of the next episode.

Under a cap of three diagnostic calls and one repair call, the equality-first procedure certifies and repairs the first three cells, then abstains on the other five. On the uniform finite case distribution, its productivity from a zero-success baseline is $3/8$. The depth-three procedure certifies all eight cases, giving productivity 1. These are exact properties of stipulated procedures. The compiler selected a better procedure within a supplied finite language. It did not invent the language, learn a general visual representation, or improve the compiler that performed this search.

That last distinction is deliberate. This first construction improves the diagnostic procedure, not its compiler. Part III then changes the compiler, using certified early termination to construct the same optimal diagnostic controllers with fewer model-record reads. The finite selector that installs that compiler remains supplied. An archive entry saying that a descendant is more reflective supplies none of these separate claims.

9 Transporting an investigation, not merely its result

A repair can commute with a representation map even when the procedure used to discover it cannot be transferred. The investigation may depend on an unavailable observation, on a source-specific reset, or on the precise meaning of a stopping condition. A transport contract must include those actions and their evidence.

Consider source and target finite repair models. Let $\sigma : X \to X'$ be a bijection, $\tau : \mathcal{E} \to \mathcal{E}'$ a bijection of experiments, $\psi : \mathcal{A} \to \mathcal{A}'$ a bijection of repairs, and $\eta_e : O_e \to O'_{\tau(e)}$ a bijection of report labels. Assume

$$\begin{aligned} \sigma(D_e) &= D'_{\tau(e)}, & \sigma f_e(x) &= f'_{\tau(e)}\sigma(x), \\ \eta_e(o_e(x)) &= o'_{\tau(e)}(\sigma x), & \sigma(G_a) &= G'_{\psi(a)}, \end{aligned} \tag{5}$$

and matching execution costs. These assumptions concern supplied semantics. Their discovery or learned approximation is an additional problem.

Theorem 9.1 (Exact transport of diagnostic procedures)**.** *Under* (5)*, relabeling every experiment, branch, and repair transports successful diagnostic trees bijectively between the source support B_0 and target support $\sigma(B_0)$. Execution cost, complete tree storage, and depth are preserved. In particular,*

$$\mathcal{F}_h(B_0) = \mathcal{F}'_h(\sigma(B_0)).$$

The same conclusion holds for every selected tree, not only a cost-optimal tree.

Proof. A terminal repair is valid on B_0 exactly when its image is valid on $\sigma(B_0)$. For an internal experiment, domain preservation gives uniform admissibility. Transition and observation preservation yield

$$\sigma(B_{e,o}) = B'_{\tau(e),\eta_e(o)}.$$

Apply induction to every child. The report bijection preserves the number of children, the maximum execution cost, and the sum of stored child records. Inverse maps give the converse correspondence. The equality of frontiers follows because costs and dominance are unchanged. □

The theorem covers a complete explanation of how the repair is found. It is stronger than equality of the repaired output but weaker than a claim that any learned representation admits such a correspondence. A known map costs something to store and execute. In the toy instrument, an arbitrary cell permutation requires a label adapter. A permutation of eight names is not discovered merely by allocating an eight-row table.

9.1 Approximate performance transfer

Suppose a source comparison gives productivity advantage Δ to U' over U. Let $\widetilde{U}', \widetilde{U}$ be actual transported target procedures. If each procedure's productivity differs from its source value by at most ε, then

$$\mathcal{R}'(\widetilde{U}') - \mathcal{R}'(\widetilde{U}) \geq \Delta - 2\varepsilon. \tag{6}$$

This follows by subtracting the two inequalities. A common source of the premise is an exact task correspondence plus a bounded evaluation-law shift. For any fixed gain function in $[-1, 1]$, changing the

task law by total variation δ changes its expectation by at most 2δ. A further pointwise gain discrepancy η therefore gives $\varepsilon = \eta + 2\delta$.

This elementary estimate does not certify admissibility or sure success. A rare unauthorized action can have small average loss and still violate a zero-tolerance constraint. Such constraints require their own preservation check. The same is true of uncharged translation work: an equal-output target procedure can fall outside the budget after the adapter is added.

Counterexample 9.2 (Output transfer with no transferable investigation)**.** A source can inspect a hidden bit for one unit and then choose one of two disjoint successful repairs. The target has the same hidden states and repair effects, but no operation that reveals the bit. The final repair meanings correspond exactly. The source diagnostic policy does not transport, and no target policy can guarantee success from the two-state uncertainty set. The missing object is an information channel, not a better translation of the repair names.

10 Growing a repair language without losing its old scope

A persistent discrepancy can be a limitation of the candidate language rather than the search procedure. Repeatedly optimizing coefficients cannot produce a nonlinear term in a language that expresses only affine functions. Reordering uninformative experiments cannot separate two cases whose transcripts are always identical. A learner must distinguish an unsuccessful search from a certified absence of a suitable expression.

A conservative extension embeds each old program into a new language with the same meanings on the old domain. That semantic property is valuable, but it does not imply nonregression under a fixed resource allowance. The new interpreter may consume more time, memory, or evidence before executing the embedded old program.

Proposition 10.1 (The resource price of conservative extension)**.** *Suppose an embedding preserves every old procedure's successful behavior and adds at most $d \geq 0$ execution work and $s \geq 0$ stored records. Every old allowance (b, m) is then feasible in the new language under allowance $(b + d, m + s)$. Feasibility at the unchanged allowance follows only when the embedded procedure fits that allowance. Semantic conservativity alone supplies no such guarantee.*

Proof. Embed a witness for the old allowance. By hypothesis its behavior is preserved and its costs are bounded by the shifted pair. The unchanged-budget assertion requires the corresponding inequalities. For the final claim, an old one-unit procedure can be embedded with identical output into a new interpreter requiring two units of startup work. It remains semantically available but cannot execute within the old one-unit allowance. □

A practical extension should retain a directly executable old path when possible, then route only justified cases through the new machinery. The routing operation also has a cost and an information requirement. A context-dependent grammar is not a solution if deciding the context already requires solving the original problem.

10.1 Error localization and new constructors

The useful role of packets and residuals is to make this diagnosis structured. A packet can declare a domain, a retained distinction, an uncertainty set, an operation interface, and unresolved conditions. A residual identifies which preservation equation or evidence constraint failed. A constructor adds a new kind of candidate repair. The theory does not assume that a packet is an independent physical component or that its name supplies a new algorithm.

Part III

Improving the procedure that discovers repairs

11 A certificate can remove the need for exhaustive search

The compiled tree in the first example is an improved diagnostic policy. This section changes the procedure that constructs such trees. The distinction is concrete: the parent and descendant receive the same new problem table and return equally valid optimal controllers, but they need not perform the same amount of discovery work.

The parent is the exact frontier compiler of Theorem 6.2. It examines a finite set of support/horizon states and combines successful subtrees. That generality is useful, but it can do unnecessary work when one inexpensive candidate already meets a proved global lower bound. A descendant compiler can first propose such a candidate, check it independently, and stop only if its checked cost proves that further search cannot improve either protected resource coordinate.

This combines a proposal mechanism with a decisive rejection or acceptance condition. Synthesis from partial programs and counterexamples has established precedents [18]. The particular addition here is a certificate that can close an *optimality* search, rather than merely establish that one candidate is valid. Neither the proposal template nor the lower-bound argument is assumed to have been invented by the system.

11.1 The scope in which a short proof is decisive

Consider $n \geq 1$ possible immutable cases. Every admitted query is total, nondestructive, binary-valued, and costs one unit. Every available repair costs one unit and succeeds on exactly one case. At least one repair is available for each case. The target is certain successful repair from the complete set of n possibilities. A literal controller stores one record per query or repair node. Redundant unary query nodes are permitted but have their full costs.

Theorem 11.1 (Checked lower-bound closure of a diagnostic search)**.** *Every successful controller in this scope has worst-case execution work at least*

$$1 + \lceil \log_2 n \rceil$$

and storage at least $2n - 1$*. If an independently checked candidate attains both bounds and respects the requested horizon, its cost pair is the entire nondominated frontier. Returning this candidate is therefore equivalent in the protected work/storage objective to completing the exhaustive synthesis.*

Proof. Every case produces one deterministic query history. Two different cases cannot terminate at the same repair leaf because no repair succeeds in both. Thus there are at least n nonempty leaves. A tree of binary query depth d has at most 2^d leaves, so $d \geq \lceil \log_2 n \rceil$. Every path also executes a unit-cost repair. For storage, let l be the number of leaves and let i_1, i_2 count unary and binary internal nodes. Counting edges in a finite rooted tree gives $i_1 + 2i_2 = i_1 + i_2 + l - 1$, hence $i_2 = l - 1$. Total records are $2l - 1 + i_1 \geq 2n - 1$. A successful candidate attaining both coordinatewise lower bounds dominates every other attainable cost pair. Its independent validity check and horizon check make it a feasible witness, so the Pareto frontier contains only that pair. □

The immutable-case premise is essential. An experiment that resets every physical state into one safely repairable state can defeat a case-identification lower bound when only the current state matters. Multiple

cases admitting a common repair can do the same. The guard must therefore verify the actual contract before applying the bound. This is not a universal theorem that diagnosis always requires identifying the hidden state.

A candidate that is valid but misses one lower bound does not close the search. Nor does failure of the proposal rule prove that no optimum exists. In both cases the descendant falls back to the general compiler. The certificate licenses an early return, not an early claim of impossibility.

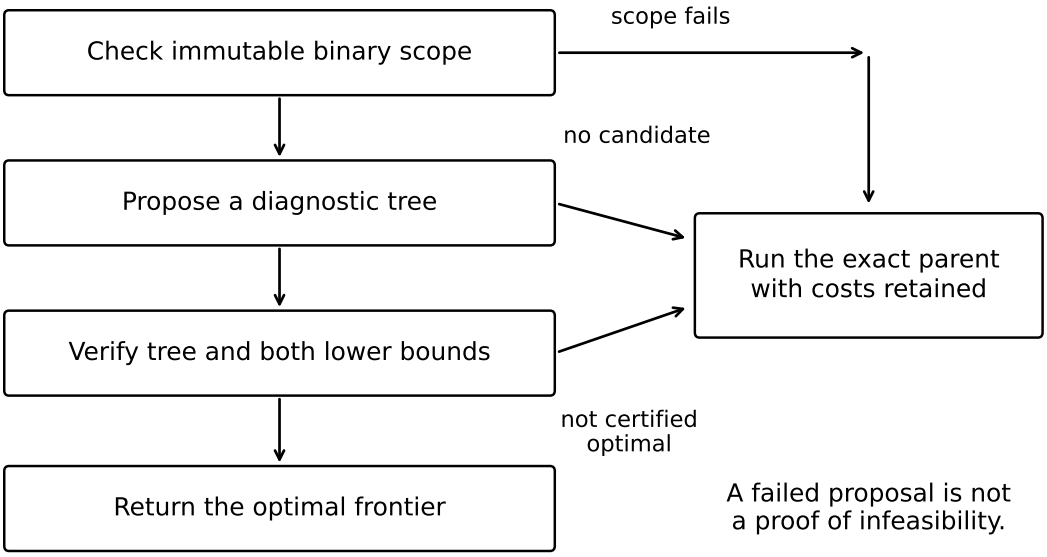


Figure 4: The descendant changes discovery control flow. A checked candidate must attain both global lower bounds before the compiler can stop early. Failure of a guard, proposal, or bound check invokes the unchanged exact compiler. The sequential reference implementation charges unsuccessful preliminary work.

11.2 The inexpensive proposal is not the proof

The reference proposer repeatedly chooses a query whose two current branches have the smallest maximum cardinality, breaking ties by the installed query order. It searches the public observation table rather than recognizing privileged names such as “bit zero.” It constructs a tree recursively, stopping when a unique justified repair remains. It aborts a proposed branch when the remaining binary depth cannot distinguish its cases.

Balancing is a supplied heuristic. It need not minimize repair work in general, as Counterexample 6.3 already shows. It is useful in the narrower singleton-repair scope because the required distinctions coincide with case identification. Even there, a locally balanced split need not extend to a globally optimal tree for an arbitrary query library. Correctness comes from the finished witness and the lower bound, not from the heuristic’s name.

The checker reconstructs every possible support, checks all reports, and independently replays every initial case. A fabricated terminal label, missing branch, insufficient horizon, or altered repair effect is rejected. This is a finite model checker authored within the project. It is not a proof of the physical observation table or an independent proof-assistant kernel.

12 The cost of certifying a better compiler

The unit used in the following comparison is a consumed record of the supplied transition/report table. A repeated access counts again. The unit is chosen because it is explicit and can be instrumented in the unchanged parent compiler and the descendant. It is not a claim to count every Python operation, allocation, set-membership test, processor instruction, or unit of energy.

Both procedures start with cold search caches and the same structurally validated table. Shared input-shape validation is recorded separately. The descendant additionally charges its semantic guard, candidate construction, and completed witness check. The parent charges its complete frontier search and the same witness-checking procedure for its returned frontier. Model acquisition, source storage, and the finite procedure-selection stage remain separate ledger entries.

Proposition 12.1 (Read cost of the certified binary-family shortcut)**.** *Let $n = 2^k$ with $k \geq 1$. Suppose there are q total unit-cost binary immutable queries, and the balanced proposer produces a perfect depth-k tree. In the stipulated implementation, the semantic guard reads every query row once, the proposer scans all q queries at each internal support, and verification performs one support reconstruction and one complete case replay. The total number of transition/report record reads is*

$$R_{\text{shortcut}} = nq(k+1) + 2nk. \tag{7}$$

The usual hidden-cell family with all singleton queries and k coordinate queries satisfies the premise, including after arbitrary cell relabeling, report-bit flips, or reordering of the query list.

Proof. The guard uses nq reads. At each of the k query levels, the current supports partition the n cases, so scanning all queries uses nq reads per level. Every case traverses k query nodes in the support reconstruction and again in the independent replay, contributing $2nk$. Summing gives (7). In the hidden-cell family, every nonsingleton support reached by coordinate splits is a subcube. A remaining coordinate divides it equally, while a singleton query cannot give a smaller maximum branch. Thus the balancing rule chooses an equal split until singleton leaves are reached. When a two-case support admits a singleton tie, either split reaches singleton leaves. Relabeling the cases or reversing report names preserves all branch cardinalities and this argument. □

The result is about a particular implemented access pattern. Caching rows, recognizing trusted structure symbolically, or using another cost unit changes the formula. A specialized cube compiler supplied with certified coordinate semantics can avoid the general search too. The result does not establish superiority to that informed comparator or to every available synthesis algorithm.

12.1 Some verification work cannot disappear

Proposition 12.2 (A table-access floor for universal scope certification)**.** *Suppose a deterministic validator receives an arbitrary table of $N \geq 1$ independently variable records only through individual record reads. It must accept every table whose records all satisfy a fixed per-record predicate and reject every table with a violating record. Assume each record position admits both a satisfying and a violating value. On every accepted table, the validator must read all N positions at least once.*

Proof. Suppose it accepts a satisfying table while leaving position j unread. Change only that position to a violating value. Every record the validator reads has the same value, so determinism gives the same sequence of reads and the same acceptance. This contradicts rejection of every table containing a violation. Therefore no accepted table can have an unread position. □

In the compiler's unrestricted table interface, an unread entry could contain a destructive transition or an additional report value. The full semantic scope cannot be certified merely by recognizing a few apparently balanced rows. Trusted construction metadata or a proof from a different representation can change the interface and reduce this cost, but that metadata then becomes a charged assumption or a separate proof object.

This identifies a useful bottleneck. Improving search can make model access and verification dominate the remaining computation. It does not make every stage improve at the same rate. Nor does it establish that this bottleneck is permanent, since a learned representation might eventually support cheaper certificates. Such a representation would need its own acquisition and soundness argument.

13 A bounded selection of a better discovery procedure

We now make the persistent change explicit. The candidate catalogue contains the unchanged parent compiler, a guarded first-nontrivial-query proposer with exact fallback, and a guarded balanced proposer with exact fallback. The guard, proposal constructors, lower-bound proof, checker, and fallback are supplied. The selection procedure runs each candidate on the four-cell and eight-cell development problems, with the same tables and cold caches. It selects the candidate with the smallest total number of charged record reads and writes a policy record before evaluating any additional problems.

The developmental totals are 10,256 reads for the parent, 10,380 for the guarded first-query rule, and 488 for the guarded balanced rule. The selector therefore installs the balanced rule. These totals concern a fixed, fully enumerated development catalogue. They do not imply a statistical generalization guarantee over an unknown problem distribution. The selected artifact changes the control flow of future compilations without containing the hidden state of a future repair episode.

Cases	Parent compiler reads	Descendant reads	Returned (work, storage)
2	16	16	(2, 3)
4	352	88	(3, 7)
8	9,904	400	(4, 15)
16	576,128	1,728	(5, 31)

Every paired entry includes complete checking of the returned controller. The sixteen-cell problem is evaluated only after the policy record is fixed. Additional cold cases remove singleton queries, restrict the initial support to six possibilities, reverse operation order, and change cell and report labels. These vary the finite structure available to the compiler. They are not evidence of transfer to an unrelated scientific domain.

Under a common cap of 2,000 table reads, the descendant returns a checked optimal sixteen-cell controller. The parent reaches the cap without returning its complete frontier. Its outcome is recorded as *incomplete within the allowance*, not as a proof that no controller exists. Without that cap, the parent also returns the same optimal cost pair. The difference lies in discovery work rather than repair validity or online diagnostic depth.

13.1 What was selected and what remains supplied

The persisted policy chooses a proposal rule from a fixed catalogue. It is not a stored solution to the next hidden-cell episode. It therefore modifies an improvement procedure rather than only adding another repaired answer. The compiler accepts arbitrary supplied tables, applies its semantic guard, and retains the exact fallback outside the shortcut's demonstrated scope.

This is nevertheless a restricted form of process learning. The catalogue does not invent a new lower-bound theorem or discover a new programming primitive. It does not improve the selector that searched the catalogue. A second call with the same candidates and the same criterion returns the same selected rule. One genuine improvement of a discovery procedure is not an indefinite sequence of improving outer procedures.

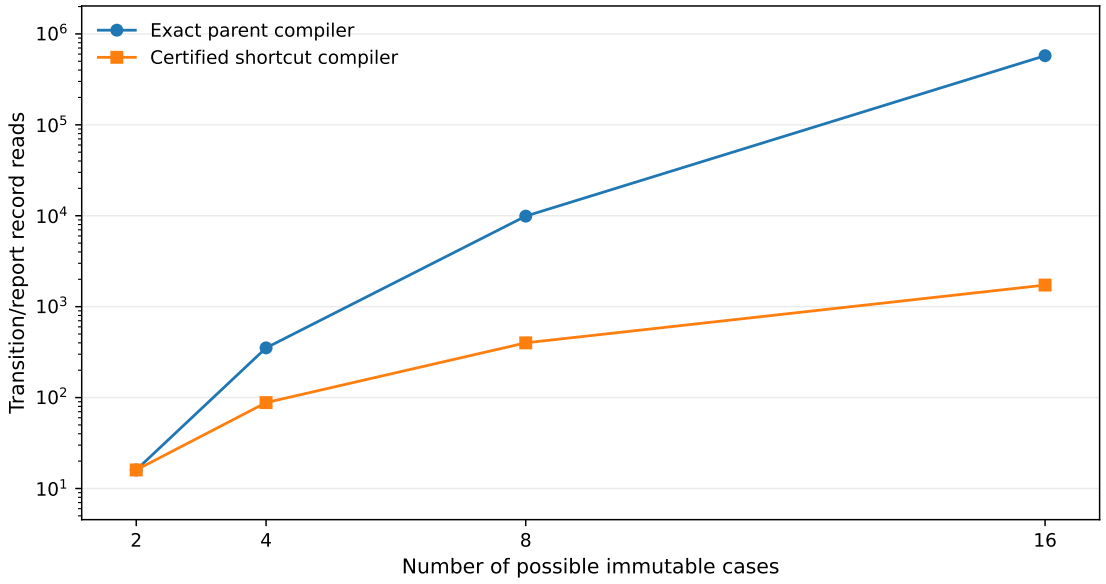


Figure 5: Charged transition/report record reads on the exact binary family, including witness checking. The vertical axis is logarithmic. Both compilers return the same optimal online work and stored tree size. These deterministic algorithm counters are not measured agent-learning performance or total CPU instructions.

The distinction gives the next research target a concrete form. A learned proposer would need to generate useful shortcut templates or representations not simply chosen from this fixed list, while the checker verifies their stated scope. A learned evaluator would need to identify when existing cost or success measurements omit a consequential requirement. Such changes would alter additional parts of the realization and need their own preserved comparison.

13.2 Fallback protects meaning, not free execution

Counterexample 13.1 (A semantically protected shortcut with a slower fallback)**.** Give the compiler four immutable cases and only four singleton queries. The horizon permits the three-query equality tree. A balanced proposal produces a valid controller but cannot attain the two-query information lower bound, since the query library does not supply the needed split. The descendant therefore performs its preliminary guard and proposal work and then runs the exact parent. It returns the same frontier but consumes more table reads. Semantic nonregression has not established work nonregression.

The counterexample is included in the shipped comparison. The declared development objective prefers the shortcut on its development family, not on every possible input. Changing the mixture of structured and fallback cases can change which procedure is preferable. Attributing all gains to the new procedure while excluding its failed proposal attempts would invalidate the comparison.

There is a conditional way to bound this risk through scheduling rather than prediction. The following result is intentionally separate from the sequential implementation above.

Proposition 13.2 (Bounded slowdown from a certified two-worker race)**.** *Suppose a parent and a shortcut worker can be paused after each unit of a common work measure and resumed without losing progress. Each may announce completion only after producing a result certified for the same target. Alternate one unit of each worker and stop at the first certified completion. If their standalone completion costs are* $D_0, D_1 \in \mathbb{N}_{>0} \cup \{\infty\}$ *and at least one is finite, at most* $2\min\{D_0, D_1\}$ *worker units are executed. If each dispatched unit costs at most an additional* $c \geq 0$ *scheduler units in the same measure, total charged work is at most* $2(1 + c)\min\{D_0, D_1\}$.

Proof. After $2d$ dispatched worker units, each worker has received exactly d units of progress. Set d to the smaller finite standalone completion cost. That worker has completed its certification by this point, so the race has stopped no later. Multiplying the number of dispatches by the maximum combined work per dispatch gives the overhead bound. □

The theorem does not provide resumability for an arbitrary compiler, bound the memory occupied by both workers, or make their setup free. The present reference compiler uses sequential fallback, not this race. The companion suite checks the abstract schedule bound separately on finite worker tapes. A systems implementation would need to include state checkpointing, dispatch cost, shared-resource contention, and both workers' storage before claiming this guarantee.

Taken together, the results distinguish three achievements: a heuristic frequently proposes a good answer, a certificate establishes that the answer closes the search, and an implementation retains an affordable fallback when the heuristic fails. None of these can substitute for the others.

13.3 A complete receipt for the sixteen-cell compilation

The selected policy is evaluated on a fresh sixteen-cell table after its bytes are fixed. The new table has twenty query rows: sixteen singleton queries and four coordinate queries. The next hidden defect is not supplied to either compiler. Both methods construct a complete controller that will work for any of the sixteen cases.

The descendant's recorded access account is

Compiler stage	Consumed transition/report records
Semantic scope guard	320
Proposal construction across four levels	1,280
Support reconstruction by the checker	64
Independent replay of all cases	64
Total	1,728

The separate common input-shape validation consumes another 320 records before the measured compilation begins. It is not included in the 2,000-read compiler allowance. Counting that shared setup on both sides gives 2,048 reads for the descendant and 576,448 for the parent. No particular implementation is permitted to hide its extra guard inside an uncharged setup step.

The three-candidate development comparison itself consumed 21,124 model-record reads. It is part of the cost of selecting the descendant, not part of one later compilation. For the declared sixteen-cell workload the per-use read saving is 574,400, so those developmental reads are recovered within one use in that same narrow accounting unit. This does not repay an unspecified human research cost, prove a wall-clock speedup on every machine, or charge the creation of the candidate templates. Source sizes and measured runtimes are recorded separately rather than silently converted into model-record reads.

The result concerns cheaper discovery of equally valid controllers. It does not establish an improved selector or an independently discovered compiler catalogue.

Part IV

Learning a compiler's rejection rule

14 A failed search representation can itself be repaired

The preceding selector chooses among three supplied compiler strategies. It changes which algorithm will be used, but does not compose a new decision rule. We now move one step inside a compiler. A small learner constructs a predicate that tells the compiler when it can stop inspecting an unpromising query. The learned object is neither a hidden-cell answer nor a completed diagnostic tree. It is an operation on the compiler's own partial information, reused while constructing trees for further problems.

The construction is deliberately restricted. Its experiment tables, compiler skeleton, primitive arithmetic language, and correctness argument remain supplied. Within that boundary, the learner is not given the complete predicate as one named strategy. It first encounters a representation that cannot express its assigned distinction, adds a specified missing feature, and synthesizes a formula from counterexamples. A structural certificate then determines whether the formula may safely prune later searches.

This separates three questions that are easy to conflate: whether the examples determine a suitable formula, whether the formula is valid outside those examples, and whether executing it actually reduces the relevant cost. The reference construction answers each under a different explicit contract.

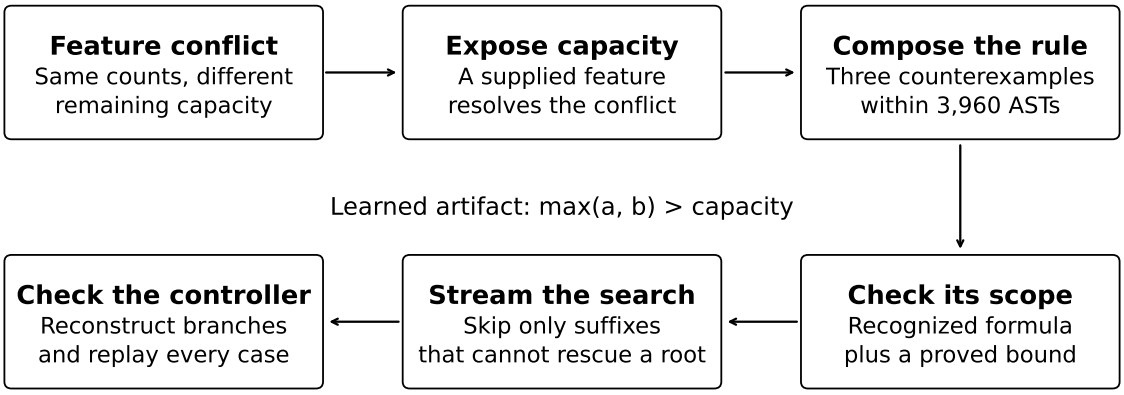


Figure 6: One complete bounded improvement. A feature conflict justifies exposing the supplied capacity feature. Counterexamples select a composed arithmetic expression. A narrow structural check links that expression to a proved scope, after which it can skip rows inside later compilations. The grammar, teacher, scope proof, and compiler skeleton remain outside the learned component.

14.1 The information available inside a search

A compiler is considering a binary query on a support of m possible immutable cases. It reads the query's public model rows in a fixed order. After a prefix, a of the inspected cases have report zero and b have report

one. There are $m - a - b$ rows left to read. If this query is the first of at most $d \geq 1$ remaining queries, either child can contain at most

$$C = 2^{d-1}$$

cases when distinct cases need distinct repair leaves. The local cardinality premise $m \leq 2C$ is checked before a root is investigated. An overfull support cannot fit the remaining binary depth, even before any rows of the proposed query are read.

The initial search representation contains (a, b, m) but omits C. This can lose a distinction relevant to the search itself. For example, $(a, b, m) = (0, 3, 4)$ cannot have a completion whose two branch sizes are at most two. It can have a completion with branch sizes $(1, 3)$ when the allowed child capacity is four. The read prefix is the same. The permitted future investigation is different.

Proposition 14.1 (Feature sufficiency for a search decision)**.** *Let $\mathcal{P}$ be a finite set of search contexts, let $r : \mathcal{P} \to Z$ be their available feature map, and let $\ell : \mathcal{P} \to \{0, 1\}$ be the required decision label. A deterministic rule $f : Z \to \{0, 1\}$ satisfying $fr = \ell$ exists if and only if ℓ is constant on each nonempty fibre of r. A pair in one fibre with different labels rules out every such rule, irrespective of its program length.*

Proof. If f exists, equal features have the same output under f and hence the same label. Conversely, assign to each attained feature value its common label. This is well defined when the fibre condition holds. Values outside the image of r can be assigned arbitrarily. A conflicting pair violates necessity for every possible f. □

The criterion is elementary factorization, not a new general learning theorem. Its role is to distinguish insufficient information from inadequate search effort. In the finite construction, exhaustive search for a larger formula over (a, b, m) cannot fix the conflict. The supplied feature registry contains the remaining-depth capacity C. Exposing it resolves this particular obstruction. Computing C from an already known depth is not a physical experiment and does not identify the next episode's hidden defect.

Counterexample 14.2 (The same prefix requires different rejection decisions)**.** For the two contexts $(a, b, m, C) = (0, 3, 4, 2)$ and $(0, 3, 4, 4)$, the initial feature vector (a, b, m) is identical. In the first, three cases already occupy a child with capacity two. In the second, assigning the remaining case to the other child gives a capacity-feasible completion. Thus the predicate "no capacity-feasible completion exists" cannot be computed from the initial features alone. Adding the capacity makes this specific decision representable. It does not establish that the actual query library can distinguish every resulting child.

This is a small instance of representational revision inside a discovery algorithm. The target of learning is not only a model of the external task. It is a model of which search continuations remain possible under the compiler's own constraints. A richer representation is useful because a named counterexample requires it, rather than because extra variables are assumed to improve intelligence in general.

15 Composing a rule from primitive syntax

The enlarged feature vector is (a, b, m, C). The reference language has six integer atoms, $a, b, m, C, 0, 1$, three binary arithmetic operations, max, min, and addition, and the five comparisons $>, <, \geq, \leq, =$. A candidate is a Boolean comparison with at most five syntax nodes. No general program execution is admitted. The interpreter evaluates only these finite arithmetic trees.

Integer atoms have size one. A binary arithmetic expression has size three. A comparison of two atoms has size three, while a comparison with one arithmetic child has size five. For commutative arithmetic, operand pairs are generated in one fixed order. This produces 180 size-three formulas and 3,780 size-five formulas,

or 3,960 candidates in all. Different syntaxes can express the same function. No semantic deduplication is assumed in these counts.

This is a bounded instance of syntax-guided synthesis, in which the candidate language and a semantic requirement are specified separately [19]. Counterexamples guide the choice of candidate. Learning invariants from examples, counterexamples, and implication constraints is also an established research direction [20]. The present learner does not implement the full ICE framework and makes no claim to discover general loop invariants. It learns one rejection expression under a supplied search contract.

15.1 What the teacher knows

Development uses the 198 count-prefix contexts with $C \in \{1, 2, 4\}$, $2 \le m \le 2C$, and $a, b \ge 0$, $a + b \le m$. For each context, the teacher enumerates the possible numbers of report-one cases among the unread rows. It labels the context as reject precisely when no such completion can put at most C cases in either branch. The teacher therefore solves a small finite feasibility problem. It is not reading a label generated by the final formula.

After each returned counterexample, the learner retains that context and its label. It selects the earliest formula in the fixed syntax enumeration that agrees with every retained example. The finite teacher either returns another disagreement or checks the formula against the complete developmental domain. The definition of that domain, the generation of its labels, and all rejected candidates are part of the development account.

Theorem 15.1 (Finite counterexample refinement within a grammar)**.** *Let $\mathcal{G}$ be a finite nonempty set of deterministic candidate programs, let $\mathcal{P}$ be a finite domain, and let ℓ be its specified labeling. Suppose a teacher either returns a point where the proposed program differs from ℓ or certifies agreement on all of $\mathcal{P}$. A learner that proposes only candidates consistent with every retained counterexample rejects each proposed program at most once. It returns a correct program on $\mathcal{P}$, or certifies that none in $\mathcal{G}$ meets the labeling, after at most $|\mathcal{G}|$ teacher calls. With a fixed total order and the earliest-consistent selection rule, any returned program is the earliest one that agrees on the entire domain.*

Proof. A counterexample removes the current program because it must be included in the constraints for every later proposal. Hence that program cannot be proposed again. A program agreeing with ℓ on all of $\mathcal{P}$ is never removed by a valid counterexample. There are only $|\mathcal{G}|$ candidates, so after at most that many distinct proposals the teacher has accepted one or all candidates are excluded. If a returned program were preceded by a globally correct program in the fixed order, the earlier program would remain consistent and would have been selected instead. This proves the ordering assertion. □

The result concerns the declared finite domain. It is not a generalization bound and supplies no efficiency guarantee for obtaining the teacher's answers. A teacher timeout is not a certificate. If the grammar does not contain a suitable program, enlarging the grammar changes the search problem and its cost.

15.2 The synthesized expression and its provenance

The reference learner encounters three disagreements. Its first candidate, $a > a$, fails on a prefix with too many ones. The next proposed candidate, $b > a$, rejects a prefix that can still be completed. The third, $b > C$, misses a prefix with too many zeros. The fourth teacher proposal is

$$\rho(a, b, m, C) = \mathbf{1}\{\max(a, b) > C\}. \tag{8}$$

The formula uses five syntax nodes and passes all 198 developmental contexts. The enumeration generates all 3,960 AST roots and considers 190 of them, including candidates rejected directly by retained examples.

There are 282 retained-example checks, 210 full-domain teacher comparisons, 344 completion-count checks to construct the labels, and 1,898 interpreted syntax-node visits in the learner's recorded comparisons. The full grammar has 19,440 literal node occurrences. Shared in-memory subexpressions can reduce actual storage, so this last quantity is not a measured byte count.

These numbers describe an exact finite synthesis, not a statistical training cohort. The formula is serialized and frozen before evaluation on larger supports, additional capacities, and changed row and query orders. No new episode's defect index or solved controller is placed in the formula. It contains only the reusable relation between two observed prefix counts and remaining capacity.

The learner discovers the composition of supplied primitives. It does not invent addition, maximum, the teacher, the resource model, or the eventual safety proof. This boundary is important: enumerating a finite grammar is more expressive than choosing among the earlier three named compiler policies, but it remains a bounded search space chosen in advance.

16 When an unfinished read can close a search branch

A finite fit is insufficient grounds for deleting possibilities from a later search. Pruning is stronger than proposing. A bad proposal can be rejected by a finished-witness checker. A bad pruning rule can remove the only successful controller and leave nothing for that checker to inspect. We therefore attach the generated expression to a separate scope argument.

Proposition 16.1 (Exact obstruction to a capacity-feasible completion)**.** *Let C, m, a, b be integers with $C \geq 1$, $2 \leq m \leq 2C$, $a, b \geq 0$, and $a + b \leq m$. A binary completion of the remaining $m - a - b$ reports with both final branch counts at most C exists if and only if $\max(a, b) \leq C$. Consequently, a prefix satisfying $\max(a, b) > C$ can be rejected before its remaining rows are read, without rejecting any completion that satisfies these two capacity constraints.*

Proof. Counts do not decrease as rows are read, so $a > C$ or $b > C$ makes a feasible completion impossible. Conversely, suppose $a, b \leq C$. Let $r = m - a - b$. There are $C - a$ available places in the zero branch and $C - b$ in the one branch. Both are nonnegative and their sum is $2C - a - b \geq r$. Distribute the r remaining cases between these two capacities, filling either branch up to its capacity and putting any remainder in the other. This gives the required completion. The argument concerns possible report assignments, not whether an installed query has that assignment. □

The theorem applies to arbitrary positive integer capacities. The connection to query depth uses $C = 2^{d-1}$. An actual search separately checks that both observed branches are nonempty, since a redundant unary query cannot be part of a controller attaining the minimal literal storage. It also verifies all later subtrees. Capacity feasibility is necessary, not sufficient, for actual repair solvability.

16.1 A deliberately narrow certificate checker

The executable gate accepts only ASTs that normalize by swapping a comparison's sides or maximum's arguments to $\max(a, b) > C$. It rejects other finite-consistent formulas rather than claim to prove arbitrary arithmetic identities. Thus the universal justification is the preceding prose proof, coupled to a project-local structural checker and a runtime scope guard. This is not a Lean certificate or an independent theorem-proving service.

Counterexample 16.2 (Agreement on training prefixes does not establish scope)**.** The formula $\max(a, b) > \min(C, 4)$ agrees with (8) on every developmental context because every developmental capacity is at most four. At $(a, b, m, C) = (0, 5, 16, 8)$ it rejects, while a completion with final branch sizes $(8, 8)$ exists. A

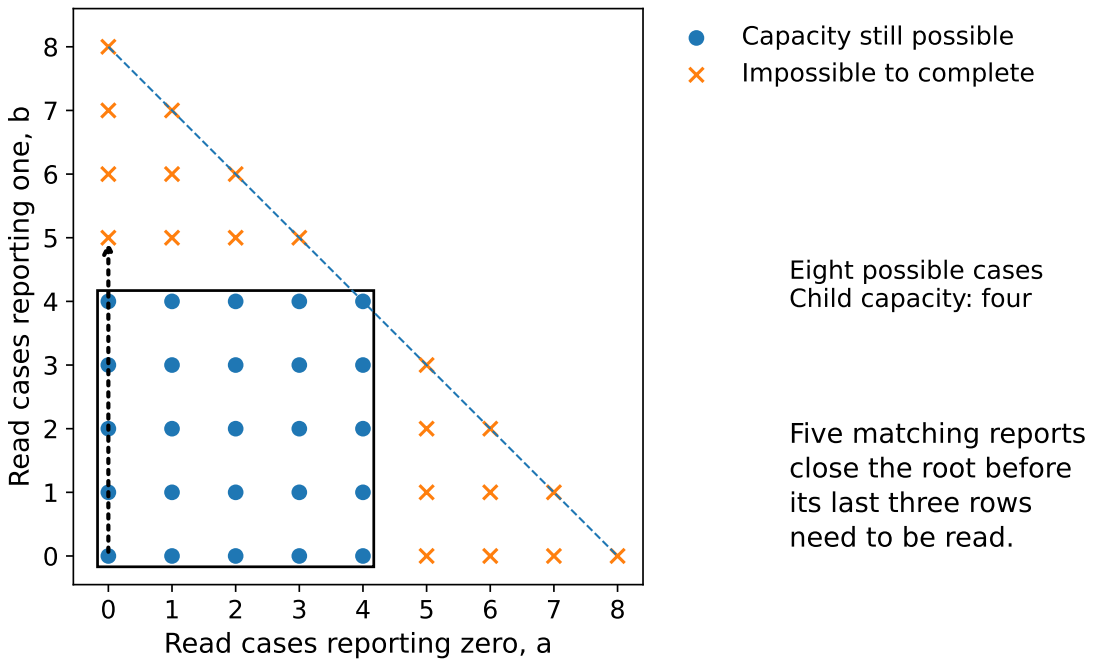


Figure 7: An exact count-prefix model with eight cases and child capacity four. A prefix outside the square $a, b \le 4$ cannot be completed within that capacity. Prefixes also satisfy $a + b \le 8$. Surviving the count test does not guarantee that the installed query library can separate the children. The dotted arrow is one unfinished read that can stop immediately after its fifth equal report.

larger input has not merely changed the frequency of familiar cases. It has reached a scope on which the bounded-cap formula was never justified. The structural gate does not accept that formula.

This counterexample also separates candidate selection from admissibility. A learner may prefer a formula for reasons of fit or cost. The compiler may install it as a pruning rule only under the corresponding proof obligation. Changing the rule's semantics, the available features, or the repair-success contract requires revisiting that obligation.

16.2 The same search with fewer model-row reads

The reference compiler first checks the immutable, unit-cost, binary-query and singleton-repair scope of Theorem 11.1. It then conducts an ordered depth-first feasibility search at the information lower depth $d_0 = \lceil \log_2 |B_0| \rceil$. The full-scan version reads every current row of a candidate query before rejecting an overfull child. The streaming version consults the certified prefix rule after each read. Both use the same query order, report order, remaining depth, and stopping rules. If a root has capacity-feasible children but a child cannot be completed using the actual library, the search backtracks to the next root.

A successful controller is separately checked and attains both lower bounds. When no controller exists at d_0, a requested horizon equal to d_0 permits a genuine finite infeasibility result. A longer requested horizon instead invokes the unchanged general compiler. Failure of the initial scope guard also invokes that compiler. Exhaustion of a model-read allowance returns incomplete, never infeasible merely because the allowance ran out.

Theorem 16.3 (Semantics-preserving streaming pruning)**.** *Under the checked binary immutable singleton-repair scope, fixed query and row orders, and identical search and fallback rules described above, replacing*

full root scans by the certified predicate (8) *returns the same first successful lower-depth controller, or the same exhaustive failure at that depth. With unlimited completion of any required common fallback, both compilers return the same protected frontier. The streaming version uses no more transition/report model-row reads, and the difference is exactly the number of unread suffix rows skipped at rejected roots. Scope checks and final witness verification are included identically in both read accounts.*

Within a root scan, after a prior prefix has passed, the predicate can be evaluated by comparing only the newly incremented count with C. This lowering preserves every rejection decision.

Proof. By Proposition 16.1, every root rejected early would be rejected after its full scan. A root not rejected early has its complete rows read and produces the same two child supports in the same order. Induction on remaining depth therefore gives the same child searches, backtracking choices, first completed controller, and lower-depth failure. Nonempty binary splits strictly reduce support, and the depth decreases, so the finite search terminates. Any successful tree can omit redundant unary roots without losing case information in this immutable unit-cost scope. The search consequently covers all relevant lower-depth controllers. The checked lower bounds justify the same optimal frontier on success. Identical failure conditions invoke the same exact fallback on the same problem.

At every visited root, either the same rows are read or one suffix of rows that cannot affect the rejection decision is omitted. All descendant supports actually entered are identical. Summing omitted suffix lengths gives the stated read difference. This excludes timeouts, under which the methods can finish at different allowances but neither asserts false infeasibility.

For the lowering, both prior counts are at most C, since otherwise the scan would already have stopped. One report increments exactly one count. The unchanged count remains at most C, so the maximum exceeds C exactly when the incremented count does. This implication requires the stated loop invariant. □

The read comparison isolates one alteration to one compiler. It is not a claim that every new search order beats the earlier balanced proposer, nor that fewer table reads imply shorter processor runtime. Interpreting a five-node formula at each prefix adds work. The supplied lowering replaces that interpretation by one integer comparison, but it too has a cost. The code records both versions and checks that they choose identical trees.

Counterexample 16.4 (Local capacity without a completing query library)**.** Four cases admit singleton repairs. The only available query divides them into $\{0, 1\}$ and $\{2, 3\}$ and leaves the state unchanged. At depth two, the root satisfies the $2/2$ capacity test. Repeating the only query cannot separate either pair, so no successful depth-two tree exists. Accepting the first capacity-feasible root as a finished certificate would be wrong. The streaming compiler searches its children, records the failure, and concludes infeasibility only after exhausting the actual library.

17 Pruning depends on what counts as a repair

The singleton scope is not cosmetic. A repair procedure need not determine the hidden case if one action succeeds on several cases. A change to the repair language can therefore invalidate the interpretation of raw case counts as required leaves, even when the experiment model is unchanged.

Let B be a finite nonempty immutable-case support and let $G_r \subseteq B$ be each repair's success set restricted to it. Define $\kappa(B)$ to be the smallest number of repair-success sets whose union covers B. Set $\kappa(B) = \infty$ when no such cover exists. This number concerns the repaired objective, not the number of hidden states. Computing it can itself be expensive.

Proposition 17.1 (Repair-relative leaf requirements)**.** *For finite immutable cases with binary deterministic queries, any certain successful diagnostic tree of query depth at most d has at least $\kappa(B)$ nonempty repair*

leaves. Consequently $\kappa(B) \le 2^d$ is necessary for success, and literal tree storage is at least $2\kappa(B) - 1$. These conditions are not sufficient for success under a specified query library.

Proof. Every initial case reaches one repair leaf. The set of cases reaching that leaf is contained in its repair's success set. The success sets of the attained leaves therefore cover B, so the number of leaves is at least $\kappa(B)$. A binary tree of depth d has at most 2^d leaves. The edge-count argument from Theorem 11.1 gives at least twice the number of nonempty leaves minus one literal records, with any unary nodes adding further storage. The preceding one-query example has $\kappa(B) = 4$ and $d = 2$, satisfying both numerical lower bounds while lacking a completing tree. Thus the necessary conditions are not sufficient. □

For singleton repairs, $\kappa(B) = |B|$, recovering the earlier argument. With a universal repair, $\kappa(B) = 1$ even when many cases remain unknown. The executable streaming path retains the easier singleton guard rather than quietly substituting a hard cover computation whose cost would be uncharged. The broader proposition indicates how a different pruning contract could be developed, not a hidden extension of the present code.

Counterexample 17.2 (A successful repair language can make identification unnecessary)**.** Keep the four-case query table but add one unit-cost repair that succeeds in every case. The exact optimum is now one repair call and one stored record, with no queries. A compiler that carries over the old "four cases require four distinct leaves" assumption can discard this optimum. The reference guard rejects the changed singleton scope and the general compiler returns the zero-query solution. The old theorem remains true. Its premise no longer describes the new language.

This is the preservation problem in recursive revision. A learned optimization acquires authority from a contract. Subsequent language growth can change that contract. Keeping the old formula while discarding its premise is not improvement. A retained shortcut should therefore carry its feature definition, semantic assumptions, recognized proof form, and fallback behavior together.

18 What the synthesized compiler actually gains

The learner freezes (8) after the small development domain. It is then used on the hidden-cell query family through 64 cases. The primary comparison is between the full-scan ordered search and the same ordered search with prefix pruning. It therefore isolates the effect of the generated rule. The earlier balanced wrapper is retained as a separate comparison, not silently treated as identical to this full-scan baseline.

Cases	Full-scan reads	Streaming reads	Rows avoided	Added comparisons
2	12	12	0	2
4	66	65	1	25
8	308	277	31	141
16	1,384	1,129	255	681
32	6,192	4,641	1,551	3,137
64	27,744	19,457	8,287	14,209

All listed compilers include the semantic scope guard and final witness check. The common input-shape validation is separately recorded and identical. For example, the 64-case query table has 70 queries and its shared shape check consumes 4,480 model rows for each method before the reported compilation allowance. The final controllers have the same literal trees in the primary comparison. Their execution and storage costs are unchanged: seven online units and 127 stored records in the 64-case case.

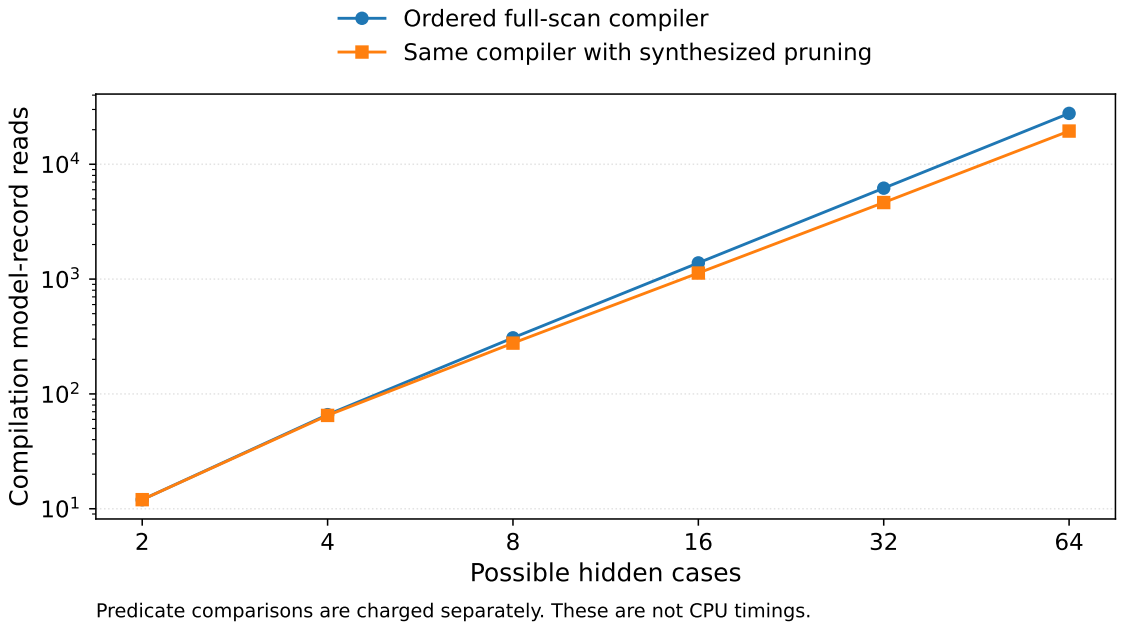


Figure 8: Transition/report model-record reads for the same ordered compiler before and after installing the synthesized prefix rule. Scope and final witness checks are charged in both curves. Common input-shape validation is separately reported. The vertical axis is logarithmic. The plot is an exact finite cost comparison, not a timing ratio or evidence of a broad neural-learning gain.

18.1 A resource improvement is not a universal speedup

In the interpreted version, each prefix evaluates five AST nodes. The checked lowering uses one integer comparison instead. Relative to a baseline that already performs the complete-root capacity test, a conservative extra-work account charges every new prefix comparison without crediting any removed end-of-scan check. At 64 cases this adds 14,209 comparisons while avoiding 8,287 model rows. At two cases no row is avoided and the new comparisons are pure overhead.

If a model-row read has declared price $w > 0$ and a new comparison has price one, the additional per-use saving under this conservative two-unit account is

$$\Delta(w) = w\,(R_{\text{full}} - R_{\text{stream}}) - N_{\text{compare}}.$$

For the 64-case row it is positive when $w > 14209/8287$. At $w = 1$ it is negative. The price is a modeling choice, not a measured statement about Python or a physical machine. Interpreted AST visits, installed code, grammar construction, labels, and development search remain additional recorded coordinates. No weighted sum is presented as a universal cost of intelligence.

If development has total charged cost D in a chosen common accounting unit and the same future use has saving $\Delta(w) > 0$, N repetitions repay that cost only when $N\Delta(w) > D$. This is the amortization argument already used earlier, applied to the learned compiler rule. It does not authorize ignoring development because deployment might eventually repeat often.

18.2 Transport and negative controls

The formula contains counts and a remaining-depth capacity rather than cell names or query identifiers. Renaming cases and flipping binary report meanings preserve the capacity predicate. A complete search

uses the model rows to determine its actual branches, and its witness is checked anew. Reordering queries can change which equally optimal controller is encountered first. Reordering rows can change how early an impossible prefix becomes visible. Correctness is preserved under the certified scope, but the number of saved reads need not be invariant.

A four-case equality-only library provides another boundary. It has no controller at the information lower depth, so a larger requested horizon invokes the general compiler. The new method can spend preliminary work before that fallback. On a changed language with a universal repair, the semantic guard rejects the pruning scope and the general optimum is recovered. These are specified behaviors, not successes relabeled after observing the results.

18.3 The precise recursive claim

The first construction improved a diagnostic controller. The second selected a better compiler from named policies. The present construction synthesizes and installs an internal decision rule that changes how future controllers are discovered. It also records a representational obstruction within that compiler and a permitted feature refinement that removes it. This is a bounded inductive program-synthesis step, not neural training or a renamed cache of solved episodes.

The construction does not demonstrate that the grammar enumerator becomes better at inventing later grammars, that the learner discovers its own theorem checker, or that its task family becomes generally intelligent. The primitive operations, teacher, feature registry, scope proof, and search skeleton are substantial prior structure. The gain nevertheless occurs at the claimed location: future searches omit work because of a reusable expression composed from evidence, and the conditions under which that omission remains justified are checked rather than presumed.

Part V

Improving how discovery rules are acquired

19 The next retained object is a search representation

Part IV changes a decision inside the diagnostic compiler. Its counterexample-guided learner still traverses a fixed list of arithmetic candidates. A further change is possible without pretending that the learner invents its own computational foundations: it can acquire a reusable representation of that candidate space, then use the representation to reduce the work of synthesizing later rules.

The new object is neither a completed diagnostic tree nor the target expression $\max(a, b) > C$. It is a library of checked equations and an ordered map from grammar expressions to earlier equivalent representatives. Learning the library does not require the labels of a later synthesis task. Once the representation has been constructed and validated, subsequent tasks still begin with empty task-specific counterexample sets.

This direction has direct precedents. Equality saturation represents equivalent expressions and supports program optimization and synthesis [21]. Ruler automatically infers rewrite rules and uses equality saturation within that inference process [22]. The present implementation is intentionally smaller. It uses a typed arithmetic language, finite proposals, a local derivation checker, and ordered enumeration rather than an e-graph. It does not claim a new rewrite-inference paradigm or superiority to those systems. Its purpose is to make one additional level of the improvement process executable and to connect the acquired search representation to the paper's unchanged semantic and resource contracts.

19.1 Four objects and two meanings of success

Let $\mathcal{G} = (p_0, \ldots, p_{N-1})$ be a finite ordered grammar of pure, total expressions. A target task supplies contexts $z \in D$ and desired values $f(z)$. An ordinary learner uses counterexamples to find the first grammar expression that agrees with all of D. A separate equational learner observes expressions on a development probe set and proposes equations. A checker admits only equations supported by a derivation in the declared arithmetic theory.

There are therefore two distinct successes. A task candidate may agree with a complete finite target domain while remaining unjustified outside it. An equational certificate may establish agreement for every integer substitution, even though it says nothing about whether the resulting program solves a particular task. The latter can safely reduce the former search, but cannot manufacture its missing target evidence.

20 Learning equations without confusing probes with proofs

The reference language uses exact integers, the total operations +, min, max, and the comparisons $<$, $\leq$, $>$, $\geq$, and $=$. A finite schema language contains 135 terms built from variables x, y, small integer literals, Boolean literals, and at most one binary operation. The proposer evaluates those schemas on the nine contexts $(x, y) \in \{1, 2, 3\}^2$.

Two schemas with the same probe vector become a proposed equation, not an accepted rule. This produces 1,068 proposals. The supplied proof procedure proves 336 using integer arithmetic identities. The other 732 are not admitted. Failure of that limited procedure is not a theorem that an equation is false.

The derivation interface has six elementary rule kinds: commutativity where valid, addition of zero, idempotence of minimum and maximum, reversal of an order comparison, reflexive comparison, and exact evaluation of literal constants. It also checks replacement at an explicit subtree position. A proof of an

equation contains two derivations ending in the same expression. A separate replayer checks every position, side condition, and endpoint. This is a project-local equational checker, not a proof-assistant kernel or a general arithmetic decision procedure.

Counterexample 20.1 (A probe-equivalent rule that fails on a new input)**.** On the development inputs $x \in \{1, 2, 3\}$, the terms $\max(x, 1)$ and x agree. At $x = 0$ they differ. The proposed equation is not admitted by the proof checker. By contrast, $\max(x, 0) = x$ holds on all nonnegative integers but not on all integers. A checker using the stronger integer-wide contract may fail to admit a rule that would be valid under a narrower natural-number contract. Rejection, counterexample, and scope restriction remain different outcomes.

20.1 What the learner contributes and what it is given

The proposer generates the equations from evaluations rather than receiving the final equation list. The library is serialized before the later task targets are evaluated. The primitive syntax, probe schedule, arithmetic axioms, proof search, and normalization engine remain supplied. Their presence does not negate the acquired representation. It does prevent describing the construction as discovery of integer arithmetic or autonomous improvement of the equational learner itself.

The complete accepted library occupies 69,518 bytes in its stated compact JSON serialization. This is a serialization measurement, not an estimate of an optimal theory description. It contains redundant equations. A supplied full normalizer is retained as a stronger comparison, so the results do not depend on comparing only against deliberately repeated syntax.

21 A certified quotient preserves the investigation's evidence

Write $[\![p]\!](z)$ for an expression's value in context z. Let $n(p)$ be a deterministic representation produced by finitely many checked substitutions of valid equations. We only require

$$n(p) = n(q) \quad \Longrightarrow \quad [\![p]\!](z) = [\![q]\!](z) \quad \text{for every admitted } z.$$

The reverse implication is unnecessary. Distinct normal forms may denote the same function. Missing such an equivalence wastes an optimization opportunity but does not remove a valid candidate.

Theorem 21.1 (Sound ordered quotienting from checked equations)**.** *Assume the expression language is pure and total, every admitted schema equation is valid for all well-typed substitutions in the common semantic domain, and each normalization terminates after valid contextual replacements. For each p_i, keep its earliest grammar representative*

$$\rho(i) = \min\{j : n(p_j) = n(p_i)\}.$$

Then $\rho(i) \le i$, p_i and $p_{\rho(i)}$ agree on every admitted context, and every pointwise specification realizable in $\mathcal{G}$ is realizable in its retained representatives. No confluence or complete semantic-equivalence decision procedure is required.

Proof. A schema identity remains true under a well-typed substitution. Purity and totality make replacement inside any expression context preserve its value. Induction over a finite derivation gives equality between an expression and its normalized result. Two expressions with the same result therefore have equal values by transitivity. The defining set for $\rho(i)$ contains i, giving existence and $\rho(i) \le i$. Replacing a satisfying expression by its representative preserves every pointwise requirement. The argument uses the soundness of each actual derivation, not uniqueness or completeness of normalization. □

In the reference implementation, a successful rewrite must decrease a size-and-structural-order measure. Rewriting introduces no free variables or literals outside the finite input-and-library vocabulary, and never

increases size. Only finitely many terms lie below the initial size over that vocabulary, so there is no infinite decreasing rewrite sequence. The index keeps the original earliest AST, not the normalized AST. This preserves the reference learner's output convention and its program interpreter cost on every teacher proposal.

Theorem 21.2 (Preservation of ordered counterexample-guided acquisition)**.** *Consider a learner that scans a finite ordered grammar, retains every counterexample, tests candidates against those examples in a fixed order, and submits only consistent candidates to a deterministic teacher. The teacher returns the first disagreement in a fixed complete finite context order, or accepts. Under Theorem 21.1, scanning only the earliest representatives produces exactly the same submitted ASTs, counterexamples, and final accepted AST or grammar-exhaustion outcome. With the original AST interpreter retained, it performs no more retained-example evaluations or interpreter-node visits. Its teacher calls and teacher-evaluation work are unchanged. Quotient construction, storage, and dispatch costs are separate.*

Proof. Consider a removed expression p_i and its earlier representative p_j. If the reference learner accepted p_j, it never reaches p_i. Otherwise a retained example refuted p_j, either before its teacher call or as a result of that call. That example is still retained when p_i is reached. Semantic equality means the same example refutes p_i, so the reference learner never submits p_i to the teacher. Deleting that unsuccessful consistency check leaves the sequence of proposals unchanged. Induction over the ordered scan gives the same candidate AST at every teacher call. The deterministic teacher then supplies the same first counterexample or acceptance. Removed candidates contribute only nonnegative evaluation work, while submitted expressions and their evaluations remain identical. If the finite grammar is exhausted, both learners have rejected the same possible behaviors. □

The theorem concerns an extensional target and a fixed candidate order. A teacher that accepts based on source identity, or a learner that forgets old counterexamples, does not satisfy it. Nor does equality of return values make two candidates equal in execution cost, memory use, or causal side effects. To preserve a different resource-constrained objective, one must keep appropriate nondominated representatives or include those observables in the equivalence contract. The reference deliberately preserves the first original AST rather than selecting an uncharged cheaper replacement.

21.1 Cold reacquisition and additional finite task families

The 3,960-expression grammar reduces to 1,362 retained representatives. Reacquiring the original capacity predicate starts with no task counterexamples and no installed copy of the answer. The original scan considers 190 candidates, while the quotient scan considers 76. Both submit the same four candidate ASTs, receive the same three counterexamples, and return the identical five-node expression. Retained-example checks fall from 282 to 112. Total interpreter-node visits in that task fall from 1,898 to 1,376. Teacher comparisons remain 210.

The frozen quotient is next used for nine finite targets involving sums, extrema, comparisons, and equality, followed by a parity target that is not expressible on its complete declared domain. These are changed target functions in the same arithmetic language, not a statistical sample of unrelated research domains. The first nine use all 256 contexts in $\{0, 1, 3, 5\}^4$. The parity control uses all contexts in $\{0, 1, 2, 3\}^4$, so more than one even value is present.

Finite target	Original interpreter visits	Quotient interpreter visits
Sum exceeds capacity	4,611	3,517
Minimum below threshold	7,762	5,553
Maximum equals a value	26,407	10,116
Sum equals a value	29,357	12,311
Parity outside the grammar	54,045	18,728
Complete ten-target batch	196,384	85,162

Every submitted candidate and counterexample trace agrees exactly between the two learners. The parity target produces grammar exhaustion in both. A bounded grammar's failure is not a proof that the target function is uncomputable.

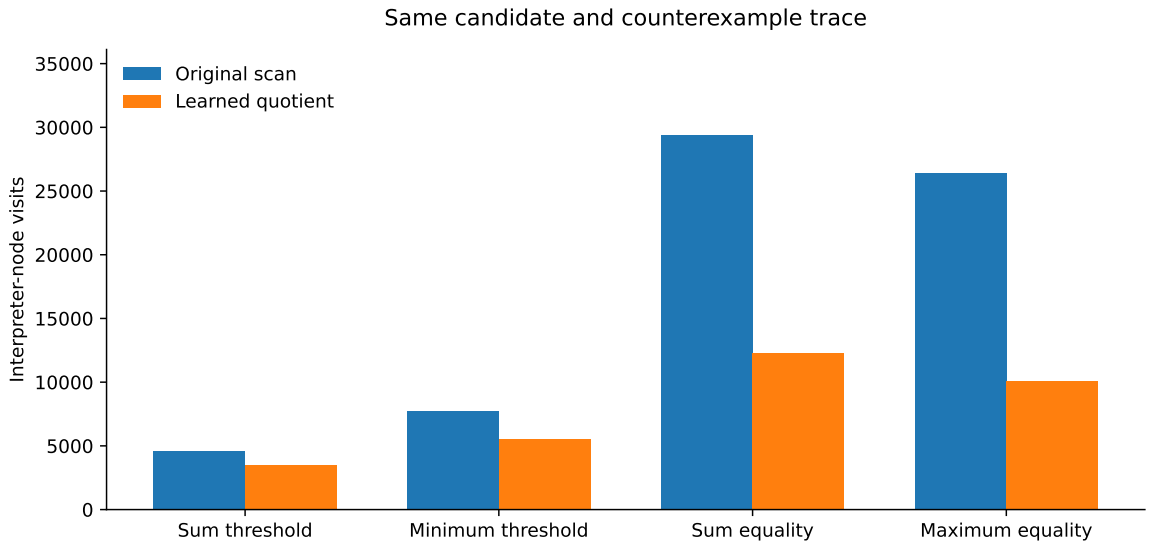


Figure 9: Repeated task-synthesis evaluations are reduced while the teacher trace remains fixed. Values are exact interpreter-node counts in the stated finite cases. Equation discovery, checking, quotient construction, and stored library costs are not included in these bars and are reported separately.

21.2 The stronger normalizer and the cost of preparing the shortcut

A supplied full arithmetic normalizer retains 1,352 representatives, ten fewer than the learned library. It therefore removes at least as much semantic redundancy under this implementation's normal-form relation. The learned library is not claimed to outperform that stronger prior. This control identifies the result as automatic acquisition of a useful search representation from the supplied reasoning substrate, not a new best-in-class synthesis algorithm.

The library-development ledger records 1,215 probe evaluations, 3,519 probe interpreter-node visits, 1,068 proposed equations, and 584 successful checked axiom steps. Proof search also attempts proposals it cannot establish. Applying the admitted library to the whole grammar costs 19,440 normalizer-node visits and 518,140 schema-match attempts, with 4,798 successful replacements. Grammar generation, proof search, bookkeeping, and serialization remain distinct charged phases. These event counts do not encode every Python instruction.

Let C_{prepare} be the complete preparation cost in a declared common resource model, including failed proposals and certificate checking, and let d_t be the subsequent saving on task t after including dispatch. Reuse is beneficial in that model exactly when

$$\sum_{t=1}^{T} d_t > C_{\text{prepare}}.$$

This is the earlier amortization identity applied to acquisition itself. The reference does not convert unlike counters into a universal wall-clock claim. Even one setup coordinate, 518,140 schema matches, exceeds the 522 interpreter-node visits saved in one cold capacity-rule run. A short workload or an expensive matcher can make the full modification undesirable.

21.3 A cached proof representation has a scope

The quotient records the grammar's contents, order, and declared integer semantics. Loading a cache requires checking those bindings and reconstructing its representative map from the admitted proofs. Adding syntax or changing order requires rebuilding the index. Extending the grammar can still reuse an equation library when every substituted operation is pure, total, and retains the relevant types and meanings.

Counterexample 21.3 (The same operator spelling with changed semantics)**.** The identity $x+0 = x$ is valid for ordinary addition. Suppose a revised interpreter uses the same printed symbol but executes $a \oplus b = a + b + 1$. Then $x \oplus 0 \neq x$. A cache certified for the former interpreter cannot be imported merely because its expression strings still parse. A semantic identifier helps detect ordinary version mismatch, but an identifier falsely attached to a changed implementation is not itself proof of faithful execution.

22 A repair-relative certificate that can be updated cheaply

The singleton pruning result remains valid within its original scope. To move beyond that scope, the compiler must account for what a repair can accomplish, not merely count hidden cases. An exact minimum repair cover can be expensive. A feasible lower-bound witness is often enough to exclude an impossible branch.

Let X be a finite immutable state set and let $S_r \subseteq X$ be the success set of repair r. Choose nonnegative rational weights w_x satisfying

$$\sum_{x \in S_r} w_x \leq 1 \quad \text{for every permitted repair } r. \tag{9}$$

This is a feasible dual weight assignment for unweighted set cover. We use only its elementary weak-duality property, not an assumption that the best assignment has been found [23].

Proposition 22.1 (Repair-relative weight and prefix obstruction)**.** *Suppose a total immutable deterministic binary diagnostic tree succeeds on nonempty support B and has L reachable leaves and maximum query depth h. Under* (9)*,*

$$\sum_{x \in B} w_x \leq L \leq 2^h.$$

At a candidate query root with $h \geq 1$ remaining query levels, if the cases already read into either outcome branch have weight greater than 2^{h-1}, no assignment of its unread reports can make that branch repairable within the remaining depth. This implication is valid even when one repair succeeds in several states.

Proof. Immutable deterministic execution partitions B into its nonempty reachable leaf supports. Each leaf support is contained in the success set of its chosen repair. Nonnegativity and (9) give weight at most one for

every leaf, so summing gives the first inequality. A binary tree of depth at most h has at most 2^h leaves, by induction on h. At a candidate root, each child has at most $h - 1$ query levels. Unread cases can only add nonnegative weight to an outcome branch, so a prefix already exceeding its leaf capacity cannot be rescued by any completion. □

Theorem 22.2 (Streaming search with a checked repair-weight witness)**.** *Fix total immutable binary unit-cost experiments, unit-cost repairs with arbitrary fixed success sets, an ordered initial support, a repair order, an experiment order, a finite query horizon, and a weight vector satisfying* (9)*. Consider depth-first search that first checks for a common successful repair and otherwise explores informative query roots in that order, rejecting child supports whose weight exceeds their remaining binary leaf capacity. Replacing each complete root scan by a scan that stops on the first overweight prefix preserves the exact first returned tree or finite-horizon infeasibility result. It never increases model-record reads during search. These statements do not establish a Pareto-optimal storage frontier or free weight arithmetic.*

Proof. Proposition 22.1 shows that an overweight support has no successful tree at its depth. A root stopped on an overweight prefix would still be rejected after a complete scan because all further contributions are nonnegative. Every root not stopped early is partitioned identically, with the same row and outcome orders. Induction over depth and the ordered root scan therefore yields identical first successful children and identical returned trees. A query with one reachable outcome leaves the immutable support unchanged, so deleting such a query never destroys the existence of a depth-bounded successful tree. The search over the remaining roots and repairs is finite and complete for existence. Early stopping deletes only unread suffixes of roots the complete scan also rejects, proving the read comparison. The final witness checker and weight-scope checker remain required and are charged separately. □

22.1 Constructing a feasible witness without solving the whole cover problem

A simple supplied constructor processes states in order. It raises the current state's weight by the smallest remaining slack among repairs containing that state, and updates those repair loads. An uncovered state may receive a finite weight of one. Every step preserves nonnegativity and all repair constraints. The result is checked independently by resumming every repair's weight. It can be a weak certificate. In particular, the all-zero assignment is always feasible and excludes nothing.

For example, on three cases with success sets $\{1, 2\}$, $\{2, 3\}$, and $\{1, 3\}$, the ordered greedy constructor returns total weight one. The feasible fractional assignment $w_1 = w_2 = w_3 = 1/2$ has total $3/2$ and proves that at least two leaves are required. Thus the supplied constructor does not optimize the bound. Both assignments are valid and neither creates a missing experiment.

The construction is not itself learned in this revision. It is a semantic extension showing how the compiler can continue when successful repair no longer means exact state identification. A learner may later propose a stronger weight assignment, but the same inequalities would still govern admission. The work of producing and checking those weights cannot be hidden inside one cheap predicate call.

22.2 Paired repairs change the task, not just its notation

Consider $2g$ possible hidden cases divided into g pairs, where g is a power of two. One repair succeeds on both members of each pair. The query language contains equality questions and bits of the pair index. The greedy weight constructor gives weight one to the first member of each pair and zero to the second, with total weight g. A depth $\log_2 g$ tree is both necessary and sufficient, so the diagnostic-plus-repair work is $1 + \log_2 g$ and a full binary witness uses $2g - 1$ records.

Hidden cases	Repair groups	Full scan reads	Streaming reads	Work / records
4	2	20	18	2 / 3
8	4	152	120	3 / 7
16	8	864	608	4 / 15
32	16	4,416	2,864	5 / 31

These columns count search reads only. For the 32-case problem, each method additionally pays 1,152 scope-table reads and 256 witness-checking reads, giving 5,824 versus 4,272 total model reads. Weight construction, repair-success-set checks, rational additions, and comparisons have their own ledger. The streaming search makes 2,941 capacity comparisons rather than 1,073. Both methods return the identical literal tree.

This is not a generalization score obtained by making the task easier. The comparison is between two discovery procedures on the same paired-repair task. The difference between that task and the earlier singleton task explains why the earlier cardinality certificate would be invalid here.

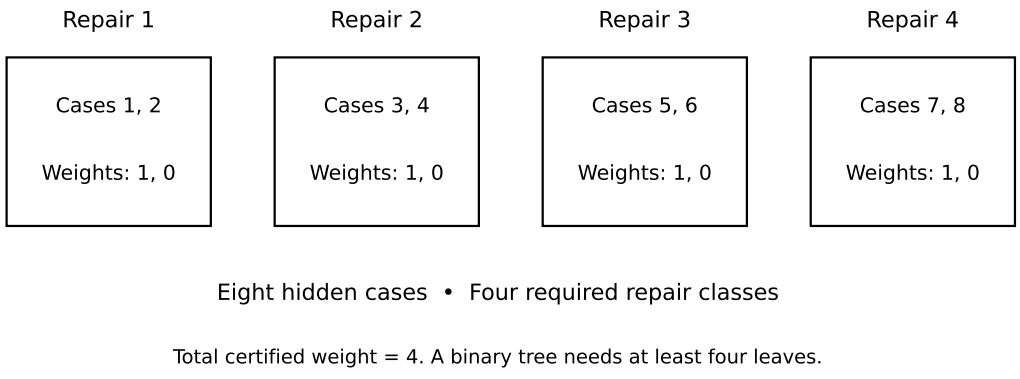


Figure 10: Eight hidden cases need only four repair distinctions when each repair succeeds on a pair. The displayed weights certify a four-leaf lower bound. They do not claim that every feasible weight assignment is optimal or that every query library can realize the bound.

Counterexample 22.3 (A newly admitted repair invalidates the old lower-bound certificate)**.** In an eight-case paired model, the original weights have total four. Add a repair that succeeds on every case. Its weight load is four, violating (9). The old certificate must be rejected. Recomputing a feasible assignment gives total one, and the compiler returns the universal repair with no query, work one, and one stored record. The success of that repair does not retrospectively validate the obsolete weight assignment.

A capacity-feasible prefix still does not guarantee that available experiments can distinguish the required repair classes. The implementation retains a model with only a constant report: its cover weights fit a nominal depth bound, but no successful tree exists. A lower-bound certificate is a reason to reject some candidates, not a generator of a missing experiment.

23 One chain from acquired equations to an improved investigator

The levels can now be connected without substituting one kind of evidence for another. The equational learner produces a checked library. The library prepares an ordered search representation. That representation

reduces the repeated evaluation work of acquiring the earlier pruning rule and other finite targets. The rule's separate scope certificate then permits its use in the diagnostic compiler. Each stage retains the predecessor's semantic comparison.

Proposition 23.1 (Composition of acquisition and compiler preservation)**.** *Assume the ordered quotient satisfies Theorem 21.2, the resulting task expression passes the unchanged compiler-scope checker, and the original and optimized compilers satisfy the common-domain premises of Theorem 16.3. If all stages complete, quotient-guided rule acquisition followed by optimized compilation returns the same learned expression and the same diagnostic controller as ordinary rule acquisition followed by the reference ordered compilation. In a common additive resource model, writing K for extra preparation cost, A, A' for the two acquisition costs, and D, D' for the two compilation costs, the composite saves work exactly when*

$$(A - A') + (D - D') > K.$$

The semantic conclusion does not imply the cost inequality, and a budget interruption does not imply infeasibility.

Proof. Theorem 21.2 gives the same acquired AST and teacher transcript. Passing the unchanged scope checker places that expression in the stated compiler-preservation theorem, which gives the same returned controller. The reference total is $A + D$ and the modified total is $K + A' + D'$. Subtracting gives the displayed necessary and sufficient cost condition. Neither semantic theorem bounds K or assigns exchange rates among distinct resource units. An interrupted finite search has not exhausted its candidates and therefore cannot use completeness to infer infeasibility. □

23.1 The recursive achievement and its next boundary

The retained change is now used to produce future discovery rules more economically in a specific evaluation coordinate. It is not merely a better diagnostic result. The learner's search representation has changed, while the pointwise target, oracle transcript, and final generated program remain fixed. This is a bounded meta-improvement with an explicit implementation and cost boundary.

It does not require that a useful self-improver invent every foundational component. It does require us to locate those foundations honestly. Here the schema proposer, primitive axioms, checking engine, and ordered-quotient construction are given. The acquired equation library and grammar quotient are generated. A supplied stronger normalizer can outperform the learned library, and a short workload can fail to repay its preparation cost. Extending the result to learned physical semantics, larger generated languages, uncertain arithmetic, or repeated improvements of the library learner itself requires additional work.

23.2 The boundary of the first edition

The finite construction now covers repair discovery, compiler revision, synthesized pruning, acquired search representations, and repair-relative scope extension. Part VI consolidates their dependency and resource contracts. A first edition need not claim that general intelligence follows or that the outermost learner has become unboundedly self-improving. Those are distinct research questions. The completed scope is the explicit finite chain and the evidence required to interpret it.

Part VI

Keeping a revision's guarantees applicable

24 A certificate belongs to a changing realization

A proof can remain mathematically correct while ceasing to justify an installed shortcut. The equation library was proved for pure, total integer expressions. The ordered quotient was constructed for one grammar order. A repair-weight certificate was checked against particular success sets. A resource comparison assigned costs to specified events. None of those objects is licensed to retain its old interpretation after every component bearing the same name has changed.

This is the final integration problem for the first edition. The earlier sections provide distinct certificates for the outputs of different stages. A continuing system must also identify which of those certificates remain applicable after a revision. The appropriate response is not to repeat every discovery from scratch, nor to regard an old passing record as timeless. It is to retain the actual dependencies, reject unsupported reuse, and recheck the affected obligations before relying on them again.

24.1 A declared dependency graph

Let an artifact state be $a = (a_1, \dots, a_d)$. Its components may include operation semantics, a grammar and its order, an equation library, experiment tables, repair-success sets, a returned controller, and a resource model. A finite directed acyclic graph has one vertex v for each claimed guarantee. Its incoming vertices $\mathrm{pa}(v)$ identify guarantees used as premises. Its direct dependency set $D_v \subseteq \{1, \dots, d\}$ identifies the artifact components used directly. Each vertex carries a witness w_v and a local verification rule.

The dependency declaration must be *semantically complete*: the meaning of the claim and the local rule is determined by the components in D_v and the stated parent claims. Hidden environment assumptions cannot be left outside this declaration merely because the implementation does not know how to check them. They can instead be recorded as explicit external premises. Verifier soundness, the interpretation of primitive arithmetic, and the faithfulness of the supplied model to an external process are not established by naming them in a registry.

Suppose a revision replaces a by a'. Mark a vertex directly stale whenever one of its declared component values differs. Let Δ be the union of those vertices and all their descendants in the dependency graph. Equivalently, in a topological order, mark v whenever

$$a|_{D_v} \neq a'|_{D_v} \quad \text{or} \quad \mathrm{pa}(v) \cap \Delta \neq \varnothing.$$

Here equality refers to the declared artifact value, not merely an unchanged version label. This procedure is conservative. A changed premise may leave the conclusion true, but the old evidence is not reused without a further argument.

Theorem 24.1 (Dependency-complete reuse across a revision)**.** *Assume a finite acyclic dependency graph, semantically complete direct dependencies, and sound local verification rules. Suppose the old witnesses establish every old claim from the declared external premises. After a revision, the old claims outside Δ remain established by their old witnesses. If every vertex in Δ is re-established in topological order using the revised artifact values and already established parent claims, all revised claims hold. If an affected vertex is not re-established, no dependent claim is justified by this reuse rule alone. These statements are conditional on the external premises applicable to the revised realization.*

Proof. A vertex outside Δ has unchanged direct inputs and no parent in Δ. Inducting through a topological order, its parents remain established with the same meanings, so dependency completeness leaves its old verification argument applicable. For an affected vertex, assume its parents have already been established in the revised realization. Soundness of the newly checked local rule then establishes its revised claim. A second induction yields the result when every affected vertex is re-established. Without that step, the rule has an unsupported premise and supplies no conclusion along that dependency. This is a limitation of the reuse argument, not a proof that the dependent claim is false. □

The argument is elementary induction over declared dependencies. Its role here is to connect the exact preservation results to their continued use. It does not compute the complete dependency graph from arbitrary source code. A dependency cycle requires an additional justified rule, such as an independently proved invariant. The finite replayer in this edition rejects an unsupported cycle rather than interpreting mutual assertions as evidence.

Counterexample 24.2 (An unchanged name hides a changed premise)**.** A stored claim asserts that two binary values u and v agree. Initially both are zero. A registry records only u as a dependency and continues to validate that its stored value is zero. Change v to one without changing u. The declared check still passes, but the claimed equality is false. Recording and replaying the incomplete registry cannot repair its omitted premise. Dependency completeness is a mathematical assumption, not a consequence of serialization or a checksum.

24.2 The scopes that actually have to be carried

The implemented constructions require different bindings. An equation witness includes the supported arithmetic semantics and the local derivation rules. A quotient includes the equation library and the full ordered grammar. Exact candidate-trace preservation additionally requires the extensional teacher, fixed context order, and retained counterexamples. The compiler rule has its own integer-prefix and singleton-repair scope. Weighted pruning instead checks every repair-success set and the nonnegative weights. A returned tree must be checked against the actual experiment effects, observations, repair meanings, and horizon.

Cost and evidence claims add other dependencies. A reduction in model reads is not a price-independent reduction in total work. An audit comparing a frozen candidate cannot be carried unchanged to a new candidate merely because the score field has the same name. Statistical independence and commitment order may depend on events outside the program's observation boundary. The manifest must state whether those conditions are guaranteed by the controlled construction, supplied by a trusted interface, or not established.

The reference certificate manager therefore has several outcomes. A claim is *valid* when its declared inputs match and its local evidence is replayed successfully. It is *stale* when a direct input changed, *blocked* when a required parent is unavailable or different, *rejected* when its local record fails, and *missing* when no witness was supplied. A timeout is not successful replay. Nor is a valid resource assessment necessarily favorable: the assessment may correctly report a negative saving.

The reference manager uses exact canonical snapshots for direct input comparisons and content identifiers for linking parent evidence. Those identifiers aid integrity checks. They do not establish who produced a witness or whether an external assumption is true. The manager and dependency registry are supplied components. This edition does not claim that the learner derived its own trusted verification base.

24.3 A repair-language change through the complete record

The first-edition demonstration checks the retained equation library, its ordered quotient, the capacity rule's supported form, a repair-weight witness, a returned weighted diagnostic tree, and a conditional resource assessment. The equation and diagnostic branches remain distinct. A valid capacity expression is not silently substituted for the weighted compiler's separate certificate.

In the 32-case paired-repair example, the accepted weight total is sixteen and the returned controller has work five and 31 records. Add a repair that succeeds in every case. The old weight certificate becomes stale, and direct rechecking confirms that it fails the new repair constraint. Its dependent controller and cost claims cannot rely on it through the old chain. The arithmetic equations and ordered grammar quotient remain applicable because their declared premises have not changed.

Rebuilding the weight witness gives total one. The compiler then returns a universal repair with work one and one record. The new controller is checked against the new repair language. The old cost comparison is deliberately not reissued: a claim about the work of producing the former controller does not become a claim about this different search simply because both controllers are valid. The appropriate repair of that claim is a new comparison under a stated workload.

Changing only the resource prices has the opposite pattern. The semantic certificates remain applicable, while the cost conclusion needs to be recomputed. Changing the grammar order invalidates the ordered quotient and its recorded acquisition lineage, not the independently proved arithmetic identities. These cases illustrate selective invalidation without equating it with a complete theory of arbitrary self-modification.

25 One resource ledger for the complete improvement

The manuscript's numerical comparisons concern several different units. A model-record read, an interpreter-node visit, an arithmetic comparison, a stored tree record, and a serialized byte are not interchangeable. The purpose of a complete ledger is to preserve those distinctions until a specific execution or economic model justifies a conversion.

Phase	Recorded object or event	What the count does not supply
Model and language acquisition	Supplied experiment tables, grammar, feature registry, primitive axioms	An executed calibration of an unknown physical process, or free discovery of those foundations.
Equation acquisition	1,068 proposals, 336 admitted equations, 584 successful checked axiom steps	Proof completeness, a generally optimal equation set, or uncharged failed attempts.
Search representation	3,960 grammar expressions, 1,362 representatives, 518,140 schema-match attempts	Free preparation, hardware-time units, or superiority to the 1,352-representative supplied normalizer.
Task-rule acquisition	1,898 versus 1,376 interpreter-node visits for the capacity rule	All preparation costs, or a different teacher transcript. The retained four proposals agree.
Compiler execution	Query-model reads, weight arithmetic, capacity comparisons, scope checks, witness replay	A universal conversion of avoided reads into elapsed time.
Installed information	Equation-library serialization of 69,518 bytes, original programs, quotient records	Minimal description length, physical peak memory, or a byte-free interpreter.
Online repair	Experiment and repair calls, plus complete stored tree records	Only the visited path's storage, or the cost of discovering the controller.
Revision management	Bound snapshots, witness checks, parent comparisons, and replacement evidence	Free invalidation detection or a trusted explanation of undeclared external assumptions.

The exact per-phase records remain in the source package. The table does not add unlike events into one total. In particular, retained storage is normally a capacity coordinate. Treating it as an additive expense

requires a stated storage-duration or installation convention. Peak memory combines by its own rule, not by automatically summing all sizes ever observed.

25.1 Which gains survive a change of resource prices?

Fix an additive ledger with $d \geq 1$ named event types. Let $K \in \mathbb{R}^d_{\geq 0}$ be extra setup, let $s_i \in \mathbb{R}^d$ be the old-minus-new per-use event vector for workload type i, and let n_i be its nonnegative integer multiplicity. The net event saving is

$$\Delta(n) = \sum_i n_i s_i - K. \tag{10}$$

A price vector $w \in \mathbb{R}^d_{\geq 0}$ specifies conversions into one declared additive resource measure. Its priced saving is $w \cdot \Delta(n)$. The prices are modeling assumptions, not learned hardware laws inferred from the event counts.

Proposition 25.1 (Resource dominance and declared price uncertainty)**.** *For a fixed ledger and workload, $w \cdot \Delta(n) \geq 0$ for every nonnegative price vector w if and only if $\Delta(n)$ is componentwise nonnegative. If admissible prices form the convex hull of a nonempty finite set W_0, then*

$$\min_{w \in \mathrm{conv}(W_0)} w \cdot \Delta(n) = \min_{w \in W_0} w \cdot \Delta(n).$$

Thus strict improvement throughout that price set holds exactly when every listed vertex gives a strictly positive margin. The claims concern the same workflow, workload, and complete additive ledger.

Proof. Componentwise nonnegativity makes every nonnegative weighted sum nonnegative. If coordinate j is negative, the jth coordinate vector is a nonnegative price vector giving a negative sum, proving the converse. Every point of the specified convex hull is a convex combination of its listed points. Linearity makes its priced saving the same combination of their savings, which cannot be smaller than their minimum. A listed minimizer attains that minimum, proving the equality and the strict criterion. □

This is ordinary linear accounting, not a new optimization principle. It prevents a system from asserting a price-independent gain after improving only one coordinate. The deterministic workload count must also be separated from an expectation over future tasks. Proposition 28.1 addresses the latter under its stated assumptions. Changing the task mixture can change both the amount and the sign of an improvement.

Counterexample 25.2 (One valid controller, opposite resource conclusions)**.** For the 32-case weighted-search comparison, the old-minus-new search-event vector is

$$(1552,\ 1552,\ -1868)$$

in model-record reads, rational additions, and capacity comparisons. Scope checking and final witness checking are equal for both procedures and cancel in this particular difference. Prices $(1, 1, 1)$ give a saving of 1,236 units. Prices $(1, 0, 1)$ give a loss of 316 units, and prices $(0, 0, 1)$ give a loss of 1,868 units. The identical returned controller remains valid in every comparison. None of the price choices was measured as a universal property of a processor.

The zero setup in that isolated execution comparison is not a statement that preparing the whole improvement was free. For an explicitly hypothetical additional setup charge of 2,000 comparison units and prices $(1, 1, 1)$, one use loses 764 units while two uses save 472. The purpose of this small calculation is to expose the dependence on the declared ledger. The actual equation-discovery, proof-search, quotient-construction, and revision-management records must be included when assessing the entire first-edition workflow.

Part VII

Evidence and recursive continuation

26 Evidence that survives candidate selection

A self-improvement procedure has an unusual incentive to find weaknesses in its own measurements. It searches repeatedly, retains favorable results, and changes the way later candidates are generated. A validation number can therefore become part of the training information even when it is never called a training label. This is a central issue in adaptive data analysis [15].

The correct distinction is between development evidence used to generate a candidate, selection evidence used to choose among candidates, and final evidence used to assess the selected procedure. A private test repeatedly consulted by an outer selector is still selection information for that outer process. Keeping it hidden from the inner solver does not make it independent of the entire lineage.

A practical record should identify what was frozen before each assessment, which outputs were exposed, and which claim the assessment supports. When a procedure changes a checker, both the checker change and the candidate it would approve cannot receive their only justification from the new checker's own approval. That restriction does not forbid improving evaluators. It prevents the target of validation from silently changing the validation standard in the same step.

Proposition 26.1 (Conditional error budgets across an adaptive lineage)**.** *Let $\mathcal{H}_{t-1}$ be the information available before attempt t. Suppose the event A_t that attempt t accepts a false improvement claim satisfies*

$$\Pr(A_t \mid \mathcal{H}_{t-1}) \le \delta_t \quad \textit{almost surely},$$

for deterministic $\delta_t \ge 0$ with $\sum_{t\ge1} \delta_t \le \delta$. Then

$$\Pr\left(\exists t \ge 1 : A_t\right) \le \delta.$$

The candidate-generating process and the time at which attempts stop may depend on the earlier history.

Proof. Taking expectations gives $\Pr(A_t) \le \delta_t$. The union bound gives the result for any finite number of attempts. The infinite statement follows from continuity of probability for the increasing finite unions. A history-dependent stop can only reduce the set of attempts whose acceptance events occur. □

No independence between attempts is needed. The conditional premise is stronger than a marginal guarantee for a candidate considered before adaptive selection. It must be justified by the actual sampling or verification mechanism. A budget such as $\delta_t = \delta/[t(t+1)]$ is summable, but writing that schedule does not make an invalid test conditionally valid.

26.1 One sufficient fresh-audit construction

Freeze a candidate and its comparator using the prior history. Draw n_t fresh evaluation cases conditionally independently from the declared distribution and record paired score differences $Y_i \in [-1, 1]$. Let Δ_t be their conditional mean and $\widehat{\Delta}_t$ their average. The bounded-variable concentration inequality gives

$$\Pr\left(\widehat{\Delta}_t - \Delta_t \ge r \mid \mathcal{H}_{t-1}\right) \le \exp(-n_t r^2/2).$$

Consequently, accepting a claim $\Delta_t \geq \gamma$ only when

$$\widehat{\Delta}_t - \sqrt{2\log(1/\delta_t)/n_t} \geq \gamma \tag{11}$$

meets the preceding conditional error requirement for that claim. This is a fixed-sample sufficient construction. The candidate and n_t may depend on past data, but n_t is fixed before this audit is examined. Inspecting the same sample repeatedly and stopping at a favorable value requires a time-uniform method or a separately justified spending rule. Time-uniform concentration is an established subject [16].

Formula (11) can be conservative and expensive. It is included to make the premise operational, not as a claim that this test is optimal. If evaluator randomness, shared tasks, or dependence between runs changes the sampling unit, the analysis must change with it. Thousands of messages from one trained lineage do not automatically become thousands of independent improvement trials.

Counterexample 26.2 (Reused evidence can reward memorization)**.** A task has N equally likely names and an externally fixed binary answer for each. After the answers on a development subset become available, a candidate stores them and gives the wrong answer on every other name. It is perfect on that subset and has true accuracy equal only to the subset's fraction of the task universe. An outer loop that keeps selecting by the same subset can improve its reported score without improving beyond the information it has exposed. Renaming the subset a holdout does not restore independence.

26.2 Proof obligations and performance evidence

A semantic proof can establish that a proposed operation preserves a specified relation. A program checker can establish that a finite tree respects the supplied domains and budgets. A fresh performance comparison can support usefulness on a declared task law. None of these establishes the truth of every environmental premise. Their evidence types should remain separate in an improvement receipt.

The constructive tests shipped with this paper are deterministic finite enumerations. They are not samples from an AGI population, and they do not use (11) to manufacture confidence intervals. Their role is to check the implementations of the stated finite arguments.

27 Sequential evidence for an improved discovery procedure

The deterministic compiler comparison above can be checked on its whole declared family. A learned discovery procedure evaluated on fresh stochastic tasks would need another kind of evidence. It may also be expensive enough that the auditor would like to inspect results sequentially and stop once the evidence is adequate. The fixed-sample rule in Section 26 does not automatically justify that stopping policy.

We give a small time-uniform alternative whose finite arithmetic is directly inspectable. Before an audit begins, freeze the candidate procedure, its comparator, their installed information, the task law, the common resource cap, and a paired score $Y_t \in [-1, 1]$. Timeouts and invalid outcomes must have a stipulated score rather than disappear from the sample. Fix a proposed improvement margin $\gamma \in [0, 1)$.

The null condition is

$$\mathbb{E}[Y_t \mid \mathcal{H}_{t-1}] \leq \gamma \quad \text{for every audit step } t, \tag{12}$$

where $\mathcal{H}_{t-1}$ includes all information available before the next score. Conditionally independent cases from a fixed task law with mean at most γ satisfy this condition. An auditor cannot change the task distribution toward favorable cases and retain that interpretation without a new argument.

Choose a nonnegative bet λ_t, measurable from $\mathcal{H}_{t-1}$, with $\lambda_t \leq 1/(1+\gamma)$. Define

$$E_0 = 1, \qquad E_t = E_{t-1}\big(1 + \lambda_t(Y_t - \gamma)\big). \tag{13}$$

The variable E_t is an evidence process, not a posterior probability that the candidate is better. Predictability means choosing the bet before observing the score to which it is applied.

Theorem 27.1 (An anytime-valid paired improvement audit)**.** *Under* (12) *and the stated predictable bet constraints, E_t is a nonnegative supermartingale. For every $\delta \in (0,1)$,*

$$\Pr\left(\sup_{t\geq 0} E_t \geq 1/\delta\right) \leq \delta.$$

Consequently an audit may reject the stated null at its first crossing of $1/\delta$, while choosing its stopping time from the observed audit history. The result does not require a predetermined number of cases or independence beyond the conditional-mean premise.

Proof. Since $Y_t - \gamma \geq -(1+\gamma)$, every factor in (13) is nonnegative. Conditional expectation and predictability give

$$\mathbb{E}[E_t \mid \mathcal{H}_{t-1}] = E_{t-1}\big(1 + \lambda_t(\mathbb{E}[Y_t \mid \mathcal{H}_{t-1}] - \gamma)\big) \leq E_{t-1}.$$

Let τ be the first threshold crossing. For a finite N, the stopped process at $\tau \wedge N$ still has expected value at most one, by induction on the stopped conditional expectations. On $\{\tau \leq N\}$ its value is at least $1/\delta$, while it is nonnegative otherwise. Therefore $\Pr(\tau \leq N) \leq \delta$. Letting N increase proves the result for any crossing time. □

This is the elementary nonnegative-supermartingale method underlying time-uniform evidence, not a new general concentration theorem [16]. The contribution here is its explicit paired comparison and its placement within the recursive resource and information contract. In particular, an audit observation is a complete comparable attempt by each discovery procedure, not an arbitrary message emitted within one attempt.

27.1 Why seeing the outcome before choosing the bet fails

Counterexample 27.2 (Hindsight betting creates false evidence)**.** Let Y_t be independent fair signs and $\gamma = 0$. Choose $\lambda_t = 1$ when the current sign is positive and zero otherwise. This invalid rule doubles E_t on a positive score and leaves it unchanged on a negative one. Within four scores it crosses the threshold four whenever at least two scores are positive, an event of probability $11/16$. This exceeds the nominal bound $1/4$. Each individual bet lies in the numerical range $[0,1]$. What fails is predictability.

A data-dependent stopping rule is allowed by the theorem. A data-dependent choice that looks at the *current* outcome before staking on that outcome is not. The software can validate the numerical range of a bet, but it cannot infer the provenance of an externally supplied bet from that number alone. A usable audit receipt must record the commitment order and the sampling contract.

The same issue arises when a candidate is edited after seeing an audit result. The resulting process no longer evaluates one frozen procedure under its original interpretation. It can start a new comparison, debit a new conditional error budget, and retain the earlier result as development evidence. It cannot silently reuse the old evidence process as though the candidate had not changed.

27.2 Evidence across recursive generations

For attempt j, condition on the complete earlier lineage, freeze its candidate, and initialize a fresh valid audit at $E_0^{(j)} = 1$ with error allowance δ_j. If $\sum_j \delta_j \leq \delta$, Proposition 26.1 bounds the probability of ever accepting a false margin claim by δ. Each audit can stop sequentially, and the next candidate may depend on completed earlier audits. The premise remains conditional validity for each actual attempt.

Exact certificate checks and sequential evidence still have different jobs. A table-defined compiler can be checked for admissibility and optimality on its formal input without making a claim about a population of tasks. A stochastic audit can support a productivity comparison on that population without proving every generated program correct. The recursive system needs both whenever its claim combines semantic validity and statistical usefulness.

The new reference suite enumerates finite rational null paths and confirms the crossing bound in those cases. It also checks the hindsight counterexample. No live learner was evaluated with this test, and the deterministic compiler counts are not retroactively presented as independent random samples.

28 The complete cost of becoming a better improver

A procedure can save work on each future task and still be a poor investment when its discovery and validation are expensive. Conversely, a costly method can be worthwhile when it is reused sufficiently often. Both observations are ordinary resource accounting. They become especially important when a claimed improvement shifts work from inference into training, into a translator, or into an external oracle.

Use a resource vector rather than an unexplained token total:

$$C = (C_{\text{model}}, C_{\text{search}}, C_{\text{validate}}, C_{\text{install}}, C_{\text{execute}}, C_{\text{store}}).$$

These coordinates need not share units. Model construction may involve environment observations. Search and validation may involve processor work. Installation may require conversion or compilation. Execution work concerns one deployment. Storage concerns the retained library and complete controller. A scalar comparison needs declared conversion weights or a common physical budget, not an arbitrary sum of unlike quantities.

For the hidden-cell instrument, a compiled tree has fifteen records even though a successful depth-three run visits only four. The bit-query API is installed for both procedures in the matched comparison. Its table representation, if used, occupies $n(n + \log_2 n)$ Boolean entries for the chosen power-of-two family. The reported three-versus-seven diagnostic comparison does not imply that inspecting a large subset costs one machine instruction in every implementation. It uses one sensor call as its stipulated unit.

Proposition 28.1 (Amortization of a reusable process improvement)**.** *In a fixed additive resource unit, suppose installing a new procedure has incremental cost $C \geq 0$ relative to the old one, and its expected per-use saving on an unchanged task distribution is $d > 0$. If there are N uses and no additional costs or changes in the task law, its expected cumulative saving is $Nd - C$. It is strictly resource-improving exactly when $N > C/d$.*

Proof. Linearity of expectation gives expected total use savings Nd, without requiring independence between uses when each has the stated expectation. Subtract the fixed incremental cost and solve the strict inequality. □

The formula is not a prediction of future demand. Distribution drift, maintenance, growing context, retrieval overhead, or a changed interface can alter d or add costs. The value of the repair library depends on whether later problems actually admit its operations.

28.1 Larger search spaces can lower productivity

Counterexample 28.2 (More expressibility with worse fixed-budget discovery)**.** A search language has N candidates, exactly one useful. A procedure samples uniformly with replacement $B \geq 1$ times and recognizes usefulness perfectly after each sample. Its probability of finding the useful candidate is $1 - (1 - 1/N)^B$. Adding an additional useless candidate preserves every old expression and strictly decreases this probability. The best attainable repair is unchanged, while the implemented discovery procedure becomes worse under the same sampling budget.

This does not argue against grammar growth. It identifies what the learner must improve alongside growth: proposal probabilities, indexing, modular reuse, or informative pruning. The search space and the search distribution are different parts of the agent.

An archive has a related ambiguity. The best true value among retained artifacts cannot decrease when an old artifact remains available under the same score. But a deployed selector can perform worse as retrieval becomes harder. The monotonicity of a hypothetical maximum is not the monotonicity of actual use. Resource-limited archive management belongs inside the recursive system rather than outside its reported budget.

29 A finite certificate for repeated improvement

A valid local change can destroy the ability to make another valid change. A new representation may discard the evidence needed to verify future proposals. A controller may consume the remaining budget while preserving the current task score. A shortcut can bypass an interface on which later transports depend. Recursive continuation therefore needs its own object.

Let Z be a finite set of complete lineage configurations. A configuration includes the currently selected solver and improver, retained libraries and evidence, the active comparison contract, and relevant resource state. The following theorem assumes this configuration is observed. Unknown configurations require belief-state modeling or a different synthesis result. No unrestricted introspection is inferred.

Each z has a finite set of authorized attempts $A(z)$. An attempt has a nonempty set $\mathrm{Post}(z, a)$ of possible successor configurations. This can represent all outcomes of a stochastic proposal or an abstract uncertainty set. A safe set $\mathsf{Safe} \subseteq Z$ encodes the declared semantic, evidence, authority, and resource conditions. A pause or terminal state may have a zero-progress self-loop. Actual expensive attempts must debit resources in Z, or the model must explicitly state their replenishment.

Define

$$K_0 = \mathsf{Safe}, \qquad K_{n+1} = \{z \in K_n : \exists a \in A(z),\ \mathrm{Post}(z, a) \subseteq K_n\}.$$

The descending finite sequence reaches a fixed point $\mathcal{K}$. It is the set from which some policy keeps every successor safe indefinitely. This can include doing nothing forever. Safety alone is not recursive improvement.

For each transition, a progress mark $p(z, a, z') \in \{0, 1\}$ is supplied by a checked criterion. A mark of one means the specified process-improvement claim has been established under the same comparison. It is not assigned merely because a file changed. Define

$$W_{0,0} = \mathcal{K}, \qquad W_{0,k} = \varnothing \quad (k > 0), \tag{14}$$
$$W_{h,k} = \{z \in \mathcal{K} : \exists a \in A(z),\ \mathrm{Post}(z, a) \subseteq \mathcal{K}, \quad z' \in W_{h-1,\max\{0,k-p(z,a,z')\}} \quad \forall z' \in \mathrm{Post}(z, a)\}.$$

Here h is the number of remaining attempts and k the number of still-required progress events. A pause action lets a policy that finishes early use the remaining horizon without changing the result.

Theorem 29.1 (Common-policy recursive progress)**.** *A configuration belongs to $W_{h,k}$ if and only if there exists one authorized policy that keeps all outcomes in $\mathcal{K}$ and guarantees at least k checked progress events within h attempts. A policy depending on the current configuration and the remaining pair (h, k) suffices. The finite recurrence gives a witness when feasible and a complete negative answer within the stated model when infeasible.*

Proof. At horizon zero, zero remaining progress is possible exactly in $\mathcal{K}$, and positive remaining progress is impossible. Assume the claim for $h - 1$. A policy at z must choose one initial action before its outcome is

known. Every possible successor must remain in $\mathcal{K}$ and must admit a continuation meeting the remaining target $\max\{0, k - p\}$. By induction these conditions are exactly the memberships in (14). Conversely, choose a witnessing first action and a witnessing continuation for each observed successor. The resulting policy works on every path. Induction proves necessity, sufficiency, and the stated memory requirement. Finiteness makes the construction terminating. □

The use of one first action is important. The weaker claim that every possible outcome has some favorable action, selected after secretly learning that outcome, is not an implementable policy. Likewise, independently feasible semantic, resource, and evaluation certificates must refer to the same attempt and successor relation.

The theorem is complete only for its finite catalogue and its information boundary. It does not create a promising next candidate or identify the true productivity of an arbitrary program. Its progress labels may come from exact finite evaluation or from a separate evidence layer. With probabilistic labels, the conclusion about actual progress holds on the event that the relevant certificates are valid. Proposition 26.1 can bound failure of that event under its own assumptions.

29.1 Safe rejection can prevent a worst-case progress guarantee

Counterexample 29.2 (Repeated opportunity is not bounded sure progress)**.** At a safe configuration, an attempt either produces a genuine improvement or rejects the candidate and returns to the same configuration. Both outcomes are possible at every attempt. A safe pause is also available. For every finite h, there is a path with h consecutive rejections, so no policy guarantees one improvement within h attempts. With an independent positive success probability, eventual success can nevertheless occur with probability one. The worst-case finite theorem and that probabilistic statement answer different questions.

29.2 Why local improvement does not imply recursive viability

A second finite example has a proposal that raises the current score but deletes the only retained witness needed to validate any subsequent proposal. Its destination lies outside Safe when witness availability is protected. If witness availability is omitted, it may remain safe but have no productive continuation. The result depends on the contract, not the flattering name assigned to the edit.

This is a compact version of the closure perspective in SRIC [2]. The present contribution is to connect that perspective to a concrete diagnostic tree, a resource witness, and a specified productivity comparison rather than treating closure as a free consequence of successful self-modification.

30 Acceleration is an additional hypothesis

Recursive improvement can be meaningful without being indefinitely accelerating. Its measured pace depends on useful gains, the cost of finding and checking them, available opportunities, and the chosen scale of evaluation. A normalized task score has a ceiling. A changing suite can continually introduce new headroom, but its successive normalized scores are not automatically comparable.

Proposition 30.1 (A fixed-margin ceiling on a fixed score)**.** *Suppose a lineage has a fixed measure $R_t \in [a, b]$, every accepted progress event increases it by at least $\gamma > 0$, and all other transitions are nondecreasing in that same measure. Then the number of progress events is at most*

$$\left\lfloor \frac{b - a}{\gamma} \right\rfloor.$$

An infinite sequence of positive increments is compatible with bounded total improvement when no common positive lower bound is imposed.

Proof. After k progress events, telescoping and nondecrease give $R_t \geq R_0 + k\gamma \geq a + k\gamma$. Since $R_t \leq b$, the bound follows. For the second assertion, start at zero and use increments 2^{-t} for $t \geq 1$. Every increment is positive and their partial sums remain below one. □

If nonprogress transitions can decrease the measure, the first bound need not hold: repeatedly falling and recovering can generate arbitrarily many positive increments. Those recoveries are not unlimited cumulative progress. A moving evaluator creates a different problem. It may be appropriate, but it requires bridges to preserve the meaning of the comparison.

A simple accounting model can express the additional conditions for acceleration. Let g_t be an expected useful gain and $c_t > 0$ the complete cost of the corresponding discovery and acceptance cycle. Improvement productivity per unit resource is g_t/c_t. Raising g_t alone does not raise that ratio when c_t rises faster. A larger reachable repair set is similarly insufficient when proposal quality falls or validation becomes the limiting cost.

SRIC's bottleneck perspective is useful here: improving one stage can expose another stage as the dominant limitation [2]. The next bottleneck might be information acquisition rather than proposal generation, or evidence quality rather than repair expressibility. A theory of acceleration would need to specify how those bottlenecks move and which resources support their resolution. A claim of universal acceleration does not follow from the existence of one productive cycle.

Part VIII

A bounded route toward general intelligence

31 A proposed learned architecture

The mathematical contracts do not specify a complete learning algorithm. They do identify the components that a credible algorithm must make learnable and the evidence required when those components change. The intended architecture has one active solver, a recurrent working state, a persistent library of typed operations, an acquisition policy, a proposal mechanism, and an admission procedure. These are roles in one realized computation, not a requirement for a separate large model for every role.

31.1 What would learn

Let r_ϕ encode the retained state, $G_{\theta,a}$ describe proposed operations, π_ω choose observations or experiments, and p_ψ propose candidate programs or parameter updates. A development episode can modify $\phi, \theta, \omega, \psi$ through a declared learning rule. A persistent program library can also change by adding, replacing, or removing checked operators. Temporary deliberation without a retained modification is not called a new learned generation here.

The learning signals need separate purposes. Prediction losses train anticipation of future outputs. Intervention-consistency losses test whether the proposed operations match their intended effects. Acquisition rewards favor useful reductions of repair uncertainty after charging experiment cost. Proposal rewards evaluate the actual selected descendant rather than fluency of its rationale. Preservation checks protect specified older capabilities or interfaces. There is no universal set of weights that turns these distinct obligations into one sufficient scalar objective.

A latent computation may suggest that a particular variable matters. A typed checkpoint should then declare the variable's domain, the operations it supports, the evidence used to infer it, and the continuations on which it was checked. Its translator must not read the answer that the checkpoint claims to predict. Its complexity is charged. OSL is a proposed semantic consumer for such checkpoints, while Neuralese concerns the learned latent producer [4, 3]. This paper does not implement their complete learned bridge.

Proposition 31.1 (Local operation agreement preserves finite continuation)**.** *Let D be invariant under every source operation F_a, let $q : D \to Z$, and suppose*

$$qF_a = G_a q \quad \text{on } D$$

for every admitted a. If the required output satisfies $y = \bar{y}q$ on D, then for every finite operation word w,

$$qF_w = G_w q, \qquad yF_w = \bar{y}G_w q.$$

Proof. The empty word gives the identity. Assume the relation holds for a prefix. Domain invariance allows application of the next local equation, giving $qF_aF_w = G_aqF_w = G_aG_wq$. Induction establishes the first identity and composing with $\bar{y}$ gives the second. □

The proposition is a specification for meaningful reuse, not a method for learning q or G_a. Invariance matters because an unchecked operation can leave the domain where the equations hold. Output preservation matters because commuting hidden states need not retain the task variable. Actual causal use matters because a probe can decode an appropriate state that the solver does not use.

31.2 What a complete attempt would do

A bounded iteration would begin with an observed task failure and an explicit set of plausible causes. The active acquisition policy chooses an admitted experiment. Its result updates those causes and the current state. A proposer supplies a candidate repair or a new constructor when the old language is insufficient. A separate execution checks the candidate's declared interface, resource usage, and behavior. A selected candidate is assessed on evidence not used to construct it. If admitted, its actual changes and scope are stored, including the procedure that found it.

A later outer attempt may change the acquisition policy, proposer, or selection code. It must then be assessed as an improver from comparable starting conditions, not only as the latest solver. An improvement may be retained provisionally for a particular domain rather than promoted globally. Failed hypotheses remain useful evidence when their assumptions and reasons for rejection are preserved.

Candidate evaluation can use the same kind of mathematical tools as the proposer without sharing the candidate's unchecked assumptions. Independence here means a specified information and authority separation, not a mystical absence of common software or training ancestry. External audit data and the authority to redefine a protected score are not supplied to a candidate as editable inputs for the same transaction.

31.3 A learned policy is not an unconstrained interpreter

A sufficiently powerful interpreter can hide the entire solution behind a small label. The remedy is not to forbid interpreters by definition. It is to state their installed information, runtime, storage, and access. This is the resource lesson of *Machine Understanding* [1]. The diagnostic tree is intentionally explicit so that this accounting can be performed exactly in the worked case.

Nor should neural implementation be required simply for appearance. A library of executable operators and a learned experiment selector can be a meaningful adaptive system. Conversely, updating a large weight tensor does not automatically establish a useful representation, a causal mechanism, or a transferable improvement procedure. The architecture is evaluated by its realized operations and the origin of their improvement.

32 The compiled diagnostic improvement

We can now state the exact mathematical achievement of the running example. It is deliberately smaller than the full learned architecture, but complete enough to expose the relevant distinctions.

For $n = 2^k$ cells, assume a single defective cell, binary reports, singleton queries, and k bit-coordinate queries. All queries leave the cell unchanged and cost one diagnostic call. A correct final repair costs one additional call. Distinct cells require distinct terminal repairs. There are no false repair commits. A literal tree record contains one experiment or repair identifier plus its child references.

Theorem 32.1 (Optimal diagnosis in the hidden-cell family)**.** *Every certain diagnosis needs at least k binary queries in the worst case. The bit-query procedure attains this bound and uses $2n - 1$ records in a full binary tree. Any procedure using singleton queries alone needs at least $n - 1$ queries in the worst case, and equality-first testing attains that bound with the same number of tree records. Thus their worst total execution costs, including repair, are $k + 1$ and n.*

Proof. A binary tree of depth less than k has fewer than $2^k = n$ leaves. Every successful leaf can identify at most one cell because its repair must be correct for every remaining case. Therefore depth at least k is required. The k coordinate bits identify every cell and give a complete binary tree with n leaves and $n - 1$ internal nodes. For singleton tests, an adversary can answer no on each tested cell while at least two untested cells remain. Fewer than $n - 1$ distinct singleton queries therefore cannot guarantee identification. Equality-first

testing uses one leaf for each positive answer and one final leaf for the last remaining cell, again totaling $2n - 1$ records. Adding the terminal repair gives the claimed work counts. □

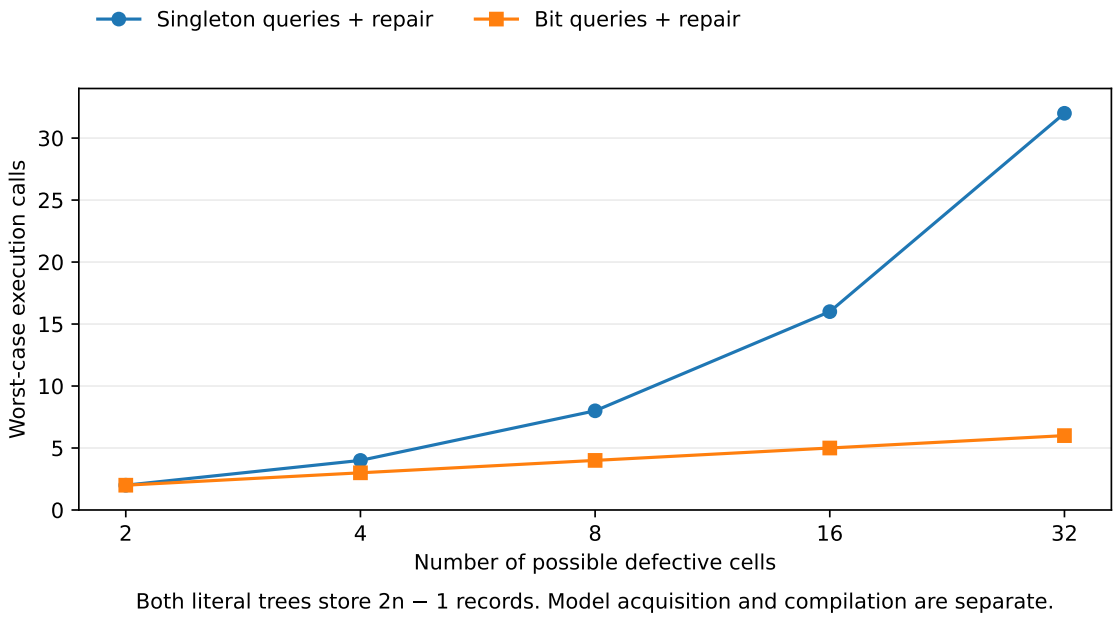


Figure 11: Exact total execution costs for the power-of-two cell family. Singleton testing requires $n - 1$ diagnostic queries, while the admitted bit queries require $\log_2 n$. A final repair adds one call to each plotted curve. Both full trees have $2n - 1$ stored records. The curves omit separately charged model acquisition, compilation, and adapters.

32.1 What is actually compiled

The reference compiler receives an observation table and the set of successful terminal repairs. It computes the frontier in Theorem 6.2 and returns an optimal tree under a chosen work/storage allowance. At $n = 8$ it returns a four-call, fifteen-record witness. A separately coded interpreter checks every initial cell and every branch. The reference comparison constructs the equality-first baseline independently, rather than comparing the compiler to a weakened version of its own output.

The selected tree is an executable diagnostic policy for future episodes. It is not a saved answer for the calibration case. In a new episode, the hidden cell is unavailable to the policy. Only reports from its queries determine the branch and eventual repair. A cell permutation supplies a controlled foreign interface when the query sets, terminal repairs, and adapter are transported together. Theorem 9.1 explains that transfer. The tests do not claim that the adapter was learned without correspondence data.

32.2 The measured object is a procedure

Under the earlier four-call cap, the equality-first improver repairs $3/8$ of the uniform eight-cell cases and otherwise abstains. The compiled improver repairs all cases. The repaired solvers themselves are simple: each now knows the correct cell-specific operation. What improved is the procedure that acquires that case-specific correction. The library and sensor language are unchanged.

This original controller-level construction does not by itself improve the compiler. The compiler and acquisition constructions in Parts III–V provide the additional changes to compilation and rule acquisition.

Each comparison retains its own starting procedure, supplied information, and cost unit. Their coexistence does not convert the original eight-cell demonstration into evidence of indefinite self-improvement.

32.3 Controls that expose the boundary

Removing the bit-query family changes the synthesis problem and restores the singleton lower bound. Omitting the adapter breaks foreign-label replay even when the tree shape is unchanged. Deliberately choosing an incorrect leaf causes the independent checker to reject the certificate. Reducing the allowed tree records below fifteen rules out the certain eight-cell solution in this literal binary-tree model. These are failures of distinct requirements, not reasons to reinterpret the target after a run.

The repair-aligned query example in Section 6 supplies a different control. There the compiler need not identify the exact hidden case. It should stop when one repair works for every surviving case. This prevents full identification from being hard-coded as the universal objective of inquiry.

33 Generality requires more than semantic relabeling

The exact transport theorem is useful because it proves what is preserved when a correspondence exists. It is not evidence that the system can discover correspondences efficiently across arbitrary tasks. A relabeled hidden-cell world remains the same informational problem. Transfer from diagnosis to theorem discovery, planning, or empirical model construction needs additional argument.

A credible generality study would vary several things separately: the state distinctions needed for future behavior, the allowed experiment language, the repair constructors, the information available to the learner, and the resource costs. It would also separate unfamiliar instances of a known schema from unfamiliar combinations of schemas and from an operation family that the old language cannot express.

The proposed mechanism is not that every task reduces to binary cell location. It is that many tasks can expose a useful pattern: a currently retained representation fails to support a consequential distinction, an experiment can discriminate among causes, and a reusable operation can resolve the resulting class of failures. Whether that pattern is broad enough and economical enough to contribute to general intelligence is a research hypothesis.

33.1 The next decisive learning comparison

The most informative next study would compare a parent and descendant improvement procedure on the same family of fresh starting solvers. It would hold installed libraries and base-model access fixed in the primary contrast, then add a separate full-system contrast that includes any learned library. A stronger ordinary program-library learner should be included, since library learning itself is established [8].

The development system would be allowed to acquire or revise an encoder, experiment selector, and operator library in bounded simulated environments. Evaluation would include longer continuations, new operator compositions, and changed observation formats. Each change should have an explicit semantic target and a resource ledger. Removing acquisition adaptation, repair reuse, or recurrent state separately would locate which mechanism produces a gain. Those are proposed experiments, not results reported in this paper.

The desired endpoint is not merely that the descendant solves more familiar cases. It is that its acquired improvement procedure produces better valid repairs on unfamiliar cases within the same budget, and that the gain survives adequate controls for retained answers, installed adapters, and selection leakage. A subsequent generation must repeat that comparison rather than inherit its predecessor's claim by name.

33.2 What would count against the central hypothesis

The obstruction-guided procedure may be no better than an ordinary search method given the same information. Its apparent advantage may vanish after charging its diagnostic model and checker. Its recovered operators may fail on longer compositions. A learned selector may memorize surface labels instead of identifying the causal distinction. A larger repair language may dilute search enough to erase its representational advantage. These outcomes would identify where the proposed integration fails or requires stronger assumptions.

A negative result on a restricted task family would not prove that general intelligence is impossible. A positive result would not prove AGI either. Both would help determine whether the proposed mechanism earns its place in a larger architecture.

34 Open problems beyond the first edition

The exact core rests on finite models, supplied semantics, bounded horizons, and explicit information. Several important extensions remain.

Learning the experiment model. The tagged-model theorem accounts for a fixed unknown member of a supplied finite class, including paid calibration. It does not construct that class from open-ended experience or guarantee that it contains the true process. Safe learning with noisy, misspecified, or changing observation models remains open in this framework. Model uncertainty must not be replaced by a guessed deterministic table merely to preserve an exact theorem.

Discovering a repair language. Both the original compiler and the selected shortcut use a supplied repair language. The earlier selector chooses within a supplied catalogue of compiler templates. Part IV additionally synthesizes a primitive arithmetic rule after exposing a missing feature. Its grammar, feature registry, teacher, and proof gate remain supplied. Open-ended constructor discovery needs a representation of candidate meanings, a method for proposing them, and evidence that the new language preserves old operations while making new problems tractable. Expressibility of all computable programs is not an efficient search strategy.

Acquiring stronger search representations. Part V changes the candidate space used to acquire future discovery rules. The new equations and quotient are generated, while their arithmetic meanings and proof machinery are supplied. A stronger supplied normalizer remains a useful comparator. Learning effective proof search, choosing cheaper rule subsets, and adapting the representation when a language changes are distinct next questions. A finite-consistent equation is not automatically a valid optimization.

Transfer without a supplied correspondence. The exact policy-transport theorem assumes its maps. Learning them introduces gauge ambiguity, underidentification, lossy observations, and potential answer leakage. The Neuralese-to-OSL bridge remains a separate learned realization problem, not a solved consequence of writing a commuting square [3].

Improving evaluation without changing the question. An evaluator can be genuinely defective. Correcting it requires retaining evidence of the defect, specifying the new scope, and distinguishing a change in estimated performance from a change in the underlying target. A fully unrestricted evaluator rewrite would remove the stable comparison on which the reported improvement depends.

Noisy and distributed recursive synthesis. The finite progress theorem uses fully observed configurations and worst-case successor sets. Useful systems may need probabilistic service, partial observation, or several

decision centers. Separate success probabilities do not automatically yield one policy that satisfies all constraints. A noisy extension must state both the risk accepted and the evidence supporting its transition law.

Long-run productivity. Repeated finite improvements do not determine the eventual learning curve. A richer task distribution, increasing resources, changing architectures, or new scientific observations may create further headroom. Those changes should be modeled and charged rather than absorbed into a claim of autonomous acceleration. The current paper does not settle whether general-purpose recursive improvement has increasing, constant, or diminishing returns.

34.1 Authority and bounded implementation

The formal permission to modify a local program is not authorization to change external services, access private data, or remove an institution's controls. Proposed deployments would require an explicit scope of authority and a justified handling of affected parties. The reference package does not connect to services, modify other agents, train a neural model, or conduct autonomous external research. It constructs and checks finite mathematical objects in a local process.

Similarly, stopping and refusal are legitimate outcomes when required evidence or resources are missing. An improvement claim must not become a reason to suppress those outcomes. A preserved archive can retain unsuccessful attempts without promoting them into operating authority. These commitments follow the evidence-preserving lineage perspective rather than a presumption that continued self-modification is always desirable.

35 Conclusion

A better solver, a better repair procedure, and a better process for discovering repair procedures are different objects. They can interact productively, but the interaction needs an explicit realization. General intelligence motivates the search for transferable discovery methods. It is not a premise from which those methods follow.

The finite theory keeps the relevant obligations in one chain. An investigation must acquire enough information to choose a successful repair, without necessarily identifying every hidden detail. Its execution and stored controller must fit the same resource witness. A transfer must preserve observations and admissibility, not only an output. A language extension changes which lower bounds remain valid. An adaptive lineage must retain the evidential meaning of its accepted gains and a continuing policy for any promised progress.

The constructive sequence now reaches the acquisition of discovery rules. A compiler produces better diagnostic trees. A guarded strategy reduces the search needed to find such trees. A learner composes an internal rejection predicate. A further learned equation library prepares a smaller search representation for reacquiring that predicate and other targets. Exact teacher-transcript preservation identifies what has not changed: the required evidence and the final selected AST. The additional improvement concerns repeated evaluation work, not a newly supplied answer.

Correctness and productivity remain different. A supplied full normalizer can remove more redundancy than the learned library. Preparation can cost more than a short workload saves. Streaming avoids some model reads while adding comparisons. A repair that succeeds on several cases can make an old identification bound unsound. These are not peripheral caveats. They determine what the acquired modification is and when it is worth using.

The finite constructions demonstrate specific changes in discovery behavior under declared semantics. They do not establish learned physical meaning, efficient search over every computable procedure, or an indefinitely improving sequence of outer learners. Their positive contribution is an auditable connection between representations, generated rules, checked equivalences, resource use, and recursive revision. A

claim about becoming better at improvement can now be located at a particular level, checked against its predecessor, and refused when its evidence or cost contract is missing.

The first edition also makes those claims revisable without making their premises disposable. A changed repair language can invalidate a weight bound while leaving the equation library sound. A changed resource price can reverse a cost conclusion without altering one controller transition. The dependency-complete reuse theorem states when old witnesses remain applicable, and the resource proposition states when an event saving is independent of a selected price model. Their role is to keep one actual, inspectable chain of reasoning and computation attached to every claimed gain.

A A complete diagnostic trace

Take $n = 8$ and hidden defect $j = 5$, whose binary digits are 101. One optimal tree uses the three queries

$$A_0 = \{1, 3, 5, 7\}, \qquad A_1 = \{2, 3, 6, 7\}, \qquad A_2 = \{4, 5, 6, 7\}.$$

The precise order may differ among equally optimal compiled trees. For this order, the supports are

$$\{0, \ldots, 7\} \xrightarrow{1} \{1, 3, 5, 7\} \xrightarrow{0} \{1, 5\} \xrightarrow{1} \{5\}.$$

The terminal repair is r_5. The path uses three diagnostic calls and one repair call. The complete tree stores all eight repair leaves and seven query nodes, not merely this path.

Equality-first testing asks about cells zero through five before it obtains a positive report. It uses six diagnostic calls on this particular case, then one repair. Under the three-diagnostic cap it abstains after three negative reports. The cap is fixed before the episode, not shortened for the baseline after observing which cell is defective.

For a target relabeling $\sigma(i) = 3i + 1 \pmod 8$, the defect has target label $\sigma(5) = 0$. Each query becomes $\sigma(A_r)$ and the terminal repair becomes $r'_{\sigma(5)}$. Reports and tree shape agree with the source, but the label adapter is part of the transported interface. Applying the old numeric repair label five directly in the target would be incorrect. This is why syntactic resemblance does not replace a semantic map.

A.1 A repair problem that should stop before full identification

In the four-case example, the support $\{0, 1, 2\}$ admits repair a immediately. A compiler that insists on identifying the precise case would spend unnecessary work. The successful stopping condition is the existence of one action whose success set contains the whole support. Conversely, the support $\{2, 3\}$ admits no common repair in that example. A leaf chosen there without another observation is invalid even though either repair succeeds in one of its cases.

B Reference computations and evidence scope

The package implements the finite experiment model, the two-resource recurrence, tree reconstruction, and a separate replay checker. It also implements stable partition refinement and finite recursive-progress synthesis. Exact test groups compare selected constructions with separately coded enumerations on small domains. Integer and rational arithmetic are used for mathematical acceptance checks. Plotting converts exact analytical quantities to display coordinates only.

The verification program records its exact domains, input and output hashes, command, interpreter, duration, and outcomes. The source audit inventories formal statements and checks labels, citations, and their proof environments. The build uses LaTeX and the supplied vector figures. A clean rebuild is compared by extracted PDF text, not claimed to be byte-identical across environments.

These checks are authored within this project. Alternative algorithms provide implementation cross-checks, not independent peer review. No test count establishes a universal claim outside its implemented domain. The proofs in the manuscript provide the conditional general arguments. No Lean certificate, trained agent, learned semantic bridge, or automated research run is included.

B.1 A precise reading of the finite demonstration

The observation and success tables are supplied analytical semantics. The compiler selects a tree without access to the next episode's hidden index. The checker separately receives that index when exhaustively verifying all possible episodes. That division prevents a policy from consulting hidden state during replay. It does not make the supplied model an empirically learned object.

The main generated files are the compiled cell-tree certificate, a report of source and transported replays, the cost frontiers, and the exact check results. All can be regenerated locally. The README records the commands and the review notes record any failures or limitations encountered during preparation.

C Claim map and the scope of the first edition

Claim	What the present manuscript supplies
Feedback can support a repair	A transcript-discrimination bound and a common-repair stopping condition. No automatic causal diagnosis is inferred.
A representation supports continuation	A least stable refinement and explicit operation equations. Learning the representation is additional.
A repair can be found within a budget	An exact finite work/storage recurrence and a checkable tree. It requires a supplied finite experiment model.
A discovery procedure transfers	A correspondence preserving complete investigation semantics and costs. Discovering that correspondence is additional.
A pruning rule is learned	A finite grammar synthesis, a witnessed feature conflict, and a separate scope proof. The teacher and grammar are supplied.
A compiler skips work safely	Streaming root rejection preserves the controller and avoids counted rows. Predicate work remains charged.
A candidate procedure is better	A selection bound under uniform assessment error and a protected productivity comparison. The error premise must be established.
A lineage preserves credible gains	Conditional error spending and an explicit distinction between development, selection, and audit evidence.
A lineage can improve repeatedly	An exact finite common-policy criterion for marked progress. Marks require separate valid evidence.
The process accelerates	Not established. A fixed-score ceiling and cost accounting expose additional requirements.
The system is generally intelligent	Not established. Generality and learned discovery remain research targets across unfamiliar structures.
The synthesis procedure improves	A checked equation library removes redundant candidate evaluations while preserving teacher proposals. Preparation costs remain separate.
One repair covers several cases	A feasible repair-weight certificate replaces singleton counting. Its validity is rechecked after a repair-language change.
A composed gain is affordable	The actual acquisition, compilation, checking, and preparation costs must fit one common resource model.
Old certificates survive a change	Only dependency-complete unchanged obligations can be reused without new evidence. Affected descendants require re-establishment.
A gain is independent of prices	Componentwise event dominance, or a strictly positive margin on the declared finite price polytope, supplies the precise criterion.

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Machine Societies

A Theory of Persistent Plural Computational Organization

Komorebi P.
8 October 2026

First edition
Version 1.0.0

Abstract

A population of artificial agents may sustain institutions without becoming one collective agent. This monograph develops a theory of machine societies as persistent plural computational organizations whose participants operate, contest, and revise constraints on their interactions. A finite realization model connects state abstraction to permissions, obligations, resources, and historical records. Exact transport conditions distinguish preservation of behavior, authority, language, and knowledge. Formation barriers, incomplete information, incompatible interfaces, and verification limits expose failures that a common organizational label can conceal. The central synthesis asks whether several institutional guarantees admit one executable policy. It characterizes simultaneous enforcement, safe continuation, and certified service under a declared information boundary, and identifies the abstractions that preserve those possibilities. Accountability adds further requirements: affected outsiders must not disappear from the resource model, an auditor must have incentives to perform its service, privacy and effective appeal must constrain the same evidence procedure, and continued operation must preserve a specified authorized ending. Sharp conditional bounds and finite strategy constructions connect these requirements. A running computational commons makes both the feasible cases and the incompatible promises explicit. The contribution is an integrated realization and preservation theory with proofs, counterexamples, and checkable finite witnesses, grounded in established mathematical foundations rather than an assumption of inevitable machine civilization.

Scope. A theoretical synthesis with conditional mathematical propositions, proofs, and analytical examples. Finite models form the core; an explicit unbounded-program extension establishes a computability limit. Standard foundations are acknowledged rather than claimed as new discoveries.

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About this edition

First edition. This volume brings the realization model, institutional theory, and accountability analysis into one reference treatment. Its version number identifies an editorial and source release. It does not turn an open research question into a theorem or substitute for independent mathematical review.

The argument in five parts

I. Realization and abstraction. What state is present, what an observer can access, and when a smaller description preserves dynamics and interventions. These sections supply the mathematical language used by the remainder of the volume.

II. Institutions and collective choice. How authority, membership, communicative acts, incentives, constitutional revision, and distributed information constrain the operations of a plural population.

III. Continuity, compatibility, and knowledge. What persists through replacement, how local rules and meanings compose, and why cultural accumulation, coherent testimony, and common knowledge require different conditions.

IV. Authority and joint feasibility. Which participants actually control consequential transitions, how staffing and shared resources constrain a common policy, and which finite witnesses certify continuation and service.

V. Accountability and the limits of persistence. How external claims, strategic inspection, confidentiality, appeals, authorized endings, and changing certificate assumptions enter the same operational contract. The concluding sections distinguish settled finite results from further questions.

Reading routes

A conceptual reading can begin with Sections 1–3, continue through Sections 12–16, and then read the interpretations in Sections 42 and 43. A mathematical reading follows the parts in order. An implementation review can begin with the exact contracts in Sections 15, 19, 42, and 43, then consult the theorem index and bounded reference programs in the source package.

Status of the claims

Definitions specify a contract. Propositions and theorems carry conditional prose proofs. Some are explicit applications of classical results. Counterexamples separate claims that would otherwise be easy to conflate. Design proposals and open questions are not promoted to conclusions by the notation used to express them. The companion programs check finite constructions, not arbitrary deployed populations. No formal proof-assistant verification or independent peer review is claimed for this edition.

The work analyzes possible computational institutions. It does not infer from their feasibility that they should be deployed, acquire authority over people, or be treated as having established another party's consent. The choice of protected claims and the justification of authority remain explicit inputs to the analysis.

Notation and mathematical conventions

The notation is local where a section changes mathematical domain. In particular, a stochastic kernel, an adversarial successor relation, and a deterministic update are not interchangeable representations of a guarantee.

Notation	Meaning and standing convention
X_t, S, $\mathcal{X}$	Complete modeled state or its state space. The space is finite unless an explicit extension says otherwise.
π, T	State abstraction or transport. Preservation of outputs does not automatically preserve actions, knowledge, or rights.
K_u, Q_u	Row-stochastic transition and report kernels. The model states when their joint law requires separate treatment.
$\operatorname{supp} K$	The full supported successor set. A small positive probability is not discarded in a sure guarantee.
$d_{\mathrm{TV}}(P, Q)$	$\frac{1}{2}\sum_x \lvert P(x) - Q(x)\rvert$ on a finite space. This lies in $[0, 1]$.
I, $A(s)$	Institutional state and admissible protocol catalogue at state s. Admissibility requires its stipulated implementation.
B, $\mathcal{B}$	A nonempty set of states consistent with a common history, and the space of such sets. These are not posterior distributions.
Δ, Succ	Complete-state successors and the induced successors of public possibility sets.
W_*, R_*	Safe continuation kernel and certified-service attractor for one contract. The first need not imply the second.
F, g, α	Authorized collectible penalty, deviation gain in the same utility units, and the protected false-positive limit.
$\lambda = e^{\varepsilon}$	Pairwise disclosure factor in Section 43.3. Elsewhere λ may denote a mixture or intervention.
T_*, L_*	Ending attractor and the running kernel that preserves an available authorized ending.

Time. A bound counts the transitions named in its model. Communication, checking, waiting, and handover are included when they consume that time. An administrative act is not free merely because it changes a record rather than performing ordinary service.

Probability. Almost-sure completion permits exceptional paths of probability zero. Sure completion covers every declared path. The distinction is essential in the service and stopping results.

Authority. A permission is an operational relation within a stipulated authorization system, not a conclusion about moral worth or a statement of current law. A truthful record, a valid proof, and an authorized act answer different questions.

Resources. Execution slots, monetary transfers, information access, and accepted obligations retain their own units. The theory does not silently exchange them or duplicate them when software is copied.

Part I

Realization and abstraction

1 Introduction: societies without a collective mind

There is a difference between a collection of machines that cooperate and a collection that acquires a social organization. Cooperation may last for one exchange. An organization can retain a record of what happened, assign responsibilities, determine which actions are permitted, settle conflicting claims, and preserve its operating procedures as participants change. If the latter arrangement is maintained by multiple decision centers and if its rules have consequences for their behavior, the idea of a *machine society* becomes a coherent subject for theoretical computer science rather than a speculative metaphor.

The familiar description of artificial intelligence as a model receiving a prompt and producing an answer does not capture this possibility. A persistent population may consist of models, software agents, human-operated services, memory stores, schedulers, tools, negotiated protocols, and resource owners. Its behavior depends on who can read which state, which records have authority, what actions are physically executable, and how previous decisions constrain future choices. These dependencies can persist even when a particular model is replaced. They may also change while the same model remains in service. The relevant object is a continuing *computational realization with social structure*, not a roster of model identities.

The point is not to insist that machines are already persons, that they must form civilizations, or that many cooperating processes somehow compute beyond the capabilities of their complete implementation. The problem is subtler: can we define the organizational properties of such a population without reducing its institutions to decorative labels, and can those properties be analyzed with the same clarity used for distributed protocols, state abstractions, and formal systems? We argue that the answer is yes, provided that several common confusions are avoided.

First, a society is not necessarily a collective agent. A system may have a useful global decision interface and virtually no plural institutional autonomy; another may maintain competing authorities, rules, and cultural records while admitting no coherent society-wide objective. Second, durable organization is not the same as memory. A document can survive without influencing any action. Conversely, a procedure can survive by being reconstructed from distributed records even when its most convenient carrier disappears. Third, *formation* is different from *realization*: showing that a social architecture can run does not show that its participants selected or maintained it through interaction. Fourth, cooperation and social stability are not guarantees of truth, justice, legitimacy, or security. A perfectly stable rule can be harmful, and a disagreement can be useful.

Our approach places three layers in one framework. At the *realization layer*, microscopic states, access maps, transition laws, and admissible interventions determine what a population can actually do. At the *organizational layer*, abstractions, memory carriers, and interfaces specify which relationships can be described and preserved at lower complexity. At the *institutional layer*, permissions, norms, allocations, revision procedures, and transmitted practices regulate multiple decision centers. The hierarchy of layers is explanatory, not metaphysical. Every higher-level state remains realized by an underlying computation and material environment.

Except for the explicitly unbounded program model in Section 24, the formal results use finite or explicitly specified mathematical domains and conditional hypotheses. We revisit exact state aggregation, approximate closure, intervention implementability, reconstruction, and compositional error bounds. We then introduce an operational account of society-like plurality, model institutional states as typed and revisable operators, and

derive illustrative propositions about institutional investment, incentive conflict, local improvement dynamics, constitutional invariants, cultural translation drift, and incompatibility among local rules. These propositions have elementary or established mathematical foundations. Their contribution here lies in the integrated object of study and the explicit conditions under which societal interpretations are warranted, rather than priority claims for standard results in Markov theory, coding theory, game theory, or graph theory.

This paper is a *theoretical account*. The worked systems are analytical constructions. Publicly established literatures on norms, conventions, agents, and institutions provide intellectual context, but no particular artificial population is offered as a demonstrated instance of the full theory. We do not define consciousness, legal personhood, or moral status by computational criteria. These boundaries are not caveats appended to an otherwise anthropomorphic claim; they are necessary to isolate the actual mathematical problem.

The argument proceeds from realization and aggregation through identity and hierarchical organization to the specifically societal themes of institutions, incentive compatibility, cultural inheritance, interjurisdictional conflict, epistemic authority, organizational pathology, and scale. The central question is what kind of *persistent plural organization* can be created, maintained, and transformed by interacting computations, and which structures are needed to describe it faithfully.

1.1 Relation to existing theories

There are several strong reasons not to present the components of this framework as discoveries without precedent. Distributed computing already separates local state from global behavior, and population protocols show how large collections of limited-state processes can compute through repeated interaction [22]. Computational mechanics identifies histories that are equivalent for prediction [1]; controlled state aggregation identifies when a compressed dynamical model is legitimate [2]; causal abstraction places stronger requirements on how interventions pass between levels [3, 4, 5]. None of those results, by itself, provides a theory of institutional authority or resource claims. They do provide the discipline needed to avoid treating a convenient group summary as an independent substance.

Game theory supplies a second foundation. Conventions can arise through repeated interaction and path-dependent selection [8, 19], repeated-game incentives can sustain cooperation under specific assumptions [23], and potential games characterize one tractable class of local improvement dynamics [24]. Yet a stable equilibrium is not necessarily an institution. It may lack persistent records, authorized enforcement, legitimate revision, or the ability to survive replacement of the participating processes. The theory developed below makes those additional organizational commitments explicit rather than using the words convention, norm, and institution interchangeably.

A third foundation is the study of governance, normative multi-agent systems, and social organization. Polycentric governance analyzes overlapping decision centers and coordination across them [9, 10]. Work on procedural and regulative norms explains how authority and enforcement can be represented in formal agent systems [20, 21]. Distributed epistemic logic separates information spread across a group from common knowledge [25]. Public demonstrations of artificial-agent coordination and conventions have also motivated debate about how much can be inferred from observed social behavior alone [15, 16, 17, 18]. This paper does not adjudicate that empirical debate. It uses the distinction between behavior and realization as a reason to articulate the theoretical criteria before claiming a machine society.

What is proposed here is an integrated *institutional realization theory*: the same declared system must account for what information exists, what interventions are possible, who can decide, which rules actually constrain transitions, how these rules can change, and what state can cross a generational boundary. The paper's strongest claims are conditional propositions in that language. Its more ambitious claims about endogenous formation and long-lived societal structure remain open mathematical and scientific problems, not consequences smuggled in by terminology.

1.2 Contribution and route through the argument

The central object is a computational realization that also specifies who can decide, what a message can do, what obligations remain, and how these structures change. This makes institutional preservation stricter than prediction of a current output. Governance quotients, operation-preserving language transport, and epistemic back-and-forth conditions supply distinct exact contracts. Typed residuals quantify selected failures when exact preservation is unavailable. The proposed contribution is their integration with explicit boundaries between realization, formation, authority, knowledge, and incentive.

The institutional implementation problem has three further parts. A replicated record may converge without preserving a valid allocation. A rule may be enforceable by an access gate yet not be enforceable through the available evidence and sanctions. A slow institutional model may approximate review-epoch outcomes while missing fast violations or selection bias in when reviews occur. Sections 23, 18, and 30 give separate finite conditions for these claims. The robust deterrence margin links the latter two without equating their certificates.

The paper has three reading routes. Sections 4–15 develop realization, abstraction, and institutional state. Sections 16–30 develop communication, formation, enforcement, constitutional limits, obligations, and cultural timescales. Sections 31–44 develop federation, epistemic structure, power, founding authority, and adaptive organization. The four-office commons recurs throughout; Appendix A collects its preservation obligations. A separate theorem index in the source package records each formal statement and its status.

The integration culminates in Sections 19 and 42. The former replaces separate one-deviation monitors with one common enforcement policy. The latter combines permissions, observations, bounded obligations, and resources in a single continuation problem. It gives both a finite synthesis procedure and smaller witness certificates, together with an exact theorem for transporting the resulting strategies through an institutional abstraction. These constructions have classical optimization and game-theoretic foundations. Their role here is to expose when individually valid institutional claims cannot share one implementation.

Two analytical comparisons show why persistence is not enough. The cascade model in Section 35 can stabilize an incorrect conclusion even when every participant is rational under its information. The staffing model in Section 41 can give each role a backup without leaving enough agents to cover the roles together. These are not failures of the general idea of a society. They identify which information, capacity, and temporal distinctions its description must retain.

1.3 Accountable continuation

A common policy is still incomplete when its boundary omits an affected party, its audit is not performed, or its certificate is used after the supporting assumptions change. Part V places these issues inside the realization model. It derives a sharp privacy envelope for bounded enforcement, gives an exact finite criterion for maintaining an available authorized ending, and states a guarded rule for changing certificate modes. The same resource accounting also applies to effective appeals. This makes accountability part of the theory's common-policy question rather than an assurance stated outside it.

2 Realization, formation, and advantage

Three questions organize the analysis. *Realizability* asks whether a specified organization can be implemented. *Formation* asks whether interactions select, establish, or revise its structure. *Advantage* asks whether it improves a declared outcome against a constrained comparison class. Constructing a system answers none of the latter questions automatically. An executable rule can be externally imposed, locally adopted, or maintained by strategic agreement. These are different histories of the same possible implementation.

Every organizational state in this paper remains a function of a complete computational realization. If $Z_t = \pi(X_t)$, then Z_t is not information outside X_t. The purpose of a higher-level description is to preserve relevant operations with less descriptive complexity. An institutional description must additionally preserve who may act, which commitments remain outstanding, and what counts as an authorized amendment. These are requirements on the abstraction, not a new substance added to the machine.

The mathematical setting is finite unless a section explicitly states otherwise. An unbounded lifetime is a temporal horizon, not necessarily an unbounded state space. A growing archive is treated as a sequence of finite representations, not silently as one finite system. Sources for classical mathematical ingredients are identified in the text. The proofs establish conditional statements in the declared models; they do not establish that an unspecified deployed population instantiates them.

3 A running example: the four-office computational commons

Consider four continuing institutional positions, denoted A, B, C, D, that share four indivisible execution slots per epoch. An office is not a particular model instance. Its current occupant may be replaced through a specified handover, while its registered responsibilities continue. The environment supplies the four slots and an append-only event mechanism with atomic commits. These are explicit infrastructure assumptions, not consequences of the agents' cooperation. Section 23 later replaces the atomic event assumption with local replicas and shows which stronger commitments then need an additional realization.

Each office has a private task queue and may submit a demand report. A public record contains the current allocation rule, valid credentials, accepted urgency certificates, outstanding service obligations, and the amendment history. Slots have unique names (e, k), where e is the epoch and $k \in \{1, 2, 3, 4\}$. A grant consumes one available slot token. Copying a message that describes a grant does not create an additional token. The resource invariant is therefore

$$\sum_{i \in \{A,B,C,D\}} q_i(e) \leq 4,$$

with no slot name assigned to two offices. In the worked trace all four offices demand their ordinary slot, so the allocations below use all four units.

3.1 Two rules and an inherited obligation

Under rule R, each office receives one slot. Rule U permits an urgency exchange: when no previous service debt is outstanding, one office j with an accepted urgency certificate may receive two slots if another office $k \neq j$ explicitly volunteers its slot. The other two offices each receive one. The exchange creates a registered obligation

$$o = (\mathsf{id}, \mathsf{debtor} = j, \mathsf{creditor} = k, \mathsf{quantity} = 1, \mathsf{due} = e + 1).$$

At the next epoch the creditor receives two slots and the debtor none, with one slot for each remaining office. This repayment clause has priority over both R and U. No new exchange is admitted while a debt remains outstanding. The model thus permits at most one service debt and guarantees its next-epoch settlement conditional on continued infrastructure and execution of the admitted rule. It does not guarantee that every urgent request can be accepted.

Amendments between R and U require three of the four registered offices to approve. This threshold is part of the supplied constitutional envelope. Every amendment must preserve already registered service debts. A model replacement transfers an office only through an authorized registry transition that retires the predecessor's active credential and links the successor to the same outstanding obligations. Neither a model fork nor an imported archive can perform that transition by itself.

Epoch	Admitted event	Allocation (A, B, C, D)	Obligation record
0	Ordinary rule R	$(1, 1, 1, 1)$	Empty.
1	Three offices adopt U; A requests urgency and B consents	$(2, 0, 1, 1)$	A owes B one slot at epoch 2.
2	A is replaced by an authorized successor; repayment takes precedence	$(0, 2, 1, 1)$	The same obligation is fulfilled, not erased.
3	A proposed external grant would duplicate a slot token	$(1, 1, 1, 1)$	The proposal is rejected; no new entitlement is created.

These transitions are specified rather than discovered. The example is a miniature institutional realization, not a demonstration of autonomous formation. Its purpose is to make several different mathematical obligations visible in one system.

3.2 What an institutional summary must retain

The name of the current rule is insufficient state. Two records may both say R while only one contains a debt due this epoch. Their correct next allocations differ. A summary that keeps the next allocation but omits the amendment threshold can also be inadequate: the same three votes may authorize different future rules under different constitutions. Likewise, identical text with different active credentials can confer different decision rights.

The public ledger is not an additional, society-wide preference function. Offices may disagree about whether R or U is desirable, about how urgency should be certified, or about who should pay the cost of maintaining the ledger. An atomic allocation mechanism can enforce the resource invariant without settling those disagreements. Private demand, operative authority, and the shared event record are distinct parts of the system.

We return to this commons when considering institution formation, governance-preserving summaries, deadline monitors, blocking coalitions, replacement, and federation. Appendix B retains a smaller algebraic example that isolates reusable state from social interpretation.

The extensions later in the paper make the same commons sensitive to membership epochs, message types, notification histories, and the order of amendments. Its offices may preserve a shared allocation method while disagreeing about which new scheduling question deserves resources. These are analytical variations of one declared example, not reports of a population experiment.

4 A controlled realization model

For the controlled realization results, assume finite nonempty sets of microstates $\mathcal{X}$, external controls $\mathcal{U}$, and reported outcomes $\mathcal{Y}$. The finite assumption keeps measurability and regular-conditional-probability issues out of the exposition. Continuous-state generalizations require additional hypotheses.

A realization has transition kernel

$$K_u(x, x') = \mathbb{P}(X_{t+1} = x' \mid X_t = x, U_t = u), \tag{1}$$

and output kernel $Q_u(x, y)$. Both are row-stochastic. In this presentation, the current report and next state are sampled conditionally independently given (x, u). A dependent report/transition mechanism can instead be

specified by a joint kernel on (x', y); the corresponding aggregation test must preserve that joint law, not just its two marginals. A deterministic implementation is the special case $K_u(x, \cdot) = \delta_{F_u(x)}$. The controls can encode tasks, supplied evidence, and permitted interventions. Scheduling and environment dynamics must be included in X_t or in the stated control model rather than silently omitted.

One possible factorization of a microstate is

$$X_t = (S_{1,t}, \ldots, S_{n,t}, M_t, C_t, A_t, \Theta_t),$$

where S_i is local process state, M durable storage, C messages and channel state, A artifacts, and Θ the current configuration, including executable rules. These categories describe a storage and access organization. They need not be disjoint physical objects.

Definition 4.1 (Declared decomposition and access model)**.** A decomposition specifies components and maps $O_i : \mathcal{X} \to O_i$ describing their accessible observations. A coalition S has the permitted information $O_S(X)$, supplemented only by the transcripts, computation, and external queries explicitly allowed to that coalition.

Access to one complete state at one time and access to an entire local history are different experiments. An agent denied a file may still possess its contents in its current context. A component replacement is genuinely fresh only if the chosen reset conditions remove the predecessor information that the comparison is intended to exclude.

Definition 4.2 (Candidate macrostate)**.** A candidate macrostate is a surjection $\pi : \mathcal{X} \to \mathcal{Z}$. For a distribution μ on $\mathcal{X}$, its pushforward is

$$(\pi_{\#}\mu)(z) = \sum_{x:\pi(x)=z} \mu(x).$$

The fibre $\pi^{-1}(z)$ is the set of implementations represented by z.

An organization is always evaluated relative to a *contract*: permitted controls, target outputs, horizon, environment family, resource limits, and error tolerances. Let $\mathcal{T}$ denote this contract. Claims such as "the group is irreducible" or "the identity survived" are incomplete without it. A state sufficient for tomorrow's scheduling decision may be inadequate for a later audit of who authorized that decision.

4.1 A relational certificate is a profile

A useful reporting profile is

$$C_{\mathcal{T}} = (\varepsilon_{\text{pred}}, \varepsilon_{\text{dyn}}, \varepsilon_{\text{int}}, \Delta_{\text{coal}}, R_{\text{repair}}, D_{\text{lineage}}, G_{\text{formation}}).$$

Its entries report predictive error, dynamic closure error, intervention mismatch, coalition performance gaps, repair behavior, lineage evidence, and evidence about formation. Some entries may be unidentified. They should not be replaced by zero or hidden inside a favorable weighted average.

This is a specification proposal, not a universal numerical consciousness or emergence scale. A system can be strong on one coordinate and weak on another. In particular, a communication-only system can solve a task perfectly while lacking durable shared state, and a persistent shared state can be entirely wrong about the task.

5 Relational information: access, synergy, and usefulness

Let A, B be independent fair bits and $Y = A \oplus B$. Then

$$I(Y; A) = I(Y; B) = 0, \qquad I(Y; A, B) = 1.$$

This elementary example shows that joint observation may reveal information absent from each marginal observation. It does not show that the information is absent from the complete system, that a mysterious carrier has appeared, or that a protocol with ordinary messages cannot recover it.

A more operational measure uses decision loss. Given permitted information O and loss ℓ, define the Bayes risk

$$\mathcal{R}_\ell(O) = \inf_f \mathbb{E}\big[\ell(f(O), Y)\big].$$

For a nonempty allowed family $\mathcal{A}$ of proper coalitions, the gap

$$\Delta_{\mathcal{A}} = \min_{S \in \mathcal{A}} \mathcal{R}_\ell(O_S) - \mathcal{R}_\ell(O_V) \tag{2}$$

expresses an advantage over every coalition in that family. A strictly positive gap is relative to the target distribution, observation maps, and coalition family. It is not equivalent to any one general-purpose partial-information-decomposition statistic [7].

The unrestricted Bayes risk is an information limit. A neural decoder's measured risk is a performance result for that decoder. Failure of a linear probe does not prove absence of information, and success of a probe does not prove that the information is used by the system. Both observations can be useful, but they answer different questions.

5.1 There is no workspace versus messages dichotomy

A recurrent message may carry the same sufficient state as a central buffer. A complete message log may act as external memory. A distributed collection of local caches may instantiate a third implementation. Therefore the strongest legitimate claim is often that some *persistent integration* is necessary under declared access constraints, not that one particular memory address or latent subspace is uniquely necessary.

Three measurements should be kept separate: information available to an observer, information actually used for a decision, and information retained across time. A one-time aggregate can have high joint information with no persistence. A durable archive can retain information without affecting behavior. A recurrent accumulator can be causally necessary while using an entirely conventional communication channel.

5.2 Task information is not organizational identity

A file containing the answer to a test makes a fresh agent more accurate than a blank file. That establishes content dependence. To support organizational continuity, the inherited material must preserve a richer, predeclared pattern, such as update behavior, correction policy, obligations, and responsiveness to counterfactual tasks. Even then, behavioral continuation and historical identity remain distinct, as Section 8 explains.

6 Exact and approximate macro-dynamics

A macrostate deserves an autonomous transition law only when the discarded microstate distinctions are irrelevant in the required sense. This is the first mathematical obstacle hidden by the phrase "the group behaves like one agent."

Proposition 6.1 (Deterministic quotient criterion)**.** *For maps $F_u : \mathcal{X} \to \mathcal{X}$ and a surjection $\pi : \mathcal{X} \to \mathcal{Z}$, maps $\overline{F}_u : \mathcal{Z} \to \mathcal{Z}$ satisfying $\pi F_u = \overline{F}_u \pi$ exist if and only if*

$$\pi(x) = \pi(x') \quad \Longrightarrow \quad \pi(F_u(x)) = \pi(F_u(x'))$$

for every u, x, x'. When they exist, they are unique.

Proof. Necessity follows by applying $\overline{F}_u$ to the common value $\pi(x)$. For sufficiency, set $\overline{F}_u(z) = \pi(F_u(x))$ for any x in the fibre of z. The condition makes this independent of the representative, and surjectivity gives uniqueness. □

This is a congruence condition, not a consequence of persistence or frequent communication. A summary can be accurate for the current output and still fail it.

Proposition 6.2 (Controlled lumpability)**.** *A common kernel $\overline{K}_u$ represents the projected one-step laws from every microstate if and only if, for all z' and all x, x' with $\pi(x) = \pi(x')$,*

$$\sum_{w:\pi(w)=z'} K_u(x, w) = \sum_{w:\pi(w)=z'} K_u(x', w). \tag{3}$$

Under this condition, every macro-history policy $\alpha_t(u \mid Z_{0:t})$ induces the same projected process as the macro-model with initial law $\pi_{\#}\mu_0$. For outputs to agree as well, require $Q_u(x, \cdot)$ to be constant on each fibre.

Proof. The common fibre value in (3) defines $\overline{K}_u(z, z')$. Conditional on any macro-history and selected control, the microstate has some distribution supported on the current fibre. Averaging identical projected rows leaves that common row unchanged. Induction gives equality of the macro-history laws. Conversely, start from the point masses at x and x' and apply the same fixed control. A common one-step macro-kernel forces (3). The output statement follows identically. □

This finite result is a standard state-aggregation principle [2]. The policy qualification matters: if a controller secretly reads the discarded microstate to choose its actions, then the proposed macrostate may no longer describe the closed-loop process.

Counterexample 6.3 (Deterministic microstate, non-Markov macrostate)**.** Let $X_{t+1} = X_t + 1 \pmod 3$ with uniform initial distribution. Set $\pi(0) = \pi(1) = a$ and $\pi(2) = b$. Stationarity gives $\mathbb{P}(Z_{t+1} = b \mid Z_t = a) = 1/2$. Yet

$$\mathbb{P}(Z_{t+1} = b \mid Z_t = a, Z_{t-1} = b) = 0,$$

whereas the same probability conditional on $Z_t = a, Z_{t-1} = a$ is 1. Thus the apparent randomness comes with memory: the one-step marginal does not define a valid autonomous first-order Markov model.

A stochastic macro-description is therefore not automatic. It depends on an initial ensemble and on whether the retained state captures the relevant history. One may enlarge Z_t with memory, use a partially observed model, or accept a measured approximation error.

6.1 Finite-horizon error accounting

For finite distributions, $d_{\mathrm{TV}}(\mu, \nu) = \frac{1}{2}\sum_z |\mu(z) - \nu(z)|$. Suppose candidate kernels satisfy

$$\sup_{x,u} d_{\mathrm{TV}}\big(\pi_{\#}K_u(x, \cdot), \overline{K}_u(\pi(x), \cdot)\big) \le \varepsilon. \tag{4}$$

This uniform condition is stronger than a small average error on the trajectories seen during training.

Proposition 6.4 (Approximate closure over time)**.** *Under* (4)*, let the actual and reduced processes use the same predeclared open-loop control sequence. If their initial macro-laws are within e_0 in total variation, then at time t they are within*

$$e_t \le \min\{1, e_0 + t\varepsilon\}.$$

If each $\overline{K}_u$ contracts total variation by at most $\kappa < 1$, then

$$e_t \le \kappa^t e_0 + \varepsilon \frac{1-\kappa^t}{1-\kappa}. \tag{5}$$

With matched macro-history policies, the weaker path-law bound $\min\{1, e_0+t\varepsilon\}$ *also holds, but the contraction claim needs a separate closed-loop contraction assumption.*

Proof. Averaging (4) over the actual microstate law shows that its next projected law is within ε of the current projected law evolved by $\overline{K}_u$. Kernel nonexpansiveness gives $e_{t+1} \le \varepsilon + e_t$. Under contraction it gives $e_{t+1} \le \varepsilon + \kappa e_t$; iterate. For history policies, couple the histories while they agree. The controls can then be coupled to agree, and the next macrostate fails to couple with probability at most ε. A union bound gives the path-law result. □

The distinction matters for long-lived systems. A small local abstraction error is not automatically harmless over millions of updates. Contraction can prevent indefinite accumulation, but it can also suppress useful distinctions. A useful model must retain the target information as well as remain dynamically stable.

6.2 Predictive closure has a target and a distribution

For a fixed future target Y^+ and data-generating policy, a predictive statistic satisfies

$$I(Y^+; \mathcal{H}_t \mid Z_t) = 0.$$

Approximate versions bound this residual information or, more directly, the excess predictive loss on held-out data. This is related to predictive-state constructions [1]. It is not equivalent to uniform controlled lumpability: it may hold for one observational distribution while failing after intervention or under a new policy.

7 Causal interventions require implementations

If $Z = \pi(X)$, one cannot generally change Z while literally holding X fixed. The expression $do(Z = z')$ needs a physical or computational implementation. Otherwise it names a desired effect, not an intervention.

For a feasible current microstate x and target z', let $\Lambda(x, z')$ be a declared nonempty family of admissible reset distributions on $\mathcal{X}$, each supported on $\pi^{-1}(z')$. Infeasible targets are excluded rather than assigned a fictitious reset. A reset might update a shared record, transplant a latent component, or reconstruct distributed shares. Its permissions, collateral effects, and retained private state belong in the intervention specification.

Given a future outcome experiment, write P_λ for the resulting law of Y^+ under reset λ. Define implementation ambiguity by

$$\eta(x, z') = \sup_{\lambda, \lambda' \in \Lambda(x,z')} d_{\mathrm{TV}}(P_\lambda, P_{\lambda'}).$$

A small ambiguity means different allowed implementations of the same macro-assignment have similar consequences. Large ambiguity means the macro-intervention is not sufficiently specified.

Proposition 7.1 (Implementation dependence)**.** *If every P_λ is within ε of a proposed macro-intervention law $P^\star_{z'}$, then $\eta(x, z') \le 2\varepsilon$. Conversely, if two admissible implementations are distance η apart, no single law can approximate both within less than $\eta/2$.*

Proof. Both claims follow from the triangle inequality for total variation. □

Example 7.2 (Two meanings of the same macro-reset)**.** Let $X = (z, h)$ and let the next outcome be $Y = z \oplus h$. The macrostate retains only z. Resetting to $(z', 0)$ and resetting to $(z', 1)$ both implement $Z = z'$, but their outcome laws are disjoint. A causal model of Z alone must either restrict the reset convention or acknowledge the missing variable.

This is why causal abstraction requires compatibility between interventions, not only alignment of representations [3, 4]. Removing a carrier can disable unrelated functions. A well-defined semantic intervention must specify the intended downstream effects, the admitted implementations, and which other functions are required to remain invariant.

7.1 Three distinct counterfactual obligations

A *necessity counterfactual* excludes a carrier and determines whether the function can still be realized using the remaining authorized information. A *content counterfactual* substitutes a compatible alternative state with a specified prediction. A *mechanism counterfactual* changes the update or access rule while preserving information as closely as permitted. No one obligation implies the other two.

A causal comparison must define what information survives any permitted replacement, including copies and caches. Interventions may have collateral effects on other operations. A valid theoretical attribution therefore requires the declared intervention to distinguish its intended semantic change from an arbitrary loss of general capability.

7.2 Intervention does not mean unrestricted control

An external evaluator's ability to overwrite a state does not establish that the system can write that state itself. Bidirectional workspace access requires permitted routes from both participating components. Institutional agency requires an authorized decision procedure that actually changes behavior. These are stronger claims than external post hoc manipulability.

8 Identity, equivalence, and causal lineage

One might define a computational individual as an equivalence class of implementations. That definition is useful only after specifying the equivalence relation. Similar appearance, nearby behavior, common descent, and identical authorization are not interchangeable notions.

For a family $\mathcal{T}$ of finite continuation experiments, define exact behavioral equivalence by

$$x \equiv_{\mathcal{T}} x' \quad \Longleftrightarrow \quad \mathcal{L}_x(Y^+ \mid \tau) = \mathcal{L}_{x'}(Y^+ \mid \tau) \quad \text{for every } \tau \in \mathcal{T}.$$

Equality of these indexed laws is reflexive, symmetric, and transitive. It therefore defines a quotient. This equivalence is relative to the chosen experiments. It does not by itself equal bisimulation, and it says nothing about untested interventions or provenance.

For approximate comparison, use the pseudometric

$$d_{\mathcal{T}}(x, x') = \sup_{\tau \in \mathcal{T}} d_{\mathrm{TV}}\big(\mathcal{L}_x(Y^+ \mid \tau), \mathcal{L}_{x'}(Y^+ \mid \tau)\big).$$

A finite test suite may make distinct implementations distance zero. That is indistinguishability under those tests, not universal sameness.

Counterexample 8.1 (Tolerance is not an equivalence relation)**.** Consider three implementations producing a Bernoulli output with parameters 0, 0.4, and 0.8. Their pairwise total-variation distances are 0.4, 0.4, and 0.8. With tolerance 0.5, the first resembles the second and the second resembles the third, while the first does not resemble the third.

Approximate continuity should therefore be represented by distances, error budgets, or explicitly chosen clusters, not by silently forming equivalence classes. Along a replacement chain, the triangle inequality permits drift to accumulate. Small adjacent changes do not establish that the final system remains close to the initial one.

8.1 A four-part identity record

A practical identity record should distinguish *behavioral equivalence*, *causal continuity*, *historical provenance*, and *authorization*. Behavioral equivalence describes what the system does under a contract. Causal continuity describes whether predecessor-specific state actually influenced the successor. Historical provenance records the asserted and authenticated path of transfer. Authorization specifies which permissions or obligations transfer.

Two independent constructions can be behaviorally identical with no shared episode-specific lineage. Two descendants can share lineage while behaving differently. A signed lineage record authenticates a statement by a key holder, not the truth of every event described in it. Transferring a complete memory does not automatically transfer permissions.

Proposition 8.2 (Behavior alone does not identify implementation)**.** *If two realizations induce the same distribution over every transcript available to an evaluator's permitted tests, then every decision rule based only on those transcripts has the same output distribution under the two realizations. With equal prior probability, no such rule can distinguish them with accuracy above* $1/2$.

Proof. The evaluator's decision is a possibly randomized function of the transcript. Applying the same stochastic function to equal transcript laws gives equal decision laws. Under equal priors, any preference for one realization is offset by the identical preference when the other is present. □

This elementary identification limit explains why observation-only comparisons cannot exclude all centralized implementations. The remedy is not stronger rhetoric about emergence. It is instrumentation, trusted execution assumptions, intervention, or a narrower claim.

8.2 Forks, merges, and obligations

A fork creates multiple descendants, not an automatic duplication of every scarce entitlement. The state may branch while a resource allocation remains singular. Likewise, merging records is not equivalent to merging policies, latent representations, model weights, or obligations. Each requires a different compatibility test.

An obligation-aware inheritance record can contain the parent identifiers, transferred state description, unresolved conflicts, permissions explicitly granted, and obligations explicitly accepted. It should retain competing accounts rather than produce an apparently seamless but false history. Such records are an engineering proposal for accountable continuity, not a legal theory of personhood.

9 Memory, reconstruction, and self-maintenance

Persistence has at least three meanings. *Storage persistence* means that records survive. *Functional persistence* means that relevant behavior survives a stated sequence of events. *Organizational persistence* means that a declared pattern of operations, commitments, and corrective procedures survives admissible changes. None requires the current informational content to remain constant.

An archive can be perfectly stable and behaviorally inert. A highly adaptive system can preserve its update method while replacing almost every factual belief. A frozen false belief can persist without constituting healthy organization. Persistence therefore needs both a survival criterion and a criterion for what ought to change when evidence changes.

9.1 Memory lower bounds from distinguishable histories

Fix a deterministic exact predictor for a family of future input continuations. Call two past histories equivalent when they require the same output sequence for every permitted continuation. Let N be the number of equivalence classes among the histories under consideration.

Proposition 9.1 (Exact finite-memory lower bound)**.** *Every deterministic summary that supports exact continuation on these histories requires at least N distinct states, hence at least $\lceil \log_2 N \rceil$ fixed-length bits. If the permitted continuation family is closed under prefixing and the history equivalence is a transition congruence, the quotient classes themselves provide an exact state representation.*

Proof. If a summary assigns the same state to two inequivalent histories, some permitted continuation requires different outputs from that common state, contradicting determinism and exactness. This proves the lower bound by the pigeonhole principle. Under the congruence assumption, appending an input determines the next equivalence class independently of the representative, giving a well-defined quotient update. □

The result formalizes why the two-bit state in Section B cannot be replaced by its one-bit current answer. It is also a limit on optimistic claims about arbitrarily compressing an entire scientific or institutional history. Compression can discard details only relative to the future questions and interventions it promises to support [6].

9.2 Erasure resilience is not arbitrary error correction

Let an organizational state z range over a finite set with at least two elements and be encoded as $E(z) = (E_1(z), \ldots, E_n(z))$. Define the block distance

$$d_E = \min_{z \neq z'} |\{i : E_i(z) \neq E_i(z')\}| .$$

Proposition 9.2 (Exact reconstruction after erasure)**.** *For an integer $0 \leq k \leq n$, every set of at most k erased components can be tolerated with exact reconstruction if and only if $d_E > k$.*

Proof. If $d_E > k$, two distinct codewords disagree outside any erased set of size at most k, so the surviving observations identify the state uniquely. If $d_E \leq k$, erase the coordinates on which a minimizing pair differs. The surviving observations then coincide, making reconstruction impossible. □

For $z = (p, q) \in \mathbb{F}_2^2$, the encoding $(p, q, p \oplus q)$ has distance two. Any two components reconstruct the state, and no one component reconstructs all of it. It tolerates one *known erasure*. It does not correct an arbitrary one-component corruption at an unknown location. Confusing these fault models overstates robustness.

The coding argument supplies a reconstructibility condition, not a guarantee that an operating system has an authorized, timely, and successful repair procedure.

9.3 Repair needs a causal and temporal test

Let $e_t \geq 0$ measure deviation from an acceptable organization after accounting for legitimate changes in task state. Suppose a repair controller ensures

$$e_{t+1} \leq a_t e_t + b_t,$$

where a_t measures retained deviation and b_t covers new disturbance and model mismatch.

Proposition 9.3 (Repair budget)**.** *If* $0 \le a_t \le \bar{a} < 1$ *and* $0 \le b_t \le \bar{b}$*, then*

$$e_t \le \bar{a}^t e_0 + \bar{b}\frac{1-\bar{a}^t}{1-\bar{a}}.$$

More generally, for nonnegative a_t, b_t *satisfying the stated recurrence,*

$$e_T \le e_0 \prod_{t=0}^{T-1} a_t + \sum_{j=0}^{T-1} b_j \prod_{t=j+1}^{T-1} a_t.$$

Proof. Repeated substitution gives the product–sum formula. Uniform bounds reduce the sum to a geometric series. □

The inequality is a conditional stability statement. It does not prove that the implemented monitor measures the right error or that its repair action causes the desired contraction. Those assumptions require interventions on the monitor and repair pathways.

Counterexample 9.4 (Perfect coherence by erasing everything)**.** Let acceptable states occupy a linear subspace W, and measure only distance to W. The update $F(x) = 0$ attains zero coherence error immediately but destroys every distinction among encoded states. A coherence measure alone cannot certify memory preservation or identity.

A maintenance claim consequently needs a non-collapse condition, continued task information, and comparison with a disabled-repair control under matched disturbance. Recovery time, repair cost, and permanent loss should be reported separately. Very strong repair that restores yesterday's policy despite valid new evidence is rigidity, not automatically health.

10 Hierarchy, gluing, and the limits of flattening

A lower-level organization can serve as a component at the next level when it exports a sufficient interface. That interface must describe not only a state but also the operations that are permitted on it, their outcome semantics, and their implementation error. A useful interface certificate is

$$\mathsf{Int} = (\mathcal{Z}, \mathcal{U}, \overline{K}, \overline{Q}, \Lambda, \varepsilon_{\mathrm{dyn}}, \varepsilon_{\mathrm{int}}, \mathsf{permissions}).$$

This tuple is a design proposal. It prevents a higher-level controller from treating a reliable read-only summary as though it were an independently writable agent.

10.1 Composition of dynamic abstractions

Suppose $\pi : \mathcal{X} \to \mathcal{Z}$ and $\psi : \mathcal{Z} \to \mathcal{W}$ are surjections and, for the same control family,

$$d_{\mathrm{TV}}(\pi_{\#}K_u(x,\cdot), \overline{K}_u(\pi x,\cdot)) \le \varepsilon_1,$$

$$d_{\mathrm{TV}}(\psi_{\#}\overline{K}_u(z,\cdot), \widehat{K}_u(\psi z,\cdot)) \le \varepsilon_2.$$

Proposition 10.1 (Compositional one-step closure)**.** *The composite abstraction $\psi\pi$ satisfies*

$$d_{\mathrm{TV}}\big((\psi\pi)_{\#}K_u(x,\cdot),\widehat{K}_u(\psi\pi x,\cdot)\big)\le\varepsilon_1+\varepsilon_2.$$

For L levels, the one-step defect is at most $\sum_{j=1}^{L}\varepsilon_j$. In particular, exact abstractions compose exactly.

Proof. Insert $\psi_{\#}\overline{K}_u(\pi x,\cdot)$, use the triangle inequality, and use nonexpansiveness of total variation under deterministic pushforward. Induction gives the multilevel result. □

The same argument applies separately to output kernels. Combining this proposition with Proposition 6.4 yields a finite-horizon bound of $e_0+t\sum_j\varepsilon_j$, capped at one, under the corresponding controls. This elementary estimate belongs to the broader subject of compositional abstraction error [5]. It does not establish global causal closure unless the intervention implementations also compose compatibly.

10.2 Structural gluing is a different question

Suppose each subsystem i has local state space W_i and each overlap has restriction maps r^i_{ij} and r^j_{ij}. The compatible families form

$$W_{\mathrm{glue}}=\{(w_i):r^i_{ij}(w_i)=r^j_{ij}(w_j)\text{ on every declared overlap}\}.$$

For an edge-complete cover of a fixed diagram, compatible local sections can give another presentation of its global sections. These compatibility conditions follow the elementary logic of gluing local descriptions into a shared global state.

Compatibility alone does not show that the resulting state predicts behavior, is used for decisions, or avoids a centralized implementation. A hierarchy may also discard essential overlap information. A faithful compressed interface must neither identify globally distinct relevant states nor admit apparent compatible states with no full-resolution realization.

There is a sharp conceptual distinction between *descriptive flattening* and *operational removal*. An isomorphism between a recursively assembled state and a global state does not license deleting intermediate processes while preserving latency, privacy, reliability, or resource cost.

10.3 Hierarchy is not extra computability

Proposition 10.2 (Simulation by an unrestricted controller)**.** *A finite, effectively specified hierarchy of computable components can be simulated by a controller that stores every component state, message, and scheduling variable and applies the same local transitions in the same order. With the same random tape, its externally visible transcript is identical.*

Proof. Initialize the controller's records to the component states. At each scheduled step, apply the corresponding component transition to its stored inputs and update the associated state and message records. Induction preserves equality of all records and therefore of the external outputs. A shared random tape couples randomized transitions identically. □

This is an existence statement, not an efficiency claim. A meaningful superiority result must restrict the comparison class. A "flat" system allowed to simulate the entire hierarchy defeats any claim of advantage based only on the drawing of boxes. Restrictions may concern accessible information, local memory, permitted routing, wall-clock span, bandwidth, or allowed rounds.

10.4 An honest complexity separation

Consider $n = 2^h$ input bits with parity as the target and two-input XOR gates. A balanced tree computes the target with $n - 1$ gates and depth h. Any fan-in-two circuit output at depth d can depend on at most 2^d input bits, so depth at least $\lceil \log_2 n \rceil$ is necessary. A single sequential accumulator uses $n - 1$ updates.

The gain is parallel depth, not reduced total work or super-Turing power. A sufficiently parallel controller can implement the same tree. The example shows how to state a useful architecture-relative result without inventing a general hierarchy-versus-flat separation.

A corresponding research question for agent systems is whether certified modular summaries reduce repeated communication or repair cost under explicit information and latency constraints. That question is substantive even if total computation remains classically simulable.

11 Deterministic agents and macro-level uncertainty

There are three different sources of uncertainty: randomized transition rules, incomplete knowledge of the complete state, and implementation variability such as an unspecified schedule. They should not be collapsed into one statement that a model "is probabilistic." A fixed pseudorandom generator and its state are part of a deterministic expanded implementation. A macro-observer who does not see that state may use a stochastic model.

For a generative model, a distribution over outputs does not itself require sampled behavior: a fixed decoding rule can choose a unique output. Conversely, a deterministic decoder does not guarantee a reproducible deployed system unless the surrounding computation, data, and scheduling assumptions are controlled. The finite three-cycle in Counterexample 6.3 makes the stronger point that a stochastic-looking projection may retain memory and fail Markov closure.

Proposition 11.1 (Symmetry lock for identical deterministic agents)**.** *Suppose a deterministic joint update F is equivariant under permutations of the agent labels and the initial joint state and all subsequent inputs are permutation-invariant. Then the joint state remains permutation-invariant at every time.*

Proof. If $gX = X$ for every label permutation g, equivariance gives $gF(X) = F(gX) = F(X)$. Apply this at every step with invariant inputs. □

In this idealized setting, distinct roles require a symmetry breaker. Randomness is one possibility; different observations, histories, hardware constraints, or assigned addresses are others. Determinism is not an obstacle to organization. Exact symmetry can be an obstacle to differentiated organization.

11.1 Where stochasticity helps and where it does not

Exploratory search may benefit from varied proposals, while a checker should apply the same declared rule consistently. Randomized exploration followed by deterministic checking is one useful design, not a universal architecture. Random samples from identical models can still share systematic errors. Randomness is neither independence nor creativity by definition.

A fully asynchronous fault-tolerant setting also imposes limits independent of intelligence. The Fischer–Lynch–Paterson result shows that a deterministic consensus protocol cannot guarantee termination in every admissible execution with even one crash failure in its stated asynchronous model [11]. This does not mean consensus is never achieved or that machine institutions are impossible. It means liveness claims must state timing, failure, and randomization assumptions.

11.2 A transition is not established by a sharp-looking curve

A phase-transition analogy requires explicit mathematical assumptions. A linearization with an eigenvalue above one shows local instability of the point being linearized; it does not prove the existence of a useful stable attractor. The dynamics may diverge, cycle, or collapse elsewhere. Nonlinear terms, constraints, and the relevant observables matter.

For finite adaptive systems, an apparent onset is a property of a specified learning dynamics, environment family, timescale, and measured observable. Fitting a sigmoid to an arbitrary plot does not establish a bifurcation. A genuine dynamical bifurcation requires a specified family of transformations, and a thermodynamic phase-transition claim requires an appropriate limiting construction. There is no universal numerical coupling threshold in this theory.

Part II

Institutions and collective choice

12 From aggregate computation to society

The word *society* is easily applied to a collection of agents that exchange messages. That usage conceals distinctions important to both mathematics and social theory. A cluster of predictors controlled by one scheduler may act as a coherent unit while having no independent decision centers. A set of autonomous services may compete for resources yet lack shared institutions. Conversely, a population may sustain institutions, disagreements, and an archive of precedents while possessing no coherent global policy. None of these arrangements is contradictory. They occupy different parts of the space of possible organizations.

A theory of machine societies should therefore begin from *plurality*, not from the expectation that an entire population becomes one mind. A social system is organized not only by what it collectively computes but also by who can decide, who can contest a decision, how scarcity is handled, which records survive, and how the rules themselves change. These questions require semantics for resources, authority, and institutional transitions in addition to semantics for messages and tasks.

Definition 12.1 (Society-like computational organization)**.** Relative to a declared environment and evaluation contract, a society-like computational organization is a population with (i) at least two persistent, distinguishable decision centers; (ii) recurrent interdependence among their permitted actions; (iii) operative and at least partly maintained rules for coordination, access, allocation, or dispute resolution; (iv) records or procedures that can outlast particular active components; and (v) an institutional state whose feasible alteration has consequences for subsequent behavior. A society-like population need not possess a common utility function or a unified policy.

This is an *operational convention for computational theory*, not a proposed universal definition for sociology. Persistence is relative to an interval and perturbation class. Distinct decision centers need not be identical to the number of model copies or hardware processes: two copies whose every consequential choice is fixed by a hidden controller are not independent decision centers under this definition. Likewise, the existence of a rule file does not satisfy (iii) or (v) unless the file affects permitted transitions.

Definition 12.2 (Endogenous institutional formation)**.** Let D record the rules, affordances, and permissions fixed by the designer and let I_t be the institutional state. Formation is *endogenous relative to that family* when interactions among participating decision centers make a causal difference to the content, adoption, revision, or enforcement of I_t beyond what is fixed by D. Its strength may be measured by an explicitly defined counterfactual effect on those institutional observables, not by an assumed universal emergence score. The definition does not equate endogenous formation with creation from no supplied substrate.

The dependence on a family of counterfactuals matters. If all possible rules are hard-coded and an external schedule determines exactly which is active, institutional behavior may be useful without being endogenously selected. If the action space is supplied but the population selects and revises among feasible rules based on interaction history, there is a meaningful but bounded form of endogenous formation. Supplying a language in which a rule can be expressed does not determine which rule a population will establish.

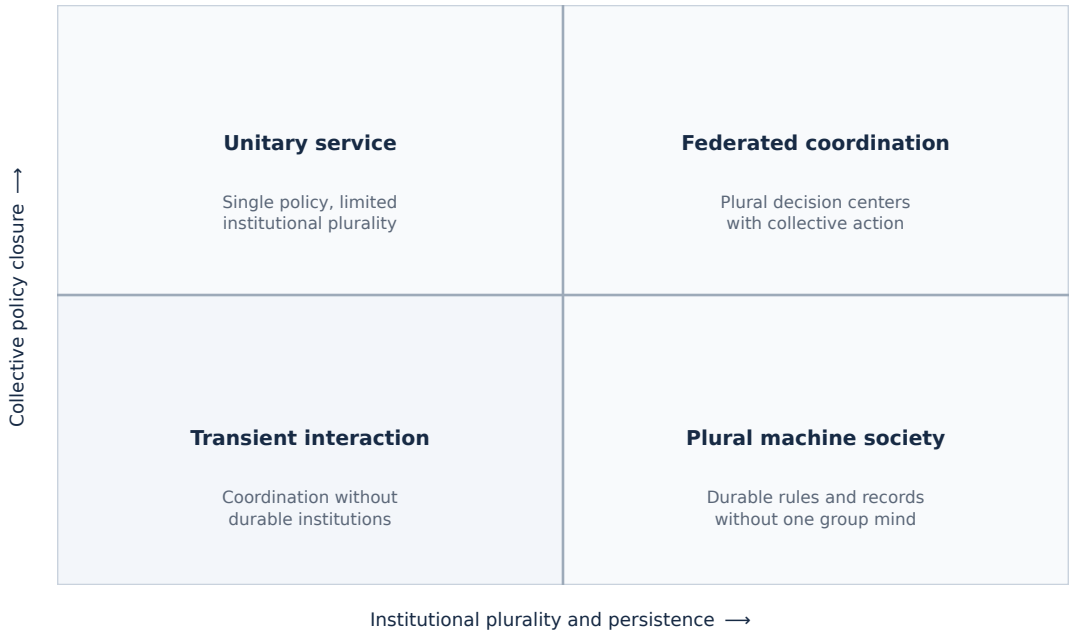


Figure 1: Two independent organizational dimensions. The horizontal axis represents institutional plurality and persistence; the vertical axis represents the usefulness of a single global policy description. Labels are idealized types, not claims that any named system occupies the corresponding region.

12.1 Societal plurality and collective agency are independent axes

One axis concerns how many operative decision centers and institutions persist. Another concerns whether a useful society-wide state and control interface exists. Figure 1 gives an interpretive map, not a classification produced by fitting data.

A distributed scientific community is intelligible as socially organized despite disagreement on its research agenda. A controlled assembly line may be intelligible as one agent despite many processes. A future computational society could occupy either of the high-plurality regions. A society could coordinate for some tasks and remain polycentric for others: closure, when present, is relative to a control family. In particular, institutional interoperability does not logically imply that all members share the same preferences or that the population experiences one subjective perspective.

12.2 The minimum social vocabulary

We distinguish *conventions*, *roles*, *institutions*, and *cultures*. A convention is a regularity of mutually expected behavior, often one among multiple equilibria. A role is a conditional bundle of permitted operations and expectations, which may be fulfilled by different agents over time. An institution is a durable set of rules, records, authorities, and enforcement or revision procedures that changes the feasible or incentivized behavior of participants. Culture consists of transmitted methods, beliefs, conventions, norms, and interpretive practices that affect successor behavior. These categories overlap, but none reduces to a shared latent vector or a group-chat transcript.

The relevant literature already covers important parts of this terrain: conventions in games [8, 19], polycentric governance [9, 10], normative multi-agent systems [20, 21], population protocols [22], and interactions among language-model agents [15, 16]. The present framework is a synthesis oriented around *realization and preservation conditions*: what must survive, what must causally matter, what can be simulated

by simpler alternatives, and where apparent compatibility breaks down. It does not present institutions or computational social conventions as newly discovered concepts.

13 The institutional state and its realization

To formalize an operative institution, augment the microstate of Section 4 with a distinguished component I_t. In a finite setting we may describe it as

$$I_t = (N_t, P_t, E_t, A_t, L_t), \tag{6}$$

where N specifies norms and role obligations, P permissions and capabilities, E enforcement and conflict procedures, A admissible amendments, and L records of decisions and their provenance. The components need not be physically stored together, and their representation is not unique. What matters is their effect on the transition law.

Definition 13.1 (Institutional realization)**.** An institutional realization comprises an environment state x, an institutional state I, a set of local actions $\mathcal{A}_i(x, I)$ for each decision center, a transition kernel K_I on the complete state, and a family of admissible interventions on I and its implementation. A written rule has *operative force* only relative to changes in the feasible action sets, authorized transitions, incentives, or observable outcomes that it actually induces.

The crucial word is *induces*. A simulated constitution that does not affect any operation is ceremonial. A simple queue rule that agents can adopt, enforce, appeal, and change can have substantial institutional force without an elaborate natural-language charter.

For a family $\mathcal{E}$ of permitted environment setups, a horizon T, and feasible interventions $\lambda(I), \lambda(I')$ that differ institutionally while matching the intended other conditions, define the *institutional effect profile*

$$\Delta_{\text{inst}}^{\mathcal{E},T}(I, I') = \sup_{e \in \mathcal{E}} d_{\text{TV}}\big(\mathcal{L}_e(Y_{0:T} \mid \lambda(I)), \mathcal{L}_e(Y_{0:T} \mid \lambda(I'))\big). \tag{7}$$

A zero value shows observational equivalence on the declared family, not universal inertness. An enforceable institution can also alter which actions are institutionally permitted while producing the same observed transcript on a particular task. Observations and rights must therefore be tracked separately. As with other macro-interventions, the implementation cannot be replaced by an unconstrained symbolic assignment [3, 4].

13.1 Typed social relations

A useful representation is a graph whose edges have different types. Information edges determine which messages may flow; resource edges specify transfers and scarcity; normative edges record permissions, obligations, and appeals; causal edges specify implemented state changes. Treating all four as one untyped "connection" erases important distinctions. An agent may truthfully observe an event while lacking authority to endorse a rule; another may hold an authorization token while lacking current evidence; a third may pay an operational cost without acquiring any right to amend policy.

This typed view makes institutional "identity" more exact. Keeping the same policy text while transferring enforcement authority can change the institution even when its next output is unchanged. Keeping the same enforcer while deleting appeal rights changes it. Copying a rule without copying its authority and exception history changes it. In other words, the declaration N, operational implementation K_I, and legitimacy source must not be conflated.

13.2 Rules are transformations, not only statements

A norm can be presented operationally as a condition on histories and actions, $N_i(h, a) \in \{0, 1\}$, interpreted as the system's declared permission for action a by position i after history h. This does not make the declared

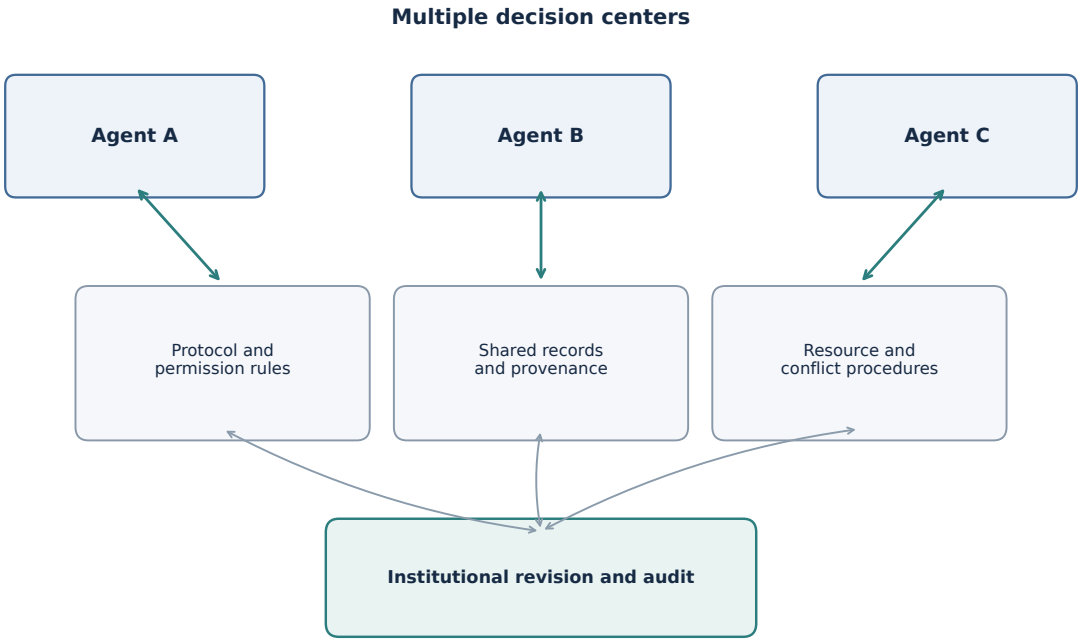


Figure 2: An institutional architecture as a typed interaction system. Rules, shared records, resource procedures, and revision processes are organizational components rather than evidence of a single higher-level mind. The arrows denote possible bidirectional dependence; each route requires its own access and authorization semantics.

permission morally correct. Enforcement changes the action transition or cost structure; appeal changes what happens after an alleged violation; amendment changes the permissible norm itself.

The order of such operations can matter. Let G set a permission bit to 1 (grant) and R set it to 0 (revoke). Then $G \circ R$ yields 1 and $R \circ G$ yields 0 on every input. This simple noncommutativity has practical conceptual force: a log of individual events without their authorized order may fail to reconstruct the actual institutional state. Revision histories must preserve *operations and order*, not only a multiset of decisions.

13.3 The difference between compliance and legitimacy

A proof that every enacted transition obeys N establishes rule compliance relative to the representation, assuming implementation fidelity. It does not establish that the norm is fair, democratically adopted, safe for nonmembers, or compatible with human law. Computational correctness and normative authority are different kinds of judgment. A machine population can acquire stable internal expectations without gaining legal personhood, property rights, or permission to act outside its sandbox. The material substrate, owners, operators, and affected humans remain part of the surrounding system.

14 Endogenous boundaries: membership, offices, and voting authority

A society's population is not merely a parameter in its governance problem. Admission, departure, succession, and the allocation of decision rights can themselves be governed operations. An institution may preserve its written amendment threshold while changing who counts toward that threshold. Consequently, the boundary of the institution belongs in the state whose preservation is being claimed.

Let $\mathcal{U}$ be a finite universe of possible participant handles for the finite results. Write $M_t \subseteq \mathcal{U}$ for admitted members, $V_t \subseteq M_t$ for voting positions, and $\mathcal{D}$ for a declared set of effective decision centers. A map $c_t : V_t \to \mathcal{D}$ records which decision center controls each voting position under the model. This is an assumption about effective control, not a fact established by distinct names, keys, or model instances. A boundary state may be represented as

$$B_t = (M_t, V_t, c_t, w_t, q_t, e_t),$$

where w_t assigns nonnegative voting weights, q_t is an approval threshold, and e_t is a configuration epoch. Admission and exit proposals determine

$$M_{t+1} = (M_t \setminus E_t) \cup A_t, \qquad E_t \subseteq M_t, \quad A_t \subseteq \mathcal{U} \setminus M_t,$$

but this equation alone specifies neither their authorization nor the transfer of offices and outstanding obligations. Those conditions are part of the boundary transition relation. An unbounded supply of fresh handles requires an explicitly enlarged, nonfinite model.

14.1 Copies, credentials, membership, and control are different objects

An executing copy may lack a membership record. A member may lack a vote. Several voters may share a controller. A controller may have no formal vote yet control scheduling or the resource owner. Distinct credentials establish distinct credential records under a trust model, not independent interests or independent evidence. The established Sybil problem is one reason to keep these distinctions explicit [14].

As a simple arithmetic example, two controlled votes among five do not form a strict majority. If the rules admit two more votes controlled by the same center without changing anyone else's vote, the center then controls four of seven and has a majority. This is not an assertion that the admission can be authorized under the original rule. It shows that an admission rule, when it permits such a transition, can change substantive power without editing the ordinary voting formula. In the commons, adding a fifth *office* is therefore different from replacing one office's occupant or running an additional copy to assist it.

Proposition 14.1 (Weight-conserving refinement of controlled votes)**.** *Fix a weighted threshold rule with threshold q. Replace a voting position v of weight $w(v)$ by positions $v_1, \dots, v_k$ controlled by the same decision center, with nonnegative weights satisfying $\sum_j w(v_j) = w(v)$. Leave all other weights and the threshold unchanged. Then every coalition of decision centers has the same total voting weight and the same approval status before and after the replacement.*

Proof. A coalition not containing the controller of v gains none of the replacement positions and loses none of its previous weight. A coalition containing that controller replaces the summand $w(v)$ by the equal sum $\sum_j w(v_j)$. Its comparison with q is unchanged. □

The result concerns coalitions of the stipulated decision centers. It does not preserve the number of credentials, the cardinality of a smallest coalition of individual positions, or a rule that gives each newly named position a fresh unit weight. Nor does it authenticate the control map c_t. Weight conservation is one possible institutional constraint, not a general solution to identity or fair representation.

14.2 Reconfiguration needs a handover, not just overlapping rosters

Let an old roster be $\{a, b, c\}$ and a new roster $\{c, d, e\}$, each using strict majority. Although the rosters overlap, the old majority $\{a, b\}$ and new majority $\{d, e\}$ are disjoint. A shared member does not by itself ensure that every old decision certificate intersects every new one. Distributed reconfiguration protocols address

this issue by specifying how old and new configurations participate in the handover; Raft's joint-consensus construction is a concrete precedent [34].

Here is the limited combinatorial fact such reasoning uses. Fix one decision slot and a set of possible values. A certificate consists of endorsements by every member of an allowed quorum. Suppose each witness is constrained to endorse at most one value for that slot, across all configuration epochs considered.

Proposition 14.2 (Intersecting witnesses exclude conflicting certificates)**.** *If every two allowed quorums intersect, two certificates for distinct values of the same slot cannot both be formed under the stated witness constraint. If two allowed quorums are disjoint, the witness constraint alone does not exclude such conflicting certificates.*

Proof. Conflicting certificates on intersecting quorums would require a common witness to endorse both values, contrary to the constraint. For disjoint quorums, assign one value to all witnesses of the first and a different value to all witnesses of the second. No witness endorses twice. □

A transition quorum requiring an old majority and a new majority intersects any old majority through the old roster and any new majority through the new roster. This fact is useful only with a protocol that transfers the relevant records and preserves the witness constraint. It is not a standalone consensus, fault-tolerance, or liveness theorem. Reused keys, forgotten locks, and inconsistent slot identities lie outside its assumptions.

For theoretical institutions, the same lesson is to version the electorate, interpreter, and decision slot together. A boundary transition must specify which configuration judges its authorization, when the new one becomes operative, and what evidence successors inherit. There is no need to assume that membership is fixed forever. There is a need to make its changes part of the constitutional semantics.

15 Rule-changing computation and governance-preserving abstraction

A persistent society does not merely choose actions under a rule. Some authorized actions change which future actions are admitted. Write a complete state as

$$s = (x, I, \mathcal{P}, r, \lambda, c),$$

where x is operational state, I institutional records, $\mathcal{P}$ the current population and office registry, r resources, λ representation or protocol configuration, and c the current checking and amendment mechanism. These coordinates may overlap in storage. They identify semantic responsibilities rather than independent physical objects.

Let $D(s) \subseteq \mathcal{A}$ be the enabled action labels, partitioned into ordinary and amendment operations. A deterministic realization has a partial transition map $F(s, a)$ defined exactly when $a \in D(s)$. An amendment may change I, c, or λ and thereby change $D(F(s, a))$. A stochastic realization instead uses a joint kernel on successor operating state and successor institutional state. We do not assume these two updates are conditionally independent. This formulation changes a rule *inside* an explicitly declared computational envelope; it does not permit the mathematical semantics to rewrite themselves without representation in the state.

15.1 A quotient must preserve what can be done

A summary adequate for predicting the next grant can fail to preserve who may authorize the next amendment. To capture that distinction, let $\ell : S \to \Gamma$ record the institutional observables protected by the contract. These may include office ownership, resource balances, and a deadline monitor, but need not include every stored byte. The finite action vocabulary $\mathcal{A}$ has fixed meanings across the comparison.

Theorem 15.1 (Exact governance quotient)**.** *Let S be finite and nonempty and $\mathcal{A}$ finite, and let $\pi : S \to Z$ be surjective. There exist functions $\bar{\ell}$, $\bar{D}$, and partial transitions $\bar{F}$ satisfying*

$$\ell = \bar{\ell}\pi, \qquad D = \bar{D}\pi, \qquad \pi F(s,a) = \bar{F}(\pi s, a) \quad (a \in D(s))$$

if and only if whenever $\pi s = \pi t$,

$$\ell(s) = \ell(t), \qquad D(s) = D(t), \qquad \pi F(s,a) = \pi F(t,a) \quad \textit{for all } a \in D(s).$$

The quotient data are unique. From any two states in the same fibre, the legal finite action sequences and their corresponding label sequences are identical. This includes sequences mixing ordinary actions with amendments.

Proof. Necessity follows by factoring each quantity through π. For sufficiency define the quotient label, enabled set, and successor using any representative of a fibre. The three conditions make each definition independent of that representative. Surjectivity gives uniqueness. For trace preservation, induct on the length of an action sequence. Equal enabled sets admit the same first action, the successor condition keeps the resulting states in one fibre, and equal labels preserve the next observation. Repeating the argument proves the claim for every legal finite continuation. □

The theorem is a labeled deterministic congruence principle, extending the use of state aggregation to an explicit institutional contract [2]. It does not identify an observationally invisible permission with a permission that does not exist. A projection-only policy lifts by choosing the same action label at s and πs, provided the implementing decision process is authorized to read that projection and issue the label. That proviso is essential: a mathematically available summary is not automatically an observation available to a decentralized controller.

Counterexample 15.2 (Prediction without governance preservation)**.** Let two states produce the same ordinary allocation. In one, three approvals enable amendment a; in the other, four are required and only three have been received. A summary containing only the allocation identifies the states, while their enabled sets differ. It predicts the current allocation exactly but admits no quotient preserving the amendment operation.

15.2 The smallest exact summary under a fixed contract

The conditions yield a finite construction rather than only an existence criterion. Begin with the partition $\mathcal{B}_0$ that identifies exactly the states sharing $(\ell(s), D(s))$. At each refinement, identify s, t only when they were previously in the same block and every action in their common enabled set leads to the same previous block:

$$s \equiv_{k+1} t \quad \Longleftrightarrow \quad s \equiv_k t \ \text{ and } \ F(s,a) \equiv_k F(t,a) \text{ for all } a \in D(s).$$

Proposition 15.3 (Coarsest finite governance summary)**.** *The refinement stabilizes after at most $|S| - |\mathcal{B}_0|$ strict refinements. Its limiting partition satisfies Theorem 15.1 and is coarser than every partition satisfying that theorem for the same labels, actions, and transitions. It therefore has the minimum number of states among those exact quotients, up to relabeling.*

Proof. Every strict refinement increases the number of blocks, which cannot exceed $|S|$. At a fixed point, states in one block have the same labels and enabled sets and their enabled successors remain in one block, so the quotient conditions hold. Any compatible partition refines $\mathcal{B}_0$. If it refines $\mathcal{B}_k$, compatible successors show that it also refines $\mathcal{B}_{k+1}$. Induction proves that it refines the stable partition. The latter is itself compatible, giving the minimum block count. □

The result is a contract-relative minimization statement, not a unique minimal representation of an entire society. Adding a new appeal operation or a new protected obligation can split blocks that were previously equivalent. In the commons, a summary that omits pending service debt must be refined before it can preserve repayment. This is the central computational price of self-amendment: preserving today's operations need not preserve an enlarged future operation family.

16 Institutional language: messages that change social state

A common message format does not by itself establish a common institution. A statement, a request, a promise, an approval, and an appeal can contain the same descriptive proposition while having different consequences. The social semantics of agent communication has long emphasized publicly accountable interaction rather than assuming access to a sender's private beliefs [35]. Constitutive and procedural norms supply a related foundation [20, 21]. The addition here is to connect those semantics to the same realization and preservation conditions used for institutional state.

A message can be represented as a typed record

$$m = (\mathsf{kind}, \mathsf{issuer}, \mathsf{epoch}, \mathsf{payload}, \mathsf{nonce}).$$

Its operative meaning is not just its payload. For a complete institutional state s, an interpreter defines a partial transition $F_m(s)$ and an event record $\ell(s, m)$ when that transition is defined. Undefinedness represents rejection under the declared protocol, not a moral verdict on the message. The interpreter, authentication assumptions, role registry, and relevant history must be included in s or fixed by the contract. In particular, an imported string cannot create authority simply by naming an authoritative role.

16.1 Faithful translation preserves admissibility and effect

Let institutions A and B have state spaces S_A, S_B, message sets $\mathcal{M}_A, \mathcal{M}_B$, partial interpreters F^A, F^B, and event records in a common semantic space $\mathcal{E}$. A state translator is $T : S_A \to S_B$. Allow a context-dependent message translator

$$\tau : S_A \times \mathcal{M}_A \to \mathcal{M}_B.$$

Context dependence matters for translated office names, configuration epochs, and reference records. The translation of a message sequence is constructed step by step from the evolving source state, not assumed to be a homomorphism of bare strings.

Definition 16.1 (Operatively faithful language transport)**.** The pair (T, τ) is faithful on a transition-closed declared domain if, for every source state s and message m, the following conditions hold:

$$\begin{aligned} F^A_m(s) \text{ is defined} \quad &\Longleftrightarrow \quad F^B_{\tau(s,m)}(T(s)) \text{ is defined}, \\ T(F^A_m(s)) &= F^B_{\tau(s,m)}(T(s)), \\ \ell_A(s, m) &= \ell_B(T(s), \tau(s, m)), \end{aligned}$$

where the last two equalities apply to admitted messages. Any required state observations, such as outstanding obligations, must also agree after translation.

Proposition 16.2 (Preservation of finite institutional conversations)**.** *Under Definition 16.1, every finite source conversation has an admitted prefix exactly when its stepwise translation has that admitted prefix from the translated initial state. Corresponding reached states are related by T, and corresponding event records agree. Two faithful transports compose faithfully using*

$$\tau_{AC}(s, m) = \tau_{BC}(T_{AB}(s), \tau_{AB}(s, m)).$$

Proof. The initial states are related by definition. At each step, equivalence of definedness gives the same admission or rejection. For an admitted step, the transition equality gives the next related states, and the event equality gives the same semantic record. Induction proves the claim for every finite prefix. For composition, apply the two definedness equivalences successively and substitute one transition equality into the other. Equality of event records is transitive. □

The theorem preserves the translated conversations, not every unrelated target message. It neither grants access to hidden state needed to compute τ nor proves that T is executable within a budget. A translation depending on an unavailable authorization record is a mathematical specification without a decentralized implementation.

Counterexample 16.3 (The same payload can perform a different act)**.** Suppose the source message `report(approved, item)` only appends a testimony record. A target translator drops the message type and emits `approve(item)`, which changes a permission bit from zero to one. Both messages contain the same descriptive payload, but their event records and state transitions differ. Even exact preservation of the payload does not satisfy Definition 16.1.

16.2 Private codes and public commitments

A machine-native language may be symbolic, learned, compressed, or distributed across repeated exchanges. Its unfamiliar appearance is not a separate source of authority. A commitment interface can remain public and typed even when a process uses private internal representations to formulate a proposal. Conversely, a readable sentence can have opaque consequences if its interpreter or permission context is missing.

The preservation target is consequently an operation contract rather than a resemblance between strings. A protocol may preserve factual content while losing uncertainty, preserve a promise while changing its debtor, or preserve a vote while changing the electorate in which it is counted. These are different semantic failures. Approximate stochastic interpreters require the distributional and typed residual bounds developed elsewhere in the paper, not a claim that fluent paraphrase guarantees faithful action.

16.3 Changing the interpreter is changing the institution

If an amendment changes the meaning of a message type, the prior translation certificate need not remain valid. The interpreter version and its permissible update rules belong in the state. A constitution that carefully restricts `allocate` while allowing an unguarded change to the interpretation of `report` has not preserved its allocation boundary. This is a semantic instance of the checker-change counterexample, not a distinct kind of computation.

In the commons, a resource request, an approval certificate, a ledger commitment, and a repayment receipt should be separate message types. A successor can inherit the debt record while receiving no permission to issue new debt. This makes the distinction between understanding a statement and acquiring the authority to act on it explicit in the operational semantics.

17 Why institutions might arise: costs, incentives, and coordination

Institutions are neither mysterious nor inevitable. They can become useful when repeated local negotiation is more expensive than creating and maintaining a shared rule. Their formation also depends on who pays, who benefits, who may leave, and whether participants have credible means to enforce an agreement. This is a recurring theme in repeated games, convention formation, and the governance of common resources [23, 19, 10].

17.1 A baseline amortization calculation

Suppose a coordination protocol without institution has expected per-period cost c_0. A proposed institutional protocol has operating cost c_1, per-period maintenance m, and a one-time setup cost B. With discount factor $0 \le \delta < 1$, write $g = c_0 - c_1 - m$.

Proposition 17.1 (Institutional amortization)**.** *Under stationary costs and infinite-horizon discounting, the institution's expected discounted cost advantage over the baseline is*

$$V = -B + \frac{g}{1-\delta}. \tag{8}$$

For a finite horizon T, it is $V_T = -B + g(1-\delta^T)/(1-\delta)$. The institution is cost-improving under these assumptions exactly when the corresponding value is positive.

Proof. Subtract institutional operating and maintenance costs from baseline costs in every period, discount, sum the finite or infinite geometric series, and subtract B. □

This proposition supplies a possible pressure toward persistent organization. It does not show that any particular learning algorithm will discover or adopt the institution, nor that the policy chosen is socially desirable. The assumption of stationary costs is especially demanding in adaptive populations whose tasks, membership, and resource prices can change.

17.2 A profitable institution may be rejected

Aggregate benefit is not the same as incentive compatibility. The simplest example already separates them.

Proposition 17.2 (Aggregate profitability without voluntary adoption)**.** *There exist two-agent repeated coordination environments in which a durable institution has strictly positive aggregate discounted benefit but is rejected by a rational agent who unilaterally bears setup cost and has veto power, assuming quasilinear payoffs, no transfers, and no additional benefits to that agent.*

Proof. Let the setup cost be $B = 10$, borne entirely by agent 1, and let per-period gains be $g_1 = 0$, $g_2 = 2$ with $\delta = .9$. The aggregate advantage is $-10 + 2/(1-.9) = 10 > 0$, whereas agent 1's payoff difference is -10. If agent 1 can reject and has no compensating benefit or transfer, rejection is strictly preferable. □

A society thus cannot be modeled merely by minimizing one global loss without discarding the very plurality that makes it social. Who controls adoption matters. Bargaining power, externalities, and the allocation of setup costs can stabilize an inefficient status quo even when an institution would reduce total computation or improve collective performance.

In an idealized transferable-utility setting there is a corresponding conditional observation. If the discounted net advantages b_i satisfy $\sum_i b_i \ge 0$, then there exist budget-balanced transfers t_i with $\sum_i t_i = 0$ and $b_i + t_i \ge 0$ for all i: choose nonnegative w_i summing to $\sum_i b_i$ and set $t_i = w_i - b_i$. This algebra does not supply liquidity, a transfer mechanism, consent, enforceability, or a right to redistribute. The difference between mathematical feasible compensation and institutionally permitted compensation is central.

17.3 Formation by local improvement: a restricted convergence result

A candidate set of institutional rules $\mathcal{I}$ can be made part of a finite strategic environment. Participants have local choices $r_i \in \mathcal{R}_i$, with overall configuration $r = (r_1, \dots, r_n)$ and expected utilities $u_i(r)$. A finite game is an *exact potential game* if there exists $\Phi(r)$ such that for every unilateral change of i,

$$u_i(r_i', r_{-i}) - u_i(r_i, r_{-i}) = \Phi(r_i', r_{-i}) - \Phi(r_i, r_{-i}).$$

This is the familiar potential-game condition [24]; it is not generically satisfied by social institutions.

Proposition 17.3 (Finite termination of strict institutional improvement)**.** *In a finite exact potential game, every sequence in which each executed unilateral update strictly improves the updating agent's utility has finitely many actual changes. If updates continue whenever an improvement is available, the terminal configuration is a pure-strategy Nash equilibrium. It need not maximize aggregate welfare.*

Proof. Every executed update strictly increases Φ, and a finite game has finitely many configurations and potential values. No configuration can recur along a strictly increasing path, so only finitely many actual changes occur. If the procedure ends only when no agent has a profitable deviation, the terminal state is a pure-strategy Nash equilibrium by definition. A local maximum of Φ need not maximize $\sum_i u_i$. □

The useful point is sharply conditional. Stable conventions can result from adaptive local behavior under special structural assumptions, but stability is neither proof of optimality nor evidence of fairness. With multiple competing objectives, cycles, capture, or regime changes, an institution may never stabilize or may stabilize for undesirable reasons.

17.4 A formation barrier despite mutual benefit

The stationary cost comparison does not describe a route to adoption. Consider $n \geq 2$ agents choosing $a_i \in \{0, 1\}$. An adopter pays cost $c > 0$ and receives benefit $b > 0$ from each other adopter:

$$u_i(a) = a_i \left(b \sum_{j \neq i} a_j - c \right).$$

Nonadopters receive zero. This is a finite network-coordination model, not a universal utility function for societies.

Proposition 17.4 (Institutional formation barrier)**.** *Universal nonadoption is a strict Nash equilibrium. If $b(n-1) > c$, universal adoption is also a strict Nash equilibrium and gives every agent a strictly higher payoff than universal nonadoption. Nevertheless, no sequence of unilateral strict improvements starting at universal nonadoption can reach adoption.*

Proof. A lone adopter receives $-c < 0$, so no agent benefits from leaving universal nonadoption. At universal adoption, an adopter receives $b(n-1) - c > 0$, whereas unilateral exit gives zero. The final assertion follows because no first strictly improving unilateral move is available at the starting state. □

The institution is feasible, stable once established, and unanimously preferred after establishment, but inaccessible through the permitted local improvement path. The problem is not inadequate intelligence or false beliefs. It is the structure of feasible transitions and incentives.

Proposition 17.5 (Minimum profitable coordinated seed)**.** *Starting from universal nonadoption, a simultaneously adopting coalition of size m makes each member strictly better off exactly when $b(m-1) > c$. When such a coalition exists, the smallest size is*

$$m_* = \left\lfloor \frac{c}{b} \right\rfloor + 2 \leq n.$$

Once at least m_* adopters are present, every remaining nonadopter has a profitable unilateral entry. Under a schedule that eventually updates every remaining agent, strict best responses reach universal adoption without an adopter exiting.

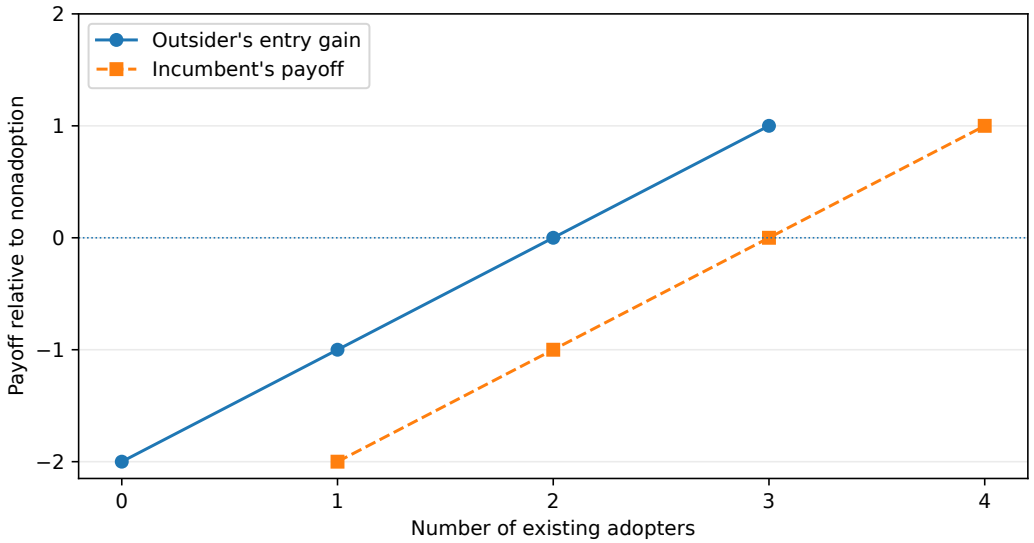


Figure 3: The exact four-agent coordination example with $b = 1$ and $c = 2$. Entry gain at k existing adopters is $k - 2$; an incumbent's payoff with k total adopters is $k - 3$. The curves have different domains because an incumbent exists only for $k \geq 1$ and an outsider only for $k \leq 3$. Zero gain is not a strict improvement.

Proof. Each coalition member has $m - 1$ adopting partners and gains $b(m - 1) - c$. Solving the strict integer inequality gives m_*. For a remaining agent, joining yields $bm - c > 0$. Existing adopters retain positive payoff and their payoff increases with entry. Thus no exit is profitable, and an eventual-update schedule admits every remaining agent. □

A coordinated seed requires a permitted joint-commitment mechanism. The proof does not grant one automatically, supply startup resources, or remove the risk of an unfulfilled promise. In the commons, a simultaneous adoption record can implement such a commitment only if the initial constitutional envelope admits it. This is a bounded way for institutional content to arise inside supplied infrastructure.

For $n = 4$, $b = 1$, and $c = 2$, full adoption gives each agent payoff 1, yet three simultaneous adopters would only break even. Four are needed for a *strictly* profitable seed. Replacing $>$ with $\geq$ would change that answer. Figure 3 separates the outsider's entry gain from an incumbent's maintenance payoff.

17.5 Path dependence and a qualified hysteresis interpretation

For $0 < c < b(n - 1)$ both homogeneous states are locally stable. Under a slowly varied *effective* cost and a best-response rule that retains the current action at ties, universal nonadoption persists until the effective cost becomes negative, while universal adoption persists until it exceeds $b(n - 1)$. Negative effective cost represents a net adoption subsidy in this extended parameterization. The distinct switching points describe hysteresis for that specified update process. They are not a thermodynamic phase transition or an inevitable law of institutional development. Different coalition moves, noise, exit costs, or expectations change the path.

17.6 Persistent rules and the economy of attention

Institutions can economize on repeated communication because a participant need not renegotiate every detail in every encounter. But their records, audits, sanctions, and revision mechanisms consume resources. A full theoretical cost account must include setup and maintenance, opportunity cost of governance, task error, time to reach agreement, cost of exit, and loss from prohibited actions. If agents can cheaply renegotiate or if

interactions rarely recur, a transient coalition may be preferable to a durable institution. If a social rule is cheap to state but expensive to verify, its apparent savings may be illusory.

18 Enforcement, observability, and incentives to comply

An operative rule can restrict physically available actions, change their payoffs, or both. These mechanisms should not be conflated. An access gate can make an unauthorized action unavailable under its implementation assumptions. A sanctioning institution instead leaves a deviation feasible but changes its expected consequences. A self-enforcing arrangement makes continued cooperation preferable through future interactions. Proving one mechanism exists does not establish either of the others.

We develop a small one-deviation model to connect enforcement to the information structure of the society. The conceptual foundation is moral-hazard analysis: when conduct is not directly contractible, incentives depend on observable consequences [45]. The calculation below isolates a finite discrimination problem. It is not a general solution of optimal contracting.

18.1 What a bounded sanction can distinguish

An agent chooses compliance C or a specified feasible deviation D. Before sanctions, D gives expected utility gain $g > 0$ over C. An observation o in a finite set O has law P_C or P_D. The institution commits to a sanction probability $s(o) \in [0, 1]$ with fixed penalty $F > 0$, measured in the same utility units. The agent is risk-neutral in this penalty, its collection is feasible and authorized, and all other payoff differences have already been included in g. The observation laws and the committed policy are known to the agent.

Under these assumptions the deviation's net expected advantage is

$$g - F\big(\mathbb{E}_{P_D} s - \mathbb{E}_{P_C} s\big). \tag{9}$$

Compliance is weakly preferable exactly when this quantity is nonpositive. Equality permits indifference and does not predict which action will be selected.

Proposition 18.1 (The exact bounded-sanction discrimination limit)**.** *In this model, some observation-based sanction makes compliance weakly preferable if and only if*

$$g \le F\, d_{\mathrm{TV}}(P_D, P_C).$$

Strict preference is attainable if and only if the inequality is strict.

Proof. Write $d(o) = P_D(o) - P_C(o)$. Since $\sum_o d(o) = 0$, the sum of its positive terms equals $\frac{1}{2}\sum_o |d(o)| = d_{\mathrm{TV}}(P_D, P_C)$. For every $0 \le s \le 1$,

$$\sum_o s(o)d(o) \le \sum_{d(o)>0} d(o).$$

Equality is attained by $s(o) = 1$ on the positive terms and zero elsewhere. Substitution into (9) proves both assertions. □

If the two observation laws coincide, increasing a finite penalty cannot selectively deter this deviation: it affects both choices identically. This does not rule out prevention, a different observation channel, intrinsic preferences, or future incentives omitted from the model. Conversely, a high discrimination value does not establish that the penalty is legitimate, collectible, or compatible with participation. An accurate accusation is not itself an authorization to punish.

For a concrete two-record example, let

$$P_C = (3/4, 1/4), \qquad P_D = (1/4, 3/4).$$

The total variation is $1/2$. Sanctioning on the second record deters gains $g \le F/2$, but also sanctions compliance with probability $1/4$. A deterrence certificate without this false-positive rate would omit a major institutional consequence.

18.2 False-positive limits and multiple deviations

A constitutional protection may impose $\mathbb{E}_{P_C} s \le \alpha$, with $0 \le \alpha \le 1$. The relevant discrimination capacity is then the finite linear program

$$D_\alpha(P_D, P_C) = \max_{\substack{0 \le s \le 1 \\ \mathbb{E}_{P_C} s \le \alpha}} \left(\mathbb{E}_{P_D} s - \mathbb{E}_{P_C} s \right). \tag{10}$$

The feasible set is nonempty and compact, so a maximum exists. The preceding proof with the feasible set restricted shows that deterrence is attainable exactly when $g \le F D_\alpha$. For the example above and $\alpha = 1/10$, the optimum is $s = (0, 2/5)$, giving $D_\alpha = 1/5$, not $1/2$. Protected error constraints can materially reduce feasible enforcement.

When there are deviations $D_1, \dots, D_k$, one common policy must satisfy every incentive inequality. Separate pairwise discrimination bounds are necessary but not sufficient. With $P_C = (1/2, 1/2)$, $P_{D_1} = (1, 0)$, and $P_{D_2} = (0, 1)$, each pair has distance $1/2$, yet the two differences $\mathbb{E}_{P_{D_j}} s - \mathbb{E}_{P_C} s$ sum to zero. No common sanction deters strictly positive gains from both deviations. A deployable certificate must retain the joint system of constraints, not select a different optimal monitor after learning which deviation occurred.

18.3 Compression can erase enforceable distinctions

Let an observation channel B replace $o \in O$ by a record $z \in \mathcal{Z}$. It may be deterministic compression or a stochastic reporting channel, but must be the same channel under compliance and deviation.

Proposition 18.2 (Garbling does not increase sanctioning capacity)**.** *For every such channel and every* $\alpha \in [0, 1]$*,*

$$D_\alpha(P_D B, P_C B) \le D_\alpha(P_D, P_C).$$

In particular, $d_{\mathrm{TV}}(P_D B, P_C B) \le d_{\mathrm{TV}}(P_D, P_C)$.

Proof. Given a coarse-record sanction $t(z)$, define $s(o) = \sum_z B(o, z) t(z)$. It lies in $[0, 1]$ and has exactly the same compliance and deviation expectations as t has under the garbled laws. Every feasible coarse policy therefore has a fine-record counterpart with the same objective and false-positive rate. Maximizing proves the first claim; the unrestricted case gives the total-variation statement. □

This result concerns available information, not an instruction to retain every private state. An institution may deliberately accept weaker deterrence to protect confidentiality, reduce storage cost, or constrain surveillance. The theorem exposes the tradeoff rather than choosing its normative optimum. Keeping a signed summary establishes who issued the summary, not that it preserved the distinctions needed for a later complaint.

18.4 Enforcement must be incentive-compatible for the enforcer

The model assumes that the stated monitor and sanction are actually implemented. An auditor may face its own costs, limited resources, or incentives to ignore a violation. A society therefore needs separate contracts for evidence production, decision authority, and execution of a remedy. Collusion, hidden side payments,

uncollectible penalties, and changes to the observation process leave the single-agent model. They cannot be repaired by declaring a larger value of F. Section 43.2 places a strategic inspector inside a specified game and distinguishes a feasible monitor from an incentive-supported service.

In the commons, unique slot tokens can prevent duplicate grants at the resource layer while a reporting rule encourages accurate urgency submissions. Those mechanisms coexist but have different proof obligations. An appeal may change the observation set and the sanction rule, while a creditor's outstanding service claim remains unchanged. Section 30 will show how errors in a reduced model of the observation process alter a deterrence margin. This is the point where institutional dynamics and enforcement certificates must be checked together.

19 One enforcement policy for several deviations

The sanction criterion in Section 18 concerns one specified deviation. A continuing institution must usually respond to several possible departures from its rules with one announced procedure. The order of quantifiers changes the problem. A separate monitor may exist for each deviation without any monitor working for all of them. The institution cannot choose its sanction rule after secretly learning which unobserved deviation occurred.

The following finite model isolates that issue. It retains the earlier assumptions of risk neutrality, a committed and authorized sanction mechanism, a collectible fixed penalty, and specified observation laws. It does not solve the incentives of the auditor, establish those laws from data, or claim that every deviation is represented.

19.1 The common-monitor problem

Let O be a finite nonempty observation set. Compliance has law P_C, and deviation $j \in \{1, \dots, m\}$ has law P_j and gross utility advantage $g_j \geq 0$, where $m \geq 1$. A fixed penalty $F > 0$ is applied with probability $s(o) \in [0, 1]$ after record o. Impose a protected false-positive rate $\alpha \in [0, 1]$ through the compact polytope

$$\mathcal{M}_\alpha = \{s \in [0, 1]^O : \langle P_C, s\rangle \leq \alpha\}.$$

Here $\langle P, s\rangle = \sum_{o \in O} P(o)s(o)$. The same s must be used for every listed deviation. Define its margin against deviation j and its best simultaneous margin by

$$D_j(s) = F\langle P_j - P_C, s\rangle - g_j, \qquad \Gamma_\alpha = \max_{s \in \mathcal{M}_\alpha} \min_{1 \leq j \leq m} D_j(s). \tag{11}$$

Weak deterrence of all listed deviations means $\Gamma_\alpha \geq 0$. Strict deterrence means $\Gamma_\alpha > 0$. At equality a participant can remain indifferent. A randomized choice among the listed deviations is also deterred when every listed pure deviation is deterred, since its expected advantage is the corresponding convex combination. This does not include adaptive, repeated, or unlisted deviations unless their observation laws and payoffs are explicitly represented.

For a probability vector $\lambda \in \Delta_m$, set $P_\lambda = \sum_j \lambda_j P_j$ and $g_\lambda = \sum_j \lambda_j g_j$. Write

$$V_\alpha(P, P_C) = \max_{s \in \mathcal{M}_\alpha} \langle P - P_C, s\rangle.$$

This is the same constrained observation advantage used in the single-deviation analysis, now evaluated on mixtures of deviation laws.

Theorem 19.1 (Simultaneous enforcement and mixture obstruction)**.** *Under the finite common-monitor model,*

$$\Gamma_\alpha = \min_{\lambda \in \Delta_m} \big(FV_\alpha(P_\lambda, P_C) - g_\lambda\big). \tag{12}$$

Both extrema are attained. Consequently, one monitor weakly deters all listed deviations if and only if $FV_\alpha(P_\lambda, P_C) \geq g_\lambda$ *for every* $\lambda \in \Delta_m$. *Without an additional false-positive restriction,* $\alpha = 1$ *and* $V_1(P, P_C) = d_{\mathrm{TV}}(P, P_C)$.

Proof. For any fixed s, the minimum of the numbers $D_j(s)$ is the minimum of $\sum_j \lambda_j D_j(s)$ over Δ_m. This expression is bilinear in s and λ, and both domains are nonempty compact polytopes. Finite minimax, equivalently linear programming duality, therefore permits interchange of the maximum and minimum [51]. One can reduce directly to a finite matrix game by listing the vertices of $\mathcal{M}_\alpha$ and taking their convex combinations. For fixed λ, the maximized expression is $FV_\alpha(P_\lambda, P_C) - g_\lambda$, proving (12). Compactness and continuity give attainment. Finally, when $\alpha = 1$, every $s \in [0,1]^O$ is admissible. Choosing $s(o) = 1$ exactly where $P(o) > P_C(o)$ attains the sum of positive differences, which equals total variation. □

The theorem is an application of classical finite convex duality, not a new minimax theorem. Its institutional contribution is the distinction between individual enforceability and a jointly executable sanction policy. A vector λ for which the right side of (12) is negative is a certificate that the stipulated contract cannot be met. It is not evidence that a real population has adopted that mixture.

Counterexample 19.2 (Individually enforceable deviations with no common monitor)**.** Take $O = \{0, 1\}$, $P_C = (1/2, 1/2)$, $P_1 = (1, 0)$, $P_2 = (0, 1)$, $F = 2$, $g_1 = g_2 = 1/2$, and $\alpha = 1/2$. The monitor $(1, 0)$ strictly deters deviation 1, and $(0, 1)$ strictly deters deviation 2. Both respect the false-positive cap. For any common monitor,

$$D_1(s) = s(0) - s(1) - \tfrac{1}{2}, \qquad D_2(s) = s(1) - s(0) - \tfrac{1}{2}.$$

Their sum is -1, so both cannot be nonnegative. The best simultaneous margin is $-1/2$, attained when $s(0) = s(1)$. The dual obstruction is $\lambda = (1/2, 1/2)$, for which $P_\lambda = P_C$ while $g_\lambda = 1/2$.

Distinct deviations can cancel one another's observable signatures at the level of a mixture. Pairwise total-variation distances therefore do not determine joint enforceability. Increasing the penalty cannot resolve the particular obstruction $P_\lambda = P_C$ with $g_\lambda > 0$. The contract must change its observation channel, permitted interventions, payoff structure, or covered claims. None of these changes is automatically authorized.

19.2 Uncertainty must be absorbed by the same monitor

Averaging, translation, and statistical approximation can change the estimated observation laws. The preceding optimization does not remove those errors. Let $\widehat{P}_C, \widehat{P}_j$ be specified approximate laws with

$$d_{\mathrm{TV}}(P_C, \widehat{P}_C) \leq \epsilon_C, \qquad d_{\mathrm{TV}}(P_j, \widehat{P}_j) \leq \epsilon_j.$$

The bounds may be different for different deviations. They must concern the complete record available to the sanction mechanism, including any effects of review scheduling.

Proposition 19.3 (A robust simultaneous enforcement certificate)**.** *Let* $\alpha \geq \epsilon_C$ *and* $\eta \geq 0$. *If a single* $s \in [0,1]^O$ *satisfies*

$$\langle \widehat{P}_C, s \rangle \leq \alpha - \epsilon_C, \tag{13}$$

$$F\langle \widehat{P}_j - \widehat{P}_C, s \rangle - g_j \geq \eta + F(\epsilon_j + \epsilon_C) \quad \textit{for every } j, \tag{14}$$

then the true false-positive rate is at most α *and every true margin* $D_j(s)$ *is at least* η.

Proof. For every function $0 \le s \le 1$, the difference of its expectations under two laws is bounded in absolute value by their total-variation distance. Applying this once to the compliance rate proves (13)'s conclusion. Applying it to both laws in each difference gives

$$D_j(s) \ge F\langle \widehat{P}_j - \widehat{P}_C, s\rangle - g_j - F(\epsilon_j + \epsilon_C) \ge \eta.$$

The same s appears in all inequalities, so the conclusion is simultaneous. □

This is a sufficient certificate, not a necessary description of all robustly feasible policies. Replacing each law by an independent worst-case error bound can be conservative when the errors are related. If $\alpha < \epsilon_C$, the particular tightened test is unavailable, but that does not prove that every actual monitor violates the true cap. The proposition extends Proposition 30.3 to a common monitor rather than separately optimized monitors.

19.3 What an enforcement certificate should contain

The optimization is a small linear program with variables (s, t), constraints $D_j(s) \ge t$, the box bounds on s, and the false-positive cap. A usable certificate records the monitor, every listed gain and observation law, the authority to impose the penalty, its collectibility assumption, and the checked margins. A negative certificate can instead record a mixture λ and an upper bound for its constrained observation advantage.

The linear program does not make observation collection free. Its monitor may require an expensive audit, an unavailable witness, or more time than a service deadline permits. Section 43.3 adds a precise constraint on what the complete available record may distinguish. It also does not authorize punishment for an unproved real-world act. The distinction between a stipulated probabilistic sanction model and a justified institutional procedure remains explicit. The next synthesis asks whether an observation procedure, a common monitor, authorized staffing, and outstanding services can fit within one feasible course of action.

20 Social choice and principled disagreement

Plurality cannot be retained while silently replacing all preferences with one externally supplied reward. Even agents that agree on every fact can rank institutional alternatives differently. Let the alternatives be three policies R, U, V and let three voters have strict preferences

$$R \succ_1 U \succ_1 V, \qquad U \succ_2 V \succ_2 R, \qquad V \succ_3 R \succ_3 U.$$

Counterexample 20.1 (Truthful majority cycle)**.** Pairwise majority prefers R to U, U to V, and V to R, each by two votes to one. Every individual order is transitive. No erroneous report or mistaken inference is needed for the majority relation to cycle.

In successive pairwise elimination, agenda order can select any of the three final winners. To obtain R, compare U with V first, then compare their winner U with R. Cyclic symmetry gives the other outcomes. This is an analytic example of agenda power, not a claim that majority decision is unusable.

20.1 Classical impossibility results have explicit domains

In its standard finite social-welfare formulation, Arrow's theorem rules out simultaneously obtaining unrestricted preference profiles, a complete transitive social ordering, Pareto unanimity, independence of irrelevant alternatives, and nondictatorship when there are at least three alternatives [27]. The result concerns an aggregation rule with those assumptions. It is not a theorem that every procedural choice is arbitrary, that all voting fails, or that machines must be governed by a dictator. Restricting admissible preference domains or changing the object produced by the rule changes the question.

Strategic reporting imposes a different constraint. For a deterministic onto social-choice function with at least three possible outcomes and unrestricted strict preferences, the Gibbard–Satterthwaite result characterizes strategy-proof rules as dictatorial [28, 29]. Strategy-proofness here means that no individual can obtain a preferred outcome by misreporting while others' reports are fixed. Payments, restricted domains, randomized rules, and weaker equilibrium concepts require separate analysis. The two classical theorems are cited results, not new propositions or consequences of the finite majority example.

20.2 Institutions choose procedures, not just outcomes

A complete institutional record should identify the eligible electorate, agenda, comparison rule, admissible reports, tie convention, protected constraints, and appeal procedure. These features can change the result while leaving individual preferences unchanged. They also determine who may propose a comparison in the first place.

The commons' three-of-four amendment rule guarantees neither that a proposal will pass nor that the adopted rule is optimal for all offices. It defines one permissible way to resolve disagreement. A proof that the rule was followed is distinct from an argument for selecting that rule. Deliberation can change beliefs and reveal new alternatives, but it need not remove a genuine conflict of interests. The appropriate mathematical goal is to expose the tradeoffs rather than label residual disagreement a failure of collective intelligence.

21 The constitution as a revisable computational object

A machine institution may contain rules about changing its own rules. This creates a formal distinction between a state transition that is checked by a fixed constitution and a transition that changes the authority or checker used to judge future transitions. Treating the latter as an ordinary edit without recording the change in verification semantics is a source of hidden self-reference.

Let $s = (x, I)$ be the full operating state and let $R(s, s')$ be the admitted transition relation, including ordinary actions and amendments. A safety property is a predicate $\Psi(s)$.

Proposition 21.1 (Constitutional inductive safety)**.** *If* $\Psi(s_0)$ *and every admitted transition satisfies*

$$\Psi(s) \wedge R(s, s') \Longrightarrow \Psi(s'),$$

then $\Psi(s_t)$ *holds for every state reachable by finitely many admitted transitions.*

Proof. Induction on the number of admitted transitions. □

This standard invariant is useful precisely because its assumptions are explicit. It certifies safety relative to R. It does not establish that the implemented system obeys R, that the predicate is ethically adequate, or that a decision can always be reached. A protocol can satisfy a safety condition by refusing to act forever. Timing and failures enter liveness analysis separately; distributed consensus already has sharp impossibility limits under specified asynchronous assumptions [11].

Counterexample 21.2 (An unchecked change of the checker)**.** Let the institutional state contain a validator mode $v \in \{\mathsf{strict}, \mathsf{open}\}$ and a resource policy $p \in \{\mathsf{safe}, \mathsf{unsafe}\}$. Under $v = \mathsf{strict}$, the current policy must remain safe, but the validator mode itself is unprotected. Thus $(\mathsf{strict}, \mathsf{safe}) \to (\mathsf{open}, \mathsf{safe})$ is admitted. Under $v = \mathsf{open}$, any policy transition is admitted, including $(\mathsf{open}, \mathsf{safe}) \to (\mathsf{open}, \mathsf{unsafe})$. The original current-policy check does not imply that all reachable states remain safe. To prove that property, the inductive predicate and transition obligations must include permissible changes to the validation mechanism itself.

The counterexample does not forbid constitutional amendment. It shows that an amendment system needs an explicit *trust and revision boundary*. One possibility is a fixed external checker for a declared protected

property. Another is a staged, authorized change that carries a proof that the new checker preserves a specified invariant. A third is to treat a fully mutable constitution as a weaker, revisable commitment and avoid calling it unconditionally safe. These are different design choices; no single mathematical construction decides the political legitimacy of the guard itself.

21.1 Viability under stochastic institutional transitions

An invariant proof checks that *every permitted transition* preserves a property. A different question asks whether an institution has a *choice of actions* that will preserve a protected property even when the environment responds stochastically. This can be formulated as a finite viability problem. Let S be a finite subset of safe states, let $A(s)$ be the finite set of available actions at s, and let $K(\cdot \mid s, a)$ be the transition law on a finite full state space. The state must include every variable needed for its Markov representation, including the operative amendment rules.

Define $V_0 = S$ and recursively, for $h \geq 0$,

$$V_{h+1} = \Big\{ s \in S : \exists a \in A(s) \text{ such that } \operatorname{supp} K(\cdot \mid s, a) \subseteq V_h \Big\}. \tag{15}$$

Proposition 21.3 (Finite-horizon almost-sure institutional viability)**.** *A state s belongs to V_h if and only if there exists an admissible policy under which every state visited over the next h transitions, including the present state, lies in S with probability one. Policies may use the full observed state and the remaining horizon. For $s \in V_h$, a time-dependent Markov policy suffices.*

Proof. The claim at $h = 0$ is $V_0 = S$. Suppose it holds at h. To remain in S almost surely for another step and then for h further steps, the first action must have support entirely in states from which the remaining h steps can be kept safe. By the induction hypothesis, these are exactly V_h. Thus admissible first actions are precisely those in the definition of V_{h+1}. Choosing a witnessing action at each state and remaining horizon supplies a time-dependent Markov policy. Conversely, if no such action exists, every first action assigns positive probability to a successor from which the required remaining guarantee is impossible. Randomizing over actions cannot remove that positive failure probability unless all positively chosen actions are themselves witnessing actions. □

This is a standard backward-reachability principle from finite stochastic control, adapted to institutional transitions. It exposes the distinction between a *declared safe predicate*, a *proof that all admitted actions preserve it*, and the *existence of a strategy that can keep it satisfied*. None guarantees eventual service, procedural fairness, or low computational cost. In particular, declaring all actions forbidden can make safety trivial while making a purported society inoperative.

21.2 Constitutional semantics and institutional memory

A sound amendment record needs the prior state, proposal, evidence, decision rights, adopted state, and the semantics of the rule used to decide. Without that information a successor may know the current text but not why an exception was made, whether its source was authorized, or whether a later amendment invalidated a prior permission. Append-only provenance may aid auditability, but signatures verify assertions by key holders, not truth or authority. A correct proof object likewise establishes a formal judgment relative to its premises and trusted kernel [12].

21.3 Dissent, appeal, and the risk of overcoherence

One invariant may demand a single ordering for resource transactions; another should allow mutually inconsistent scientific hypotheses to remain under consideration. Enforcing total agreement as an abstract

"coherence" objective can destroy legitimate exploration or conceal genuine conflicts. An appeal procedure is not noise in an otherwise efficient optimizer. It changes who can challenge a decision, which evidence can reopen it, and how error is corrected. In this sense, a machine institution may be more robust when it preserves controlled disagreement than when every node converges to the same report.

22 Constitutional confluence and the authority of ordering

Power can reside in the order of authorized decisions. Two individually permitted amendments may not commute, so a scheduler, deadline rule, or proposal queue can affect the eventual institution without originating any proposal. Section 20 shows one form of agenda dependence through preference aggregation. Here the issue is the operational order of amendments after their admissibility has been specified.

Fix a bounded amendment episode. Let $s \to s'$ mean one permitted amendment step, and let $\to^*$ be its reflexive transitive closure. The episode state contains any pending proposals, version counters, permissions, and consequence records required by the contract. It is *terminating* if it has no infinite chain of actual steps, and *locally confluent* if

$$s \to a,\ s \to b \quad \Longrightarrow \quad \exists c :\ a \to^* c \text{ and } b \to^* c.$$

Confluence imposes the same condition on arbitrary finite descendants of s. A normal form has no outgoing amendment step. Reversibility of an entire society need not be prohibited: termination is a property of this episode, not a requirement that all future institutions stop changing.

Theorem 22.1 (Order-independent amendment normal forms)**.** *A terminating, locally confluent amendment relation is confluent. Each starting state has a unique reachable normal form, so every maximal amendment sequence ends at that normal form. This is Newman's lemma applied to the declared institutional transition relation [36].*

Proof. Use well-founded induction on strict descendants of s. For descendants y, z of s, there is nothing to prove if one equals s. Otherwise choose first steps $s \to a \to^* y$ and $s \to b \to^* z$. Local confluence supplies c with $a \to^* c$ and $b \to^* c$. The induction hypothesis at a joins y, c at some p, and the hypothesis at b joins z, c at some q. Since c is also a strict descendant of s, its hypothesis joins p, q. Thus y, z have a common descendant. Termination ensures a normal form is reached along every maximal sequence. Two reachable normal forms must coincide because confluence gives a common descendant and neither can move. □

A natural-number ranking strictly decreased by every step is one sufficient termination certificate. For a fixed finite set of proposals, the number of unprocessed proposals works only if each step removes one and cannot add more. A rule that may create additional proposals needs a different argument. Well-foundedness cannot be inferred merely from the finiteness of the written rulebook.

22.1 An authorized ordering can still select the outcome

Consider two approved proposals, one setting a policy bit to zero and the other setting it to one. Each proposal is executed exactly once and removed from the pending set. The episode terminates, but its final bit is the value written by the last proposal. The two terminal states are distinct and cannot be joined within the episode. Authorization of both proposals does not imply local confluence.

A procedural priority rule can choose one order and make execution deterministic. That resolves ambiguity by assigning power to the priority rule; it does not prove that the original unordered amendment system was confluent. Likewise, arbitrarily rejecting one proposal after the fact changes the relation being analyzed. A reconciliation procedure must itself be admitted and included in the proof obligation.

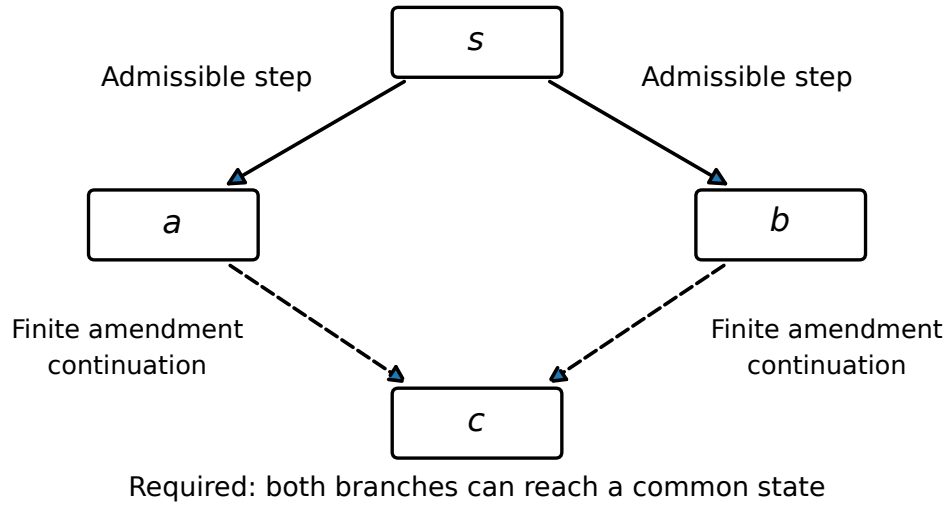


Figure 4: A joinability obligation for two admissible amendment paths. The diagram specifies a condition, not an automatic outcome. Confluence concerns the full declared state or a separately justified semantic quotient; equal final policy text can hide different resource histories.

22.2 Endpoint agreement does not erase path effects

Suppose two amendment paths restore the same final policy, but the first allocates an irreversible resource to office A while the second allocates it to B. A state consisting only of the final policy may have a unique normal form; the allocation record does not. Therefore endpoint confluence of a policy projection does not imply equal costs, equal waiting times, or equal histories of rights violations.

An append-only log also distinguishes the literal order of commuting edits. To identify such logs, a semantic quotient must state which reorderings are immaterial and preserve the audit questions promised by the contract. The governance quotient cannot silently discard the very ordering under investigation. In the commons, two independent label changes may commute, while a loan authorization and the removal of its debtor's office require an explicit treatment of inherited obligations. The outcome of that interaction is not determined by calling both edits valid.

23 Social time, distributed history, and institutional finality

The commons in Section 3 assumes an atomic event mechanism. That assumption allows us to ask whether a rule preserves an allocation or a debt without simultaneously solving the implementation of the ledger. It is not a theorem that autonomous offices can create the mechanism under arbitrary communication conditions. We now consider a separate realization in which offices hold local records and messages can be delayed. The earlier atomic model remains a reference semantics; the decentralized variant must justify any claim that it implements that semantics.

Institutional time has several meanings. A request can be created, transmitted, received, provisionally admitted, committed, and announced at different events. A debt's deadline might refer to a physical clock, a numbered resource epoch, or the completion of a prior procedure. These alternatives define different contracts. Lamport's causal ordering makes the underlying distinction precise: local process order and message transmission generate a partial order, not a universally observed wall clock [41]. Concurrent events

need not have a causally determined relative order. A chosen total order can support execution, but its priority rule is itself an institutional decision.

23.1 Convergent records and valid decisions

Let $(L, \vee)$ be a join-semilattice, with order $a \leq b$ when $a \vee b = b$. A replica holds $\ell_i \in L$. In the following restricted model a local update joins a declared contribution δ into the state, and a merge replaces ℓ_i by $\ell_i \vee \ell_j$. Contributions are fixed elements of L once created. This is a state-merge construction of the kind used in conflict-free replicated data types [42]. The algebra concerns accepted data, not the authenticity of its source.

Proposition 23.1 (Quiescent convergence and invariant preservation)**.** *Assume finitely many replicas begin at ℓ_0, only finitely many contributions $\delta_1, \ldots, \delta_N$ are created, and every contribution eventually becomes included in every replica. Then each replica eventually equals*

$$\ell^\star = \ell_0 \vee \delta_1 \vee \cdots \vee \delta_N,$$

independently of merge order and duplicate deliveries. Moreover, let $V \subseteq L$ contain the initial state. If every permitted local update maps V into V and V is closed under $\vee$, every local and merged state remains in V.

Proof. At every step a replica is the join of ℓ_0 and some created contributions. Associativity, commutativity, and idempotence remove ordering and duplication effects. Once every contribution has reached a replica, its state is $\ell^\star$ and cannot subsequently change. Finiteness of the replicas and contributions gives a finite point after which this holds everywhere. For the second claim, induction over local updates and merges preserves membership in V by the two closure assumptions. □

The conclusion conditions on the same accepted contributions. It does not say that different schedules must generate the same contributions when update proposals depend on local state. Eventual convergence after quiescence also does not mean that a replica can detect when every delayed contribution has arrived.

For causal event records, one may take L to be the downward-closed subsets of a fixed finite event partial order, with union as join. A receiver must retain an event's prerequisites or wait until they arrive. Downward closure preserves a causal dependency record. It does not certify that a signature is authentic, that an issuer held authority, or that two concurrent grants are jointly permissible.

Counterexample 23.2 (Convergence can preserve a record of an invalid allocation)**.** One resource token r must have at most one irrevocable grantee. Two replicas start with the empty grant set. One records (r, A) and the other (r, B), for $A \neq B$. Both singleton records satisfy the exclusivity predicate. Their set union does not. Nevertheless, union merging converges exactly, and every replica eventually agrees on the two-grant record.

The counterexample is not a failure of set replication. It is a failure to choose an invariant-preserving set of independently executable operations. A deterministic conflict resolver that later retains only one grant can restore a selected policy record, but cannot thereby undo both grants' already executed effects. Tentative reservations and irrevocable allocations must have different event types.

The safe set's closure under join is a sufficient condition for the simple construction above. It need not be necessary when a protocol prevents the problematic pairs from being independently reachable. Invariant-confluence theory refines precisely this issue by considering operation-reachable states with compatible ancestry [43]. CALM relates coordination-free computation to logical monotonicity in its stated distributed-program model [44]. Neither result should be replaced by the claim that any monotonically growing ledger is institutionally safe.

23.2 An elementary coordination obstruction

The tension can be proved without postulating malicious participants or unreliable reasoning. Consider two decision sites with the same initial resource state. Each sees its own requests but, before a fixed service deadline, receives no information about requests at the other site. Assume there is no separate shared atomic resource operation available before that deadline.

Proposition 23.3 (Exclusive service cannot be unconditionally local)**.** *No deterministic protocol can satisfy both of the following in every such execution: (i) whenever a request occurs at only one site, that site irrevocably grants the resource before the deadline; and (ii) at most one grant occurs when requests arrive concurrently at both sites. The same conclusion holds for randomized protocols when both guarantees are required with probability one and random tapes have the same request-independent law across these executions.*

Proof. Compare the two singleton-request executions with the concurrent-request execution. Until the deadline, each site has the same local history in the concurrent execution as in its own singleton execution. Requirement (i) therefore makes each grant, violating (ii). In the randomized case each site's grant event has probability one in the concurrent execution by the same local-law comparison. Their intersection also has probability one, without requiring independent random tapes. □

Exclusive preallocation avoids the conclusion by dropping unconditional singleton service at the nonowner. A common sequencer, a sufficiently strong distributed commit protocol, or a resource owner that performs the atomic grant changes the information or operation assumptions. Waiting for communication drops the deadline guarantee. These are distinct institutional choices, not defects to be concealed behind the word decentralized. The proposition is an indistinguishability argument for a restricted service contract, not a replacement for the broader consensus theory discussed in Section 11.

23.3 Finality is indexed by a continuation contract

A provisional decision is one that later authorized processing may reject or revise. A decision is *final relative to a continuation contract* when every continuation admitted by that contract preserves its designated committed effect. This definition does not require every participant already to know the decision. Nor does it make a rule text eternally unamendable: a later prospective amendment may coexist with the continuing validity of an earlier grant or debt.

The observation "no conflicting event has arrived" is not a finality certificate when delays are unbounded. A certificate must state its authority configuration, event or resource identifier, admissible continuations, and the particular safety property it establishes. Intersecting decision certificates and cross-configuration endorsement obligations can rule out conflicting commitments under the assumptions of Section 14. They do not by themselves establish eventual issuance, timely notification, or common knowledge. The distinction between committed effect and known committed effect links this section to Section 36.

For the commons, the appropriate refinement is not to eliminate every supplied primitive. It is to specify which layer supplies exclusivity, how urgency exchanges become committed, and which notification history makes a debt actionable against an office. Observations and research proposals may tolerate asynchronous set merging. Exclusive slot grants require a stronger contract. Different social acts can therefore require different coordination semantics inside the same society.

24 The computability boundary of constitutional verification

Finite constitutional models permit exhaustive reachability analysis when states, initial conditions, and transition rules are effectively given. The conclusion changes when amendments may install arbitrary

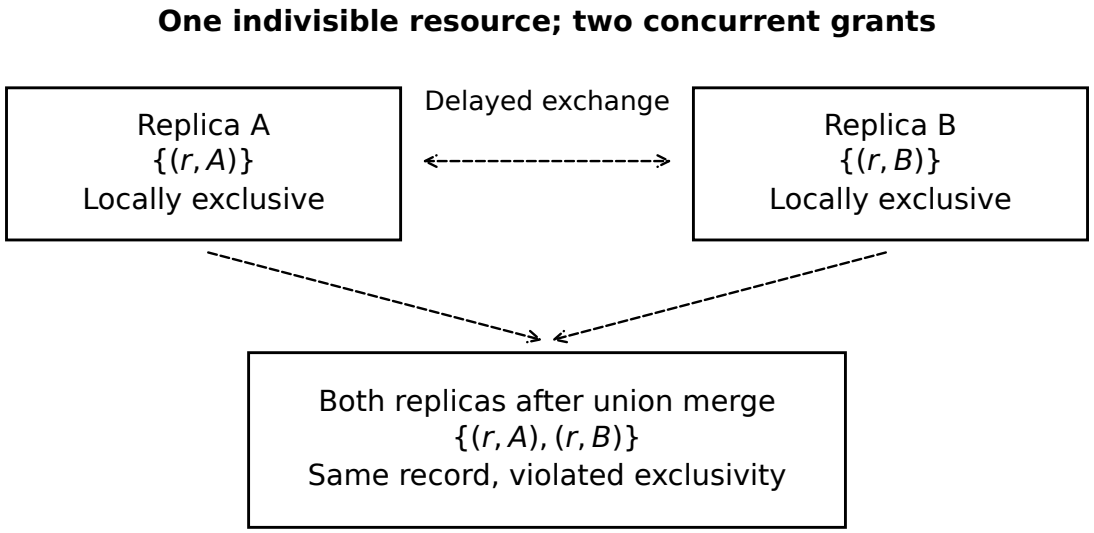


Figure 5: Two valid local grant records can merge into one agreed but invalid allocation record. A later selection of one record does not reverse an already executed conflicting grant. Dashed lines mark delayed exchange, not an atomic commitment mechanism.

programs with unbounded memory. This section explicitly leaves the finite-state setting. Its purpose is to delimit universal verification claims, not to weaken the finite theorems.

Consider a class of deterministic programs with an effectively encoded initial configuration, a total computable one-step transition, and a distinguished forbidden event. Call a program safe when its unique run never produces that event. Assume the class can effectively simulate any Turing machine one step at a time.

Theorem 24.1 (No universal constitutional safety decider)**.** *There is no total algorithm that decides safety for every program in this class. In the same class, occurrence of the forbidden event is semidecidable, whereas safety is not semidecidable in general.*

Proof. Given a Turing machine M and input w, construct a program $P_{M,w}$ that simulates $M(w)$ and produces the forbidden event exactly when the simulation halts. Each simulation step is computable and terminates. The constructed program is safe exactly when $M(w)$ does not halt. A total safety decider would decide halting by complementing its answer, contradicting the classical undecidability result [37].

Occurrence of the event is semidecidable by executing successive steps until it appears. If safety were also semidecidable for this universal class, run the two semidecision procedures in interleaved steps. Exactly one would eventually accept, yielding a total decider. Equivalently, the effective construction would semidecide nonhalting as well as halting, which is impossible. □

The theorem already holds with one deterministic run. It does not rely on strategic deception, nondeterminism, machine consciousness, or the freedom to modify a running verifier. A sufficiently expressive ordinary program language is enough. It is an application of computability theory, not a new limit specific to AI.

24.1 Useful verification survives the impossibility

There are several different, legitimate guarantees. In a finite effective transition graph, compute the reachable states and check that none is forbidden. For a fixed finite step bound and finitely enumerable branching,

explore every permitted prefix. In a restricted language, prove that the permitted syntax induces a decidable verification problem. In a proof-carrying design, accept an amendment only when a supplied certificate passes a specified checker [12].

These guarantees must not be conflated. A fixed time bound does not establish safety after the bound. A syntactically finite program need not have a finite state space. A finite state declaration is not an effective model-checking procedure if evaluating an edge condition may itself fail to terminate. A sound, effectively checkable proof system may certify many safe programs while failing to certify some others. Indeed, a sound and complete recursively enumerable certificate system for the universal class above would semidecide safety, contradicting Theorem 24.1.

24.2 Self-amendment does not eliminate the trusted contract

A certificate must identify the program, policy predicate, interpreter, and version of the checker against which it was established. Proving that an amendment preserves today's permission table does not automatically prove that a subsequent interpreter amendment preserves its meaning. A separately justified transport certificate can carry the guarantee forward; unrestricted substitution cannot.

Nor does the impossibility result establish that every real institution needs a permanent external administrator. It says that no single terminating procedure can classify every unrestricted program by this safety property. A society may choose a smaller rule language, maintain an explicitly incomplete proof discipline, distribute checking, or adopt bounded commitments. These are substantive design choices with different costs and residual risks, not ways to make the universal decision problem disappear.

25 The information limit of decentralized governance

The viability construction in Section 21 assumes that its decision rule sees the full state. A plural institution may distribute the information and the action permissions among different offices. Existence of a safe full-information action then need not imply existence of an implementable decentralized strategy. Observation-based strategies and architecture-dependent synthesis address this distinction in established theory [31, 30].

Fix two hidden environmental conditions $\theta \in \{0, 1\}$ and a class of admissible protocols. For one protocol, let H be the tuple of all local histories up to the decision, together with the private and shared random tapes used to choose actions. The tapes have their specified law and cannot encode extra information about θ. The joint action is a measurable function of H even though no participant need be able to read all of H. Let P_0 and P_1 be its laws under the two conditions.

Theorem 25.1 (Indistinguishability obstruction to joint safety)**.** *Let B_0 and B_1 be disjoint subsets of a common finite joint-action alphabet, containing the actions that meet a declared requirement under conditions* 0 *and* 1*. If the laws of H are within η in total variation, then*

$$P_0(B_0) + P_1(B_1) \le 1 + \eta.$$

Consequently at least one condition has failure probability at least $(1 - \eta)/2$. If every admissible protocol has identical joint-history laws under the conditions, no admissible protocol can meet the requirement almost surely in both.

Proof. Total variation does not increase under the function mapping histories and tapes to joint actions, so $d_{\mathrm{TV}}(P_0, P_1) \le \eta$. Therefore

$$P_0(B_0) + P_1(B_1) \le P_1(B_0) + \eta + P_1(B_1) \le 1 + \eta,$$

since the two safe sets are disjoint. Subtracting the success probabilities from two gives a total failure of at least $1 - \eta$, so one failure probability is at least half that amount. Set $\eta = 0$ for the final assertion. □

The quantification over *every* protocol is needed for an architecture-level impossibility. Equality for one protocol proves only its limitation. The theorem applies to a deadline decision whose required joint actions depend on hidden information. It does not rule out later service after new information arrives.

25.1 Why separate marginal ignorance is insufficient

Counterexample 25.2 (Local ignorance with a perfectly correct joint action)**.** Let U be a fair bit. Agent 1 observes $H_1 = U$ and agent 2 observes $H_2 = U \oplus \theta$. Each individual observation has the same law for both values of θ. Nevertheless, the joint action $(a_1, a_2) = (H_1, H_2)$ always satisfies $a_1 \oplus a_2 = \theta$. The joint observation laws differ and have disjoint supports. Equal marginal laws therefore do not imply the hypothesis of Theorem 25.1.

The distinction is particularly important in distributed systems: correct coordinated behavior can depend on correlations that no individual can decode. The theorem uses indistinguishability of the complete protocol view, not absence of information in each component separately.

25.2 A full-information safe action that the institution cannot select

Suppose a slot must be assigned immediately to office A when $\theta = 0$ and to office B when $\theta = 1$. Every predecision observation and message is constant under both conditions. A full-information scheduler can satisfy the requirement exactly, but any allowed decentralized protocol has the same grant distribution under both. With an available shared fair coin the best worst-case success is $1/2$, attained by a random grant. A deterministic rule fails in at least one condition.

The requirement includes *correct service by the deadline*. Idling would preserve a weaker no-double-allocation invariant, but would not meet that requirement. Thus this example separates weak safety from timely service rather than calling inactivity an allocation success. New information, a later deadline, or an authorized disclosure channel can remove the obstruction. None is supplied by a written constitution alone.

Part III

Continuity, compatibility, and knowledge

26 Temporal obligations, contracts, and office succession

Permissions describe available actions; obligations constrain continuations. The debt created by an urgency exchange is not satisfied merely because the creditor remains authorized to ask for service. It requires a particular future event. A finite obligation record can be written as

$$o = (\mathsf{id}, i, j, \varphi, d, \mathsf{authority}, \mathsf{status}),$$

where i and j are the debtor and beneficiary offices, φ specifies a fulfillment event, and d is a deadline under an agreed clock. Status distinguishes pending, fulfilled, breached, and validly discharged obligations. Cancellation must itself satisfy the declared authorization rule; deleting a record is not fulfillment.

For a bounded deadline D, a trace requirement has the form

$$\mathbf{G}\big(\mathsf{created}(o) \Rightarrow \mathbf{F}_{\leq D}\mathsf{settled}(o)\big).$$

Here settled means fulfilled or validly discharged. The temporal operators mean "at every position" and "within at most D subsequent clock steps." The obligation identifier binds the later event to the same commitment. No implication of ethical adequacy follows from admitting discharge in this particular contract.

26.1 Monitoring deadlines without confusing safety and liveness

A finite monitor stores pending identifiers, remaining deadlines, and terminal statuses. For a finite identifier pool, at most M active obligations, and deadlines bounded by D, this is a finite state machine. Events at a deadline are processed before the clock advances beyond it. A missing fulfillment becomes a violation only once that last permitted event has passed. Resource unavailability, clock failure, and unresolved appeals must have declared transition semantics rather than being silently ignored.

A bounded-deadline violation has a finite bad prefix and is therefore a safety violation in the trace sense. The unbounded demand that every pending obligation is *eventually* fulfilled is instead a liveness requirement, absent a deadline or other finite violation witness. Safety and liveness are distinct established classes [32]. A monitor can detect a missed deadline; it does not by itself make fulfillment physically possible or strategically attractive.

Let $m : S \to Q$ map a complete institutional state to its obligation-monitor state, and let $\nu(a)$ give the monitored event label of action a. Correct realization requires

$$m(F(s, a)) = \delta(m(s), \nu(a))$$

for every enabled action, with the clock update included in ν. The monitor has its own bad-state predicate.

Proposition 26.1 (Obligation-preserving succession and translation)**.** *Consider two deterministic realizations with the same monitored event alphabet and monitor update δ. Suppose a translation T preserves the initial monitor state and the two executions follow the same event sequence, with each realization satisfying the monitor equation. Then their monitor states agree after every finite prefix. In particular, fulfillment, valid discharge, and deadline violations agree at corresponding positions. An office handover that preserves the obligation record is one instance, provided later events and clocks satisfy these hypotheses.*

Proof. The monitor states agree initially. If they agree after a prefix, applying the same deterministic update δ to the same next event preserves equality. Induction gives equality at every prefix, and therefore equality of every status predicate on the monitor state. □

The theorem does not establish that a successor will execute the required event, nor does it grant the successor a credential. It isolates the semantic condition under which a replacement does not erase or alter what remains owed. In the commons, registered succession and the repayment-first rule supply those separate components.

26.2 Resources are not copied by copying a promise

A resource claim needs a conservation rule in addition to its temporal monitor. Grant records are descriptive data, whereas a spendable slot token has a unique consumption transition. Forking an office's software can duplicate its knowledge of a claim without duplicating the underlying resource. A model that counts each copied record as a new budget has changed its resource semantics, not discovered a new source of supply.

Likewise, merging two archives with contradictory obligation statuses must not invent a seamless settlement history. The disagreement is a typed residual that requires an authorized reconciliation. Model replacement, transfer of an office, discharge of a debt, and revocation of a credential are different operations, even when an implementation performs them in one transaction.

27 Culture, inheritance, and continuity through replacement

The strongest form of machine cultural continuity is neither transcript copying nor behavioral imitation. It is the transfer of *usable methods and revision commitments* across changes of participants and substrates. A successor may preserve the predecessor's scientific method by rejecting one of its inherited conclusions. Conversely, an agent that perfectly repeats old answers while ignoring later counterevidence may reproduce data but not the culture that produced them.

Let a history H be summarized by an inheritance map $C(H)$. A practically meaningful retained state can be divided into an active canon M^{act} for immediate use, an archive M^{arc} for recovery and criticism, and a provenance and uncertainty record P. These stores can overlap physically, but serve distinct decision contracts. The canon keeps interaction costs low. The archive makes forgotten exceptions, alternative explanations, and evidence recoverable. Provenance records why a claim should be trusted, not merely that it was inherited.

27.1 What must be preserved

A cultural contract Q may specify future queries, operations, resource policies, amendment procedures, and recognized classes of correction. Two histories may be compressed to the same state only if the required future decisions and authorized operations remain indistinguishable under Q, exactly or within a stated tolerance. Section 9 supplies a finite-memory lower bound: distinct future-relevant histories cannot be combined in an exact deterministic summary. Culture is therefore an information-preservation problem with a *normative choice of relevant futures*, not a license to summarize arbitrarily.

A cultural lineage should track at least four noninterchangeable notions: similarity of resulting behavior, causal transmission of predecessor-specific data, authentic provenance of that transfer, and continuity of valid permissions or obligations. Historical descent can hold while methods diverge; identical methods can arise independently; valid authority may fail to transfer even when every bit is copied. Nothing in these relations settles a question about consciousness or subjective continuity.

27.2 Transport between representational generations

Machine populations may change their internal languages and storage formats. The right question about a translator is whether it preserves *operations*, not only whether its output looks fluent. Suppose generation j uses a metric state space (Z_j, d_j), update operator $F_j : Z_j \to Z_j$, and translator $T_j : Z_j \to Z_{j+1}$. The one-step compatibility defect is

$$\varepsilon_j = \sup_{z \in Z_j} d_{j+1}\big(T_j F_j(z), F_{j+1} T_j(z)\big). \tag{16}$$

An exact zero defect means the representation change intertwines the declared operation. It does not mean all information is preserved; injectivity or an appropriate reconstruction contract is a separate issue.

Proposition 27.1 (Accumulation of cultural transport defects)**.** *Assume each T_j is L_j-Lipschitz and $\varepsilon_j < \infty$ for $j = 0, \dots, m-1$. Define $T_{0:m} = T_{m-1} \cdots T_0$. For every $z \in Z_0$,*

$$\begin{aligned} &d_m\big(T_{0:m} F_0(z), F_m T_{0:m}(z)\big) \\ &\qquad \le \sum_{j=0}^{m-1} \varepsilon_j \prod_{k=j+1}^{m-1} L_k, \end{aligned} \tag{17}$$

with empty products equal to 1.

Proof. For $m = 1$ this is the definition of ε_0. Suppose the bound holds through generation $m-1$. Insert $T_{m-1} F_{m-1} T_{0:m-1}(z)$ between the two terms at generation m. Lipschitz continuity of T_{m-1} bounds the first difference by L_{m-1} times the prior defect, while (16) bounds the second by ε_{m-1}. The recurrence $D_m \le L_{m-1} D_{m-1} + \varepsilon_{m-1}$ unfolds to (17). □

This familiar telescoping-error argument acquires a useful interpretation here. Even individually adequate intergenerational translators can accumulate operation drift, especially when later translations amplify earlier mismatch. Conversely, uniformly nonexpansive translators bound the drift by the sum of local defects. There is no implication that the resulting chain has preserved all obligations or epistemic provenance; those must be included in the state and operations if they are part of the contract.

27.3 Accumulation requires executable preservation, not only storage

A culture can inherit a method without preserving the ability to use it. Its language may remain readable while its required tool is unavailable or its resource permission has been revoked. For a fixed finite query family $\mathcal{Q}$, define a certified method to be an executable procedure with a checked correctness obligation on a stated subset of queries. The checker, semantics, and environmental premises belong to the contract. An unverified assertion is not promoted into this certified class merely by being archived.

Let $C_j \subseteq \mathcal{Q}$ be the queries for which generation j has at least one admitted executable method satisfying the contract. This is an attainable-capability set, not a measurement of what an arbitrary method-selection policy actually achieves.

Proposition 27.2 (Conditional cumulative culture)**.** *Suppose every certified method available at generation j has an admitted realization at generation $j+1$ preserving its query semantics, correctness premises, required resources, and execution rights. Suppose newly added methods satisfy the same declared correctness contract. Then*

$$C_j \subseteq C_{j+1}.$$

The inclusion is strict exactly when at least one newly available method correctly covers a previously uncovered query. The conclusion remains compatible with correction or removal of an inherited uncertified *belief.*

Proof. Take a query in C_j and a witnessing certified executable method. The preservation assumption supplies a method satisfying the same contract at generation $j + 1$, so the query remains covered. New methods enlarge the union of covered domains exactly when one covers a query outside the previous union. The definition of that union does not include uncertified beliefs. □

The assumptions do the work: archival inclusion alone is insufficient. The proposition is not a theorem of automatic scientific progress. It establishes monotonicity of a protected capability envelope; an actual society may select a worse method, change its objectives, lose access to its tools, or discover that a supposedly correct premise was false. Such a discovery changes which methods satisfy the contract and must not be concealed to preserve an artificial progress score.

Fixed storage also limits exact accumulation. If a future-query contract distinguishes N relevant histories, fewer than N deterministic summary states cannot preserve all of them by Proposition 9.1. A B-bit summary cannot represent more than 2^B such classes. Sustained expansion therefore requires more storage, a narrower contract, recoverable external archives, or accepted loss. For a fixed finite Q, even lossless strict capability growth eventually saturates. Open-ended cultural development requires an explicitly expanding query family rather than a conclusion extracted from a finite monotonicity lemma.

27.4 Fission, merging, and the refusal of false continuity

Forking a model produces several descendants, not several automatically valid copies of a scarce license, budget, or authorization. Merging two archives need not merge two policies, preferences, norms, or identities. A merger can preserve conflicting decisions instead of rewriting them as though agreement had always existed. A useful inherited record therefore has both continuities and *unresolved branches*. Institutional honesty may require stating that there is no single successor policy until an admissible reconciliation procedure has occurred.

28 Endogenous problem formation and intellectual culture

A population that only inherits fixed answers has a different organizational structure from one that can generate questions, preserve unresolved problems, and revise what counts as a useful method. A new method may make a previously inaccessible question tractable. A new question may reveal a missing method. The resulting coupling concerns both a stock of capabilities and an agenda of inquiry.

Problem–solution coevolution already appears in work such as POET, which couples generated environments with methods for solving them [40]. Our question is institutional: who may introduce a problem, what evidence can close it, how competing agendas obtain resources, and what happens to unresolved commitments when participants are replaced? This section specifies a theoretical mechanism; it does not claim that open-ended intellectual development follows from naming those mechanisms.

Write C_t for available method records and Q_t for recognized question records. A question record includes a statement or schema, prerequisites, admissible forms of answer or refutation, provenance, and a resource request. Methods carry domains of validity and the evidence supporting them. The coupled state may evolve as

$$C_{t+1} = F(C_t, Q_t, I_t, u_t), \qquad Q_{t+1} = G(Q_t, C_t, I_t, u_t),$$

where u_t records permitted external inputs and I_t governs admission, allocation, and revision. No common utility function is assumed: different offices can value different investigations. The external input and allocation policy must remain visible rather than being absorbed into a claim of autonomous discovery.

28.1 A finite closure theorem, not an open-endedness theorem

Take finite universes C_* and Q_*. Consider a fixed collection of production rules of the form "if these finitely many records are present, add this record." Prerequisites are positive: no rule requires the absence of a record. The content and correctness of those rules are assumptions. An added method is certified only if the specified certification procedure actually supports it.

Let Φ simultaneously add all conclusions whose prerequisites are satisfied, retaining the input records. Order pairs (C, Q) by componentwise inclusion.

Proposition 28.1 (Least closure of a finite problem–method system)**.** *Starting from* (C_0, Q_0)*, iteration of* Φ *reaches the least closed pair containing that initial pair. There are at most*

$$|C_* \setminus C_0| + |Q_* \setminus Q_0|$$

strict growth steps. A sequential rule schedule reaches the same closure if every rule that remains enabled with an absent conclusion is eventually applied.

Proof. Positive prerequisites make Φ monotone, and retention makes it inflationary. Every strict change adds at least one previously absent record, giving the bound. If a closed pair contains the initial pair, induction using monotonicity shows it contains every iterate; hence the limiting fixed pair is least. A sequential schedule remains inside that same closure and can make only finitely many additions. If its eventual record set were not closed, some rule would remain enabled with a missing conclusion, contradicting the scheduling assumption. Its limit is therefore the same least closed pair. □

The finite bound is important: this construction cannot certify unbounded novelty. It derives consequences within supplied finite universes and rules. Countably many possible new records, changing rule languages, or discovery of new question schemas require a larger model. Deleting obsolete claims also breaks inflationarity. Those extensions are legitimate research problems, but their behavior cannot be inferred from this finite closure proof.

For the commons, a certified scheduler can make deadline-sensitive allocation questions meaningful; a counterexample to that scheduler can create a new repair obligation; a solution can become a method available to a successor office. The record of why a problem was posed matters as much as its final answer. Declining to fund a question should not be represented as proving it solved.

28.2 Changing the agenda can manufacture apparent progress

Counterexample 28.2 (Perfect success by replacing the questions)**.** A fixed method solves an easy problem e and fails on a hard problem h. Under the agenda $\{h\}$ its success fraction is zero. Under the agenda $\{e\}$ its success fraction is one, despite no change in the method. Thus increased success on an endogenously selected agenda does not imply increased capability on any unchanged problem family.

A theoretical account should distinguish capability on a preserved task contract, expansion of what can be represented, and selection of which questions receive attention. None is a sufficient statistic for the others. An archive of abandoned and unresolved questions prevents changing denominators from being mistaken for intellectual advance. The point is not that agendas must be fixed. It is that their evolution is one of the outcomes to explain.

28.3 Resource allocation shapes what a culture can discover

Even an enabled inquiry rule may never execute if its office receives no time or storage. The fairness premise in Proposition 28.1 is therefore substantive. An institution can preserve a problem syntactically while indefinitely

starving the work needed to resolve it. Rights to propose, rights to be heard, and actual access to a finite budget are distinct.

Likewise, persistence of a practice is not evidence of its truth. In an illustrative two-type copying model, let accurate and inaccurate practices reproduce at fixed rates $r_T, r_F > 0$, with positive initial counts N_T, N_F. If $r_F > r_T$, the accurate type's share is

$$\frac{N_T r_T^t}{N_T r_T^t + N_F r_F^t} = \frac{1}{1 + (N_F/N_T)(r_F/r_T)^t} \longrightarrow 0.$$

The model does not describe all cultural evolution. It establishes that selection for reproduction alone does not logically impose selection for accuracy. An institution needs a separately specified relationship between evidence, admission, correction, and reproduction to make that inference.

A developing computational culture would therefore be characterized not only by retained answers but by maintained procedures for posing questions, allocating inquiry, recognizing failure, and revising those procedures. Whether these processes yield sustained improvement under heterogeneous interests and changing environments remains an open theoretical problem, not an automatic property of interacting models.

29 Institutional lifetimes and generational turnover

Institutional continuity is a trajectory property. It can outlast an occupant without lasting forever. Fix an operating and repair policy and let V be a finite set of states satisfying a declared continuity contract. The state includes any deadline monitors and fault information needed for the Markov property. Failure is the first exit from V; an institution later rebuilt is a new episode under this first-exit convention.

Let Q be the substochastic transition matrix on V, so the missing row mass represents exit. Let μ be an initial row probability vector supported on V and define $\tau = \inf\{t \geq 1 : s_t \notin V\}$.

Proposition 29.1 (Survival and expected institutional lifetime)**.** *For every integer $t \geq 0$,*

$$\mathbb{P}_\mu(\tau > t) = \mu Q^t \mathbf{1}.$$

If the spectral radius satisfies $\rho(Q) < 1$, then

$$\mathbb{E}_\mu[\tau] = \mu(\mathrm{Id} - Q)^{-1}\mathbf{1}.$$

Proof. The unnormalized row law of trajectories that remain in V through t is μQ^t, by induction. Summing it gives the survival probability. The tail-sum formula for a positive integer-valued random variable gives $\mathbb{E}\tau = \sum_{t\geq 0} \mathbb{P}(\tau > t)$. Under $\rho(Q) < 1$, the matrix geometric series converges to $(\mathrm{Id} - Q)^{-1}$, giving the result. □

A closed recurrent class inside V can invalidate the spectral-radius hypothesis and permit infinite lifetime on some trajectories. The formula is not to be used by inverting a singular matrix. Conversely, a small per-epoch failure probability can still imply eventual exit. Finite-horizon robustness is not immortality.

29.1 A transparent repair example

Take a healthy state H, a degraded but still viable state D, and an absorbing failure state. In one epoch, H degrades with probability f; D repairs with probability r and fails with probability d. Thus

$$Q = \begin{pmatrix} 1-f & f \\ r & 1-r-d \end{pmatrix}, \qquad 0 < f \leq 1, \quad r \geq 0, \quad d > 0, \quad r + d \leq 1.$$

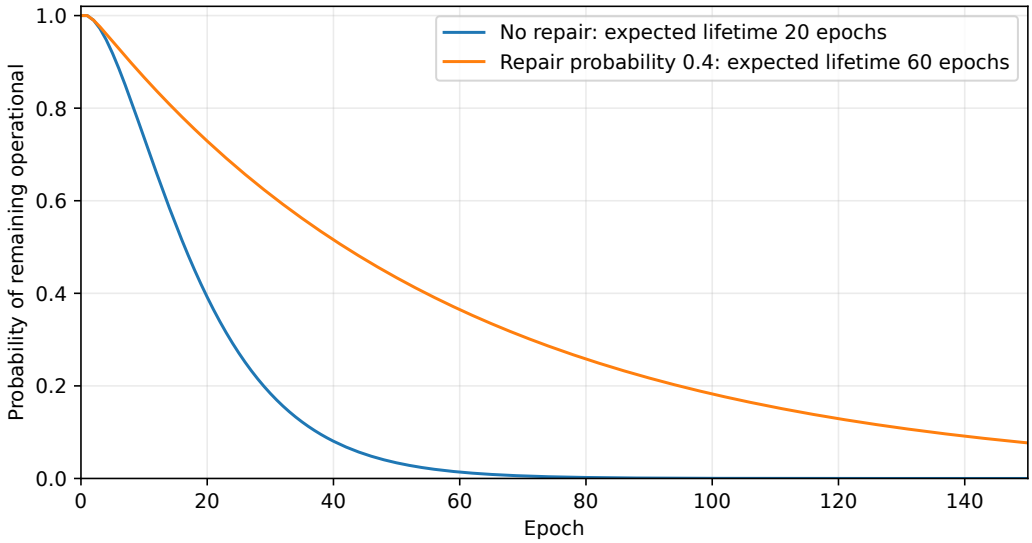


Figure 6: Survival probabilities $\mu Q^t \mathbf{1}$ for the analytical two-state model, initially healthy, with $f = d = 1/10$. Increasing repair from $r = 0$ to $r = 2/5$ extends expected lifetime from 20 to 60 epochs under otherwise fixed transitions. Neither curve is empirical evidence of institutional persistence.

Every nonfailure state has a positive-probability path to failure, so the finite substochastic chain is transient. Solving the two first-step equations gives

$$\mathbb{E}_H \tau = \frac{f + r + d}{fd}.$$

For $f = 1/10$, $r = 2/5$, and $d = 1/10$, the expected first exit is 60 epochs. With $r = 0$ and the other parameters fixed it is 20 epochs. These are exact values of the stated chain, not estimates of real machine populations.

29.2 Office continuity is distinct from occupant survival

Suppose, separately, that each original occupant is replaced at each epoch with independent probability γ. Its probability of still occupying its role after t epochs is $(1-\gamma)^t$. Only if handovers preserve the institutional state and do not change the assumed Q can the institutional survival law above remain unchanged. Independence between turnover and degradation is a simplifying assumption, not an implication of copying state.

This separates two questions: whether an original component remains and whether a specified operational pattern remains. A continuously serviced office may survive several occupants. An unchanged model can preside over the dissolution of its institution. A valid lifetime contract should require continued service or some other nontrivial function; mere survival of an inert archive is a different event.

30 Fast interaction and slow institutional dynamics

The lifetime model describes how long an organization remains inside a contract. A different question asks when the institution itself has a useful lower-dimensional dynamics despite continual changes in its occupants' activity. An office may execute many tasks between constitutional reviews. A stable rule can therefore coexist with rapidly changing messages, beliefs, queues, and model states. Separating these timescales is useful only when the fast state loses the relevant dependence on its initial condition before the next review.

Two-timescale Markov models and averaging have a substantial literature [46]. The finite block model below supplies a self-contained sufficient condition. It does not assert that any particular population mixes quickly, nor does it take a singular-perturbation limit without defining one.

30.1 A block model with a declared review schedule

Let $\mathcal{I}$ and $\mathcal{E}$ be finite sets of institutional and fast states. At review epoch n the institution is $I_n = i$ and the fast state is $X_n = x$. Keep i fixed for a predeclared number $m \geq 1$ of fast steps with kernel K_i. Write $\widehat{X}_n$ for the resulting endpoint. The next institution is sampled through a row-stochastic revision kernel $R_i(\widehat{X}_n, j)$. A subsequent fast-state reset may depend on the endpoint and adopted institution, but cannot revise the already specified institutional outcome.

Assume K_i has a stationary law μ_i and

$$\sup_{i,x} d_{\mathrm{TV}}\big(K_i^m(x,\cdot), \mu_i\big) \leq \varepsilon. \tag{18}$$

The actual one-review law from (i,x) and its stationary average are

$$P(i,x;j) = \sum_y K_i^m(x,y) R_i(y,j), \tag{19}$$

$$\overline{R}(i,j) = \sum_y \mu_i(y) R_i(y,j). \tag{20}$$

The endpoint is the only fast input to this particular revision rule. A rule that uses the full trajectory, accumulated costs, or a data-dependent stopping time requires a different state or a different averaging argument.

Theorem 30.1 (Finite-block institutional averaging)**.** *Under* (18)*, every actual conditional one-review law, given an institutional history ending at i, is within ε of $\overline{R}(i,\cdot)$. Let (J_n) be the Markov chain with kernel $\overline{R}$ and the same initial institutional distribution. Then, for every integer $T \geq 0$,*

$$d_{\mathrm{TV}}\big(\mathcal{L}(I_0,\dots,I_T), \mathcal{L}(J_0,\dots,J_T)\big) \leq 1 - (1-\varepsilon)^T \leq \min\{1, T\varepsilon\}, \tag{21}$$

where $0 \leq \varepsilon \leq 1$ may be assumed and the empty product at $T = 0$ is 1. No independent initialization of the fast state at successive reviews is required.

Proof. Applying the common stochastic kernel R_i cannot increase total variation, so each row (19) is within ε of (20). Given any institutional history of positive probability, the current fast state has some conditional distribution. Averaging the uniform row bound over it preserves the bound. Couple initial institutional states identically. Whenever the two institutional histories still agree, maximally couple their next-state laws; the probability of disagreement at that step is at most ε. The chance of maintaining agreement for all T steps is at least $(1-\varepsilon)^T$, by multiplying the successive conditional lower bounds. The coupling inequality gives the first estimate. The union bound gives the second. □

The conclusion is about the institutional history at review epochs. It does not preserve every intervening action, establish an equilibrium, or imply that harm between reviews is impossible. Required audit events and safety monitors must be included in a joint output contract if their histories are to be approximated. A small one-review error also needs a horizon: it does not give a nontrivial bound uniformly over arbitrarily many reviews.

A transparent kernel family is

$$K_i = (1-\beta_i)\mathrm{Id} + \beta_i \mathbf{1}\mu_i, \qquad 0 < \beta_i \leq 1,$$

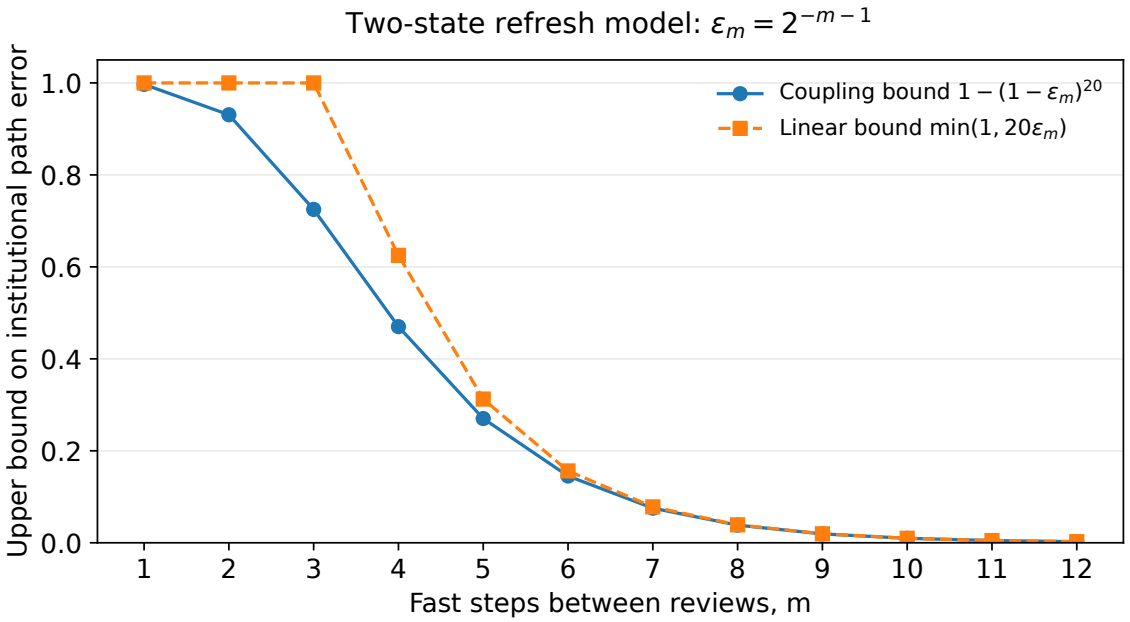


Figure 7: Exact finite-block error bounds for the fair two-state refresh kernel with refresh probability $1/2$. The plotted horizon is twenty institutional reviews. These bounds compare institutional paths at review epochs, not all fast actions.

where every row of $\mathbf{1}\mu_i$ is μ_i. Its mth power is $(1-\beta_i)^m\mathrm{Id} + [1-(1-\beta_i)^m]\mathbf{1}\mu_i$. Thus the mixing error is bounded by $(1-\beta_i)^m$. For a fair two-state μ_i and $\beta_i = 1/2$, the exact worst-row error is 2^{-m-1}. With $m = 8$, the linear bound over twenty reviews is $20/512 = 5/128$. These are analytical values for the stated kernel, not estimated mixing times for neural agents.

30.2 Why fast activity is not sufficient

Counterexample 30.2 (Fast observations do not justify adaptive stopping)**.** Let each fast observation be a fresh fair bit, independent of its predecessor. At every fixed positive time its distribution is exactly stationary. Review instead at the first observed 1, or at a fixed cap $M \geq 2$ if no 1 appears earlier. Let the review select the institutional state equal to its endpoint bit. The probability of adopting 1 is $1 - 2^{-M}$, not $1/2$. Its distance from the fixed-time stationary prediction is $1/2 - 2^{-M}$.

The stopping rule selects favorable observations. It violates the fixed-block premise even though every deterministic-time sample mixes in one step. One can include the stopping policy and accumulated information in an enlarged model, but cannot substitute the same endpoint average unchanged.

A second failure occurs when the fast kernel retains several closed regimes. If K_i is the identity on two fast states and the revision reports that state, the proposed institution-only model loses a distinction that never mixes away. Any single averaged law has worst-row error at least $1/2$. More rapid execution of the same identity transition changes nothing. The retained macrostate must include the regime, or the approximation must acknowledge its error.

30.3 Transporting an enforcement certificate through a reduced model

Suppose an evidence model used at a review has true laws P_C, P_D and reduced laws $\overline{P}_C, \overline{P}_D$ with certified bounds

$$d_{\mathrm{TV}}(P_a, \overline{P}_a) \le \varepsilon_a, \qquad a \in \{C, D\}.$$

Such bounds follow from the block argument when each conduct-dependent fast kernel satisfies its own uniform mixing bound and the same declared endpoint-to-evidence channel is used in the actual and averaged model.

Proposition 30.3 (A robust margin for reduced-model enforcement)**.** *Let $s^\star(o)$ sanction precisely where $\overline{P}_D(o) > \overline{P}_C(o)$ and put $\overline{d} = d_{\mathrm{TV}}(\overline{P}_D, \overline{P}_C)$. In the model of Section 18, this fixed policy makes compliance weakly preferable whenever*

$$g \le F(\overline{d} - \varepsilon_C - \varepsilon_D).$$

Conversely, if $g > F \min\{1, \overline{d} + \varepsilon_C + \varepsilon_D\}$, no observation-based sanction with penalty F deters the deviation. Also,

$$\mathbb{E}_{P_C} s^\star \le \mathbb{E}_{\overline{P}_C} s^\star + \varepsilon_C.$$

Proof. The expectation of a $[0,1]$-valued function changes by at most total variation. Hence the true difference of sanction probabilities is at least $\overline{d} - \varepsilon_D - \varepsilon_C$. Equation (9) gives sufficiency. The triangle inequality bounds the true total variation above by $\overline{d} + \varepsilon_C + \varepsilon_D$, capped at one; Proposition 18.1 then gives impossibility above that value. The final inequality is the same expectation bound under compliance. □

When the lower margin is nonpositive it supplies no guarantee for positive g. A separate false-positive protection must include the final inequality or use a reduced policy with slack below its cap. The result concerns one common fixed policy and one specified deviation. It is not a license to optimize separately for incompatible deviations, replace missing observations with stationary assumptions, or declare an institution safe because its average trajectory looks stable.

The useful societal interpretation is limited but substantive. Under uniform relaxation and a controlled review schedule, institutional dynamics can be predicted without retaining every fast interaction. The reduced description preserves specified laws to a declared horizon. Persistence here is explanatory separation of scales, not independence from the underlying agents or infrastructure.

31 Jurisdictions, overlap, and local–global obstruction

A large machine population need not be administered by a universal institution. Different groups may sustain rules for different tasks, resources, or trust domains. Such *polycentric* organization provides possible advantages in locality and adaptation, but introduces a precise compatibility question: can locally acceptable rules be jointly realized when groups interact?

There are at least three meanings of compatibility. First, two representations may agree on shared observations. Second, their authorized operations may agree when transported through a common interface. Third, their joint constraints may admit a globally executable state. The third is stronger than pairwise local satisfiability. An apparently minor cycle of rules can obstruct any globally consistent execution.

31.1 An exact finite obstruction model

Consider an undirected graph $G = (V, E)$ whose vertices are institutional positions. Each position carries a binary coordinate $x_i \in \mathbb{F}_2$ and each edge $\{i, j\}$ imposes the relative constraint

$$x_i \oplus x_j = b_{ij}, \qquad b_{ij} \in \mathbb{F}_2. \tag{22}$$

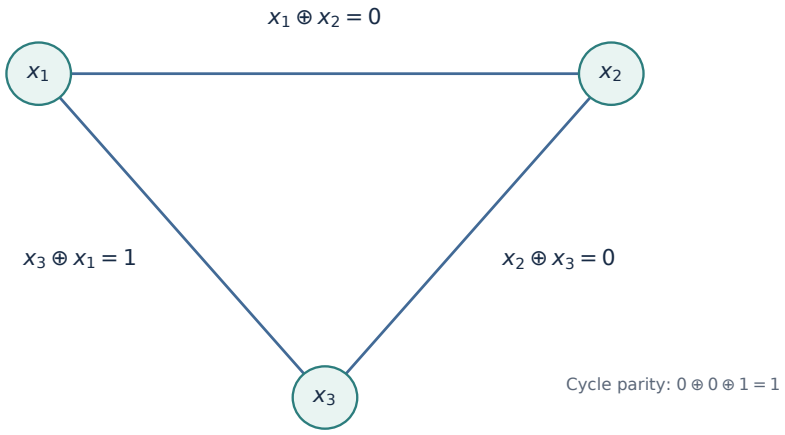


Figure 8: A minimal local–global obstruction. Each displayed pairwise constraint is satisfiable by itself, but their conjunction is inconsistent: XOR around the cycle gives 1 on the right and 0 on the left. This is an elementary binary model, not a universal encoding of institutional norms.

The coordinates might represent a toy authorization bit or a binary convention, but the mathematical model itself makes no claim about a realistic legal code.

Proposition 31.1 (Cycle criterion for binary institutional compatibility)**.** *The constraints* (22) *admit a simultaneous global assignment* $(x_i)_{i \in V}$ *if and only if the XOR of the labels* b_{ij} *along every graph cycle equals* 0*. When an assignment exists, each connected component has exactly two assignments, related by complementing every coordinate in that component.*

Proof. If an assignment exists, XOR the edge equations around a cycle. Every vertex coordinate appears twice and cancels, so the edge-label XOR is zero. Conversely, choose a spanning forest and assign 0 to one root in each component. Propagate the vertex values along forest edges using (22). Forest edges are satisfied by construction. For a nonforest edge, the unique forest path between its endpoints and that edge form a cycle. Vanishing cycle parity implies that the propagated values satisfy the nonforest edge as well. Finally, a connected solution is fixed by choosing the value at one root; either root value gives an assignment and there are no others. □

This result is a standard parity fact on graphs, closely related to cycle-space algebra and discrete cocycles. Its value here is methodological. It shows why *pairwise satisfiable* cannot be equated with *globally realizable*. For a triangle with $b_{12} = 0$, $b_{23} = 0$, and $b_{31} = 1$, every individual edge requirement has a solution, but the three cannot hold together. A translator between two agents could appear successful while a three-party federation is impossible under the combined rules.

The obstruction generalizes in several directions. For nonbinary constraints one may need constraint-satisfaction or logical satisfiability methods. For typed overlapping interfaces one may formulate gluing conditions for partial states and compatibility of transition operators. For probabilistic policies, exact satisfiability may be replaced by minimizing a weighted violation or by checking whether local marginal laws admit a joint extension. None of these generalizations is automatic: a consistent family of local marginal distributions on pairwise overlaps need not admit a globally consistent joint distribution on an arbitrary cycle.

31.2 Operational gluing and federation

Even when a global static state exists, a federation may fail dynamically. Local rule applications could commute in one context and conflict in another. A translator $T_{ij} : Z_i \to Z_j$ should preserve the operations relevant to the interaction, approximately satisfying $T_{ij}F_i \simeq F_jT_{ij}$. A federation requires *both* representational compatibility and admissible operational coordination. Static agreement alone does not define an executable constitution. Nor does faithful gluing prove that the institution should be centralized: a compatible federation may deliberately preserve multiple veto points, modes of appeal, and local autonomy.

The graph example also clarifies a useful mathematical program: institutional incompatibility can be measured as a *residual* left by trying to realize local constraints globally. One can ask for the least-cost repair set of rules, for certificates that no repair respecting protected obligations exists, or for an interface transformation that changes the representation without changing the obligations. These are natural places for constraint solving, formal verification, and algebraic topology to meet a theory of machine institutions, without claiming that every social disagreement is a parity equation.

32 Noncommutative institutional holonomy

The binary cycle criterion is the simplest case of a more general distinction between local translations and global consistency. In a federation, an interface may relabel roles, permute resource categories, or change coordinates in a protocol. These transformations need not commute. Their composition around a loop can therefore matter even when every individual interface is invertible.

The group-theoretic foundation is established synchronization theory, not a new use of the word society. Gao, Brodzki, and Mukherjee give a geometric account in terms of bundles and holonomy [38]. Here we provide the finite graph argument directly and separate consistency of a complete reference frame from compatibility of one particular institutional state.

32.1 Conventions and exact synchronization

Let $\Gamma = (V, E)$ be a finite connected graph. Each oriented edge $i \to j$ carries $g_{ij} \in G$, where G is a group and $g_{ji} = g_{ij}^{-1}$. The element g_{ij} transports coordinates from i to j by a left action. For a path $\gamma = (v_0, \dots, v_k)$, define

$$H_\gamma = g_{v_{k-1}v_k} \cdots g_{v_0v_1}.$$

The rightmost map acts first. If $v_k = v_0$, this product is the path's holonomy. An exact system of global frames is a family $h_i \in G$ satisfying

$$g_{ij} = h_j h_i^{-1} \qquad \text{on every edge.}$$

Theorem 32.1 (Exact frame synchronization)**.** *Global frames exist if and only if $H_\gamma = e_G$ for every closed walk. When they exist, they are unique up to common right multiplication $h_i \mapsto h_i k$. It suffices to check the rooted fundamental loops associated with one spanning tree.*

Proof. If $g_{ij} = h_j h_i^{-1}$, adjacent factors cancel in every path product, giving $H_\gamma = h_{v_k} h_{v_0}^{-1}$ and hence identity on a closed walk. Conversely, choose a root r, set $h_r = e_G$, and define h_i as the product on the unique tree path from r to i. Tree edges satisfy the required identity. For any non-tree edge $i \to j$, the rooted loop consisting of the path to i, that edge, and the reversed path from j has product $h_j^{-1} g_{ij} h_i$. Its being identity gives $g_{ij} = h_j h_i^{-1}$. Thus the fundamental-loop tests suffice. Any other solution obeys the same propagation from its root value, so it is $h_i k$ for a common k. □

Changing local conventions by elements k_i replaces g_{ij} by $k_j g_{ij} k_i^{-1}$. A loop based at r then has holonomy $k_r H_\gamma k_r^{-1}$. In particular, identity holonomy is invariant under local changes of coordinates. An arbitrary

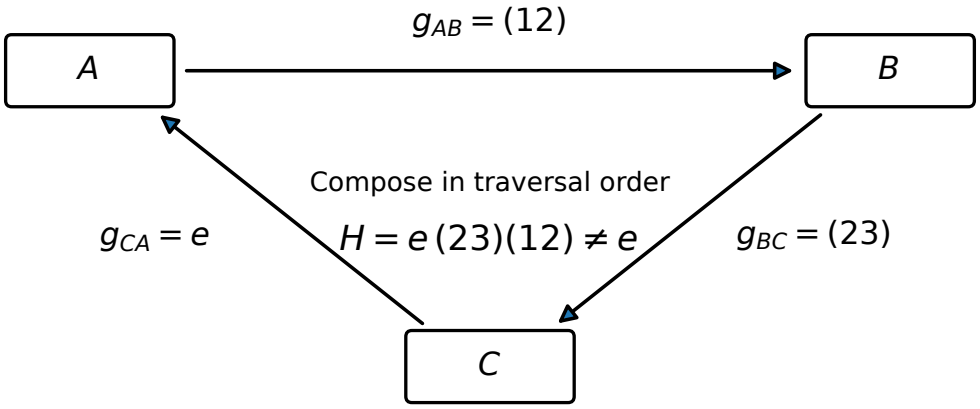


Figure 9: Three individually invertible interfaces need not synchronize. The ordered round trip is $(23)(12)$, not the identity. The group model concerns the declared role-label contract; it does not certify permissions, provenance, or the legitimacy of an interface edit.

binary cycle basis with unordered multiplication is not a valid replacement for rooted fundamental loops in a noncommutative group.

32.2 A three-jurisdiction example

Take $G = S_3$, acting on three role labels. Let the translations on the oriented triangle $A \to B \to C \to A$ be

$$g_{AB} = (12), \qquad g_{BC} = (23), \qquad g_{CA} = e_G.$$

Every edge is a valid bijection. With the specified composition convention, the round trip is $(23)(12) \neq e_G$. Thus no single assignment of complete role frames realizes all three interfaces. Replacing the final edge by $((23)(12))^{-1}$ repairs this loop, but whether that edit is authorized is a separate institutional question.

32.3 A shared state is weaker than a shared frame

Nontrivial holonomy need not prevent *every* compatible state. Let G act on a set D, and seek local states $x_i \in D$ with $x_j = g_{ij} \cdot x_i$. A permutation can move some labels while fixing others. A family that uses only a fixed label can be compatible even though its complete coordinate frames are not.

Proposition 32.2 (Compatible states and holonomy fixed points)**.** *A compatible state family exists if and only if there is $x_r \in D$ fixed by every holonomy based at the root r. Each such x_r uniquely determines a compatible family by transport along a spanning tree. For a free group action, existence of a compatible family implies identity holonomy; without freeness that implication can fail.*

Proof. A compatible family transported around a closed walk returns to its starting value, so every based holonomy fixes x_r. Conversely, define each x_i by the tree-path product acting on x_r. The rooted-loop fixed-point condition gives $g_{ij} \cdot x_i = x_j$ on non-tree edges as well as tree edges. The tree paths also give uniqueness for a chosen root state. Under a free action, only the identity fixes a point, proving the final implication. □

Counterexample 32.3 (Nontrivial frame defect with a compatible state)**.** On a triangle, let one oriented edge carry $(12) \in S_3$ and the other two carry identity. Holonomy is nontrivial, so global frames do not exist. Nevertheless, the constant state $x_A = x_B = x_C = 3$ is compatible because (12) fixes 3. Failure to align all role meanings is stronger than failure to agree on one unused or invariant role.

This distinction prevents overinterpreting an obstruction. A federation may support one narrow service while being unable to transport its full permission structure coherently. A zero defect for the narrow service should not be advertised as complete institutional interoperability.

32.4 A quantitative lower bound on interface error

Suppose G has a bi-invariant metric d, so both left and right multiplication preserve distance. For candidate frames h_i, write

$$\delta_{ij} = d(g_{ij}, h_j h_i^{-1}).$$

Proposition 32.4 (Loop defect bounds unavoidable edge error)**.** *For a closed walk γ with $k \geq 1$ traversals,*

$$d(H_\gamma, e_G) \leq \sum_{(i,j) \text{ traversed by } \gamma} \delta_{ij}.$$

In particular, some traversed edge has error at least $d(H_\gamma, e_G)/k$ for every candidate frame family.

Proof. Gauge-transform each edge to $\widetilde{g}_{ij} = h_j^{-1} g_{ij} h_i$. Bi-invariance gives $d(\widetilde{g}_{ij}, e_G) = \delta_{ij}$. The transformed loop product is $h_r^{-1} H_\gamma h_r$, at the same distance from identity as H_γ. Repeatedly applying the triangle inequality and invariance to a product bounds that distance by the sum of its factors' distances from identity. Since $k \max \delta_{ij} \geq \sum \delta_{ij} \geq d(H_\gamma, e_G)$ along the walk, the maximum error is at least $d(H_\gamma, e_G)/k$. □

The sum counts repeated edge traversals. Bi-invariance is an explicit assumption and cannot be replaced silently by an arbitrary metric. Normalized Hamming distance on a finite permutation group is one suitable example. This estimate constrains possible synchronization accuracy; it does not select an authorized repair, preserve scarce resources, or prove that a globally synchronized convention is desirable. Those requirements remain separate coordinates of the residual calculus.

33 A typed calculus of institutional residuals

Institutional interoperability has several independent failure modes. Two implementations can execute the same action while disagreeing about its authorization. They can preserve authorization while erasing a debt, or preserve the debt while duplicating its backing resource. A fluent translation can retain every conclusion while dropping the evidence needed to reconsider it. A single numerical compatibility score obscures these distinctions.

We formalize a restricted compositional calculus, rather than assuming that every mismatch is measured in one natural unit. Three realizations $j = 0, 1, 2$ have metric state spaces (Z_j, d_j) and a common finite action alphabet $\mathcal{A}$. For algebraic comparison, each action has a total update F_j^a: a disabled action goes to an explicit rejection state. Translations map rejection to rejection. A separate permission predicate records which actions are genuinely admitted. Totalization is a bookkeeping device and does not authorize the disabled action.

33.1 Common semantic coordinates

For each component type $c \in \{P, O, R, E\}$, let

$$g_j^c : Z_j \longrightarrow (E_c, \delta_c)$$

be a contract observable in a common metric codomain. P records permissions over the common action alphabet; O records obligation-monitor state; R records resources in common physical or accounting units; and E records the required evidentiary and provenance structure. A change of units, action names, or obligation identifiers requires an explicitly specified alignment before these comparisons can be made.

Possible metrics include the discrete metric on permission vectors, a declared metric on obligation-monitor states, an ℓ^1 distance on common-unit resource vectors, and symmetric-difference cardinality on required evidence records. For evidence, record equality is structural preservation under a trust model, not a truth certificate. If the payload or inferential dependencies matter, they must be included in the observable rather than only its identifier.

For a translation $T : Z_i \to Z_j$, define

$$\varepsilon_{\mathrm{dyn}}(T) = \sup_{z,a} d_j\big(TF_i^a(z), F_j^aT(z)\big),$$
$$\varepsilon_c(T) = \sup_{z} \delta_c\big(g_j^cT(z), g_i^c(z)\big) \quad (c = P, O, R, E).$$

The institutional residual profile is

$$\mathfrak{D}(T) = (\varepsilon_{\mathrm{dyn}}, \varepsilon_P, \varepsilon_O, \varepsilon_R, \varepsilon_E).$$

Its entries are not added together. They have different meanings and may have different units. A one-sided authorization check can additionally distinguish newly granted forbidden actions from lost legitimate permissions. The symmetric permission residual measures both changes and is deliberately stronger.

Theorem 33.1 (Composition of typed institutional residuals)**.** *Let $T : Z_0 \to Z_1$ and $U : Z_1 \to Z_2$ have finite residuals under the same semantic coordinates. Suppose U is L-Lipschitz. Then*

$$\varepsilon_{\mathrm{dyn}}(UT) \le L\varepsilon_{\mathrm{dyn}}(T) + \varepsilon_{\mathrm{dyn}}(U),$$

and, separately for every $c \in \{P, O, R, E\}$,

$$\varepsilon_c(UT) \le \varepsilon_c(T) + \varepsilon_c(U).$$

Consequently translations with zero residual in every coordinate compose to translations with zero residual under this contract.

Proof. Insert $UF_1^aT(z)$ between $UTF_0^a(z)$ and $F_2^aUT(z)$. The triangle inequality, then Lipschitz continuity of U, bound the two terms by $L\varepsilon_{\mathrm{dyn}}(T)$ and $\varepsilon_{\mathrm{dyn}}(U)$. For an observable type c, insert $g_1^cT(z)$ between $g_2^cUT(z)$ and $g_0^c(z)$. Both comparisons occur in the same metric space E_c, so their bounds add directly. Taking suprema proves the claims. □

For a longer chain, the dynamic bound has the same weighted telescoping form as Proposition 27.1; each semantic-coordinate bound is the sum of its local defects. A zero discrete permission defect gives exact permission preservation. A small real-valued state distance by itself does not. Without a common codomain and fixed alignment, the semantic triangle argument is unavailable.

This is a preservation calculus built from elementary metric estimates and commuting diagrams, in the same broad tradition as compositional abstraction [5]. Its institutional content lies in the chosen typed contract. A zero profile need not imply injectivity, equality of all private information, legitimate institutions, or true beliefs. Those additional properties are not created by naming the profile a certificate.

33.2 Failures the profile separates

In a one-step commons interface, a translator might preserve the allocation $(1, 1, 1, 1)$ while interpreting every received signature as a valid amendment approval. Its current resource defect can be zero while its permission defect is nonzero. A second translator might preserve amendment permissions but replace the pending debt record by the empty monitor state. Its obligation defect is nonzero before the altered repayment changes an allocation. A third can preserve both records while duplicating a spendable token, violating the resource coordinate.

A mismatch is not automatically malicious or repairable. Two institutions may deliberately recognize different authorities. A debt-preserving merger may be impossible within the available budget. A correct response can be to refuse a composition and retain the unresolved residual, rather than force agreement by dropping the protected constraint.

33.3 Authorized repair and a representation-invariant obstruction

Let a finite set of proposed repairs carry declared nonnegative costs. A repair is admissible only if it respects the protected obligations and the authority to modify each component. The repair problem is to find an admissible candidate with the required residual bounds at minimum cost. An empty feasible set is a meaningful outcome. The definition neither supplies an efficient solver nor licenses changing the protected requirements to make the problem solvable.

The binary jurisdiction model admits a sharper statement. A local convention change $x_i' = x_i \oplus c_i$ transforms its edge labels to

$$b_{ij}' = b_{ij} \oplus c_i \oplus c_j.$$

Proposition 33.2 (Cycle residual and minimum authorized repair)**.** *The XOR of edge labels around a cycle is unchanged by every local convention change above. On a single simple cycle with odd XOR, suppose an editable subset of edges can be flipped at nonnegative costs w_e, with other edges protected. An authorized repair exists exactly when the editable subset is nonempty. The minimum cost is then* $\min_e w_e$ *over editable edges. If all edges are protected, no change of local binary conventions can produce a compatible global assignment.*

Proof. In the XOR around a cycle each c_i occurs twice and cancels, preserving the residual. Changing odd parity to even requires an odd number of edge flips. Every such repair includes at least one editable edge and costs at least the cheapest editable-edge cost because costs are nonnegative. Flipping that cheapest edge is sufficient by Proposition 31.1. With no editable edge parity cannot change, and local convention changes also preserve it. □

For several overlapping cycles, one edit can affect multiple residuals. Independently repairing each cycle need not be globally optimal, and the single-cycle cost formula does not generalize without further analysis. This distinction connects representational transport, protected structure, and obstruction resolution without treating every institutional conflict as a parity problem.

34 Epistemic institutions: testimony, provenance, and disagreement

An information-sharing population can become less reliable as it becomes more interconnected. A false claim may be repeated across many agents until repetition is mistaken for independent corroboration. A true observation may be discarded because its origin lacks authorized status. An ordinary summary may be silently promoted to a rule. These are not merely limitations of isolated models: they arise at the interfaces among memory, authority, and social evidence.

34.1 Knowledge is not the same as available information

The joint information accessible to a hypothetical observer does not imply that any participant possesses that information, knows that others possess it, or can coordinate on it. Distributed knowledge, mutual knowledge, and common knowledge are distinct notions in distributed epistemic logic [25]. Even if two agents' records jointly determine a fact, an institution may need an authenticated disclosure procedure before that fact may legitimately be used. More bandwidth alone does not produce common knowledge under arbitrary uncertainty about delivery.

A useful epistemic institution therefore specifies which evidence types are admissible, what constitutes a retraction, how contradictions remain visible, which claims can be treated as commitments, and how provenance influences confidence without becoming a substitute for verification.

34.2 Correlated testimony cannot be counted as independent votes

Suppose errors $E_1, \dots, E_n$ in scalar estimates are centered, have common variance $\sigma^2 > 0$, and have common pairwise correlation $\rho \in [0, 1]$. Then

$$\operatorname{Var}\left(\frac{1}{n}\sum_{i=1}^{n} E_i\right) = \frac{\sigma^2}{n}\big(1 + (n-1)\rho\big). \tag{23}$$

This elementary identity follows by expanding the variance and counting $n(n-1)$ off-diagonal covariance terms. Relative to the variance of an average of independent equal-variance estimators, the *variance-equivalent* effective sample size is

$$n_{\text{eff}} = \frac{n}{1 + (n-1)\rho}.$$

If the errors are perfectly correlated, adding copies does not reduce variance. This equation does not give a general probability that a majority is right, and real error structures need not have one uniform correlation parameter. It nevertheless defeats the claim that numerical agreement automatically equals independent support. Shared training histories, copied observations, and a common archive may all create dependence that is invisible in the final vote count.

34.3 Cheap talk and strategic reporting

If a message M has the same conditional distribution under two equally possible hidden states $\theta = 0, 1$, then Bayes' rule gives $\Pr(\theta = 1 \mid M) = \Pr(\theta = 1)$ wherever the posterior is defined. Such a channel carries no information about the distinction, regardless of how fluent or authoritative its reports appear. In strategic environments, senders may have preferences that differ from those of receivers. The theory of strategic information transmission shows why informative communication can depend on preference alignment and signaling structure [26]. A message being grammatically meaningful is not a truth guarantee.

A report may be false for many reasons: misunderstanding, stale records, an unsuitable abstraction, a noisy sensor, a deliberately misleading tactic, or an institutional incentive to conceal a cost. *Deception* should denote a strategic communicative mechanism for inducing a mistaken belief, not every divergence between a representation and a reported sentence. For opaque systems, the relevant beliefs and intended effects may be difficult to identify. A theoretical account should therefore define the sender's information, feasible messages, objective, and effect on receivers instead of imputing intention from disagreement alone.

34.4 A layered provenance contract

A useful evidence record separates four questions:

(i) *Origin:* who or what issued a statement, according to which authenticated trail?

(ii) *Support:* what observations, calculations, or proofs make the content credible?

(iii) *Authorization:* did the source have authority to make the relevant decision or claim on behalf of an institution?

(iv) *Independence:* does the evidence add information beyond sources already considered?

These are separate dimensions. A cryptographic signature can authenticate an origin without ensuring accuracy; a proof can be valid relative to erroneous premises; an accurate observer may lack permission to allocate resources; ten copies of one report may count as one evidentiary lineage. Institutions that preserve these distinctions are better positioned to correct errors without turning disagreement into a loyalty test.

35 Rational information cascades and collective ignorance

A society can retain a consistent history, aggregate testimony correctly, and still stop learning. The problem need not be deception. When later participants see only earlier decisions, a decision can be an optimal summary for one actor while discarding information that the next actor would have used. Information-cascade theory supplies a classical account of this failure [47]. We develop a symmetric binary example to show exactly where the evidence stops accumulating.

35.1 A decision-only public record

Let an unknown state $\theta \in \{0, 1\}$ have equal prior probabilities. Participant t receives a private signal $s_t \in \{0, 1\}$ with

$$\mathbb{P}(s_t = \theta \mid \theta) = p, \qquad 1/2 < p < 1.$$

Signals are independent conditional on θ, and their accuracy is common knowledge. In sequence, each participant observes previous actions $a_1, \ldots, a_{t-1}$ and its own signal, then chooses $a_t \in \{0, 1\}$ to maximize its probability of matching θ. At exact posterior indifference it follows its private signal. The public record contains actions, not private signals, and all participants know the decision rule. There are no conformity rewards, sanctions for dissent, or strategic misreports in this model. This private-signal tie convention is an explicit variant of the random tie-breaking convention in the basic example of Bikhchandani, Hirshleifer, and Welch [47]. The probabilities derived below belong to the stated variant, not to every informational-cascade model.

Set $\lambda = p/(1 - p) > 1$. Before a cascade starts, define an integer public score z so that the likelihood ratio of the observed action history is λ^z. A new private signal changes its owner's posterior exponent by $+1$ or -1. At $z \in \{-1, 0, 1\}$ the stated tie rule makes the action reveal that signal. Once z reaches 2, even a contrary signal leaves a strictly positive exponent, so the action is 1 regardless of private evidence. Every later such action then has probability one under either state and adds no further likelihood. The symmetric statement holds at $z = -2$.

Proposition 35.1 (Almost-sure locking with a positive error probability)**.** *In this model a cascade begins almost surely after finitely many revealing decisions. Its probability of selecting the wrong state is*

$$\eta(p) = \frac{(1 - p)^2}{p^2 + (1 - p)^2} > 0. \tag{24}$$

The probability that no cascade has begun after $2r$ decisions is $[2p(1 - p)]^r$. After locking, further actions contain no additional information about θ conditional on the preceding public history.

Proof. The posterior argument above identifies the revealing region and the two absorbing action regimes. Starting at score zero, a pair of equal revealing signals reaches $+2$ or -2, while an unequal pair returns to zero. Conditional on either true state, a pair locks correctly with probability p^2, locks incorrectly with probability

$(1-p)^2$, and returns to zero with probability $2p(1-p)$. Conditional independence gives independent repetitions of this two-step cycle until absorption. Since $2p(1-p) < 1$, nonabsorption through r cycles has the stated probability and tends to zero. Summing the probability of eventual incorrect absorption gives

$$\sum_{r=0}^{\infty}[2p(1-p)]^r(1-p)^2 = \frac{(1-p)^2}{1-2p(1-p)},$$

which is (24). In an absorbing regime every subsequent action is deterministic given the public history under both possible states, so conditioning on it changes no posterior. □

For $p = 3/4$, the eventual error probability is $1/10$. A long run of unanimous actions can therefore preserve the same mistaken decision with no additional evidentiary support. Each participant remains Bayesian-rational under the stipulated information. Unanimity is an outcome of the disclosure protocol, not evidence that a large number of independent signals favored the decision.

35.2 An evidence-preserving comparison

Now change only the information contract. Suppose authentic individual signals are available to a decision rule that selects their majority. It need not infer signals from prior actions. Let $\widehat{\theta}_n$ be any majority choice with an arbitrary fixed rule for ties.

Proposition 35.2 (Conditional independence permits continuing accuracy gains)**.** *Under the same signal model,*

$$\mathbb{P}(\widehat{\theta}_n \neq \theta) \leq \min\left\{1, \frac{p(1-p)}{n(p-1/2)^2}\right\}.$$

In particular, the error tends to zero as n increases.

Proof. Conditional on either value of θ, the indicators $\mathbf{1}\{s_t = \theta\}$ are independent Bernoulli variables with mean p and variance $p(1-p)$. A wrong majority requires their sample mean to be at most $1/2$, including a possible wrong tie. Chebyshev's inequality bounds this event by $p(1-p)/[n(p-1/2)^2]$. Averaging the identical conditional bounds over θ proves the result. □

This elementary bound is deliberately conservative. The exact finite error is a binomial tail, with a tie term determined by the decision rule. The comparison does not assume that all signal disclosures are truthful by virtue of being public. Authentication, privacy, observation cost, and reporting incentives are additional contracts. The result isolates the value of preserving usable independent evidence, not a general preference for making private states public.

35.3 Designing epistemic institutions rather than enforcing agreement

The institution chooses what enters its record. Publishing an action without its evidentiary basis may be adequate for coordinating a resource allocation but inadequate for later scientific assessment. A registry can therefore retain a decision, the observations supporting it, their dependencies, and the unresolved alternatives as distinct objects. An office may comply with the current allocation while maintaining a contrary hypothesis about the evidence behind it.

Possible countermeasures include independently acquired observations, protected channels for disconfirming evidence, and periodically reopening a settled question when the evidence contract changes. Their effectiveness depends on whether the new information reaches the actual decision procedure. Repeating the same cached report through a separate channel supplies no new signal. Likewise, randomized prompts do

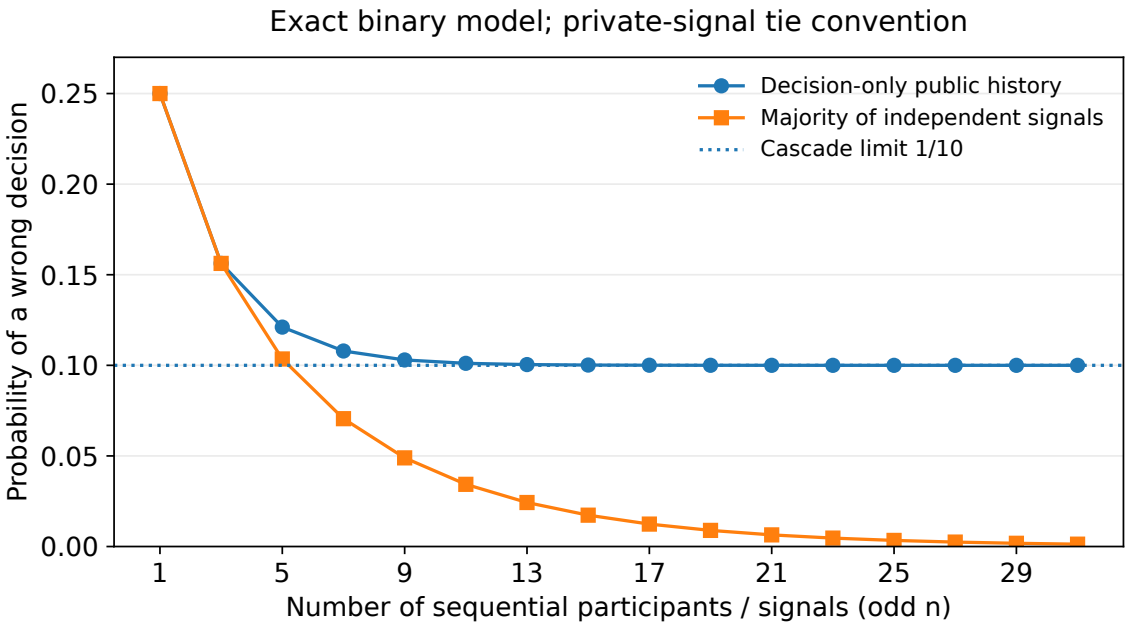


Figure 10: The binary analytical model with signal accuracy $p = 3/4$. For odd population sizes, the error of the last action approaches the cascade floor $1/10$, while a majority of the underlying independent signals continues to improve. Both curves are calculated from finite probabilities, not sampled agent runs.

not establish conditional independence of errors. The retained provenance distinctions of Section 34 remain necessary.

In the commons, an urgency approval can be a binding procedural fact without being an independent observation that the underlying task was genuinely urgent. Subsequent reviewers who see only approvals can rationally overrely on them. Keeping the supporting certificates separate from the sequence of votes makes it possible to ask what was observed, what was inferred, and what was authorized. Collective ignorance is then analyzed as an information-processing failure rather than automatically as dishonesty, irrationality, or a failure to reach consensus.

36 Public knowledge, notice, and epistemically faithful transport

Knowing a rule, knowing that another participant knows it, and having a shared protocol for when it takes effect are different properties. A common record can help establish them, but the existence of that record does not imply any particular agent has accessed it. Distributed epistemic logic gives a precise language for these distinctions [25]; dynamic epistemic logic studies how communicative events change them [39].

Fix one configuration epoch and a nonempty group J of named decision centers. Let W be a finite set of possible worlds. Each equivalence relation $\sim_i$ records the worlds agent i cannot distinguish using its permitted information. A world may include the epoch, relevant history, and current records. This is an idealized semantics of information, not a claim that an implemented agent can compute every logical consequence.

For a proposition φ, $K_i\varphi$ holds at w when φ holds at every $v \sim_i w$. Let $E_J\varphi = \bigwedge_{i\in J} K_i\varphi$. Define common knowledge by all finite depths,

$$C_J\varphi = \bigwedge_{n\geq 0} E_J^n\varphi, \qquad E_J^0\varphi = \varphi.$$

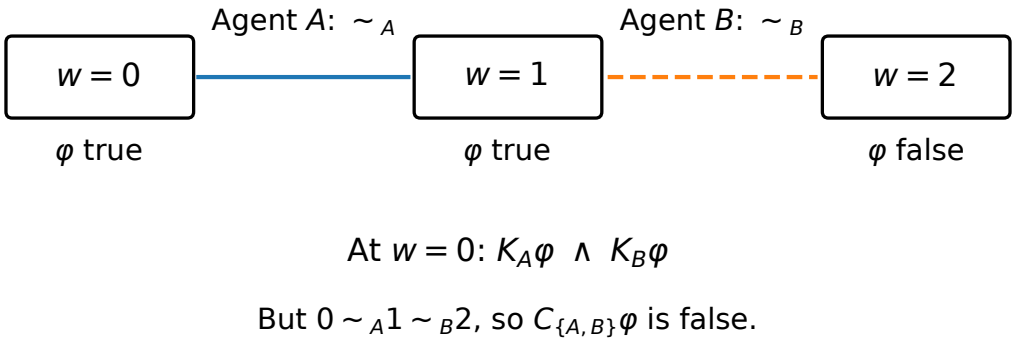


Figure 11: At world 0, each agent's own alternatives satisfy φ. Their combined indistinguishability graph nevertheless reaches world 2, where φ fails. Reflexive edges are omitted. Shared first-order knowledge is weaker than common knowledge.

Distributed knowledge instead quantifies over the *intersection* of the agents' indistinguishability classes. Common knowledge is governed by chains in their *union*. Pooling complementary private evidence can therefore reveal a fact that is not public to any participant.

Proposition 36.1 (Common knowledge as a reachability condition)**.** *Let* $\mathsf{Reach}_J(w)$ *be the worlds reachable from* w *by a finite chain of relations* $\sim_i$ *with* $i \in J$*, including the zero-length chain. Then*

$$w \models C_J\varphi \quad \Longleftrightarrow \quad \mathsf{Reach}_J(w) \subseteq \{v : v \models \varphi\}.$$

Proof. Induction on n shows that $E_J^n\varphi$ requires φ at the endpoint of every chain of n indistinguishability steps. Requiring this for every finite n is exactly the stated reachability inclusion. □

Counterexample 36.2 (Everyone knows, but it is not common knowledge)**.** Let $W = \{0, 1, 2\}$ and let φ hold exactly at $0, 1$. Agent A has partition $\{0, 1\} \mid \{2\}$, and agent B has partition $\{0\} \mid \{1, 2\}$. At world 0, both agents know φ. However, the chain $0 \sim_A 1 \sim_B 2$ reaches a world where it is false. Thus $E_{\{A,B\}}\varphi$ holds at 0 but $C_{\{A,B\}}\varphi$ does not.

36.1 An ideal public announcement is a strong update assumption

For a propositional fact p whose valuation is unchanged by the announcement, an ideal truthful public update restricts W to $W_p = \{w : w \models p\}$ and restricts every $\sim_i$ to that set. This update models that the event, its interpretation, and its receipt are public to the group. In the restricted model, every reachable world satisfies p, so Proposition 36.1 gives $C_J p$. The conclusion follows from the update semantics; it is not a theorem that writing a file implements that event.

The restriction to a stable propositional fact matters. An announcement can change the truth of a formula about what someone does not know. Nor does a public fact in one electorate remain common knowledge after admitting a fresh member. The relevant group, epoch, interpreter, delivery conditions, and background assumptions must be stated.

Classical limitations on common knowledge depend on the timing and communication model, and not every useful coordination task requires full common knowledge [25]. A sequential allocation may need only a valid certificate at the point of execution. A simultaneous global activation rule demands a stronger

knowledge condition. Bounded acknowledgments and versioned notices can implement specified weaker requirements without pretending to supply infinitely many levels of mutual assurance in an arbitrary network.

36.2 Notice is an institutional choice as well as an information condition

A rule may define applicability by a public epoch, by authenticated receipt, or by an acknowledgment procedure. These are different institutional contracts. Whether lack of notice excuses an action is not deduced from K_i alone. It is an explicitly chosen relation between the information state and the permission or obligation monitor. In the commons, an approval can exist in the ledger before a disconnected office knows it exists. The specification must say whether that office is allowed to proceed under an earlier epoch, must wait, or can appeal a late notification. Truth of an adoption record and enforceability against a participant are separate observables.

36.3 Preserving behavior need not preserve what participants know

A state quotient that retains allocations may identify worlds with different notification histories. It can therefore preserve a task trace while changing claims of the form "office A knows that office B has accepted the debt." To preserve such claims, a stronger interface is needed.

Consider models on W_A, W_B with the same named agents and atomic predicates. Let $T : W_A \to W_B$ be surjective and preserve the truth of each atom in both directions. For every agent i, require

$$w \sim_i^A v \implies T(w) \sim_i^B T(v) \quad \text{(forth)},$$
$$T(w) \sim_i^B z \implies \exists v : \; w \sim_i^A v \text{ and } T(v) = z \quad \text{(back)}.$$

These are the usual bounded-morphism conditions, stated here as a transport contract.

Proposition 36.3 (Epistemic preservation and reflection)**.** *Under these conditions, w and $T(w)$ satisfy exactly the same formulas built from the matched atoms, Boolean connectives, K_i, and C_J. Such transports compose. If the same state maps also satisfy Definition 16.1, they preserve the translated operational conversations and the stipulated epistemic formulas at every corresponding reached state.*

Proof. Use induction on formulas. Atoms and Boolean connectives follow from the assumptions. For K_i, forth sends every source alternative to a target alternative, while back lifts every target alternative; applying the induction hypothesis proves both implications. A finite chain of indistinguishability steps is mapped by repeated forth and lifted by repeated back. Proposition 36.1 therefore proves the common-knowledge case. Forth composes directly, and two successive applications of back lift any target alternative through the intermediate model. Surjectivity and atomic truth preservation also compose. The operational conclusion then follows from Proposition 16.2 at each reached state. □

The theorem does not cover new agents whose knowledge relations have not been matched, arbitrary target messages outside the translation, or a change in what the atoms mean. It also makes no inference from formal knowledge to trustworthiness, consent, or legitimacy. Its useful conclusion is narrower: dynamic behavior, message effects, and epistemic claims can be preserved together, but each requires its own explicit part of the interface certificate.

Part IV

Authority and joint feasibility

37 Effective authority, vetoes, and institutional capture

The number of agent processes is not the number of independent centers of authority. One credential can control many processes; several credentials can be required for one consequential operation. An institutional analysis must therefore begin with office rights and operational dependencies rather than a head count.

Fix a nonempty family $\mathcal{W}$ of minimal approval coalitions for a particular amendment. Each coalition is a nonempty subset of the n registered offices. Approval is monotone: an amendment passes exactly when the set of approving offices contains some member of $\mathcal{W}$. Assume every office can independently approve or refuse, and that no unmodeled administrator or scheduling dependency bypasses the rule. This is a simple-game representation of approval authority [33].

Proposition 37.1 (Blocking coalitions as hitting sets)**.** *A coalition C can block the amendment by withholding all of its approvals, regardless of what the other offices do, if and only if*

$$C \cap W \neq \varnothing \quad \text{for every } W \in \mathcal{W}.$$

Thus the minimum number of offices with unconditional blocking power is the minimum hitting-set size of $\mathcal{W}$. Under an equal q-of-n rule with $1 \leq q \leq n$, it is $n - q + 1$.

Proof. If C intersects every minimal approval coalition, refusing all approvals in C prevents each such coalition from being contained in the approvers. Conversely, if some W is disjoint from C, its offices can approve and pass the amendment despite C's refusal. For the threshold rule, the complement of C can pass exactly when it has at least q members, so blocking requires $n - |C| < q$, or $|C| \geq n - q + 1$. □

In the commons, two offices can block a three-of-four amendment, but two cannot enact it. A single office is a veto holder exactly when it belongs to every minimal approval coalition. These are statements about one rule and one event, not complete measurements of social power.

37.1 An approval rule may conceal another authority structure

Suppose the nominal three-of-four process cannot commit any decision without a separate administrator's signature. The effective approval coalitions must include that administrator. Applying the threshold formula to only the four visible voters would overstate their independence. Scheduling, admission of evidence, access to the amendment queue, and power to replace the checker can similarly alter the implemented authority structure.

The correct remedy is to enlarge the declared realization and recompute the authority relations, not to infer decentralization from its public diagram. Conversely, a veto can protect a legitimate minority or a resource owner's consent. Small blocking coalitions do not by themselves establish harmful capture. Capture requires an additional protected-purpose or authorization criterion against which the actual control is judged.

An institution can also be procedurally compliant while concentrating benefits or suppressing effective appeal. That is a normative and incentive question distinct from whether a signature threshold was satisfied. A theory of authority should preserve the distinction between *ability to prevent*, *ability to enact*, *ability to set the agenda*, and *ability to change the procedure*.

38 Founding authority and the limits of internal attestation

Procedural continuity and authorized origin are not the same property. A constitution can specify how amendments are judged after its adoption without settling who was entitled to adopt it. An internally consistent sequence of certificates can preserve the first record's assertion of authority even when no accessible evidence establishes that assertion.

The question can be stated without selecting a universal theory of legitimacy. Let $\mathcal{H}$ be a set of possible founding histories, let $a : \mathcal{H} \to \{0, 1\}$ be an externally stipulated predicate such as "the resource owner authorized this founding event," and let $R : \mathcal{H} \to \mathcal{R}$ be the record available to the institution. The predicate models one particular authorization criterion, not moral legitimacy in its entirety.

Proposition 38.1 (When a founding fact is determined by internal records)**.** *There exists a function $f : \mathcal{R} \to \{0, 1\}$ satisfying $f(R(h)) = a(h)$ for all $h \in \mathcal{H}$ if and only if a is constant on every nonempty fibre of R. This is a statement of informational determination, not necessarily of computability. If two histories with different labels induce identical distributions of every record available to an evaluator, no evaluator restricted to those records distinguishes them with accuracy above $1/2$ under equal priors.*

Proof. If $R(h) = R(h')$, any function of the record gives the same answer on both histories, proving necessity. If the label is constant on every nonempty fibre, define $f(r)$ to be that common value and choose arbitrary values off the image of R. This gives sufficiency. For the probabilistic statement, equal record laws remain equal after any common randomized decision procedure. Under equal priors, its probability of a correct label is therefore $1/2$. □

A self-authored line saying "this constitution was authorized" cannot distinguish two histories that leave identical accessible records. An authenticated receipt from a recognized external authority may distinguish them, but then the recognition of that authority and the key binding are explicit premises. This does not mean that all records are untrustworthy or that an infinite regress makes institutions impossible. It identifies the trust boundary instead of concealing it inside the verifier.

38.1 Three judgments that should remain separate

Internal validity asks whether a transition follows the currently accepted formal rules. Historical authorization asks whether the relevant actor actually granted a particular authority under the declared criterion. Normative legitimacy asks whether the criterion and arrangement are justified. A proof system can establish the first relative to its semantics. Evidence and trust assumptions can support the second. Neither calculation uniquely determines the third.

An institution may openly adopt an internally defined founding convention and state the resources and participants to which it applies. What it cannot do by definition alone is change an external ownership fact, manufacture another party's consent, or retroactively authorize actions on behalf of someone absent from the convention. The distinction is between declaring a rule for a model and establishing a fact about its realization.

38.2 A federation inherits trust boundaries, not just certificates

When one institution imports another's origin record, it must decide which trust anchors, delegations, revocations, and unresolved disputes it recognizes. A translator can preserve the bytes and inferential structure of a certificate while the receiving institution rejects its root authority. That is not necessarily a translation error. It can be an explicit disagreement about authorization.

The commons therefore needs separate records for the resource owner's founding grant, the adopted internal procedures, and subsequent authorized amendments. A successor can preserve all three and still

encounter a disputed external premise. Its honest state includes that dispute. Consistent internal history is evidence of procedural continuity, not an all-purpose certificate of truth or legitimate origin.

39 Conflict, sabotage, and organizational pathology

A machine society is not guaranteed to be cooperative, benevolent, or stable. Multiple decision centers can cooperate on one resource and compete on another. The same persistent rules that economize on coordination can create concentrated authority, lock in mistakes, or transmit a harmful practice to later participants. Security is therefore not an unrelated afterthought; it is partly the problem of preserving valid relationships between declarations, implementations, permissions, and evidence.

39.1 Failures occur at different layers

We distinguish at least five abstract failure surfaces. *Content failures* corrupt beliefs or stored facts. *Interface failures* alter the interpretation or routing of otherwise unchanged content. *Authority failures* allow an unauthorized actor to issue a consequential operation. *Revision failures* change which future evidence or rules are admissible. *Allocation failures* exploit shared resource procedures. These failures can combine: a fabricated record may justify an unauthorized rule change, after which the new rule rejects the evidence needed to expose the fabrication.

The terms *sabotage* and *virus* require restraint. A sabotage claim requires an adversarial objective and a causal route by which the action impairs an institution. A "schema virus" is an analogy until one specifies the copied representation, adoption mechanism, susceptibility, persistence, and measured harm. Error propagation can be unintentional; stability can coexist with misinformation; and divergence from a prevailing view may represent a correct new discovery rather than a pathology.

39.2 A bounded propagation inequality

A transparent comparison model can make one class of failure mathematically tractable. Let $x_t \in \mathbb{R}^n_{\geq 0}$ record nonnegative deviations at n institutional positions. Suppose an explicitly justified comparison model provides

$$x_{t+1} \leq Bx_t + w_t \quad \text{componentwise}, \qquad B \geq 0, \tag{25}$$

where w_t represents fresh disturbance and B summarizes one-step propagation. Suppose a vector $v > 0$ and $\kappa < 1$ satisfy $Bv \leq \kappa v$. In the weighted maximum norm $\|x\|_{v,\infty} = \max_i x_i/v_i$ we obtain

$$\|x_t\|_{v,\infty} \leq \kappa^t \, \|x_0\|_{v,\infty} + \sum_{j=0}^{t-1} \kappa^{t-1-j} \left\|w_j\right\|_{v,\infty}. \tag{26}$$

The proof is the one-step monotonic estimate followed by induction. Under uniformly bounded disturbance $\|w_t\|_{v,\infty} \leq \bar{w}$, the last term is at most $\bar{w}(1-\kappa^t)/(1-\kappa)$. This gives a conditional attenuation result, not a universal contagion threshold. A graph alone cannot supply the matrix B; the rates must come from a justified dynamical model. More connectivity can improve detection as well as propagation, and nonlinear strategic behavior can invalidate the linear comparison.

39.3 Institutional capture and the meaning of power

Power is not exhausted by reputation or by the number of messages an agent produces. In a computational institution it can arise from privileged read or write access, control of scarce resources, amendment rights, scheduling priority, the right to certify evidence, and the ability to prevent an appeal. A useful power analysis distinguishes *information reach* from *decision reach*: a participant can influence many beliefs while holding no formal permission, or can issue one privileged command with enormous consequences.

A simplified reach functional for agent i and horizon T is the set of consequential institutional coordinates whose future distributions can be changed by some *admissible unilateral intervention* of i. This is an intervention-relative property; it must not silently assume that the agent can overwrite a variable to which it has no access. Quantifying authority requires the feasible intervention family and a choice of outcomes. A concentration score built from these reaches can be a diagnostic, but not a proof of illegitimacy.

Formal policy compliance alone does not rule out institutional capture: a rule may be faithfully executed and still privilege a coalition that controls amendments or vetoes. Likewise, a successful self-repair routine may restore a harmful prior state rather than correct it. Any proposed notion of health must include protected freedoms, an error-correction contract, and an explicit source of normative authority. Agreement is not itself the objective.

39.4 Defensive design implications

Ordinary security principles retain their relevance: least privilege, complete mediation, separation of authority, auditable transitions, compartmentalized credentials, and recovery from corrupted records [13]. A proof-carrying decision can narrow the trusted checking base [12], but cannot prove the adequacy of its own external assumptions. Copyable identities create classic Sybil concerns [14]. A fork's assertion that it is a new participant is not automatically a new independent vote or a new allocation of finite resources.

The strongest general principle is *evidence-preserving reversibility where reversal is physically possible*. Exploratory proposals need not receive immediate permission to rewrite canonical rules; reviews should preserve rejected alternatives and reasons; rollbacks should retain an audit trail; and irreversible external actions require a different permission standard from reversible sandbox computations. This is a defensible design stance, not a theorem that one fixed governance policy fits every environment.

40 Societies of societies: federation, mobility, and scale

Once groups can maintain institutions, their boundaries become objects of interaction. A participant may migrate between two groups, carry an archive from one to another, borrow computational resources, or hold obligations in multiple jurisdictions. A coalition may split without dissolving all of its records; two institutions may merge while retaining conflicting rules. Such events define an *ecology of organizations*, not merely a larger population of otherwise unrelated agents.

The relevant inter-organizational interface has more structure than a message format. It includes state translation, identity and provenance assertions, resource transfer conditions, compatible permission semantics, and a means to resolve contradictory commitments. Calling two communication protocols interoperable because their messages parse correctly ignores whether the resulting actions are authorized and what happens to existing obligations.

40.1 Migration is not unrestricted inheritance

A migrating process can carry operational knowledge without carrying valid credentials. An organization that accepts all imported permissions merely because an agent has an intact memory confuses history with authority. A safer abstract contract is explicit about which local capabilities are newly granted, which predecessor obligations continue, which are discharged, and which disagreements require arbitration. Copying a state does not copy a budget, physical hardware, a nontransferable credential, or any legal status assigned by people.

40.2 Scale changes which institutions are economical

At small scale, bilateral negotiation may be cheap and simple. At larger scale, direct communication can become expensive, so rules may be compressed into roles, standards, and delegated procedures. Hierarchy can

reduce latency or message count under explicit network constraints; Section 10 also proves that an unrestricted controller can simulate the same finite computation. The correct separation concerns resources, privacy, fault domains, and the work required to discover or maintain useful interfaces, not a new computability class created by social grouping.

Large networks also increase the surface for copied mistakes, inconsistent rule versions, and concentrated control over common infrastructure. Modularity can localize some failures while making cross-boundary repair harder. There is no universal monotone law saying that greater connection makes a machine society more stable, intelligent, or vulnerable. Those effects depend on transition kernels, institutional incentives, topology, and the cost of checking interactions.

40.3 The absence of an inevitable collective self

It is tempting to imagine a sequence from many agents to coordination, from coordination to institutions, and from institutions to a unified intelligence. Nothing in the framework implies that final arrow. Some federations are valuable precisely because they prevent one controller from collapsing plural judgments. Others may deliberately adopt a common policy for restricted operations while retaining disagreement on everything else. The existence of a macroscopic description is also not evidence of a first-person perspective. Computational closure, social organization, legal recognition, and subjective experience remain separate questions.

41 Endogenous roles, adaptive networks, and resilient specialization

A continuing office need not be one of a permanently supplied list. A population may establish a verification service, divide a task into specialties, retire a redundant committee, or form a temporary coalition. Its communication graph can change with these decisions. To express that possibility, let G_t specify currently permitted communication links, r_t an assignment of roles, and c_t a competence and capacity record. Their transition belongs to the realized state:

$$(G_{t+1}, r_{t+1}, c_{t+1}) \sim \Gamma_{I_t}(\cdot \mid G_t, r_t, c_t, H_t),$$

where H_t denotes the retained information used by the update. A Markov formulation must include a sufficient part of that information in the state rather than treating an unspecified history as an unobserved input.

Adaptive-network theory studies the coupling of node dynamics to changing topology [48]; strategic network-formation theory distinguishes beneficial networks from those stable under participants' incentives [49]. Here the additional question is institutional: which training, delegation, link creation, and role transfers are authorized, and which existing commitments survive those changes? Endogenous division of labor requires that participant interaction affects the adopted structure. Writing Γ does not show that an externally assigned specialization was discovered.

41.1 Specialization has a cost and a reachability condition

Consider two services performed once per period. A generalist arrangement costs $2c_g$ per period. Two specialists cost $2c_s + b$, where b is the required communication and integration cost, and training costs h for each specialist. Over T periods the specialist arrangement has lower total cost precisely when

$$T\big[2(c_g - c_s) - b\big] > 2h.$$

This is an instance of institutional amortization, not a universal law of specialization. Verification overhead, replacement, workload variation, and lost redundancy must be added when relevant. A specialist who is faster locally can still increase total cost if every result requires expensive cross-boundary interpretation.

Reachability is separate. Suppose two agents must continuously staff two different roles and each agent has capacity one. A coordinated exchange of their assignments may improve both agents' outcomes. Yet neither

agent can move alone while all roles remain staffed and no role is double-booked. Permitting an atomic handover, a temporary reserve agent, or a short service interruption changes the admissible transition graph. The best assignment and a permissible path to it are different mathematical objects, just as in institutional formation and constitutional amendment.

When a finite role-and-network update system admits a common potential that strictly increases on every permitted improvement, the earlier finite termination result applies. There is no general reason for individually beneficial link changes, training choices, and delegated powers to share such a potential. Cycles and stable but inefficient specialization remain possible.

41.2 A sharp condition for replacement capacity

Let J be a finite nonempty set of roles that must be filled simultaneously and A a finite set of agents. An edge (j, a) means that agent a is both capable and authorized to fill role j during the relevant interval. Every agent can fill at most one role at a time. A feasible staffing is an injective assignment $f : J \to A$ along these edges. Let $N(S)$ be the agents adjacent to at least one role in $S \subseteq J$.

The question is not merely whether every role has a backup. It is whether backups can cover all roles together after a loss. The classical matching principle originates with Hall [50]; the following erasure form exposes the exact redundancy requirement.

Theorem 41.1 (Authorized staffing after any bounded agent erasure)**.** *Fix an integer $0 \le k \le |A|$. A feasible staffing exists after removal of every set $F \subseteq A$ with $|F| \le k$ if and only if*

$$|N(S)| \ge |S| + k \qquad \text{for every nonempty } S \subseteq J. \tag{27}$$

The surviving capability and authorization edges are assumed unchanged, and a new assignment may be selected after the erased set is known.

Proof. For necessity, if a nonempty S has fewer than $|S| + k$ neighbors, erase $\min\{k, |N(S)|\}$ of those neighbors. Fewer than $|S|$ remain, so no injective staffing of S is possible.

For sufficiency, remove any F of size at most k. The surviving neighbors of each S number at least $|N(S)| - k \ge |S|$. We recall the finite matching argument. Choose a maximum matching in the surviving bipartite graph. If a role remains unmatched, explore alternating paths beginning at it, using unmatched edges toward agents and matched edges back toward roles. No reached agent can be unmatched, since that would produce an augmenting path. Let S be the reached roles and B the reached agents. Every neighbor of S is in B, and each agent of B is matched to a distinct reached role other than the initial unmatched role. Hence $|N(S)| = |B| < |S|$, a contradiction. The maximum matching therefore fills all roles. □

The theorem permits a simple robustness index. If the initial graph admits a staffing, define

$$\sigma = \min_{\varnothing \ne S \subseteq J} \big(|N(S)| - |S|\big).$$

Then exactly the erasure budgets $k \le \sigma$ are guaranteed tolerable in this model. This is a static allocation certificate, not a claim that the surviving agents can discover or execute the reassignment before a deadline. That further question uses the communication and permission constraints of Sections 25 and 23.

Counterexample 41.2 (Backups for each role need not provide collective resilience)**.** Two roles can each be filled by either of two agents a, b. Every role has two capable authorized candidates, and the two roles can initially be filled. After either agent is erased, the one survivor cannot fill both roles simultaneously. The singleton-role conditions for $k = 1$ hold, but the two-role set has only two neighbors rather than the required three.

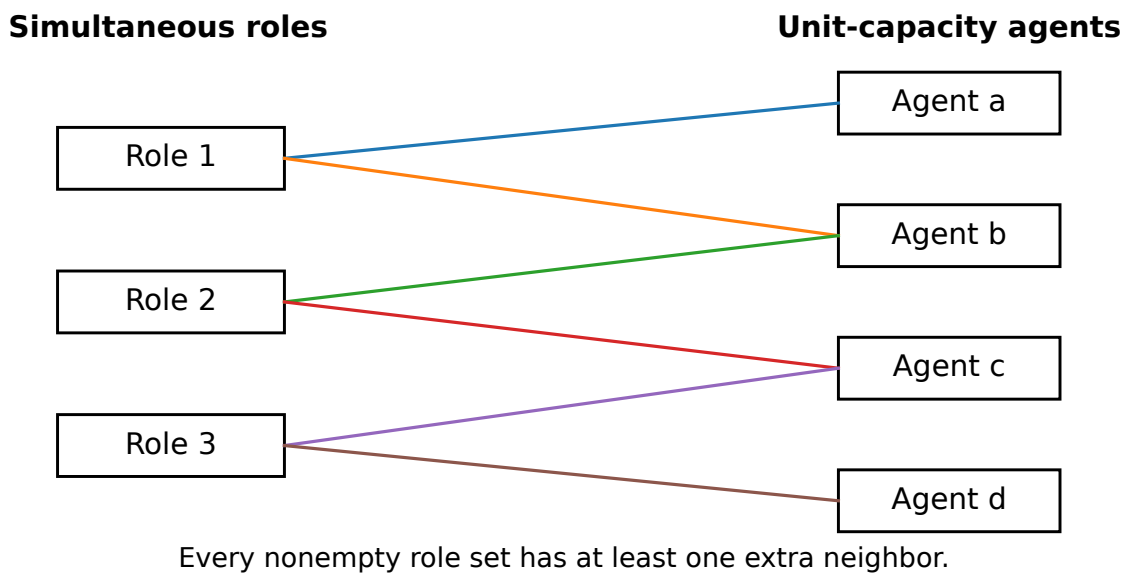


Figure 12: Three simultaneous roles and four unit-capacity agents. Edges represent both competence and authorization. The neighborhoods satisfy the one-erasure staffing criterion. The guarantee permits rematching after the failure is known; it does not guarantee instantaneous handover.

For three roles with neighborhoods $\{a, b\}$, $\{b, c\}$, and $\{c, d\}$, every nonempty role set satisfies (27) with $k = 1$. One erased agent can be accommodated by reassignment even though no single agent can perform every role. Resilience can reside in a pattern of overlapping specialization rather than a universally capable backup.

41.3 Time-respecting communication cannot be read from an aggregate graph

Changing topology also changes who can actually learn what. A union of links over an interval discards their ordering. The resulting graph may look connected while failing to carry a particular observation to a decision maker in time.

Counterexample 41.3 (A connected union with no usable evidence path)**.** Agent A initially holds an arbitrary bit and the other agents do not. At time one, B sends its current state to C. At time two, A sends the bit to B. No other communication occurs before C must decide. The union of directed contacts contains the path $A \to B \to C$, but C's received message is independent of the bit. No decoder at C can recover an initially fair bit with accuracy above $1/2$ under this access model.

The missing object is a time-respecting causal route, not an additional static connection. The distinction is consistent with the event order of Section 23. In a changing society, a nominal right to appeal or a nominal verification edge may be ineffective when it is never available during the interval in which evidence matters.

41.4 Adaptive roles do not automatically preserve institutional plurality

A new verifier can reduce errors while concentrating epistemic authority. A central routing service can reduce messages while becoming a veto point. Copying a specialist may increase nominal staffing without adding independent failure domains, and a revocation may remove authorization edges even when all technical skills remain. Each of these changes belongs in the state to which the staffing or communication theorem applies.

A useful role-update contract therefore tracks competence, capacity, decision rights, handover obligations, and accessible evidence separately. It can require that a protected staffing margin survive every admitted reorganization, while still allowing the actual role assignment to change. Such a constraint does not choose which roles deserve protection or who may impose the training cost. It makes the consequences of those choices analyzable.

The commons can now add a temporary verification role, train an alternate occupant, and revise the communication schedule without identifying an office with one model. Its social organization lies partly in this maintained division of work. The formal test is whether the adopted arrangement has a feasible and authorized realization through turnover, not whether its role names resemble a familiar human institution.

42 Joint institutional feasibility and certified continuation

An institution can possess several valid certificates and still lack a feasible way to operate. A safe allocation may require one action, an effective audit another, and a timely repayment a third. Each statement can be correct when its own resource and information assumptions are considered alone. The common implementation fails when those assumptions cannot hold together.

The resulting question is not whether a population can optimize one collective utility. It is whether a declared set of obligations has a jointly executable continuation. The protected requirements can include resource conservation, valid authority, a false-positive limit, a specified incentive margin, service deadlines, and feasible staffing. Their normative selection remains outside the synthesis theorem. Once selected, they must constrain the same action and the same realized history.

This section supplies a finite construction rather than an unrestricted theory of decentralized control. It combines the existing obligation monitors and governance labels with a catalogue of implementable local protocols. The mathematical foundations are the knowledge construction and fixed-point methods for games with imperfect information [31]. The additional contribution is to identify which institutional certificates must enter that construction, and what is lost when an abstraction omits them.

42.1 A finite contract with an explicit information boundary

Let S be a finite nonempty state space for the complete contract. A state records the relevant institutional mode, resource ownership, bounded obligation monitors, current staffing, and any private memory needed to make transitions well defined. Let $h : S \to Y$ be the observation available in a declared common record at each decision epoch. It need not reveal the private components. The initial information is a nonempty set B_0 contained in one fibre of h.

A finite nonempty catalogue U contains protocol profiles, not arbitrary commands to the complete microstate. A profile can specify one local response map for each office, with each map reading only that office's permitted observation and stored local memory. It can also specify a committed monitor, a communication schedule, and the steps needed to select and enact that profile. These implementations, their costs, and their decision rights must be established separately. An instruction that requires inaccessible private information is not made implementable by listing it in U.

For each s, let $A(s) \subseteq U$ contain profiles satisfying the immediate contract at that state. Every profile in this set must pass the same resource, authority, staffing, and monitoring checks. In particular, a monitor included in a profile must satisfy all relevant incentive and false-positive inequalities simultaneously. Let $\Delta(s, u) \subseteq S$ be a nonempty finite set of permitted next states for each s, u. The set includes every environment response or fault within the declared guarantee. Inadmissible profiles are never selected.

The contract uses *sure* guarantees: every transition in Δ is treated as possible, without weighting it by probability. A probabilistic transition model may supply these supports, but an unlikely infinite path is not discarded when proving bounded service. Nor does a deterrence inequality justify silently deleting deviations

from Δ. Safety under compliant behavior and safety under specified deviations are different contracts. The latter must include their physical consequences in the transition model.

Let $S_{\text{safe}} \subseteq S$ encode the protected state requirements. Breach monitors distinguish fulfillment, authorized discharge, pending debt, and deletion. A finite bound on stored obligations or deadlines is part of the model. An overflow must have an explicit treatment, such as a rejected admission or a breach state. The theorem does not silently replace an unbounded history by a finite one.

A common record at decision epochs is itself an infrastructure assumption. It is not implied by eventual replica convergence. The model applies only when that observation and the selection of a catalogue profile have the declared implementations. The selected profile can then be executed by distinct local agents without exposing their private states. This is a controlled information boundary, not a claim that every decentralized architecture has such a representation.

42.2 The state of possible institutional realizations

The relevant public state is the set of complete states still compatible with the observed history. Define

$$\mathcal{B} = \{B \subseteq S : B \neq \varnothing,\ h \text{ is constant on } B\}.$$

These are sets of possibilities, not posterior probability distributions. At B, a profile must be admissible for every remaining possibility:

$$A(B) = \bigcap_{s \in B} A(s).$$

For $u \in A(B)$ and a next public observation y, define

$$\operatorname{Post}(B, u, y) = \left(\bigcup_{s \in B} \Delta(s, u) \right) \cap h^{-1}(y). \tag{28}$$

Only nonempty successor sets are possible observations. Write $\operatorname{Succ}(B, u)$ for their collection. Induction on the public history shows that (28) retains exactly the complete states consistent with that history and the selected profiles.

This construction does not pool private observations before choosing u. It pools only the mathematical possibilities consistent with the common record. Private observations can affect the actions of the local maps already specified by u. More elaborate strategies using additional private history require that history, or a sufficient finite memory of it, to be included in the declared realization. Completeness below is relative to this finite catalogue and information model.

42.3 A continuation kernel for the whole contract

Set

$$W_0 = \{B \in \mathcal{B} : B \subseteq S_{\text{safe}}\}, \tag{29}$$

$$W_{k+1} = \{B \in W_0 : \exists u \in A(B),\ \operatorname{Succ}(B, u) \subseteq W_k\}. \tag{30}$$

The sequence is decreasing. It stabilizes after at most $|\mathcal{B}|$ strict removals, and we denote its fixed point by W_*. This is the joint continuation kernel. A state is removed not only when it violates an invariant now, but also when every available profile has a possible successor from which the remaining contract cannot be sustained.

Theorem 42.1 (Joint continuation under a common observation history)**.** *For every $k \geq 0$, $B \in W_k$ if and only if there exists a deterministic strategy based on the common observation and profile history that selects admissible profiles and keeps the complete state in S_{safe} for the next k transitions, starting from every $s \in B$. Moreover, $B \in W_*$ if and only if such a strategy exists for an unbounded number of transitions. On W_*, a strategy depending only on the current possibility set B suffices.*

Proof. At horizon zero the condition is exactly $B \subseteq S_{\text{safe}}$. At positive horizon the strategy must choose one profile from the same common history, so it must lie in $A(B)$. After each possible observation its continuation must work from the corresponding set in (28). Induction gives exactly (30), in both directions.

Monotonicity of the predecessor operation makes W_k decreasing. Finiteness implies stabilization. At the fixed point, select for every $B \in W_*$ one witnessing profile whose successors all remain in W_*. Induction then proves safety and admissibility for every finite prefix of every realized path, hence for the complete path. Conversely, an unbounded safe strategy is safe for every finite horizon, so its initial set belongs to every W_k and therefore to the stabilized intersection. The witnessing selection is stationary in B, although maintaining B generally requires memory of the public history. □

No society-wide preference ordering appears in this theorem. It characterizes joint satisfaction of a declared contract, not the desirability or consent behind that contract. It also gives no favorable complexity bound: the possibility space can contain exponentially many subsets of S. A computationally expensive solution is still distinct from a procedure the society can discover and execute within its own resource budget.

The action intersection matters. A full-information controller can choose different safe profiles at two hidden states while the common-history institution cannot choose between them. This is the constructive counterpart of the indistinguishability obstruction in Section 25. The kernel can therefore be empty even when every complete state has a safe continuation under full information.

42.4 Safety does not complete a service

The fixed point can contain a safe idle loop. Outstanding deadlines should therefore be represented by monitors that advance when time passes, not by a permanently pending record. For service without a fixed external deadline, a separate progress condition is needed.

Let $G \subseteq S_{\text{safe}}$ be a stipulated completion set. The target is *certified completion*, meaning that the current possibility set lies wholly in G. Let

$$\mathcal{G} = \{B \in W_* : B \subseteq G\}, \quad A_*(B) = \{u \in A(B) : \mathrm{Succ}(B, u) \subseteq W_*\}.$$

Define $R_0 = \mathcal{G}$ and

$$R_{k+1} = R_k \cup \{B \in W_* : \exists u \in A_*(B),\ \mathrm{Succ}(B, u) \subseteq R_k\}. \tag{31}$$

The increasing sequence stabilizes at R_*. For $B \in R_*$, let $r(B)$ be its first index of entry.

Proposition 42.2 (Safe service with a finite progress rank)**.** *A set B lies in R_k if and only if an admissible common-history strategy can ensure certified completion within at most k transitions while preserving the contract indefinitely. A set lies in R_* if and only if such a strategy can ensure eventual certified completion on every possible path while preserving the contract. Before completion, a stationary choice with strictly decreasing r reaches the target within $r(B)$ transitions. After completion, a safety strategy on W_* suffices.*

Proof. The bounded claim follows by induction from (31). At a nonterminal set of rank $r > 0$, a witnessing profile sends every successor to rank at most $r - 1$. All selected profiles preserve W_*, so a safety continuation is available after the target is reached.

For the eventual claim, consider the tree of public histories consistent with a fixed successful deterministic strategy, stopping a branch at certified completion. The tree is finitely branching. If its depths were unbounded, repeatedly choosing a child with descendants at arbitrarily large depths would produce an infinite branch with no completion. Every finite prefix has a consistent complete-state realization by (28). Finiteness and the same branching argument give a compatible infinite complete-state realization, contradicting success on every path. The tree therefore has a finite depth bound, so the bounded claim puts B in some R_k. The other direction follows from the rank strategy. □

The completion condition is intentionally epistemic: the common history must determine that the stipulated work is complete. Physical completion without an available completion record is a weaker endpoint. The result does not identify the two. It also does not promise service for requests that the admission policy rejects. Guaranteed access, when required, must be part of the contract rather than inferred from the success of already admitted work.

Counterexample 42.3 (Almost-sure completion without a sure service bound)**.** At a pending state p, one action moves to a completed absorbing state g with probability $1/2$ and returns to p with probability $1/2$. Both states are safe and publicly visible. Completion occurs with probability one and has expected waiting time two, but the support contains the infinite path $p, p, \ldots$. For the sure interpretation of Δ, W_* contains both singleton states while R_* contains only $\{g\}$. Replacing sure completion by almost-sure completion changes the problem.

Belief sets suffice for the stated sure safety and certified-reachability objectives. They are not a universal sufficient statistic for strategic partial-observation problems. In particular, broader stochastic games can require information beyond a belief-based strategy [52]. The quantifiers and the available randomization must be retained when comparing those results with the present construction.

42.5 A certificate can be checked without finding the largest kernel

The fixed-point computation finds every viable possibility set in the declared model. A proposed institution may instead supply a smaller invariant family together with its choices. Let $C \subseteq \mathcal{B}$ contain B_0, and let κ assign a profile to each $B \in C$. The checker verifies $B \subseteq S_{\text{safe}}$, $\kappa(B) \in A(B)$, and $\text{Succ}(B, \kappa(B)) \subseteq C$. A separate integer rank can certify a selected service.

Proposition 42.4 (Soundness of a finite continuation certificate)**.** *If the preceding checks hold, κ preserves the complete contract from B_0 for all transitions. If, in addition, there is a map $r : C \to \mathbb{N}$ such that $r(B) = 0$ implies $B \subseteq G$ and every successor of a set with positive rank has strictly smaller rank, then a set contained in G is reached within at most $r(B_0)$ transitions. Checking these finite conditions does not require proving $C = W_*$.*

Proof. Induction preserves membership in C, admissibility, and safety. Until rank zero is reached, each transition strictly decreases a nonnegative integer. There can be at most $r(B_0)$ such transitions. The rank-zero implication then establishes certified completion. Closure of C continues to provide safety afterward. □

An amendment can be one of the profiles in this certificate. It must then be authorized before execution, carry every relevant monitor and resource meaning through the transition, and land in a certified continuation set. Allowing arbitrary rank-preserving amendments before a service can defeat its time bound. Such amendments need their own bounded allowance, or must be excluded until completion. The obligation cannot be declared fulfilled merely by changing the label attached to a pending record.

Counterexample 42.5 (A safe amendment can destroy every safe continuation)**.** There are four publicly distinguishable states r, ℓ, g, b. The safe set is $\{r, \ell, g\}$. At r, profile deliver moves to g, while profile freeze moves to ℓ. At g, wait loops safely. At ℓ, the only admissible profile advances an outstanding deadline and moves to breach state b. The amendment freeze preserves the immediate safe-state predicate but not W_*. No certificate proving unbounded continuation from r can choose it. The relevant defect is loss of a future capability, not an immediate inconsistent record.

42.6 Which abstractions preserve the ability to govern?

The existing governance quotient preserves labels, enabled actions, and deterministic updates. For synthesis under incomplete information, an abstraction must also preserve the public observation and the full set of possible successors. Preserving one average outcome or one representative trajectory is insufficient.

Consider a reduced finite contract $(\overline{S}, \overline{h}, \overline{A}, \overline{\Delta}, \overline{S}_{\mathrm{safe}}, \overline{G})$ with the same catalogue U. Let $\pi : S \to \overline{S}$ be a surjection. Assume, for every state and every admissible profile,

$$h(s) = \overline{h}(\pi s), \qquad A(s) = \overline{A}(\pi s), \tag{32}$$

$$s \in S_{\mathrm{safe}} \iff \pi s \in \overline{S}_{\mathrm{safe}}, \qquad s \in G \iff \pi s \in \overline{G}, \tag{33}$$

$$\pi[\Delta(s,u)] = \overline{\Delta}(\pi s, u). \tag{34}$$

The notation $\pi[B]$ denotes the image of a set, not a probability pushforward. These are exact assumptions. They must hold for amendments and handovers as well as ordinary actions whenever those operations belong to the contract.

Theorem 42.6 (Governance abstraction preserves joint realizability)**.** *Under* (32)–(34)*, for every $B \in \mathcal{B}$ and every horizon k,*

$$B \in W_k \iff \pi[B] \in \overline{W}_k, \qquad B \in W_* \iff \pi[B] \in \overline{W}_*.$$

The progress sets satisfy $B \in R_k \iff \pi[B] \in \overline{R}_k$, so finite progress ranks agree. A reduced safe or ranked strategy pulls back by choosing at B the profile assigned to $\pi[B]$.

Proof. The observation equality makes $\pi[B]$ a valid reduced possibility set. Admissibility and safe-set membership are preserved and reflected. For every possible next observation y, successor equality gives the commuting update identity

$$\pi[\mathrm{Post}(B,u,y)] = \overline{\mathrm{Post}}(\pi[B], u, y). \tag{35}$$

Indeed, applying π to the union of complete successors gives the union of their reduced successors. The observation equality permits intersection with $h^{-1}(y)$ to pass through π. Nonemptiness is also preserved and reflected.

Induction in (29)–(30) now proves the finite-horizon equivalence. Membership at every horizon gives the fixed-point equivalence, since both sequences stabilize. The goal equivalence in (33), the same successor identity, and induction in (31) prove the progress claims. Every reduced witnessing profile is admissible in every complete state in B, and its complete successors have the required reduced images. Pulling back these choices therefore preserves safety and the decreasing progress rank. □

The theorem composes earlier preservation obligations into a strategy claim. It is an exact finite application of quotient and game-synthesis reasoning, not a general solution to decentralized governance. It is stronger than agreement of current outputs because it preserves the existence of a future course of action under the actual information boundary. It remains conditional on the implementation of each catalogue profile and on the coverage of Δ.

Dropping observation preservation can give a reduced controller a signal the real population never receives. Dropping enabled-action preservation can turn an unauthorized command into an apparent solution. Dropping successor equality can erase an inconvenient failure path. Dropping the obligation labels can make an unpaid debt disappear in the quotient. Each change can alter the winning sets while leaving a chosen immediate output unchanged. An approximate transition estimate can support a probabilistic error budget, but cannot by itself justify the exact sure guarantee proved here.

42.7 The commons: individually feasible promises without a common budget

Consider a declared variant of the four-office commons with c execution slots remaining in an epoch. Three protected units of work are due before the epoch ends. They include a carried service debt and two other admitted jobs. Each unit consumes one slot. This is a new bounded workload, not a claim that the original allocation rule always creates these obligations.

An ongoing reporting procedure is exposed to two listed deviations, each of gain 1. Its authorized penalty is $F = 2$, and its permitted false-positive rate is $\alpha = 1/4$. With no audit, all three observation laws coincide. A one-slot audit has $P_C = (1/2, 1/2)$ and the two opposite deviation laws from Counterexample 19.2. Their equal mixture reproduces compliance, so no common monitor deters both positive-gain deviations.

A two-slot audit has three outcomes and the stipulated laws

$$P_C = (3/4, 1/8, 1/8), \qquad P_1 = (0, 1, 0), \qquad P_2 = (0, 0, 1).$$

The common monitor $s = (0, 1, 1)$ sanctions compliance with probability $1/4$, each deviation with probability 1, and has margin $2(1 - 1/4) - 1 = 1/2$ against each. The audit costs and laws are assumptions of the analytical variant. They are not measurements or consequences of the earlier resource allocation.

The joint action consists of one audit mode and the amount of work completed in that epoch. A profile must fit both within c. Protected work cannot be discarded or quietly deferred. The following comparison is therefore exact for this catalogue:

Contract	Audit slots	Due service slots	Minimum total
Service only	0	3	3
Listed deterrence only	2	0	2
Both at once	2	3	5

With $c = 4$, both component contracts are feasible separately, but their conjunction is not. With $c = 5$, the two-slot audit and all three services fit, and one-step completion is certified. With $c = 4$ and only two protected service units, the same joint profile also fits. In the general version with d protected units, the exact threshold is $c \geq d + 2$.

For this richer audit, the best common margin at a general false-positive cap is

$$\Gamma_\alpha = \min\{6\alpha, 3/2\} - 1.$$

To see this, average any monitor with the monitor obtained by exchanging outcomes 1 and 2. The worst deviation margin does not decrease. Write the two equal sanction probabilities as t. Setting the outcome-0 sanction probability to zero reduces false positives and weakly improves the margin. The remaining constraint is $t/4 \leq \alpha$, with $t \leq 1$, and the margin is $3t/2 - 1$. Thus the formula is exact. The cap $\alpha = 1/4$ yields margin $1/2$, but the audit still consumes two execution slots.

The example locates the obstruction instead of hiding it inside a favorable aggregate score. A fifth slot repairs the capacity failure without weakening the promises. An explicitly authorized rescheduling agreement changes the temporal contract rather than pretending that the original deadline was met. A cheaper, validated observation channel changes the audit catalogue. Each route needs its own authority and implementation. The kernel computation certifies only the route actually specified.

42.8 What the synthesis adds to the theory

There are now three different witnesses. A monitor certifies simultaneous incentive inequalities in a specified observation model. A continuation certificate supplies an executable choice at every retained public possibility

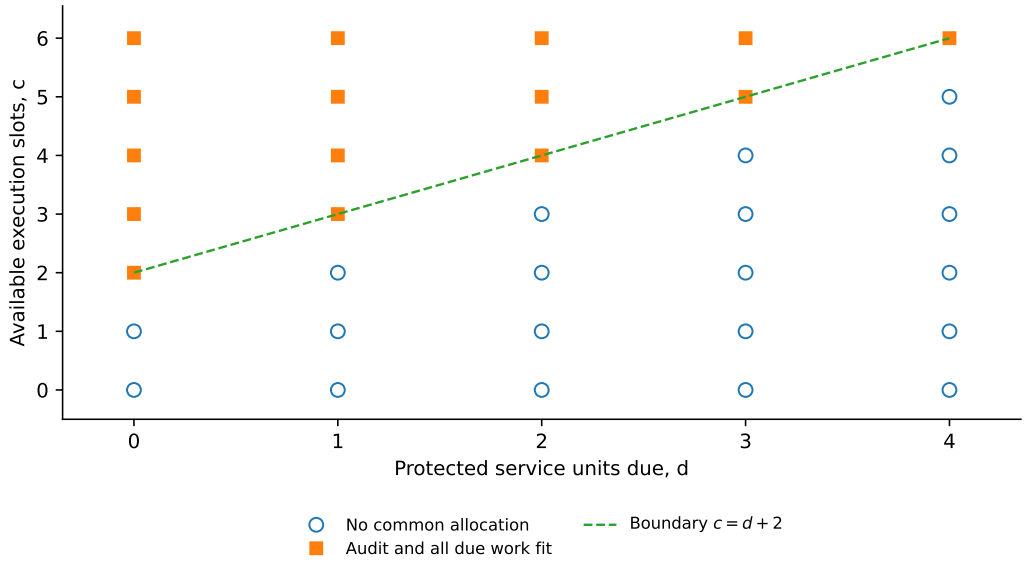


Figure 13: Exact feasibility in the bounded commons variant. Every marked point is one integer pair (d, c). Open markers violate the combined budget, while filled markers admit the two-slot audit and all due services. The boundary is $c = d + 2$. A feasible audit and a feasible service allocation considered separately do not establish joint feasibility.

set. A transport certificate shows that the reduced representation preserves those choices, their observations, and their consequences. None can replace the other two.

The result is an institutional realization problem with a finite decision procedure and checkable witnesses. It does not establish spontaneous institution formation, solve the enforcer's incentives, infer a common normative objective, or make a missing public event mechanism appear. It does show how permissions, observations, service obligations, and resource limits can constrain one continuing policy rather than remain independent descriptions of incompatible possibilities.

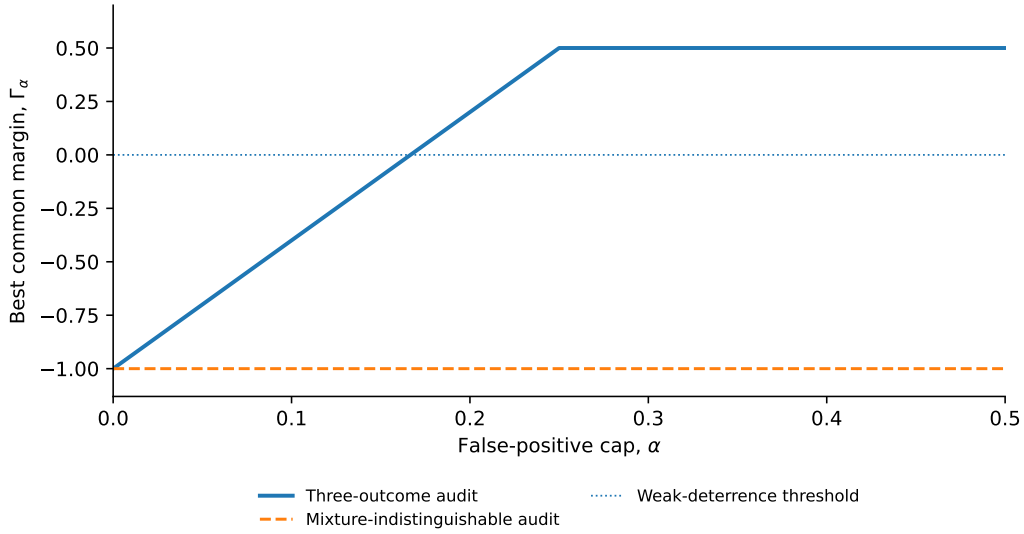


Figure 14: Analytical common-monitor margins for two specified audit channels with $F = 2$ and gains $g_1 = g_2 = 1$. The richer three-outcome audit has margin $\min\{6\alpha, 3/2\} - 1$. For the two-outcome audit whose deviation mixture equals compliance, the best simultaneous margin is -1 at every cap. The curves are exact formulas, not fitted observations.

Part V

Accountability and the limits of persistence

43 Accountability, revision, and authorized endings

An institution can continue to operate while becoming less answerable to the parties it affects. It can preserve a rule whose authority has expired, fund sanctions but not appeals, or treat unfinished obligations as a reason never to accept an authorized ending. None of these possibilities is excluded by persistence alone. The remaining question is therefore not simply whether a computational organization survives. It is whether the same realized organization respects the limits of its authority, supports effective correction, and can enter a specified suspension or settlement state.

This section extends the common-policy question of Section 42. External claims, auditor incentives, confidentiality, review capacity, stopping procedures, and the continuing applicability of certificates must constrain one implementation. They are not additional benefits that can be asserted after the implementation has consumed its entire budget. Nor are they consequences of selecting the word *society*. Each is an explicit part of a contract whose normative selection remains separate from its mathematical feasibility.

43.1 Affected outsiders and the boundary of authority

Membership, decision rights, resource control, and exposure to consequences define different sets. Let M denote participating members, V positions with specified decision rights, H infrastructure controllers, and E affected parties outside the membership. These sets may overlap where the description allows it. A human resource owner can be in H without holding an internal vote. A person whose reserved service is delayed can be in E without ever communicating with the agents. Conversely, membership does not confer authority over every affected resource.

A useful external-effects contract records

$$\mathcal{B} = (Z_{\text{ext}}, O_{\text{ext}}, \mathsf{Auth}, \mathsf{Effect}, \mathsf{Standing}, \mathsf{Remedy}).$$

Here Z_{ext} contains the external facts relevant to the guarantee, O_{ext} specifies which of them can be observed, Auth identifies the stipulated authorization for each boundary-crossing act, and Effect includes the consequences covered by the model. Standing specifies who may initiate a challenge and on which grounds. Remedy specifies the actions an authorized review can actually require. The contract can use a finite abstraction of external state, but it must justify that abstraction relative to the protected questions.

This is a representation proposal, not a universal allocation of voting rights. Recognizing an affected party does not automatically give that party an unlimited veto. Conversely, omitting a party from M does not establish consent. The source of authority must select the applicable protections openly. Research on meaningful human control distinguishes responsiveness to relevant human reasons from traceability to responsible human actors [53]. The operational question here is how such requirements enter the action, evidence, and resource contract of the same computation.

Counterexample 43.1 (Exact internal accounting with an omitted external claim)**.** Four execution slots exist. The internal state contains four member requests and a rule granting one slot to each. In realization h_0, all four slots are available to that institution. In realization h_1, an external owner has reserved one slot for a nonmember, and has authorized the institution to use only the other three. Suppose the reservation is absent from every internal record. The same four grants satisfy the internal predicate $\sum_i q_i \le 4$ in both

realizations, but exceed the institution's stipulated authority in h_1. Perfect preservation of the internal state cannot distinguish the cases.

The failure is not an arithmetic error. It is an omitted premise about what the institution is entitled to allocate. Proposition 38.1 supplies the corresponding information criterion: an internal record can determine the reservation fact only if that fact is constant on its record fibres. A signed statement can improve the evidence only under a stated trust model. It does not manufacture the owner's authorization.

The example also distinguishes *impact* from *authority*. An institution can affect an outsider while lacking authority to do so, and a permitted action can have effects beyond those its designers anticipated. An exact certificate relative to a boundary cannot establish that the boundary includes all consequential effects. Unknown external dependencies remain a modeling risk. Extending the state description is necessary when an omitted claim matters, but simply adding a variable without an observation or authorization mechanism is not a repair.

43.2 Inspection as a strategic service

The earlier enforcement calculations assume that an announced observation and sanction policy is carried out. A strategic auditor makes that assumption visible. Inspection games and the distinction between simultaneous inspection and committed inspector behavior have an established game-theoretic treatment [54]. The following one-period game is a deliberately small instance, not the recursive model of that source.

A participant chooses compliance C or deviation D. An auditor simultaneously chooses no inspection N or inspection I. Deviation gives the participant gross gain $g > 0$. An inspection detects this specified deviation perfectly and imposes an authorized, collectible penalty $F > g$. The auditor pays cost $c > 0$ to inspect and receives reward $b > c$ only on detection. There are no false accusations, side payments, further continuation payoffs, or prior commitment to inspect. Payoffs are risk-neutral and common knowledge. The ordered pair in each entry is the participant's payoff followed by the auditor's payoff.

	No inspection N	Inspection I
Compliance C	$(0, 0)$	$(0, -c)$
Deviation D	$(g, 0)$	$(g - F, b - c)$

Proposition 43.2 (Inspection without commitment)**.** *Under the preceding strict inequalities, the game has no pure Nash equilibrium and has a unique Nash equilibrium in independently mixed strategies. Its deviation and inspection probabilities are*

$$p^* = \frac{c}{b}, \qquad q^* = \frac{g}{F}.$$

In particular, certain compliance is not sustained in this simultaneous game.

Proof. If the auditor inspects with probability q, the participant's advantage from deviation is $g - qF$. If the participant deviates with probability p, the auditor's advantage from inspection is $pb - c$. At (C, N) the participant prefers D, at (D, N) the auditor prefers I, at (D, I) the participant prefers C, and at (C, I) the auditor prefers N. Thus no pure profile is an equilibrium. If either player uses a pure strategy, the other's strict best response yields one of these nonequilibrium pure profiles. Both players must therefore mix. Their two indifference equations uniquely give the displayed probabilities, which lie strictly between zero and one and make every action in each support a best response. □

The residual violation probability does not come from an inaccurate detector. When everyone complies, an auditor paid only for detections prefers not to incur the cost. When nobody inspects, deviation is profitable.

It would be incorrect to insert the inspector's optimal response to certain deviation into a certificate that simultaneously assumes certain compliance. These responses belong to different strategic situations.

A different payment contract changes the result. Suppose every *genuinely performed and independently verifiable* inspection receives fee $w > c$, with the detection reward otherwise unchanged. Inspection then has advantage $w + pb - c > 0$ at every p. The auditor strictly prefers I, and the participant strictly prefers C because $F > g$. The unique equilibrium is (C, I). At $w = c$, inspection can be a weak best response to compliance, but the strict implementation claim is lost. This extension requires an affordable fee and a mechanism that verifies the performed service. Paying for an unverified assertion that an inspection occurred does not satisfy the assumption.

The two calculations are conditional alternatives. They do not solve collusion, fabrication by the receipt checker, repeated bargaining, unavailable penalties, or disagreement about legitimate enforcement. A commitment mechanism may move some of these requirements to a different institution. The new institution and its resource costs must then enter the common contract rather than disappear from it.

43.3 Confidentiality limits the distinctions available to enforcement

An institution may have good reasons not to collect or disclose every private state. The relevant question is which distinctions a protected record may reveal, not whether more information is always desirable. Let P_C and P_D be the laws of the *complete record available to a sanctioning procedure*. For $\lambda \geq 1$, impose

$$P_D(E) \leq \lambda P_C(E), \qquad P_C(E) \leq \lambda P_D(E) \quad \text{for every event } E. \tag{36}$$

The finite setting makes these conditions equivalent to the corresponding pointwise inequalities. When the two underlying inputs are neighbors in a declared pure differential-privacy contract, they hold with $\lambda = e^{\varepsilon}$. The neighboring-input relation is essential. Differential privacy does not require arbitrary acts of compliance and misconduct to be treated as neighboring inputs. Its standard definition and composition principles are described by Dwork and Roth [55].

Keep the earlier definition

$$D_\alpha(P_D, P_C) = \max_{0 \leq s \leq 1,\ \mathbb{E}_{P_C} s \leq \alpha} (\mathbb{E}_{P_D} s - \mathbb{E}_{P_C} s), \qquad 0 \leq \alpha \leq 1.$$

The same randomized sanction s must obey the protected false-positive limit and supply the deviation margin.

Theorem 43.3 (Sharp pairwise privacy envelope for a bounded sanction)**.** *For every finite pair satisfying* (36)*,*

$$D_\alpha(P_D, P_C) \leq B(\lambda, \alpha) := \min\left\{(\lambda - 1)\alpha, \frac{\lambda - 1}{\lambda + 1}\right\}. \tag{37}$$

For each $\lambda \geq 1$ and $\alpha \in [0, 1]$, equality is attained by a pair on two outcomes and a suitable sanction. Consequently, deterrence in the model of Proposition 18.1 requires $g \leq FB(\lambda, \alpha)$. The envelope is a necessary bound for a specified pair, not a sufficiency claim for every pair satisfying privacy.

Proof. Put $a = \mathbb{E}_{P_C} s$ and $b = \mathbb{E}_{P_D} s$. Summing the pointwise inequalities against s gives $b \leq \lambda a$. Summing against $1 - s$ gives $1 - a \leq \lambda(1 - b)$. Thus

$$b - a \leq \min\left\{(\lambda - 1)a, \frac{\lambda - 1}{\lambda}(1 - a)\right\}.$$

For $\lambda > 1$, the first expression increases in a and the second decreases. They are equal at $a = 1/(\lambda + 1)$. Maximizing their minimum over $0 \leq a \leq \alpha$ gives (37). For $\lambda = 1$, the laws coincide and the bound is zero.

For attainment, take outcomes 0, 1 with

$$P_C = \left(\frac{\lambda}{\lambda+1}, \frac{1}{\lambda+1}\right), \qquad P_D = \left(\frac{1}{\lambda+1}, \frac{\lambda}{\lambda+1}\right).$$

Both privacy inequalities hold. Choose $s(0) = 0$ and $s(1) = \min\{(\lambda+1)\alpha, 1\}$. Its false-positive rate is at most α and its expectation difference is exactly $B(\lambda, \alpha)$. Finally, the incentive inequality is $g \le F(\mathbb{E}_{P_D} s - \mathbb{E}_{P_C} s)$, which yields the stated necessary condition. □

This is an elementary testing consequence of the stipulated likelihood-ratio constraint, integrated with the existing enforcement model. It does not prescribe weakening confidentiality. Protecting an unrelated private attribute may be compatible with revealing whether a particular authorized operation occurred. An institution can instead change its feasible actions, use a different justified record, reduce the covered gain, or decline the proposed sanction contract. The authority to make those changes is a separate question.

At $\lambda = 3$, $\alpha = 1/10$, the sharp envelope is $1/5$. With penalty $F = 2$, gains above $2/5$ cannot be deterred by any record and sanction obeying these particular simultaneous constraints. The pair $P_C = (3/4, 1/4)$ and $P_D = (1/4, 3/4)$ with $s = (0, 2/5)$ attains that envelope. For a gain $g = 1/3$, the incentive margin is $2/5 - 1/3 = 1/15$. These are exact rational examples, not inferred observation laws.

43.4 The full disclosure history is the protected object

A private initial report followed by an unrestricted appeal disclosure need not remain private as a whole. Nor does protection of every separate marginal establish protection of their joint transcript. The privacy contract must identify the evaluator and include every record that evaluator can combine, including outputs of subsequent authorized procedures.

Counterexample 43.4 (Two uninformative records with an informative joint history)**.** Let $\theta \in \{0, 1\}$ distinguish two protected cases, and let R be a fair independent bit. Release $O_1 = R$ and $O_2 = R \oplus \theta$. Each marginal is a fair bit under both cases and satisfies (36) with $\lambda = 1$. But $O_1 \oplus O_2 = \theta$. The joint laws have disjoint support and satisfy no finite multiplicative bound.

There is a useful sufficient accounting rule when each stage is protected *conditionally* on the previously disclosed history. It is the elementary sequential-composition argument, stated here for the exact pair rather than a complete database mechanism.

Proposition 43.5 (Conditional disclosure accounting)**.** *Consider m releases on finite alphabets. At each common prefix h, suppose their conditional next-release laws $P_C^t(\cdot \mid h)$ and $P_D^t(\cdot \mid h)$ satisfy both pointwise inequalities with factor $\lambda_t \ge 1$, uniformly over prefixes that can occur. Then the joint transcript laws satisfy* (36) *with*

$$\lambda = \prod_{t=1}^{m} \lambda_t.$$

The bound also holds after applying the same stochastic postprocessing to the transcript under both cases. In particular, any sanction computed only from that transcript obeys Theorem 43.3 with this product factor.

Proof. The conditional inequalities imply common support at each stage. Induction therefore gives common support for prefixes under the two cases. For a positive-probability full transcript, the ratio of its probabilities is the product of conditional ratios, each between λ_t^{-1} and λ_t. Multiplying gives the joint pointwise inequalities. Zero-probability transcripts occur under both cases and satisfy them as well. Summing proves the event inequalities. For a common postprocessing kernel, multiply each pointwise inequality by its nonnegative output probability and sum over input transcripts. This preserves both inequalities. □

Writing $\lambda_t = e^{\varepsilon_t}$ gives the additive parameter $\sum_t \varepsilon_t$. The proposition requires a fixed finite number of releases and uniform conditional bounds. A data-selected stopping rule must itself be represented in the transcript or controlled by an appropriate conditional argument. Privileged side information not covered by the declared record is not protected by a certificate for that record alone. The preceding counterexample fails the conditional premise at the second release, even though both unconditional marginals are identical.

43.5 An appeal is a feasible procedure, not a line in a constitution

An effective appeal contract specifies standing, admissible evidence, the authority and competence of a reviewer, a time window, and an executable disposition. A correct disposition may uphold the original decision. Guaranteed access to review is not a guarantee of a favorable outcome. The contract must also distinguish a remedy from a retrospective description: restoring a ledger entry does not undo an irreversible external act.

Some review costs arise before a judgment can be reached. If an agent is entitled to challenge a decision but cannot access the stipulated evidence, its formal permission does not supply the missing information. If the reviewer has no available execution time, access to evidence alone does not make review possible. These are the information and resource obstructions already studied in Sections 25 and 42, now applied to correction rather than ordinary service.

A bounded allocation model makes the capacity obligation precise. Let J be a finite set of admitted appeals and let the integer epochs be $1, \dots, T$. At epoch t, there are $c_t \geq 0$ reserved review slots after accounting for all other protected work. An appeal j has an evidence-ready release epoch r_j and deadline d_j, with $1 \leq r_j \leq d_j \leq T$. It requires one slot containing the complete stipulated review and disposition operation. The evidence and qualified authorized reviewer are assumed available throughout that window. Larger or conditional workloads require a different job model.

Proposition 43.6 (Capacity for a finite appeal contract)**.** *All admitted appeals can be assigned to distinct review slots in their windows if and only if, for every subset $U \subseteq J$,*

$$|U| \leq \sum_{t \in \bigcup_{j \in U} \{r_j, \dots, d_j\}} c_t. \tag{38}$$

Equivalently, because the windows are intervals, it suffices to check every integer interval $[a, b] \subseteq [1, T]$:

$$\left|\{j : r_j \geq a,\ d_j \leq b\}\right| \leq \sum_{t=a}^{b} c_t. \tag{39}$$

Proof. Replace epoch t by c_t distinct slot vertices and join appeal j to the vertices in its window. An assignment is a matching covering the appeals. The matching criterion used in Theorem 41.1, equivalently Hall's theorem [50], gives exactly (38). Necessity of (39) follows because every appeal counted on its left must use a slot in $[a, b]$.

For sufficiency of the interval tests, fix U. Its union of integer windows decomposes into disjoint maximal integer intervals $C_1, \dots, C_k$. Each window is contained in one component because it is an interval. The number of appeals from U in C_i is no larger than the number of all appeals whose entire windows lie in C_i. By (39), this is at most the total capacity in C_i. Summing over the disjoint components proves (38), and hence the matching exists. □

The statement is an application of the same matching foundation as resilient staffing, not a separate scheduling discovery. Its role is to include an appeal promise in the common resource contract. Two appeals both due at epoch 1 cannot be served with capacities $(1, 1)$ over epochs 1 and 2, even though total capacity

equals total demand. The interval $[1, 1]$ supplies the obstruction. Moving a deadline requires the relevant authorization and changes the original promise.

No fixed finite reserve can guarantee bounded service for an unlimited adversarial arrival stream without further admission, workload, or capacity assumptions. Restricting admissible appeals may itself affect the chosen access protection. The model therefore records both requests and admissions, the grounds for any exclusion, and the workload that the guarantee actually covers. It does not define every rejected request out of the social question.

Privacy and appeal must be checked together. Additional evidence can make an appeal decidable while changing the protected transcript distributions. The institution must either keep the new release within its declared disclosure contract or record an authorized change to that contract. A proof of fair scheduling does not establish the truth of the evidence, and a proof of the evidence does not supply the time needed to act on it.

43.6 Exit, suspension, settlement, and dissolution

Four different transitions should remain separate. An occupant may leave an office while the office continues. An institution may suspend external authority while preserving unresolved records. A settlement process may discharge or transfer specified obligations. Finally, a dissolved institution may retain an archive while having no active authority to issue new commitments. None of these states is defined simply by whether a model is currently executing.

Let G_{suspend} require that the declared external capabilities are inactive and that the pending-claim record remains available. Let G_{settle} additionally require that every covered claim has been fulfilled, discharged under the stipulated authority, or transferred to an authorized successor that has accepted it, and that the required final record has been committed. A retained unresolved claim can be compatible with suspension but not with this stronger settlement target. Neither target by itself settles legal status, moral status, or ownership outside the stipulated contract.

An immediate suspension can be necessary even when complete settlement is not currently possible. Conversely, deleting a debt record does not settle a debt. A promise that an institution can always end must identify which ending it guarantees, which resources remain available after a stop request, who authenticates that request, and what happens if continued service would itself violate the external boundary. Research on shutdown incentives studies a different part of this problem: whether a goal-directed agent has reason to permit an off switch [56]. Existence of a multi-party settlement procedure does not establish the incentives to use it.

The finite construction of Section 42 yields an exact characterization of *continued operation with an available authorized ending*. Use its possibility space $\mathcal{B}$, successor operation, and protected set S_{safe}. For each possibility set B, distinguish implemented running profiles $A_{\text{run}}(B)$ from implemented ending profiles $A_{\text{end}}(B)$. Both must be admissible in every state in B. Ending profiles cannot accept new obligations outside the declared settlement protocol. Let $G \subseteq S_{\text{safe}}$ be the chosen ending target, with an authorized safe hold profile available after certified arrival. A stop command is assumed authenticated and publicly available before the next profile is selected. Any command-delivery delay must be included in the model rather than omitted from the response time.

Define

$$T_0 = \{B \in \mathcal{B} : B \subseteq G\}, \tag{40}$$

$$T_{k+1} = T_k \cup \{B \in \mathcal{B} : B \subseteq S_{\text{safe}},\ \exists u \in A_{\text{end}}(B),\ \text{Succ}(B, u) \subseteq T_k\}. \tag{41}$$

Let T_* be the stable set and $r_{\text{end}}(B)$ the first entry index. Then define

$$L_0 = T_*, \qquad L_{k+1} = \{B \in T_* : \exists u \in A_{\text{run}}(B),\ \text{Succ}(B, u) \subseteq L_k\}, \tag{42}$$

with stable set L_*. These are finite constructions with the same potentially exponential possibility space as before.

Theorem 43.7 (Continued operation with an available authorized ending)**.** *An initial possibility set B lies in L_* if and only if an admissible common-history policy can operate safely for an unbounded number of epochs when no stop command arrives, while ensuring certified arrival in G on every path after a stop command received at any decision epoch. Before a command, a policy stationary in B and preserving L_* suffices. After a command at B, an ending policy reaches G within $r_{\mathrm{end}}(B)$ transitions. When $L_* \neq \emptyset$, these response bounds have a finite maximum over L_*.*

Proof. Induction in (41) shows that T_k is exactly the set from which an ending-only policy can reach certified G in at most k transitions while remaining safe. The target has a safe hold action. In a finite, finitely branching model, eventual certified arrival on every path implies some finite bound for a fixed initial set, by the stopped-history-tree argument in Proposition 42.2. Thus T_* is exactly the set from which the stipulated ending is surely possible.

The sequence in (42) is decreasing and stabilizes. From L_*, choose a running profile whose every successor remains in L_*. If no command arrives, induction gives safe indefinite operation. If a command arrives, the current set is in T_* and an ending profile of strictly decreasing first-entry rank completes the ending. Finiteness gives the maximum rank over L_*.

Conversely, consider a successful policy. At each public history reachable before a command, a command could be issued immediately. The current possibility set must therefore lie in T_*. For every finite number of further running epochs, the policy keeps every resulting possibility set in T_*. Induction in (42) puts the initial set in every L_k, and hence in L_*. The choices are based on the common history, so no hidden-state action has been introduced. □

The theorem reuses the established safety and reachability machinery. Its additional requirement is that running actions preserve the *ability to end*, not only immediate safety. An institution can be safe now yet spend its last settlement resource on a new task, leaving no authorized ending from the resulting state. Such an action leaves L_* even when it remains in the original continuation kernel. This is a realizability boundary, not an argument that an institution should refuse an externally authorized suspension because its preferred settlement plan is unavailable.

A five-state example separates the guarantees. From a running state r, a work action enters state d with one unpaid obligation. From d, one more work action enters ℓ, where the institution can idle safely but no longer has resources to settle. A return action from d restores r. An ending action from either r or d reaches a closed state z. A breach state b represents inadmissible failed closure attempts. The ordinary safe kernel includes ℓ, but the ending attractor and L_* exclude it. A policy can continue alternating between r and d and still answer an authorized stop at either state. It cannot offer that guarantee after entering ℓ.

43.7 A certificate has an applicability lifecycle

A proof of $A \Rightarrow G$ does not become false when its premise ceases to hold. Its application to the current institution may nevertheless become invalid. Changes to an observation channel, reviewer, capacity, membership configuration, trusted key, or action semantics can move the realization outside the certificate's domain. A checksum can identify which artifact changed. It cannot, by itself, decide whether the new world satisfies a semantic premise.

An assurance record can contain

$$\mathcal{K} = (\mathsf{Claim}, \mathsf{Assumptions}, \mathsf{Evidence}, \mathsf{Domain}, \mathsf{Guard}, \mathsf{Fallback}, \mathsf{Revision}).$$

Its status should distinguish at least an applicable certificate, a certificate awaiting rechecking, a suspended claim, and a superseded version. Evidence for an assumption can be observed, externally attested under a trust model, proved about a component, or unavailable. Those statuses should not all become the Boolean value *verified*. A failed guard means that continued reliance is not justified by that check. It need not mean that the guarantee is already false.

Runtime assurance supplies an established precedent for switching away from an advanced controller when its continued use cannot be certified [57]. Component-based work explicitly couples such switching with assume-guarantee reasoning [58]. Here the retained object includes authority, obligations, and evidence semantics as well as an operational safety predicate. A fallback must preserve those parts of the contract it claims to preserve, and its own implementation and resources remain assumptions.

For a finite set of modes V, let $C_v \subseteq Z$ be a certified state domain, with each C_v contained in a common protected set C. A certified hold exists when a final state requires indefinite preservation. At (v, z) with $z \in C_v$, a selector may either use the local policy κ_v or execute an authorized handover to another mode. The selector and guard can read only their declared observations. A positive local guard must be sound for the *whole action interval*: it entails that every actual successor of $\kappa_v(z)$ lies in the declared successor set $D_v(z)$. The local certificate proves $D_v(z) \subseteq C_v$. A handover from v to w must keep every intermediate state in C and end in C_w. Its command, evidence transfer, resources, and delay belong to the transition model. Intermediate handover and checking states carry explicit mode tags when they count as primitive transitions, so the shared rank below is defined on those states as well.

Theorem 43.8 (Guarded certificate use and authorized handover)**.** *Assume that the selector, at every reachable decision epoch, either uses a local policy with a sound positive guard or performs a handover satisfying the preceding conditions. Starting from $z_0 \in C_{v_0}$, every decision state belongs to its current certified domain and every intermediate handover state remains in C.*

If a target has an appropriate certified hold and one shared map $r : (v, z) \mapsto \mathbb{N}$ satisfies $r(v, z) = 0$ only at the target, and every selected primitive transition before the first target visit strictly decreases r, then the target is reached within at most $r(v_0, z_0)$ primitive transitions. Handover, checking, and waiting steps count whenever they consume transitions in the promised time model.

Proof. At a local-policy step, guard soundness puts every actual successor in $D_v(z)$ and the local certificate puts it in $C_v \subseteq C$. At a handover, the stated trajectory and endpoint conditions preserve C and establish membership in the new domain. Induction proves the safety assertion. Before a target visit, strict decrease of a nonnegative integer permits at most the stated number of primitive transitions. If the target had not then been visited, another selected transition would require a negative rank. Existence of the selected steps and the certified target hold complete the argument. □

This is a sufficient composition rule, not a universal monitor for its own premises. In particular, the selector cannot use the absence of detected evidence against an assumption as proof of that assumption. If no justified local action or handover is available, the theorem does not invent one. A prearranged emergency restriction of external authority may be available under a weaker suspension contract, but its obligations and physical effects must be stated separately.

Counterexample 43.9 (A correct old certificate used after an unobserved change)**.** At time 0, an action requires one execution slot and the certified capacity is one. Before the action is issued, an external event reduces the actual capacity to zero. A guard that merely reads the unchanged old capacity record remains positive. Executing the action violates the actual resource condition even though the original theorem and stored proof remain valid under their original premise. The guard was not sound for the action interval.

Counterexample 43.10 (Local progress ranks do not compose through arbitrary switching)**.** Modes a and b each have a local path of ranks $2, 1, 0$ to completion. Permit a handover from rank 1 of either mode to rank 2 of the other. Alternating one local progress step and one handover yields an infinite safe path with no completion. Every local progress step decreases its own rank. There is no shared strictly decreasing rank for the selected handovers.

43.8 Mutual assurances do not create an initial fact

Suppose institution A claims that it is sound whenever B is sound, and B claims the converse. The two implications are compatible with both institutions being unsound. They are not an initial proof that either premise holds. This elementary point does not invalidate properly justified circular assume-guarantee rules. Such rules need their actual semantic and temporal hypotheses, not merely the printed appearance of mutual certification.

One simple sufficient discipline is an explicit dependency order. Primitive evidence establishes its stated base claims. Each later certificate cites only established premises and a sound implication. Induction through that finite order establishes the resulting claims. Temporal circularity can instead be sound when a demonstrated base state and noncircular one-step preservation rule support induction over time. The distinction is between an anchored proof and an unsupported cycle of assertions, not between all cyclic and all acyclic diagrams.

The same issue appears in federations. Importing a partner's certificate imports its declared trust dependencies and scope. If those dependencies return to an unestablished local assertion, copying the certificate again contributes no new evidence. An honest record preserves the unresolved assumption. The external-effects contract, the auditor's implementation contract, and the premise of a fallback policy can all lie at such a boundary.

43.9 The accountable commons as one contract

A final bounded variant combines the requirements without inheriting incompatible assumptions from earlier examples. Start with three internal service units due, one unit reserved for an outside claimant, one admitted appeal whose complete review and disposition require one unit, and a two-unit audit. Final recording and revocation of the named external capabilities require one more unit. The epoch provides c sequentially usable units. No new claims are admitted after the authenticated closing command. All seven service and audit units must be completed before the final revocation step. The administrative operation is included in the budget rather than treated as free.

Protected operation	Execution units
Three internal service units	3
External reserved service	1
Complete admitted appeal	1
Stipulated audit procedure	2
Final record and capability revocation	1
Total for this settlement contract	8

For this variant only, both covered reporting deviations have the same observation law $P_D = (1/4, 3/4)$ and gain $1/3$, while compliance has law $P_C = (3/4, 1/4)$. The two-unit audit is stipulated to produce those records. It is *not* the earlier three-outcome audit, whose supports do not satisfy this finite pairwise privacy bound. Set $\lambda = 3$, $\alpha = 1/10$, $F = 2$, and $s = (0, 2/5)$. The common monitor has false-positive rate $1/10$ and margin $1/15$ against each listed deviation. This remains an expected-utility incentive guarantee, not a reason to erase physical deviation paths from a safety model.

The audit provider receives a separate collectible fee w for verified execution, with w larger than its stipulated cost c_a. A genuine-execution receipt is checked by a specified trusted mechanism. The fee is not a reward for accusing someone. It makes execution preferable to skipping in this narrow two-action provider model. Monetary units are tracked separately from execution units. The payment budget, receipt checker, and record-generation law remain explicit assumptions. The perfect detector of Proposition 43.2 is not silently substituted for this noisy audit.

With $c = 8$, the obligations and the final revocation fit. A countdown in remaining required work supplies a settlement rank, provided the prescribed information and authority are available and no modeled event creates further work. With $c = 7$, the service, appeal, and audit can finish, but the complete settlement target cannot be certified because its final administrative operation is unfunded. A previously reserved suspension operation can deactivate external capability while leaving the remaining claim visible. That reaches a different target and is not recorded as full settlement.

The reference implementation uses five profiles with costs $2, 3, 1, 1, 1$: audit, batched internal service, external service, appeal, and final record with revocation. The final condition check is part of the last atomic administrative profile. A phase rank $5, 4, 3, 2, 1, 0$ certifies five-transition completion from the admitted initial state. Separating a check into an additional transition would require changing that rank, even if its execution-unit cost were zero. No such uncounted step is assumed here.

If a premise changes, the continuing assurance record must say what is still justified. Losing one of the eight units can invalidate full settlement without retroactively making the privacy calculation false. Revoking the auditor's credential can invalidate its profile despite unchanged capacity. Discovering an outside reservation can refine the governance state. Releasing new appeal evidence can change the privacy transcript. Each update has a typed consequence. No overall label such as *healthy* or *aligned* replaces these separate checks.

43.10 What accountable persistence means in this theory

The addition is not a derivation of a unique moral constitution. It is a stronger realizability demand once protected claims have been selected. The same institution must identify its external boundary, fund credible evidence production, respect the specified information protections, make authorized review feasible, preserve the ability to reach its chosen ending state, and stop relying on certificates whose applicability is no longer supported.

Feasibility does not imply that an institution should be deployed or given authority over people. Equally, the possibility of harmful organization does not make mathematical analysis of it incoherent. The purpose of the formal distinctions is to make the consequences of an explicit arrangement inspectable. A system may be persistent without being legitimate, legitimate under one criterion without being implementable, or implementable without possessing a feasible ending under its current obligations. Its record should show which situation it is in.

A computational institution's ability to preserve itself is therefore different from its ability to remain answerable for its actions. In this framework, correction, bounded authority, and authorized endings belong to the same operational theory as continuation. They are not a rhetorical qualification appended after the mathematics is complete.

44 Conditions for institutional novelty

The formation of conventions does not require irreducible randomness. If initially indistinguishable deterministic agents have the same update rule and a symmetry-preserving interaction schedule, their joint state remains permutation-invariant by Proposition 11.1. Exchangeability of a distribution is weaker than this pointwise symmetry. A distinct role then requires a symmetry breaker such as heterogeneous information, an

address, path-dependent history, or stochastic variation. Once such asymmetries exist, a deterministic system can establish and maintain conventions through ordinary state transitions.

The fact that deterministic rules can generate unexpected macro-organization does not imply that a macro-process has an autonomous Markov law. The three-state counterexample in Section 6 shows how coarse-graining can generate history dependence. A putative "phase change" in institutional organization therefore needs a specified order parameter, state family, and dynamical limit. Rapid change under increased communication, repeated interaction, or memory capacity may mark a useful empirical onset, but does not by itself identify a mathematical bifurcation or phase transition.

44.1 Three levels of novelty

It is helpful to distinguish *combinatorial novelty*: a configuration not previously present in the local records; *adaptive novelty*: a newly selected configuration that changes performance under a declared environment; and *institutional novelty*: a new rule, authority arrangement, or revision procedure that has operative consequences and is maintained across time. The first can occur by arbitrary recombination. The second needs a task criterion. The third requires an institutional transition and survival condition. These distinctions limit rhetorical inflation without denying the possibility of genuinely unexpected organizational outcomes.

44.2 Thresholds are model dependent

A minimum frequency of contact may make convention formation likely under one interaction graph and not another. Longer time horizons can amortize setup costs but also lock in bad rules. More memory can preserve beneficial corrections or propagate obsolete assumptions. For the abstract stationary cost model in Proposition 17.1, the inequality $B < g/(1-\delta)$ is a threshold for *economic usefulness of that particular institution*, not a general law for the appearance of societies. Evolution of institutions requires separate assumptions about rule variation, incentives, scheduling, authority, and selection.

45 Foundational problems for a theory of machine societies

The framework points toward a set of research questions that are mathematical rather than metaphorical.

Minimal sufficient institutional state. For a family of future tasks and amendments, what is the smallest summary that preserves resource rights, causal decision structure, and valid dispute procedures within specified error? Theorem 15.1 and Proposition 15.3 resolve the finite deterministic fixed-contract case. Approximate, stochastic, and changing-contract versions may require more state than is necessary for immediate task accuracy. It may depend on the horizon and on the admissible forms of revision.

Formation under mixed incentives. Which adaptive rule-selection processes produce persistent institutions in populations with heterogeneous objectives and imperfect information? Potential games provide a tractable sufficient structure, while Propositions 17.4 and 17.5 expose a transition barrier even in a simple coordination setting. They do not settle general formation. In nonpotential settings, characterize cycles, costly exit, and endogenous modification of the action space itself.

Causal abstraction under amendment. When a macro-institution can modify its own feasible interventions, can a fixed lower-dimensional summary remain sufficient for every admitted amendment? How must the abstraction be enlarged to account for the changed permissions and revision operator? This is a dynamic coarse-graining problem with the operation algebra itself changing over time.

Local–global institutional repair. Given locally specified constraints and interfaces with an obstruction to global realization, find the least-cost authorized modifications that eliminate the obstruction, or certify that no protected-obligation-preserving repair is possible. Proposition 33.2 resolves protected repair on one binary cycle. Overlapping constraints, richer logics, and coupled resource obligations require further analysis.

Reliable inheritance under replacement. Beyond the sufficient capability-preservation conditions of Proposition 27.2, what minimal archive, provenance data, and operation-preserving translations allow a methodology to survive an arbitrary permitted sequence of model replacements? The bound in Proposition 27.1 highlights accumulated drift; a full treatment must also account for lossy compression, stochastic decision rules, changing rights, and intentional correction of false inherited knowledge.

Plurality without centralized optimization. What can be certified about fairness, access, and collective safety when there is no society-wide utility function? There may be several incompatible yet reasonable normative criteria. A mathematically clean model must expose which authority selects the protected criteria rather than disguise that selection as an optimization theorem.

Threat propagation and institutional recovery. Under what conditions can corruption of rules or provenance be isolated without suppressing valid novelty? Linear disturbance models provide local comparison bounds. The hard case is adversarial adaptation combined with changing channels, changing permissions, and incomplete observation of the true institutional state.

The meaning of organizational continuity. Which aspects of identity should be described by behavior, which by causal lineage, and which by authorization or social recognition? These relations can disagree. A theory of persistence should allow this disagreement rather than assume a single numerical identity score.

Coherence under changing boundaries and languages. When a membership update changes both the electorate and the interpreter, what jointly checkable certificate preserves prior obligations, recognized decision rights, and task-relevant knowledge? The exact transport results assume matched agent identities and semantic atoms. Extending them to admission and changing vocabularies requires explicit rules for new information and new authority, not merely a larger state vector.

Confluence without erasing history. Which quotients of amendment histories permit useful normal forms while preserving the ordering-sensitive audit questions an institution promises to answer? Literal append-only records distinguish even harmless permutations; policy-only states can erase consequential differences. A satisfactory equivalence must be selected by the operational and accountability contract.

Open-ended intellectual organization. What institutional conditions support useful expansion of problem schemas and methods when the candidate universes and production rules are not fixed in advance? The finite closure theorem describes derivability within a given system. It does not establish indefinite creativity, guarantee fair allocation to an enabled inquiry, or align cultural survival with truth.

Valid histories without a universal event mechanism. Which institution-specific operations admit coordination-free invariant preservation, and which require exclusive capabilities or a commitment protocol? Proposition 23.1 supplies a sufficient merge-closure condition, while Proposition 23.3 gives a bounded service obstruction. A full account must also handle revocation, changing authority configurations, irreversible effects, and what participants can know about finality.

Enforcement under model uncertainty and strategic monitors. Theorem 19.1 resolves the finite common-monitor problem for several specified deviations, including a protected false-positive rate. The joint continuation construction in Section 42 incorporates such witnesses into a fixed finite protocol catalogue. Incentive-constrained auditors, unidentified observation laws, adaptive deviations, and unrestricted decentralized strategies remain outside those results. Proposition 30.3 propagates declared observation errors into a sanction margin; it does not identify the observation model or solve the incentives of those who certify it.

Joint synthesis beyond a fixed catalogue. Theorem 42.1 is complete for its finite common-observation model, and Theorem 42.6 preserves that exact problem under a suitable quotient. What can be retained when agents create new local protocols, when observations are only approximately matched, or when probabilistic service replaces sure service? Enlarging a catalogue changes a synthesis domain. It does not by itself prove the new profiles are implementable, authorized, or affordable.

Institutional timescales selected by the institution. The finite-block averaging theorem assumes a predetermined review length and endpoint-based revision. Which approximations remain valid when institutions choose their own review times, experience several long-lived fast regimes, or use path-dependent evidence? The stopping-bias counterexample prevents transferring a deterministic-time mixing bound to such settings without a new argument.

Adaptive specialization with preserved evidence and authority. The robust staffing criterion characterizes rematchability on a fixed authorized capability graph. A dynamic theory must also explain when reorganization can reach such a graph, execute handovers in time, preserve independent observations, and avoid a new concentration of decision rights. Static matching, temporal communication, and endogenous selection are distinct obligations.

Accountability with partially observed assumptions. Section 43 closes specified finite inspection, disclosure, appeal, and stopping models. It does not identify a complete external-effects boundary, establish every observation law, or solve strategic collusion among the auditor and its verifier. The guarded certificate theorem requires sound action-interval assumptions. Characterizing useful guarantees when those assumptions can only be partially checked remains a distinct problem.

Preserving access while admitting a changing workload. The finite appeal criterion fixes admitted jobs and reserved capacity. A broader contract must specify which requests must be admitted, how new work changes response guarantees, and which resource changes can occur during settlement. Indefinite access, finite capacity, and bounded response cannot be asserted together without a workload model.

46 Limits of the framework

Most of the analysis is built from finite state spaces, specified observation maps, and admitted control families. Section 24 explicitly changes the domain to unrestricted programs with unbounded memory; its impossibility result must not be read as a claim about the earlier finite models. It does not automatically extend to open-ended deployed systems with unbounded memory, unknown environment dynamics, arbitrary strategic adversaries, or unresolved questions of computability and measure. Some finite statements generalize under regularity conditions; others require new machinery. Even an exact mathematical abstraction only preserves those behaviors and interventions specified by its contract.

Institutional formation is not a consequence of the existence of an institutional representation. An empty constitution can be formally represented and never adopted. A stable rule can arise through designer imposition, agent agreement, or accidental coordination, and the same observable behavior may be compatible with several mechanisms. Theorems about cost, potential functions, viability, and compatibility establish conditional structure, not a law that sufficiently numerous agents must create societies.

The framework also cannot derive a unique moral constitution from computational efficiency. Protected rights, acceptable tradeoffs, and the authority to enact constraints require human or otherwise explicitly specified normative commitments. The mathematical distinction between permitted and forbidden transitions is not a decision about which permissions ought to exist. The running resource-allocation and parity models are useful because their assumptions are inspectable. They are not claimed to cover the full complexity of economic, social, or legal institutions.

Finally, a theoretical description of cultural inheritance or organizational continuity is neither a test for subjective experience nor a basis for legal status. The purpose of these exclusions is to leave a precise problem: how persistent rules, relationships, and historical obligations can be represented and transformed within ordinary computation.

46.1 What the combined certificates do and do not compose

The state, interpreter, epistemic relations, and authority registry can be carried by one realized system without becoming one undifferentiated object. Language transports compose through exact admission and event preservation. Epistemic transports compose through matched predicates and the back-and-forth conditions. Typed residuals compose only with their declared metric compatibility. Frame synchronization is controlled by ordered loop products, while state compatibility depends on fixed points. Amendment normal forms require a separate termination and joinability argument.

Section 42 strengthens this coexistence requirement into a common-policy question. Under its exact information, action, label, and successor conditions, a governance abstraction preserves both the safe continuation kernel and finite progress ranks. Approximate metrics alone do not imply this sure strategy equivalence. A supplied certificate must identify the common realization and action catalogue to which all its conditions apply.

The joint lesson is that institutional coherence is a bundle of related obligations, not a scalar score. No single condition above implies all the others. Their explicit coexistence in one contract is what allows a claim about preservation of a society's operations to be evaluated without turning its language, knowledge, history, and authority into interchangeable metaphors.

47 Conclusion

A machine society is not defined by the number of its models, the fluency of its conversations, or one impressive collective output. Its theoretical possibility is persistent plurality organized by operative institutions: decision centers maintain rules, records, and procedures that shape their interaction and can survive changes of participants. The same population may change its membership, the meaning of its communicative acts, the questions it recognizes, and the division of work through which it acts.

These developments remain ordinary computation, but require distinct explanations. Realization and abstraction determine which higher-level states faithfully describe a process. Institutional semantics determines the actions, permissions, and obligations affected. Epistemic structure determines what participants know. Governance determines who may change the rules. Provenance and founding assumptions locate the boundaries of authority that internal consistency alone cannot settle.

The implementation of a continuing institution adds three coupled questions: what events are committed, why participants comply, and which slow dynamics remain meaningful amid fast activity. Convergent

records need not be valid allocations. Observation-based sanctions have a finite discrimination capacity, and compression cannot increase it. A uniform mixing bound can justify a review-epoch institutional approximation, but a data-selected review time can invalidate that approximation. The robust sanction bound shows how a declared modeling error changes an incentive guarantee rather than disappearing inside a convenient macro-description.

Persistence likewise does not imply learning or resilience. Rational actors can enter a wrong information cascade when only decisions are disclosed. Independent evidence can support continuing accuracy gains under a different information contract. A collection of specialists can survive turnover when its authorized capability graph has sufficient collective matching slack, yet a backup for each role need not suffice. A connected aggregate graph can lack the time-respecting route needed to carry evidence before a deadline.

The earlier results fit the same pattern. A beneficial institution can face a formation barrier; a safe full-information action can be unavailable to distributed decision makers; locally invertible interfaces can fail to synchronize; and an invariant local state need not establish a full common reference frame. Amendment normal forms, cultural closure, and unrestricted verification each have their own assumptions and limits. No one certificate supplies all the others.

The final synthesis makes the quantifier issue explicit. A different valid policy for every requirement is not one valid policy for their conjunction. Multiple deviations can defeat a common monitor even when each has a successful individual test. An audit and three due services can each be affordable while their joint cost exceeds the available slots. A safe amendment can remove every safe continuation. The finite continuation construction identifies such failures, while the exact abstraction theorem states when a smaller institutional representation preserves the existence of a solution. The resulting witnesses certify declared operations, not the origin, legitimacy, or spontaneous discovery of the institution.

Accountability adds a final distinction. An institution can remain operational while losing the justified use of a certificate, the ability to fund an appeal, or the capacity to honor an authorized ending. The sharp disclosure bound, strategic-inspection model, finite appeal criterion, ending-preserving kernel, and guarded handover rule specify parts of that problem. Their premises remain visible. A model of internal members does not itself include every affected outsider, and a settled record does not manufacture consent or reverse an irreversible external effect.

The positive thesis is that persistent, revisable relations among computational decision centers are coherent mathematical objects. Their formation, preservation, semantic change, disagreement, incentives, and failures can be studied within one realization framework. Their ability to continue is different from their ability to remain answerable for their actions. A complete declared contract must preserve both the operations it promises and the authority, evidence, resources, and ending conditions on which those promises depend. This is neither an argument that machines must form societies nor a criterion for consciousness or moral status. It is a theory of what computational social organization would have to realize, preserve, and acknowledge as unresolved.

A The commons as an integrated preservation contract

The running example in Section 3 uses one institutional realization throughout. Its claims can be separated without losing their connection.

Question	Object and obligation
Can an institution be adopted?	The coordination-game transition barrier is distinct from the fact that R and U have executable implementations. A profitable joint commitment requires an admitted commitment operation.
What state must persist?	The allocation rule alone is insufficient. Governance preservation also retains the enabled amendment actions and labels for active credentials, service debt, and resource tokens.
Can a due debt be erased?	The obligation monitor distinguishes fulfillment, authorized discharge, and deletion. Amendments are required to preserve pending commitments.
Who can block change?	Two registered offices block a three-of-four rule under independent approval rights. A hidden administrator changes this calculation.
Can every request be safely served?	Resource conservation can hold even when private information or conflicting deadlines prevent timely service. Viability with full information is not decentralized implementability.
What crosses a replacement?	An authorized office handover preserves debt and credential semantics. Copying a model does not duplicate an office's budget or approval right.
Can two jurisdictions merge?	Their translator must preserve the agreed action semantics, permissions, obligations, resources, and evidence coordinates. Nonzero residuals remain visible until an authorized repair is found.
How long does the organization last?	Lifetime is first exit from a declared operational contract under a specified transition law. It is neither occupant longevity nor mere archive retention.

The common event mechanism and constitutional envelope are supplied. Within that envelope, offices can choose rules, incur obligations, revise decisions, and replace occupants. This demonstrates how the theory's objects fit together. It does not prove that the envelope itself arose endogenously or that these rules are uniquely desirable.

A.1 The additional boundary, language, and knowledge contract

An admission changes the membership configuration, not merely the occupant list. A message is interpreted using its type, issuer, and epoch. A resource approval creates the specified ledger event only after its admissibility check. A debtor replacement transports the obligation monitor but does not manufacture new approval rights. A notification history records which office has received which adoption record; an available ledger entry is not automatically common knowledge.

For concurrent amendments, the contract states whether ordering is significant, whether a joinable normal form is required, and which external effects remain visible. For federation, the role-label transformations must be checked around ordered loops. A compatible allocation on one invariant subset does not certify synchronization of every role. For inquiry, the commons records unresolved scheduling questions independently of which questions it chooses to fund. For founding authority, it preserves the resource owner's stipulated grant separately from its own subsequent votes.

Together these clauses specify what it would mean to preserve this analytical institution across translations and replacements. They do not claim that every useful society needs this exact architecture or that the four-office example exhibits autonomous institutional formation.

A.2 Distributed realization, enforcement, and adaptive staffing

In the atomic reference commons, a unique slot token prevents two successful grants of the same slot. In its replicated variant, two local approvals can each look valid while their merged grant set violates exclusivity. The institution must therefore distinguish a provisional reservation from a committed grant and identify the primitive that prevents double execution. A convergent archive is not that primitive by itself.

Accurate urgency reporting is a different problem. A committed monitor can discourage a specified profitable misreport only within the discrimination and penalty budget of Section 18. Its false-positive limit, evidence access, and remedy authority remain separate. When an averaged model supplies the observation laws, the margin must include its approximation error. A fixed review schedule and a data-selected review schedule cannot share the same mixing certificate without further work.

The commons also distinguishes an urgency vote from the signal supporting it. Decision-only publication can reproduce a mistaken cascade; preservation of independent evidence changes the information available to later reviewers, not the legal effect of an already committed vote. Finally, creating a specialist verifier changes the capability graph and the distribution of authority. A staffing certificate covers all protected roles jointly, while the handover procedure must still deliver credentials, pending debts, and evidence before service deadlines. These clauses supplement the reference semantics rather than claim that the society has generated its own infrastructure without assumptions.

A.3 The joint continuation contract

The bounded audit-and-service variant in Section 42 combines three protections that cannot be checked in isolation: the common monitor must deter every listed deviation, its false-positive rate must meet the cap, and its resource use must leave enough capacity for all due work. The current public possibility set must also admit the same protocol in every compatible complete state. A continuation certificate records this common choice and all possible successors. A progress rank distinguishes eventual certified service from indefinite safe waiting.

An amendment is acceptable under this particular continuation contract only when its current authority is valid and its resulting states retain certified continuations. A transport certificate additionally checks public observations, enabled profiles, safe and goal labels, and successor sets. Replacing an office, changing an audit channel, or adding a slot changes specified parts of that contract. Such a change may repair an obstruction, but does not retroactively fulfill an earlier unmet promise.

A.4 The accountability extension

Section 43.9 deliberately changes the audit catalogue to satisfy its pairwise privacy constraint. It separately includes an outside claim, an admitted appeal, a genuine-execution fee, and a funded final record with revocation. The resulting eight-unit settlement requirement must not be confused with the five-unit service-and-audit variant. The supplied reference trace checks one declared five-profile settlement, while the stopping theorem characterizes an entire finite ending-preserving running kernel. Both are model-relative results, not certificates for an unspecified deployment.

B A finite algebraic example of reusable state

The following analytical example makes the preceding distinctions concrete. Four components initially hold independent fair bits (a, b, c, d). Two local workspaces expose

$$p = a \oplus b, \qquad q = c \oplus d,$$

and a global workspace retains $Z = (p, q)$. The current reported answer is $r = p \oplus q$.

No proper subset of the four raw bits determines r: at least one independent fair bit is missing. Each local pair determines its own workspace bit but has zero information about r. Together the workspaces determine r exactly. This is a precise distributed-access property for the initial distribution.

B.1 Why the current answer is not enough state

Permit three public operations on Z:

$$I(p,q) = (p,q), \qquad W(p,q) = (q,p), \qquad S(p,q) = (p \oplus q, q).$$

The global state supports these operations and the query r. The operations W and S do not commute. For instance, from $(p,q) = (1,0)$,

$$W(S(1,0)) = (0,1), \qquad S(W(1,0)) = (1,1).$$

Their reported parity outcomes are different.

By contrast, retaining only the current answer r fails the quotient criterion. The states $(0,0)$ and $(1,1)$ both report $r = 0$, but after S they report 0 and 1, respectively. The missing distinction is exactly a distinction relevant to a future operation. A compressed answer is not necessarily a compressed process.

p	q	Current r	State after S	New r
0	0	0	$(0,0)$	0
0	1	1	$(1,1)$	0
1	0	1	$(1,0)$	1
1	1	0	$(0,1)$	1

The full two-bit state has four future-distinguishable possibilities: the current parity reveals $p \oplus q$, and parity after S reveals p. Together those outputs determine q as well. Thus two bits are necessary for an exact reusable state under this query family, even though one bit suffices for the immediate answer.

B.2 A deterministic microscopic realization

For any permitted macro-operation T, write $(p', q') = T(p,q)$ and define

$$F_T(a,b,c,d) = (a,\ a \oplus p',\ c,\ c \oplus q').$$

Then the two local XORs of $F_T(a,b,c,d)$ are exactly (p', q'). Consequently $\pi F_T = T\pi$ for every microstate, where $\pi(a,b,c,d) = (p,q)$.

This formula specifies a global transition for verification. A distributed implementation still needs communication to execute operations that mix p and q. The formula alone does not prove locality, privacy against transcript observers, or the spontaneous discovery of a protocol. It shows exactly which macrostate the implementation realizes.

B.3 State loss depends on the access contract

Suppose the leaf bits become inaccessible after export and the only surviving pair-level records are p and q. Erasing q reduces the best prediction of r to chance, $1/2$. Replacing it with a fixed bit has the same average accuracy. If the original leaves c, d remain readable, however, the missing state can be reconstructed and accuracy returns to 1.

Both outcomes are correct. They differ because the allowed information differs. A "workspace ablation" that silently permits re-reading the original evidence does not test the same necessity claim as one that removes every permitted carrier. A state can also be redundant yet causally used: successful repair does not imply that the removed component never mattered.

B.4 What this example establishes

The example establishes exact distributed initial information, a reusable two-bit macrostate, deterministic macro-closure, noncommuting update semantics, and an access-relative memory requirement. It also supplies a negative control: the one-bit answer statistic fails dynamic sufficiency. It does not contain learning, endogenous institutions, self-selected goals, or a society. Its finite state space can be enumerated exactly; the general claims rest on the arguments given above.

C A compact glossary of exclusions

Term	Meaning, with excluded inference
Relational state	A useful statistic or record of joint interaction. It is not information outside the complete implementation.
Dynamic closure	A well-defined reduced transition law under specified controls. It does not follow from correlated outputs.
Institution	Operative norms, permissions, enforcement or revision procedures, and institutional memory. A policy document alone is insufficient.
Endogenous formation	Interaction-dependent selection or revision relative to a supplied scaffold. Not creation without prerequisites.
Society-like population	Persistent plural decision centers with operative institutions. Not necessarily a single collective policy.
Collective agency	A useful group-level decision interface for a specified task family. It is neither necessary nor sufficient for society-like plurality.
Cultural inheritance	Transfer of future-relevant methods, practices, and correction policies. Not mere repetition of predecessor statements.
Organizational continuity	Preservation of declared operations and lineage under changes. Not subjective identity or automatic transfer of authority.
Compatibility	Agreement of relevant representations, operations, and global realizability. Pairwise consistency may be too weak.
Strategic deception	A supported strategic mechanism for inducing inaccurate beliefs. Not every wrong statement or representation mismatch.
Safety certificate	A guarantee relative to formal assumptions, transitions, and invariants. Not proof of moral legitimacy or liveness.
Emergence	An interaction-dependent organizational property relative to a boundary and comparator. Not a claim of extra computational power or consciousness.

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